

## Lecture 1 — Introduction

### Introduction

This course studies Classical Mechanics and its application to aerospace systems. Particle motion in Classical Mechanics is governed by Newton's laws and is sometimes referred to as Newtonian Mechanics. The motion of extended rigid bodies is analyzed by application of Newton's law to multi-particle systems. These laws are empirical in that they combine observations from nature and some intuitive concepts. Newton's laws of motion are not self evident. For instance, in Aristotelian mechanics before Newton, a force was thought to be required in order to maintain motion. Much of the foundation for Newtonian mechanics was laid by Galileo at the end of the 16th century. Newton, in the middle of the 17th century, stated the laws of motion in the form known and used today. Shortly thereafter, he formulated the law of universal attraction. This led to a complete theory with which he was able to explain many observed phenomena, in particular the motion of the planets. Nevertheless, these laws still left many unanswered questions at that time, and it was not until later years that the principles of classical mechanics were deeply studied and rationalized. In the eighteenth century, there were many contributions in this direction, such as the principle of virtual work by Bernoulli, D'Alembert's principle and the theory of rigid body dynamics developed by Euler. In the nineteenth century, Lagrange and later Poisson, Hamilton and Jacobi developed the so called analytical or rational mechanics and gave the theory of Newtonian mechanics a much richer mathematical structure. Classical Mechanics has limitations and breaks down where more modern theories such as relativity and quantum mechanics, developed in the twentieth century, are successful. Newtonian mechanics breaks down for systems moving at speeds comparable with the speed of light, and also fails for systems of dimensions comparable to the size of the atom. Nevertheless, for practical engineering applications, Newtonian mechanics provides a good model of reality. In fact, it is difficult to find examples in aerospace where Newtonian mechanics is not adequate. The most notable perhaps are the relativistic corrections that need to be made for modeling satellite communications.

### Particles, Rigid Bodies and “Real Bodies”

In this course real bodies will be idealized either as particles or as rigid bodies.

A *particle* is a body which has mass, but has negligible dimensions. When the dimensions of a body are unimportant to the description of its motion, it may be idealized as a particle.

A *rigid body* is a body that has a finite size but does not deform. This is a useful approximation when the deformation of a body is negligible compared to the overall motion. For instance, an aircraft may be approximated as a rigid body when considering the behavior of the aircraft along its flight path, even though under some specific conditions the deflection of the wing tips may be considerable. In describing the motion of a rigid body, one must specify not only its position but also its orientation.

Finally, real bodies have a finite size and are always deformable under loading. In some situations it will be required to consider the deformation of the body when considering its dynamic behavior, but this is outside the scope of this course, except for an introductory treatment of vibration and waves.

## Newton's Laws

Newton's three laws of motion are:

1. *A particle in isolation moves with constant velocity.*

A particle in isolation means that the particle does not interact with any other particle. Constant velocity means that the particle moves along a straight line with constant speed. This includes being at rest. It turns out that the motion (e.g. velocity and acceleration) observed depends on the reference frame used. Therefore, the above law cannot be verified in all reference frames. The reference frames for which this law is satisfied are called *inertial reference frames*. In some sense, Newton's first law postulates that inertial reference frames exist.

2. *The acceleration of a particle relative to an inertial reference frame is equal to the force per unit mass applied to the particle.*

In other words, if  $\mathbf{F}$  represents the (vector) sum of all forces acting on a particle of mass  $m$ , any inertial observer will see that the particle has an acceleration  $\mathbf{a}$  which is given by,

$$\mathbf{F} = m\mathbf{a} . \tag{1}$$

This equation introduces two new concepts: force and mass. Precise definitions for these concepts are not easy even though everyone has some intuition about both force and mass. Forces result when bodies interact. Forces are vectors with magnitude and direction. They are measured by comparison, e.g. to the weight of a standard mass, or to the deformation of a spring. It is assumed that the concept of force is absolute and does not depend on the observer. Once force is defined, mass can be defined as the constant of proportionality between force and acceleration. Mass is a scalar quantity. One could think of mass as the resistance of bodies to a change in motion. That is, a given force applied to a body with small mass will produce a large acceleration, whereas the same force on a body of large mass will produce a small acceleration.

Equation (10) is a vector equation. This means that the force and the acceleration always have the same direction and the ratio of their magnitudes is  $m$ . It turns out that this equation is the basis for most engineering dynamics.

3. *The forces of action and reaction between interacting bodies are equal in magnitude and opposite in direction*

This law makes explicit the fact that a force is the result of interaction between bodies. This law is clearly satisfied when the bodies are in contact and in static equilibrium. The situation for bodies in motion interacting at a distance, e.g. electromagnetic or gravity interactions, is a little bit more complicated. Electromagnetic signals travel at a finite speed and therefore there is a time delay whenever two bodies interact at a distance. Unfortunately, Newton did not foresee such a situation, and in these cases, Newton's third law breaks down. For practical purposes and for most engineering applications, however, the error made by assuming that these interactions are instantaneous, and, hence, assuming that Newton's third law is applicable, is negligible.

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**Note**

**Units**

Two systems of units are in common usage: the *International System*, also called SI, and the *English System*. The international system is the most widely used system for science and engineering. The English system, however, is still in widespread use within the United States engineering community. The SI units are *meter* (m), *kilogram* (kg), and *second* (s) for length, mass and time, respectively. Acceleration is measured in  $\text{m/s}^2$ , and force is measured in  $\text{kg}\cdot\text{m}/\text{s}^2$ , which is also called a *Newton* (N) i.e.  $1\text{N} = 1\text{Kg}\cdot\text{m}/\text{s}^2$ . In the English system, the units of length, mass and time are *foot* (ft), *slug*, and *second* (s), respectively. Acceleration is measured in  $\text{ft/s}^2$ , and force is measured in  $\text{slug}\cdot\text{ft}/\text{s}^2$ , also called *pound* (lb), i.e.  $1\text{pound} = 1\text{slug}\cdot\text{ft}/\text{s}^2$ . The following are the conversion factors:

$$\begin{aligned} 1 \text{ ft} &= 0.348 \text{ m} \\ 1 \text{ slug} &= 14.5 \text{ kg} \\ 1 \text{ N} &= 0.224 \text{ lb} \end{aligned}$$


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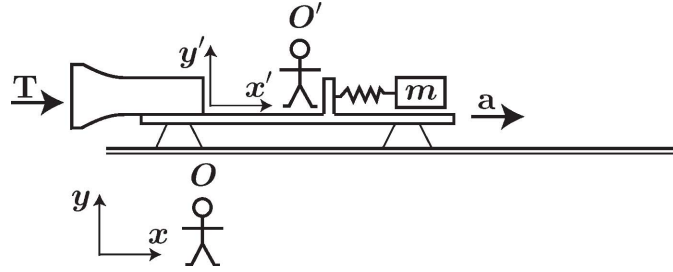
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**Example**

**Inertial vs. Non-inertial observers**

This example is meant to illustrate the fact that Newton's second law is not necessarily satisfied for accelerating, non-inertial observers.

Consider a rocket sled which can move on a horizontal track as shown. An inertial observer  $O$  is fixed on the ground and an observer  $O'$  is on the sled. An accelerometer is mounted on the sled. This consists of a known proof mass  $m$ , whose horizontal motion relative to the sled is constrained by a spring. Assume friction between the mass and the sled is negligible. This means that the only mechanism to exert a horizontal force on the mass is through the spring.



Consider two situations:

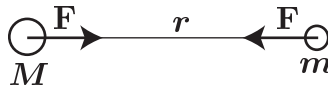
- The engine is off,  $\mathbf{T} = \mathbf{0}$ ,  $\mathbf{a} = 0$ , the mass is at rest and the spring is uncompressed. Both observers  $O$  and  $O'$  agree in all their measurements.
- The engine is on,  $\mathbf{T}$  is a constant non-zero force, the sled has an acceleration  $\mathbf{a}$ , and the spring is compressed and exerting a force  $\mathbf{F}$  on the mass. Both observers are able to measure the force exerted by the spring by reading on a scale how much the spring deforms. For observer  $O'$ , the mass is not moving (assume that the initial transient oscillation is gone and the spring has settled to a fixed position relative to the sled). For the observer on the ground the mass is accelerating with acceleration  $\mathbf{a}$ . Since observer  $O$  is an inertial observer, he or she is able to verify that  $\mathbf{F} = m\mathbf{a}$ . On the other hand, the observer on the sled,  $O'$ , measures a force  $\mathbf{F}$ , but observes a zero acceleration. Hence, the observations of observer  $O'$  do not satisfy Newton's second law.

## Law of Universal Attraction

The law of universal attraction was proposed by Newton shortly after formulating the laws of motion. The law postulates that the force of attraction between any two particles, of masses  $M$  and  $m$ , has a magnitude,  $F$ , given by

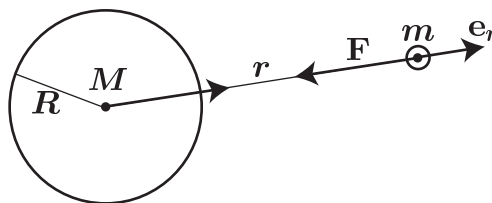
$$F = G \frac{Mm}{r^2} \quad (2)$$

where  $r$  is the distance between the two particles, and  $G = 6.673(10^{-11}) \text{ m}^3/(\text{kg} \cdot \text{s}^2)$  is the universal constant of gravitation which is determined according to experimental evidence. The direction of the force is parallel to the line connecting the two particles.



The law of gravitation stated above is strictly valid for point masses. One would expect that when the size of the masses is comparable to the distance between the masses one would observe deviations to the above law. It turns out that if the mass  $M$  is distributed uniformly over a sphere of radius  $R$ , the force on a mass  $m$ , *outside*  $M$ , is still given by (2), with  $r$  being measured from the sphere's center.





## Weight

The gravitational attraction from the earth to any particle located near the surface of the earth is called the weight. Thus, the weight,  $\mathbf{W}$ , of a particle of mass  $m$  at sea level is given by

$$\mathbf{W} = -G \frac{M_e m}{R_e^2} \mathbf{e}_r = -g_0 m \mathbf{e}_r = m \mathbf{g}_0 .$$

Here,  $M_e \approx 5.976 \times 10^{24}$  kg and  $R_e \approx 6.371 \times 10^6$  m, are the mass and radius of the earth, respectively, and  $\mathbf{g}_0 = -(GM_e/R_e^2) \mathbf{e}_r$ , is the *gravitational* acceleration vector at sea level. The average value of its magnitude is  $g_0 = 9.825$  m/s<sup>2</sup>.

The variation of the gravitational attraction with altitude is easily determined from the gravitational law. Thus, the weight at an altitude  $h$  above sea level is given by

$$\mathbf{W} = -G \frac{M_e m}{(R_e + h)^2} \mathbf{e}_r = -g_0 \frac{R_e^2}{(R_e + h)^2} m \mathbf{e}_r = m \frac{R_e^2}{(R_e + h)^2} \mathbf{g}_0 .$$

It turns out that the earth is not quite spherical and so the weight does not exactly obey the inverse-squared law. The magnitude of the gravitational acceleration,  $g_0$ , at the poles and at the equator, is slightly different. In addition, the earth is also rotating. This introduces an inertial centrifugal force which has the effect of reducing the vertical component of the weight. These effects will be revisited later in the course.

## References

- [1] R. Dugas, *A History of Mechanics*, Dover, 1988.

## ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition  
1/1, 1/2, 1/3, 1/4, 1/5 (Effect or Altitude only), 1/6, 1/7, 3/2

## Lecture 2 — Vectors

Physical laws can often be written in a simple form using vectors and vector operations. Since Dynamics makes extensive use of vectors, it is useful to review some of their important properties.

### Vectors

For most purposes one can think of a *vector* as a mathematical representation of a physical entity which has both *magnitude* and *direction*. Examples of physical vectors are forces, moments, and velocities. Geometrically, a vector can be represented as an arrow. The length of the arrow represents its magnitude.

Unless indicated otherwise, it is assumed that parallel translation does not change a vector. Vectors satisfying this property are called *free vectors*. Thus, two vectors are equal if and only if they point in the same direction and have equal length.

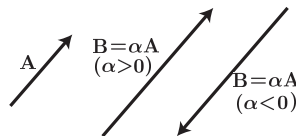
Vectors are usually typed in boldface and scalar quantities appear in lightface italic type, e.g. the vector quantity  $\mathbf{A}$  has magnitude, or modulus,  $A = |\mathbf{A}|$ . In handwritten text, vectors are often expressed using the arrow, or underbar notation, e.g.  $\vec{A}$ ,  $\underline{A}$ .

### Vector Algebra

Here is an overview of useful operations which are defined for free vectors.

#### Multiplication by a scalar

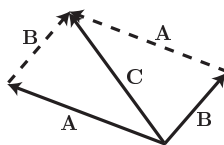
Multiplying a vector  $\mathbf{A}$  by a scalar  $\alpha$ , results in a vector  $\mathbf{B} = \alpha\mathbf{A}$ , which has magnitude  $B = |\alpha|A$ . The vector  $\mathbf{B}$  is parallel to  $\mathbf{A}$  and points in the same direction if  $\alpha > 0$ . For  $\alpha < 0$ , the vector  $\mathbf{B}$  is parallel to  $\mathbf{A}$  but points in the opposite direction (antiparallel).



Multiplying an arbitrary vector,  $\mathbf{A}$ , by the inverse of its magnitude,  $(1/A)$ , results in a *unit vector* which is parallel to  $\mathbf{A}$ . There exist several common notations to denote a unit vector, e.g.  $\hat{\mathbf{A}}$ ,  $\mathbf{e}_A$ , etc. Thus,  $\hat{\mathbf{A}} = \mathbf{A}/A = \mathbf{A}/|\mathbf{A}|$ , and  $\mathbf{A} = A \hat{\mathbf{A}}$ ,  $|\hat{\mathbf{A}}| = 1$ .

## Vector addition

Vector addition has a simple geometrical interpretation. To add vector  $\mathbf{B}$  to vector  $\mathbf{A}$ , place the tail of  $\mathbf{B}$  at the head of  $\mathbf{A}$ . The sum is a vector  $\mathbf{C}$  from the tail of  $\mathbf{A}$  to the head of  $\mathbf{B}$ . Thus,  $\mathbf{C} = \mathbf{A} + \mathbf{B}$ . The same result is obtained if the roles of  $\mathbf{A}$  and  $\mathbf{B}$  are reversed. That is,  $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ . This commutative property is illustrated below with the parallelogram construction.

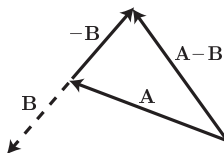


It can easily be verified that vector sum has the property of association, that is,

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}).$$

## Vector subtraction

Since  $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$ , subtracting  $\mathbf{B}$  from  $\mathbf{A}$ , is equivalent to multiplying  $\mathbf{B}$  by  $-1$  and then adding.

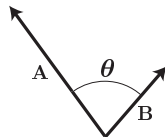


## Scalar product (Dot product)

The scalar product involves two vectors and results in a scalar quantity. The scalar product between two vectors  $\mathbf{A}$  and  $\mathbf{B}$ , is denoted by  $\mathbf{A} \cdot \mathbf{B}$ , and is defined as

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta.$$

Here  $\theta$ , is the angle between the vectors  $\mathbf{A}$  and  $\mathbf{B}$  when they are drawn with a common origin.



Note that since  $\cos \theta = \cos(-\theta)$ , it makes no difference which vector is considered first when measuring the angle  $\theta$ . Hence,  $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$ . If  $\mathbf{A} \cdot \mathbf{B} = 0$ , then either  $A = 0$  and/or  $B = 0$ , or,  $\mathbf{A}$  and  $\mathbf{B}$  are orthogonal, that is,  $\cos \theta = 0$ . Also note that  $\mathbf{A} \cdot \mathbf{A} = A^2$ . If one of the vectors is a unit vector, say  $B = 1$ , then  $\mathbf{A} \cdot \hat{\mathbf{B}} = A \cos \theta$ , is the projection of vector  $\mathbf{A}$  along the direction of  $\hat{\mathbf{B}}$ .

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**Exercise**

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Using the definition of scalar product, derive the Law of Cosines which says that, for an arbitrary triangle with sides of length  $A$ ,  $B$ , and  $C$ ,

$$C^2 = A^2 + B^2 - 2AB \cos \theta.$$

Here,  $\theta$  is the angle opposite side  $C$ . *Hint : associate to each side of the triangle a vector such that  $\mathbf{C} = \mathbf{A} - \mathbf{B}$ , and expand  $C^2 = \mathbf{C} \cdot \mathbf{C}$ .*

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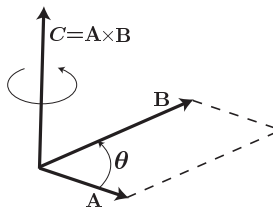
**Vector product (Cross product)**

The vector product operation involves two vectors  $\mathbf{A}$  and  $\mathbf{B}$ , and results in a new vector  $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ . The magnitude of  $\mathbf{C}$  is given by

$$C = AB \sin \theta ,$$

where  $\theta$  is the angle between the vectors  $\mathbf{A}$  and  $\mathbf{B}$  when drawn with a common origin. To eliminate ambiguity between the two possible choices,  $\theta$  is always taken as the angle smaller than  $\pi$ . One can easily show that  $C$  is equal to the area enclosed by the parallelogram defined by  $\mathbf{A}$  and  $\mathbf{B}$ .

The vector  $\mathbf{C}$  is orthogonal to both  $\mathbf{A}$  and  $\mathbf{B}$ , i.e. it is orthogonal to the plane defined by  $\mathbf{A}$  and  $\mathbf{B}$ . The direction of  $\mathbf{C}$  is determined by the *right-hand* rule as shown.



From this definition, it follows that

$$\mathbf{B} \times \mathbf{A} = -\mathbf{A} \times \mathbf{B} ,$$

which indicates that vector multiplication is *not* commutative (but anticommutative). If  $\mathbf{A} \times \mathbf{B} = \mathbf{0}$ , then either  $\mathbf{A}$  and/or  $\mathbf{B}$  are zero, or  $\mathbf{A}$  and  $\mathbf{B}$  are parallel, although not necessarily pointing in the same direction. Thus,  $\mathbf{A} \times \mathbf{A} = \mathbf{0}$ .

Having defined vector multiplication, it might appear natural to define vector division. One might try to define “ $\mathbf{A}$  divided by  $\mathbf{B}$ ”, as a vector  $\mathbf{C}$  such that  $\mathbf{A} = \mathbf{B} \times \mathbf{C}$ . There are a number of immediate difficulties with this definition. In particular, if  $\mathbf{A}$  is not perpendicular to  $\mathbf{B}$ , the vector  $\mathbf{C}$  does not exist. Moreover, if  $\mathbf{A}$  is perpendicular to  $\mathbf{B}$ , then there are an infinite number of vectors that satisfy  $\mathbf{A} = \mathbf{B} \times \mathbf{C}$ . To see that, assume that  $\mathbf{C}$  satisfies  $\mathbf{A} = \mathbf{B} \times \mathbf{C}$ . Then, any vector  $\mathbf{D} = \mathbf{C} + \beta \mathbf{B}$ , for any scalar  $\beta$ , also satisfies  $\mathbf{A} = \mathbf{B} \times \mathbf{D}$ , since  $\mathbf{B} \times \mathbf{D} = \mathbf{B} \times (\mathbf{C} + \beta \mathbf{B}) = \mathbf{B} \times \mathbf{C} = \mathbf{A}$ . Vector division, therefore, is not a well defined operation.

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**Exercise**

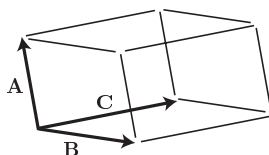
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Show that  $|\mathbf{A} \times \mathbf{B}|$  is the area of the parallelogram defined by the vectors  $\mathbf{A}$  and  $\mathbf{B}$ , when drawn with a common origin.

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**Triple product**

Given three vectors  $\mathbf{A}$ ,  $\mathbf{B}$ , and  $\mathbf{C}$ , the triple product is a scalar given by  $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ . Geometrically, the triple product can be interpreted as the volume of the three dimensional parallelepiped defined by the three vectors  $\mathbf{A}$ ,  $\mathbf{B}$  and  $\mathbf{C}$ .



It can be easily verified that  $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$ .

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**Exercise**

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Show that  $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$  is the volume of the parallelepiped defined by the vectors  $\mathbf{A}$ ,  $\mathbf{B}$ , and  $\mathbf{C}$ , when drawn with a common origin.

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**Double vector product**

The double vector product results from repetition of the cross product operation. A useful identity here is,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}.$$

Using this identity it is easily verified that the double cross product is not associative, that is,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}.$$

**Vector Calculus**

Vector differentiation and integration follow standard rules. If a vector is a function of, say time, then its derivative with respect to time is also a vector. Similarly, the integral of a vector is a vector.

**Derivative of a vector**

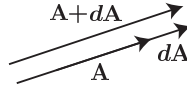
Consider a vector  $\mathbf{A}(t)$  which is a function of, say, time. The derivative of  $\mathbf{A}$  with respect to time is defined as,

$$\frac{d\mathbf{A}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{A}(t + \Delta t) - \mathbf{A}(t)}{\Delta t}. \quad (1)$$

A vector has magnitude and direction, and it changes whenever either of them changes. Therefore the rate of change of a vector will be equal to the sum of the changes due to magnitude and direction.

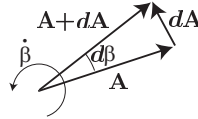
### Rate of change due to magnitude changes

When a vector only changes in magnitude from  $\mathbf{A}$  to  $\mathbf{A} + d\mathbf{A}$ , the rate of change vector  $d\mathbf{A}$  is clearly parallel to the original vector  $\mathbf{A}$ .



### Rate of change due to direction changes

Consider the situation where only the direction of the vector changes, while the magnitude stays constant. This is illustrated in the figure where a vector  $\mathbf{A}$  undergoes a small rotation. From the sketch, it is clear that if the magnitude of the vector does not change,  $d\mathbf{A}$  is perpendicular to  $\mathbf{A}$  and as a consequence, the derivative of  $\mathbf{A}$  must be perpendicular to  $\mathbf{A}$ . (Note that in the picture  $d\mathbf{A}$  has a finite magnitude and therefore,  $\mathbf{A}$  and  $d\mathbf{A}$  are not exactly perpendicular. In reality,  $d\mathbf{A}$  has infinitesimal length and we can see that when the magnitude of  $d\mathbf{A}$  tends to zero,  $\mathbf{A}$  and  $d\mathbf{A}$  are indeed perpendicular).



An alternative, more mathematical, explanation can be derived by realizing that even if  $\mathbf{A}$  changes but its modulus stays constant, then the dot product of  $\mathbf{A}$  with itself is a constant and its derivative is therefore zero.  $\mathbf{A} \cdot \mathbf{A} = \text{constant}$ . Differentiating,

$$d\mathbf{A} \cdot \mathbf{A} + \mathbf{A} \cdot d\mathbf{A} = 2\mathbf{A} \cdot d\mathbf{A} = 0,$$

which shows that  $\mathbf{A}$ , and  $d\mathbf{A}$ , must be orthogonal.

Suppose that  $\mathbf{A}$  is instantaneously rotating in the plane of the paper at a rate  $\dot{\beta} = d\beta/dt$ , with no change in magnitude. In an instant  $dt$ ,  $\mathbf{A}$ , will rotate an amount  $d\beta = \dot{\beta}dt$  and the magnitude of  $d\mathbf{A}$ , will be

$$dA = |d\mathbf{A}| = A d\beta = A \dot{\beta} dt.$$

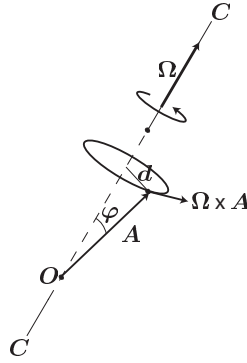
Hence, the magnitude of the vector derivative is

$$\left| \frac{d\mathbf{A}}{dt} \right| = A\dot{\beta}.$$

In the general three dimensional case, the situation is a bit more complicated because the rotation of the vector may occur around a general axis. The instantaneous rotation of  $\mathbf{A}$  can be expressed in terms of an angular velocity  $\boldsymbol{\Omega}$  (recall that the angular velocity vector is aligned with the axis of rotation and the direction of the rotation is determined by the right hand rule), then the derivative of  $\mathbf{A}$  with respect to time is simply,

$$\left( \frac{d\mathbf{A}}{dt} \right)_{\text{constant magnitude}} = \boldsymbol{\Omega} \times \mathbf{A}. \quad (2)$$

To see that, consider a vector  $\mathbf{A}$  rotating about the axis  $C - C'$  with an angular velocity  $\boldsymbol{\Omega}$ . The derivative will be the velocity of the tip of  $\mathbf{A}$ . Its magnitude is given by  $l\Omega$ , and its direction is both perpendicular to  $\mathbf{A}$  and to the axis of rotation. Note that  $\boldsymbol{\Omega} \times \mathbf{A}$  has the right direction, and the right magnitude since  $l = A \sin \varphi$ .

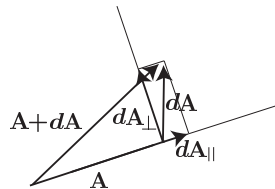


Expression (2) is also valid in the more general case where  $\mathbf{A}$  is rotating about an axis which does not pass through the origin of  $\mathbf{A}$ . A rotation about an arbitrary axis can always be written as a rotation about a parallel axis plus a translation, and translations do not affect the magnitude not the direction of a vector.

The general expression for the derivative of a vector (1) can be expressed as

$$\frac{d\mathbf{A}}{dt} = \left( \frac{d\mathbf{A}}{dt} \right)_{\text{constant direction}} + \left( \frac{d\mathbf{A}}{dt} \right)_{\text{constant magnitude}} = \left( \frac{d\mathbf{A}}{dt} \right)_{\text{constant direction}} + \boldsymbol{\Omega} \times \mathbf{A}.$$

Note that  $(d\mathbf{A}/dt)_{\text{constant direction}}$  is parallel to  $\mathbf{A}$  and  $\boldsymbol{\Omega} \times \mathbf{A}$  is orthogonal to  $\mathbf{A}$ . The figure below shows the general differential of a vector, which has a component which is parallel to  $\mathbf{A}$ ,  $d\mathbf{A}_{\parallel}$ , and a component which is orthogonal to  $\mathbf{A}$ ,  $d\mathbf{A}_{\perp}$ . The magnitude change is given by  $dA_{\parallel}$ , and the direction change is given by  $d\mathbf{A}_{\perp}$ .



## Rules for Vector Differentiation

Vector differentiation follows similar rules to scalars regarding vector addition, multiplication by a scalar, and products. In particular, for any vectors  $\mathbf{A}$ ,  $\mathbf{B}$ , and any scalar  $\alpha$ ,

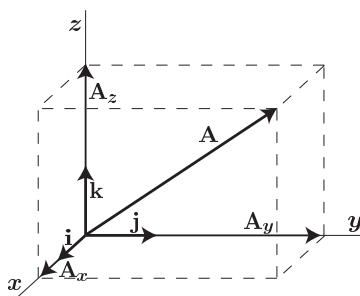
$$\begin{aligned} d(\alpha\mathbf{A}) &= d\alpha\mathbf{A} + \alpha d\mathbf{A} \\ d(\mathbf{A} + \mathbf{B}) &= d\mathbf{A} + d\mathbf{B} \\ d(\mathbf{A} \cdot \mathbf{B}) &= d\mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot d\mathbf{B} \\ d(\mathbf{A} \times \mathbf{B}) &= d\mathbf{A} \times \mathbf{B} + \mathbf{A} \times d\mathbf{B}. \end{aligned}$$

## Components of a Vector

As illustrated above, it is possible to define several operations involving vectors without ever introducing a reference frame. This is a rather important concept which explains why vectors and vector equations are so useful to express physical laws, since these, must be obviously independent of any particular frame of reference.

In practice however, reference frames need to be introduced at some point in order to express, or measure, the direction and magnitude of vectors.

Consider a *right-handed* set of axes  $xyz$ , defined by three mutually orthogonal unit vectors  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$  ( $\mathbf{i} \times \mathbf{j} = \mathbf{k}$ ). Since the vectors  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$  are mutually orthogonal they form a basis. The projections of  $\mathbf{A}$  along the three  $xyz$  axes are the *components* of  $\mathbf{A}$  in the  $xyz$  reference frame.



The components of  $\mathbf{A}$  can be written using the scalar product:

$$A_x = \mathbf{A} \cdot \mathbf{i}, \quad A_y = \mathbf{A} \cdot \mathbf{j}, \quad A_z = \mathbf{A} \cdot \mathbf{k}.$$

The vector  $\mathbf{A}$ , can thus be written as a sum of the three vectors along the coordinate axis which have magnitudes  $A_x$ ,  $A_y$ , and  $A_z$  and using matrix notation, as a column vector containing the component magnitudes.

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k} = \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}.$$



## Vector operations in component form

The vector operations introduced above can be expressed in terms of the vector components in a rather straightforward manner. For instance, if  $\mathbf{A} = \mathbf{B}$ , this implies that the projections of  $\mathbf{A}$  and  $\mathbf{B}$  along the  $xyz$  axes are the same, and therefore, this is equivalent to three scalar equations e.g.  $A_x = B_x$ ,  $A_y = B_y$ , and  $A_z = B_z$ . Regarding vector summation, subtraction and multiplication by a scalar, if  $\mathbf{C} = \alpha\mathbf{A} + \beta\mathbf{B}$ , then

$$C_x = \alpha A_x + \beta B_x, \quad C_y = \alpha A_y + \beta B_y, \quad C_z = \alpha A_z + \beta B_z.$$

## Scalar product

Since  $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1$  and  $\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0$ , the scalar product of two vectors can be written as

$$\mathbf{A} \cdot \mathbf{B} = (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}) \cdot (B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}) = A_xB_x + A_yB_y + A_zB_z.$$

Note that,  $\mathbf{A} \cdot \mathbf{A} = A^2 = A_x^2 + A_y^2 + A_z^2$ , which is consistent with Pythagoras' theorem.

## Vector product

Here,  $\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = \mathbf{0}$  and  $\mathbf{i} \times \mathbf{j} = \mathbf{k}$ ,  $\mathbf{j} \times \mathbf{k} = \mathbf{i}$ , and  $\mathbf{k} \times \mathbf{i} = \mathbf{j}$ . Thus,

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}) \times (B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}) \\ &= (A_yB_z - A_zB_y)\mathbf{i} + (A_zB_x - A_xB_z)\mathbf{j} + (A_xB_y - A_yB_x)\mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}. \end{aligned}$$

## Triple product

The triple product  $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$  can be expressed as the following determinant

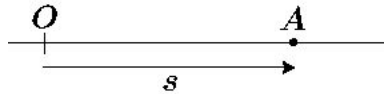
$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

which clearly is equal to zero whenever the vectors are linearly dependent (if the three vectors are linearly dependent they must be co-planar and therefore the parallelepiped defined by the three vectors has zero volume).

## Lecture 3 — Rectilinear Motion and Degrees of Freedom

### Rectilinear Motion

Rectilinear motion refers to the simple motion of a particle along a straight line. The position of particle  $A$  at any instant can be specified by the coordinate  $s$  with origin at some fixed point  $O$ .



The instantaneous velocity is

$$v = \frac{ds}{dt} = \dot{s} , \quad (1)$$

where the “dot” notation indicates a time derivative, i.e.  $\dot{()}\equiv d()/dt$ . Here, a positive  $v$  means that the particle is moving in the direction of increasing  $s$ , whereas a negative  $v$  indicates that the particle is moving in the opposite direction. The acceleration is

$$a = \frac{dv}{dt} = \dot{v} = \frac{d^2s}{dt^2} = \ddot{s} . \quad (2)$$

The above expression can be used to compute speed and acceleration if  $s$  and/or  $v$  are given as a function of  $t$ , i.e.  $s(t)$  and  $v(t)$ . More often, however, the acceleration is known, and the velocity and the position will have to be determined by integration.

### Determining velocity from acceleration

**From  $a(t)$**

If  $a(t)$  is given, then the velocity can be determined by simple integration of Equation (18),

$$v(t) = v_0 + \int_{t_0}^t a(t) dt . \quad (3)$$

Here,  $v_0$  is the velocity at time  $t_0$ , which is determined by the initial conditions.

**From  $a(v)$** 

If the acceleration  $a(v)$  is given as a function of velocity, then Equation (18) still applies, but in this case one can solve for the time as a function of velocity,

$$\frac{dv}{dt} = a = a(v) \quad (4)$$

$$\frac{dv}{a(v)} = dt \quad (5)$$

$$t(v) = t_0 + \int_{v_0}^v \frac{dv}{a(v)} . \quad (6)$$

Once the relationship  $t(v)$  has been obtained, one can in principle solve for the velocity to obtain  $v(t)$ . A typical example in which the acceleration is known as a function of velocity is when aerodynamic drag forces are present. Drag forces cause an acceleration which opposes the motion and is typically of the form  $a(v) \propto v^2$  (the sign “ $\propto$ ” means *proportional to*, that is,  $a(v) = \kappa v^2$  for some  $\kappa$  which is not a function of velocity).

**From  $a(s)$** 

When the acceleration is given as a function of  $s$ , a combination of Equations (17) and (18) can be used to solve the problem. From the chain rule,

$$a = \frac{dv}{dt} = \frac{dv}{ds} \frac{ds}{dt} = v \frac{dv}{ds}, \quad (7)$$

one obtains

$$a \, ds = v \, dv . \quad (8)$$

Integrating this equation

$$\int a \, ds = \int v \, dv \quad (9)$$

results in  $v$  as a function of  $s$ ,

$$v^2(s) = v_0^2 + 2 \int_{s_0}^s a(s) \, ds . \quad (10)$$

where  $v_0$  is the velocity of the particle at point  $s_0$ . This used the fact that

$$\int_{v_0}^v v \, dv = \frac{v^2}{2} - \frac{v_0^2}{2} .$$

A classic example of an acceleration dependent on the spatial coordinate  $s$  is that induced by a deformed linear spring. In this case, the acceleration is of the form  $a(s) \propto s$ .

when the acceleration is constant, any of the above expressions can be employed, resulting in

$$v = v_0 + a(t - t_0), \quad \text{or} \quad v^2 = v_0^2 + 2a(s - s_0) .$$

If  $a = g$ , this reduces to the familiar

$$v = v_0 + g(t - t_0), \quad \text{or} \quad v^2 = v_0^2 + 2g(s - s_0) .$$

### Determining the position from the velocity

Once velocity is known, position can be found by integrating  $ds = v dt$  from Equation (17). Thus, if the velocity is known as a function of time, then

$$s_0 + \int_{t_0}^t v(t) dt . \quad (11)$$

If the velocity is known as a function of position, then

$$\frac{ds}{dt} = v(s) \quad (12)$$

$$ds = v(s) dt \quad (13)$$

$$t = t_0 + \int_{s_0}^s \frac{ds}{v(s)} , \quad (14)$$

where  $s_0$  is the position at time  $t_0$ .

It is worth pointing out that Equation (8) can also be used to derive an expression for  $v(s)$  given  $a(v)$ ,

$$s - s_0 = \int_{s_0}^s ds = \int_{v_0}^v \frac{v}{a(v)} dv . \quad (15)$$

This equation can be used whenever Equation (6) is applicable, and gives  $v(s)$  instead of  $t(v)$ .

For the case of constant acceleration  $a$ , either (11) or (14) can be used to obtain

$$s = s_0 + v_0(t - t_0) + \frac{1}{2}a(t - t_0)^2 .$$

In many practical situations, it may not be possible to carry out the above integrations analytically in which case, numerical integration is required. Usually, numerical integration will also be required when either the velocity or the acceleration depend on more than one variable, e.g.  $v(s, t)$  or  $a(s, v)$ .

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#### Example

#### *Reentry, Ballistic Coefficient, Terminal Velocity*

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##### Terminal Velocity

Terminal velocity occurs when the acceleration becomes zero and the velocity becomes constant. Consider an air-dropped payload starting from rest. The force on the body is a combination of gravity and air drag and has the form

$$F = mg - \frac{1}{2}\rho v^2 C_D A \quad (16)$$

Applying Newton's law and solving for the acceleration  $a$ ,

$$a = g - \frac{1}{2}\rho v^2 \frac{C_D A}{m} = g \left(1 - \left(\frac{v}{v_{\text{terminal}}}\right)^2\right) \quad (17)$$

where  $v_{\text{terminal}}$  is the terminal velocity. The quantity  $\frac{m}{C_D A}$  is known as the Ballistic Coefficient and characterizes the combined effect of body shape and mass on acceleration; it is an important parameter in the study of reentry. Unlike many coefficients that appear in aerospace problems, the Ballistic Coefficient is not non-dimensional, but rather has units of mass/length<sup>2</sup>. Also, in some applications, the ballistic coefficient is defined as the inverse,  $B = \frac{C_D A}{m}$ , so it pays to be careful in its application. Equation (17) then becomes

$$a = g - \frac{1}{2}\rho v^2 / \beta \quad (18)$$

Terminal velocity occurs when the force of gravity equals the drag on the object resulting in zero acceleration. This balance gives the terminal velocity as

$$v_{\text{terminal}} = \sqrt{\frac{2gm}{C_D A \rho}} = \sqrt{\frac{2g\beta}{\rho}} \quad (19)$$

For the Earth, atmospheric density at sea level is  $\rho = 1.225 \text{ kg/m}^3$  (assumed constant for this example, but variation of  $\rho$  with altitude is important for applications like reentry of space vehicles). Typical values of  $\beta$  range from  $\beta = 1$  (Assuming  $C_D = 0.5$ , a tennis ball has  $\beta = 35$ .) to  $\beta = 1000$  for a reentry vehicle. As an example, consider a typical case of  $\beta = 225$ , where the various parameters then give the following expression for the acceleration

$$a = g - 0.002725 v^2 \text{ m/s}^2.$$

Here  $g = 9.81 \text{ m/s}^2$  is the acceleration due to gravity and  $v$  is the downward velocity. It is clear from this expression that initially the acceleration will be  $g$ . Therefore, the velocity will start to increase and keep increasing until  $a = 0$ , at which point the velocity will stay constant. The terminal velocity is then given by,

$$0 = g - 0.002725 v_{\text{terminal}}^2 \quad \text{or} \quad v_{\text{terminal}} = 60 \text{ m/s}.$$

To determine the velocity as a function of time, the acceleration can be re-written introducing the terminal velocity as  $a = g(1 - (v/v_{\text{terminal}})^2)$  and then using expression (6) to obtain

$$t = \frac{1}{g} \int_{v_0}^{v_f} \frac{dv}{(1 - (v/v_{\text{terminal}})^2)} \quad (20)$$

where  $t$  is the time required for the object to attain a final velocity  $v_f$  given an initial velocity  $v_0$ . There are several possibilities: The object could be dropped with an initial velocity of  $v_0 = 0$ ; the object could have an initial velocity less than the terminal velocity  $v_0 < v_{\text{terminal}}$  and thus would speed up during its fall; or the object could have an initial velocity greater than  $v_{\text{terminal}}$ , and thus slow down to terminal velocity. (This

complexity is embedded in the evaluation of the log function.) For the case  $v_0 = 0$ ,

$$t = \frac{1}{g} \int_0^{v_f} \frac{dv}{1 - (v/v_{\text{terminal}})^2} = \frac{1}{g} \int_0^v \left( \frac{1/2}{(1 + (v/v_{\text{terminal}}))} + \frac{1/2}{(1 - (v/v_{\text{terminal}}))} \right) dv = \frac{v_{\text{terminal}}}{2g} \ln \left( \frac{v_{\text{terminal}} + v}{v_{\text{terminal}} - v} \right).$$

Solving for  $v$ ,

$$v_f = v_{\text{terminal}} \frac{e^{2gt/v_{\text{terminal}}} - 1}{e^{2gt/v_{\text{terminal}}} + 1} \text{ m/s}.$$

We can easily verify that for large  $t$ ,  $v_f = v_{\text{terminal}}$ . We can also find out how long it takes for the payload to reach, say, 95% of the terminal velocity,

$$t = \frac{v_{\text{terminal}}}{2g} \ln \frac{1.95}{0.05} = 11.21 \text{ s}.$$

To obtain an expression for the velocity as a function of the traveled distance for the case  $v_0 = 0$ , Equation (15) gives

$$s = \frac{1}{g} \int_0^v \frac{v dv}{1 - (v/v_{\text{terminal}})^2} = -\frac{v_{\text{terminal}}^2}{2g} \ln(1 - (v/v_{\text{terminal}})^2).$$

Solving for  $v$ ,

$$v = v_{\text{terminal}} \sqrt{1 - e^{-2gs/v_{\text{terminal}}^2}} \text{ m/s}.$$

We see that for, say,  $v = 0.95v_{\text{terminal}}$ ,  $s = 427.57 \text{ m}$ . This is the distance traveled by the payload in 11.21s, which can be compared with the distance that would be traveled in the same time if air resistance were neglected,  $s_{\text{no drag}} = gt^2/2 = 615.75 \text{ m}$ .

Results for the other cases, general speeding up or slowing down of an object, can be found by suitable manipulate and integration of equation (15).

### Example

### Spring-mass system

Consider a mass allowed to move without friction on a horizontal slider and subject to the force exerted by a linear spring. Initially the system is in equilibrium (no force on the spring) at  $s = 0$ . Suddenly, the mass is given a velocity  $v_0$  and then the system is left free to oscillate. We know that the effect of the spring is to cause an acceleration to the body, opposing the motion, of the form  $a = -\kappa s$ , where  $\kappa > 0$  is a constant.



Using Equation (18) results in

$$v^2 = v_0^2 - \kappa s^2.$$

The displacement can now be obtained using expression (14),

$$t = \int_0^s \frac{ds}{\sqrt{v_0^2 - \kappa s^2}} = \frac{1}{\sqrt{\kappa}} \arcsin \frac{\sqrt{\kappa} s}{v_0},$$

which gives,

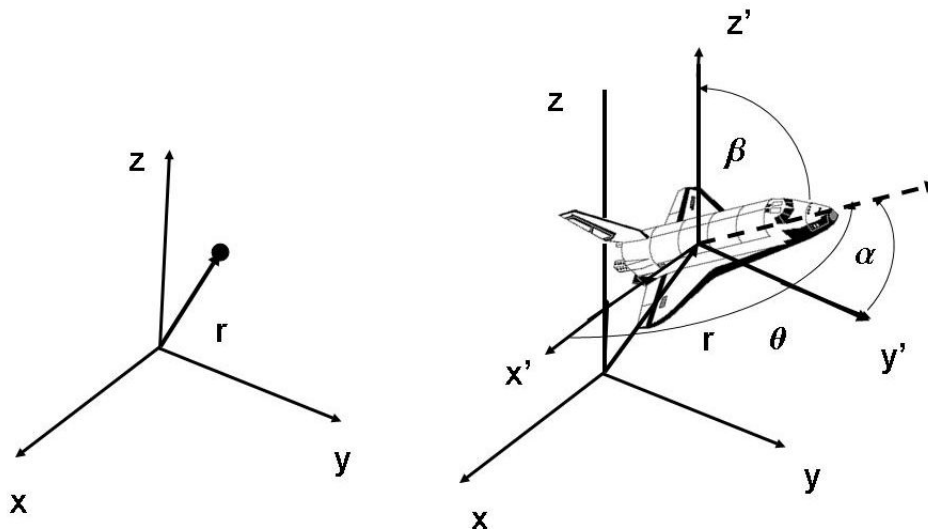
$$s = \frac{v_0}{\sqrt{\kappa}} \sin \sqrt{\kappa} t .$$

Finally, the velocity as a function of time is simply,  $v = v_0 \cos \sqrt{\kappa} t$ . We recognize this motion as that of an undamped harmonic oscillator.

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## Degrees of Freedom

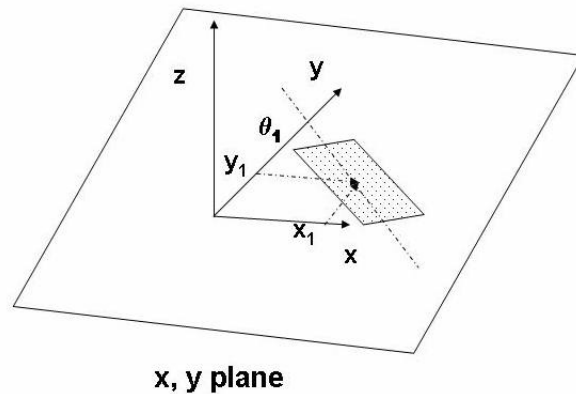
Degree of freedom is the number of independent spatial coordinates that must be specified to determine the position(s) and orientation(s) of a body or system of bodies. For a point mass in Euclidean space, only three positional coordinates are required. For an extended rigid body, such as an aircraft, three position coordinates and three angular coordinates are required to completely specify position and orientation.



## Kinematic Constraints

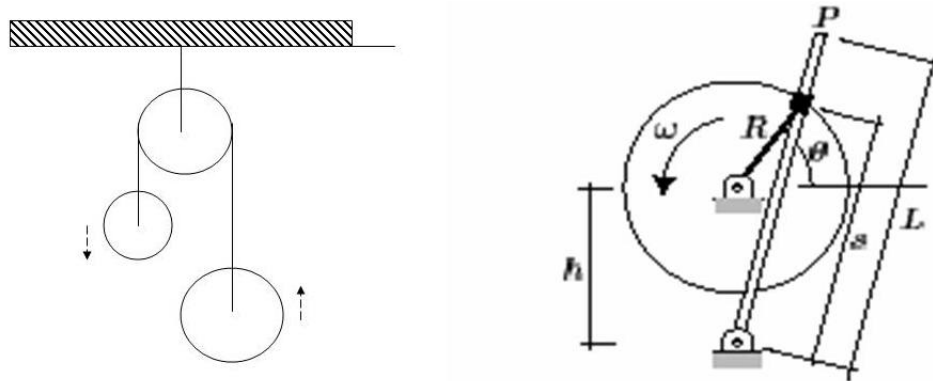
In many situations the number of independent coordinates will be reduced below this number, either because the number of spacial dimensions is reduced or because there are relationships specified among the spatial coordinates. When setting up problems for solution it is useful to think of these relationships as constraints.

For example, if a point mass is constrained to move in a plane (two dimensions) the number of spatial coordinates necessary to describe its motion is two. If instead of being a point mass, this body has extended dimensions, such as a flat plate confined to a plane, it requires three coordinates to specify its position and orientation: two position coordinates and one angular coordinate.



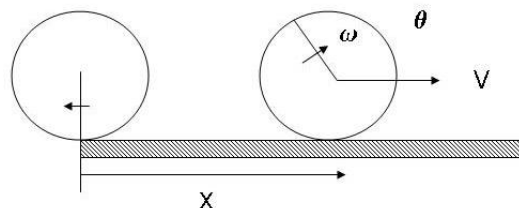
If a particle is confined to move on a curve in either two or three dimensions, such as a bead moving on a wire, the number of independent coordinates necessary to describe its motion is one.

Another source of constraints on the motion of particles is connections between them. For example, the two particle connected by a cable passing over a pulley are constrained to move in equal and opposite directions. More complex arrangements are possible and can be analyzed using these ideas. Two gears in contact are constrained to move together according to their individual geometry.



### SINGLE DEGREE OF FREEDOM SYSTEMS

A cylinder rolling on a plane is constrained in two ways. Contact with the plane reduces the two-dimensional motion to one spatial coordinate along the plane, and the constraint of rolling provides a relationship between the angular coordinates and the spatial position, resulting in a single degree of freedom system.

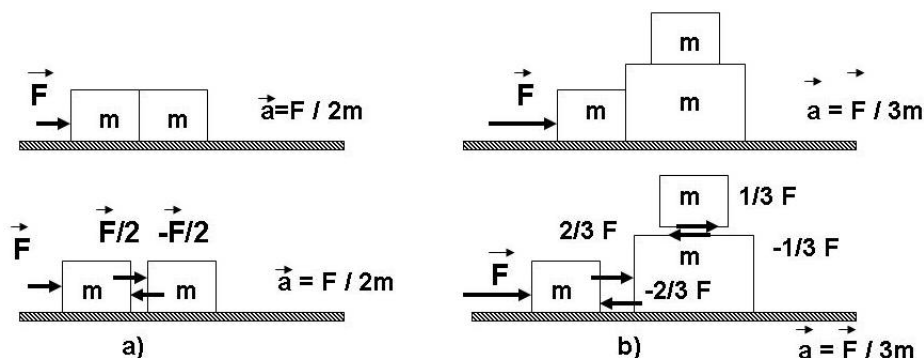




## Internal Force-Balance Constraints

Another type of constraint occurs when considering a system of particles and the necessary force balance that occurs between the parts. These constraints follow directly from Newton's third law: the force of action and reaction between two bodies are equal in magnitude and opposite in direction.

Consider the systems shown in **a)** and **b)**.



### Internal Force Balance Constraints

System **a)** consists of two masses  $m$  in contact resting on a frictionless plane in the presence of gravity. A force  $\vec{F}$  is applied to mass 1, and it is obvious that the two masses will accelerate at  $a = F/(2m)$ . Examining the two masses separately allows one to compute what internal force must exist between them to cause the motion. It is clear that each mass feels a net force of  $F/2$ , since its acceleration is  $a = F/(2m)$ . This net force arises because between the two masses there is an equal and opposite force  $F/2$  acting across the interface. Another way to look at this is that the interface between the bodies is a “body” of zero mass, and therefore can have no net force acting upon it otherwise its acceleration would be infinite.

System **b)** is a bit more complex, primarily because the forces between one mass and the mass above it are shear forces and must be supplied by friction. Assuming that the friction coefficient is large enough to accelerate the three masses an equal amount given by  $a = F/(3m)$ , by the reasoning discussed, the force balance is as sketched in **b)**: equal and opposite normal forces  $F 2/3$  on the vertical surfaces, and equal and opposite shear forces  $F 1/3$  on the horizontal surfaces.

## References

- [1] R. Dugas, *A History of Mechanics*, Dover, 1988.

### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition  
1/1, 1/2, 1/3, 1/4, 1/5 (Effect or Altitude only), 1/6, 1/7, 3/2

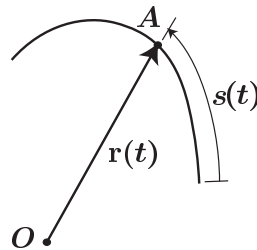
## Lecture 4 — Curvilinear Motion in Cartesian Coordinates

### Kinematics of curvilinear motion

*Dynamics* studies motion and the forces that cause, or are generated as a result of, motion. *Kinematics* describes motions irrespective of the forces that produce them. A thorough understanding of kinematics is important before moving on to *kinetics*, in which forces are also considered.

### Position vector and Path

Curvilinear motion considers the general situation of a particle moving in a three dimensional space. To locate the position of a particle in space requires an origin point,  $O$ , whose location is known. The position of a particle  $A$ , at time  $t$ , can then be described in terms of the *position vector*,  $\mathbf{r}$ , joining points  $O$  and  $A$ . In general, this particle will not be still, but its position will change in time. Thus, the position vector will be a function of time, i.e.  $\mathbf{r}(t)$ . The curve in space described by the particle is called the *path*, or *trajectory*.



The *path* or *arc length coordinate*,  $s$ , measures the distance traveled by the particle along the curved path. Note that  $\mathbf{r}(t)$  is a vector while  $s(t)$  is a scalar. For the previously-considered case of rectilinear motion, the arc length coordinate and the coordinate,  $s$ , are the same.

The path coordinate enables an alternative representation of the motion of the particle. Assume that  $\mathbf{r}$  is known as a function of  $s$ , i.e.  $\mathbf{r}(s)$ , and that the path coordinate is known as a function of time  $t$ , i.e.  $s(t)$ . The *speed* at which the particle moves on the path can then be calculated simply as  $v = \dot{s} \equiv ds/dt$ . The rate of change of speed is  $a_t = \ddot{s} = d^2s/dt^2$ . However, since  $\mathbf{a}(t)$  is a vector, the normal component  $a_n$  is needed to complete this description.

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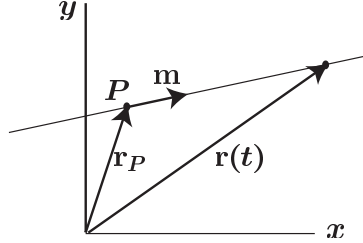
### Example

### Motion along a straight line in 2D

In this example the position vector is referred to a fixed cartesian coordinate system. Consider for illustration purposes two particles that move along a line defined by a point  $P$  and a *unit* vector  $\mathbf{m}$ . Further assume

## REFERENCES

that at  $t = 0$ , both particles are at point  $P$ . The position vector of the first particle is given by  $\mathbf{r}_1(t) = \mathbf{r}_P + \mathbf{m}t = (r_{Px} + m_x t)\mathbf{i} + (r_{Py} + m_y t)\mathbf{j}$ , whereas the position vector of the second particle is given by  $\mathbf{r}_2(t) = \mathbf{r}_P + \mathbf{m}t^2 = (r_{Px} + m_x t^2)\mathbf{i} + (r_{Py} + m_y t^2)\mathbf{j}$ .

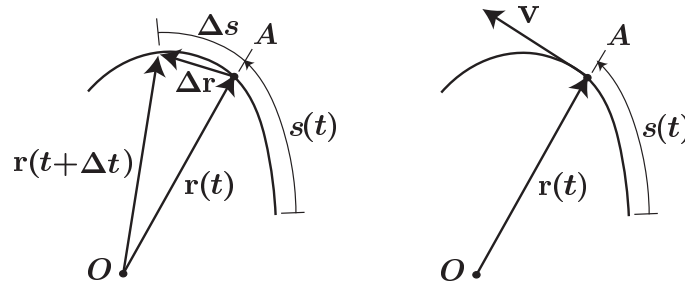


Clearly the path for these two particles is the same, but the speed at which each particle moves along the path is different. This is seen clearly by parameterizing the path with the path coordinate,  $s$ . That is,  $\mathbf{r}(s) = \mathbf{r}_P + \mathbf{m}s = (r_{Px} + m_x s)\mathbf{i} + (r_{Py} + m_y s)\mathbf{j}$ . It is straightforward to verify that  $s$  is indeed the path coordinate, i.e. the distance between two points  $\mathbf{r}(s)$  and  $\mathbf{r}(s + \Delta s)$  is equal to  $\Delta s$ . The two motions introduced earlier simply correspond to two particles moving according to  $s_1(t) = t$  and  $s_2(t) = t^2$ , respectively. Thus,  $\mathbf{r}_1(t) = \mathbf{r}(s_1(t))$  and  $\mathbf{r}_2(t) = \mathbf{r}(s_2(t))$ .

It turns out that in many situations, no expression for the path as a function of  $s$  will be available. Still, it is possible to obtain the speed directly from  $\mathbf{r}(t)$  without the need for an arc length parametrization of the trajectory.

## Velocity Vector

Consider the positions of the particle at two different times  $t$  and  $t + \Delta t$ , where  $\Delta t$  is a small increment of time. Let  $\Delta \mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$ , be the displacement vector as shown in the diagram.



The average velocity of the particle over this small increment of time is

$$\mathbf{v}_{ave} = \frac{\Delta \mathbf{r}}{\Delta t},$$

which is a vector whose direction is that of  $\Delta \mathbf{r}$  and whose magnitude is the length of  $\Delta \mathbf{r}$  divided by  $\Delta t$ . If  $\Delta t$  is small, then  $\Delta \mathbf{r}$  will become tangent to the path, and the modulus of  $\Delta \mathbf{r}$  will be equal to the distance the particle has moved on the curve  $\Delta s$ .

The instantaneous velocity vector is given by

$$\mathbf{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} \equiv \frac{d\mathbf{r}(t)}{dt} \equiv \dot{\mathbf{r}}, \quad (1)$$

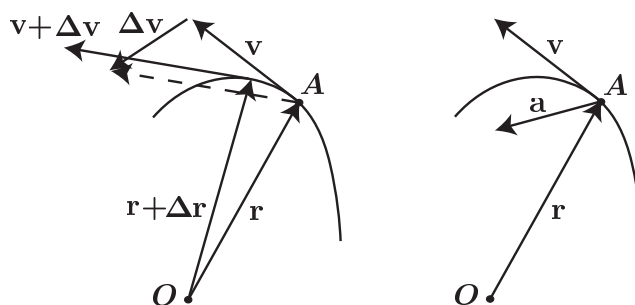
and is *always* tangent to the path. The magnitude, or speed, is given by

$$v = |\mathbf{v}| = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} \equiv \frac{ds}{dt} \equiv \dot{s}.$$

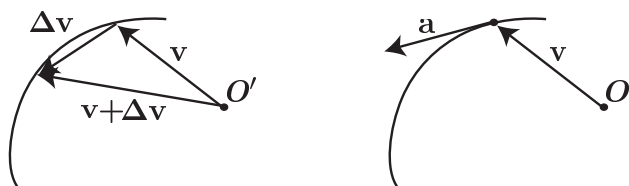
## Acceleration Vector

The acceleration vector can be defined in an analogous manner. Particle  $A$  occupies position  $\mathbf{r}(t)$  at time  $t$ , and has velocity  $\mathbf{v}(t)$ . At time  $t + \Delta t$ , it has position  $\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \Delta \mathbf{r}$ , and velocity  $\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \Delta \mathbf{v}$ . Considering an infinitesimal time increment, the acceleration vector is defined as the derivative of the velocity vector with respect to time,

$$\mathbf{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}}{\Delta t} \equiv \frac{d\mathbf{v}}{dt} = \frac{d^2 \mathbf{r}}{dt^2}. \quad (2)$$



Note that the acceleration vector reflects changes of velocity in both magnitude and direction. The acceleration vector will, in general, not be tangent to the trajectory (in fact it is only tangent when the velocity vector does not change direction). A sometimes useful way to visualize the acceleration vector is to translate the velocity vectors, at different times, such that they all have a common origin, say,  $O'$ . Then, the heads of the velocity vector will change in time and describe a curve in space called the *hodograph*. The acceleration vector is tangent to the hodograph at every point.




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### Note

### Unit tangent and arc-length parametrization

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The unit tangent vector to the curve can be simply calculated as

$$\mathbf{e}_t = \mathbf{v}/v.$$

It is clear that the tangent vector depends solely on the geometry of the trajectory and not on the speed at which the particle moves along the trajectory. That is, the geometry of the trajectory determines the tangent vector, and hence the direction of the velocity vector. How fast the particle moves along the trajectory determines the magnitude of the velocity vector. This is clearly seen by considering the arc-length parametrization of the trajectory  $\mathbf{r}(s)$ . Then, applying the chain rule for differentiation,

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d\mathbf{r}}{ds} \frac{ds}{dt} = \mathbf{e}_t v \quad (3)$$

$$\dot{s} = v \quad (4)$$

$$\frac{d\mathbf{r}}{ds} = \mathbf{e}_t. \quad (5)$$

The fact that the modulus of  $d\mathbf{r}/ds$  is always unity reflects that the distance traveled along the path  $\mathbf{r}(s)$  per unit of  $s$  is, in fact, unity. This is not surprising since by definition the distance between two neighboring points is  $ds$ , i.e.  $|d\mathbf{r}| = ds$ .

## Cartesian Coordinates

When working with fixed cartesian (i.e. rectangular) coordinates, vector differentiation takes a particularly simple form. Since the vectors  $\mathbf{i}$ ,  $\mathbf{j}$ , and  $\mathbf{k}$  do not change, the derivative of a vector  $\mathbf{A}(t) = A_x(t)\mathbf{i} + A_y(t)\mathbf{j} + A_z(t)\mathbf{k}$ , is simply  $\dot{\mathbf{A}}(t) = \dot{A}_x(t)\mathbf{i} + \dot{A}_y(t)\mathbf{j} + \dot{A}_z(t)\mathbf{k}$ . That is, the components of the derivative vector are simply the derivatives of the components.

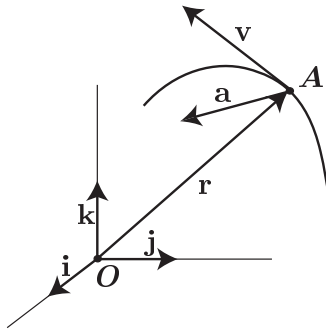
Thus, if position, velocity, and acceleration are referred to a fixed cartesian coordinate system, then

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} \quad (6)$$

$$\mathbf{v}(t) = \frac{d\mathbf{r}(t)}{dt} = v_x(t)\mathbf{i} + v_y(t)\mathbf{j} + v_z(t)\mathbf{k} = \dot{x}(t)\mathbf{i} + \dot{y}(t)\mathbf{j} + \dot{z}(t)\mathbf{k} = \dot{\mathbf{r}}(t) \quad (7)$$

$$\mathbf{a}(t) = \frac{d\mathbf{v}(t)}{dt} = a_x(t)\mathbf{i} + a_y(t)\mathbf{j} + a_z(t)\mathbf{k} = \dot{v}_x(t)\mathbf{i} + \dot{v}_y(t)\mathbf{j} + \dot{v}_z(t)\mathbf{k} = \dot{\mathbf{v}}(t) \quad (8)$$

since  $\frac{d(\cdot)}{dt} = 0$  for  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$ .



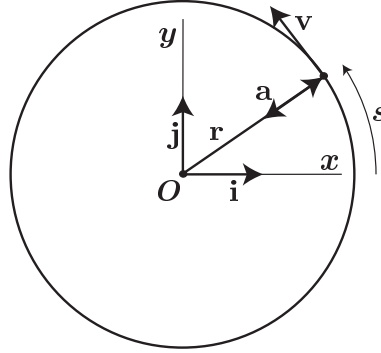
Here, the speed is  $v = \sqrt{v_x^2 + v_y^2 + v_z^2}$ , and the magnitude of the acceleration is  $a = \sqrt{a_x^2 + a_y^2 + a_z^2}$ . The advantages of cartesian coordinate systems are that they are simple to use, and that if  $\mathbf{a}$  is constant, or a

function of time only, each component of the acceleration and velocity can be integrated independently.

**Example****Circular Motion**

Consider motion of a particle along a circle of radius  $R$  at a constant speed  $v_0$ . The parametrization of a circle in terms of arc length is

$$\mathbf{r}(s) = R \cos\left(\frac{s}{R}\right)\mathbf{i} + R \sin\left(\frac{s}{R}\right)\mathbf{j} .$$



Since  $v_0$  is constant,  $s = v_0 t$ . Thus,

$$\mathbf{r}(t) = R \cos\left(\frac{v_0 t}{R}\right)\mathbf{i} + R \sin\left(\frac{v_0 t}{R}\right)\mathbf{j} .$$

The velocity is

$$\mathbf{v}(t) = \frac{d\mathbf{r}(t)}{dt} = -v_0 \sin\left(\frac{v_0 t}{R}\right)\mathbf{i} + v_0 \cos\left(\frac{v_0 t}{R}\right)\mathbf{j} ,$$

which clearly has a constant magnitude  $|\mathbf{v}| = v_0$ . The acceleration is

$$\mathbf{a}(t) = \frac{d\mathbf{v}(t)}{dt} = -\frac{v_0^2}{R} \cos\left(\frac{v_0 t}{R}\right)\mathbf{i} - \frac{v_0^2}{R} \sin\left(\frac{v_0 t}{R}\right)\mathbf{j} .$$

Note that the acceleration is perpendicular to the path (in this case it is parallel to  $\mathbf{r}$ ), since the velocity vector changes direction, but not magnitude.

One can also verify that, from  $\mathbf{r}(s)$ , the unit tangent vector,  $\mathbf{e}_t$ , could be computed directly as

$$\mathbf{e}_t = \frac{d\mathbf{r}(s)}{ds} = -\sin\left(\frac{s}{R}\right)\mathbf{i} + \cos\left(\frac{s}{R}\right)\mathbf{j} = -\sin\left(\frac{v_0 t}{R}\right)\mathbf{i} + \cos\left(\frac{v_0 t}{R}\right)\mathbf{j} .$$

**Example****Motion along a helix**

The equation  $\mathbf{r}(t) = R \cos t \mathbf{i} + R \sin t \mathbf{j} + h t \mathbf{k}$  defines the motion of a particle moving on a helix of radius  $R$ , and pitch  $2\pi h$ , at a constant speed. The velocity vector is given by

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = -R \sin t \mathbf{i} + R \cos t \mathbf{j} + h \mathbf{k} ,$$

and the acceleration vector is given by

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = -R \cos t \mathbf{i} + -R \sin t \mathbf{j}.$$

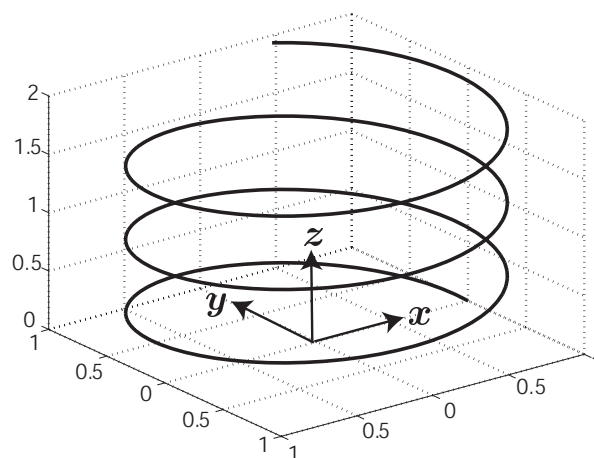
The speed at which the particle moves is simply the modulus of the velocity vector,

$$v = |\mathbf{v}| = \sqrt{R^2 \sin^2 t + R^2 \cos^2 t + h^2} = \sqrt{R^2 + h^2}.$$

The equation of the path in terms of the arc-length coordinate is simply

$$ds = |d\mathbf{r}| = v dt = \sqrt{R^2 + h^2} dt.$$

Integrating,  $s = s_0 + \sqrt{R^2 + h^2} t$ , where  $s_0$  corresponds to the path coordinate of the particle at time zero. Substituting  $t$  in terms of  $s$ , results in an expression for the position vector in terms of the arc-length coordinate. In this case,  $\mathbf{r}(s) = R \cos(s/\sqrt{R^2 + h^2})\mathbf{i} + R \sin(s/\sqrt{R^2 + h^2})\mathbf{j} + hs/\sqrt{R^2 + h^2}\mathbf{k}$ . The figure below shows the particle trajectory for  $R = 1$  and  $h = 0.1$ .




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**Example**
**Ballistic Motion**

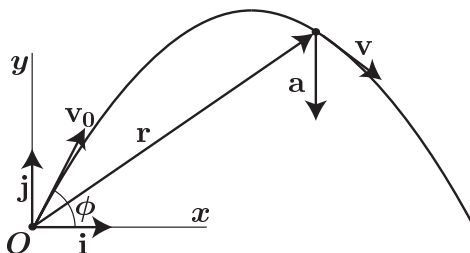
Consider the free-flight motion of a projectile which is initially launched with a velocity  $\mathbf{v}_0 = v_0 \cos \phi \mathbf{i} + v_0 \sin \phi \mathbf{j}$ . If air resistance is neglected, the only force on the projectile is the weight, which causes the projectile to have a constant acceleration  $\mathbf{a} = -g\mathbf{j}$ . In component form this equation can be written as  $dv_x/dt = 0$  and  $dv_y/dt = -g$ . Integrating and imposing initial conditions results in

$$v_x = v_0 \cos \phi, \quad v_y = v_0 \sin \phi - gt.$$

Note that the horizontal velocity is constant. A further integration yields the trajectory

$$x = x_0 + (v_0 \cos \phi) t, \quad y = y_0 + (v_0 \sin \phi) t - \frac{1}{2}gt^2,$$

which is recognized as the equation of a parabola.



The maximum height,  $y_{mh}$ , occurs when  $v_y(t_{mh}) = 0$ , which gives  $t_{mh} = (v_0/g) \sin \phi$ , or

$$y_{mh} = y_0 + \frac{v_0^2 \sin^2 \phi}{2g}.$$

The range,  $x_r$ , can be obtained by setting  $y = y_0$ , which gives  $t_r = (2v_0/g) \sin \phi$ , or,

$$x_r = x_0 + \frac{2v_0^2 \sin \phi \cos \phi}{g} = x_0 + \frac{v_0^2 \sin(2\phi)}{g}.$$

If one wants to maximize the range  $x_r$ , for a given velocity  $v_0$ , then  $\sin(2\phi) = 1$ , or  $\phi = 45^\circ$ .

Finally, note that if one wants to model a more realistic situation and include aerodynamic drag forces of the form, say,  $-\kappa v^2$ , then it would not be possible to solve for  $x$  and  $y$  independently, and this would usually require numerical integration.

#### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

2/1, 2/3, 2/4



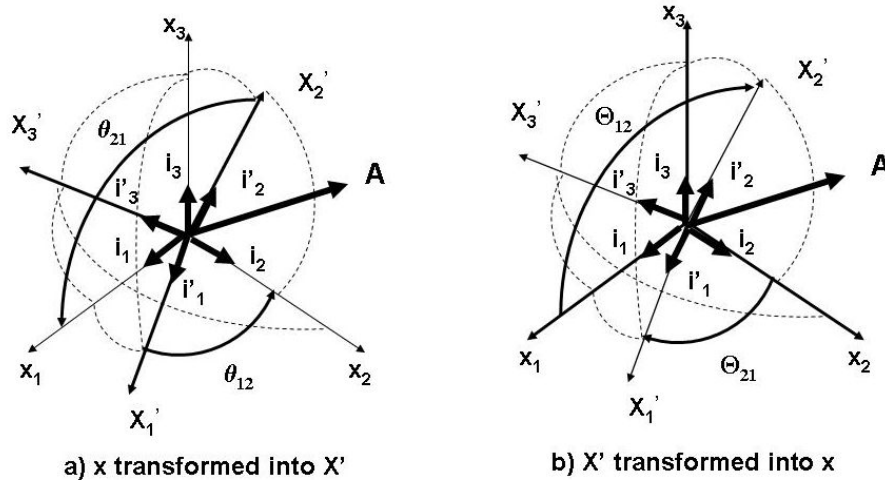
## Lecture 5 — Coordinate Transformations

### Vector Transformations

In many problems, multiple coordinate systems are used to describe different vector quantities. Vector operations written in component form only hold once all the vectors involved are described with respect to the same frame. This section shows how the components of a vector are transformed when the reference frame is changed.

Consider two different orthogonal, right-handed reference frames  $x_1, x_2, x_3$  and  $X'_1, X'_2, X'_3$ . A vector  $\mathbf{A}$  in coordinate system  $\mathbf{x}$  can be transformed to coordinate system  $\mathbf{X}'$  by considering the 9 angles that define the relationships between the two systems. (It turns out that only three of these angles are independent).

The figures below show the vector  $\mathbf{A}$ , the  $\mathbf{x}$  and  $\mathbf{X}'$  coordinate systems, the unit vectors  $\mathbf{i}_1, \mathbf{i}_2, \mathbf{i}_3$  of the  $\mathbf{x}$  system and the unit vectors  $\mathbf{i}'_1, \mathbf{i}'_2, \mathbf{i}'_3$  of the  $\mathbf{X}'$  system. a) focuses on the transformation of coordinates from  $\mathbf{x}$  to  $\mathbf{X}'$ , while b) focuses on the reverse transformation from  $\mathbf{X}'$  to  $\mathbf{x}$ .



In the  $\mathbf{x}$  coordinate system, the vector  $\mathbf{A}$  can be written as

$$\mathbf{A} = A_1 \mathbf{i}_1 + A_2 \mathbf{i}_2 + A_3 \mathbf{i}_3, \quad (1)$$

or, when referred to the frame  $\mathbf{X}'$ , as

$$\mathbf{A} = A'_1 \mathbf{i}'_1 + A'_2 \mathbf{i}'_2 + A'_3 \mathbf{i}'_3. \quad (2)$$

And since the vector  $\mathbf{A}$  must remain the same regardless of the coordinate transformation,

$$\mathbf{A} = A_1 \mathbf{i}_1 + A_2 \mathbf{i}_2 + A_3 \mathbf{i}_3 = A'_1 \mathbf{i}'_1 + A'_2 \mathbf{i}'_2 + A'_3 \mathbf{i}'_3, \quad (3)$$

The components of the vector  $\mathbf{A}$  in the transformed system can be found by simply taking the dot product of this equation with the desired unit vector in the  $\mathbf{X}'$  system, e.g.

$$A'_j = A_1 \mathbf{i}'_j \cdot \mathbf{i}_1 + A_2 \mathbf{i}'_j \cdot \mathbf{i}_2 + A_3 \mathbf{i}'_j \cdot \mathbf{i}_3 \quad (4)$$

where  $A'_j$  is the  $j$ th component of  $\mathbf{A}$  in the  $\mathbf{X}'$  system. Repeating this operation for each component of  $\mathbf{A}$  in the  $\mathbf{X}'$  system results in the matrix form for  $\mathbf{A}$

$$\begin{bmatrix} A'_1 \\ A'_2 \\ A'_3 \end{bmatrix} = \begin{bmatrix} \mathbf{i}'_1 \cdot \mathbf{i}_1 & \mathbf{i}'_1 \cdot \mathbf{i}_2 & \mathbf{i}'_1 \cdot \mathbf{i}_3 \\ \mathbf{i}'_2 \cdot \mathbf{i}_1 & \mathbf{i}'_2 \cdot \mathbf{i}_2 & \mathbf{i}'_2 \cdot \mathbf{i}_3 \\ \mathbf{i}'_3 \cdot \mathbf{i}_1 & \mathbf{i}'_3 \cdot \mathbf{i}_2 & \mathbf{i}'_3 \cdot \mathbf{i}_3 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix}.$$

The above expression is the relationship that expresses how the components of a vector in one coordinate system relate to the components of the *same* vector in a different coordinate system.

Referring to the figure,  $\mathbf{i}'_j \cdot \mathbf{i}_i$  is equal to the cosine of the angle between  $\mathbf{i}'_j$  and  $\mathbf{i}_i$  which is  $\theta_{ji}$ . In particular,  $\mathbf{i}'_2 \cdot \mathbf{i}_1 = \cos \theta_{21}$  while  $\mathbf{i}'_1 \cdot \mathbf{i}_2 = \cos \theta_{12}$ . These angles are in general not equal. Therefore, the components of the vector  $\mathbf{A}$  are transformed from the  $\mathbf{x}$  coordinate system to the  $\mathbf{X}'$  system through the transformation

$$A'_j = A_1 \cos \theta_{j1} + A_2 \cos \theta_{j2} + A_3 \cos \theta_{j3}. \quad (5)$$

where the coefficients relating the components of  $\mathbf{A}$  in the two coordinate systems are the various direction cosines of the angles between the coordinate directions.

The above relations for the transformation of  $\mathbf{A}$  from the  $\mathbf{x}$  to the  $\mathbf{X}'$  system can be written in matrix form:

$$\mathbf{A}' = \begin{bmatrix} A'_1 \\ A'_2 \\ A'_3 \end{bmatrix} = \begin{bmatrix} \cos(\theta_{11}) & \cos(\theta_{12}) & \cos(\theta_{13}) \\ \cos(\theta_{21}) & \cos(\theta_{22}) & \cos(\theta_{23}) \\ \cos(\theta_{31}) & \cos(\theta_{32}) & \cos(\theta_{33}) \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix}. \quad (6)$$

Of course the vector  $\mathbf{A}$  is unchanged by the transformation. The symbol  $\mathbf{T}$  represents the transformation matrix from  $\mathbf{x}$  to  $\mathbf{X}'$ ,

$$\mathbf{A}' = \mathbf{T} \mathbf{A}, \quad (7)$$

where  $\mathbf{T}$  is the transformation matrix.

Now consider the reverse process that transforms the vector  $\mathbf{A}'$  from the  $\mathbf{X}'$  system to the  $\mathbf{x}$  system.

$$\mathbf{A} = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix} = \begin{bmatrix} \cos(\Theta_{11}) & \cos(\Theta_{12}) & \cos(\Theta_{13}) \\ \cos(\Theta_{21}) & \cos(\Theta_{22}) & \cos(\Theta_{23}) \\ \cos(\Theta_{31}) & \cos(\Theta_{32}) & \cos(\Theta_{33}) \end{bmatrix} \begin{bmatrix} A'_1 \\ A'_2 \\ A'_3 \end{bmatrix}. \quad (8)$$

Comparing the coordinate transformations shown in a) and b),  $\cos \theta_{12} = \cos \Theta_{21}$ , and therefore the matrix element of magnitude  $\cos \theta_{12}$  which appears in the 12 position in the transformation matrix from  $\mathbf{x}$  to  $\mathbf{X}'$  now appears in the 21 position in the matrix which transforms  $\mathbf{A}$  from  $\mathbf{X}'$  to  $\mathbf{x}$ . This pattern is repeated for all off-diagonal elements. The diagonal elements remain unchanged since  $\cos \theta_{ii} = \cos \Theta_{ii}$ . Thus the matrix which transposes the vector  $\mathbf{A}$  in the  $\mathbf{X}'$  system back to the  $\mathbf{x}$  system is the transpose of the original transformation matrix,

$$\mathbf{A} = \mathbf{T}^T \mathbf{A}'. \quad (9)$$

where  $\mathbf{T}^T$  is the transpose of  $\mathbf{T}$ . Since transforming  $\mathbf{A}$  from  $\mathbf{x}$  to  $\mathbf{X}'$  and back to  $\mathbf{x}$  results in no change, the matrix  $\mathbf{T}^T$  is also  $\mathbf{T}^{-1}$ , the inverse of  $\mathbf{T}$

$$\mathbf{A} = \mathbf{T}^T \mathbf{T} \mathbf{A} = \mathbf{T}^{-1} \mathbf{T} \mathbf{A} = \mathbf{I} \mathbf{A} = \mathbf{A}. \quad (10)$$

where  $\mathbf{I}$  is the identity matrix

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (11)$$

This is a remarkable and useful property of transformation matrices, which is not true in general for all matrices.

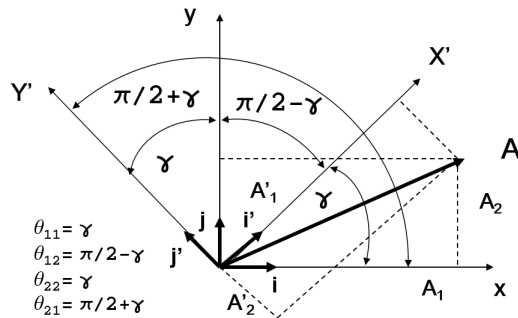
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### Example

### Coordinate transformation in two dimensions

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Consider the change of coordinates between two reference frames  $x, y$ , and  $X', Y'$ , as shown in the diagram.



The angle between  $\mathbf{i}$  and  $\mathbf{i}'$  is  $\gamma$ . Therefore,  $\mathbf{i} \cdot \mathbf{i}' = \cos \gamma$ . Similarly,  $\mathbf{j} \cdot \mathbf{i}' = \cos(\pi/2 - \gamma) = \sin \gamma$ ,  $\mathbf{i} \cdot \mathbf{j}' = \cos(\pi/2 + \gamma) = -\sin \gamma$ , and  $\mathbf{j} \cdot \mathbf{j}' = \cos \gamma$ . Finally, the transformation matrix  $\mathbf{T}$  is

$$\mathbf{T} = \begin{bmatrix} \cos(\theta_{11}) & \cos(\theta_{12}) \\ \cos(\theta_{21}) & \cos(\theta_{22}) \end{bmatrix} = \begin{bmatrix} \cos \gamma & \sin \gamma \\ -\sin \gamma & \cos \gamma \end{bmatrix},$$

and we can write,

$$\begin{bmatrix} A'_1 \\ A'_2 \end{bmatrix} = \mathbf{T} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}.$$

and

$$\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \mathbf{T}^T \begin{bmatrix} A'_1 \\ A'_2 \end{bmatrix}.$$

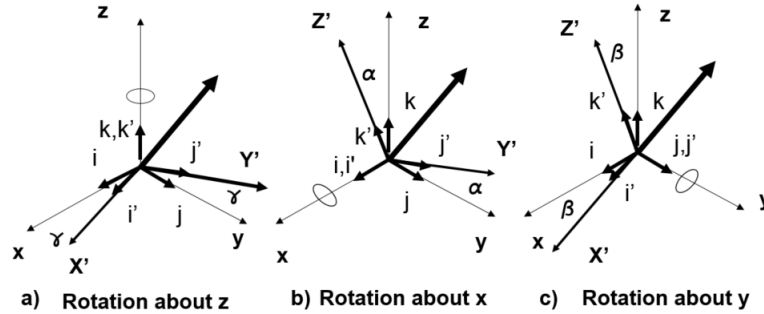
Therefore,

$$A'_1 = A_1 \cos \gamma + A_2 \sin \gamma \quad (12)$$

$$A'_2 = -A_1 \sin \gamma + A_2 \cos \gamma. \quad (13)$$

For instance, we can easily check that when  $\gamma = \pi/2$ , the above expressions give  $A'_1 = A_2$ , and  $A'_2 = -A_1$ , as expected.

An additional observation can be made. If in three dimensions, we rotate the  $x, y, z$  coordinate system about the  $z$  axis, as shown in **a)** leaving the  $z$  component unchanged,



the transformation matrix becomes

$$\mathbf{T} = \begin{bmatrix} \cos(\theta_{11}) & \cos(\theta_{12}) & 0 \\ \cos(\theta_{21}) & \cos(\theta_{22}) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \gamma & \sin \gamma & 0 \\ -\sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Analogous results can be obtained for rotation about the  $x$  axis or rotation about the  $y$  axis as shown in **b)** and **c)**.

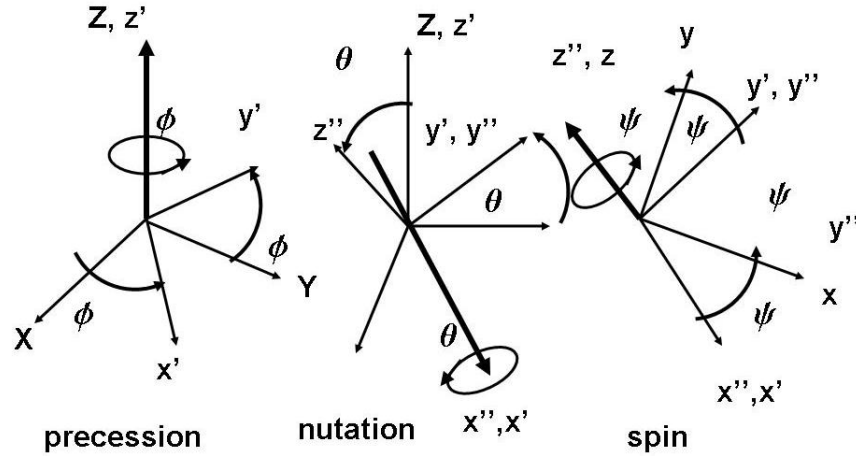
## Sequential Transformations; Euler Angles

The general orientation of a coordinate system can be described by a sequence of rotations about coordinate axes. One particular set of such rotations, called Euler Angles, leads to a description particularly convenient for describing the motion of a three-dimensional rigid body in general spinning motion, call Euler angles.

## REFERENCES

We shall treat this topic in Lecture 28. For now, we examine how this rotation fits into our general study of coordinate transformations. A coordinate description in terms of Euler angles is obtained by the sequential rotation about different axes as shown in the figure; the order of rotations makes a difference.

To develop the description of this motion, we use a series of transformations of coordinates. The final result is shown below. This is the coordinate system used for the description of motion of a general three-dimensional rigid body such as a top described in body-fixed axis. To identify the new position of the coordinate axes as a result of angular displacement through the three Euler angles, we go through a series of coordinate rotations.



First, we rotate from an initial  $X, Y, Z$  system into an  $x', y', z'$  system through a rotation  $\phi$  about the  $Z, z'$  axis.

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \mathbf{T}_1 \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}.$$

The resulting  $x', y'$  coordinates remain in the  $X, Y$  plane, while  $Z$  and  $z'$  remain co-linear. Then, we rotate about the  $x', x''$  axis into the  $x'', y'', z''$  system through an angle  $\theta$ . The  $x''$  axis remains coincident with the  $x'$  axis. The axis of rotation for this transformation is called the "line of nodes". The plane containing the  $x'', y''$  coordinate is now tipped through an angle  $\theta$  relative to the original  $X, Y$  plane.

$$\begin{bmatrix} x'' \\ y'' \\ z'' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \mathbf{T}_2 \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}.$$

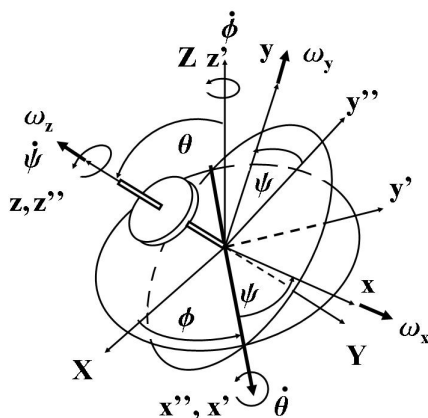
And finally, we rotate about the  $z'', z$  system through an angle  $\psi$  into the  $x, y, z$  system. The  $z''$  axis is called the spin axis. It is coincident with the  $z$  axis.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x'' \\ y'' \\ z'' \end{bmatrix} = \mathbf{T}_3 \begin{bmatrix} x'' \\ y'' \\ z'' \end{bmatrix}.$$

The final coordinate system used to describe the position of the body is shown below. The angle  $\psi$  is called

the spin; the angle  $\phi$  is called the precession; the angle  $\theta$  is called the nutation. The total transformation is given by

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = T_3 T_2 T_1 \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}.$$



Euler angles are not always defined in exactly this manner, either the notation or the order of rotations can differ. The particular transformation used in any example should be clearly described.

## References

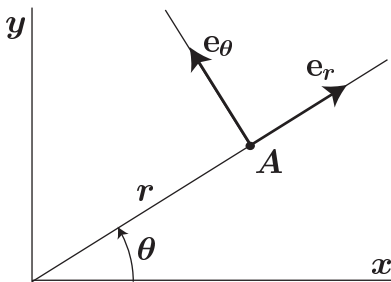
- [1] J.B. Marion and S.T. Thornton, *Classical Dynamics of Particles and Systems*, Harcourt Brace, 1995.
- [2] D. Kleppner and R.J. Kolenkow, *An Introduction to Mechanics*, McGraw Hill, 1973.

## Lecture 6 — Polar, Cylindrical, and Spherical Coordinate Systems

This lecture looks at some common non-Cartesian systems of coordinates. It presents polar coordinates in two dimensions and cylindrical and spherical coordinates in three dimensions. These systems are particularly useful for certain classes of problems.

### Polar Coordinates ( $r - \theta$ )

In polar coordinates, the position of a particle  $A$  is determined by the value of the radial distance to the origin,  $r$ , and the angle that the radial line makes with an arbitrary fixed line, such as the  $x$  axis. Thus, the trajectory of a particle will be determined if  $r$  and  $\theta$  are known as functions of  $t$ , i.e.  $r(t), \theta(t)$ . The directions of increasing  $r$  and  $\theta$  are defined by the orthogonal unit vectors  $\mathbf{e}_r$  and  $\mathbf{e}_\theta$ .

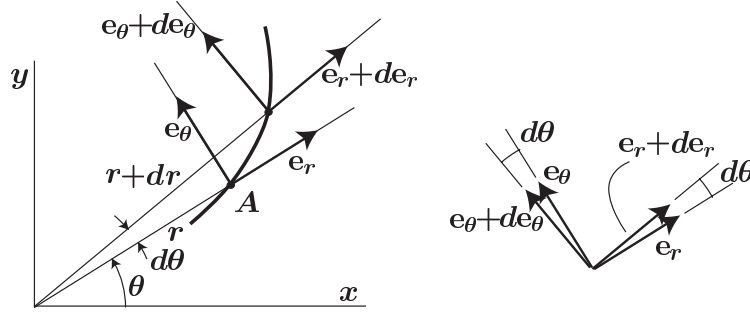


The position vector of a particle has a magnitude equal to the radial distance, and a direction determined by  $\mathbf{e}_r$ . Thus,

$$\mathbf{r} = r\mathbf{e}_r . \quad (1)$$

Since the vectors  $\mathbf{e}_r$  and  $\mathbf{e}_\theta$  are clearly different from point to point, their variation will have to be considered when calculating velocity and acceleration.

Over an infinitesimal interval of time  $dt$ , the coordinates of point  $A$  will change from  $(r, \theta)$ , to  $(r + dr, \theta + d\theta)$  as shown in the diagram.



The velocity is

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r. \quad (2)$$

The derivatives of vectors in polar coordinates, depend on  $\frac{d\mathbf{e}_r}{dt}$  and  $\frac{d\mathbf{e}_\theta}{dt}$ . Using the chain rule,

$$\frac{d\mathbf{e}_r}{dt} = \frac{d\mathbf{e}_r}{d\theta} \frac{d\theta}{dt} + \frac{d\mathbf{e}_r}{dr} \frac{dr}{dt} \quad (3)$$

$$\frac{d\mathbf{e}_\theta}{dt} = \frac{d\mathbf{e}_\theta}{d\theta} \frac{d\theta}{dt} + \frac{d\mathbf{e}_\theta}{dr} \frac{dr}{dt} \quad (4)$$

Note that the vectors  $\mathbf{e}_r$  and  $\mathbf{e}_\theta$  do not change when the coordinate  $r$  changes, because  $dr$  is a change in length in the direction  $\mathbf{e}_r$ , and the orientations of  $\mathbf{e}_r$  and  $\mathbf{e}_\theta$  remain unchanged. Thus,  $d\mathbf{e}_r/dr = \mathbf{0}$  and  $d\mathbf{e}_\theta/dr = \mathbf{0}$ . On the other hand, when  $\theta$  changes to  $\theta + d\theta$ , the vectors  $\mathbf{e}_r$  and  $\mathbf{e}_\theta$  are rotated by an angle  $d\theta$ . From the diagram,  $d\mathbf{e}_r = d\theta\mathbf{e}_\theta$ , and  $d\mathbf{e}_\theta = -d\theta\mathbf{e}_r$ . This is because their magnitudes in the limit are equal to the unit vector as radius times  $d\theta$  in radians. Dividing through by  $d\theta$ ,

$$\frac{d\mathbf{e}_r}{d\theta} = \mathbf{e}_\theta, \quad \text{and} \quad \frac{d\mathbf{e}_\theta}{d\theta} = -\mathbf{e}_r.$$

Multiplying these expressions by  $d\theta/dt \equiv \dot{\theta}$ ,

$$\frac{d\mathbf{e}_r}{d\theta} \frac{d\theta}{dt} \equiv \frac{d\mathbf{e}_r}{dt} = \dot{\theta}\mathbf{e}_\theta, \quad \text{and} \quad \frac{d\mathbf{e}_\theta}{dt} = -\dot{\theta}\mathbf{e}_r. \quad (5)$$

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#### Note

#### Alternative calculation of the unit vector derivatives

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An alternative, more mathematical, approach to obtaining the derivatives of the unit vectors is to express  $\mathbf{e}_r$  and  $\mathbf{e}_\theta$  in terms of their cartesian components along  $\mathbf{i}$  and  $\mathbf{j}$ ,

$$\begin{aligned} \mathbf{e}_r &= \cos\theta\mathbf{i} + \sin\theta\mathbf{j} \\ \mathbf{e}_\theta &= -\sin\theta\mathbf{i} + \cos\theta\mathbf{j} . \end{aligned}$$

Differentiating,

$$\begin{aligned} \frac{d\mathbf{e}_r}{dr} &= 0, & \frac{d\mathbf{e}_r}{d\theta} &= -\sin\theta\mathbf{i} + \cos\theta\mathbf{j} \equiv \mathbf{e}_\theta \\ \frac{d\mathbf{e}_\theta}{dr} &= 0, & \frac{d\mathbf{e}_\theta}{d\theta} &= -\cos\theta\mathbf{i} - \sin\theta\mathbf{j} \equiv -\mathbf{e}_r . \end{aligned}$$



## Velocity vector

Differentiating expression (1) with respect to time,

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r. \quad (6)$$

Using expression (5),

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta. \quad (7)$$

Here,  $v_r = \dot{r}$  is the *radial velocity* component,  $v_\theta = r\dot{\theta}$  is the *circumferential velocity* component, and  $v = \sqrt{v_r^2 + v_\theta^2}$  is the speed. The radial component is the rate at which  $\mathbf{r}$  changes magnitude, or stretches, and the circumferential component is the rate at which  $\mathbf{r}$  changes direction, or swings.

## Acceleration vector

Differentiating again with respect to time gives the acceleration,

$$\mathbf{a} = \dot{\mathbf{v}} = \frac{d\mathbf{v}}{dt} = \frac{d(\dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r)}{dt} = \frac{d(\dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta)}{dt} = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta. \quad (8)$$

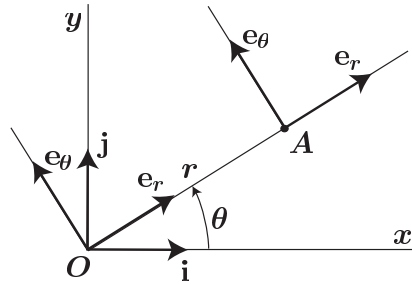
Using the Equation (5),

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta, \quad (9)$$

where  $a_r = (\ddot{r} - r\dot{\theta}^2)$  is the *radial acceleration* component, and  $a_\theta = (r\ddot{\theta} + 2\dot{r}\dot{\theta})$  is the *circumferential acceleration* component. Also,  $a = \sqrt{a_r^2 + a_\theta^2}$  is the magnitude of acceleration.

## Change of basis

In many practical situations, it will be necessary to transform the vectors expressed in polar coordinates to cartesian coordinates and vice versa.

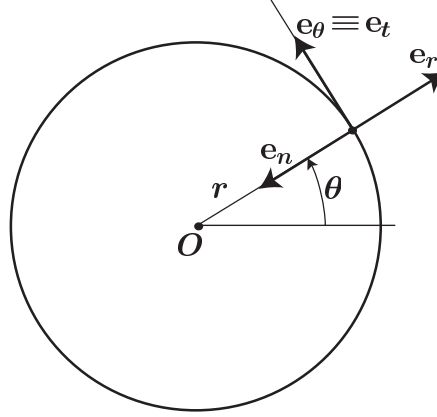


Since we are dealing with free vectors, we can translate the polar reference frame for a given point  $(r, \theta)$ , to the origin, and apply a standard change of basis procedure. This will give, for a generic vector  $\mathbf{A}$ ,

$$\begin{bmatrix} A_r \\ A_\theta \end{bmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{bmatrix} A_x \\ A_y \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} A_x \\ A_y \end{bmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{bmatrix} A_r \\ A_\theta \end{bmatrix}.$$

**Example****Circular motion**

Consider as an illustration the motion of a particle in a circular trajectory having angular velocity  $\omega = \dot{\theta}$ , and constant angular acceleration  $\alpha = \dot{\omega}$ .



In polar coordinates, the equation of the trajectory is

$$r = R = \text{constant}, \quad \theta = \omega_0 t + \frac{1}{2} \alpha t^2.$$

The velocity components are

$$v_r = \dot{r} = 0, \quad \text{and} \quad v_\theta = r\dot{\theta} = R(\omega_0 + \alpha t) = v,$$

and the acceleration components are

$$a_r = \ddot{r} - r\dot{\theta}^2 = -R(\omega_0 + \alpha t)^2 = -\frac{v^2}{R}, \quad \text{and} \quad a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = R\alpha = a_t,$$

where  $a_r \equiv -a_n$ , and  $a_\theta \equiv a_t$ .

In cartesian coordinates, the position is

$$x = R \cos(\omega_0 t + \frac{1}{2} \alpha t^2), \quad y = R \sin(\omega_0 t + \frac{1}{2} \alpha t^2),$$

the velocity is

$$v_x = -R(\omega_0 + \alpha t) \sin(\omega_0 t + \frac{1}{2} \alpha t^2), \quad v_y = R(\omega_0 + \alpha t) \cos(\omega_0 t + \frac{1}{2} \alpha t^2),$$

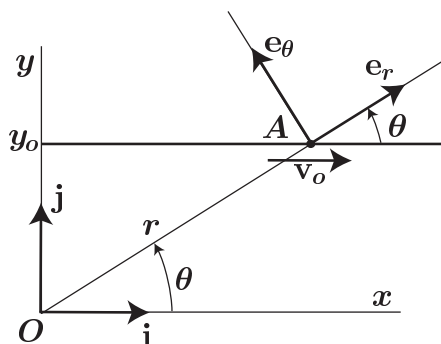
and the acceleration is

$$a_x = -R(\omega_0 + \alpha t)^2 \cos(\omega_0 t + \frac{1}{2} \alpha t^2) - R\alpha \sin(\omega_0 t + \frac{1}{2} \alpha t^2), \quad a_y = -R(\omega_0 + \alpha t)^2 \sin(\omega_0 t + \frac{1}{2} \alpha t^2) + R\alpha \cos(\omega_0 t + \frac{1}{2} \alpha t^2).$$

Note that for this problem, the result is much simpler when expressed in polar coordinates.

**Example****Motion on a straight line**

This example considers the problem of a particle moving with constant velocity  $v_0$ , along a horizontal line  $y = y_0$ .



Assuming that at  $t = 0$  the particle is at  $x = 0$ , the trajectory and velocity components in cartesian coordinates are simply,

$$\begin{aligned} x &= v_0 t & y &= y_0 \\ v_x &= v_0 & v_y &= 0 \\ a_x &= 0 & a_y &= 0 \end{aligned} .$$

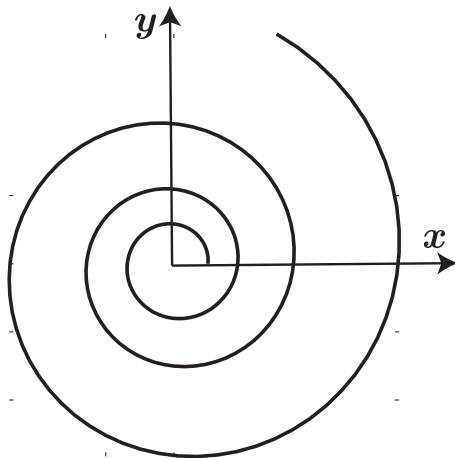
In polar coordinates,

$$\begin{aligned} r &= \sqrt{v_0^2 t^2 + y_0^2} & \theta &= \tan^{-1}\left(\frac{y_0}{v_0 t}\right) \\ v_r &= \dot{r} = v_0 \cos \theta & v_\theta &= r\dot{\theta} = -v_0 \sin \theta \\ a_r &= \ddot{r} - r\dot{\theta}^2 = 0 & a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 \end{aligned} .$$

Here, the expressions obtained in cartesian coordinates are simpler than those obtained using polar coordinates. It is also reassuring that the acceleration in both the  $r$  and  $\theta$  direction, calculated from the general two term expression in polar coordinates, work out to be zero as it must for constant velocity-straight line motion.

**Example****Spiral motion (Kelppner/Kolenkow)**

A particle moves with  $\dot{\theta} = \omega = \text{constant}$  and  $r = r_0 e^{\beta t}$ , where  $r_0$  and  $\beta$  are constants.



We shall show that for certain values of  $\beta$ , the particle moves with  $a_r = 0$ .

$$\begin{aligned}\mathbf{a} &= (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta \\ &= (\beta^2 r_0 e^{\beta t} - r_0 e^{\beta t} \omega^2)\mathbf{e}_r + 2\beta r_0 \omega e^{\beta t}\mathbf{e}_\theta\end{aligned}$$

If  $\beta = \pm\omega$ , the radial part of  $\mathbf{a}$  vanishes. It seems quite surprising that when  $r = r_0 e^{\beta t}$ , the particle moves with zero radial acceleration. The error is in thinking that  $\ddot{r}$  makes the only contribution to  $a_r$ ; the term  $-r\dot{\theta}^2$  is also part of the radial acceleration, and cannot be neglected.

The paradox is that even though  $a_r = 0$ , the radial velocity  $v_r = \dot{r} = r_0 \beta e^{\beta t}$  is increasing rapidly in time. In polar coordinates

$$v_r \neq \int a_r(t) dt ,$$

because this integral does not take into account the fact that  $\mathbf{e}_r$  and  $\mathbf{e}_\theta$  are functions of time.

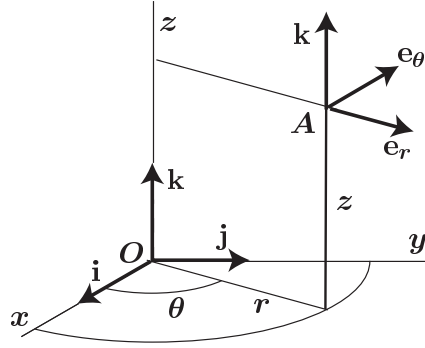
## Equations of Motion

In two dimensional polar  $r\theta$  coordinates, the force and acceleration vectors are  $\mathbf{F} = F_r \mathbf{e}_r + F_\theta \mathbf{e}_\theta$  and  $\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta$ . Thus, in component form,

$$\begin{aligned}F_r &= m a_r = m(\ddot{r} - r\dot{\theta}^2) \\ F_\theta &= m a_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) .\end{aligned}$$

## Cylindrical Coordinates ( $r - \theta - z$ )

Polar coordinates can be extended to three dimensions in a straightforward manner. By simply adding the  $z$  coordinate, which is then treated in a cartesian like manner, every point in space is determined by the  $r$  and  $\theta$  coordinates of its projection in the  $xy$  plane, and its  $z$  coordinate.



The unit vectors  $\mathbf{e}_r$ ,  $\mathbf{e}_\theta$  and  $\mathbf{k}$ , expressed in cartesian coordinates, are,

$$\begin{aligned}\mathbf{e}_r &= \cos \theta \mathbf{i} + \sin \theta \mathbf{j} \\ \mathbf{e}_\theta &= -\sin \theta \mathbf{i} + \cos \theta \mathbf{j}\end{aligned}$$

and their derivatives are

$$\dot{\mathbf{e}}_r = \dot{\theta} \mathbf{e}_\theta, \quad \dot{\mathbf{e}}_\theta = -\dot{\theta} \mathbf{e}_r, \quad \dot{\mathbf{k}} = \mathbf{0}.$$

The kinematic vectors can now be expressed relative to the unit vectors  $\mathbf{e}_r$ ,  $\mathbf{e}_\theta$  and  $\mathbf{k}$ . Thus, the position vector is

$$\mathbf{r} = r \mathbf{e}_r + z \mathbf{k},$$

and the velocity,

$$\mathbf{v} = \dot{r} \mathbf{e}_r + r \dot{\theta} \mathbf{e}_\theta + \dot{z} \mathbf{k},$$

where  $v_r = \dot{r}$ ,  $v_\theta = r \dot{\theta}$ ,  $v_z = \dot{z}$ , and  $v = \sqrt{v_r^2 + v_\theta^2 + v_z^2}$ . Finally, the acceleration becomes

$$\mathbf{a} = (\ddot{r} - r \dot{\theta}^2) \mathbf{e}_r + (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) \mathbf{e}_\theta + \ddot{z} \mathbf{k},$$

where  $a_r = \ddot{r} - r \dot{\theta}^2$ ,  $a_\theta = r \ddot{\theta} + 2 \dot{r} \dot{\theta}$ ,  $a_z = \ddot{z}$ , and  $a = \sqrt{a_r^2 + a_\theta^2 + a_z^2}$ .

*Note that when using cylindrical coordinates,  $r$  is not the modulus of  $\mathbf{r}$ . This is somewhat confusing, but it is consistent with the notation used by most books. Whenever we use cylindrical coordinates, we will write  $|\mathbf{r}|$  explicitly, to indicate the modulus of  $\mathbf{r}$ , i.e.  $|\mathbf{r}| = \sqrt{r^2 + z^2}$ .*

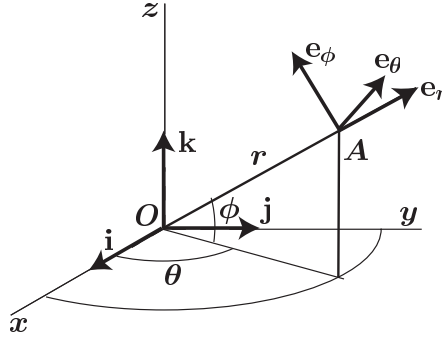
## Equations of Motion

In cylindrical  $r\theta z$  coordinates, the force and acceleration vectors are  $\mathbf{F} = F_r \mathbf{e}_r + F_\theta \mathbf{e}_\theta + F_z \mathbf{e}_z$  and  $\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta + a_z \mathbf{e}_z$ . Thus, in component form,

$$\begin{aligned}F_r &= m a_r = m (\ddot{r} - r \dot{\theta}^2) \\ F_\theta &= m a_\theta = m (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) \\ F_z &= m a_z = m \ddot{z}.\end{aligned}$$

## Spherical Coordinates ( $r - \theta - \phi$ )

Spherical coordinates utilize two angles and a distance to specify the position of a particle.



The unit vectors written in cartesian coordinates are,

$$\begin{aligned} \mathbf{e}_r &= \cos \theta \cos \phi \mathbf{i} + \sin \theta \cos \phi \mathbf{j} + \sin \phi \mathbf{k} \\ \mathbf{e}_\theta &= -\sin \theta \mathbf{i} + \cos \theta \mathbf{j} \\ \mathbf{e}_\phi &= -\cos \theta \sin \phi \mathbf{i} - \sin \theta \sin \phi \mathbf{j} + \cos \phi \mathbf{k} \end{aligned}$$

The derivation of expressions for the velocity and acceleration follow easily once the derivatives of the unit vectors are known. In three dimensions, the geometry is somewhat more involved, but the ideas are the same. Here, we give the results for the derivatives of the unit vectors,

$$\dot{\mathbf{e}}_r = \dot{\theta} \cos \phi \mathbf{e}_\theta + \dot{\phi} \mathbf{e}_\phi, \quad \dot{\mathbf{e}}_\theta = -\dot{\theta} \cos \phi \mathbf{e}_r + \dot{\theta} \sin \phi \mathbf{e}_\phi, \quad \dot{\mathbf{e}}_\phi = -\dot{\phi} \mathbf{e}_r - \dot{\theta} \sin \phi \mathbf{e}_\theta,$$

and for the kinematic vectors,

$$\begin{aligned} \mathbf{r} &= r \mathbf{e}_r \\ \mathbf{v} &= \dot{r} \mathbf{e}_r + r \dot{\theta} \cos \phi \mathbf{e}_\theta + r \dot{\phi} \mathbf{e}_\phi \\ \mathbf{a} &= (\ddot{r} - r \dot{\theta}^2 \cos^2 \phi - r \dot{\phi}^2) \mathbf{e}_r \\ &\quad + (2\dot{r} \dot{\theta} \cos \phi + r \ddot{\theta} \cos \phi - 2r \dot{\theta} \dot{\phi} \sin \phi) \mathbf{e}_\theta \\ &\quad + (2\dot{r} \dot{\phi} + r \dot{\theta}^2 \sin \phi \cos \phi + r \ddot{\phi}) \mathbf{e}_\phi. \end{aligned}$$

## Equations of Motion

Finally, in spherical  $r\theta\phi$  coordinates, we write  $\mathbf{F} = F_r \mathbf{e}_r + F_\theta \mathbf{e}_\theta + F_\phi \mathbf{e}_\phi$  and  $\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta + a_\phi \mathbf{e}_\phi$ . Thus,

$$\begin{aligned} F_r &= m a_r = m (\ddot{r} - r \dot{\theta}^2 \cos^2 \phi - r \dot{\phi}^2) \\ F_\theta &= m a_\theta = m (2\dot{r} \dot{\theta} \cos \phi + r \ddot{\theta} \cos \phi - 2r \dot{\theta} \dot{\phi} \sin \phi) \\ F_\phi &= m a_\phi = m (2\dot{r} \dot{\phi} + r \dot{\theta}^2 \sin \phi \cos \phi + r \ddot{\phi}). \end{aligned}$$

## Application Examples

We will look at some applications of Newton's second law, expressed in the different coordinate systems that have been introduced. Recall that Newton's second law

$$\mathbf{F} = m\mathbf{a} , \quad (10)$$

is a vector equation which is valid for inertial observers.

In general, we will be interested in determining the motion of a particle given that we know the external forces. Equation (10), written in terms of either velocity or position, is a differential equation. In order to calculate the velocity and position as a function of time we will need to integrate this equation either analytically or numerically. On the other hand, the reverse problem of computing the forces given motion is much easier and only requires direct evaluation of (10). It is also common to have mixed type problems, in which we know some components of the force and some components of the acceleration. The goal is then to determine the remaining unknown terms.

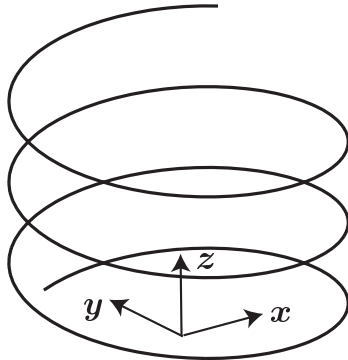
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### Example

### Aircraft flying on a helix

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A 10,000 lb aircraft is descending on a cylindrical helix. The rate of descent is  $\dot{z} = -10\text{ft/s}$ , the speed is  $v = 211\text{ ft/s}$ , and  $\dot{\theta} = 3^\circ \approx 0.05\text{rad/s}$ . This is standard for gas turbine powered aircraft. We want to know the force on the aircraft and the radius of curvature ( $\rho$ ) of the path.



We have,

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta + \dot{z}\mathbf{e}_z = v\mathbf{e}_t$$

Since,  $r = R$ ,  $\dot{r} = 0$ . Therefore,  $211 = \sqrt{(0.05R)^2 + 10^2}$ , or  $R = 4,215\text{ ft}$ . For the acceleration,

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta + \ddot{z}\mathbf{e}_z = \dot{v}\mathbf{e}_t + \frac{v^2}{\rho}\mathbf{e}_n ,$$

where  $\mathbf{e}_t$  is the unit vector tangential to the trajectory,  $\mathbf{e}_n$  is the unit vector normal to the trajectory in the direction of radius of curvature, and  $\rho$  is the radius of curvature. Then, considering only the non-zero terms,

$$\mathbf{a} = -R\dot{\theta}^2\mathbf{e}_r = \frac{v^2}{\rho}\mathbf{e}_n .$$

We see that  $\mathbf{e}_n = -\mathbf{e}_r$ , and that,

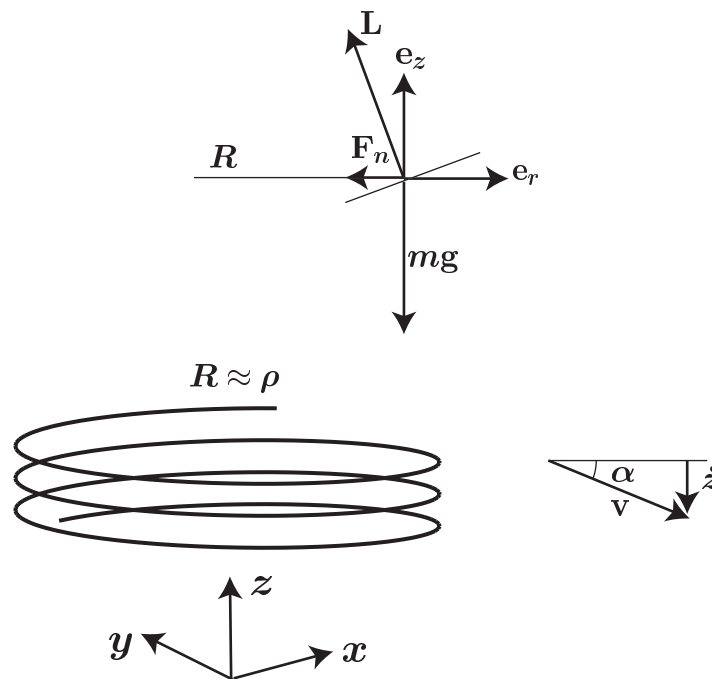
$$a = (0.05)^2 4,215 = 10.54 \text{ ft/s}^2 = \frac{v^2}{\rho}, \quad \rho = \frac{211}{10.54} = 4,225 \text{ ft} .$$

The normal force on the aircraft is

$$F_n = ma_n = \frac{10,000}{32} 10.54 = 3,273 \text{ lb} ,$$

and finally, the lift,  $\mathbf{L}$ , is

$$\mathbf{L} = -3,273 \mathbf{e}_r + 10,000 \mathbf{e}_z \text{ lb} .$$



The angle of descent  $\alpha$  is calculated as  $\sin \alpha = -\dot{z}/v$ , or,  $\alpha = -2.72^\circ$ . This angle is sometimes called the *pitch* of the helix.

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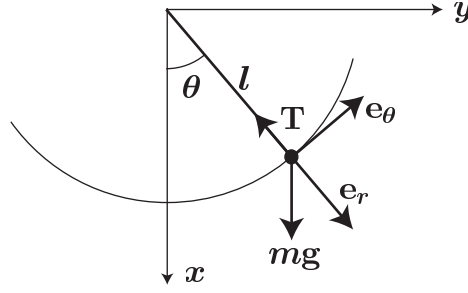
**Example**


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**Pendulum**

Now consider a simple pendulum consisting of a mass,  $m$ , suspended from a string of length  $l$  and negligible mass.





We can formulate the problem in polar coordinates, and noting that  $r = l$  (constant), write for the  $r$  and  $\theta$  components of acceleration  $\mathbf{a}$  and force  $\mathbf{F}$ ,

$$a_r = \ddot{r} - r\dot{\theta}^2 \quad (11)$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} \quad (12)$$

$$F_r = mg \cos \theta - T \quad (13)$$

$$F_\theta = -mg \sin \theta \quad (14)$$

to express the components of Newton's law (noting that  $\dot{r} = 0$  and  $\ddot{r} = 0$ )

$$\begin{aligned} mg \cos \theta - T &= -ml\dot{\theta}^2 \\ -mg \sin \theta &= ml\ddot{\theta} , \end{aligned} \quad (15)$$

where  $T$  is the tension on the string resulting in

$$\ddot{\theta} + g/l \sin \theta = 0 \quad (16)$$

a nonlinear differential equation. If we restrict the motion to small oscillations, we can approximate  $\sin \theta \approx \theta$ , and the  $\theta$ -equation becomes

$$\ddot{\theta} + \frac{g}{l} \theta = 0 .$$

Integrating, we obtain the general solution,

$$\theta(t) = C_1 \cos(\sqrt{\frac{g}{l}} t) + C_2 \sin(\sqrt{\frac{g}{l}} t) ,$$

where the constants  $C_1$  and  $C_2$  are determined by the initial conditions. Thus, if  $\theta(0) = \theta_{\max}$ ,

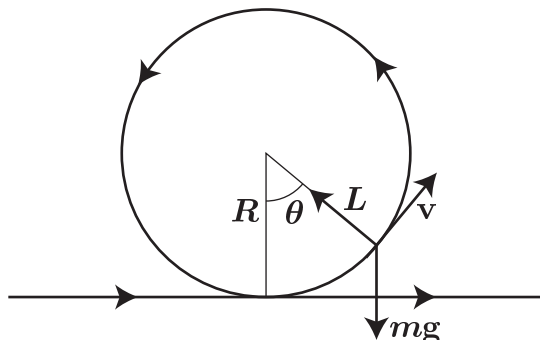
$$\theta(t) = \theta_{\max} \cos(\sqrt{\frac{g}{l}} t) .$$

### Example

### Aircraft flying a perfect loop (Hollister)

Consider an aircraft flying a perfect loop, i.e. a circle in the vertical plane. Assume that the engine thrust

exactly cancels the aerodynamic drag so that the lift and gravity are the only unbalanced forces on the aircraft. This assumption makes the problem into the same dynamical model used in the previous example.



Since the lift,  $\mathbf{L}$ , is perpendicular to the flight path, we have that the force on the aircraft, in normal and tangential components, is

$$\mathbf{F} = -mg \sin \theta \mathbf{e}_t + (L - mg \cos \theta) \mathbf{e}_n .$$

Thus,

$$\begin{aligned} a_t &= \dot{v} = r\ddot{\theta} = -g \sin \theta \\ a_n &= \frac{v^2}{R} = \frac{L}{m} - g \cos \theta . \end{aligned} \quad (17)$$

Since,  $v dv = a_t ds = a_t R d\theta = -Rg \sin \theta d\theta$ . Thus, integrating,

$$v^2 = v_0^2 + 2Rg(\cos \theta - 1) , \quad (18)$$

where  $v_0$  is the velocity at the bottom of the loop when  $\theta = 0$ . To be able to go over the top we need  $v > 0$  when  $\theta = \pi$ . This means that we need  $v_0 > 2\sqrt{Rg}$ .

Note that for  $v_0 < 2\sqrt{Rg}$ , we can calculate the maximum angle the aircraft can reach,  $\theta_{max}$ . Setting  $v = 0$  when  $\theta = \theta_{max}$ ,

$$\theta_{max} = \cos^{-1}\left(1 - \frac{v_0^2}{2Rg}\right) .$$

The necessary lift,  $L$ , can be calculated as a function of  $\theta$ . From (17) and (18),

$$\frac{L}{m} = \frac{v^2}{R} + g \cos \theta = \frac{v_0^2}{R} + 3g \cos \theta - 2g .$$

Thus, in order for  $\theta$  to go from 0 to  $\pi$ , the aircraft has to have a range of lift capability that extends over  $5g$ . It turns out that most aircraft do not have this capability and consequently do not fly perfect loops.

#### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

2/6, 2/7, 3/5

## Lecture 7 — Intrinsic Coordinates

In many problems, it is useful to use a coordinate system that aligns with both the velocity vector of the particle — which is of course tangent to the particle's trajectory — and the normal to this trajectory, forming a pair of orthogonal unit vectors. The unit vectors aligned with these two directions also define a third direction, called the binormal, which is normal to both the velocity vector and the normal vector. This can be obtained from these two unit vectors by taking their cross product. Such a coordinate system is called an *intrinsic* coordinate system.

### Tangential, Normal and Binormal components.

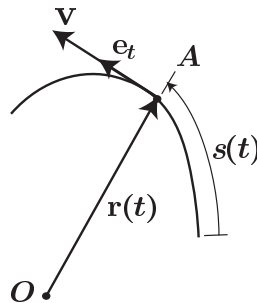
We follow the motion of a point using a vector  $\mathbf{r}(t)$  whose position along a known curved path is given by the scalar function  $s(t)$  representing the arc length along the curve. We obtain the velocity  $\mathbf{v}$  from the time rate of change of the vector  $\mathbf{r}(t)$  following the particle

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d\mathbf{r}}{ds} \frac{ds}{dt} = \dot{s} \frac{d\mathbf{r}}{ds} = \dot{s} \mathbf{e}_t \quad (1)$$

We identify the scalar  $\dot{s}$  as the speed, the scalar magnitude of the velocity  $\mathbf{v}$ , and  $\frac{d\mathbf{r}}{ds} = \mathbf{e}_t$  as the unit vector tangent to the curve at the point  $s(t)$ . Therefore we have

$$\mathbf{v} = v \mathbf{e}_t, \quad (2)$$

where  $\mathbf{r}(t)$  is the position vector,  $v = \dot{s}$  is the speed,  $\mathbf{e}_t$  is the unit tangent vector to the trajectory, and  $s$  is the path coordinate along the trajectory.



The unit tangent vector can be written as,

$$\mathbf{e}_t = \frac{d\mathbf{r}}{ds} . \quad (3)$$

The acceleration vector is the derivative of the velocity vector with respect to time. Since  $\frac{d\mathbf{r}}{ds}$  depends only on  $s$ , using the chain rule we can write

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \ddot{s} \frac{d\mathbf{r}}{ds} + \dot{s} \frac{d}{dt} \left( \frac{d\mathbf{r}}{ds} \right) = \ddot{s} \frac{d\mathbf{r}}{ds} + \dot{s}^2 \frac{d^2\mathbf{r}}{ds^2} = \dot{v} \frac{d\mathbf{r}}{ds} + v^2 \frac{d^2\mathbf{r}}{ds^2} = \dot{v} \mathbf{e}_t + v^2 \frac{d^2\mathbf{r}}{ds^2} . \quad (4)$$

The second derivative  $\frac{d^2\mathbf{r}}{ds^2}$  is another vector property of the path. We shall see that it is related to the local radius of curvature  $\rho$ :  $\frac{d^2\mathbf{r}}{ds^2} = \frac{\mathbf{e}_n}{\rho}$ .

Taking the time derivative of Equation (2), an alternate expression can be written in terms of the unit vector  $\mathbf{e}_t$  as

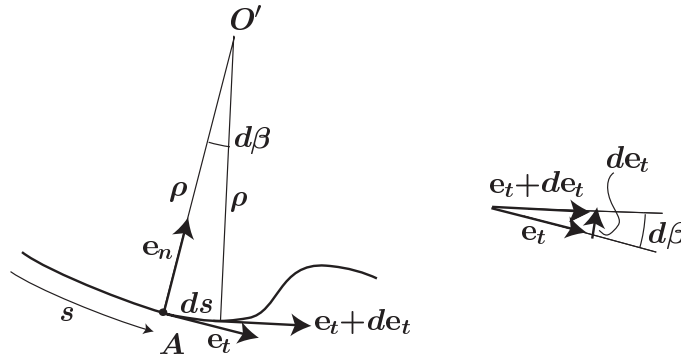
$$\mathbf{a} = \frac{dv}{dt} \mathbf{e}_t + v \frac{d\mathbf{e}_t}{dt} . \quad (5)$$

The vector  $\mathbf{e}_t$  is the local unit tangent vector to the curve which changes from point to point. Consequently, the time derivative of  $\mathbf{e}_t$  will, in general, be nonzero.

The time derivative of  $\mathbf{e}_t$  can be written as,

$$\frac{d\mathbf{e}_t}{dt} = \frac{d\mathbf{e}_t}{ds} \frac{ds}{dt} = \frac{d\mathbf{e}_t}{ds} v . \quad (6)$$

In order to calculate the derivative of  $\mathbf{e}_t$ , we note that, since the magnitude of  $\mathbf{e}_t$  is constant and equal to one, the only changes that  $\mathbf{e}_t$  can have are due to rotation, or *swinging*.



When we move from  $s$  to  $s + ds$ , the tangent vector changes from  $\mathbf{e}_t$  to  $\mathbf{e}_t + d\mathbf{e}_t$ . The change in direction can be related to the angle  $d\beta$ .

The direction of  $d\mathbf{e}_t$ , which is perpendicular to  $\mathbf{e}_t$ , is called the *normal* direction. On the other hand, the magnitude of  $d\mathbf{e}_t$  will be equal to the length of  $\mathbf{e}_t$  (which is one), times  $d\beta$ . Thus, if  $\mathbf{e}_n$  is a *unit normal vector* in the direction of  $d\mathbf{e}_t$ , we can write

$$d\mathbf{e}_t = d\beta \mathbf{e}_n . \quad (7)$$

From the figure, we see that the magnitude of  $ds$  is equal to  $\rho d\beta$ : so that  $\frac{d\beta}{ds} = \frac{1}{\rho}$

Dividing by  $ds$  yields,

$$\frac{d\mathbf{e}_t}{ds} = \frac{d\beta}{ds} \mathbf{e}_n = \kappa \mathbf{e}_n = \frac{1}{\rho} \mathbf{e}_n. \quad (8)$$

Here,  $\kappa = d\beta/ds$  is a local property of the curve, called the *curvature*, and  $\rho = 1/\kappa$  is called the *radius of curvature*. Combining equation (5) with  $\frac{d\mathbf{e}_t}{dt} = v \frac{d\mathbf{e}_t}{ds}$  (6) and  $\frac{d\mathbf{e}_t}{ds} = \frac{1}{\rho} \mathbf{e}_n$  (8) we have

$$\mathbf{a} = \frac{dv}{dt} \mathbf{e}_t + v \frac{d\mathbf{e}_t}{dt} = \frac{dv}{dt} \mathbf{e}_t + v^2 \frac{d\mathbf{e}_t}{ds} = \frac{dv}{dt} \mathbf{e}_t + \frac{v^2}{\rho} \mathbf{e}_n \quad (9)$$

Thus the acceleration in local intrinsic coordinates is the time rate of change of **speed**  $v$  in the tangential direction plus the **speed** squared divided by the local radius of curvature in the direction normal to the path  $s(t)$ .

(Note that in the picture, the sizes of  $d\mathbf{e}_t$ ,  $ds$ , and  $d\beta$  are exaggerated for illustration purposes and actually represent the changes in the limit as  $ds$  (and also  $dt$ ) approach zero.)

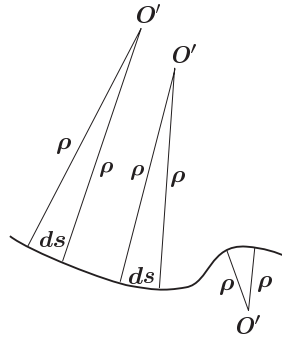
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### Note

### Curvature and radius of curvature

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We consider here two tangent vectors  $\mathbf{e}_t$  and  $\mathbf{e}_t + d\mathbf{e}_t$ , separated by a small  $ds$  and having an angle between them of  $d\beta$ . If we draw perpendiculars to these two vectors, they will intersect at a point, say,  $O'$ . Because the two lines meeting at  $O'$  are perpendicular to each of the tangent vectors, the angle between them will be the same as the angle between  $\mathbf{e}_t$  and  $\mathbf{e}_t + d\mathbf{e}_t$ ,  $d\beta$ . The point  $O'$  is called the *center of curvature*, and the distance,  $\rho$ , between  $O'$  and  $A$  is the radius of curvature. Thus, from the sketch, we have that  $ds = \rho d\beta$ , or  $d\beta/ds = \kappa = 1/\rho$ .



Intuitively, we can see that each infinitesimal arc,  $ds$ , can be represented by a circle segment of radius  $\rho$  having its center at the center of curvature. It is clear that both the radius of curvature and the center of curvature are functions of  $s$ , and consequently they change from point to point.

There are two limiting cases which are of interest. When the trajectory is a circle, the center of curvature does not change and coincides with the center of the circle, and the radius of curvature is equal to the radius of the circle. On the other hand, when the trajectory is a straight line, the curvature is zero, and the radius of curvature is infinite. Note also, that, in this case, the derivative of  $\mathbf{e}_t$  is always zero, and the normal direction is not defined.

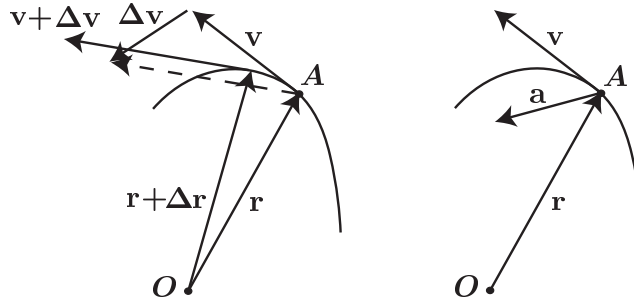
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Going back to expression (6), we have that

$$\frac{d\mathbf{e}_t}{dt} = \frac{d\beta}{ds} v \mathbf{e}_n = \dot{\beta} \mathbf{e}_n = \frac{v}{\rho} \mathbf{e}_n . \quad (10)$$

Finally, we have that the acceleration, can be written as

$$\mathbf{a} = \frac{dv}{dt} \mathbf{e}_t + \frac{v^2}{\rho} \mathbf{e}_n = a_t \mathbf{e}_t + a_n \mathbf{e}_n . \quad (11)$$

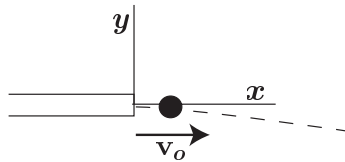


Here,  $a_t = \dot{v}$ , is the *tangential* component of the acceleration, and  $a_n = v^2/\rho$ , is the *normal* component of the acceleration. Since  $a_n$  is the component of the acceleration pointing towards the center of curvature, it is sometimes referred to as *centripetal* acceleration. When  $a_t$  is nonzero, the velocity vector changes magnitude, or *stretches*. When  $a_n$  is nonzero, the velocity vector changes direction, or *swings*. The modulus of the total acceleration can be calculated as  $a = \sqrt{a_t^2 + a_n^2}$ .

---

### Example

A ball is ejected horizontally from the tube with a speed  $v_0$ . The only acceleration on the ball is due to gravity. We want to determine the radius of curvature of the trajectory just after the ball is released.



The simplest way to determine the radius of curvature is to note that, initially, the only nonzero component of the acceleration will be in the normal direction, i.e.  $a_n = g$ . Thus, from  $a_n = v_0^2/\rho$ , we have that,

$$\rho = \frac{v_0^2}{g} .$$

Alternatively, we can obtain an equation for the trajectory of the form  $y = f(x)$  and use expression (12) to calculate the curvature. The trajectory is given as,

$$\begin{aligned} x &= v_0 t \\ y &= -\frac{1}{2} g t^2 . \end{aligned}$$

Thus, eliminating  $t$ , we have

$$y = -\frac{g}{2v_0^2}x^2.$$

At  $x = 0$ ,  $dy/dx = 0$ ,  $d^2y/dx^2 = -g/v_0^2$ , and the above expression gives,  $\rho = v_0^2/g$ , as expected.

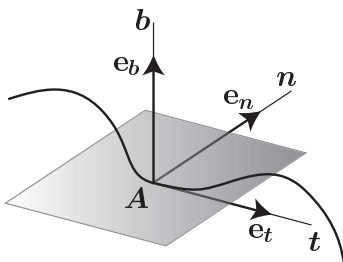
### Note

### Relationship between $s$ , $v$ and $a_t$

The quantities  $s$ ,  $v$  and  $a_t$  are related in the same manner as the quantities  $s$ ,  $v$  and  $a$  for rectilinear motion. In particular we have that  $v = \dot{s}$ ,  $a_t = \dot{v}$ , and  $a_t ds = v dv$ . This means that if we have a way of knowing  $a_t$ , we may be able to integrate the tangential component of the motion independently. We will be exploiting these relations in the future.

The vectors  $\mathbf{e}_t$  and  $\mathbf{e}_n$ , and their respective coordinates  $t$  and  $n$ , define two orthogonal directions. The plane defined by these two directions, is called the *osculating* plane. This plane changes from point to point, and can be thought of as the plane that locally contains the trajectory (Note that the tangent is the current direction of the velocity, and the normal is the direction into which the velocity is changing).

In order to define a right-handed set of axes we need to introduce an additional unit vector which is orthogonal to  $\mathbf{e}_t$  and  $\mathbf{e}_n$ . This vector is called the *binormal*, and is defined as  $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$ .



At any point in the trajectory, the position vector, the velocity and acceleration can be referred to these axes. In particular, the velocity and acceleration take very simple forms,

$$\begin{aligned}\mathbf{v} &= v\mathbf{e}_t \\ \mathbf{a} &= \dot{v}\mathbf{e}_t + \frac{v^2}{\rho}\mathbf{e}_n.\end{aligned}$$

The difficulty of working with this reference frame stems from the fact that the orientation of the axis depends on the trajectory itself. The position vector,  $\mathbf{r}$ , needs to be found by integrating the relation  $d\mathbf{r}/dt = \mathbf{v}$ ,

$$\mathbf{r} = \mathbf{r}_0 + \int_0^t \mathbf{v} dt,$$

where  $\mathbf{r}_0 = \mathbf{r}(0)$  is given by the initial condition.

We note that, by construction, the component of the acceleration along the binormal is always zero.

When the trajectory is planar, the binormal stays constant (orthogonal to the plane of motion). However, when the trajectory is a space curve, the binormal changes with  $s$ . It can be shown (see note below) that the

derivative of the binormal is always along the direction of the normal. The rate of change of the binormal with  $s$  is called the *torsion*,  $\tau$ . Thus,

$$\frac{d\mathbf{e}_b}{ds} = -\tau \mathbf{e}_n \quad \text{or,} \quad \frac{d\mathbf{e}_b}{dt} = -\tau v \mathbf{e}_n.$$

We see that whenever the torsion is zero, the trajectory is planar, and whenever the curvature is zero, the trajectory is linear.

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**Note**
**Calculation of the radius of curvature and torsion for a trajectory**


---

In some situations the trajectory will be known as a curve of the form  $y = f(x)$ . The radius of curvature in this case can be computed according to the expression,

$$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}. \quad (12)$$

This expression is not hard to derive. Try it! Since  $y = f(x)$  defines a planar curve, the torsion  $\tau$  is zero. On the other hand, if the trajectory is known in parametric form as a curve of the form  $\mathbf{r}(t)$ , where  $t$  can be time, but also any other parameter, then the radius of curvature  $\rho$  and the torsion  $\tau$  can be computed as

$$\rho = \frac{(\dot{\mathbf{r}} \cdot \dot{\mathbf{r}})^{3/2}}{\sqrt{(\dot{\mathbf{r}} \cdot \dot{\mathbf{r}})(\ddot{\mathbf{r}} \cdot \ddot{\mathbf{r}}) - (\dot{\mathbf{r}} \cdot \ddot{\mathbf{r}})^2}},$$

where  $\dot{\mathbf{r}} = d\mathbf{r}/dt$ , and  $\ddot{\mathbf{r}} = d^2\mathbf{r}/dt^2$  and

$$\tau = \frac{(\dot{\mathbf{r}} \times \ddot{\mathbf{r}}) \cdot \ddot{\ddot{\mathbf{r}}}}{|\dot{\mathbf{r}} \times \ddot{\mathbf{r}}|^2} \quad (13)$$

where  $\ddot{\ddot{\mathbf{r}}} = d^3\mathbf{r}/dt^3$ .

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## Equations of Motion in Intrinsic Coordinates

Newton's second law is a vector equation,  $\mathbf{F} = m\mathbf{a}$ , which can now be written in intrinsic coordinates.

In tangent, normal and binormal components, *tnb*, we write  $\mathbf{F} = F_t\mathbf{e}_t + F_n\mathbf{e}_n$  and  $\mathbf{a} = a_t\mathbf{e}_t + a_n\mathbf{e}_n$ . We observe that the positive direction of the normal coordinate is that pointing towards the center of curvature. Thus, in component form, we have

$$\begin{aligned} F_t &= m a_t = m \dot{v} = m \ddot{s} \\ F_n &= m a_n = m \frac{v^2}{\rho} \end{aligned}$$

Note that, by definition, the component of the acceleration along the binormal direction,  $\mathbf{e}_b$ , is always zero, and consequently the binormal component of the force must also be zero. This may seem surprising, at first, but recall that the tangent and normal directions are determined by the motion, and, hence, we can say that the motion “chooses” the binormal direction to be always orthogonal to the applied force. In other words,



if we apply a force to a particle, the particle will experience an acceleration which is parallel to the force. The normal direction is chosen so that the acceleration vector is always contained in the plane defined by the tangent and the normal. Thus, the binormal is always orthogonal to the external force.

Intrinsic coordinates are sometimes useful when we are dealing with problems in which the motion is constrained, such as a car on a roller coaster. The geometry of the trajectory is known, and, therefore, the directions of the tangent, normal and binormal vectors are also known. In these cases it may be possible to integrate the component of the equation of motion along the tangential direction (especially if there is no friction), and then calculate, a-posteriori, the reaction force using the normal component of the equation of motion.

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**Note (optional)**
**Frenet formulae**


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The Frenet formulae give us the variations of the unit vectors  $\mathbf{e}_t$ ,  $\mathbf{e}_n$  and  $\mathbf{e}_b$  with respect to the path coordinate  $s$ . The first formula

$$\frac{d\mathbf{e}_t}{ds} = \frac{1}{\rho} \mathbf{e}_n ,$$

has already been defined. Now, since  $\mathbf{e}_b$  is a unit vector,  $d\mathbf{e}_b/ds$  will be orthogonal to  $\mathbf{e}_b$ . Hence, it will be of the form,

$$\frac{d\mathbf{e}_b}{ds} = b_t \mathbf{e}_t + b_n \mathbf{e}_n .$$

If we perform the dot product of this expression with  $\mathbf{e}_t$ , we obtain

$$b_t = \frac{d\mathbf{e}_b}{ds} \cdot \mathbf{e}_t = -\frac{d\mathbf{e}_t}{ds} \cdot \mathbf{e}_b = -\frac{1}{\rho} \mathbf{e}_n \cdot \mathbf{e}_b = 0 .$$

The second equality follows from the fact that the derivative of  $\mathbf{e}_t \cdot \mathbf{e}_b$  is zero, i.e.  $d\mathbf{e}_t \cdot \mathbf{e}_b + \mathbf{e}_t \cdot d\mathbf{e}_b = 0$ . Therefore, only  $b_n$  is nonzero. Defining  $b_n = -\tau$ , we obtain the third Frenet formula

$$\frac{d\mathbf{e}_b}{ds} = -\tau \mathbf{e}_n .$$

Finally, the second formula can be obtained in a similar manner (we leave the details as an exercise) and gives,

$$\frac{d\mathbf{e}_n}{ds} = -\frac{1}{\rho} \mathbf{e}_t + \tau \mathbf{e}_b ,$$

or, multiplying by  $v$ ,

$$\frac{d\mathbf{e}_n}{dt} = -\frac{v}{\rho} \mathbf{e}_t + \tau v \mathbf{e}_b . \quad (14)$$

As we move along  $s$ , the osculating plane (and hence  $\mathbf{e}_b$ ) may rotate, making the curve non planar. As an example, an aeroplane may be rolling as it flies along  $\mathbf{e}_t$ . The derivative of  $\mathbf{e}_b$  is in the direction opposite to  $\mathbf{e}_n$  if the rotation is in the direction of a right hand screw, and this is taken as the positive direction for the torsion.

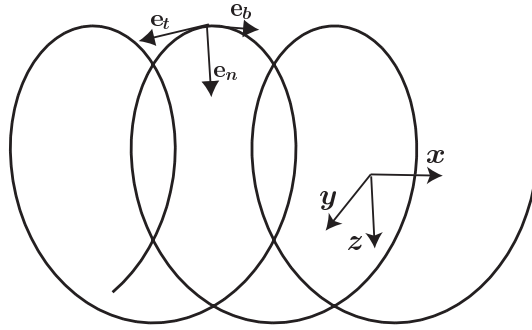
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**Example****“Simplified” Aircraft Kinematics (W. M. Hollister)**

The flight of an aircraft through the sky is an example of curvilinear motion. Think of  $\mathbf{e}_t$  as the roll axis aligned with the velocity vector of the aircraft. Think of  $\mathbf{e}_b$  as being the pitch axis. The lift is then directed along  $\mathbf{e}_n$ . The roll rate of the aircraft can be interpreted as  $\tau v$ , and the pitch rate as  $v/\rho$ . In order to turn the aircraft out of the vertical plane, it is necessary to rotate the direction of the lift  $\mathbf{e}_n$  so that there is a component of acceleration out of the vertical plane. Neglecting gravity, the velocity vector along  $\mathbf{e}_t$  determines where the aircraft is going, and the lift along  $\mathbf{e}_n$  determines where the velocity vector is going. The roll rate determines how the lift vector will be rotated out of the osculating plane. As shown by equation (14), the direction of the lift vector is changed by rolling  $\tau v$ , as well as pitching  $v/\rho$ .

Consider the following example. An aircraft follows a spiral path in the sky while doing a barrel roll. The coordinates are given below, where  $v_0 = 194$  ft/s,  $\omega = 0.4$  rad/s, and  $h = 125$  ft are constants.

$$\begin{aligned}x &= v_0 t \\y &= h \cos \omega t \\z &= h \sin \omega t\end{aligned}$$



We have,

$$\begin{aligned}\mathbf{v} &= v_0 \mathbf{i} - h\omega \sin \omega t \mathbf{j} + h\omega \cos \omega t \mathbf{k} \\v &= \sqrt{v_0^2 + h^2 \omega^2} = 200 \text{ ft/s} \approx v_0 \\ \mathbf{a} &= -h\omega^2 \cos \omega t \mathbf{j} - h\omega^2 \sin \omega t \mathbf{k} \\a &= h\omega^2 = 20 \text{ ft/s}^2\end{aligned}$$

Since  $\dot{v} = 0$ ,  $\mathbf{a} = (v^2/\rho)\mathbf{e}_n$ , or,

$$h\omega^2 = \frac{v^2}{\rho}, \quad \rho = \frac{v^2}{h\omega^2} = 2000 \text{ ft}$$

,

$$\begin{aligned}\mathbf{e}_t &= \frac{v_0}{v} \mathbf{i} - \frac{h\omega}{v} \sin \omega t \mathbf{j} + \frac{h\omega}{v} \cos \omega t \mathbf{k} \\ \mathbf{e}_n &= -\cos \omega t \mathbf{j} - \sin \omega t \mathbf{k} \\ \mathbf{e}_b &= \frac{h\omega}{v} \mathbf{i} + \frac{v_0}{v} \sin \omega t \mathbf{j} - \frac{v_0}{v} \cos \omega t \mathbf{k}\end{aligned}$$

Finally,

$$\frac{d\mathbf{e}_b}{dt} = \frac{v_0\omega}{v} \cos \omega t \mathbf{j} + \frac{v_0\omega}{v} \sin \omega t \mathbf{k}$$

which corresponds to a roll rate of

$$\tau v = \frac{v_0\omega}{v} \approx \omega = 0.4 \text{ rad/s}$$

### Example

### Aircraft flying an “imperfect” loop (Hollister)

It is more common to fly a loop keeping the normal acceleration,  $a_n$ , approximately constant at, say,  $ng$  ( $n \sim 3 - 4$ ).

Let  $\beta$  be the flight path angle which the velocity vector (or tangent) makes with the horizontal, and let  $\rho$  be the radius of curvature of the path. Then,

$$\begin{aligned} a_n &= \frac{v^2}{\rho} = v\dot{\beta} = ng \\ a_t &= -g \sin \beta . \end{aligned}$$

From,  $v dv = a_t ds$ , with  $ds = \rho d\beta$ , we have,

$$v dv = a_t \rho d\beta = (-g \sin \beta) \left( \frac{v^2}{ng} \right) d\beta ,$$

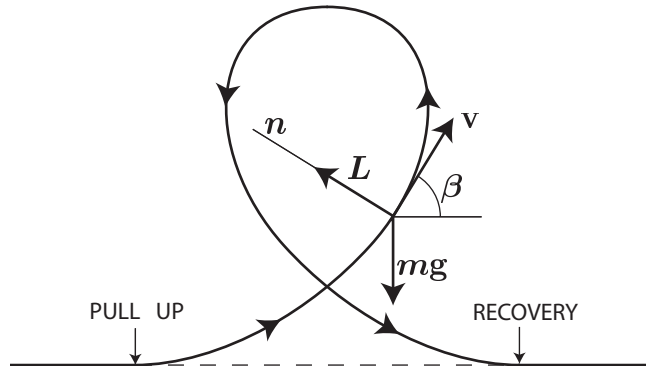
or, integrating,

$$\frac{n dv}{v} = -\sin \beta d\beta , \quad n \ln v|_{v_0}^v = \cos \beta|_0^\beta , \quad v = v_0 e^{-\frac{1}{n}(1-\cos \beta)} ,$$

and

$$\rho = \frac{v_0^2}{ng} e^{-\frac{2}{n}(1-\cos \beta)} . \quad (15)$$

A sketch of this path is shown in the figure below.

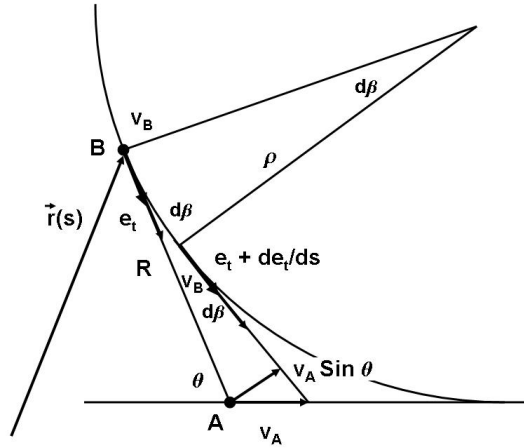


In going over the top of the imperfect loop, the aircraft does not go as high or lose as much velocity as it does going over the perfect loop. Unlike the perfect loop case however, the aircraft does need to pull up before, and recover after, the point of maximum altitude (see diagram). Note that the solution of this example and the previous one would have been rather difficult using rectangular coordinates. Note also that the form of

the solution given by equation (29) is rather unusual, e.g. the radius of curvature is given as a function of the attitude angle. A possible way to plot the trajectory starting from an initial position and velocity would be to first determine  $\beta$ , and then draw a small circle segment with the appropriate  $\rho$ . After calculating the new position, the process would be repeated to draw the entire trajectory.

**Example****Pursuit Problem**

A classic problem in particle kinematics is the pursuit problem. This has an elegant history in mathematics and practical applications in air combat. At its most basic, we have a "target" **A** traveling on a known curve at constant speed. The target is pursued by **B** whose strategy is to turn such that her velocity vector is always pointed at the target.



We take the simplest version of this problem where the velocities of the two vehicles are equal:  $v_A = v_B$  and the evading vehicle travels on a straight line in the positive  $x$  direction. Analyzing this problem using intrinsic coordinates can give us some insight into the motion, into the formalism of tangential and normal unit vectors and the role of radius of curvature of the path. We examine this problem at the instant where the distance between **A** and **B** is given by  $R$  and the angle between their directions of travel is  $\theta$ . (i.e. if the particles are moving in the same direction,  $\theta = 0$ .) We allow a time  $dt$  to elapse and examine the resulting geometry. Although the figure shows a rather large  $dt$  (to allow us to see the motion), we are interested in a limiting process as  $dt \rightarrow 0$ .

From the figure, we see that in a time  $dt$ , the vector  $\mathbf{R}$  between the two points **A** and **B** changes direction by an angle  $d\beta = v_A \sin \theta / R dt$ . Therefore the direction of the tangential vector also changes by  $d\beta$ . From our earlier analysis the change of the tangential vector due to a change in  $ds$  is given by

$$\frac{de_t}{ds} = e_n \frac{d\beta}{ds} = \frac{1}{\rho} e_n \quad (16)$$

Therefore, using the chain rule, the magnitude of radius of curvature at the point  $\mathbf{r}(t)$  is given by

$$\frac{1}{\rho} = \frac{d\beta}{ds} = \frac{\frac{d\beta}{dt}}{\frac{ds}{dt}} \quad (17)$$

Now,  $ds/dt$  is the scalar velocity of the point  $B$  along the pursuit curve, while  $d\beta/dt$  is the rate at which the angle  $\beta$  changes as a result of the motion of point  $A$ .

$$\frac{ds}{dt} = v_B \quad (18)$$

$$\frac{d\beta}{dt} = \frac{v_A \sin \theta}{R} \quad (19)$$

so that the radius of curvature at the point  $A$  is given by

$$1/\rho = \frac{v_A \sin \theta}{v_B R} \quad (20)$$

Given this, the acceleration of particle  $B$  is easily found as  $a = v_B^2/\rho$ .

#### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition  
2/5, 3/5 (normal and tangential coordinates only)

## Lecture 8 — Relative Motion using Translating Axes

The previous kinematics lectures described particle motion as it would be seen by an observer standing still at a fixed origin. This type of motion is called *absolute* motion. In many situations of practical interest, we describe the motion of bodies while we are simultaneously moving with respect to a fixed reference frame. There are many examples where such situations occur. For example, if we try to track the motion of aircraft in the airspace using satellites, it makes sense to first consider the motion of the aircraft relative to the satellite and then combine this motion with the motion of the satellite relative to the earth's surface. In this lecture we will introduce the ideas of *relative* motion analysis.

### Types of observers

We will consider four different types of observers (or reference frames) depending on their motion with respect to a fixed frame:

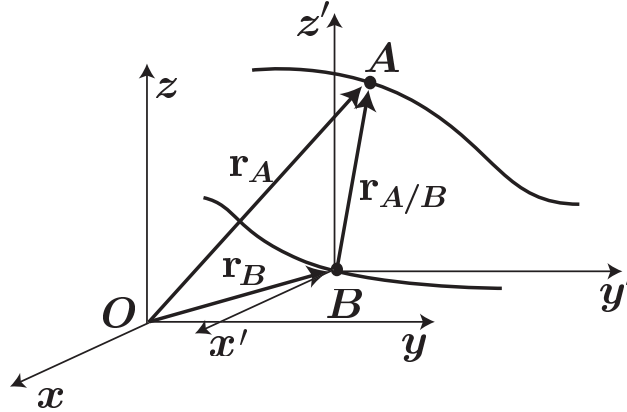
- observers who do not accelerate or rotate, i.e. those who at most have constant velocity.
- observers who accelerate but do not rotate
- observers who rotate but do not translate
- observers who accelerate and translate

This lecture considers relative motion involving observers of the first two types, and defers the study of relative motion involving rotating frames to the next lecture.

### Relative motion using translating axes

Consider two particles,  $A$  and  $B$ , in curvilinear motion along two different paths. Their motion can be described with respect to a fixed reference frame  $xyz$  with origin  $O$  and unit vectors  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$ , as before. The motion relative to this frame is called absolute. The position of particle  $A$  is given by  $\mathbf{r}_A(t)$  and the position of particle  $B$  is given by  $\mathbf{r}_B(t)$ ; both vectors are defined with respect to the fixed reference frame  $O$ . In addition, it is useful in many problems to ask "how would  $B$  describe the motion of  $A$  and how would this description be translated to the fixed inertial coordinate system  $O$ ?"

To answer this question, consider another *translating* reference frame attached to particle  $B$ ,  $x'y'z'$ , with unit vectors  $\mathbf{i}'$ ,  $\mathbf{j}'$  and  $\mathbf{k}'$ . Translating means that the angles between the axes  $xyz$  and  $x'y'z'$  do not change during the motion. Although in the figure the axes  $xyz$  are shown parallel to the axes  $x'y'z'$ , it should be clear that one could have non-parallel **translating** axes.



The position vector  $\mathbf{r}_{A/B}$  defines the position of  $A$  with respect to point  $B$  in the reference frame  $x'y'z'$ . The subscript notation “ $A/B$ ” means “ $A$  relative to  $B$ ”. The positions of  $A$  and  $B$  relative to the absolute frame are given by the vectors  $\mathbf{r}_A$  and  $\mathbf{r}_B$ , respectively. Thus,

$$\mathbf{r}_A = \mathbf{r}_B + \mathbf{r}_{A/B} .$$

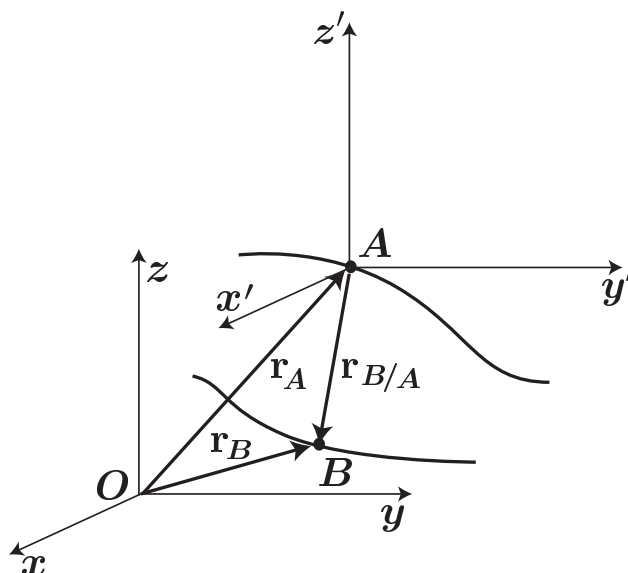
Differentiating this expression with respect to time results in

$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \dot{\mathbf{r}}_{A/B} \quad \text{or} \quad \mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B} ,$$

which relates the absolute velocities  $\mathbf{v}_A$  and  $\mathbf{v}_B$  to the relative velocity of  $A$  as observed by  $B$ . Differentiating again gives an analogous expression for the accelerations,

$$\ddot{\mathbf{r}}_A = \ddot{\mathbf{r}}_B + \ddot{\mathbf{r}}_{A/B} \quad \text{or} \quad \mathbf{a}_A = \mathbf{a}_B + \mathbf{a}_{A/B} .$$

We will now reverse the roles of  $A$  and  $B$ , asking “how would  $A$  describe the motion of  $B$ ?” We now attach the reference frame  $x'y'z'$  to  $A$ , and then we can observe  $B$  from  $A$ . The relative motion of  $B$  as seen by  $A$  is now denoted  $\mathbf{r}_{B/A}$ , the position of  $B$  as seen by  $A$ .



Using the same arguments as before,

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}, \quad \mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}, \quad \mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}.$$

Comparing these expressions with those above,

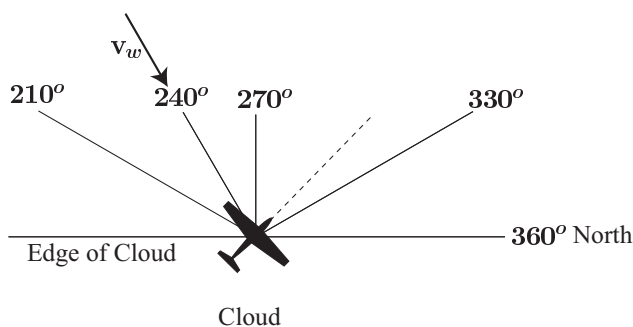
$$\mathbf{r}_{B/A} = -\mathbf{r}_{A/B}, \quad \mathbf{v}_{B/A} = -\mathbf{v}_{A/B}, \quad \mathbf{a}_{B/A} = -\mathbf{a}_{A/B}.$$

One important observation is that, whenever the moving system, say  $A$ , has a constant velocity relative to the fixed system,  $O$ , the acceleration seen by the two observers is the same, i.e. if  $\mathbf{a}_A = \mathbf{0}$ , then  $\mathbf{a}_B = \mathbf{a}_{B/A}$ . This broadens the application of Newton's second law to systems which have a constant absolute velocity.

### Example

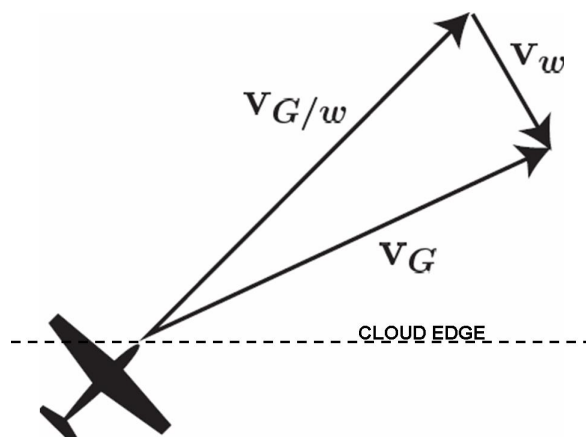
### Glider in cross wind

Consider a glider flying above the edge of a cloud which is aligned North/South ( $360^\circ$ ). The glider is flying horizontally, and the cloud is stationary with respect to the ground. At the altitude of the glider flight, there is a wind velocity,  $\mathbf{v}_w$ , of magnitude 52 knots coming from the direction  $240^\circ$ , as shown in the sketch.



The glider, on the other hand, is flying at a speed  $\mathbf{v}_{G/w}$  relative to the moving air mass and at an orientation given by his compass heading.



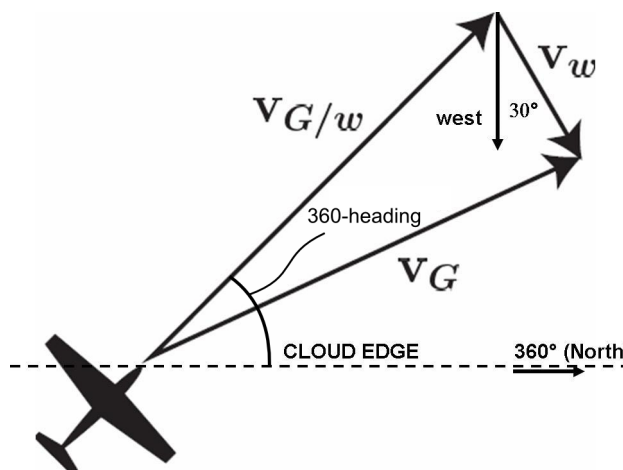


In order to determine the speed of the glider with respect to the ground, we consider a reference frame moving with the wind speed and write

$$\mathbf{v}_G = \mathbf{v}_w + \mathbf{v}_{G/w},$$

For these conditions, to balance the effect of wind speed and glider motion so that the glider has no east-west ground velocity and therefore stays at the edge of the cloud, we require

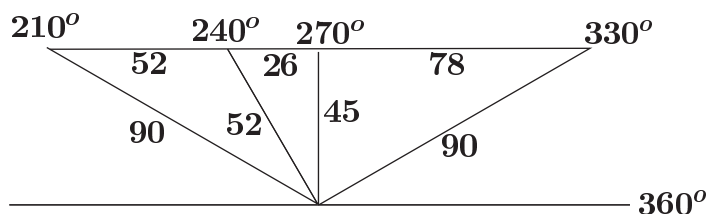
$$v_{G/w} \sin(360^\circ - \text{heading}) = v_w \cos(30^\circ) \quad (1)$$



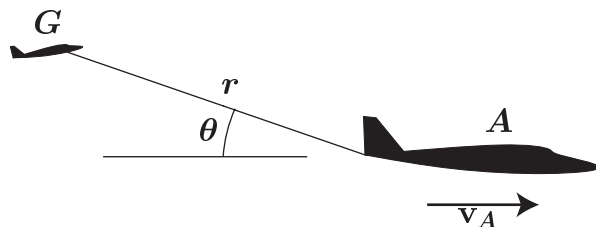
Using this diagram, by adding and subtracting the vector components of the various velocities, we can ask questions such as: if the glider is flying relative to the wind ( $v_{G/w}$ ) at a speed of 90 knots, what angle must it make with the ground—its absolute course heading—to stay at the edge of the cloud? If the glider flies with a heading of  $330^\circ$  and a speed (relative to the wind - i.e. airspeed) of 90 knots, it will stay at the edge of the cloud with a ground speed of  $104 = (26 + 78)$  knots towards the north;

By inspection of the diagram below, we can also verify the following situations: If the glider flies with a heading of  $210^\circ$  at 90 knots, the glider will stay at the edge of the cloud with a ground speed of 52 knots towards the south; if the glider flies into the wind at  $240^\circ$ , with a speed equal to 52 knots, it will remain stationary with respect to the ground; finally, the lowest speed at which the glider can fly to stay at the edge

of the cloud is 45 knots and this will be possible for a heading of  $270^\circ$ . The ground speed in this case will be 26 knots towards the north.

**Example****Aircraft towing glider (Meriam)**

Airplane  $A$  is flying horizontally with a constant speed of 200km/h, and is towing a glider  $B$ . The glider is gaining altitude. The tow cable has a length  $r = 60\text{m}$ , and  $\theta$  is increasing at a constant rate of 5 degrees per second. We want to determine the magnitude of the velocity,  $\mathbf{v}_G$ , and the acceleration,  $\mathbf{a}_G$ , of the glider for the instant when  $\theta = 15^\circ$ .



The velocity of the glider will be  $\mathbf{v}_G = \mathbf{v}_A + \mathbf{v}_{G/A}$ . The velocity of the glider relative to  $A$  is best expressed using a polar coordinate system because the tow cable length  $r$  is fixed and the motion of the glider relative to the tow plane can only change in  $\theta$ . Thus, we write,

$$\mathbf{v}_{G/A} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta .$$

Since the length of the cable is constant,  $v_r = \dot{r} = 0$ , and  $\dot{\theta} = 5 \frac{\text{deg}}{\text{sec}} (2\pi\text{rad}/360\text{deg}) = 0.0873 \text{ rad/sec}$ , which gives  $v_\theta = r\dot{\theta} = 5.236 \text{ m/s}$ . We can now transform the velocity vector to cartesian coordinates to obtain,

$$\mathbf{v}_{G/A} = 5.236\mathbf{e}_\theta = 5.236 \sin 15^\circ \mathbf{i} + 5.236 \cos 15^\circ \mathbf{j} = 1.3552\mathbf{i} + 5.0576\mathbf{j} \text{ m/s} ,$$

which, when summed with  $\mathbf{v}_A$ , gives,

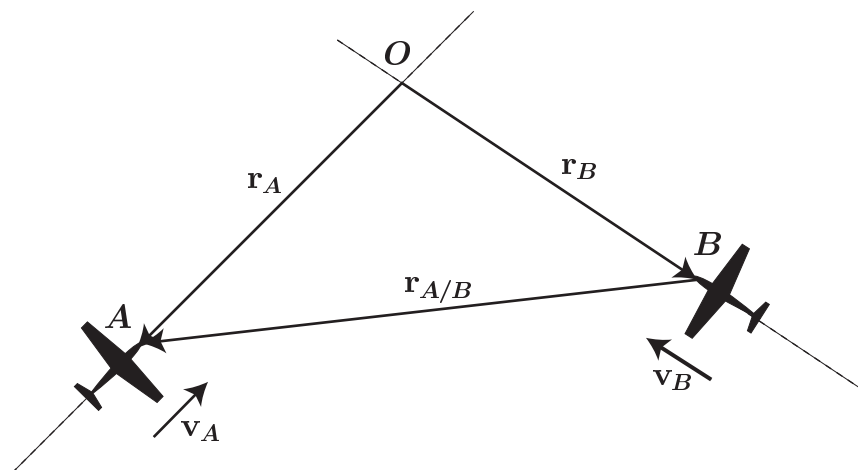
$$\mathbf{v}_G = ((200/3.6) + 1.3552)\mathbf{i} + 5.0575\mathbf{j} \text{ m/s}, \quad v_G = 55.91 \text{ m/s} = 204.9 \text{ km/h} .$$

For the acceleration  $\mathbf{a}_G = \mathbf{a}_{G/A}$ , since  $\mathbf{a}_A = \mathbf{0}$ . We also know that  $\ddot{\theta} = 0$ , and, obviously,  $\ddot{r} = 0$ , thus,

$$\mathbf{a}_{G/A} = -r\dot{\theta}^2\mathbf{e}_r = -0.457\mathbf{e}_r \text{ m/s}^2, \quad a_G = -0.457 \text{ m/s}^2 .$$

**Example****Aircraft collision avoidance**

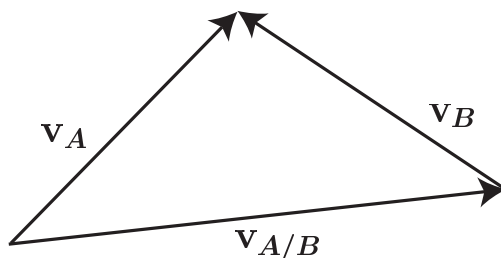
Consider two aircraft,  $A$  and  $B$ , flying horizontally at the same level with constant velocities  $\mathbf{v}_A$  and  $\mathbf{v}_B$ . Aircraft  $B$  is capable of determining by measurement the relative position and velocity of  $A$  relative to  $B$ , i.e.,  $\mathbf{r}_{A/B}$  and  $\mathbf{v}_{A/B}$ . We want to know whether it is possible to determine from this measurement whether the two aircraft are on a collision course.



Take the origin  $O$  at the hypothetical point of collision where the two trajectories intersect. If collision is to occur, the time taken for both aircraft to reach  $O$  will be the same. Therefore,

$$\frac{r_A}{v_A} = \frac{r_B}{v_B}.$$

Now consider the velocity triangle formed by  $\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B}$ .

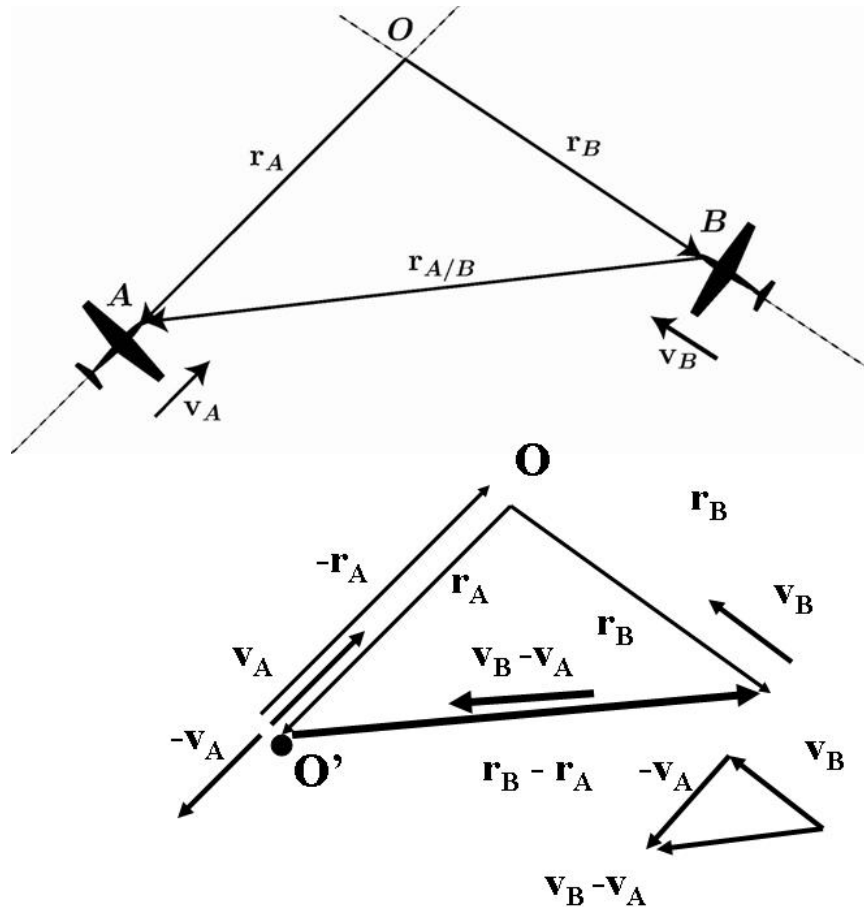


Clearly,  $\mathbf{v}_A$  is parallel to  $\mathbf{r}_A$ , and  $\mathbf{v}_B$  is parallel to  $\mathbf{r}_B$ . In addition, if collision is to occur,  $\mathbf{v}_{A/B}$  will be parallel to  $\mathbf{r}_{A/B}$ , since the position and velocity triangles will, in this case, be similar. We see, therefore, that the condition for the two planes to collide is that the relative velocity between them is parallel to the relative position vector. If aircraft  $B$  uses a polar coordinate system (common for radar measurements), then the relative position of  $A$  with respect to  $B$  will be of the form  $\mathbf{r}_{A/B} = r\mathbf{e}_r$ . Therefore, the velocity also has to be of that form, which implies that  $v_\theta = r\dot{\theta} = 0$ , or  $\theta = \text{constant}$ . That is, the angle at which  $B$  sees  $A$  (sometimes called the bearing angle) should not change.

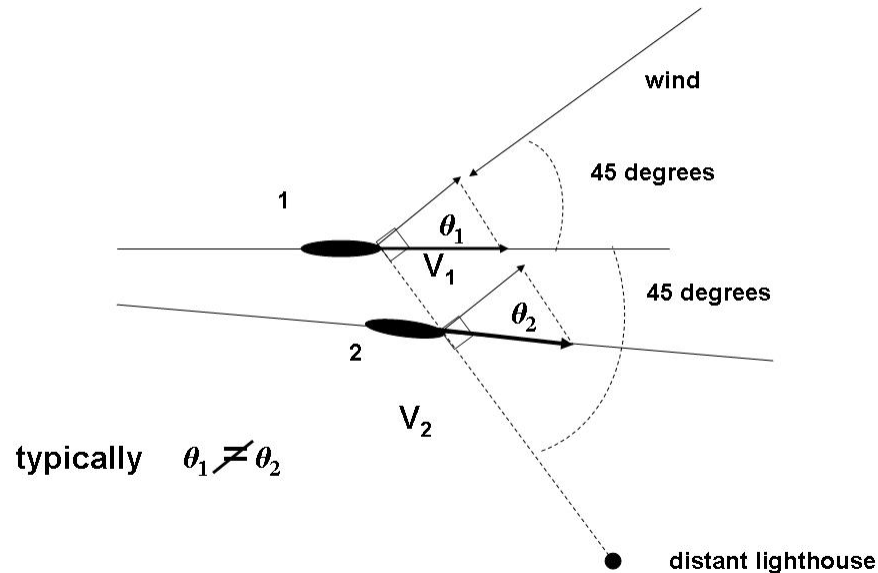
If we move into the coordinate system centered at the point  $O'$  following aircraft  $A$  (see diagram below), we can see directly what  $A$  "sees" by constructing the vectors of displacement and velocity relative to  $A$ . If  $A$ 's

## REFERENCES

position and velocity are  $r_A$  and  $v_A$  relative to coordinate center  $O$ , and  $B$ 's position and velocity relative to  $O$  are  $r_B$  and  $v_B$ , then relative to  $O'$ , the instantaneous position of  $A$ , the position and velocity vectors of  $B$  are  $r_B - r_A$  and  $v_B - v_A$ . It is clear from the figure that if  $v_B - v_A$  is parallel to  $r_B - r_A$ , a collision will occur—if  $B$  is moving towards  $A$ .

**Example****Velocity made good to windward**

In sailboat racing, the most important variable is the component of velocity made by the sailboat in the direction of the wind. In most races, there is a 'weather mark' located directly upwind. Typically sailboats can sail at an angle of  $45^\circ$  relative to the wind direction. Consider sailboat 1 making a velocity vector  $v_1$  at an angle of  $45^\circ$  with respect to the wind in the fixed coordinate system – he knows his angle relative to the wind from compass measurements. He is racing sailboat 2 who makes a velocity vector  $v_2$  at an angle  $\theta_2$  to the wind, neither of which is known to the skipper of sailboat 1.



The skipper of sailboat 1 would like to know whether he is traveling faster in the direction of the wind than sailboat 2. He sees a distant lighthouse at an angle of  $45^\circ$  from his direction of travel. He lines up his opponent with this stationary lighthouse. How can he use this observation to determine whether he is winning the race to the weather mark (which is directly upwind)?

The instantaneous observation of relative velocity in the direction of the wind can be determined by whether sailboat 2 pulls ahead of the line between the lighthouse and a fixed point on sailboat 1. Typically,  $\theta_1$  is not equal to  $\theta_2$ . Sailboat 1 can observe the sign of  $V_2 \cos \theta_2 - V_1 \cos \theta_1$  but not these quantities individually. However, that combination, which he can observe by sighting, is the difference in the components of velocities of the two sailboats in the direction of the wind. Therefore, if sailboat 2 drops behind the line of sight to the distant lighthouse, sailboat 1 is winning the race. However, if sailboat 2 pulls ahead of the line of sight to the distant lighthouse, sailboat 2 is winning the race.

#### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

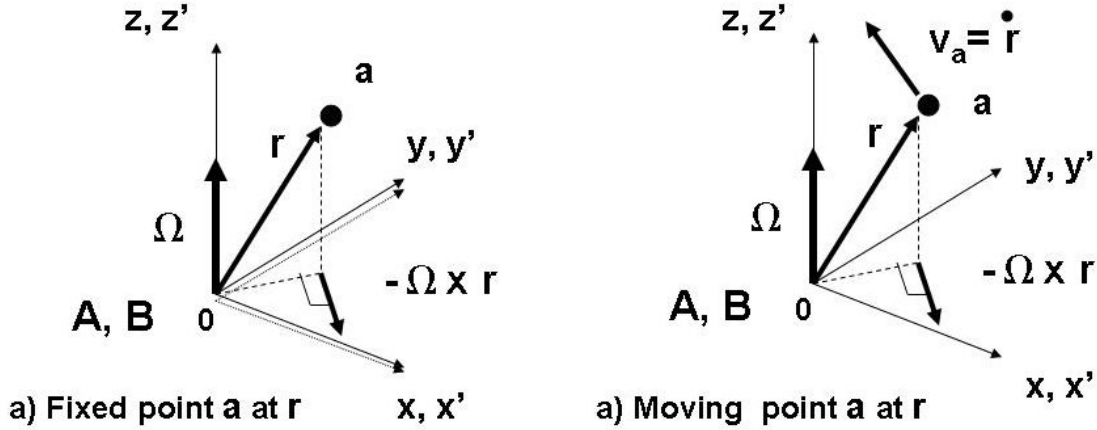
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## Lecture 9 — Relative Motion using Rotating Axes

The previous lecture covered relative motions experienced by two observers translating with respect to each other. This lecture extends this to another type of observer: one who rotates relative to a stationary observer. Although governed by the same equations, it helps to distinguish two cases: first is a rotating observer that observes a phenomena that a stationary observer would categorize as non-accelerating motion, such as a fixed object or one moving at constant velocity. The rotating observer might be a child riding on a merry-go-round or play-ground turntable. Such a rotating observer would observe the unforced motion as tracing a curved trajectory with changing velocity and therefore as accelerating motion. The transformation from the rotating into the fixed system must show that the observed motion was non-accelerating and thus force-free by Newton's law. We should also be able to describe the motion seen by the rotating observer.

The second case is where a physical phenomena is constrained to take place on a platform that is rotating, such as a mass-spring system fixed to a turntable. In this case, the motion, the forces and the frequency of oscillation will be affected by the rotation and will differ from the system behavior in an inertial coordinate system. Our approach must demonstrate this difference and provide the tools for analyzing the system dynamics in the rotating frame.

As a matter of illustration, consider a simple situation in which a point  $\mathbf{a}$  at rest with respect to the fixed observer  $A$  located at the origin of the coordinates  $x, y, z$ , point  $O$ , is also observed by a rotating observer,  $B$ , who is also located at point  $O$ . The  $x', y', z'$  coordinate system used by  $B$  is *instantaneously aligned* with  $x, y, z$  but rotating with angular velocity  $\Omega$ . In a), the coordinate system  $x, y, z$  and  $x', y', z'$  are shown slightly offset for emphasis.



Consider first the case shown in a), in which the point **a** located at position **r** does not move in inertial space. Observer *A* will observe  $\mathbf{v}_a = 0$  for the point **r**. Now consider the same situation observed by *B*. In the rotating coordinate system, *B* will observe that the point **r**, which is fixed in inertial space, appears to move backwards due to the rotation of *B*'s coordinate system,

$$\mathbf{v}_{a/B} = -\boldsymbol{\Omega} \times \mathbf{r}. \quad (1)$$

To reconcile these two observations, it is clear that to obtain the velocity in inertial coordinates, *A* must correct for the "artificial" velocity seen by *B* due to the rotation of *B*'s coordinate system by adding  $\boldsymbol{\Omega} \times \mathbf{r}$  to the velocity reported by *B*.

$$\mathbf{v}_{a/A} = \mathbf{v}_{a/B} + \boldsymbol{\Omega} \times \mathbf{r} \quad (2)$$

In this case, we obtain the rather trivial result that

$$\mathbf{v}_{a/A} = 0 = -\boldsymbol{\Omega} \times \mathbf{r} + \boldsymbol{\Omega} \times \mathbf{r}. \quad (3)$$

If as is shown in b) the point **a** is moving with velocity  $\mathbf{v}_{a/A} = \dot{\mathbf{r}}_A$  then we have the more general result

$$\mathbf{v}_{a/A} = \mathbf{v}_{a/B} + \boldsymbol{\Omega} \times \mathbf{r}. \quad (4)$$

Another way to obtain this expression is directly from Coriolis theorem relating the derivative of a vector seen by an observer in a rotating coordinate system to that seen in inertial space.

$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \boldsymbol{\Omega} \times \mathbf{r}$$

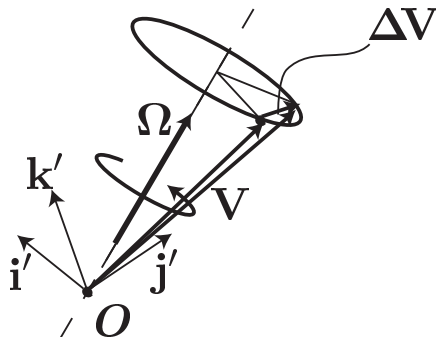
In order to apply this relationship, it is not necessary that the vector  $\boldsymbol{\Omega}$  be constant in time.  $\boldsymbol{\Omega}$  may be changing in magnitude or direction.

We have now completed the analysis of the position vector and the velocity vector in a co-located inertial and rotating coordinate system. To determine the acceleration, we again discuss the general formula for the

time derivative of a vector in a rotating frame, Coriolis Theorem.

## Time derivative of a fixed vector in a rotating frame

We consider the reference frame of  $x'y'z'$  rotating with an angular velocity  $\Omega$  with respect to a fixed frame  $xyz$  with which it is *instantaneously aligned*. We consider the more general case where  $\Omega$  is not aligned with the  $z'$  axis. Let  $\mathbf{V}$  be *any* vector, which is *constant* relative to the frame  $x'y'z'$ . That is, the vector components in the  $x'y'z'$  frame do not change, and, as a consequence,  $\mathbf{V}$  rotates as if it were rigidly attached to the frame.



In the absolute frame, the time derivative will be equal to

$$\frac{d\mathbf{V}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{(\mathbf{V} + \Delta\mathbf{V}) - \mathbf{V}}{\Delta t} = \Omega \times \mathbf{V} ,$$

which can be interpreted as the velocity of the tip of vector  $\mathbf{V}$ . The above expression applies to any vector which is rigidly attached to the frame  $x'y'z'$ .

## Time derivative of a vector in a rotating frame: Coriolis' theorem

Now let  $\mathbf{V}$  be an arbitrary vector (e.g. position, velocity), which is allowed to change in both the fixed  $xyz$  frame and the rotating  $x'y'z'$  frame. If we now consider the time derivative of  $\mathbf{V}$ , as seen by the fixed frame, we have

$$\dot{\mathbf{V}}_{xyz} = \frac{d\mathbf{V}}{dt} = (\dot{\mathbf{V}})_{x'y'z'} + \Omega \times \mathbf{V} . \quad (5)$$

Here,  $(\dot{\mathbf{V}})_{x'y'z'}$  is the time derivative of the vector  $\mathbf{V}$  as seen by the rotating frame. The expression (5) above is known as *Coriolis' theorem*. Given an arbitrary vector, it relates the derivative of that vector as seen by a fixed frame with the derivative of the same vector as seen by a rotating frame. Symbolically, we can write,

$$(\dot{\phantom{\mathbf{V}}})_{xyz} = (\dot{\phantom{\mathbf{V}}})_{x'y'z'} + \Omega \times (\phantom{\mathbf{V}}) .$$

We will often omit the notation  $(\phantom{\mathbf{V}})_{xyz}$  when the derivative is taken with respect to the fixed frame.



## Summary: Relationships between position, velocity and acceleration vectors from rotating to non-rotating coordinate systems

We will now systematically apply Coriolis theorem to determine the relationship between the position, velocity and acceleration vectors as seen by observer  $A$  and  $B$ .

### Position vector

Since we have defined our coordinate system as having a common origin  $O$  and being *instantaneously aligned*, the position vector  $\mathbf{r}$  seen by both the fixed observer  $A$  and the rotating observer  $B$  at this instant are the same.

$$\mathbf{r}_{a/A} = \mathbf{r}_{a/B} \quad (6)$$

Note that this is an instantaneous concept, and therefore it is immaterial whether the observer  $B$  is rotating or not. In other words, the above expression is valid at any given instant. Of course, if we choose to express  $\mathbf{r}_{a/A}$  in  $xyz$  components, and  $\mathbf{r}_{a/B}$  in  $x'y'z'$  components, then we will have to make sure that the proper coordinate transformation is done before the components of the vectors are added. Also, since the orientation of  $x'y'z'$  changes in time, this transformation will depend on the instant considered.

### Velocity vector

Differentiating (6) with respect to time, we have,

$$\begin{aligned} \mathbf{v}_{a/A} = \dot{\mathbf{r}}_{a/A} &= \dot{\mathbf{r}}_{a/B} + \boldsymbol{\Omega} \times \mathbf{r}_{a/B} \\ &= \mathbf{v}_{a/B} + \boldsymbol{\Omega} \times \mathbf{r}_{a/B} . \end{aligned} \quad (7)$$

Here, we have used Coriolis' theorem (5) to calculate the derivative of  $\mathbf{r}_{a/A}$ . In the above expression,  $\mathbf{v}_{a/A}$  is the velocity of  $a$  relative to the fixed frame. The term  $\mathbf{v}_{a/B}$  is the velocity of  $a$  measured by the rotating observer,  $B$ . Finally,  $\boldsymbol{\Omega}$  is the angular velocity of the rotating frame, and  $\mathbf{r}_{a/B}$  is the relative position vector of  $a$  with respect to  $B$ .

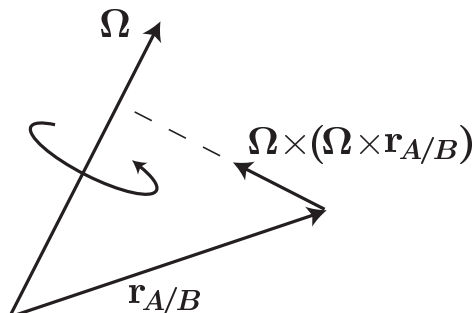
### Acceleration vector

Differentiating (17) once again,

$$\begin{aligned} \mathbf{a}_{a/A} = \dot{\mathbf{v}}_{a/A} &= \dot{\mathbf{v}}_{a/B} + \boldsymbol{\Omega} \times \mathbf{v}_{a/B} \\ &+ \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{a/B} + \boldsymbol{\Omega} \times \mathbf{v}_{a/B} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{a/B}) \\ &= \mathbf{a}_{a/B} + 2\boldsymbol{\Omega} \times \mathbf{v}_{a/B} + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{a/B} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{a/B}) . \end{aligned} \quad (8)$$

The term  $(\mathbf{a}_{a/B})$  is the acceleration of  $A$  measured by an observer  $B$  that rotates with the axes  $x'y'z'$  *instantaneously aligned* with  $x, y, z$ . The term  $2\boldsymbol{\Omega} \times (\mathbf{v}_{a/B})$  is called Coriolis' acceleration. The term  $\dot{\boldsymbol{\Omega}} \times \mathbf{r}_{a/B}$  is due to the change in  $\boldsymbol{\Omega}$ , and, finally, the term  $\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{a/B})$  is called centripetal acceleration. It can be easily seen that this acceleration always points towards the axis of rotation, and is orthogonal to  $\boldsymbol{\Omega}$ .

(when trying to show this, you may find the vector identity,  $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$ , introduced in Lecture 2, applied to  $\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{a/B}) = (\boldsymbol{\Omega} \cdot \mathbf{r}_{a/B})\boldsymbol{\Omega} - (\boldsymbol{\Omega} \cdot \boldsymbol{\Omega})\mathbf{r}_{a/B}$ , useful).




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**Note**
 $\dot{\boldsymbol{\Omega}}_A$  vs.  $(\dot{\boldsymbol{\Omega}})_B$ 


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When writing the derivative of the angular acceleration vector,  $\dot{\boldsymbol{\Omega}}$ , we did not specify whether the derivative was taken with respect to the fixed observer,  $A$ , or with respect to the rotating observer,  $B$ . This may seem a little bit sloppy at first, but, in fact, it turns out that it does not really matter. The time derivatives of vectors which are parallel to  $\boldsymbol{\Omega}$  are the same for both observers. This can be easily seen if we go back to Coriolis' theorem (5) and apply it to  $\boldsymbol{\Omega}$ . That is,

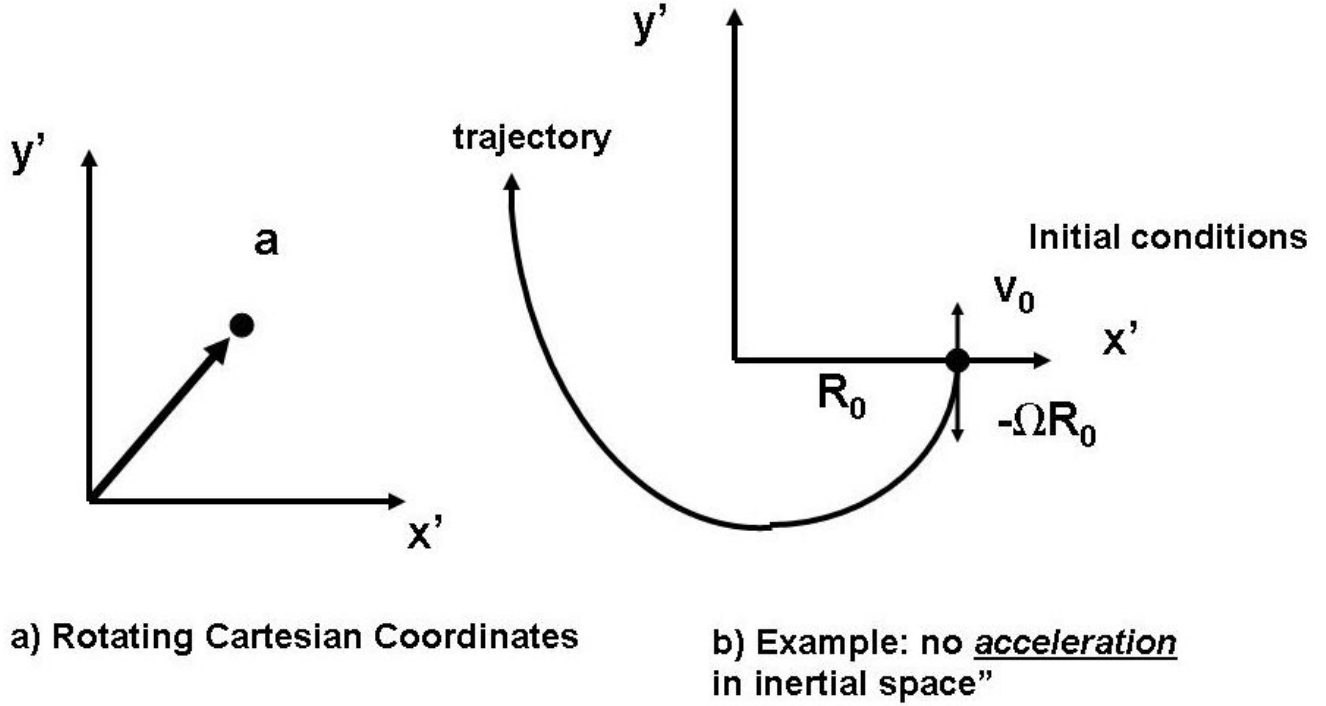
$$\dot{\boldsymbol{\Omega}}_A = (\dot{\boldsymbol{\Omega}})_B + \boldsymbol{\Omega} \times \boldsymbol{\Omega} = (\dot{\boldsymbol{\Omega}}) .$$

since  $\boldsymbol{\Omega} \times \boldsymbol{\Omega} = 0$ , (cross product).

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## Acceleration in Rotating Cartesian Coordinates

For later application to problems involving orbital motion (and other applications), we now derive the expression for acceleration in inertial space "seen" by the rotating observer  $B$  in the cartesian coordinates shown in a). It is this expression for acceleration which must be used in the application of Newton's Law  $\mathbf{F} = m\mathbf{a}$  in a rotating coordinate system.



Without loss of generality, we take  $\boldsymbol{\Omega}$  to be aligned with the  $z, z'$  axis. We have also allowed  $\boldsymbol{\Omega}$  to be a function of time although in most applications  $\boldsymbol{\Omega}$  will be constant. The position vector is  $\mathbf{r} = x'\mathbf{i}' + y'\mathbf{j}' + z'\mathbf{k}'$  and  $\boldsymbol{\Omega} = \Omega\mathbf{k}$ . The components of acceleration are

$$a_x = \frac{d^2 x'}{dt^2} - x'\Omega^2 - y'\frac{d\Omega}{dt} - 2\Omega\frac{dy'}{dt} \quad (9)$$

$$a_y = \frac{d^2 y'}{dt^2} - y'\Omega^2 + x'\frac{d\Omega}{dt} + 2\Omega\frac{dx'}{dt} \quad (10)$$

$$a_z = \frac{d^2 z'}{dt^2} \quad (11)$$

Note that since the  $\boldsymbol{\Omega}$  vector is aligned with the  $z$  axis, the rotation of the coordinate system does not add additional terms to the acceleration in the  $z$  direction. The various terms in the  $x, y'$  equations are the Coriolis and centripetal accelerations expressed in cartesian coordinates plus the effects of time-varying  $\boldsymbol{\Omega}$ . These are the expressions for acceleration that must be used in applying Newton's Law to motion in a rotating cartesian coordinate system. Of significant importance is the setting of boundary conditions. We need 4 boundary conditions: the  $x', y'$  position of the particle; and the  $x', y'$  components of velocity *as seen by the rotating observer*. Since the fixed and rotating coordinates are instantaneously aligned in our analysis, the position seen by both inertial and rotating observers is the same.

To determine the velocity boundary conditions, we distinguish between two cases: a) the observer and b) the actor.

- a) the observer: If the phenomena takes place in inertial space, and the rotating observer simply observes the motion, then the velocity of the particle in inertial space is  $\mathbf{v}_{a/A} = \mathbf{v}_0$ , and the initial position is  $\mathbf{r} = \mathbf{R}_0$ . In this case, the boundary condition on velocity *seen by the rotating observer* must be corrected to reflect the effect of his motion so that in the rotating coordinate system  $\mathbf{v}_{a/B} = \mathbf{v}_0 - \boldsymbol{\Omega} \times \mathbf{R}_0$ .
- b) the actor: If the rotating observer actually caused the motion he is observing, such as an astronaut tossing a cheeseburger to another astronaut, then the boundary condition to be applied to particle velocity is that observed (or caused) by the "actor",  $\mathbf{v}_{a/B} = \mathbf{v}_0$ .

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**Note**      **Example: Constant velocity motion observed by an observer rotating at constant  $\Omega$**

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We now consider a particle moving at constant velocity as shown in b) and consider what trajectory the rotating observer  $B$  would observe. We take  $\boldsymbol{\Omega}$  to be constant in time as well. For a particle moving with constant velocity, the motion observed by the rotating observer satisfies equation (9) and (10) with  $a_x = 0$  and  $a_y = 0$ ; for simplicity we will consider two-dimensional planar motion.

$$0 = \frac{d^2 x'}{dt^2} - x' \Omega^2 - 2\Omega \frac{dy'}{dt} \quad (12)$$

$$0 = \frac{d^2 y'}{dt^2} - y' \Omega^2 + 2\Omega \frac{dx'}{dt} \quad (13)$$

Of significant importance is the setting of boundary conditions. We need 4 boundary conditions: the  $x$   $y$  position of the particle; and the  $x$ ,  $y$  components of velocity *as seen by the rotating observer*. Since the fixed and rotating coordinates are instantaneously aligned in our analysis, the position seen by both inertial and rotating observers is the same.

These equations have a closed form solution. For initial conditions in the rotating frame  $x(0), y(0), x'(0)$  and  $y'(0)$ , the closed form solution is

$$x(t) = (x(0) + t * y(0) - \Omega t x'(0)) \cos(\Omega t) + (\Omega t x(0) + x'(0) + t y'(0)) \sin(\Omega t) \quad (14)$$

$$y(t) = (\Omega t x(0) + x'[0] + t y'(0)) \cos(\Omega t) - (x(0) + t y(0) - t \Omega x'(0)) \sin(\Omega t). \quad (15)$$

We consider the simple case shown in b): a particle located at  $x(0) = R_0, y(0) = 0$  with velocity  $v_x = 0$  and  $v_y = v_0$  relative to inertial space or  $y'(0) = v_0 - \Omega * R_0$  in the rotating coordinate system. If  $v_0$  is less than  $\Omega R_0$ , then the particle observed by the rotating observer  $B$  would actually appear to move backwards along a trajectory as shown in b).

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**ADDITIONAL READING**

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

5/7, 7/60

## Lecture 10 — Linear Momentum, Impulse, and Collisions

In this lecture, we will consider the equations that result from integrating Newton's second law,  $\mathbf{F} = m\mathbf{a}$ , in time. This will lead to the principle of linear impulse and momentum. This principle is very useful when solving problems in which we are interested in determining the global effect of a force acting on a particle over a time interval.

### Linear Momentum

We consider the curvilinear motion of a particle of mass  $m$  under the influence of a force  $\mathbf{F}$ . Assuming that the mass does not change, we have from Newton's second law,

$$\mathbf{F} = m\mathbf{a} = m \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(m\mathbf{v}) .$$

The case where the mass of the particle changes with time (e.g. a rocket) will be considered later on in this course. The *linear momentum* vector,  $\mathbf{L}$ , is defined as

$$\mathbf{L} = m\mathbf{v} .$$

Thus, an alternative form of Newton's second law is

$$\mathbf{F} = \dot{\mathbf{L}} , \tag{1}$$

which states that the total force acting on a particle is equal to the time rate of change of its linear momentum.

### Principle of Linear Impulse and Momentum

Imagine now that the force considered acts on the particle between time  $t_1$  and time  $t_2$ . Equation (1) can then be integrated in time to obtain

$$\int_{t_1}^{t_2} \mathbf{F} dt = \int_{t_1}^{t_2} \dot{\mathbf{L}} dt = \mathbf{L}_2 - \mathbf{L}_1 = \Delta \mathbf{L} . \tag{2}$$

Here,  $\mathbf{L}_1 = \mathbf{L}(t_1)$  and  $\mathbf{L}_2 = \mathbf{L}(t_2)$ . The term

$$\mathbf{I} = \int_{t_1}^{t_2} \mathbf{F} dt = \Delta \mathbf{L} = (m\mathbf{v})_2 - (m\mathbf{v})_1,$$

is called the *linear impulse*. Thus, the linear impulse on a particle is equal to the linear momentum change  $\Delta \mathbf{L}$ . In many applications, the focus is on an impulse modeled as a large force acting over a small time. But in fact, this restriction is unnecessary. All that is required is to be able to perform the integral  $\int_{t_1}^{t_2} \mathbf{F} dt$ . If the force is a constant  $\mathbf{F}$ , then  $\Delta \mathbf{L} = \int_{t_1}^{t_2} \mathbf{F} dt = \mathbf{F}(t_2 - t_1)$ . If the force is given as a function of time, then  $\Delta \mathbf{L} = \int_{t_1}^{t_2} \mathbf{F}(t) dt$

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**Note**
**Units of Impulse and Momentum**


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Linear impulse and momentum have the same units. In the SI system they are  $\text{N} \cdot \text{s}$  or  $\text{kg} \cdot \text{m/s}$ , whereas in the English system they are  $\text{lb} \cdot \text{s}$ , or  $\text{slug} \cdot \text{ft/s}$ .

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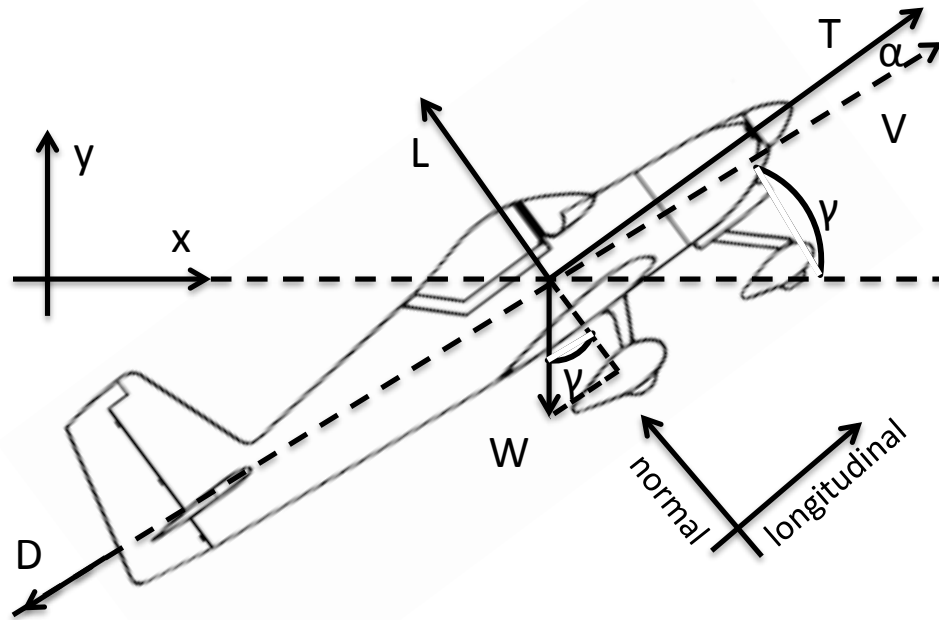


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**Example (MK)**
**Average Drag Force**


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The pilot of a 90,000-lb airplane which is originally flying horizontally at a speed of 400 mph cuts off all engine power and enters a glide path as shown where  $\beta = 5^\circ$ . After 120 s, the airspeed of the plane is 360 mph. We want to calculate the magnitude of the time-averaged drag force.



**Figure 1: Forces acting on an aircraft in a flight inclined at a flight path  $\gamma$  to the horizon**

Aligning the  $x$ -axis with the flight path, we can write the  $x$  component of equation (2) as follows

$$\int_0^{120} (W \sin \beta - D) dt = L_x(120) - L_x(0) .$$

The time-averaged value of the drag force,  $\bar{D}$ , is

$$\bar{D} = \frac{1}{120} \int_0^{120} D dt .$$

Therefore,

$$(W \sin \beta - \bar{D})120 = m(v_x(120) - v_x(0)) .$$

Substituting and applying the appropriate unit conversion factors we obtain,

$$(90,000 \sin 5^\circ - \bar{D})120 = \frac{90,000}{32.2}(360 - 400)\frac{5280}{3600} \rightarrow \bar{D} = 9,210 \text{ lb} .$$

## Impulsive Forces

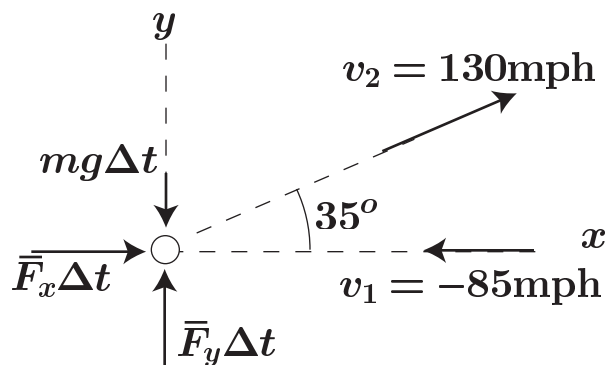
We typically think of impulsive forces as being forces of very large magnitude that act over a very small interval of time, but cause a significant change in the momentum. Examples of impulsive forces are those generated when a ball is hit by a tennis racquet or a baseball bat, or when a steel ball bounces on a steel plate. The table below shows typical time intervals over which some of these impulses occur.

|                                    | Time interval $\Delta t$ [s] |
|------------------------------------|------------------------------|
| Racquet hitting a tennis ball      | 0.005 – 0.05                 |
| Bat hitting a baseball             | 0.01–0.02                    |
| Golf club hitting a golf ball      | 0.001                        |
| Pile Driver                        | 0.01 – 0.02                  |
| Shotgun                            | 0.001                        |
| Steel ball bouncing on steel plate | 0.0002                       |

### Example (MK)

### Baseball Bat

A baseball is traveling with a horizontal velocity of 85 mph just before impact with the bat. Just after the impact, the velocity of the  $5\frac{1}{8}$  oz. ball is 130 mph, at an angle of  $35^\circ$  above the horizontal. We want to determine the horizontal and vertical components of the average force exerted by the bat on the baseball during the 0.02 s impact.



If we consider the time interval between the instant the baseball hits the bat,  $t_1$ , and the instant after it leaves the bat,  $t_2$ , the forces responsible for changing the baseball's momentum are gravity and the contact



forces exerted by the bat. We can use the  $x$  and  $y$  components of equation (2) to determine the average force of contact. First, consider the  $x$  component,

$$mv_{x1} + \int_{t_1}^{t_2} F_x dt = mv_{x2} .$$

Inserting numbers and using appropriate conversion factors, we obtain

$$-\frac{5.125/16}{32.2}(85\frac{5280}{3600}) + \bar{F}_x(0.02) = \frac{5.125/16}{32.2}(130\frac{5280}{3600} \cos 35^\circ) \rightarrow \bar{F}_x = 136.7 \text{ lb} .$$

For the  $y$  direction, we have

$$mv_{y1} + \int_{t_1}^{t_2} F_y dt = mv_{y2} ,$$

which gives,

$$0 + \bar{F}_y(0.02) - \frac{5.125}{16}(0.02) = \frac{5.125/16}{32.2}(130\frac{5280}{3600} \sin 35^\circ) \rightarrow \bar{F}_y = 54.7 \text{ lb} .$$

We note that  $mg\Delta t = 0.0064$  lb-s, whereas  $\bar{F}_y(\Delta t) = 1.094$  lb - s. Thus,  $mg\Delta t$  is 0.59% of the total impulse. We could have safely neglected it. In this case we would have obtained  $\bar{F}_y = 54.4$  lb.

## Conservation of Linear Momentum

We see from equation (1) that if the resultant force on a particle is zero during an interval of time, then its linear momentum  $\mathbf{L}$  must remain constant. Since equation (1) is a vector quantity, we can have situations in which only some components of the resultant force are zero. For instance, in Cartesian coordinates, if the resultant force has a non-zero component in the  $y$  direction only, then the  $x$  and  $z$  components of the linear momentum will be conserved since the force components in  $x$  and  $z$  are zero.

Consider now two particles,  $m_1$  and  $m_2$ , which interact during an interval of time. Assume that interaction forces between them are the only unbalanced forces on the particles. Let  $\mathbf{F}$  be the interaction force that particle  $m_2$  exerts on particle  $m_1$ . Then, according to Newton's third law, the interaction force that particle  $m_1$  exerts on particle  $m_2$  will be  $-\mathbf{F}$ . Using expression (2), we will have that  $\Delta\mathbf{L}_1 = -\Delta\mathbf{L}_2$ , or  $\Delta\mathbf{L} = \Delta\mathbf{L}_1 + \Delta\mathbf{L}_2 = \mathbf{0}$ . That is, the changes of momentum of particles  $m_1$  and  $m_2$  are equal in magnitude and opposite in sign, and the total momentum change equals zero. Recall that this is true if the only unbalanced forces on the particles are the interaction forces. The more general situation in which external forces can be present will be considered in future lectures.

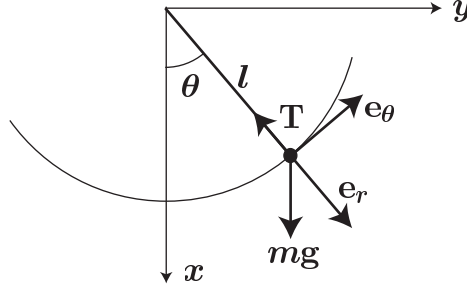
We note that the above argument is also valid in a componentwise sense. That is, when two particles interact and there are no external unbalanced forces along a given direction, then the total momentum change along that direction must be zero.

### Example

### Ballistic Pendulum

The ballistic pendulum is used to measure the velocity of a projectile by observing the maximum angle  $\theta_{\max}$

to which the box of sand with the embedded projectile swings. Find an expression that relates the initial velocity of a projectile  $v_0$  of mass  $m$  to the maximum angle  $\theta_{\max}$  reached by the pendulum. The mass of the sand box is  $M$  and the length of the pendulum is  $L$ .



We consider the equation of conservation of linear momentum along the horizontal direction. The initial momentum of the projectile is  $mv_0$ . Since the sand box is initially at rest its momentum is zero. Just after the projectile penetrates into the box, the velocity of the sand box and the projectile are the same. Therefore, if  $v_1$  is the velocity of the sand box (with the embedded projectile) just after impact, we have from conservation of momentum,

$$mv_0 = (M + m)v_1 .$$

After impact, the problem reduces to that of a simple pendulum. The only force doing any work is gravity and therefore we can apply the principle of conservation of work and energy. At the point when  $\theta$  is maximum, the velocity will be zero. From energy conservation we finally obtain,

$$\frac{1}{2}(M + m)v_1^2 = (M + m)gh_{\max} ,$$

and since  $h = L(1 - \cos \theta)$ , we have

$$\theta_{\max} = \cos^{-1} \left( 1 - \left( \frac{m}{M + m} \right)^2 \frac{v_0^2}{2Lg} \right) .$$

Note that energy is not conserved in the collision. The initial kinetic energy of the system is the kinetic energy of the projectile  $T_0 = \frac{1}{2}mv_0^2$  (taking the reference height as zero). After the collision the kinetic energy of the system is

$$T_1 = \frac{1}{2}(m + M)v_1^2. \quad (3)$$

Since  $v_1 = v_0 \frac{m}{(m+M)}$  we have

$$T_1 = \frac{1}{2} \frac{m^2}{(M + m)} v_0^2 \quad (4)$$

which is less than  $T_0$  for  $M > 0$ .

## Collisions

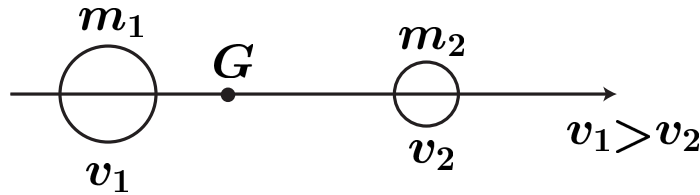
We consider now, the situation of two isolated particles colliding. The only forces on the particles are due to their mutual interaction. When the velocity vector of the two particles is parallel to the line joining the two particles, we say that we have one-dimensional collisions, otherwise we say that the collision is oblique.

### 1D Collisions

Here, we are dealing with rectilinear motion and therefore the velocity vector becomes a scalar quantity. In order to understand the collision process we consider five different stages.

#### I) Before Impact

Let particle 1, of mass  $m_1$ , occupy position  $x_1$  and travel with velocity  $v_1$  along the direction parallel to the line joining the two particles. And let particle 2, of mass  $m_2$ , occupy position  $x_2$  and travel with speed  $v_2$  also in the same direction. We assume that  $v_1 > v_2$  so that collision will occur.



We can introduce the position of the center of mass

$$x_G = (m_1 x_1 + m_2 x_2) / m, \quad (5)$$

where  $m = m_1 + m_2$ . The velocity of the center of mass is then given by

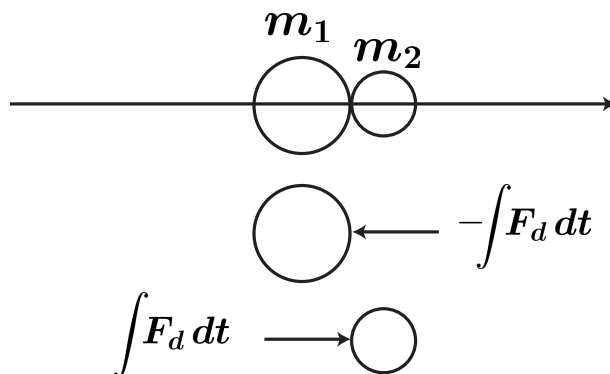
$$v_G = (m_1 v_1 + m_2 v_2) / m. \quad (6)$$

Note that since conservation of momentum requires  $(m_1 v_1 + m_2 v_2) = v_G = (m_1 v'_1 + m_2 v'_2) = v'_G$ , the velocity of the center of mass remains  $v_G$  unchanged by the collision process between the particles since  $v_G = v'_G$ . If we define the relative velocity  $g = v_1 - v_2$  we can express  $v_1$  and  $v_2$  as a function of  $v_G$  and  $g$  as,

$$\begin{aligned} v_1 &= v_G + \frac{m_2}{m} g \\ v_2 &= v_G - \frac{m_1}{m} g. \end{aligned}$$

#### II) Deformation

The particles establish contact and the force between them  $F_d$  increases until the instant of maximum deformation.



### III) Maximum Deformation

The contact force is at its maximum and the two particles travel at the same velocity  $v_G$ . Thus the deformation force has slowed  $m_1$  down to a velocity of  $v_G$  and sped up  $m_2$  to a velocity of  $v_G$ .

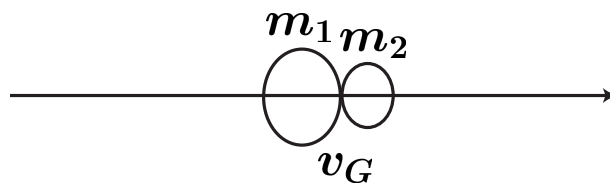
The impulse applied to particle 1 and 2 from the deformation force equals the change in momentum in the deformation process.

For particle 1

$$-\int F_d dt = m_1 v_G - m_1 v_1 \quad (\text{deformation phase})$$

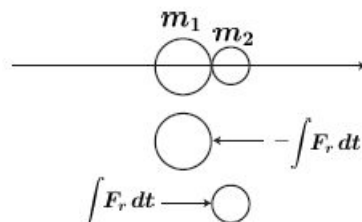
and for particle 2

$$\int F_d dt = m_2 v_G - m_2 v_2 \quad (\text{deformation phase})$$



### IV) Restoration

The contact force  $F_r$  decreases and the particles move apart.



The impulse applied to particle 1 and 2 from the restoring force equals the change in momentum in the restoration process. After the restoration process the velocity of  $m_1$  is  $v'_1$  and the velocity of  $m_2$  is  $v'_2$ . For particle 1

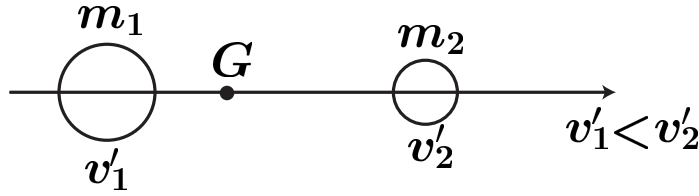
$$-\int F_r dt = m_1 v'_1 - m_1 v_G \quad (\text{restoration phase})$$

and for particle 2

$$\int F_r dt = m_2 v'_2 - m_2 v_G \quad (\text{restoration phase})$$

#### V) After Impact

The particles travel with a constant velocity  $v'_1$  and  $v'_2$ .



From momentum conservation, the total momentum before and after impact should remain the same, i.e.  $m_1 v_1 + m_2 v_2 = m_1 v'_1 + m_2 v'_2$ , and therefore the velocity of the center of mass will remain unchanged  $v'_G = v_G$ , thus  $m v_G = m_1 v'_1 + m_2 v'_2$ . If we define  $g' = v'_1 - v'_2$  as the relative velocity of the two particles after impact, then using the definition of the velocity of the center of mass we can write

$$v'_1 = v_G + \frac{m_2}{m} g' \quad (7)$$

$$v'_2 = v_G - \frac{m_1}{m} g' . \quad (8)$$

and confirm as an identity

$$v'_1 = \frac{m_1 v'_1 + m_2 v'_2}{m_1 + m_2} + \frac{m_2}{m} (v'_1 - v'_2) \quad (9)$$

$$v'_2 = \frac{m_1 v'_1 + m_2 v'_2}{m_1 + m_2} + \frac{m_1}{m} (v'_1 - v'_2). \quad (10)$$

#### Coefficient of Restitution

We define the coefficient of restitution as the ratio between the restoration and deformation impulses.

$$e = \frac{\int F_r dt}{\int F_d dt}$$

For physically acceptable collisions  $0 < e < 1$ . The value of  $e = 1$  corresponds to an *elastic* collision, whereas the value of  $e = 0$  corresponds to a totally *inelastic* collision in which the restoration impulse is equal to zero.

We can consider each particle separately and set the impulse on the particle equal to the change of linear momentum

Since for particle 1

$$\begin{aligned} - \int F_d dt &= m_1 v_G - m_1 v_1 && \text{(deformation phase)} \\ - \int F_r dt &= m_1 v'_1 - m_1 v_G && \text{(restoration phase)} \end{aligned}$$

Thus,  $e = (v'_1 - v_G)/(v_G - v_1)$ , or  $(e + 1)v_G = v'_1 + ev_1$ .  
and for particle 2

$$\begin{aligned} \int F_d dt &= m_2 v_G - m_2 v_2 && \text{(deformation phase)} \\ \int F_r dt &= m_2 v'_2 - m_2 v_G && \text{(restoration phase)} \end{aligned}$$

Thus,  $e = (v'_2 - v_G)/(v_G - v_2)$ , or  $(e + 1)v_G = v'_2 + ev_2$ .

Combining the above expressions, we have

$$e = \frac{\int F_r dt}{\int F_d dt} = -\frac{v'_1 - v'_2}{v_1 - v_2} = -\frac{g'}{g},$$

which expresses the coefficient of restitution as the ratio of the relative velocities after and before the collision.

Finally, if we know  $e$  we can use equations (7) and (8) to obtain

$$\begin{aligned} v'_1 &= v_G - \frac{m_2}{m} eg \\ v'_2 &= v_G + \frac{m_1}{m} eg. \end{aligned}$$

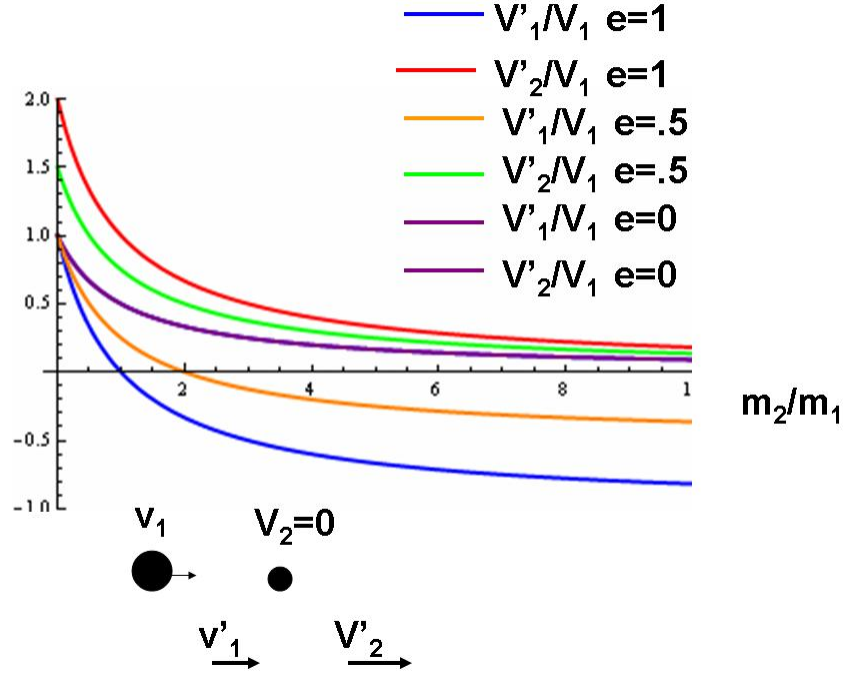
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### Example

### Simple Collisions

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As an example, we consider a simple case of a mass  $m_2$  at rest and a mass of  $m_1$  traveling at velocity  $v_1$  headed for a collision with  $m_2$ . After the collision, the velocities of the two masses will in general be non-zero. We consider the cases  $e = 1, .5, \text{ and } 0$ , there is a range of coefficient of restitution between 1, perfectly elastic, to zero, perfectly inelastic. The result for the final velocities of  $m_1$  and  $m_2$ , normalized by the initial velocity  $v_1$ , are given by the equations above and shown in the figure.



For  $e = 0$ , there is only one curve since the particles stick together:  $v'_1/v_1 = v'_2/v_1$ . For the elastic collision, note the result for equal masses:  $v'_1 = 0$  and  $v'_2/v_1 = 1$

## Energy

We can now look at the kinetic energy before and after the impact. The kinetic energy before impact is,

$$\begin{aligned}
 T &= \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \\
 &= \frac{1}{2}m_1(v_G^2 + 2\frac{m_2}{m}gv_G + \frac{m_2^2}{m^2}g^2) + \frac{1}{2}m_2(v_G^2 - 2\frac{m_1}{m}gv_G + \frac{m_1^2}{m^2}g^2) \\
 &= \frac{1}{2}mv_G^2 + \frac{1}{2}\frac{m_1m_2}{m}(v_1 - v_2)^2 = \frac{1}{2}mv_G^2 + \frac{1}{2}\mu g^2.
 \end{aligned}$$

Here we introduce the notation  $\mu = (m_1m_2)/m$ ; this is often referred to as the reduced mass of the pair. The reduced mass approaches 1/2 of each mass when the two masses are similar, but it approaches the lighter mass if the masses are dissimilar. The kinetic energy after impact will be

$$\begin{aligned}
 T' &= \frac{1}{2}mv_G^2 + \frac{1}{2}\frac{m_1m_2}{m}(v'_1 - v'_2)^2 \\
 &= \frac{1}{2}mv_G^2 + \frac{1}{2}\frac{m_1m_2}{m}g'^2 \\
 &= \frac{1}{2}mv_G^2 + \frac{1}{2}\mu e^2 g^2.
 \end{aligned}$$

since  $g' = eg$  and  $\frac{m_1 m_2}{m} = \mu$ . Therefore, the kinetic energy lost during the collision is

$$\Delta E = T - T' = \frac{1}{2} \mu (1 - e^2) g^2,$$

which is zero for  $e = 1$  (elastic collision), and maximum when  $e = 0$ .

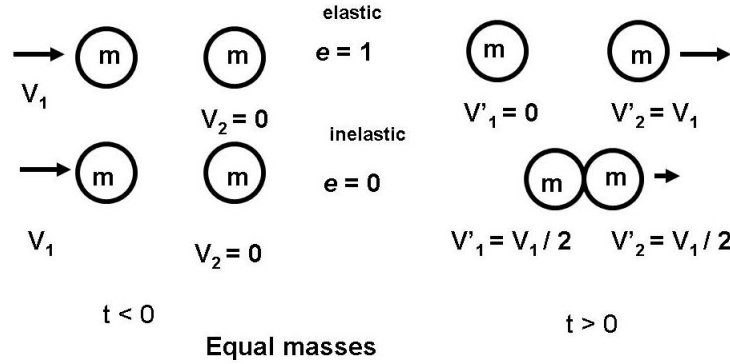
### Momentum exchange

We can also look at the momentum which is exchanged between particle 1 and particle 2,

$$M_{12} = m_1 v_1 - m_1 v'_1 = m_1 (v_G + \frac{m_2}{m} g - v_G + \frac{m_2}{m} eg) = (1 + e) \mu g.$$

This, is maximum when  $e = 1$  (elastic collision) and minimum when  $e = 0$ .

It is particularly illuminating to examine a special case of collisions, that for which the two masses are equal.



We consider two cases shown in the figure. In both cases one particle is at rest, the other approaching with velocity  $v_1$ . In one case, we assume an elastic collision,  $e = 1$ ; in the other a completely inelastic collision,  $e = 0$ . In the first case, kinetic energy is conserved so it is obvious—although you can work out the algebra—that the first particle completely transfers its momentum to the second particle, where upon it continues to the right with the original velocity  $v'_2 = v_1$ . In the second case of a completely inelastic collision, there is no rebounding force and the particles stick and move together. Conservation of momentum requires the  $v'_2 = v_1/2$ . Kinetic energy is not conserved.

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### Example

### Bouncing Ball

If a ball, initially at rest, is released on a flat surface from a height  $h$  and it rebounds to a height  $h'$ , the coefficient of restitution is given by

$$e = -\frac{v'}{v} = \sqrt{\frac{h'}{h}},$$

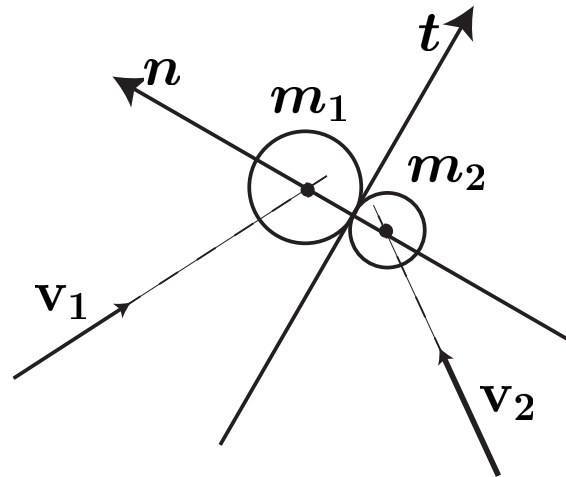
where  $v$  and  $v'$  are the velocities before and after impact.

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## Oblique Collisions

In the case of oblique collisions, we consider the instant of impact and define the normal direction,  $n$ , along the line that connects the two mass centers, and the tangential direction,  $t$ , along the line tangent to the surfaces at the point of contact.



We write the equations of conservation of momentum in the tangent and normal directions. Since the force of contact is assumed to act along the normal direction, the conservation of linear momentum along the tangential direction implies

$$\begin{aligned} m_1(v_1)_t &= m_1(v'_1)_t &\Rightarrow & (v'_1)_t = (v_1)_t \\ m_2(v_2)_t &= m_2(v'_2)_t &\Rightarrow & (v'_2)_t = (v_2)_t \end{aligned}$$

In the normal direction, we solve a 1D collision problem, that is,

$$\begin{aligned} m_1(v_1)_n + m_2(v_2)_n &= m_1(v'_1)_n + m_2(v'_2)_n \\ e &= -\frac{(v'_1)_n - (v'_2)_n}{(v_1)_n - (v_2)_n} \end{aligned}$$

which determine  $(v'_1)_n$  and  $(v'_2)_n$ .

## ADDITIONAL READING

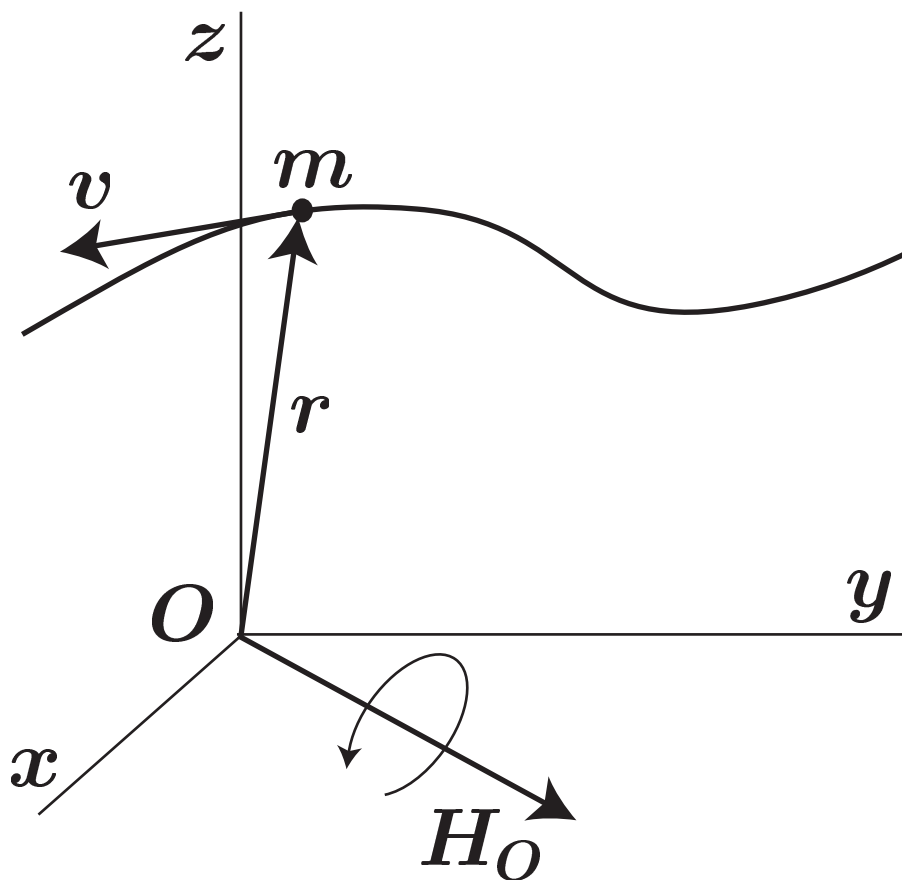
J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition  
3/8, 3/9

## Lecture 11 — Angular Momentum and Impulse

In addition to the equations of linear impulse and momentum considered in the previous lecture, there is a parallel set of equations that relate the angular impulse and momentum.

### Angular Momentum

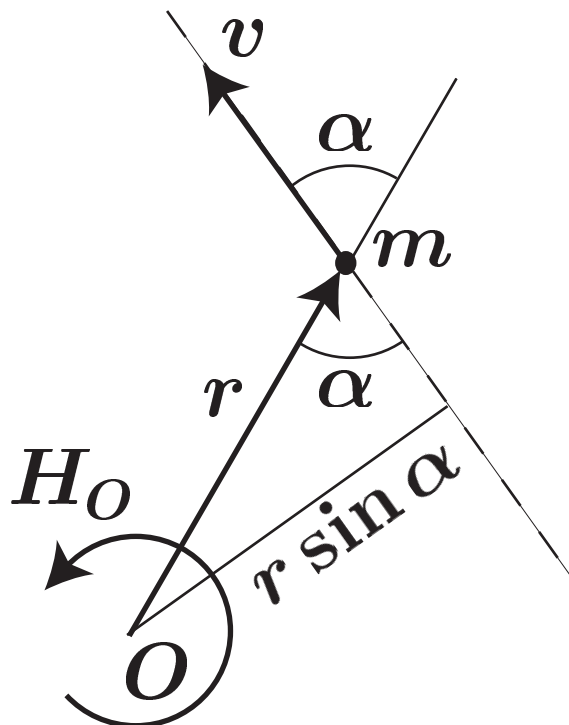
We consider a particle of mass  $m$ , with velocity  $\mathbf{v}$ , moving under the influence of a force  $\mathbf{F}$ . The angular momentum about point  $O$  is defined as the “moment” of the particle’s linear momentum,  $\mathbf{L}$ , about  $O$ .



Thus, the particle's angular momentum is given by,

$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v} = \mathbf{r} \times \mathbf{L} . \quad (1)$$

The units for the angular momentum are  $\text{kg}\cdot\text{m}^2/\text{s}$  in the SI system, and  $\text{slug}\cdot\text{ft}^2/\text{s}$  in the English system.



It is clear from its definition that the angular momentum is a vector which is perpendicular to the plane defined by  $\mathbf{r}$  and  $\mathbf{v}$ . Thus, on some occasions it may be more convenient to determine the direction of  $\mathbf{H}_O$  from the right hand rule, and its modulus directly from the definition of the vector product,

$$H_O = mvr \sin \alpha ,$$

where  $\alpha$  is the angle between  $\mathbf{r}$  and  $\mathbf{v}$ .

In other situations, it may be convenient to directly calculate the angular momentum in component form. For instance, using a right handed cartesian coordinate system, the components of the angular momentum are calculated as

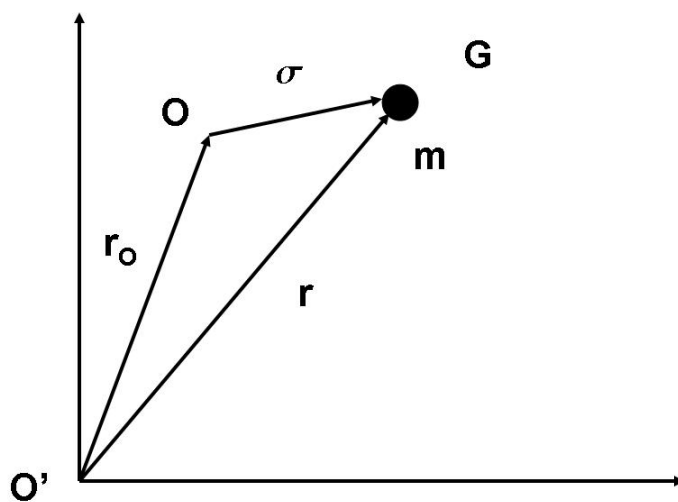
$$\mathbf{H}_O = H_x \mathbf{i} + H_y \mathbf{j} + H_z \mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ mv_x & mv_y & mv_z \end{vmatrix} = m(v_z y - v_y z) \mathbf{i} + m(v_x z - v_z x) \mathbf{j} + m(v_y x - v_x y) \mathbf{k} .$$

Similarly, in cylindrical coordinates we have

$$\mathbf{H}_O = H_r \mathbf{e}_r + H_\theta \mathbf{e}_\theta + H_z \mathbf{k} = \begin{vmatrix} \mathbf{e}_r & \mathbf{e}_\theta & \mathbf{k} \\ r & 0 & z \\ mv_r & mv_\theta & mv_z \end{vmatrix} = -mv_\theta z \mathbf{e}_r + m(v_r z - v_z r) \mathbf{e}_\theta + mv_\theta r \mathbf{k} .$$

### Rate of Change of Angular Momentum; Choice of Coordinates

We now derive the rate of change of angular momentum about an arbitrary point O, a point which is not necessarily fixed nor the center of mass. We will show that these latter two points have special properties that make them ideal choices for our coordinate system,



We now want to examine how the angular momentum changes with time for an arbitrary choice of coordinates. We center our coordinates at the point O. We then reference the position of the point O to the fixed point O' as  $\mathbf{r}_O$ . The point mass is located a distance  $\sigma$  from the coordinate center O at the center of mass  $\mathbf{G}$ . The angular momentum about the point O is then

$$\mathbf{H}_O = \sigma \times m\dot{\sigma} \quad (2)$$

where  $\sigma = \mathbf{r} - \mathbf{r}_O$ . Taking the time derivative of  $\mathbf{H}_O$  with respect to time, we have

$$\dot{\mathbf{H}}_O = \dot{\sigma} \times m\dot{\sigma} + \sigma \times m\ddot{\sigma} \quad (3)$$

Since  $\dot{\sigma} \times \dot{\sigma} = 0$  and  $\ddot{\sigma} = \ddot{\mathbf{r}} - \ddot{\mathbf{r}}_O$  we have

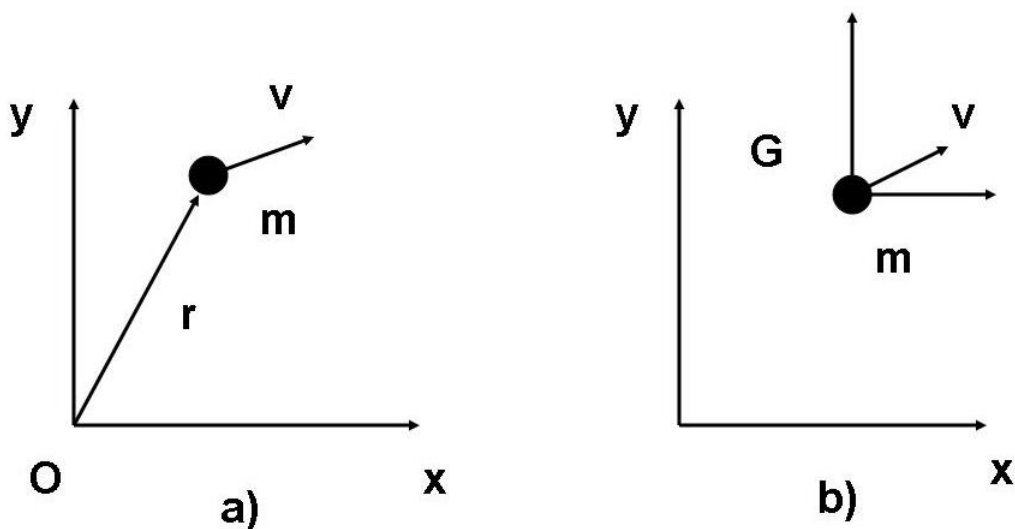
$$\dot{\mathbf{H}}_O = \sigma \times m\ddot{\mathbf{r}} - \sigma \times m\ddot{\mathbf{r}}_O \quad (4)$$

Since  $\mathbf{r}$  is a vector in inertial space, from Newton's law have  $m\ddot{\mathbf{r}} = \mathbf{F}$  resulting in

$$\dot{\mathbf{H}}_O = \boldsymbol{\sigma} \times \mathbf{F} + m\ddot{\mathbf{r}}_O \times \boldsymbol{\sigma} \quad (5)$$

where  $\boldsymbol{\sigma}$  is the vector from the center of our choice of coordinates to the point mass  $m$  located at the center of mass  $G$  and  $\mathbf{F}$  is the force acting on  $m$  so that  $\boldsymbol{\sigma} \times \mathbf{F}$  is  $\mathbf{M}$ , the moment about  $O$  due to  $\mathbf{F}$ . To obtain the simplest statement of conservation of angular momentum, we now examine two cases where the second term in  $\dot{\mathbf{H}}_O$  goes to zero. The first is a fixed point  $O'$  for which  $\ddot{\mathbf{r}}_O$  is zero; the second point is the center of mass  $G$  for which  $\boldsymbol{\sigma}$  goes to zero.

### Rate of Change of Angular Momentum About a Fixed Point $O$



The angular momentum about the fixed point  $O$  is

$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v} \quad (6)$$

Taking a time derivative of expression (2), we have

$$\dot{\mathbf{H}}_O = \dot{\mathbf{r}} \times m\mathbf{v} + \mathbf{r} \times m\dot{\mathbf{v}} .$$

Here, we have assumed that  $m$  is constant. If  $O$  is a fixed point, then  $\ddot{\mathbf{r}}_O = 0$  and  $\dot{\mathbf{r}} = \mathbf{v}$  and  $\dot{\mathbf{r}} \times m\mathbf{v} = 0$ . Thus, we end up with,

$$\dot{\mathbf{H}}_O = \mathbf{r} \times m\dot{\mathbf{v}} = \mathbf{r} \times m\mathbf{a} .$$

Applying Newton's second law to the right hand side of the above equation, we have that  $\mathbf{r} \times m\mathbf{a} = \mathbf{r} \times \mathbf{F} = \mathbf{M}_O$ , where  $\mathbf{M}_O$  is the moment of the force  $\mathbf{F}$  about point  $O$ . The equation expressing the rate of change of angular momentum is then written as

$$\mathbf{M}_O = \dot{\mathbf{H}}_O . \quad (7)$$

We note that this expression is valid whenever point  $O$  is fixed. The above equation is analogous to the equation derived in the previous lecture expressing the rate of change of linear momentum. It states that the rate of change of linear momentum about a fixed point  $O$  is equal to the moment about  $O$  due to the resultant force acting on the particle. Since this is a vector equation, it must be satisfied for each component independently. Thus, if the force acting on a particle is such that the component of its moment along a given direction is zero, then the component of the angular momentum along this direction will remain constant. This equation is a direct consequence of Newton's law. It will not give us more information about the momentum of a particle, but a clever choice of coordinates may make angular momentum easier to apply in any given case.

### Rate of Change of Angular Momentum About the Center of Mass

For a particle, the angular momentum about the center of mass is zero. We examine the expression for the rate of change of angular momentum

$$\dot{\mathbf{H}}_O = \dot{\mathbf{r}} \times m\mathbf{v} + \mathbf{r} \times m\dot{\mathbf{v}}. \quad (8)$$

Since the coordinate system moves with the particle, both  $\dot{\mathbf{r}}$  and  $\dot{\mathbf{v}}$  are zero. This is true even if the coordinate system is not inertial, in contrast to the application of Newton's law for linear momentum. Therefore for a mass point, the rate of change of angular momentum is zero in a coordinate system moving with the point. This implies that no moments can be applied to a mass point in a coordinate system moving with the particle. This is a rather trivial result for a mass point. However, this result will be extended to bodies of finite size resulting in a statement of conservation of angular momentum in an accelerating coordinate system fixed at the center of mass.

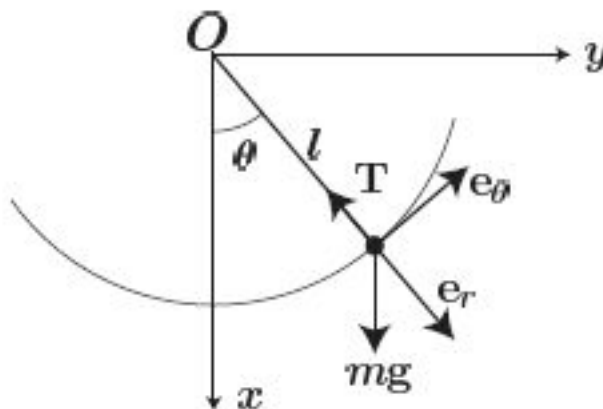
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#### **Example**

#### **Pendulum**

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Here, we revisit the simple pendulum problem introduced in lecture L6 and re-derive the pendulum equation using equation (7).



There are two forces acting on the suspended mass: the string tension and the weight. The string tension,  $\mathbf{T}$ , is parallel to the position vector  $\mathbf{r}$  and therefore its moment about  $O$  is zero. On the other hand, the weight creates a moment about  $O$  which is  $\mathbf{M}_O = -lmg \sin \theta \mathbf{k}$ .

The angular momentum is given by

$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v} = l\mathbf{e}_r \times ml\dot{\theta}\mathbf{e}_\theta = ml^2\dot{\theta} \mathbf{e}_r \times \mathbf{e}_\theta = ml^2\dot{\theta}\mathbf{k}.$$

Therefore, the  $z$  component of equation (7) gives

$$ml^2\ddot{\theta} = -lmg \sin \theta ,$$

or,

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0 ,$$

which is precisely the same equation as the one derived in lecture L5.

## Principle of Angular Impulse and Momentum

Equation (7) gives us the instantaneous relation between the moment and the time rate of change of angular momentum. Imagine now that the force considered acts on a particle between time  $t_1$  and time  $t_2$ . Equation (7) can then be integrated in time to obtain

$$\int_{t_1}^{t_2} \mathbf{M}_O dt = \int_{t_1}^{t_2} \dot{\mathbf{H}}_O dt = (\mathbf{H}_O)_2 - (\mathbf{H}_O)_1 = \Delta \mathbf{H}_O . \quad (9)$$

Here,  $(\mathbf{H}_O)_1 = \mathbf{H}_O(t_1)$  and  $(\mathbf{H}_O)_2 = \mathbf{H}_O(t_2)$ . The term

$$\int_{t_1}^{t_2} \mathbf{M}_O dt ,$$

is called the *angular impulse*. Thus, the angular impulse on a particle is equal to the angular momentum change.

Equation (9) is particularly useful when we are dealing with impulsive forces. In such cases, it is often possible to calculate the integrated effect of a force on a particle without knowing in detail the actual value of the force as a function of time.

## Conservation of Angular Momentum

We see from equation (2) that if the moment of the resultant force on a particle is zero during an interval of time, then its angular momentum  $\mathbf{H}_O$  must remain constant.

Consider now two particles  $m_1$  and  $m_2$  which interact during an interval of time. Assume that interaction forces between them are the only unbalanced forces on the particles that have a non-zero moment about a fixed point  $O$ . Let  $\mathbf{F}$  be the interaction force that particle  $m_2$  exerts on particle  $m_1$ . Then, according to Newton's third law, the interaction force that particle  $m_1$  exerts on particle  $m_2$  will be  $-\mathbf{F}$ . Using expression (9), we will have that  $\Delta(\mathbf{H}_O)_1 = -\Delta(\mathbf{H}_O)_2$ , or  $\Delta\mathbf{H}_O = \Delta(\mathbf{H}_O)_1 + \Delta(\mathbf{H}_O)_2 = \mathbf{0}$ . That is, the changes in angular momentum of particles  $m_1$  and  $m_2$  are equal in magnitude and of opposite sign, and the total angular momentum change equals zero. Recall that this is true only if the unbalanced forces, those with non-zero moment about  $O$ , are the interaction forces between the particles. The more general situation in which external forces can be present will be considered in future lectures.

We note that the above argument is also valid in a componentwise sense. That is, when two particles interact and there are no external unbalanced moments along a given direction, then the total angular momentum change along that direction must be zero.

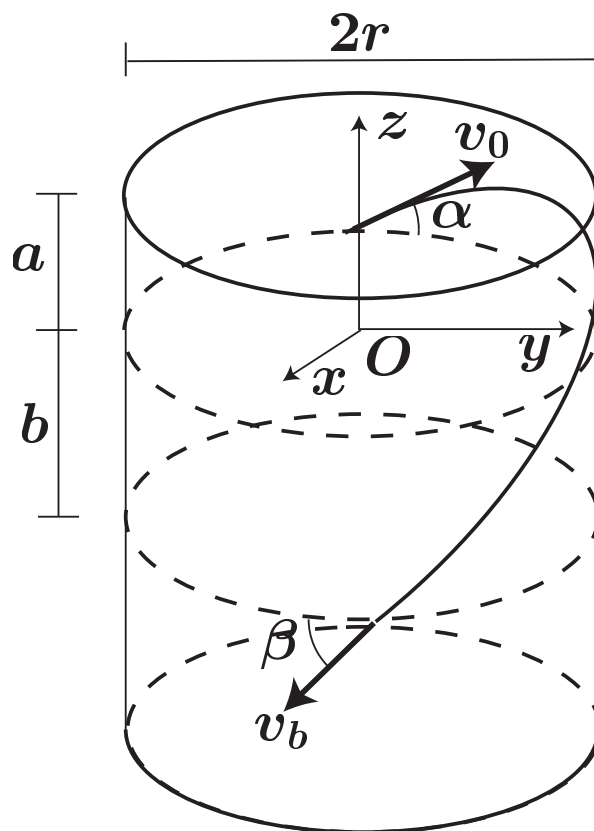
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### Example

### Ball on a cylinder

A particle of mass  $m$  is released on the smooth inside wall of an open cylindrical surface with a velocity  $\mathbf{v}_0$  that makes an angle  $\alpha$  with the horizontal tangent. The gravity acceleration is pointing downwards. We want to obtain : i) an expression for the largest magnitude of  $\mathbf{v}_0$  that will prevent the particle from leaving the cylinder through the top end, and ii) an expression for the angle  $\beta$  that the velocity vector will form with the horizontal tangent, as a function of  $b$ .





The only forces on the particle are gravity and the normal force from the cylinder surface. The moment of these forces about  $O$  (or, in fact, about any point on the axis of the cylinder) always has a zero component in the  $z$  direction. That is,  $(M_O)_z = 0$ . To see that, we notice that for any point on the surface of the cylinder,  $\mathbf{r}$  and  $\mathbf{F}$  are always contained in a vertical plane that contains the  $z$  axis. Therefore, the moment must be normal to that plane. Since the moment has zero component in the  $z$  direction,  $(H_O)_z$  will be constant. Thus, we have that

$$(H_O)_z = rmv_0 \cos \alpha = \text{constant} .$$

For part i), we consider the trajectory for which the velocity is horizontal when  $z = a$  and let  $(\mathbf{v}_0)_{\text{limit}}$  be the initial velocity that corresponds to this trajectory. It is clear that for any trajectory for which  $\mathbf{v}_0$  has a larger magnitude than  $(\mathbf{v}_0)_{\text{limit}}$ , the particle will leave the cylinder through the top end. Thus, for the limit trajectory we have, from conservation of energy

$$\frac{1}{2}m(v_0)_{\text{limit}}^2 = \frac{1}{2}mv_a^2 + mga ,$$

and from conservation of angular momentum

$$(H_O)_z = rm(v_0)_{\text{limit}} \cos \alpha = rmv_a .$$

Here,  $v_a$  is the magnitude of the velocity for the limit trajectory when  $z = a$ . Eliminating  $v_a$  from these

equations we finally arrive at,

$$(v_0)_{\text{limit}} = \sqrt{\frac{2ga}{1 - \cos^2 \alpha}} .$$

Therefore, for  $v_0 \leq (v_0)_{\text{limit}}$  the mass will not leave the cylinder through the top end.

For part ii), we also consider conservation of energy

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_b^2 - mgb$$

and conservation of angular momentum,

$$rmv_0 \cos \alpha = rmv_b \cos \beta .$$

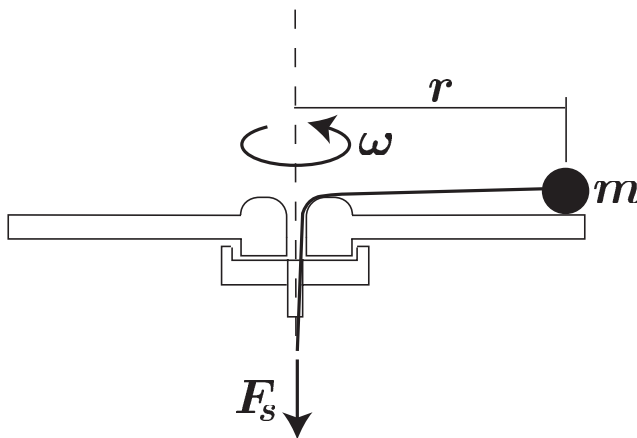
Eliminating,  $v_b$  from these two expressions we obtain,

$$\beta = \cos^{-1} \left( \frac{\cos \alpha}{\sqrt{1 + 2gb/v_0^2}} \right) .$$

### Example

### Spinning Mass

A small particle of mass  $m$  and its restraining cord are spinning with an angular velocity  $\omega$  on the horizontal surface of a smooth disk, shown in section. As the force  $\mathbf{F}_s$  is slowly increased,  $r$  decreases and  $\omega$  changes. Initially, the mass is spinning with  $\omega_0$  and  $r_0$ . Determine : i) an expression for  $\omega$  as a function of  $r$ , and ii) the work done on the particle by  $\mathbf{F}_s$  between  $r_0$  and an arbitrary  $r$ . Verify the principle of work and energy.



The component of the moment of the forces acting on the particle is zero along the spinning axis. Therefore, the vertical component of the angular momentum will be constant. For i), we have

$$mr_0v_0 = mrv, \quad v_0 = \omega_0r_0, \quad v = \omega r \quad \rightarrow \quad \omega = \frac{r_0^2\omega_0}{r^2} .$$

For ii), we first calculate the force on the string

$$F_s = -m \frac{v^2}{r} = -m \frac{r^2 \omega^2}{r} = -m \frac{r_0^4 \omega_0^2}{r^3} .$$

The work done by  $F_s$ , will be

$$W = \int_{r_0}^r F_s dr = -m r_0^4 \omega_0^2 \int_{r_0}^r \frac{dr}{r^3} = m r_0^4 \omega_0^2 \frac{1}{2} \left( \frac{1}{r^2} - \frac{1}{r_0^2} \right) .$$

The energy balance implies that

$$T_0 + W = T .$$

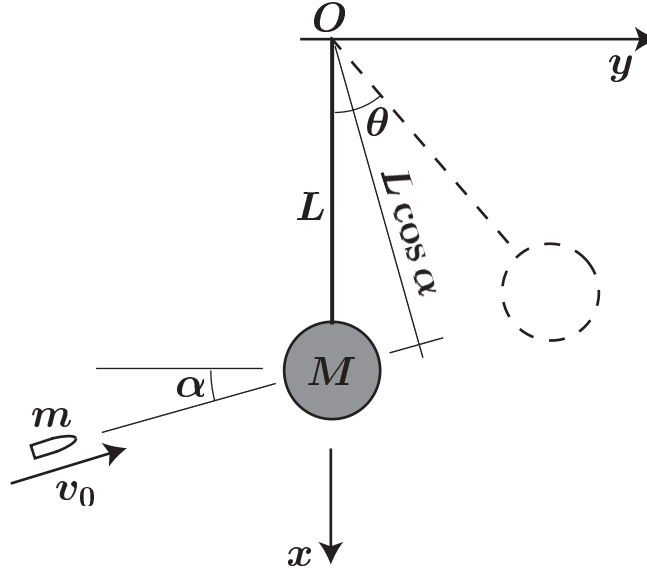
This expression can be directly verified since,

$$\underbrace{\frac{1}{2} m (\omega_0 r_0)^2}_{T_0} + \underbrace{\frac{1}{2} m (\omega_0 r_0)^2 \left( \frac{r_0^2}{r^2} - 1 \right)}_W = \underbrace{\frac{1}{2} m (\omega r)^2}_T .$$

### Example

### Ballistic Pendulum

We consider a pendulum consisting of a mass,  $M$ , suspended by a *rigid* rod of length  $L$ . The pendulum is initially at rest and the mass of the rod can be neglected. A bullet of mass  $m$  and velocity  $v_0$  impacts  $M$  and stays embedded in it. We want to find out the angle  $\theta_{\max}$  reached by the pendulum. The angle that the velocity vector  $v_0$  forms with the horizontal is  $\alpha$ .



Because the rod is assumed to be rigid, we can expect that when the bullet impacts the mass, there will be an impulsive reaction that the rod will exert on the bullet. If we use the principle of linear impulse and momentum, it will be necessary to solve for this impulsive force. An alternative approach that simplifies the

problem considerably is to use the principle of angular impulse and momentum. We consider the angular momentum about point  $O$  of the particles  $m$  and  $M$  just before and after the impact. The only external forces acting on the two particles are gravity and the reaction from the rod. It turns out that gravity is not an impulsive force and therefore its effect on the total angular impulse, over a very short time interval, can be safely neglected (it turns out that in this case, the moment about  $O$  of the gravity forces at the time of impact is also zero). On the other hand, we can expect the reaction from the rod to be large. However, the moment about  $O$  of this reaction is zero, since it is directed in the direction of the rod. Therefore, we have that during impact, the  $z$  component of the angular momentum is conserved.

The angular momentum before impact will be,

$$[(H_O)_z]_1 = L \cos \alpha \, m v_0 .$$

After impact, the velocity  $\mathbf{v}_1$  has to be horizontal. Thus, the angular momentum will be

$$[(H_O)_z]_2 = L(M + m)v_1 .$$

Equating these two expressions we get,

$$v_1 = \frac{m}{M + m} v_0 \cos \alpha .$$

After impact, the system is conservative, and the maximum height can be easily obtained from conservation of energy,

$$\frac{1}{2}(M + m)v_1^2 = (M + m)gL(1 - \cos \theta_{\max}) .$$

Thus,

$$\theta_{\max} = \cos^{-1} \left( 1 - \left( \frac{m}{(M + m)} \right)^2 \frac{v_0^2 \cos^2 \alpha}{2gL} \right) .$$

#### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

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## Lecture 12 — Conservation Laws for Systems of Particles

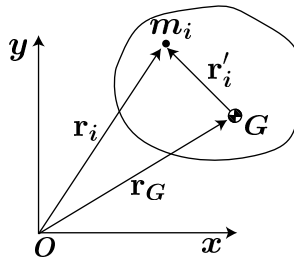
In this lecture, we will revisit the application of Newton's second law to a system of particles and derive some useful relationships expressing the conservation of angular momentum. We also specialize these results to two-dimensional rigid bodies.

### Center of Mass

Consider a system made up of  $n$  particles. A typical particle,  $i$ , has mass  $m_i$ , and, at the instant considered, occupies the position  $\mathbf{r}_i$  relative to a frame  $xyz$ . We can then define the center of mass,  $G$ , as the point whose position vector,  $\mathbf{r}_G$ , is such that,

$$\mathbf{r}_G = \frac{1}{m} \left( \sum_{i=1}^n m_i \mathbf{r}_i \right) . \quad (1)$$

Here,  $m$  is the total mass of the system given by  $m = \sum_{i=1}^n m_i$ .



It is important to note that the center of mass is a property of the system and does not depend on the reference frame used. In particular, if we change the location of the origin  $O$ ,  $\mathbf{r}_G$  will change, but the absolute position of the point  $G$  within the system will not. Often, it will be convenient to describe the motion of particle  $i$  as the motion of  $G$  plus the motion of  $i$  relative to  $G$ . To this end, we introduce the relative position vector,  $\mathbf{r}'_i$ , and write,

$$\mathbf{r}_i = \mathbf{r}_G + \mathbf{r}'_i . \quad (2)$$

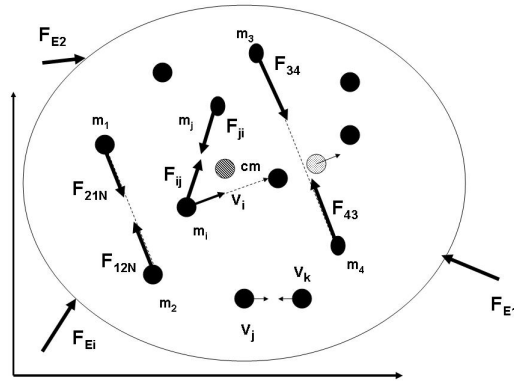
It follows immediately, from the definition of the center of mass (1) and the definition of the relative vector  $\mathbf{r}'_i$  (2), that,

$$\sum_{i=1}^n m_i \mathbf{r}'_i = \sum_{i=1}^n m_i (\mathbf{r}_i - \mathbf{r}_G) = \sum_{i=1}^n m_i \mathbf{r}_i - m \mathbf{r}_G = 0 . \quad (3)$$

This result will simplify our later analysis.

## Forces

In order to derive conservation laws for our system, we isolate it a little more carefully, identify what mass particles it contains and what forces act upon the individual particles.



We will consider two types of forces acting on the particles :

**External** forces arising outside the system. We will denote the resultant of all the **external** forces acting on the system as  $\sum_{i=1}^N \mathbf{F}_{Ei} = \mathbf{F}$ .

**Internal** forces due to pairwise particle interactions. Let  $\mathbf{f}_{ij}$  denote the force that particle  $j$  exerts on particle  $i$  directed along the line joining the two particles  $i$  and  $j$ . This force could arise from gravitation attraction or from internal force due to the connections between particles. It could also arise from collisions between individual particles that, as we have seen, produce equal and opposite impulsive forces that conserve momentum.

By Newton's third law, these internal forces act in pairwise equal and opposite directions. Therefore,  $\mathbf{f}_{ij} = -\mathbf{f}_{ji}$ , where  $\mathbf{f}_{ji}$  is the force that particle  $i$  exerts on particle  $j$  along the line joining the particles. The total internal force on particle  $i$  is then

$$\sum_{j=1, j \neq i}^n \mathbf{f}_{ij} ,$$

and, if we sum over all particles, we have

$$\sum_{i=1}^n \sum_{j=1, j \neq i}^n \mathbf{f}_{ij} = \mathbf{0} ,$$

since, for every force,  $\mathbf{f}_{ij}$ , there is an equal and opposite force,  $\mathbf{f}_{ji}$  for both normal and tangential forces. (The notation  $j \neq i$  means to exclude  $j=i$  in the sum since of course the particle induces no net force upon itself.)

## Conservation of Linear Momentum

The linear momentum of the system is defined as,

$$\mathbf{L} = \sum_{i=1}^n m_i \mathbf{v}_i . \quad (4)$$

From equation (2), we have that  $\mathbf{v}_i = \dot{\mathbf{r}}_i = \dot{\mathbf{r}}_G + \dot{\mathbf{r}}'_i$ , which, combined with the above equation, gives,

$$\mathbf{L} = \sum_{i=1}^n m_i (\dot{\mathbf{r}}_G + \dot{\mathbf{r}}'_i) = \sum_{i=1}^n m_i \mathbf{v}_G + \frac{d}{dt} \left( \sum_{i=1}^n m_i \mathbf{r}'_i \right) = m \mathbf{v}_G , \quad (5)$$

since by the definition of center of mass  $\sum_{i=1}^n m_i \mathbf{r}'_i = \mathbf{0}$ . We now consider the time variation of the linear momentum. If we assume that the reference frame  $xyz$  is inertial, then, starting from equation (4), we have,

$$\dot{\mathbf{L}} = \sum_{i=1}^n m_i \mathbf{a}_i = \sum_{i=1}^n (\mathbf{F}_i + \sum_{j=1, j \neq i}^n \mathbf{f}_{ij}) = \sum_{i=1}^n \mathbf{F}_i = \mathbf{F} , \quad (6)$$

where  $\mathbf{F}$  is the sum of all external forces acting on the system. Since the sum of the internal forces balance when summed over  $i$  and  $j$ , we are left with only the summation over the external forces. Thus, for a system of particles, we have that,

$$\dot{\mathbf{L}} = \mathbf{F} . \quad (7)$$

Note that, from equation (5), we can also write  $\dot{\mathbf{L}} = m \mathbf{a}_G$ . This is a powerful result. Note that the center of mass is in general not fixed to a particular particle but is a point in space about which the individual particles move.

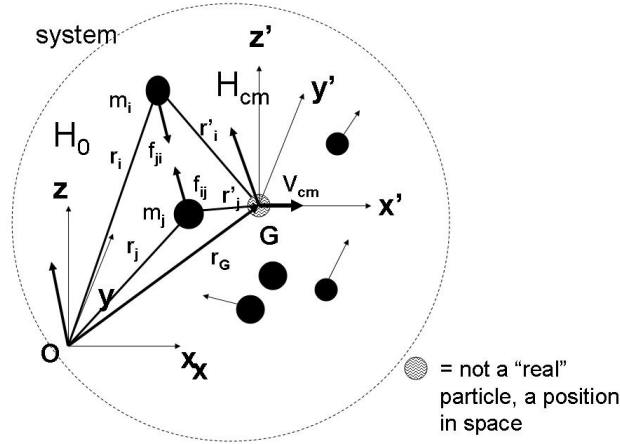
These ideas also describe the conservation of linear momentum under external and internal collisions. Since individual internal collisions between particles in the system conserve momentum, the sum of their interactions also conserves momentum. If we consider an external particle imparting momentum to the system, it could be treated as an external impulse. Conversely we can consider the particle about to collide to be a part of the system, and include its momentum as part of total system momentum, which is then conserved by Newton's law.

## Conservation of Angular Momentum

Since the angular momentum is defined with respect to a point in space, we will consider two cases, using a different reference point for each case: 1) conservation of angular momentum about a fixed (or more generally a non-accelerating) point  $O$ ; and 2) conservation of angular momentum about the center of mass, which could be accelerating in inertial space.

The figure shows the system: the coordinate system  $x, y, z$  about the fixed point  $O$ , the coordinate system  $x', y', z'$  about the center of mass,  $G$ , the internal forces between particles, the angular momentum  $\mathbf{H}_O$  about the point  $O$  and the angular momentum  $\mathbf{H}_G$  about the center of mass, point  $G$ , which in general are not equal.

### Conservation of Angular Momentum about a Fixed Point $O$



The angular momentum of a system of particles about a fixed point,  $O$ , is the sum of the angular momentum of the individual particles,

$$\mathbf{H}_O = \sum_{i=1}^n (\mathbf{r}_i \times m_i \mathbf{v}_i) . \quad (8)$$

The time variation of  $\mathbf{H}_O$  can be written as,

$$\dot{\mathbf{H}}_O = \sum_{i=1}^n (\dot{\mathbf{r}}_i \times m_i \mathbf{v}_i) + \sum_{i=1}^n (\mathbf{r}_i \times m_i \dot{\mathbf{v}}_i) = \mathbf{0} + \sum_{i=1}^n (\mathbf{r}_i \times (\mathbf{F}_i + \sum_{j=1, j \neq i}^n \mathbf{f}_{ij})) = \sum_{i=1}^n (\mathbf{r}_i \times \mathbf{F}_i) + \sum_{i=1}^n M_i . \quad (9)$$

where we replace  $m\dot{\mathbf{v}}_i$  by the sum of the forces acting on particle  $i$ :  $m\dot{\mathbf{v}}_i = (\mathbf{F}_i + \sum_{j=1, j \neq i}^n \mathbf{f}_{ij})$ .  $\sum_{i=1}^n M_i$  is the sum of any external moments that act on the system. The term  $(\dot{\mathbf{r}}_i \times m_i \mathbf{v}_i)$  in equation (9) is zero; since  $\dot{\mathbf{r}}_i = \mathbf{v}_i$ , the two vectors are parallel and their cross-product is zero. In the second term we may write  $\mathbf{r}_i \times \mathbf{f}_{ij} + \mathbf{r}_j \times \mathbf{f}_{ji} = (\mathbf{r}_i - \mathbf{r}_j) \times \mathbf{f}_{ij} = \mathbf{0}$  since the forces are aligned with  $(\mathbf{r}_i - \mathbf{r}_j)$ , and their values are equal and opposite, their cross product with  $(\mathbf{r}_i - \mathbf{r}_j)$  is zero, and therefore, the internal forces have no net effect on the total angular momentum change of the particle system. Therefore

$$\dot{\mathbf{H}}_O = \sum_{i=1}^n (\mathbf{r}_i \times m_i \dot{\mathbf{v}}_i) = \sum_{i=1}^n (\mathbf{r}_i \times \mathbf{F}_i) + \sum_{i=1}^n M_i . \quad (10)$$

When evaluating the moments which act to change the angular momentum from equation (13), we see that the sum of the internal moments is zero so that the only moment which acts to change the angular momentum is the moments created by the external forces about the point  $O$  plus any external moments applied to the



system.

Thus, we have that

$$\dot{\mathbf{H}}_O = \mathbf{M}_O , \quad (11)$$

where  $\mathbf{M}_O = \sum_{i=1}^n (\mathbf{r}_i \times \mathbf{F}_i) + \sum_{i=1}^n M_i$  is the total moment, about  $O$ , due to the applied external forces plus any external moments.

## Conservation of Angular Momentum about $G$

The angular momentum about the center of mass  $G$  is given by,

$$\mathbf{H}_G = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}_i) . \quad (12)$$

Taking the time derivative of equation (2), we obtain

$$\mathbf{v}_i = \dot{\mathbf{r}}_i = \dot{\mathbf{r}}_G + \dot{\mathbf{r}}'_i = \mathbf{v}_G + \mathbf{v}'_i . \quad (13)$$

Inserting this expression into equation 12, we obtain

$$\mathbf{H}_G = \sum_{i=1}^n (\mathbf{r}'_i \times m_i (\dot{\mathbf{r}}_G + \dot{\mathbf{r}}'_i)) = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \dot{\mathbf{r}}_G) + \sum_{i=1}^n (\mathbf{r}'_i \times m_i \dot{\mathbf{r}}'_i) = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}'_i) , \quad (14)$$

since  $\sum_{i=1}^n (\mathbf{r}'_i \times m_i \dot{\mathbf{r}}_G) = -\dot{\mathbf{r}}_G \times \sum_{i=1}^n m_i \mathbf{r}'_i = \mathbf{0}$  (see equation 3). We note that equations (12) and (14) give us alternative representations for  $\mathbf{H}_G$ . Equation (12) is called the *absolute* angular momentum (since it involves absolute velocities,  $\mathbf{v}_i$ ), whereas equation (14) is called the *relative* angular momentum (since it involves velocities,  $\mathbf{v}'_i$ , relative to  $G$ ). When  $G$  is chosen to be the origin for the relative velocities, both the absolute and relative angular momentum are identical. In general, the absolute and relative angular momentum with respect to an arbitrary point are not the same.

We can now go back to equation (12) and consider the time variation of  $\mathbf{H}_G$ ,

$$\dot{\mathbf{H}}_G = \sum_{i=1}^n (\dot{\mathbf{r}}'_i \times m_i (\mathbf{v}_G + \mathbf{v}'_i)) + \sum_{i=1}^n (\mathbf{r}'_i \times m_i \dot{\mathbf{v}}'_i) = \mathbf{0} + \sum_{i=1}^n (\mathbf{r}'_i \times \mathbf{F}_i) + \sum_{i=1}^n M_i . \quad (15)$$

In the above equation, the term  $\dot{\mathbf{r}}'_i \times m_i \mathbf{v}'_i$  is clearly zero, and  $\sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}_G) = -\mathbf{v}_G \times \sum_{i=1}^n m_i \mathbf{r}'_i = -\mathbf{v}_G \times d(\sum_{i=1}^n m_i \mathbf{r}'_i)/dt = \mathbf{0}$ . Thus, we have that

$$\dot{\mathbf{H}}_G = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \dot{\mathbf{v}}'_i) = \mathbf{M}_G . \quad (16)$$

Here,  $\mathbf{M}_G = \sum_{i=1}^n (\mathbf{r}'_i \times \mathbf{F}_i) + \sum_{i=1}^n M_i$ , is the total moment, about  $G$ , of the applied external forces plus any external moments. Note that external forces in general produce unequal moments about  $O$  and  $G$  while applied external moments (torques) produce the same moment about  $O$  and  $G$ .

The above expression is very powerful and allows us to solve, with great simplicity, a large class of problems

in rigid body dynamics. Its power lies in the fact that it is applicable in very general situations: In the derivation of equation (16), we have made no assumptions about the motion of the center of mass,  $G$ . That is, equation (16) is valid even when  $G$  is *accelerated*. We have implicitly assumed that the reference frame used to describe  $\mathbf{r}'_i$  in equation 13 is non-rotating with respect to the fixed frame  $xyz$  (otherwise, we would have written  $\dot{\mathbf{r}}'_i = \mathbf{v}'_i + \boldsymbol{\omega}' \times \mathbf{r}'_i$ , with  $\boldsymbol{\omega}'$ , the angular velocity of the frame considered). It is not difficult to show that equation (16) is still valid if the reference frame rotates, provided the angular velocity is *constant*. If the reference frame rotates with a constant angular velocity, the angular momentum will differ from that of equations (12) and (14) by a constant, but equation 16 still will be valid.

Finally, by combining equations 3 and 12, the angular momentum about a fixed point,  $O$ , can be expressed as a function of the angular momentum about the center of mass, as,

$$\mathbf{H}_O = \mathbf{r}_G \times m\mathbf{v}_G + \mathbf{H}_G. \quad (17)$$

Just as we could incorporate collisions in our statement of conservation of linear momentum, we can incorporate collision in our statement of conservation of angular momentum. Collisions conserve both linear and angular momentum. Just as changes in linear momentum result for linear impulses, changes in angular momentum result from angular impulses.

## Kinetic Energy for Systems of Particles

Here, we derive the expression for the kinetic energy of a system of particles that will be used in the following lectures. A typical particle,  $i$ , will have a mass  $m_i$ , an absolute velocity  $\mathbf{v}_i$ , and a kinetic energy  $T_i = (1/2)m_i\mathbf{v}_i \cdot \mathbf{v}_i = (1/2)m_iv_i^2$ .

The total kinetic energy of the system,  $T$ , is simply the sum of the kinetic energies for each particle,

$$T = \sum_{i=1}^n T_i = \sum_{i=1}^n \frac{1}{2}m_iv_i^2.$$

It is convenient to decompose the velocity of each particle,  $\mathbf{v}_i$ , into the velocity of the center of mass,  $\mathbf{v}_G$ , and the velocity relative to the center of mass,  $\dot{\mathbf{r}}'_i$ . Then,

$$T = \sum_{i=1}^n \frac{1}{2}m_i(\mathbf{v}_G + \dot{\mathbf{r}}'_i) \cdot (\mathbf{v}_G + \dot{\mathbf{r}}'_i) = \sum_{i=1}^n \frac{1}{2}m_i(v_G^2 + 2\mathbf{v}_G \cdot \dot{\mathbf{r}}'_i + \dot{r}'_i{}^2) = \frac{1}{2}mv_G^2 + \sum_{i=1}^n \frac{1}{2}m_i\dot{r}'_i{}^2,$$

since  $\mathbf{v}_G \cdot \sum_{i=1}^n m_i\dot{\mathbf{r}}'_i = 0$ , and  $\sum_{i=1}^n m_i = m$ . Thus, we see that the kinetic energy of a system of particles equals the kinetic energy of a particle of mass total  $m$  moving with the velocity of the center of mass, plus the kinetic energy due to the motion of the particles relative to the center of mass,  $G$ . We have said nothing about the conservation of energy for a system of particles. As we shall see, that depends upon the details of internal interactions and the work done by the external forces.

We will now particularize the conservation principles presented in the previous lectures to the case in which the system of particles considered is a 2D rigid body.

## Conservation of Angular Momentum for 2D Rigid Body

In subsequent Lectures, we will apply conservation of angular momentum for a general system in three dimensions. However, the dynamics of a 2D rigid body can easily be incorporated into our study of particle motion. We therefore specialize these results to the case of a 2D rigid body. The equations describing the general motion of a rigid body follow from the conservation laws for systems of particles. Since the general motion of a **2D rigid body** can be determined by three parameters (e.g.  $x$  and  $y$  coordinates of position, and a rotation angle  $\theta$ ), we will need to supply three equations. Conservation of linear momentum yields one vector equation, or two scalar equations. The additional condition is conservation of angular momentum. We saw that there are several ways to express conservation of angular momentum. In principle, they are all equivalent, but, depending on the problem situation, the use of a particular form may greatly simplify the problem. The best choices for the origin of coordinates are: 1) the center of mass  $G$ ; 2) a fixed point  $O$ .

### Conservation of Angular Momentum about the Center of Mass

When considering a 2D rigid body, the velocity of any point relative to  $G$  consists of a pure rotation and, therefore, can be expressed as

$$\mathbf{v}'_i = \boldsymbol{\omega} \times \mathbf{r}'_i,$$

where  $\boldsymbol{\omega}$  is the angular velocity vector perpendicular to the plane of motion. Both  $\mathbf{v}'_i$  and  $\mathbf{r}'_i$  are in the plane of the motion. These two equations can be combined to give,

$$\mathbf{H}_G = \sum_{i=1}^n (\mathbf{r}'_i \times m_i (\boldsymbol{\omega} \times \mathbf{r}'_i)) = \boldsymbol{\omega} \sum_{i=1}^n m_i r_i'^2 = \boldsymbol{\omega} \int_m r'^2 dm \quad (18)$$

For a continuous body, the sum over the mass points is replaced by an integral.  $\int_m r'^2 dm$  is defined as the mass moment of inertia  $I_G$  about the center of mass. Here, we have used the vector identity,  $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$ , and imposed the fact that  $\mathbf{r}'_i$  and  $\boldsymbol{\omega}$  are perpendicular for 2D planar bodies. Thus, for a 2-D rigid body, the conservation law for angular momentum about the center of mass,  $G$  is

$$I_G \boldsymbol{\alpha} = \mathbf{M}_G, \quad (19)$$

where  $I_G = \int_m r'^2 dm$ ,  $\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}$  and  $\mathbf{M}_G$  is the total moment about  $G$  due to external forces and external moments. Although equation (7) is a vector equation,  $\boldsymbol{\alpha}$  and  $\mathbf{M}_G$  are always perpendicular to the plane of motion, and, therefore, equation (7) only yields one scalar equation. The moment of inertia,  $I_G$ , can be interpreted as a measure of the body's resistance to changing its angular velocity as a result of applied external moments. The moment of inertia,  $I_G$ , is a scalar quantity. It is a property of the solid which indicates the way in which the mass of the solid is distributed relative to the center of mass. For example, if most of the mass is far away from the center of mass,  $r'_i$  will be large, resulting in a large moment of inertia. The dimensions of the moment of inertia are  $[M][L^2]$ .

## Conservation of Angular Momentum about a fixed point $O$

If the fixed point  $O$  is chosen as the origin, a similar result is obtained. Since for a 2D rigid body the velocity in the coordinate system fixed at the point  $O$  is

$$\mathbf{v}_i = \boldsymbol{\omega} \times \mathbf{r}_i ,$$

conservation of angular momentum gives

$$\mathbf{H}_O = \sum_{i=1}^n (\mathbf{r}_i \times m_i (\boldsymbol{\omega} \times \mathbf{r}_i)) = \boldsymbol{\omega} \sum_{i=1}^n m_i r_i^2 = \boldsymbol{\omega} \int_m r^2 dm = \boldsymbol{\omega} I_O \quad (20)$$

where  $I_O$  is the moment of inertia about the point  $O$ . In general,  $I_O \neq I_G$ . The conservation law for angular momentum about the fixed point  $O$  is then

$$I_O \boldsymbol{\alpha} = \mathbf{M}_O, \quad (21)$$

where  $I_O = \int_m r^2 dm$ ,  $\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}$  and  $\mathbf{M}_O$  is the total applied moment due to external forces and moments (torques). Also it is important to point out that both the angular velocity  $\boldsymbol{\omega}$  and the angular acceleration  $\boldsymbol{\alpha}$  are the same for any point on a rigid body:  $\boldsymbol{\omega}_G = \boldsymbol{\omega}_O$ ,  $\boldsymbol{\alpha}_G = \boldsymbol{\alpha}_O$ .

Most textbooks on dynamics have tables of moments of inertia for various common shapes: cylinders, bars, plates. See Meriam and Kraige, *Engineering Mechanics*, DYNAMICS (Appendix B) for more examples.

## Radius of Gyration

It is common to report the moment of inertia of a rigid body in terms of the radius of gyration,  $k$ . This is defined as

$$k = \sqrt{\frac{I}{m}} ,$$

and can be interpreted as the root-mean-square of the mass element distances from the axis of rotation.

Since the moment of inertia depends upon the choice of axis, the radius of gyration also depends upon the choice of axis. Thus we write

$$k_G = \sqrt{\frac{I_G}{m}} ,$$

for the radius of gyration about the center of mass, and

$$k_O = \sqrt{\frac{I_O}{m}} ,$$

for the radius of gyration about the fixed point  $O$ .

## Parallel Axis Theorem

We will often need to find the moment of inertia with respect to a point other than the center of mass. For instance, the moment of inertia with respect to a given point,  $O$ , is defined as

$$I_O = \int_m r^2 dm .$$

Assuming that  $O$  is a fixed point,  $\mathbf{H}_O = I_O \boldsymbol{\omega}$ . If we know  $I_G$ , then the moment of inertia with respect to point  $O$ , can be computed easily using the parallel axis theorem. Given the relations  $r^2 = \mathbf{r} \cdot \mathbf{r}$  and  $\mathbf{r} = \mathbf{r}_G + \mathbf{r}'$ , we can then write,

$$I_O = \int_m r^2 dm = \int_m (r_G^2 + 2\mathbf{r}_G \cdot \mathbf{r}' + r'^2) dm = r_G^2 \int_m dm + 2\mathbf{r}_G \cdot \int_m \mathbf{r}' dm + \int_m r'^2 dm = mr_G^2 + I_G ,$$

since  $\int_m \mathbf{r}' dm = \mathbf{0}$ .

From this expression, it also follows that the moment of inertia with respect to an arbitrary point is **minimum** when the point coincides with  $G$ . Hence, the **minimum** value for the moment of inertia is  $I_G$ .

## Equations of Motion for a 2D rigid Body.

Now that we have developed the equation governing conservation of angular momentum for a 2D rigid body in planar motion, we can state the governing equations for this three degree of freedom system.

The conservation of linear momentum yields the vector equation,

$$m\mathbf{a}_G = \mathbf{F} , \quad (22)$$

where  $m$  is the body mass,  $\mathbf{a}_G$  is the acceleration of the center of mass, and  $\mathbf{F}$  is the sum of the external forces acting on the body.

Conservation of angular momentum requires

$$\dot{\mathbf{H}}_G = \mathbf{M}_G = \dot{\boldsymbol{\omega}} I_G = \boldsymbol{\alpha} I_G \quad (23)$$

where  $I_G$  is the moment of inertia about the center of mass;  $\boldsymbol{\omega}$  is the angular velocity, whose vector direction is perpendicular to the  $x, y$  coordinate system; and  $\boldsymbol{\alpha}$  is the angular acceleration. For impulsive forces, we write that the change in angular momentum is equal to the time integral of the applied moments, whether or not these moments are impulsive (but we have to be able to do the time integral, so impulsive forces are easier).

A body fixed at a point  $O$  is a single degree of freedom system. Therefore, only one equation is required, conservation of angular momentum about the point  $O$ .

$$\dot{\mathbf{H}}_O = \mathbf{M}_O = \dot{\boldsymbol{\omega}} I_O = \boldsymbol{\alpha} I_O \quad (24)$$

## Impulsive Motion

Just as we applied the continuous form of Newton's law for a particle to the motion resulting from impulsive forces and collisions, we can extend these result to impulsive linear and angular motion resulting from impulsive forces and moments.

$$\mathbf{H}_{G2} - \mathbf{H}_{G1} = \int_{t_1}^{t_2} \sum_{i=1}^n (\mathbf{r}'_i \times \mathbf{F}_i) . \quad (25)$$

where  $\int_{t_1}^{t_2} (\sum_{i=1}^n \mathbf{r}'_i \times \mathbf{F}_i) = \mathbf{A}_I$  is the **angular impulse**, the integral over time of the sum of the moments acting about the center of mass. A similar result holds for moments taken about the point  $O$ .

$$\mathbf{H}_{O2} - \mathbf{H}_{O1} = \int_{t_1}^{t_2} \sum_{i=1}^n (\mathbf{r}_i \times \mathbf{F}_i) . \quad (26)$$

where  $\int_{t_1}^{t_2} (\sum_{i=1}^n \mathbf{r}_i \times \mathbf{F}_i) = \mathbf{A}_I$

## Kinetic Energy for a 2D Rigid Body

We specialize the result of the analysis of kinetic energy for a system of particles about the center of mass for the case of a two-dimensional rigid body for which the velocity of a particle is given by

$$\mathbf{v}'_i = \boldsymbol{\omega} \times \mathbf{r}'_i . \quad (27)$$

Applying this to the general form for from Equation

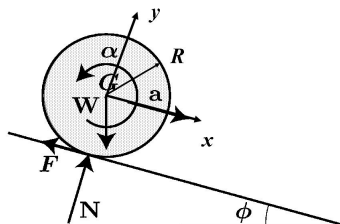
$$T = \sum_{i=1}^n \frac{1}{2} m_i (\mathbf{v}_G + \dot{\mathbf{r}}'_i) \cdot (\mathbf{v}_G + \dot{\mathbf{r}}'_i) = \frac{1}{2} m v_G^2 + \frac{1}{2} \boldsymbol{\omega} \int_m r'^2 dm = \frac{1}{2} m v_G^2 + \frac{1}{2} \omega^2 I_G. \quad (28)$$

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### Example

### Cylinder on a ramp

Let us consider a uniform cylinder of weight  $W$  and radius  $R$  rolling without slipping down a ramp of angle  $\phi$ .



We consider the fixed reference frame,  $x, y$ , instantaneously located at the center of mass as shown in the figure. The equations of motion, 22 and 23, are, in this case,

$$\begin{aligned} m\ddot{x}_G &= W \sin \phi - F \\ m\ddot{y}_G &= N - W \cos \phi \\ I_G \alpha &= -FR \end{aligned}$$

In these equations,  $W = mg$ , but the normal force,  $N$ , and the friction force,  $F$ , are unknown. These two additional unknowns can be determined if we provide two additional kinematic conditions. First, we have that  $\ddot{y}_G = 0$ , from which we can determine  $N$  as

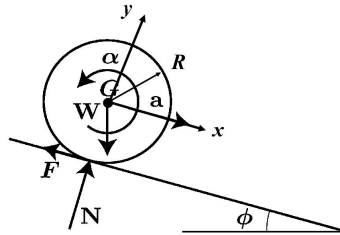
$$N = W \cos \phi .$$

Second, since the cylinder rolls without sliding, we have that  $\ddot{x}_G = -R\alpha$ . Solving for  $\ddot{x}_G$ , we obtain

$$\ddot{x}_G = \frac{g \sin \phi}{1 + (I_G/mR^2)} , \quad (29)$$

and  $F = (I_G \ddot{x}_G)/R^2$ . For the uniform cylinder, we have that  $I_G = mR^2/2$  and  $\ddot{x}_G = (2g \sin \phi)/3$ . If instead of having a uniform disc, we had a uniform ring with all the mass concentrated at the rim, then  $I_G = mR^2$  and  $\ddot{x}_G = (g \sin \phi)/2$ . Also, if there was no friction,  $F$  would be zero, the cylinder would not rotate, and the acceleration would be that of a sliding mass point  $\ddot{x}_G = g \sin \phi$ .

We now consider that the total cylinder mass is located as a mass point in the center of the cylinder. Then the moment of inertia about  $G$  is zero. The cylinder will roll down the ramp but it requires no moment to cause this motion.



Writing the equations about the center of mass we obtain the result that since the moment of inertia is zero, the moment about  $G$  must be zero. Since both  $g$  and  $N$  are directed towards the center of mass, the only moment comes from the friction  $F$ ; therefore,  $F$  must be zero.

## Rotation about a Fixed Axis

For cases in which there is a fixed point in the body, the motion of the body can be described with a single parameter (e.g. the rotation angle). In principle, we could still consider equations (22) and (23) and use the

kinematic conditions to enforce that the motion of the fixed point is zero. Alternatively, the analysis is often simplified if we consider the conservation of angular momentum about the fixed point directly. In this case, we have,

$$I_O \alpha = M_O . \quad (30)$$

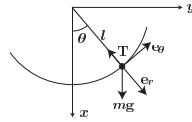
where  $I_O$  is the moment of inertia about **the fixed point**,  $O$ ,  $\alpha$  is the angular acceleration, and  $M_O$  is the sum of external moments about  $O$ .

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**Example**
**Compound Pendulum**


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An example of a rigid body rotating about a fixed axis is the compound pendulum. A compound pendulum is a rigid body hinged, without friction, about a horizontal axis offset from its center of mass, and acted upon by its own weight as an external force. The angular velocity of the pendulum about its pivot point is  $\omega = \dot{\theta}$ ; the angular acceleration of the pendulum about its pivot point is  $\alpha = \ddot{\theta}$ . As noted before, every mass point in the 2D solid body of the pendulum has the identical angular velocity and angular acceleration,  $\omega$  and  $\alpha$ .



It is convenient to apply the conservation on angular momentum about the point  $O$  because the unknown reaction forces at the support contribute no moment about  $O$ . The conservation of angular momentum about  $O$  results in

$$I_O \ddot{\theta} = M_O = -mgL \sin \theta ,$$

or,

$$\ddot{\theta} = -\frac{g}{I_O/mL} \sin \theta .$$

Comparing it with the equation for a simple pendulum from Lecture 10, we see that the motion of a compound pendulum is identical to the motion of a simple pendulum of *equivalent* length,  $L_{equiv}$ ,

$$L_{equiv} = \frac{I_O}{mL} .$$

We now apply conservation of angular momentum about the center of mass to determine the reaction force  $\mathbf{F}$ . We see that gravity produces no moment about  $G$  and the only moments are produced by reaction forces at the attachment point. The tension  $\mathbf{T}$  at the attachment point produces no moment about the center of mass. The force  $\mathbf{F}$  produces a moment  $M_G = FL$  which must equal  $I_G \ddot{\theta}$ . We remark on our earlier observation from Lecture 10 that if the pendulum is a mass point, the moment of inertia about the center of mass is zero. Therefore, there can be no moment exerted by the attachment point about the center of mass. Therefore there can be no reaction in the direction normal to the direction from the pivot to the center of mass—which would produce a moment. Therefore the force is aligned with the direction of the tension  $\mathbf{T}$  as in a string. We now apply Newton's law about the center of mass in the direction of  $\mathbf{T}$ , balancing the



tension and a component of the gravitational force with the centripetal acceleration to determine  $T$ .

$$-T + mg \cos \theta = -mL\dot{\theta}^2 \quad (31)$$

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#### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition  
4/1, 4/2, 4/3 (kinetic energy expression only), 4/4, 4/5 (momentum only) 6/1, 6/2, 6/3, 6/4, 6/5 (angular  
momentum)

## Lecture 13 — Work and Energy

So far we have used Newton's second law  $\mathbf{F} = m\mathbf{a}$  to establish the instantaneous relation between the sum of the forces acting on a particle and the acceleration of that particle. Once the acceleration is known, the velocity (or position) is obtained by integrating the expression of the acceleration (or velocity). Newton's law and the conservation of momentum give us a vector description of the motion of a particle in three dimensions.

There are two situations in which the cumulative effects of unbalanced forces acting on a particle are of interest to us. These involve:

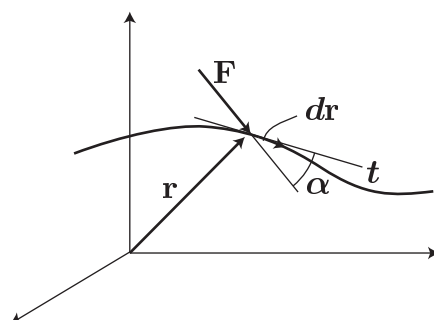
- a) forces acting along the trajectory. In this case, integration of the forces with respect to the *displacement* leads to the principle of *work* and *energy*.
- b) forces acting over a time interval. In this case, integration of the forces with respect to the *time* leads to the principle of *impulse* and *momentum*.

It turns out that in many situations, these integrations can be carried beforehand to produce equations that relate the velocities at the initial and final integration points. In this way, the velocity can be obtained directly, thus making it unnecessary to solve for the acceleration. We shall see that these integrated forms of the equations of motion are very useful in the practical solution of dynamics problems.

### Mechanical Work

Consider a force  $\mathbf{F}$  acting on a particle that moves along a path. Let  $\mathbf{r}$  be the position of the particle measured relative to the origin  $O$ . The work done by the force  $\mathbf{F}$  when the particle moves an infinitesimal amount  $d\mathbf{r}$  is defined as

$$dW = \mathbf{F} \cdot d\mathbf{r} . \quad (1)$$



That is, the work done by the force  $\mathbf{F}$ , over an infinitesimal displacement  $d\mathbf{r}$ , is the dot product of  $\mathbf{F}$  and  $d\mathbf{r}$ . It follows that the work is a *scalar* quantity. Using the definition of the dot product, we have that  $dW = \mathbf{F} \cdot d\mathbf{r} = F ds \cos \alpha$ , where  $ds$  is the modulus of  $d\mathbf{r}$ , and  $\alpha$  is the angle between  $\mathbf{F}$  and  $d\mathbf{r}$ . Since  $d\mathbf{r}$  is parallel to the tangent vector to the path,  $\mathbf{e}_t$ , (i.e.  $d\mathbf{r} = ds \mathbf{e}_t$ ), we have that  $\mathbf{F} \cdot \mathbf{e}_t = F_t$ . Thus,

$$dW = F_t ds , \quad (2)$$

which implies that only the tangential component of the force “does” work.

During a finite increment in which the particle moves from position  $\mathbf{r}_1$  to position  $\mathbf{r}_2$ , the total work done by  $\mathbf{F}$  is

$$W_{12} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = \int_{s_1}^{s_2} F_t ds . \quad (3)$$

Here,  $s_1$  and  $s_2$  are the path coordinates corresponding to  $\mathbf{r}_1$  and  $\mathbf{r}_2$ .

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### Note

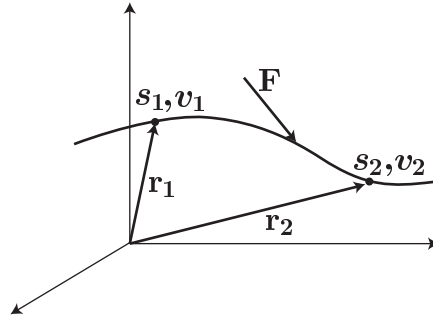
### Units of Work

In the international system, SI, the unit of work is the *Joule* (J). We have that  $1 \text{ J} = 1 \text{ N} \cdot \text{m}$ . In the English system the unit of work is the ft-lb. We note that the units of work and moment are the same. It is customary to use ft-lb for work and lb-ft for moments to avoid confusion.

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## Principle of Work and Energy

We now consider a particle moving along its path from point  $\mathbf{r}_1$  to point  $\mathbf{r}_2$ . The path coordinates at points 1 and 2 are  $s_1$  and  $s_2$ , and the corresponding velocity magnitudes  $v_1$  and  $v_2$ .



If we start from (3) and use Newton’s second law ( $\mathbf{F} = m\mathbf{a}$ ) to express  $F_t = ma_t$ , we have

$$W_{12} = \int_{s_1}^{s_2} F_t ds = \int_{s_1}^{s_2} ma_t ds = \int_{v_1}^{v_2} m \frac{dv}{ds} \frac{ds}{dt} ds = \int_{v_1}^{v_2} mv dv = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 . \quad (4)$$

Here, we have used the relationship  $a_t ds = v dv$ , which can be easily derived from  $a_t = \dot{v}$  and  $v = \dot{s}$ .

Defining the *kinetic energy*<sup>1</sup>, as

$$T = \frac{1}{2}mv^2 ,$$

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<sup>1</sup>The use of  $T$  to denote the kinetic energy is customary in dynamics textbooks.

we have that,

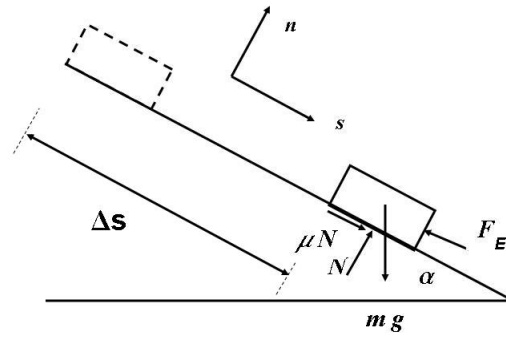
$$W_{12} = T_2 - T_1 \quad \text{or} \quad T_1 + W_{12} = T_2 . \quad (5)$$

The above relationship, known as the *principle of work and energy*, states that the mechanical work done on a particle is equal to the change in the kinetic energy of the particle.

### Example

### Block on an Incline

Consider a block resting on an incline at the lower position in the presence of gravity. The gravitational force acting in the vertical direction is  $mg$ . The block is supported on the plane by a normal force  $F_N = mg \sin \alpha$ . We desire to push the block up to the higher (dashed) position. To do this, we must apply an external force  $F_E = mg \sin \alpha + \mu mg \cos \alpha$  to overcome the forces internal to the system  $F_I$ , the component of gravity along the plane plus the friction force acting to oppose the motion:  $F_I = -(mg \sin \alpha + \mu mg \cos \alpha)$ . In this process, the normal force does no work.



The total tangential force acting on the block in its direction of motion is then

$$F_T = F_E + F_I = mg \sin \alpha + \mu mg \cos \alpha - (mg \sin \alpha + \mu mg \cos \alpha) = 0 \quad (6)$$

for a total force of zero. Under these assumptions, having started with zero velocity, the block arrives at position 2 with zero velocity. Therefore, there is no change in kinetic energy and therefore by equation (5), no work is done.

$$W_{12} = T_2 - T_1 = F_E \Delta s - F_I \Delta s = 0 \quad (7)$$

At first, this does not agree with our intuition. We certainly felt we did work in pushing the block up the plane. But the work done by the friction force and gravitational force exactly canceled this work. We also feel that having raised the block to a higher position, there is some inherent "gain" in energy which could be

collected in the future. Our intuition is correct. These issues will be considered when we consider balance and interchanges between kinetic and potential energy.

## Work Done by External and Internal Forces

We now consider that several types of forces act on the particle. For the moment we shall refer to each of these as  $\mathbf{F}_i$ . Thus the total work done on the particle may be determined by summing the individual components of work done by these individual forces  $\mathbf{F}_i$ :

$$\sum_{i=1}^n W_i = \sum_{i=1}^n \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F}_i \cdot d\mathbf{r} \quad (8)$$

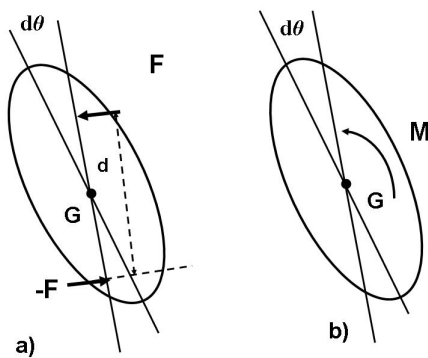
In a typical dynamical system, the various forces  $\mathbf{F}_i$  are of two types: **external** forces  $F_{i,E}$  and **internal** forces  $F_{i,I}$ . The external force results from an external actor applying an arbitrary force to the system. The external force does work and changes the energy of the system at the whim of the actor. When the external force is removed, the system may oscillate or otherwise move subject to internal forces. Since the body is rigid and the internal forces act in equal and opposite directed forces applied to the rigid body are capable of doing any work.

### Example

### Work done by Couples

If the sum of the external forces acting on the rigid body is zero, it is still possible to have non-zero work. Consider, for instance, a moment  $\mathbf{M}$  acting on a rigid body. This moment could be produced by a couple: forces  $\mathbf{F}$  equal and opposite applied at a distance  $d$  producing a moment  $M = Fd$ , shown in a); or a pure moment as shown in b). If the body undergoes a pure translation, it is clear that all the points in the body experience the same displacement, and, hence, the total work done by a couple is zero. If, on the other hand, the body experiences a rotation  $d\theta$ , then the work done by the couple or alternatively by the pure moment is

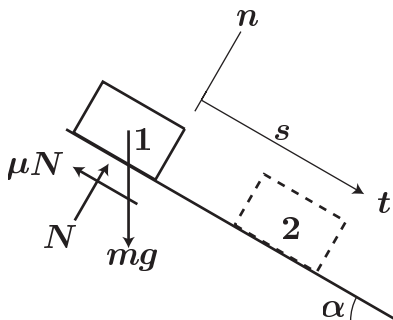
$$dW = dF d\theta = M d\theta \quad (9)$$



If  $M$  is constant, the work is simply  $W_{1-2} = M(\theta_2 - \theta_1)$ .

**Example****Block sliding down an incline**

A block is released from rest at the top of a ramp. The coefficient of kinetic friction between the surface of the ramp and the block is  $\mu$ . We want to determine the velocity of the block as a function of the distance traveled on the ramp,  $s$ .



The forces on the block are: the weight,  $mg$ , the normal force,  $N$ , and, the friction force,  $\mu N$ . We have that  $F_n = ma_n$  and  $F_t = ma_t$ . Since  $F_n = N - mg \cos \alpha$  and  $a_n = 0$ , we have  $N = mg \cos \alpha$ . Thus,  $F_t = mg \sin \alpha - \mu N = mg \sin \alpha - \mu mg \cos \alpha$ , which is constant. If we apply the principle of work and energy between the position (1), when the block is at rest at the top of the ramp, and the position (2), when the block has traveled a distance  $s$ , we have  $T_1 = 0$ ,  $T_2 = (mv^2)/2$ , and the work done by  $F_t$  is simply  $W_{12} = F_t s$ . Thus,

$$T_1 + W_{12} = T_2, \quad \text{or,} \quad mg(\sin \alpha - \mu \cos \alpha)s = \frac{1}{2}mv^2.$$

From which we obtain, for the velocity,

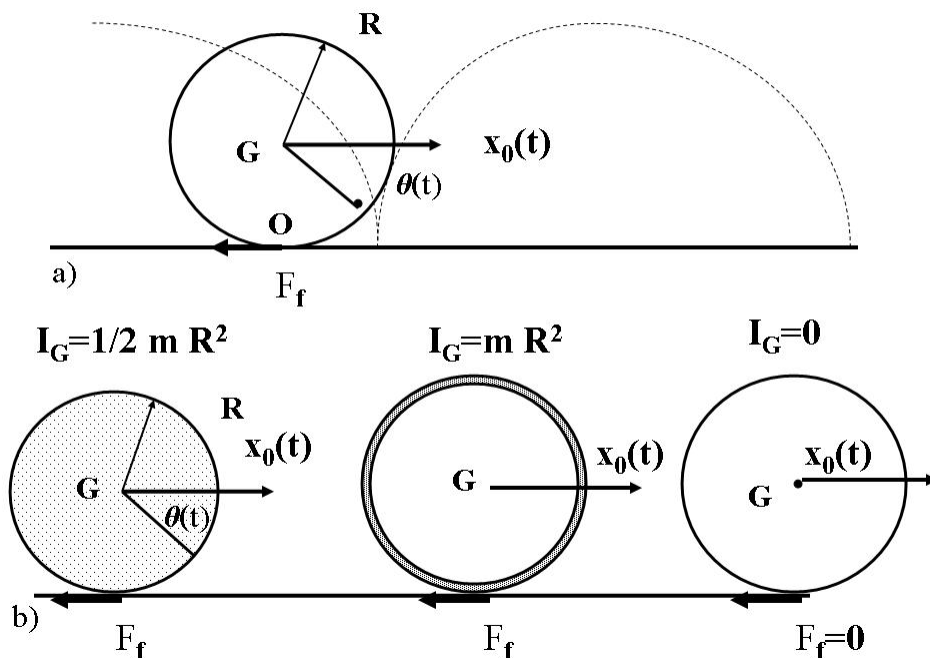
$$v = \sqrt{2g(\sin \alpha - \mu \cos \alpha)s}.$$

We make two observations: first, the normal force,  $N$ , does no work since it is, at all times, perpendicular to the path, and second, we have obtained the velocity of the block directly without having to carry out any integrations. Note that an alternative, longer approach would have been to directly use  $\mathbf{F} = m\mathbf{a}$ , and integrate the corresponding expression for the acceleration.

## Rolling Cylinder, Friction Forces, Work

The cylinder rolling without slip on a flat plane is a very basic configuration in dynamics. As noted in Lecture 2, it is a single degree of freedom system with a definite relationship between the position of the center of the cylinder,  $x_0(t)$ , and the rotation angle  $\theta(t)$ :  $\theta(t) = -x_0(t)/R$ . The kinematics of the rolling cylinder are shown in a). Consider a mass point at the edge of the cylinder. The dashed curve shows the path taken by this mass point. Of significance is the behavior near the plane. The point O is an instantaneous

center of rotation; the tangential velocity of the cylinder is zero about this point. The acceleration of the mass point is not zero, nor is its vertical velocity.



The dynamics of the rolling cylinder is shown in b). If the cylinder moves with constant velocity, nothing more need be said. However if  $\ddot{x}_0(t)$  is not zero, then  $\ddot{\theta} = \dot{\omega} = \alpha$ , the angular acceleration will not be zero. If the cylinder has a non-zero moment of inertia, the required moment can only come from friction at the contact point on the plane:  $F_f R = I_G \alpha$ , where  $I_G$  is the moment of inertia about the center of mass G. Since the point in contact with the plane is an instantaneous center of rotation—does not slide—this friction does no work. Also shown in b) are a variety of configurations of rolling cylinders. The solid cylinder has a moment of inertia  $I_G = 1/2 m R^2$ ; the cylinder whose mass is concentrated in the rim has a moment of inertia  $I_G = m R^2$ ; the cylinder whose mass point is concentrated at G has a moment of inertia of zero. Therefore, for this case no moment is required to change its angular velocity and it behaves dynamically like a mass point moving on a frictionless surface. The collision of two mass points can easily be realized by the collision of two such rolling cylinders.

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**Note**
**Cylinder rolling down a ramp**

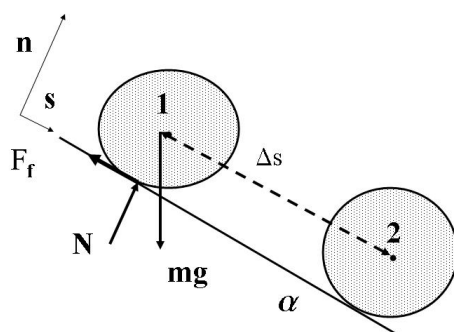
In parallel with our discussion of a block sliding down a ramp in the presence of friction, we now consider a cylinder rolling down a ramp in the presence of friction. We assume that friction force are large enough to keep the kinematic relationship between the velocity of the cylinder and the angular velocity of the cylinder intact. The cylinder is located at an initial position **1** as shown, and is at rest. It is released and rolls down to position **2** where we observe it in motion. The forces acting on the cylinder are gravity, the normal force  $\mathbf{N}$ , and the friction force  $\mathbf{F}_f$ . Although the friction force is necessary to keep the kinematic relationship between the position of the center of the cylinder and its accumulated angular rotation intact, **it does no**

**work.** Since the cylinder has no motion in the normal direction, the normal force **does no work**. Therefore the only work is done by gravity. Thus we can write

$$T_1 + W_{12} = T_2 \quad (10)$$

The initial kinetic energy is zero. The work done by gravity is  $W_{12} = mg \sin \alpha$ ; the final kinetic energy which includes both the kinetic energy due to translation of the center of mass and the kinetic energy due to rotation about the center of mass is  $T_2 = 1/2 mv_G^2 + 1/2 I_G (v/R)^2$ . Therefore the final velocity of the center of mass of the cylinder is

$$v_G = \sqrt{\frac{2g \sin \alpha \Delta s}{1 + \frac{I_G}{mR^2}}} \quad (11)$$




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### Note

### Alternative expressions for $dW$

We have seen in expression (2) that a convenient set of coordinates to express  $dW$  are the tangential- normal- binormal coordinates. Alternative expressions can be derived for other coordinate systems. For instance, we can express  $dW = \mathbf{F} \cdot d\mathbf{r}$  in:

cartesian coordinates,

$$dW = F_x dx + F_y dy + F_z dz ,$$

cylindrical (polar) coordinates,

$$dW = F_r dr + F_\theta r d\theta + F_z dz ,$$

or spherical coordinates,

$$dW = F_r dr + F_\theta r \cos \phi d\theta + F_\phi r d\phi .$$



As an illustration, let's calculate the work done by a constant force, such as that due to gravity. The force on a particle of mass  $m$  is given by  $\mathbf{F} = -mg\mathbf{k}$ . When the particle moves from position  $\mathbf{r}_1 = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$  to position  $\mathbf{r}_2 = x_2\mathbf{i} + y_2\mathbf{j} + z_2\mathbf{k}$ , work is done, and the work may be written as

$$W_{12} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = \int_{z_1}^{z_2} -mg \, dz = -mg(z_2 - z_1) .$$

## Power

In many situations it is useful to consider the rate at which a device can deliver work. The work per unit time is called the *power*,  $P$ . Thus,

$$P = \frac{dW}{dt} = \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} = \mathbf{F} \cdot \mathbf{v} .$$

The unit of power in the SI system is the *Watt* (W). We have that 1 W = 1 J /s. In the English system the unit of power is the ft-lb/s. A common unit of power is also the *horse power* (hp), which is equivalent to 550 ft-lb/s, or 746 W.

### Note

### Efficiency

The ratio of the power delivered *out of* a system,  $P_{out}$ , to the power delivered *in to* the system,  $P_{in}$ , is called the *efficiency*,  $e$ , of the system.

$$e = \frac{P_{out}}{P_{in}} .$$

This definition assumes that the energy into and out of the system flows continuously and is not retained within the system. The efficiency of any real machine is always less than unity since there is always some mechanical energy dissipated as heat due to friction forces.

## ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

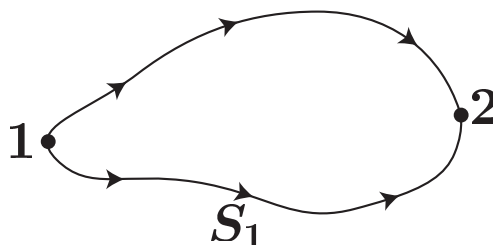
## Lecture 14 — Conservative Forces and Potential Energy

We have seen that the work done by a force  $\mathbf{F}$  on a particle is given by  $dW = \mathbf{F} \cdot d\mathbf{r}$ . If the work done by  $\mathbf{F}$  when the particle moves from any position  $\mathbf{r}_1$  to any position  $\mathbf{r}_2$  can be expressed as

$$W_{12} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = -(V(\mathbf{r}_2) - V(\mathbf{r}_1)) = V_1 - V_2, \quad (1)$$

then we say that the force is **conservative**. In the above expression, the scalar function  $V(\mathbf{r})$  is called the **potential**. It is clear that the potential satisfies  $dV = -\mathbf{F} \cdot d\mathbf{r}$  (the minus sign is included for convenience). There are two main consequences that follow from the existence of a potential:

1. The work done by a conservative force between points  $\mathbf{r}_1$  and  $\mathbf{r}_2$  is **independent of the path**. This follows from (1) since  $W_{12}$  only depends on the initial and final potentials  $V_1$  and  $V_2$  (and not on how we go from  $\mathbf{r}_1$  to  $\mathbf{r}_2$ ).
2. The work done by potential forces is **recoverable**. Consider the work done in going from point  $\mathbf{r}_1$  to point  $\mathbf{r}_2$ ,  $W_{12}$ . If we now go from point  $\mathbf{r}_2$  to  $\mathbf{r}_1$ , we have that  $W_{21} = -W_{12}$ , since the total work  $W_{12} + W_{21} = (V_1 - V_2) + (V_2 - V_1) = 0$ .



In one dimension any force which is only a function of position is conservative. That is, if we have a force,  $F(x)$ , which is only a function of position, then  $F(x) dx$  is always a perfect differential. This means that we can define a potential function as

$$V(x) = - \int_{x_0}^x F(x) dx,$$

where  $x_0$  is arbitrary.

In two and three dimensions, we might expect that any force that depends only on position,  $\mathbf{F}(\mathbf{r})$ , would be conservative. However, it turns out that, in general, this is not sufficient. In multiple dimensions, the

condition for a force field to be conservative is that it can be expressed as the gradient of a potential function. That is,

$$\mathbf{F}_C = -\nabla V .$$

This result follows from the gradient theorem, which is often called the fundamental theorem of calculus, which states that the integral

$$-\int_{\mathbf{r}_1}^{\mathbf{r}_2} \nabla V \cdot d\mathbf{r} = -(V_2 - V_1)$$

is independent of the path between  $\mathbf{r}_1$  and  $\mathbf{r}_2$ . Therefore the work done by conservative forces depends only upon the endpoints  $\mathbf{r}_2$  and  $\mathbf{r}_1$  rather than the details of the path taken between them.

$$\int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F}_C \cdot d\mathbf{r} = -\int_{\mathbf{r}_1}^{\mathbf{r}_2} \nabla V \cdot d\mathbf{r} = -(V_2 - V_1)$$

In the general case, we will deal with forces that are a combination of conservative and non-conservative forces:

$$\mathbf{F} = \mathbf{F}_C + \mathbf{F}_{NC} = -\nabla V + \mathbf{F}_{NC} .$$

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### Note

### The gradient operator, $\nabla$

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The gradient operator,  $\nabla$  (called “del”), in cartesian coordinates is defined as

$$\nabla(\ ) \equiv \frac{\partial(\ )}{\partial x} \mathbf{i} + \frac{\partial(\ )}{\partial y} \mathbf{j} + \frac{\partial(\ )}{\partial z} \mathbf{k} .$$

When operating on a scalar function  $V(x, y, z)$ , the result  $\nabla V$  is a vector, called the gradient of  $V$ . The components of  $\nabla V$  are the derivatives of  $V$  along each of the coordinate directions,

$$\nabla V \equiv \frac{\partial V}{\partial x} \mathbf{i} + \frac{\partial V}{\partial y} \mathbf{j} + \frac{\partial V}{\partial z} \mathbf{k} .$$

If we consider a particle moving due to conservative forces with potential energy  $V(x, y, z)$ , as the particle moves from point  $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$  to point  $\mathbf{r} + d\mathbf{r} = (x + dx)\mathbf{i} + (y + dy)\mathbf{j} + (z + dz)\mathbf{k}$ , the potential energy changes by  $dV = V(x + dx, y + dy, z + dz) - V(x, y, z)$ . For small increments  $dx, dy$  and  $dz$ , and  $dV$ , can be expressed, using Taylor series expansions, as

$$dV = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz = \nabla V \cdot d\mathbf{r} ,$$

where  $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$ .

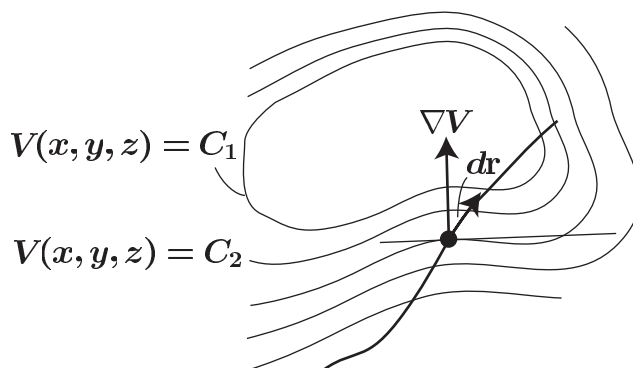
This equation expresses the fundamental property of the gradient. The gradient allows us to find the change in a function induced by a change in its variables.

If we write  $V(x, y, z) = C$ , for some constant  $C$ , this is the implicit equation of a surface, which is called a constant energy surface. This surface is made up by all the points in the  $x, y, z$  space for which the function  $V(x, y, z)$  is equal to  $C$ . It is clear that if a particle moves on a constant energy surface,  $dV = 0$ , since  $V$  is constant on that surface. Therefore, when a particle moves on a constant energy surface,  $d\mathbf{r}$  will be tangent

to that surface, and since

$$0 = dV = \nabla V \cdot d\mathbf{r} ,$$

we have that  $\nabla V$  is perpendicular to any tangent to the surface. This situation is illustrated in the picture below for the two dimensional case. Here, the constant energy surfaces are contour curves, and we can see that the gradient vector is always normal to the contour curves.



### Note

### Gradient operator in cylindrical coordinates

The gradient operator can be expressed in cylindrical coordinates by writing  $x = r \cos \theta$ ,  $y = r \sin \theta$ , and  $r = \sqrt{x^2 + y^2}$ ,  $\theta = \tan^{-1}(y/x)$ . Thus, applying the chain rule for differentiation, we have

$$\begin{aligned} \frac{\partial(\ )}{\partial x} &= \frac{\partial r}{\partial x} \frac{\partial(\ )}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial(\ )}{\partial \theta} = \cos \theta \frac{\partial(\ )}{\partial r} - \frac{\sin \theta}{r} \frac{\partial(\ )}{\partial \theta} \\ \frac{\partial(\ )}{\partial y} &= \frac{\partial r}{\partial y} \frac{\partial(\ )}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial(\ )}{\partial \theta} = \sin \theta \frac{\partial(\ )}{\partial r} + \frac{\cos \theta}{r} \frac{\partial(\ )}{\partial \theta} . \end{aligned}$$

If we note that  $\mathbf{i} = \cos \theta \mathbf{e}_r - \sin \theta \mathbf{e}_\theta$  and  $\mathbf{j} = \sin \theta \mathbf{e}_r + \cos \theta \mathbf{e}_\theta$ , we have that

$$\nabla(\ ) \equiv \frac{\partial(\ )}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial(\ )}{\partial \theta} \mathbf{e}_\theta + \frac{\partial(\ )}{\partial z} .$$

An expression for spherical coordinates can be derived in a similar manner.

## Conservation of Energy

When all the forces doing work are conservative, the work is given by (1), and the principle of work and energy derived in the last lecture,

$$T_1 + W_{12} = T_2,$$

reduces to,

$$T_1 + V_1 = T_2 + V_2$$

since

$$W_{12} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = - \int_{\mathbf{r}_1}^{\mathbf{r}_2} \nabla V \cdot d\mathbf{r} = V_1 - V_2 \quad (2)$$

Since the points  $\mathbf{r}_1$  and  $\mathbf{r}_2$  are arbitrary,

$$E = T + V = \text{constant} . \quad (3)$$

throughout the entire system.

This equation states that whenever only conservative forces act upon a system, the total energy stays constant, and that during the motion only exchanges between kinetic and potential energy occur.

In the general case, however, we will have a combination of conservative,  $\mathbf{F}_C$ , and non-conservative,  $\mathbf{F}_{NC}$ , forces. In this case, the work done by the conservative forces will be calculated using the corresponding potential function, i.e.,  $W_{12}^C = V_1 - V_2$ , and the work done by the non-conservative forces will be path dependent and will need to be calculated using the work integral. Thus, in the general case, we will have,

$$T_1 + V_1 + \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F}_{NC} \cdot d\mathbf{r} = T_2 + V_2 .$$

Note that the work done by non-conservative forces which oppose the motion is negative.

## Examples of Conservative Forces

### Gravity near the earth's surface

On a “flat earth”, the specific gravity  $\mathbf{g}$  points down (along the  $-z$  axis), so  $\mathbf{F} = -mg\mathbf{k}$ . Call  $V = 0$  on the surface  $z = 0$ , and then

$$V(z) = - \int_0^z (-mg) dz, \quad V(z) = mgz .$$

For the motion of a projectile, the total energy is then

$$E = \frac{1}{2}mv^2 + mgz = \text{constant} .$$

Since  $v_x$  and  $v_y$  remain constant, we also have  $\frac{1}{2}mv_z^2 + mgz = \text{constant}$ .

### Gravity

In a central gravity field

$$\mathbf{F} = -G \frac{Mm}{r^2} \mathbf{e}_r = -\nabla \left( -G \frac{Mm}{r} \right) ,$$

and so, taking  $V(r \rightarrow \infty) = 0$ ,

$$V = -G \frac{Mm}{r} .$$

Thus, conservation of energy requires that  $E = T + V = \text{constant}$  so that

$$\frac{1}{2}mv^2 - G\frac{Mm}{r} = \text{constant} \quad (4)$$

between any two points in an orbit in a central gravitational field.

## Spring Force

For small displacements, the force exerted by a spring in a direction opposite to its displacement is  $F = -kx$ . The elastic potential energy of the spring is the work done on it to deform it an amount  $x$ . Thus, we have

$$V = - \int_0^x -kx \, dx = \frac{1}{2}kx^2 .$$

If the deformation, either tensile or compressive, increases from  $x_1$  to  $x_2$  during the motion, then the change in potential energy of the spring is the difference between its final and initial values, or,

$$\Delta V = \frac{1}{2}k(x_2^2 - x_1^2) .$$

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### Exercise

### Non-conservative Forces

Consider a block sliding on a horizontal surface. Is the friction force conservative? Explain why. Give examples of non-conservative forces.

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### Note

### Path independent forces for which there is no potential (Optional)

The condition  $\mathbf{F} = -\nabla V$  is *sufficient*, but not *necessary* for the work to be path independent. An important example is the magnetostatic force  $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ , where the work  $\mathbf{F} \cdot d\mathbf{r} = \mathbf{F} \cdot \mathbf{v}dt$  is zero (the force is always perpendicular to  $\mathbf{v}$ ). Thus, for a general velocity-dependent electro-magnetic force,  $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ , where only  $\mathbf{B}$  is irrotational, the work is still path independent, and the total energy ( $mv^2/2 + q\phi$ ) is conserved.

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## Gravity Potential for a Rigid Body

In this case, the potential  $V_i$  associated with particle  $i$  is simply  $V_i = m_i g z_i$ , where  $z_i$  is the height of particle  $i$  above some reference height. The force acting on particle  $i$  will then be  $\mathbf{F}_i = -\nabla V_i$ . The work done on the whole body will be

$$\sum_{i=1}^n \int_{\mathbf{r}_i^1}^{\mathbf{r}_i^2} \mathbf{F}_i \cdot d\mathbf{r}_i = \sum_{i=1}^n ((V_i)_1 - (V_i)_2) = \sum_{i=1}^n m_i g ((z_i)_1 - (z_i)_2) = V_1 - V_2 \quad ,$$

where the gravity potential for the rigid body is simply,

$$V = \sum_{i=1}^n m_i g z_i = m g z_G ,$$

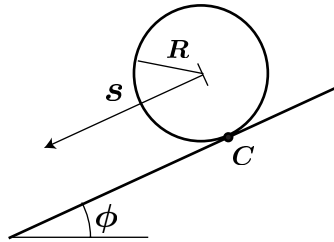
where  $z_G$  is the  $z$  coordinate of the center of mass. It's obvious but worth noting that because the gravitational potential is taken about the center of mass, the moment of inertia plays no role in determining the gravitational potential.

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**Example**
**Cylinder on a Ramp**


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We consider a homogeneous cylinder released from rest at the top of a ramp of angle  $\phi$ , and use conservation of energy to derive an expression for the velocity of the cylinder.



Conservation of energy implies that  $T + V = T_{initial} + V_{initial}$ . Initially, the kinetic energy is zero,  $T_{initial} = 0$ . Thus, for a later time, the kinetic energy is given by

$$T = V_{initial} - V = mgs \sin \phi ,$$

where  $s$  is the distance traveled down the ramp. The kinetic energy is simply  $T = \frac{1}{2} I_C \omega^2$ , where  $I_C = I_G + mR^2$  is the moment of inertia about the instantaneous center of rotation  $C$ , and  $\omega$  is the angular velocity. Thus,  $I_C \omega^2 = 2mgs \sin \phi$ , or,

$$v^2 = \frac{2gs \sin \phi}{1 + (I_G/mR^2)} ,$$

since  $\omega = v/R$ . For the general case of a cylinder with the center of mass at the center of the circle but an uneven mass distribution, we write  $T = \frac{1}{2} m(1 + k_G^2/R^2)$ , where the effect of mass distribution is captured in  $k_G$ ; the smaller  $k_G$ , the more concentrated the mass about the center of the cylinder. Then

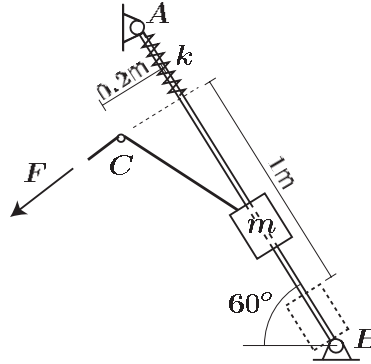
$$v^2 = \frac{2gy}{1 + k_G^2/R^2} \tag{5}$$

where  $s \sin \phi$  has been replaced by the vertical displacement  $y$  of the center of mass. This equation shows that the more the mass is concentrated towards the center of the cylinder ( $k_G$  small), a higher velocity will be reached for a given height, i.e less of the potential energy will go into rotational kinetic energy.

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**Example****Principle of Work and Energy**

The  $m = 30$  kg collar is released from rest at  $B$  and slides with negligible friction up the fixed rod inclined  $60^\circ$  from the horizontal under the action of a constant force  $\mathbf{F} = 450\text{N}$  applied to the cable. We want to calculate the required stiffness  $k$  of the spring so that its maximum deflection equals 5cm. The position of the small pulley at  $C$  is fixed.



First, we want to calculate the work done by the cable. When the collar is at  $B$ , the length of cable between the collar and the pulley is  $\sqrt{(1^2 + 0.2^2)} = 1.0198$  m. When the collar reaches its final position, the length of cable between the collar and the pulley is 0.2m. Since the force is constant, the work done by the force  $\mathbf{F}$  on the collar is simply  $W_{\text{cable}} = 450(1.0198 - 0.2) = 368.9118$  Nm. Applying the principle of work and energy between the initial position and the point of maximum spring compression (denoted by the subscript  $f$ ), we have

$$T_B + V_B + W_{\text{cable}} = T_f + V_f .$$

Here, the kinetic energy at  $B$  and  $f$  is zero since both the initial and final velocities are zero. We can arbitrarily set the potential energy at  $B$  equal to 0. The potential energy at  $f$  will be due to gravity and to the compression of the spring. Thus, we will have  $V_f = mg(1 + \delta) \sin(60^\circ) + k\delta^2/2$ . Or,

$$368.9118 = (30)(1 + 0.05) \sin(60^\circ) + \frac{1}{2}k(0.05)^2 \quad \rightarrow \quad k = 81038\text{N/m}$$

Note that when calculating the length of the cable in the final state, we have neglected the compression of the spring. This effect could easily be taken into account and the result would not differ much from the one obtained here.

## Equilibrium and Stability

If all the forces acting on the system are conservative, then the potential energy can be used very effectively to determine the equilibrium positions of a system and the nature of the stability at these positions. Let us assume that all of the forces acting on the system can be derived from a potential function,  $V$ . It is clear that if  $\mathbf{F} = -\nabla V = 0$  for some position, this will be a point of equilibrium in the sense that if the body

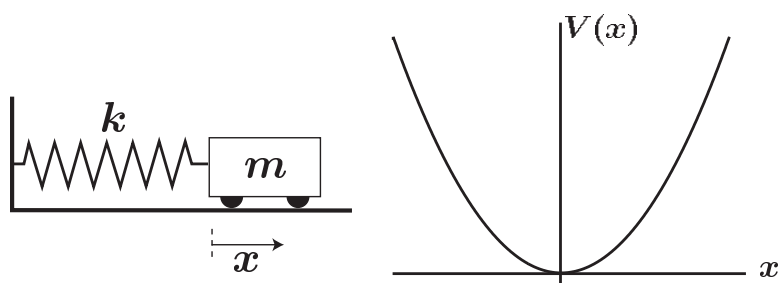


## REFERENCES

is at rest, at such a point, then no forces will act on the body, and hence on acceleration, to change the equilibrium since the result force is zero.

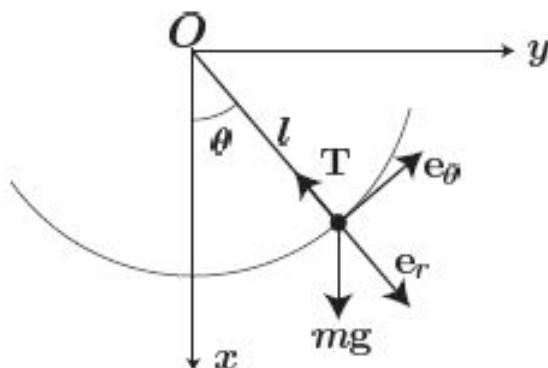
Once equilibrium has been established, the stability of the equilibrium point can be determined by examining the *shape* of the potential function. If the potential function has a *minimum*, at the equilibrium point, then the equilibrium will be stable. This means that if the potential function has a minimum at the equilibrium point, then there is no potential energy left that can be *traded* for kinetic energy. Analogously, if the potential energy is a maximum, then the equilibrium point is unstable.

Let us consider a particle under the effect of a potential force. The result  $\mathbf{F} = -\nabla V$  is useful not only for computing the force but also for computing the stability of the motion from a diagram of the potential energy. For instance, in the case of a particle attached at the end of a spring the potential energy is  $V = \frac{1}{2}kx^2$ .



At a point  $x > 0$ ,  $\nabla V = dV/dx > 0$  and so the force is negative. Similarly for  $x < 0$  the force is positive. At  $x = 0$ ,  $dV/dx = 0$  and the force is zero. We see that the force is directed towards the origin no matter which way the particle is displaced and the force is only zero at the origin. The minimum of the potential energy coincides with the equilibrium position of the particle. It is clearly a stable equilibrium, since any displacement of the particle produces a force which tends to push the particle toward its resting point.

When  $\nabla V = 0$  the system is in equilibrium. However, if this occurs at a maximum of  $V$ , the equilibrium is not stable, since a positive displacement produces a positive force that tends to increase the displacement. A pendulum of length  $L$  supporting a mass  $m$  is a good illustration of this.



If we take the potential energy to be zero at the bottom of its swing, we see that

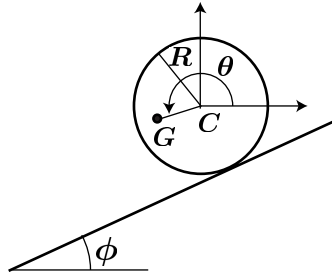
$$V(\theta) = mgL(1 - \cos \theta) .$$

## REFERENCES

The pendulum is in equilibrium for  $\theta = 0$  and  $\theta = \pi$ . However, only  $\theta = 0$  is a stable minimum since it is the only one that corresponds to a minimum. (Equilibrium requires that no force acts on the particle:  $F_\theta(\theta) = \frac{dV(\theta)}{d\theta} = 0$ .) When the potential is a function of just one variable (e.g.  $x$  or  $\theta$ ), there is a simple test that can be used to determine if the equilibrium points (i.e. points where  $dV/dx = 0$ ) are stable or unstable. This test is based on looking at the value of the second derivative of the potential at the equilibrium point. That is if  $d^2V/dx^2 > 0$  then the equilibrium point corresponds to a minimum of the potential energy and therefore, the equilibrium is **stable**. When  $d^2V/dx^2 < 0$  then the equilibrium point occurs at a maximum of the potential function and the equilibrium point is **unstable**. The test only breaks down when  $d^2V/dx^2 = 0$ . In this case, we would need to look at higher derivatives to determine the stability of the system.

**Example****Equilibrium and Stability**

A cylinder of radius  $R$ , for which the center of gravity,  $G$ , is at a distance  $d$  from the geometric center,  $C$ , lies on a rough plane inclined at an angle  $\phi$ .



Since gravity is the only external force acting on the cylinder that is capable of doing any work, we can examine the equilibrium and stability of the system by considering the potential energy function. We have  $z_C = z_{C0} - R\theta \sin \phi$ , where  $z_{C0}$  is the value of  $z_C$  when  $\theta = 0$ . We take  $d$  as the distance between the center of mass and the center of the cylinder. We then have the following expression for the gravitational potential as

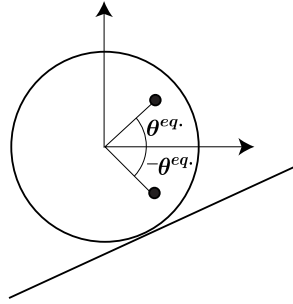
$$V = mgz_G = mg(z_C + d \sin \theta) = mg(z_{C0} - R\theta \sin \phi + d \sin \theta) .$$

The equilibrium points are given by  $\nabla V = \mathbf{0}$ , but, in this case, since the position of the system is uniquely determined by a single coordinate, e.g.  $\theta$ , we can write

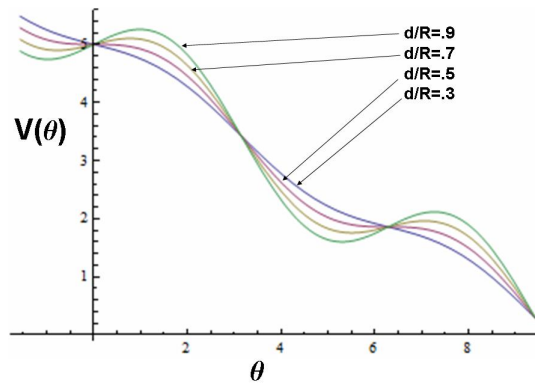
$$\nabla V = \frac{dV}{d\theta} \nabla \theta ,$$

which implies that, for equilibrium,  $dV/d\theta = mg(-R \sin \phi + d \cos \theta) = 0$ , or,  $\cos \theta = (R \sin \phi)/d$ . Since  $|\cos \theta| \leq 1$ , if  $d < R \sin \phi$ , there will be *no equilibrium* positions. On the other hand, if  $d \geq R \sin \phi$ , then  $\theta^{eq} = \cos^{-1}[(R \sin \phi)/d]$  is an equilibrium point. We note that if  $\theta^{eq}$  is an equilibrium point, then  $-\theta^{eq}$  is also an equilibrium point (i.e.  $\cos \theta = \cos(-\theta)$ ).

## REFERENCES



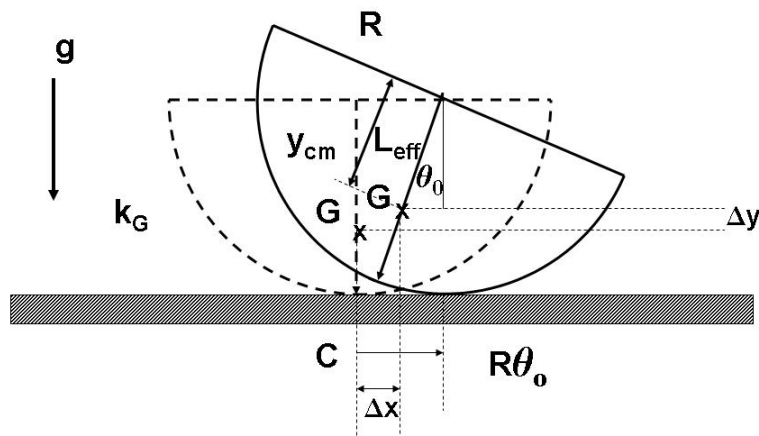
In order to study the stability of the equilibrium points, we need to determine whether the potential energy is a maximum or a minimum at these points. Since  $d^2V/d\theta^2 = -mgd \sin \theta$ , we have that when  $\theta^{eq} < 0$ , then  $d^2V/d\theta^2 > 0$  and the potential energy is a minimum at that point. Consequently, for  $\theta^{eq} < 0$ , the equilibrium is stable. On the other hand, for  $\theta^{eq} > 0$ , the equilibrium point is unstable. As an example, the figure shows the gravitational potential for the offset mass point on the cylinder as a function of  $\theta$  for various values of  $d/R$  for a fixed values of  $\phi$ :  $\phi = 30^\circ$ .



The equilibrium points are identified by the condition  $dV/d\theta = 0$ ; the stability of these equilibrium points are identified by  $d^2V/d\theta^2 > 0$  for stability and  $d^2V/d\theta^2 < 0$  for instability. Note that there are no equilibrium points for  $d/R \leq .5$ .

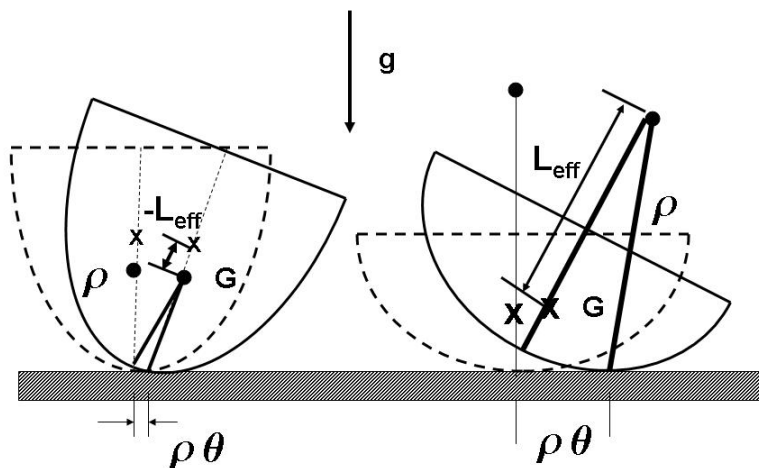
**Example*****Tipped Cylinder and Ellipse***

Consider the solid semi-circle at rest on a flat plane in the presence of gravity. At rest, it is in equilibrium since the gravitational moments balance. We consider that it tips and rolls, keeping the no-slip condition satisfied. This motion results in a vertical displacement of the center of mass,  $\Delta y$  and a horizontal displacement of the center of mass  $\Delta x$ , where  $\Delta y$  and  $\Delta x$  can be found from the geometry.



To determine the stability, we consider the change in potential energy,  $V(\theta)$  for small rotations  $d\theta$ . Only the vertical displacement of the center of mass contributes to a change in potential. If we expand the potential  $V(\theta)$  for small  $\theta$ , we will get an expression  $V(\theta) = A\theta^2$ . (Recall that for the pendulum,  $V(\theta) = mgL\theta^2/2$ .) The question of stability depends upon the sign of  $A$ . If  $A$  is positive, the system is stable; if  $A$  is negative, the system is unstable.

This system behaves like a pendulum: as it rotates and rolls about the contact point, the center of mass moves in the vertical direction, causing a change in potential energy. For a cylinder, the radius of curvature at the point of contact is equal to the radius of the cylinder. The position of the center of mass "rotates" about the center of curvature. Thus the effective length,  $L_{eff}$  of the "pendulum" is the distance between the center of mass and the center of curvature. If  $L_{eff}$  is positive the system is stable, behaving as a pendulum. We now consider the two systems shown in the figure. These are simply semi-ellipses resting on a flat plane.

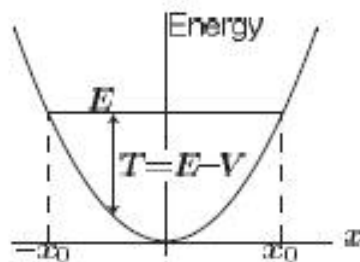


Again, the point of symmetry will be an equilibrium point since the gravitational moments will balance. But the question of stability relates to whether the center of mass moves up or down as  $\theta$  increases. We have  $V(\theta) = A\theta^2$ , with stability for  $A > 0$  and instability for  $A < 0$ . We feel instinctively, that one of these systems—the tall skinny one—could be unstable. This implies that it will not remain balanced about the equilibrium point, but will tip over.

To determine the stability of the tipped ellipse, we need to determine the relationship between the radius of curvature at the contact point relative to the position of the center of mass. If the center of mass is below the center of curvature, the system is stable with an  $L_{eff}$  equal to the vertical distance between the center of curvature and the center of mass. If the center of mass lies above the center of curvature, then  $L_{eff}$  is negative and the system is unstable.

## Energy Diagrams(KK)

Energy diagrams provide a useful way to study the motion of conservative one dimensional systems. In a conservative system, the total energy  $E$  is a constant; the motion transforms the form of the energy from kinetic to potential while keeping the total constant. For a given position of the system,  $x$ , the potential energy can be plotted,  $V(x)$ . The total energy of the system is constant, and is also shown in the diagram. Since the sum of the kinetic energy and the potential energy is a constant as the system moves in  $x$ , the kinetic energy  $T = E - V$  is easily found by inspection. Since the kinetic energy can never be negative, the motion is constrained to regions where  $V \leq E$ .



Since the system is conservative, the total energy  $E$  is constant. The kinetic energy  $T$  is greatest at the origin  $x = 0$ . As the particle goes past the origin in either direction, it is slowed by the spring and comes to a complete rest at one of the turning points  $\pm x_0$ . The particle then moves to the origin increasing its kinetic energy, and the cycle is repeated. We see that in the case of a harmonic oscillator the motion is always bounded. As  $E$  increases the turning points move farther and farther off, but the particle remains bounded. Also, note that when  $E = 0$  then the particle is at  $x = 0$  and the particle lies at rest in equilibrium.

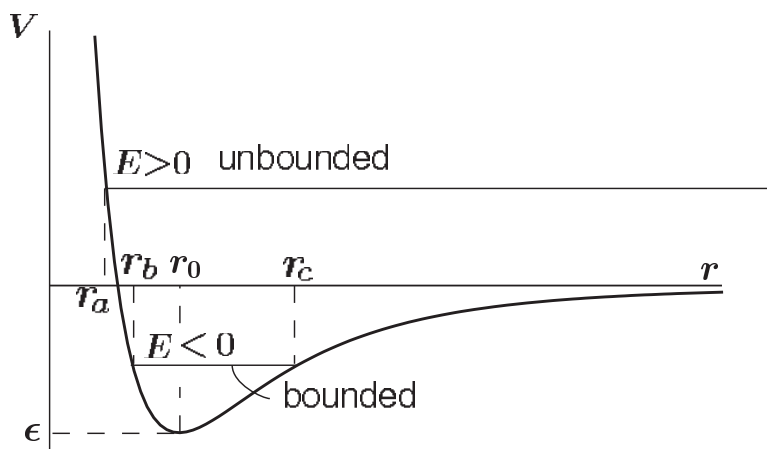
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**Example**
**van der Waals Force**


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The situation is different when the function  $V$  does not increase indefinitely with distance. Consider for instance the interaction between two atoms. At large separations the atoms attract each other weakly with the van der Waals force, which varies as  $1/r^7$ . As the atoms approach the electron clouds begin to overlap creating strong repulsive forces. The corresponding potential is given by

$$V(r) = \epsilon \left[ \left( \frac{r_0}{r} \right)^{12} - 2 \left( \frac{r_0}{r} \right)^6 \right]$$



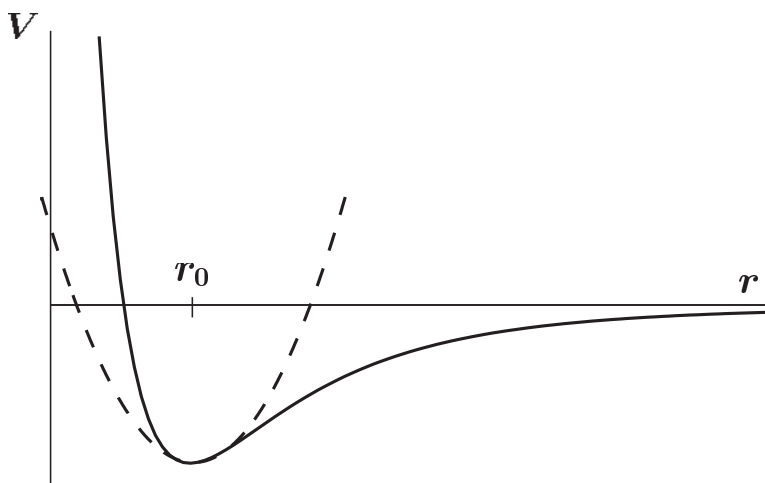
For a positive energy  $E > 0$ , the motion is unbounded and the atoms are free to fly apart. As the diagram shows, the distance of closest approach  $r_a$  does not change appreciably as  $E$  is increased. The situation is quite different for  $E < 0$ . In this case the motion is bounded for small and large separations. The atoms never approach closer than  $r_b$  and they never move apart farther than  $r_c$ . A bound system of two atoms is a molecule. If two atoms collide with positive energy they cannot form a molecule unless some means is

available for losing energy to make  $E$  negative. In general a third body is necessary to carry off the excess energy.

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## Small Oscillations in a bound system (KK)

Every bound system oscillates as a harmonic oscillator about its equilibrium position if it is perturbed from the equilibrium position by a small amount. This can be seen by noting that the minimum of the potential energy can be generally approximated by a parabola in the neighborhood of the minimum.



If the total energy is low enough so that the motion is restricted to the region where the curve is nearly parabolic, the system will behave like a harmonic oscillator. If  $V(r)$  is well behaved and has a minimum at  $r_0$ , then we can always expand it in Taylor's series about point  $r_0$ . Thus,

$$V(r) = V(r_0) + (r - r_0) \left. \frac{dV}{dr} \right|_{r_0} + \frac{1}{2}(r - r_0)^2 \left. \frac{d^2V}{dr^2} \right|_{r_0} + \dots$$

However, since at  $r_0$ ,  $dV/dr = 0$ , for sufficiently small displacements we can truncate the series and obtain,

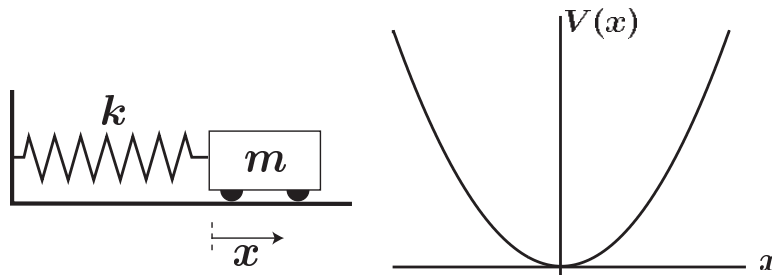
$$V(r) = V(r_0) + \frac{1}{2}(r - r_0)^2 \left. \frac{d^2V}{dr^2} \right|_{r_0} .$$

$$V(r) = \text{constant} + \frac{k(r - r_0)^2}{2} .$$

We can also identify the *effective* spring constant as  $k = d^2V/dr^2|_{r_0}$ . These ideas can be applied to many systems, identifying the oscillatory behavior by considering the behavior of the potential function near the equilibrium point, the minimum of the potential function. (The term "effective" is used to emphasize that the stiffness in a system can be due to many effects: a spring, gravity, elasticity or a combination of these effects.) The value of the constant plays no role in the dynamics of the system.

## Mass-spring System

We now consider the potential for the familiar mass-spring system, previously discussed.



For a mass spring system whose equilibrium position is  $x = 0$ , the potential function can be written

$$V(x) = \frac{kx^2}{2} .$$

where  $k$  is the spring constant. In a harmonic oscillator without damping, such as the examples discussed here, energy is conserved. As potential energy increases, kinetic energy decreases. Thus the minimum of  $V$  occurs at the maximum of  $T$ . For small amplitude motions about  $x = 0$ , both the displacement  $x$  and the velocity are sinusoidal in time:  $x(t) = A \sin(\omega t + \phi_0)$ , where  $A$  and  $\phi_0$  are determined from the initial conditions. The kinetic energy is then given by

$$T = \frac{1}{2}mv(t)^2 = \frac{1}{2}m\omega^2(A \cos(\omega t + \phi_0))^2 \quad (6)$$

where  $\omega$  is the natural frequency of oscillation, that frequency which occurs as an unforced interchange between kinetic and potential energy. The oscillation occurs symmetrically about  $x = 0$  the minimum of the potential function. Since the total energy  $E$  remains constant during the oscillation

$$E = V(x(t)) + T(x(t)) = \frac{k(A \sin(\omega t + \phi_0))^2}{2} + \frac{m\omega^2(A \cos(\omega t + \phi_0))^2}{2} , \quad (7)$$

and we obtain the result  $\omega = \sqrt{\frac{k}{m}}$ .

We also observe that in this case the maximum value of  $V(x)$  is  $V_{MAX} = \frac{kA^2}{2}$ ; this occurs at  $x = A$ . The maximum value of  $T(x)$  is  $T_{MAX} = \frac{m\omega^2 A^2}{2}$ ; this occurs at  $x = 0$ . Equating these maximum values we again obtain  $\omega = \sqrt{\frac{k}{m}}$ . Any constant added to  $V(x)$  plays no role in the dynamics. If  $V(0) = C$ , then  $V_{MAX}$  would be written  $V_{MAX} = (V(x) - V(0))_{MAX}$ , thus removing the constant from consideration.

## Pendulum

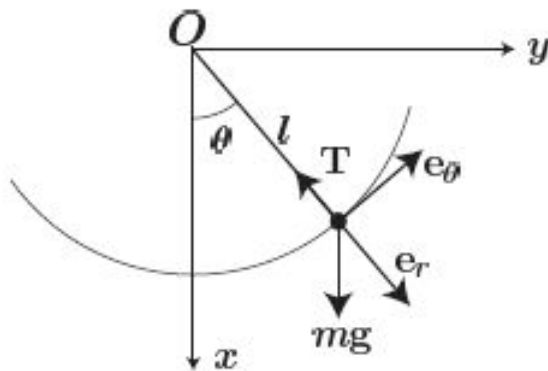
Earlier we derived the potential function for the pendulum as a function of the angle  $\theta$ . We saw that the pendulum exhibited a range of behavior from stable oscillation about  $\theta = 0$  to unstable divergence if the initial position was near  $\theta = \pi$ . The potential function for the pendulum was given by

$$V(\theta) = mgL(1 - \cos \theta) .$$



Expanding  $\cos \theta$  about  $\theta = 0$  and keeping only the terms to order  $\theta^2$ , we obtain

$$V(\theta) = mgL(1 - \cos \theta) = mgL(1 - (1 - \frac{\theta^2}{2} + \dots)) = mgL\frac{\theta^2}{2}.$$



For small amplitude motions about  $\theta = 0$ , both the displacement  $\theta$  and the velocity are sinusoidal in time. As potential energy increases, kinetic energy decreases. Thus the minimum of  $V$  occurs at the maximum of  $T$ . The total energy is constant. The maximum value of kinetic energy is given by

$$T_{MAX} = \frac{1}{2}mL^2 \left( \frac{d\theta}{dt} \right)_{MAX}^2 = \frac{1}{2}mL^2\omega^2\theta_0^2 \quad (8)$$

where  $\omega$  is the natural frequency of oscillation and  $\theta_0$  is the maximum amplitude. The oscillation occurs symmetrically about  $\theta = 0$ , the minimum of the potential function. The maximum value of the potential is  $V_{MAX} = mgL\frac{\theta_0^2}{2}$ ; this occurs at  $\theta = \theta_0$  the point of maximum amplitude where the velocity is zero. Equating the maximum value of kinetic energy to the maximum value of the potential

$$T_{MAX} = \frac{1}{2}mL^2\omega^2\theta_0^2 = V_{MAX} = mgL\frac{\theta_0^2}{2} \quad (9)$$

we obtain the result  $\omega = \sqrt{\frac{g}{L}}$ .

Since  $F_\theta = -\frac{1}{L}\frac{\partial V}{\partial \theta} = -mg\theta$ , application of Newton's Law give the familiar differential equation for the small amplitude oscillation of the pendulum about  $\theta = 0$  as

$$mL\ddot{\theta} + mg\theta = 0 \quad (10)$$

Conversely if we expand the potential for small displacement near  $\theta = \pi$ , substituting  $\theta = \pi + \alpha$ , and expanding  $V(\alpha)$  for small  $\alpha$  we obtain

$$V(\alpha) = mgL(1 - \cos(\pi + \alpha)) = mgL(1 + \cos(\alpha)) = mgL(1 + (1 - \frac{\alpha^2}{2} + \dots)) = mgL(2 - \frac{\alpha^2}{2}) \quad (11)$$

This result completely changes the dynamics of the pendulum system. Since  $F_\alpha = -\frac{1}{L}\frac{\partial V(\alpha)}{\partial \alpha} = mg\alpha$ , a positive force in the direction of motion would exist. If there was no additional restoring force, say from

a spring opposing the pendulum motion, an unstable divergence would occur. (Recall that any constant adding to the value of  $V(\alpha)$  is of no significance in determining the dynamics of a system; only the slope and higher order derivatives play a role.) Since  $F_\alpha = -dV/d\alpha = mgL\alpha$ , application of Newton's Law gives the following differential equation for the small amplitude oscillation of the pendulum about  $\theta = 0$  as

$$mL\ddot{\alpha} - mg\alpha = 0 \quad (12)$$

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**Example**


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**Oscillating Tipped Cylinder and Ellipse**


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We now consider how to determine the frequency of oscillation of the tipped ellipse previously discussed for small amplitude oscillations, the cylinder being just a special case of an ellipse. It is obvious that the semi-circle will oscillate about its center of symmetry. The change in the gravitational potential for the cylinder is given by the "pendulum" formula where  $L_{eff}$  is the distance between the center of mass and the center of curvature at the maximum displacement  $\theta_{MAX}$ ; giving  $V_{MAX} = 1/2mgL_{eff}\theta_{MAX}^2$ . For a cylinder, the radius of curvature at the contact point is the radius of the cylinder  $R$ . For the more general case of an ellipse, the change in the gravitational potential is given by the "pendulum" formula where  $L_{eff}$  is distance between the center of mass and the center of curvature at the contract point  $\rho$ :  $V_{MAX} = 1/2mgL_{eff}\theta_{MAX}^2$ . If the center of mass lies above the center of curvature, then  $L_{eff}$  is negative and the system is unstable: no oscillation occurs.

To determine the frequency, we need to identify  $T_{MAX}$  as well, the maximum value of kinetic energy. The system has both translation and rotational kinetic energy, and both will be at their maximum values when the system moves through the point of symmetry,  $\theta = 0$ .

The translational kinetic energy will be the sum of the rotational and translational kinetic energy; the sum of  $1/2mv_{CG}^2$  where  $v_{CG}$  is the maximum velocity of the center of mass as the system moves through the point of symmetry: plus the rotation kinetic energy,  $1/2I_G\dot{\theta}^2$  where  $\dot{\theta}$  is the maximum value of the angular velocity as the system moves through the point of symmetry.

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## References

- [1] M. Martinez-Sanchez, *Unified Engineering Notes*, Course 95-96.
- [2] D. Kleppner and R.J. Kolenkow, *An Introduction to Mechancis*, McGraw-Hill, 1973.

## ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

## Lecture 15 — Variable Mass Systems: The Rocket Equation

In this lecture, we consider the problem in which the mass of the body changes during the motion, that is,  $m$  is a function of  $t$ , i.e.  $m(t)$ . Although there are many cases for which this particular model is applicable, one of obvious importance to us are rockets. We shall see that a significant fraction of the mass of a rocket is the fuel, which is expelled during flight at a high velocity and thus, provides the propulsive force for the rocket.

As a simple model for this process, consider the cases sketched in a) through f) of the figure. In a) we consider 2 children standing on a stationary flat car. At  $t = 0$ , they both jump off with velocity **relative to the flat car** of  $u$ . (While this may not be a good model for children jumping off a flatcar, it is a good model for the expulsion of mass from a rocket where the mass flow from the choked nozzle occurs at a relative velocity  $u$  from the moving rocket.) In b) we consider that they jump off in sequence, each jumping with a velocity  $u$  relative to the then velocity of the flatcar, which will be different for the 2nd jumper since the car has begun to move as a result of jumper 1. For a), we have a final velocity of the flat car as

$$V_{2R} = \frac{2mu}{M + 2m} \quad (1)$$

for b) we have

$$V_{22} = \frac{mu}{M + 2m} + \frac{mu}{M + m} \quad (2)$$

where  $M$  is the mass of the flatcar, and  $m$  is the mass of the jumper. We can see the case b) gives a higher final velocity. We now consider cases with more blocks; we introduce the notation  $V_{NN}$  to be the final velocity with  $N$  blocks coming off one at a time. We use the notation  $V_{NR}$  to denote the reference velocity if the  $N$  blocks came off together. If we consider 3 blocks, we can see by inspection that

$$V_{3R} = \frac{3mu}{M + 3m} \quad (3)$$

and

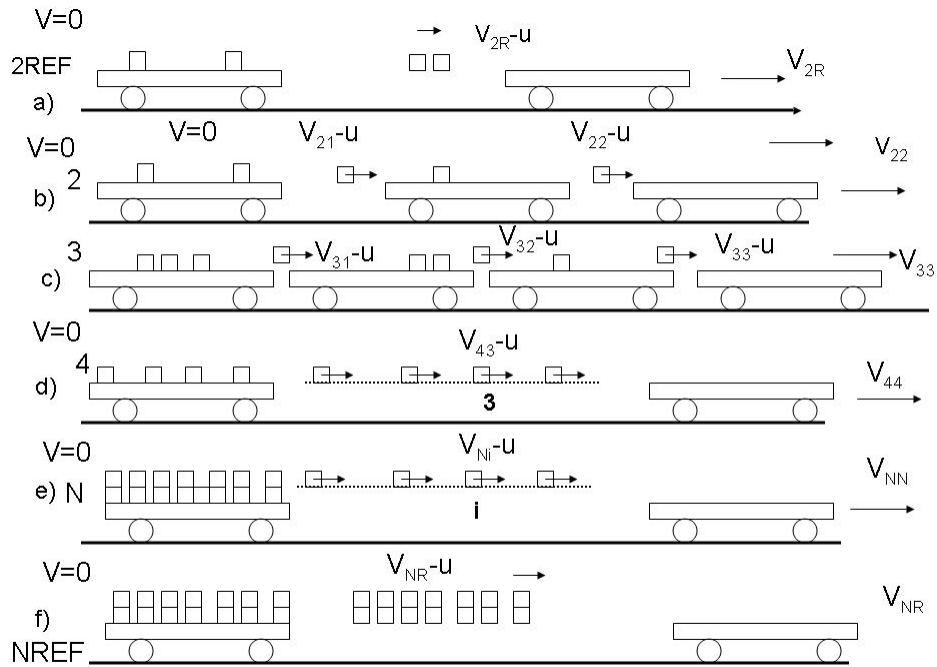
$$V_{33} = \frac{mu}{M + 3m} + \frac{mu}{M + 2m} + \frac{mu}{M + m} \quad (4)$$

While for 4 blocks

$$V_{4R} = \frac{4mu}{M + 4m} \quad (5)$$

and

$$V_{44} = \frac{mu}{M + 4m} + \frac{mu}{M + 3m} + \frac{mu}{M + 2m} + \frac{mu}{M + m} \quad (6)$$

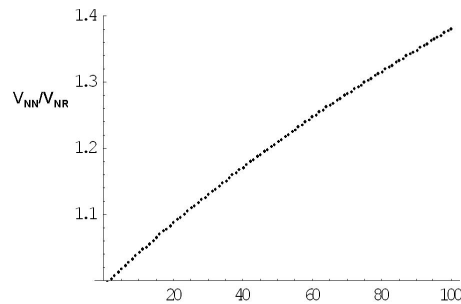


Generalizing this to arbitrary  $N$ , we can write

$$V_{NR} = \frac{Nmu}{M + 4m} \quad (7)$$

and

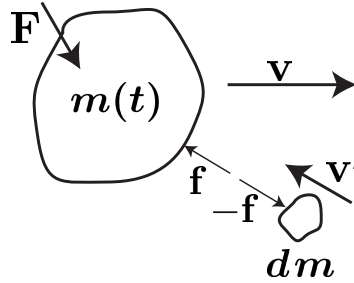
$$V_{NN} = \sum_{i=1}^N \frac{mu}{M + im} \quad (8)$$



The figure shows the results for  $V_{100,100}/V_{100R}$ . Since the 100 blocks with  $m = .01$  equal the mass  $M$ , the propellant ratio of the "rocket" is 50%. The final velocity of the cart is 40% higher than would be obtained if the masses come off together. Momentum is conserved. The sum of the momentum of the cart plus all the individual mass points is equal to zero. However, the momentum of all the particles left behind is of very little interest to us. We care only about the final velocity of the cart.

To analyze this question we must consider a system of variable mass, and the process by which it gains velocity as a result of ejecting mass. While our previous discussion provided a discrete model of this process, we now consider a continuous process, obtaining a relation for  $v(t)$  as a function of the momentum interaction between the system and the external world.

We shall start by considering a body with velocity  $\mathbf{v}$  and external forces  $\mathbf{F}$ , gaining mass at a rate  $\dot{m} = dm/dt$ . Let us look at the process of gaining a small amount of mass  $dm$ . Let  $\mathbf{v}'$  be the velocity of  $dm$  before it is captured by  $m$ , and let  $\mathbf{f}$  represent the average value of the impulsive forces that  $dm$  exerts on  $m$  during the short interval  $dt$ , in which the capturing takes place. By Newton's third law,  $dm$  will experience a force  $-\mathbf{f}$ , exerted by  $m$ , over the same  $dt$ .



We can now examine the capture process from the point of view of  $dm$  and equate the impulse,  $-\mathbf{f}dt$ , to the change in linear momentum of  $dm$ ,

$$-\mathbf{f}dt = dm(\mathbf{v} + d\mathbf{v} - \mathbf{v}'). \quad (9)$$

Here,  $\mathbf{v} + d\mathbf{v}$  is the velocity of  $m$  (and  $dm$ ) after impact. Analogously, from the point of view of  $m$ , we write

$$\mathbf{F}dt + \mathbf{f}dt = m(\mathbf{v} + d\mathbf{v}) - m\mathbf{v} = m d\mathbf{v}. \quad (10)$$

The term  $dm d\mathbf{v}$  in equation 9 is a higher order term and will disappear when we take limits. The impulse due to the contact force can be eliminated by combining equations 9 and 10,

$$\mathbf{F}dt - dm(\mathbf{v} - \mathbf{v}') = m d\mathbf{v},$$

or, dividing through by  $dt$ ,

$$m \frac{d\mathbf{v}}{dt} = \mathbf{F} - (\mathbf{v} - \mathbf{v}') \frac{dm}{dt} = \mathbf{F} + (\mathbf{v}' - \mathbf{v}) \frac{dm}{dt}. \quad (11)$$

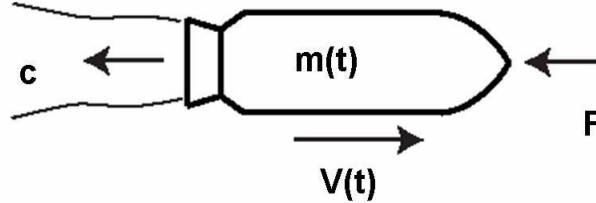
Here,  $\mathbf{v}' - \mathbf{v}$  is the velocity of  $dm$  relative to  $m$ . This expression is valid when  $dm/dt > 0$  (mass gain) and when  $dm/dt < 0$  (mass loss). If we compare this expression to the more familiar form of Newton's law for a particle of fixed mass  $m \frac{d\mathbf{v}}{dt} = \mathbf{F}$ , we see that the term  $(\mathbf{v}' - \mathbf{v})dm/dt$  is an additional force on  $m$  which is due to the gain (or loss) of mass. Equation 11 can also be written as

$$\frac{d(m\mathbf{v})}{dt} = \mathbf{F} + \mathbf{v}' \frac{dm}{dt},$$

where  $v'$  is the velocity of the captured (or expelled) mass relative to the velocity of the mass  $m$ . This shows that, for systems involving variable mass, the usual expression stating conservation of linear momentum,  $d(m\mathbf{v})/dt = \mathbf{F}$ , is only applicable when the initial (final) velocity of the captured (expelled) mass,  $\mathbf{v}'$ , is zero. The behavior of  $m(t)$  is not an unknown, but is specified according to the characteristics of the rocket. In most cases,  $\frac{dm}{dt}$  is a constant and negative. In some cases, the behavior of  $m(t)$  may be determined by a control system. In any case, it is a given quantity.

## The Rocket Equation

We consider a rocket of mass  $m$ , moving at velocity  $\mathbf{v}$  and subject to external forces  $\mathbf{F}$  (typically gravity and drag). The rocket mass changes at a rate  $\dot{m} = dm/dt$ , with a velocity vector  $\mathbf{c}$  relative to the rocket. We shall assume that the magnitude of  $\mathbf{c}$  is constant.



The velocity of the gas observed from a stationary frame will be  $\mathbf{v}' = \mathbf{v} + \mathbf{c}$ . In this frame,  $\mathbf{c}$  is a vector aligned along the flight path in a negative direction.  $\mathbf{c} = -c\mathbf{t}$ , where  $\mathbf{t}$  is the unit direction along the flight path. Thus,

$$m \frac{d\mathbf{v}}{dt} = \mathbf{F} + \mathbf{c} \frac{dm}{dt}. \quad (12)$$

The term  $\mathbf{T} = \mathbf{c} dm/dt$  is called the *thrust* of the rocket and can be interpreted as an additional force on the rocket due to the gas expulsion.

Equation 12 is a vector equation which can be projected along the direction of  $\mathbf{v}$  (tangent to the path). Thus,

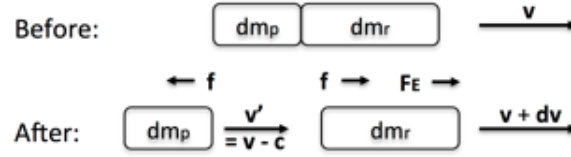
$$m \frac{dv}{dt} = F_t - c \frac{dm}{dt} = F_t + T, \quad (13)$$

where  $F_t$  is the tangential component of  $\mathbf{F}$ ,  $v$  and  $c$  are the magnitudes of  $\mathbf{v}$  and  $\mathbf{c}$  respectively, and we have assumed that  $\mathbf{c}$  is parallel and has opposite direction to  $\mathbf{v}$ . The magnitude of the thrust is  $T = -c\dot{m}$ . Note that for a rocket,  $\dot{m}$  will be negative (mass is lost). If the force  $F_t$  is known, this equation can be integrated in time to yield an expression for the velocity as a function of time.

## Alternate Formulation

Consider the scenario shown in the figure below. Before a differential element of propellant,  $dm_p$ , is burned, it is attached to the mass of the rocket,  $m_r$ , and both move with an inertial velocity of  $\mathbf{v}$ . Then, the

differential element of propellant is burned, causing a force on the rocket,  $\mathbf{f}$ , and an equal and opposite force on the propellant. After the differential element of propellant exits the nozzle at a relative velocity of  $\mathbf{c}$ , it has an inertial velocity of  $\mathbf{v}' = \mathbf{v} - \mathbf{c}$ .



From the perspective of the center of mass of the rocket, the impulse equals the change in linear momentum

$$\mathbf{F}_E dt + \mathbf{f} dt = m_r(\mathbf{v} + d\mathbf{v}) - m_r \mathbf{v} = m_r d\mathbf{v}$$

From the perspective of the differential element of propellant

$$-\mathbf{f} dt = dm_p(\mathbf{v}' - \mathbf{v})$$

Adding these two equations to cancel the equal and opposite impulses,

$$\mathbf{F}_E dt = (\mathbf{v}' - \mathbf{v}) dm_p + m_r d\mathbf{v}$$

Dividing by  $dt$  and rearranging gives

$$m_r \frac{d\mathbf{v}}{dt} = \mathbf{F}_E + (\mathbf{v} - \mathbf{v}') \frac{dm_p}{dt} = \mathbf{F}_E + \mathbf{c} \frac{dm_p}{dt}$$

From conservation of mass

$$\frac{dm_p}{dt} = -\frac{dm_r}{dt} \Rightarrow dm_p = -dm_r$$

Substituting

$$m_r \frac{d\mathbf{v}}{dt} = \mathbf{F}_E - (\mathbf{v} - \mathbf{v}') \frac{dm_r}{dt} = \mathbf{F}_E - \mathbf{c} \frac{dm_r}{dt}$$

for the differential equation governing the rocket. Given we are now only interested in the rocket, we can drop the subscripts to get

$$m_r \frac{d\mathbf{v}}{dt} = \mathbf{F}_E - \mathbf{c} \frac{dm}{dt} = \mathbf{F}_E + \mathbf{T}$$

Let us consider some simple cases:

$$1) \underline{F_t = 0}$$

If gravity and drag effects are neglected, we have,

$$m \frac{dv}{dt} = -c \frac{dm}{dt}$$

or, integrating between an initial time  $t_0$ , and a final time  $t$ ,

$$\Delta v = v - v_0 = -c(\ln m - \ln m_0) = -c \ln \frac{m}{m_0}. \quad (14)$$

Alternatively, this expression can be cast as the well known rocket equation,

$$m = m_0 e^{-\Delta v/c}, \quad (15)$$

which gives the mass of the rocket at a time  $t$ , as a function of the initial mass  $m_0$ ,  $\Delta v$ , and  $c$ . The mass of the propellant,  $m_{propellant}$ , is given by,

$$m_{propellant} = m_0 - m = m_0(1 - e^{-\Delta v/c}).$$

From the above equations, we see that for a given  $\Delta v$  and  $m_0$ , increasing  $c$  increases  $m$  (payload) and decreases  $m_{propellant}$ . Unfortunately, we can only choose  $c$  as high as the current technology will allow. Note that if  $\dot{m}$  is constant, we can write  $m(t) = m_0 + \dot{m}t$ , and substitute into equation 14 to obtain an expression for  $v$  as a function of  $t$ ,

$$v = v_0 - c \ln\left(1 + \frac{\dot{m}}{m_0} t\right) \quad .$$

Recall that according to the convention used,  $\dot{m}$  is negative as the mass decreases with time.

## 2) $F_t = -mg$

A constant gravitational field acting in the opposite direction to the velocity vector can be easily incorporated. In this case, equation 13 becomes,

$$m \frac{dv}{dt} = -mg - c \frac{dm}{dt},$$

which can be integrated to give

$$v = v_0 - c \ln \frac{m}{m_0} - gt = v_0 - c \ln \frac{m}{m_0} - g \frac{m - m_0}{\dot{m}}. \quad (16)$$

This solution assumes that  $c\dot{m} > m_0g$  at  $t=0$ . If this is not true, the rocket will sit on the pad, burning fuel until the remaining mass satisfies this requirement.

## 3) $F_t = -mg - D$

The effect of the drag force,  $D$ , is harder to quantify. It turns out that for many important applications drag effects are very small. The drag force is characterized in terms of a drag coefficient,  $C_D$ . Thus,

$$D = \frac{1}{2} \rho v^2 A C_D ,$$

where  $A$  is the cross-sectional area of the rocket. The air density changes with altitude  $z$ , and may be



approximated by

$$\rho = \rho_0 e^{-z/H} ,$$

where  $H \approx 8000$  m is the so-called “scale height” of the atmosphere, and  $\rho_0$  is the air density at sea level. It turns out that the differential equation that results for the velocity cannot be integrated explicitly and, in practice, needs to be integrated numerically. It is interesting to note, however, that the effect of drag losses is usually quite small, and it is often reasonable to ignore it in a first calculation. In order to see the importance of  $D$  versus the effect of gravity, we can estimate the value of the ratio  $D/mg$ . At conditions typical for maximum drag,  $\rho \approx 0.25$  kg/m<sup>3</sup> and  $v = 700$  m/s. Considering a rocket of 12,000 kg with a cross section of  $A = 1$  m<sup>2</sup> and  $C_D = 0.2$ , we have,

$$\frac{\rho A C_D v^2}{2mg} = \frac{0.25 \times 1 \times 0.2 \times 700^2}{2 \times 12,000 \times 9.8} = 0.021 ,$$

which indicates that the drag force is only about 2% of the gravity force.

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### Note (Optional)

### Gravity Losses [1]

Let us consider a rocket providing a constant thrust  $T = -c\dot{m}$  which is launched vertically upwards from rest. Neglecting drag but considering gravity forces, the velocity is given by expression (16) which can be re-written as

$$v = c \left( \ln \frac{1}{\mu} - \frac{1-\mu}{n} \right) , \quad (17)$$

where we have introduced  $\mu = m/m_0$  and the thrust induced acceleration measured in “gees”,  $n = -c\dot{m}/(gm_0)$ . Now,  $v = dz/dt$ , and the above equation can be integrated to give

$$z = \frac{c^2}{gn} \left( 1 - \mu \ln \frac{1}{\mu} - \mu - \frac{(1-\mu)^2}{2n} \right) . \quad (18)$$

If we consider equations (17) and (18) at burnout time (when all the propellant is consumed), the mass fraction will be  $\mu = (m_0 - m_{propellant})/m_0$ . After this time, the rocket will coast in free-flight. The total energy per unit mass at this (and all subsequent times) is

$$E = \frac{v^2}{2} + gz = c^2 \left( \frac{1}{2} \left( \ln \frac{1}{\mu} \right)^2 + \frac{1-\mu - \ln(1/\mu)}{n} \right) .$$

Since,  $0 < \mu < 1$ ,  $1 - \mu - \ln(1/\mu)$  is negative and consequently, higher  $n$  means larger final energy per unit mass  $E$ , tending to the ideal limit

$$E_{ideal} = \frac{1}{2} \left( c \ln \frac{1}{\mu} \right)^2 ,$$

which is that obtained for an impulsive start. Reducing  $n$  means applying smaller thrust for a longer time, and, as the above equation shows there is a price to be paid in the eventual energy of the payload. One way to quantify this loss is to define an equivalent impulsive velocity  $(v_0)_{eq}$  such as to achieve the energy per

unit mass  $E$  in a very short time, hence, at ground level,

$$(v_0)_{eq} = c \sqrt{\left(\ln \frac{1}{\mu}\right)^2 + 2 \frac{1 - \mu - \ln(1/\mu)}{n}},$$

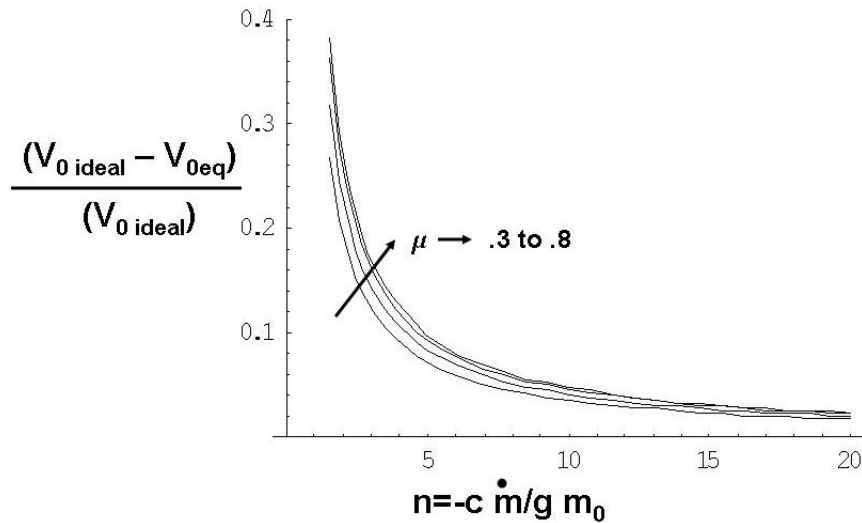
and to compare this to the ideal impulsive velocity

$$(v_0)_{ideal} = c \ln \frac{1}{\mu}.$$

Some representative fractional losses  $((v_0)_{ideal} - (v_0)_{eq})/(v_0)_{ideal}$ , are shown below,

|           | $\mu = 0.7$ | $\mu = 0.5$ | $\mu = 0.3$ |
|-----------|-------------|-------------|-------------|
| $n = 1.5$ | 0.3628      | 0.3188      | 0.2676      |
| $n = 3$   | 0.1615      | 0.1444      | 0.1235      |
| $n = 10$  | 0.0456      | 0.0410      | 0.0354      |

The table shows losses of  $\sim 15\%$  for  $n = 3$ ,  $\sim 4\%$  for  $n = 10$ . (Recall that  $n$  must be greater than 1 for immediate lift off from the launch pad.) The losses are severe for  $n$  as low as 1.5. The figure shows a graph of these losses as a function on  $n$  for  $\mu = .8, .7, .5$ , and  $.3$ .




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### Example

### Single vs. Two Stage Rockets [1]

#### Single stage

To achieve an orbital speed of  $v = 7600$  m/s, we require an *ideal*  $\Delta v$  of about 9000 m/s where the extra velocity is needed to overcome gravity and drag. Chemical rockets produce exhaust jets at velocities of  $c \approx 2500 - 4500$  m/s. Using the higher  $c$ , if we wish to place a payload in orbit with a single stage rocket,

we have a mass ratio (mass at burn out,  $m_f$ , divided by initial mass,  $m_0$ ) of

$$\frac{m_f}{m_0} = e^{-9000/4500} = e^{-2} = 0.135 \quad .$$

The mass  $m_f$  must include all components of the rocket infrastructure, including the engine, empty tank, guidance equipment, etc., as well as the payload. The mass of the propellant will be  $m_0 - m_f = (1 - 0.135)m_0 = 0.865m_0$ . If we assume that the tank plus engine are the main contributors to the weight of the rocket and weigh 10% of the propellant,  $0.087m_0$ , we have,

$$m_{payload}^{1stage} = (0.135 - 0.087)m_0 = 0.048m_0 \quad .$$

Clearly, there is not much margin here, and, in fact, single-stage-to-orbit vehicles have yet to be engineered successfully. The alternative solution is to subdivide the rocket, so that empty tanks are dropped when they are no longer needed.

### Two stages

Consider now a two stage vehicle. We shall again assume that each empty tank plus its engine weighs 10% of the propellant it carries. The required  $\Delta v$  is now subdivided into two  $\Delta v$ 's of 4500 m/s each.

#### First stage

If  $m_0$  is the initial mass and  $m_1$  is the mass after burn out, we have,

$$m_1 = e^{-4500/4500}m_0 = e^{-1}m_0 = 0.368m_0$$

The fuel burnt will be  $m_0 - m_1 = (1 - 0.368)m_0 = 0.632m_0$ , and the weight of the tank and engine to be dropped will be  $0.0632m_0$ , leaving an initial mass for the second stage,  $m_2$ , of

$$m_2 = 0.368m_0 - 0.063m_0 = 0.295m_0 \quad .$$

#### Second stage

After burn out,

$$m_f = e^{-1}m_2 = e^{-1}0.295m_0 = 0.109m_0.$$

The fuel burnt in this stage will be  $m_2 - m_f = (0.295 - 0.109)m_0 = 0.186m_0$  and the weight of the tank plus engine will be  $0.019m_0$ , leaving for the payload

$$m_{payload}^{2stage} = (0.109 - 0.019)m_0 = 0.090m_0,$$

which is still very small but about *twice* the size of the payload obtained for the single stage rocket.

### Note

### Launch Strategy

The earth rotates towards the East with a period of about 23.9 hours. This results in a tangential velocity of  $v_t = 466\text{km/sec}$ . at the equator. The corresponding value for higher latitudes is  $v_t = 466\text{km/sec} * \cos[\theta_0]$ , where  $\theta_0$  is the latitude of position on the earth. In most situations, we desire to use this tangential velocity

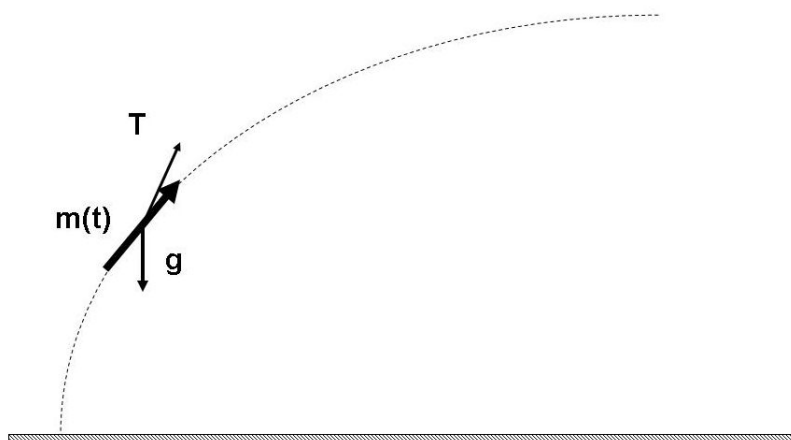
## REFERENCES

to add to the energy available to place a satellite in orbit. Therefore, most launch sites are located as close to the equator as possible and the launch occurs with a velocity component directed towards the east. The table shows the latitude of the current major international launch sites. Of interest is "Sea Launch", which leaves the US west coast with a sea-going launch platform and control ship and launches very close to the equator. Other criterion for launch site is safety; a failed launch should not put populations in danger.

|            |          |             |       |
|------------|----------|-------------|-------|
| Kennedy    | 28.5°    | Tanegashima | 30.4  |
| Vandenburg | 34.4°    | Xichang     | 28.25 |
| Baikonur   | 45.6°    | Kourou      | 5.2   |
| Sea Launch | 0°19 min |             |       |

**Note****Gravity Turn**

The model we have introduced for rocket launch is incomplete if our goal is to place a spacecraft into orbit. This requires that the flight end with a roughly horizontal velocity at orbital speed. One useful maneuver to accomplish this transition is called the gravity turn. In this maneuver, gravity acts to turn the trajectory of the rocket towards the horizontal.



This maneuver offers two main advantages over a conventional thrust-controlled trajectory where the rocket's own thrust is used to steer the vehicle. First, any thrust used in changing the rocket's direction is not being used to accelerate the vehicle into orbit, constituting a loss which can be reduced by using gravity to steer the vehicle onto its desired trajectory. Second, and more importantly, because the force of gravity is doing the steering during the initial ascent phase of the launch the ship can maintain low or even zero angle of attack. This minimizes transverse stress on the launch vehicle; allowing for a weaker, and thus lighter, launch vehicle. For a constant thrust, and a variable mass, the details of the trajectory and motion require a numerical solution.

**Note****Specific Impulse**

The specific impulse  $I_{sp}$ , with units of seconds, is often used in practice to characterize the performance of a rocket engine. The definition of the specific impulse is the magnitude of the thrust divided by the propellant weight flow rate,

$$I_{sp} = \frac{T}{\dot{m}g} = \frac{c}{g}.$$

Typical values of  $I_{sp}$  are around 300 s for solid propellants and up to 500 s for higher energy fuels.

**Example****Payload Penalty for Recoverable First Stage**

Consider a two-stage launch scenario where fuel is saved on the first stage so that it can be recovered down range. After the first stage of the rocket separates, it falls at terminal velocity through the atmosphere with extra fuel on board. Just before impact with the ground or ocean, the engines relight to decelerate it to a soft touchdown. This example calculates the cost, in terms of useful payload delivered by the second stage to Low Earth Orbit (LEO), associated with making the first stage recoverable. For the descending first stage, the velocity is given by

$$v_f = v_0 - c \ln \frac{m_f}{m_0} - g \frac{m_f - m_0}{\dot{m}} \quad (19)$$

where

$$\dot{m} = \frac{m_{p_l}}{T_l} \quad (20)$$

and  $m_{p_l}$  is the mass of the propellant for landing and  $T_l$  is the time over which this propellant is consumed. Assume that this time is short and the landing burn can be considered impulsive, making the gravity loss negligible. Also, assume that the final velocity is zero and the initial velocity is  $v_0 = -53 \text{ m/s}$ : terminal velocity. Then,

$$0 = v_0 - c \ln \frac{m_t}{m_t + m_{p_l}} \quad (21)$$

where  $m_t$  is the mass of the first stage tank and engines. The mass of the required landing propellant can then be found

$$m_{p_l} = m_t \left( e^{\frac{\Delta v_l}{c}} - 1 \right) \quad (22)$$

where  $\Delta v_l$  is the change in velocity associated with the landing ( $\Delta v_l = v_f - v_0 = 0 - (-53 \text{ m/s}) = 53 \text{ m/s}$ ). The payload of the first stage, which is the initial mass of the second stage, is the final mass of the first stage minus the mass of the landing propellant, tanks and engines:

$$m_{0_2} = m_{P/L_1} = m_{f_1} - (m_{p_l} + m_{t_1}) = m_{f_1} - (m_{p_l} + \alpha(m_p + m_{p_l})) \quad (23)$$

where the mass of the tanks and engines,  $m_t$ , is proportional to the total first stage propellant mass, which includes the propellant for landing,  $m_{p_l}$ , and the propellant for ascent,  $m_p$ .

$$m_t = \alpha(m_p + m_{p_l}) \quad (24)$$

Substituting using the rocket equation to calculate the final mass of the first stage,  $m_{f_1}$ , as a function of the initial mass of the first stage,  $m_0$ , gives

$$\frac{m_{P/L_1}}{m_0} = (1 + \alpha)e^{-\frac{\Delta v}{c}} - \alpha - \alpha(1 + \alpha) \left(1 - e^{-\frac{\Delta v}{c}}\right) \left(e^{\frac{\Delta v_l}{c}} - 1\right) \quad (25)$$

Splitting the total  $\Delta v$  for the launch evenly across the two stages,  $\Delta v = \Delta v_{tot}/2$ .

$$\frac{m_{P/L_2}}{m_{P/L_1}} \frac{m_{P/L_1}}{m_0} = \left[(1 + \alpha)e^{-\frac{\Delta v_{tot}}{2c}} - \alpha\right] \left[(1 + \alpha)e^{-\frac{\Delta v_{tot}}{2c}} - \alpha - \alpha(1 + \alpha) \left(1 - e^{-\frac{\Delta v_{tot}}{2c}}\right) \left(e^{\frac{\Delta v_l}{c}} - 1\right)\right] \quad (26)$$

If  $\alpha = 0.1$ ,  $\Delta v_{tot} = 9000 \text{ m/s}$ ,  $c = 4500 \text{ m/s}$ , and  $\Delta v_l = 53 \text{ m/s}$ , 9.27% of the initial mass makes it to orbit as useful payload mass. As shown earlier, without recovering the first stage, 9.29% made it to orbit. The payload mass penalty is negligible. However, the recovery and refurbishment costs may not. Also, this assumes that we carry exactly the right amount of propellant for the launch and the landing. This calculation did not include margin. It also assumes that the atmosphere cancels all of the initial velocity ( $\Delta v_{tot}/2$ ), except for the gravity-induced terminal velocity. In actuality, a retro-propulsive burn will need to take place to slow the first stage. Once the actual  $\Delta v_l$  is known, it can be substituted into the above equation to calculate the useful payload delivered to orbit. The real question is whether it is less expensive to build a production line to build new engines and tanks or to recover and refurbish used engines and tanks. Remember, this analysis was for downrange recovery. A fly-back booster would require much more landing propellant.

## Note

## Limits of Staging

We can explore the benefits of having more than two stages. Looking at the previous two-stage analysis, and assuming that the total required velocity change for the mission,  $\Delta v_{tot}$ , is evenly divided across the stages, the ratio of useful payload mass to initial launch mass is

$$\gamma_N = \frac{m_{P/L}}{m_0} |_N = \left[(1 + \alpha)e^{-\frac{\Delta v_{tot}}{Nc}} - 1\right]^N \quad (27)$$

where  $\gamma$  is often called the "gear ratio." Consider a scenario where  $\alpha = 0.1$ ,  $c = 4500 \text{ m/s}$ , and  $\Delta v_{tot} = 9000 \text{ m/s}$ . For one, two and three stages,  $\gamma$  is 0.049, 0.093, and 0.100, respectively. But what's the limit as the number of stages,  $N$ , increases? Define  $b = \Delta v_{tot}/c$  and rearrange the previous equation

$$\gamma_N = e^{-b} \left[(1 + \alpha) - \alpha e^{b/N}\right]^N \quad (28)$$

For  $N \gg 1$ , we can expand the exponential function to first order as

$$e^{b/N} \approx 1 + b/N \quad (29)$$

This leads to

$$\gamma_N \approx e^{-b} \left[ (1 + \alpha) - \alpha \left( 1 + \frac{b}{N} \right) \right]^N = e^{-b} \left( 1 - \alpha \frac{b}{N} \right)^N \quad (30)$$

Also for  $N \gg 1$ , we can expand the power binomial as

$$\left( 1 - \alpha \frac{b}{N} \right)^N \approx 1 - \alpha b + \frac{1}{2}(\alpha b)^2 \quad (31)$$

and finally for  $N \gg 1$ , we obtain

$$\gamma_\infty \approx e^{-b} (1 - (\alpha b) + \frac{1}{2}(\alpha b)^2) \quad (32)$$

For the values given above, retaining three terms in the binomial expansion converges to  $\gamma_\infty = 0.1107$ . Clearly, adding stages has diminishing returns, particularly considering the cost, complexity, and risk associated with adding more separation events.

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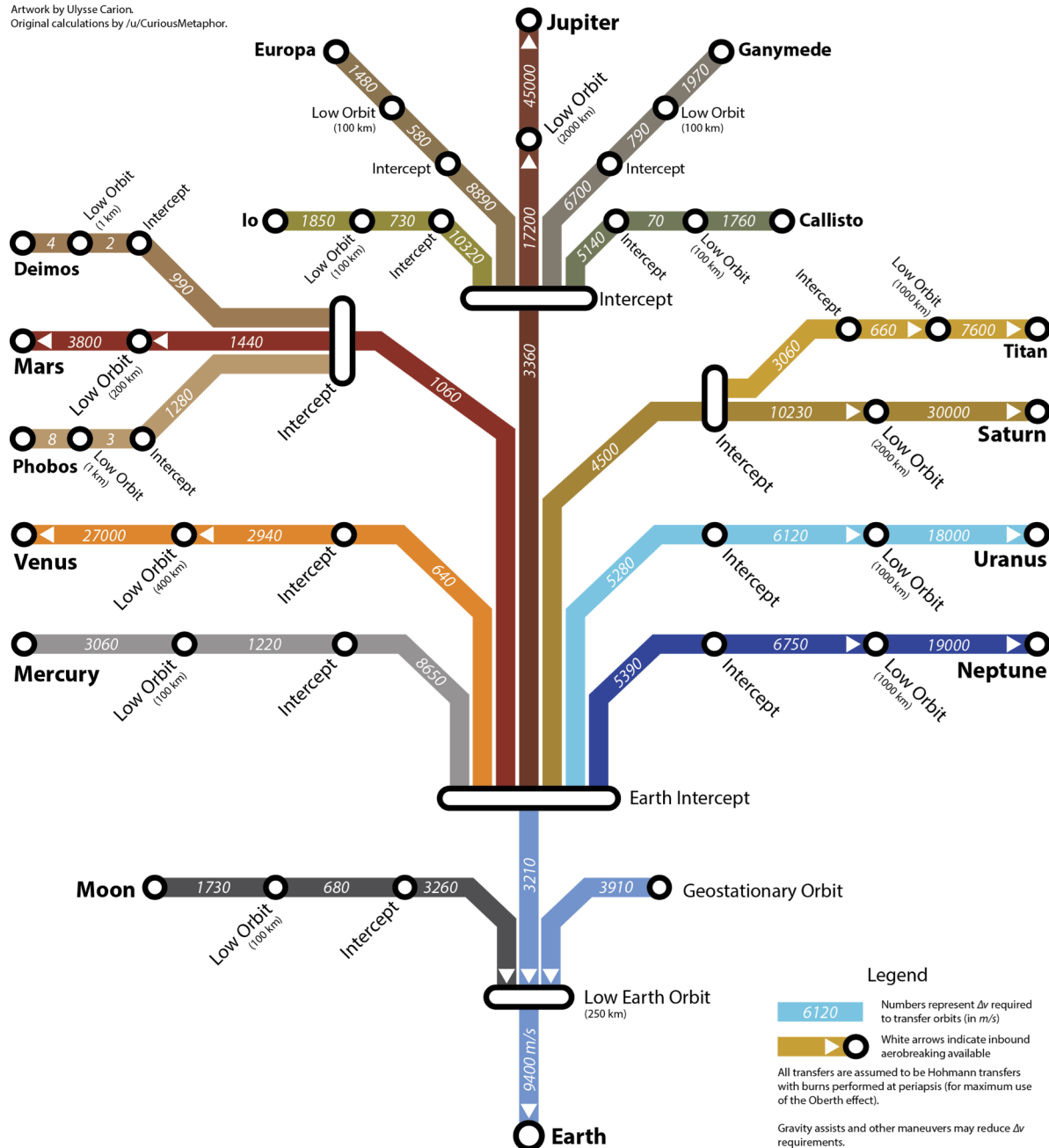
## Subway Map to the Solar System

The figure below shows approximate requirements for  $\Delta V$ , in units of meters per second, for various destinations in the solar system. Note that while some  $\Delta V$  terms will require propulsion, others may have the option to use aero-braking or aero-capture. Intercepts are locations where the magnitude of the gravitational force of the planet (or moon) is equal to that of the Sun (or parent planet). These values are only estimates and depend on the actual techniques employed (e.g., impulsive thrusts versus continuous thrusting, orbit circularization or direct departure, the use of gravity assist).

# The Solar System

## A subway map

Artwork by Ulysse Carion.  
Original calculations by /u/CuriousMetaphor.



### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

4/6, 4/7



*REFERENCES***References**

- [1] M. Martinez-Sanchez, *Unified Engineering Notes*, Course 95-96.

## Lecture 16 — Central Force Motion: Kepler's Laws

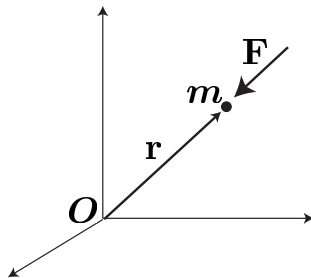
When the only force acting on a particle is always directed towards a fixed point, the motion is called *central force motion*. This type of motion is particularly relevant when studying the orbital movement of planets and satellites. The laws which govern this motion were first postulated by Kepler and deduced from observation. In this lecture, we will see that these laws are a consequence of Newton's second law. An understanding of central force motion is necessary for the design of satellites and space vehicles.



*Johannes Kepler*  
geh. am 27. Dezember 1571 in Weil der Stadt – gest. am 15. November 1630 in Regensburg.  
Phantasie und genialer Blick, Kraft und Ausdauer der Rechnung,  
haben ihm die Gesetze der Planetenbewegung erschlossen.

### Kepler's Problem

We consider the motion of a particle of mass  $m$ , in an inertial reference frame, under the influence of a fixed gravitational force,  $\mathbf{F}$ , directed towards the origin.



We will be particularly interested in the case when the force is inversely proportional to the square of the distance between the particle and the origin, such as the gravitational force. In this case,

$$\mathbf{F} = -\frac{\mu}{r^2} m \mathbf{u}_r, \quad (1)$$

where  $\mu$  is the gravitational parameter,  $r$  is the modulus of the position vector,  $\mathbf{r}$ , and  $\mathbf{u}_r = \mathbf{r}/r$ .

From Newton's law  $\mathbf{F} = m\mathbf{a}$ , we obtain the governing equation for a body attracted by a fixed gravitational force; this is often referred to as Kepler's problem.

$$m\ddot{\mathbf{r}} = -\frac{\mu}{r^2} m \mathbf{u}_r \quad (2)$$

For a fixed gravitational attractor, such as the massive sun which can be considered at rest in calculations of planetary motion, this is the governing equation. A more complex situation occurs for mutually attracting bodies such as the earth and moon whose joint influence cannot be ignored. It can be shown that, in general, Kepler's problem is equivalent to the two-body problem, in which two masses,  $M$  and  $m$ , move solely due to the influence of their mutual gravitational attraction. This equivalence is obvious when  $M \gg m$ , since, in this case, the center of mass of the system can be taken to be at  $M$ . However, even in the more general case when the two masses are of similar size, the problem can be reduced to a Kepler problem (see note below). Although most problems in celestial mechanics involve more than two bodies, many problems of practical interest can be accurately solved by just looking at two bodies at a time. When more than two bodies are involved, the problem is considerably more complicated, and, in this case, no general solutions are known. The two body problem was studied by Kepler (1571-1630) who lived before Newton was born. His interest was in describing the motion of planets around the sun. He postulated the following laws:

- 1.- The orbits of the planets are ellipses with the Sun at one focus
- 2.- The line joining a planet to the Sun sweeps out equal areas in equal intervals of time
- 3.- The square of the period of a planet is proportional to the cube of the major axis of its elliptical orbit

In this lecture, we will start from Newton's laws and verify that the above three laws can indeed be derived from Newtonian mechanics.

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#### Note

#### *Equivalence between the two-body problem and Kepler's problem*

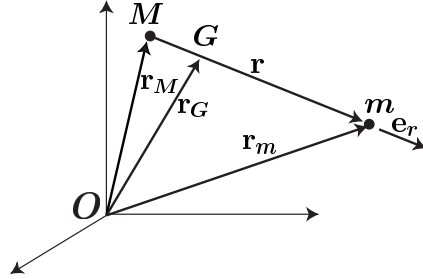
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Here we consider the problem of two isolated bodies of masses  $M$  and  $m$  which interact through gravitational attraction. Let  $\mathbf{r}_M$  and  $\mathbf{r}_m$  denote the position vectors of the two bodies relative to a fixed origin  $O$ . Thus,

the position of the center of gravity,  $G$ , of the two bodies will be

$$\mathbf{r}_G = \frac{M\mathbf{r}_M + m\mathbf{r}_m}{M + m} . \quad (3)$$

Since the two bodies are isolated, we will have, from momentum conservation, that  $\dot{\mathbf{r}}_G = \text{constant}$ , and  $\ddot{\mathbf{r}}_G = \mathbf{0}$ . Therefore, the position of the center of gravity, at all times, can be found trivially from the initial conditions.



If the position vector of  $m$  as observed by  $M$ ,  $\mathbf{r} = \mathbf{r}_m - \mathbf{r}_M$ , is known, then the position vectors of  $M$  and  $m$  could be computed as

$$\mathbf{r}_M = \mathbf{r}_G - \frac{m}{M + m} \mathbf{r} \quad (4)$$

$$\mathbf{r}_m = \mathbf{r}_G + \frac{M}{M + m} \mathbf{r}. \quad (5)$$

Therefore, the problem of determining  $\mathbf{r}_M$  and  $\mathbf{r}_m$  is equivalent to that of determining  $\mathbf{r}$ . In order to derive an equation for  $\mathbf{r}$ , we first consider the equations of motion for  $m$  and  $M$  independently. Thus, we have

$$M\ddot{\mathbf{r}}_M = G \frac{Mm}{r^2} \mathbf{u}_r, \quad m\ddot{\mathbf{r}}_m = -G \frac{Mm}{r^2} \mathbf{u}_r ,$$

where  $r = |\mathbf{r}|$ ,  $\mathbf{u}_r = \mathbf{r}/r$ , and  $G$  is the gravitational constant. If we add these two equations, the equal and opposite gravitational attraction between them drops out and the result is

$$M\ddot{\mathbf{r}}_M + m\ddot{\mathbf{r}}_m = 0 = (M + m)\ddot{\mathbf{r}}_G \quad (6)$$

This states that the acceleration of the center of mass  $\ddot{\mathbf{r}}_G$  is zero: the center of mass can at most move at constant velocity. Subtracting these two equations, we obtain the equation for the relative motion of the two masses,

$$\ddot{\mathbf{r}} = \ddot{\mathbf{r}}_m - \ddot{\mathbf{r}}_M = -G \frac{M + m}{r^2} \mathbf{u}_r ,$$

or,

$$\frac{Mm}{M + m} \ddot{\mathbf{r}} = -G \frac{Mm}{r^2} \mathbf{u}_r .$$

The above expression shows that the motion of  $m$  relative to  $M$  is in fact a Kepler problem in which the force is given by  $-GMm\mathbf{u}_r/r^2$  (this is indeed the real force), but the mass of the orbiting body ( $m$  in this case), has been replaced by the *reduced mass*,  $Mm/(M + m)$ . Note that when  $M \gg m$ , the reduced mass becomes  $m$ . However, the above expression is general and applies to general masses  $M$  and  $m$ .

Alternatively, the above expression can be written as

$$m\ddot{\mathbf{r}} = -G \frac{(M+m)m}{r^2} \mathbf{u}_r, \quad (7)$$

which is again a Kepler problem for an orbiting body of mass  $m$ , in which the gravitational parameter  $\mu$  is given by  $\mu_{eff} = G(M+m)$ . Once the solution  $\mathbf{r}$  to this somewhat "fictitious" problem has been obtained, the solution for the motion of the two bodies can be obtained directly for Equations (4) and (5) giving the position of  $\mathbf{r}_m$  and  $\mathbf{r}_M$  in terms of the center of mass  $\mathbf{r}_G$  and the solution to Equation (6)  $\mathbf{r}$ , where  $\mathbf{r}$  is the distance between the two bodies.

This is a very powerful result. Translating the equations of motion for two gravitationally-interacting bodies into their joint motion about their center of mass, which travels with constant velocity, results in boundary conditions on position and velocity for a single "fictitious" body. Solution of this "problem" will provide an orbit. As will be seen, from conservation of angular momentum, this orbit will be planar. Using this solution to describe the motion of the actual two bodies results in two planar orbits, in the same plane, in which the bodies orbit about their common center of mass.

## Equations of Motion for the Kepler Problem

The equation of motion ( $\mathbf{F} = m\mathbf{a}$ ), is

$$-\frac{\mu m}{r^2} \mathbf{u}_r = m\ddot{\mathbf{r}}.$$

Since the only force in the system is directed towards point  $O$ , the angular momentum of  $m$  with respect to the origin will be constant. Therefore, the position and velocity vectors,  $\mathbf{r}$  and  $\dot{\mathbf{r}}$ , will be in a plane orthogonal to the angular momentum vector, and, as a consequence, the motion will be planar. Using cylindrical coordinates, with  $\mathbf{u}_z$  being parallel to the angular momentum vector, we have,

$$-\frac{\mu}{r^2} \mathbf{u}_r = (\ddot{r} - r\dot{\theta}^2) \mathbf{u}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \mathbf{u}_\theta.$$

Now, we consider the radial and circumferential components of this equation separately.

### Circumferential component

We have,

$$0 = r\ddot{\theta} + 2\dot{r}\dot{\theta}.$$

Using the following identity,

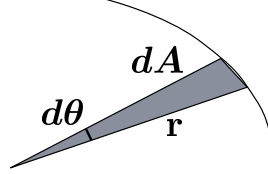
$$\frac{1}{r} \left( \frac{d}{dt} (r^2 \dot{\theta}) \right) = r\ddot{\theta} + 2\dot{r}\dot{\theta},$$

the above equation implies that

$$r^2 \dot{\theta} = h \equiv \text{constant}. \quad (8)$$

We note that the constant of integration,  $h$ , that will be determined by the initial conditions, is precisely the magnitude of the specific angular momentum vector, i.e.  $h = |\mathbf{r} \times \mathbf{v}|$ .

In a time  $dt$ , the area,  $dA$ , swept by  $\mathbf{r}$  will be  $dA = r r d\theta/2$ .



Therefore,

$$\frac{dA}{dt} = \frac{1}{2} r^2 \dot{\theta} = \frac{h}{2},$$

which proves **Kepler's second law**: *The line joining a planet to the Sun sweeps out equal areas in equal intervals of time.*

### Radial component

The radial component of the equation of motion reads,

$$-\frac{\mu}{r^2} = \ddot{r} - r\dot{\theta}^2. \quad (9)$$

Since  $-r^2 \frac{d}{dt} \left( \frac{1}{r} \right) = \dot{r}$ , and  $r^2 = h/\dot{\theta}$  from equation 2, we can write

$$\dot{r} = -\frac{h}{\dot{\theta}} \frac{d}{dt} \left( \frac{1}{r} \right) = -h \frac{d}{d\theta} \left( \frac{1}{r} \right).$$

Differentiating with respect to time,

$$\ddot{r} = -h \frac{d^2}{d\theta^2} \left( \frac{1}{r} \right) \dot{\theta} = -\frac{h^2}{r^2} \frac{d^2}{d\theta^2} \left( \frac{1}{r} \right).$$

Inserting this expression into equation 9, and using equation 2, we obtain the following differential equation for  $1/r$  as a function of  $\theta$ .

$$\frac{d^2}{d\theta^2} \left( \frac{1}{r} \right) + \frac{1}{r} = \frac{\mu}{h^2}.$$

This is a linear second order ordinary differential equation which has a general solution of the form,

$$\frac{1}{r} = \frac{\mu}{h^2} (1 + e \cos(\theta + \psi)) ,$$

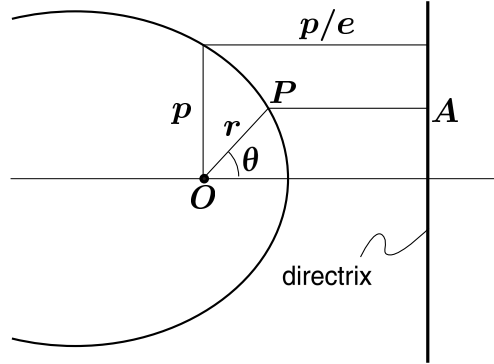
where  $e$  and  $\psi$  are two constants of integration. If we choose  $\theta$  to be zero when  $r$  is minimum, then  $e$  will be positive, and  $\psi = 0$ . The equation describing the trajectory will be

$$r = \frac{h^2/\mu}{1 + e \cos \theta} . \quad (10)$$

We shall see below that this is the equation of a conic section in polar coordinates.

## Conic Sections

Conic sections are planar curves that are defined as follows: given a line, or directrix, and a point, or focus  $O$ , a conic section is the locus of points,  $P$ , such that the ratio of the distance between the point and the focus,  $PO$ , to the distance between the point and the directrix,  $PA$ , is a constant  $e$ . That is,  $e = PO/PA$ .



Since  $PO = r$  and  $PA = p/e - r \cos \theta$ , we have

$$r = \frac{p}{1 + e \cos \theta} . \quad (11)$$

Here,  $p$  is the *parameter* of the conic and is equal to  $r$  when  $\theta = \pm 90^\circ$ . The constant  $e \geq 0$  is called the eccentricity, and, depending on its value, the conic surface will be either an open or closed curve. In particular, we have that when

- $e = 0$  the curve is a circle
- $e < 1$  the curve is an ellipse
- $e = 1$  the curve is a parabola
- $e > 1$  the curve is a hyperbola.

Comparing equation(11) which deals solely with the property of a conic section, and equation(1) which provides the solution of the motion of a point mass in a gravitational field, we can identify the properties of the conic section orbits in terms of the physical parameters of the Kepler problem. In particular, we see that the trajectory of a mass under the influence of a central force will be a conic curve with parameter

$$p = h^2/\mu. \quad (12)$$

When  $e < 1$ , the trajectory is an ellipse, thus proving **Kepler's first law**: *The orbits of the planets are ellipses with the Sun at one focus.* The point in the trajectory which is closest to the focus is called the **periapsis** and is denoted by  $r_\pi$ .

Since the angular momentum  $h$  is constant for an orbit, and the velocity at the point of closest approach  $v_\pi$  is perpendicular to the radius  $r_\pi$ , we can obtain an expression for the velocity at  $r_\pi$  from the angular momentum  $h = r_\pi v_\pi$  resulting in

$$v_\pi = \sqrt{\frac{\mu(1 + e)}{r_\pi}} \quad (13)$$

For elliptical orbits, the point in the trajectory which is farthest away from the focus is called the *apoapsis* and is denoted by  $\alpha$ . When considering orbits around the earth, these points are called the *perigee* and *apogee*, whereas for orbits around the sun, these points are called the *perihelion* and *aphelion*, respectively. Since for a hyperbolic or parabolic orbit, the radius goes to infinity,  $r \rightarrow \infty$ , at or before  $\theta \rightarrow \pi$ , there is no second crossing of the x axis at finite  $r$ , therefore  $r_\alpha \rightarrow \infty$ .

## Elliptical Trajectories

If  $a$  is the semi-major axis of the ellipse, then

$$2a = r_\pi + r_\alpha. \quad (14)$$

Using equation (10) to evaluate  $r_\pi$  and  $r_\alpha$ , (for  $\theta = 0$  and  $\theta = \pi$ ) we obtain

$$r_\pi = \frac{p}{1+e} \quad (15)$$

$$r_\alpha = \frac{p}{1-e} \quad (16)$$

resulting in

$$a = p/(1 - e^2). \quad (17)$$

and

$$r_\pi = a(1 - e) \quad (18)$$

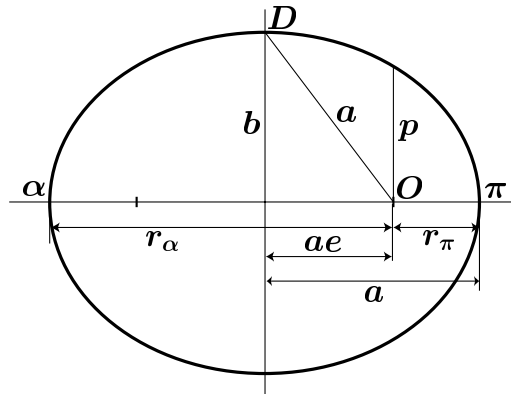
$$r_\alpha = a(1 + e) \quad (19)$$

Since  $p = h^2/\mu$ , we can also write

$$h^2 = a\mu(1 - e^2) \quad (20)$$

Also, the distance between  $O$  and the center of the ellipse will be

$$a - r_\pi = ae. \quad (21)$$





Other geometric properties of the ellipse are that the distance between point  $D$  and the directrix will be equal to  $DO/e$ , which in turn will be equal to the sum of the distance between the focus and the center of the ellipse, plus the distance between the focus and the directrix. That is,  $DO/e = ae + p/e$ . Therefore,  $DO = ae^2 + p = a$ . Hence, using Pythagoras' theorem,  $b^2 + (ae)^2 = a^2$ , the semi-minor axis of the ellipse will be  $b = a\sqrt{1 - e^2}$ .

The area of the ellipse is given by

$$A = \pi ab. \quad (22)$$

Also, since

$$dA/dt = h/2 \quad (23)$$

is a constant, we have

$$A = hT/2, \quad (24)$$

where  $T$  is the period of the orbit. Equating these two expressions and expressing  $h$  in terms of the semi-major axis as

$$h^2 = \mu p = \mu a(1 - e^2), \quad (25)$$

we have

$$T^2 = \frac{(2\pi)^2 a^3}{\mu} \quad (26)$$

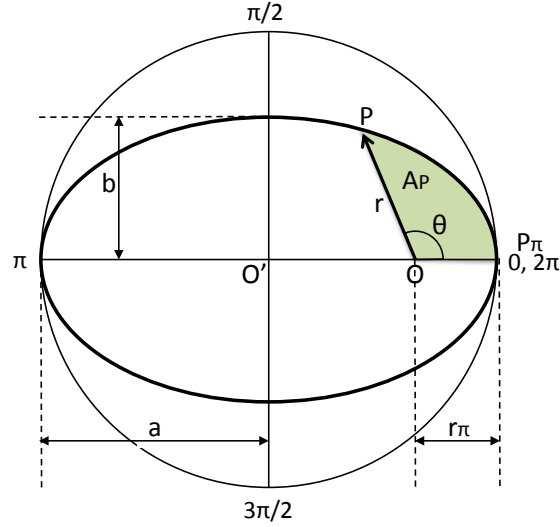
This is often written as

$$\mu = \left(\frac{2\pi}{T}\right)^2 a^3, \quad (27)$$

This result proves **Kepler's third law**: *The square of the period of a planet is proportional to the cube of the major axis of its elliptical orbit.* This result is significant: it states that the period of an orbit is a function only of its semimajor axis, and is independent of its eccentricity  $e$ .

## Time of Flight (TOF) in Elliptical Trajectories

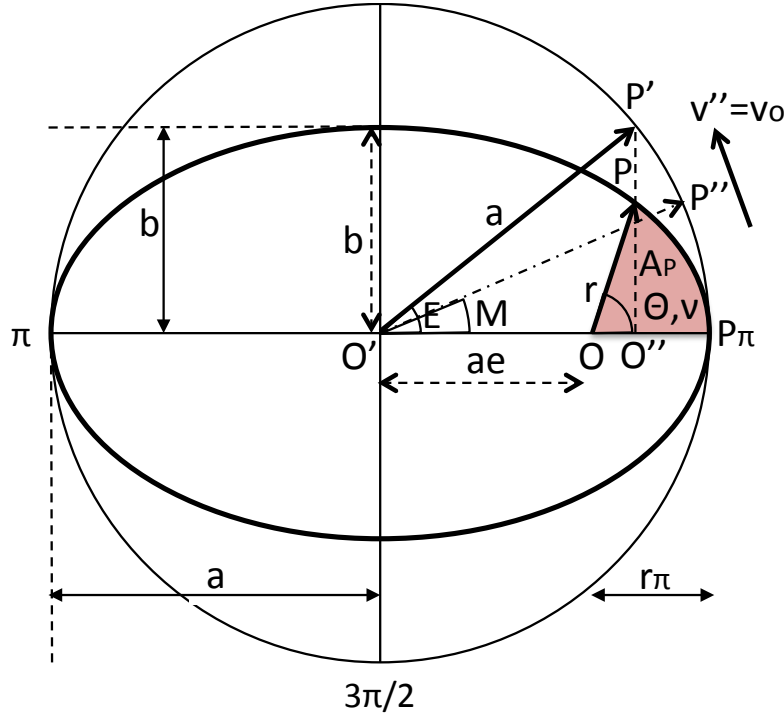
We have found  $r(\theta)$ , the prediction of the shape of the orbit. However, this solution gives us no direct information about the time behavior of the motions, such as  $\theta(t)$ . In many situations we will need to determine the time of flight between two arbitrary points along the ellipse. In order to do that, we use Kepler's second law, which states that the motion of the planet sweeps out area at a constant rate.



Referring to the figure, we see that the time required to travel between the point  $P_\pi$ , the periapsis, and an arbitrary point  $P$ ,  $t_P$ , relative to the total period  $T$  is the ratio of the curved area denoted by  $A_P$  ( $A_P$  is the sector defined by  $O$ ,  $P$ ,  $P_\pi$ ) to the total area of the ellipse. More specifically, since the total period of the orbit is  $T$  and the total area of an ellipse is  $\pi ab$ , the ratio of  $t_P$ , the time required to travel from  $P_\pi$  to  $P$ , to the period  $T$  equals the fraction that the area  $A_P$  represents of the total area of the ellipse.

$$\frac{t_P}{T} = \frac{A_P}{\pi ab}. \quad (28)$$

To find the area  $A_P$  we construct a circle of radius  $a$  with origin,  $O'$ , at the **center of the ellipse**. We identify a point  $P'$  on the circle to be in a vertical line with the point of interest  $P$ , on the ellipse intersecting the point  $O''$  on the major axis.



The various geometric quantities of the elliptical orbit have standard definitions: the position angle  $\theta$  is often called the **true anomaly**. In the set of Classical Orbital Elements (COEs), the angle  $\nu$  is used for true anomaly in place of  $\theta$ . The radial line of the circle from the origin  $O'$  to  $P'$  and the major axis of the ellipse define an angle  $E$ , which is referred to as the **eccentric anomaly**. In addition, we define a third anomaly, the **mean anomaly**  $M$ , as

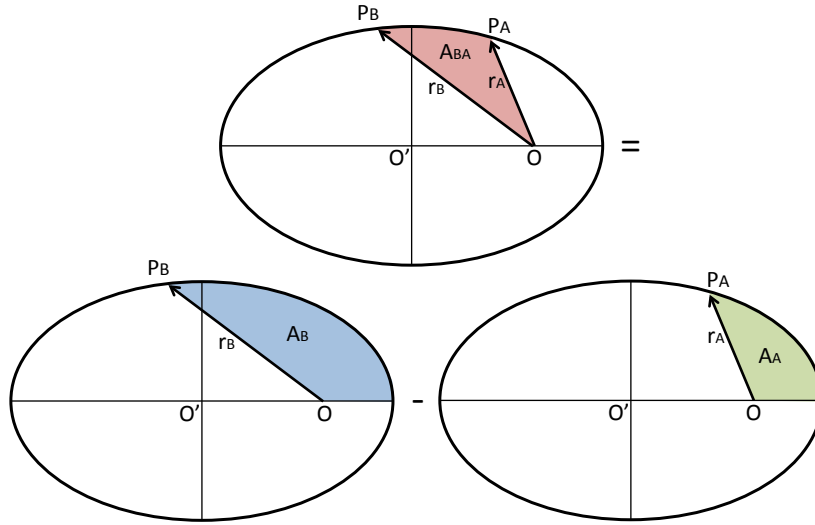
$$M_P = \frac{2\pi(t_P - T_{P_\pi})}{T} = 2\pi \frac{A_P}{\pi ab} . \quad (29)$$

Here,  $t_P - T_{P_\pi}$  is the time of flight from the perigee to the point  $P$ . Thus when a complete orbit has been traversed,  $M_P = 2\pi$ . The Mean Anomaly is shown in the figure as the vector from the origin,  $O'$  to the point  $P''$  which creates an angle  $M$  with the semi-major axis. The rotation rate of this vector, about the origin,  $O'$ , is constant thereby representing the "mean" motion.

If we want to determine the time of flight between two points  $A$  and  $B$  on the ellipse, we can use equation (29) and write

$$TOF = t_B - t_A = \frac{T}{2\pi} (M_B - M_A) = \frac{A_B - A_A}{\pi ab} T \quad (30)$$

where  $A_B - A_A$  is the area swept out between points  $A$  and  $B$ .



The mean anomaly can also be written as  $M_P = \frac{2\pi * A_P}{A_T}$ , where  $A_P$  is the area swept out up to the point  $P$ . When the area swept out equals the total area of the ellipse,  $A_T$ , the time  $t$  equals the period  $T$  and the mean anomaly  $M_\pi = 2 * \pi$ . (The subscript  $\pi$  denotes the return to the periapsis  $\pi$ .)

All that is needed now is an expression for the mean anomaly  $M$  as a function of the orbit parameters. We start by obtaining a relation between  $\theta$  and  $E$ . From simple trigonometry, we have that

$$a \cos E - r \cos \theta = ae \quad (31)$$

or noting that  $r = a(1 - e^2)/(1 + e \cos \theta)$ ,

$$\cos E = \frac{e + \cos \theta}{1 + e \cos \theta}, \quad \rightarrow \quad \cos \theta = \frac{\cos E - e}{1 - e \cos E} . \quad (32)$$

We now develop relationships between the various areas indicated on the figure, with the goal to find the formula for the area  $A_P$ , the area swept out by the vector  $\mathbf{r}$  as it travels from the periapsis  $P\pi$  to the point  $P$ .



The figure above is zoomed in on the upper right quadrant of the ellipse and circle from previous figures. The area we are trying to find,  $A_P$ , is shown in the top-left sub-figure. We label it as  $A_{O/P/P_\pi}$  since it corresponds to the area in the region defined by the vertices at the origin of gravitational force,  $O$ , the point  $P$ , and periapsis,  $P_\pi$ . As the top row of sub-figures indicates, this area is

$$A_P = A_{O/P/P_\pi} = A_{O/P/O''} + A_{O''/P/P_\pi} \quad (33)$$

The area  $A_{O''/P/P_\pi}$  can be related to the area  $A_{O''/P'/P_\pi}$  using *geometric similarity*.

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**Note**
**Geometric Similarity**


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Consider the region contained by  $O''/P'/P_\pi$ . the equations for the ellipse and the circle in cartesian coordinates are

$$\frac{x^2}{a^2} + \frac{y_e^2}{b^2} = 1 \Rightarrow y_e = b\sqrt{1 - \frac{x^2}{a^2}} \quad (34)$$

$$\frac{x^2}{a^2} + \frac{y_c^2}{a^2} = 1 \Rightarrow y_c = a\sqrt{1 - \frac{x^2}{a^2}} \quad (35)$$

By inspection

$$\frac{y_e}{y_c} = \frac{b}{a} \quad (36)$$

Now consider the area under the two curves

$$A_e = \int_{O''}^{P_\pi} y_e dx = \int_{O''}^{P_\pi} b\sqrt{1 - \frac{x^2}{a^2}} dx = \frac{b}{a} A_c \Rightarrow \frac{A_e}{A_c} = \frac{b}{a} \quad (37)$$

We can also see that

$$r \sin \theta = \frac{b}{a} \sin E = b \sin E \quad (38)$$

---

Therefore,

$$A_{O/P/P_\pi} = A_{O/P/O''} + \frac{b}{a} A_{O''/P'/P_\pi} \quad (39)$$

We can further see that the area  $A_{O''/P'/P_\pi}$  is the area of the pie section subtended by the angle  $E$  minus the area of the triangle shown on the bottom row of the figure.

$$A_{O/P/P_\pi} = A_{O/P/O''} + \frac{b}{a} (A_{O''/P'/P_\pi} - A_{O''/P'/O''}) \quad (40)$$

We can now derive these various areas

$$A_{O''/P'/P_\pi} = \frac{1}{2} E a^2 \quad (41)$$

$$A_{O'/P'/O''} = \frac{1}{2}a^2 \sin E \cos E \quad (42)$$

$$A_{O''/P'/P_\pi} = \frac{1}{2}a^2 (E - \sin E \cos E) \quad (43)$$

$$A_{O/P/O''} = \frac{1}{2}(r \sin \theta)(r \cos \theta) = \frac{1}{2}(b \sin E)(r \cos \theta) \quad (44)$$

For this last equation, we can substitute the following

$$a \cos E = ae + r \cos \theta \Rightarrow r \cos \theta = a (\cos E - e) \quad (45)$$

to find

$$A_{O/P/O''} = \frac{1}{2}ab \sin E (\cos E - e) \quad (46)$$

We can now sum these areas to find  $A_P$

$$A_P = A_{O/P/P_\pi} = \frac{1}{2}ab (E - e \sin E) \quad (47)$$

Substituting back into Equation (30), the time of flight from periapsis at  $P_\pi$  to the point  $P$  is given by

$$t_P - T_{P_\pi} = \sqrt{\frac{a^3}{\mu}} (E - e \sin E) = \sqrt{\frac{a^3}{\mu}} M_P \quad (48)$$

where

$$M_P = \sqrt{\frac{\mu}{a^3}} (t_P - T_{P_\pi}) = [E - e \sin E] \quad (49)$$

This is Kepler's Equation.

This equation is very easy to use if we want to know the time at which the satellite is at position  $\theta$ . The only thing required, in this case, is the calculation of the eccentric anomaly  $E$  using equation (32). On the other hand, if we need to find the position  $\theta$  of the satellite at a given time  $t$ , then, we need to solve Kepler's equation which is non-linear using an iterative numerical algorithm such as Newton's method.

### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

3/13 (except energy analysis)



## Lecture 17 — Central Force Motion: Orbits

In the previous lecture, we derived three basic relationships embodying Kepler's laws:

- Equation for the orbit trajectory,

$$r = \frac{h^2/\mu}{1 + e \cos \theta} \quad \underbrace{\left( = \frac{a(1 - e^2)}{1 + e \cos \theta} \right)}_{\text{elliptical orbits}}. \quad (1)$$

- Conservation of angular momentum,

$$h = r^2 \dot{\theta} = |\mathbf{r} \times \mathbf{v}|. \quad (2)$$

- Relationship between the major semi-axis and the period of an elliptical orbit,

$$\mu = \left( \frac{2\pi}{T} \right)^2 a^3. \quad (3)$$

- Time of Flight (TOF) expressions for elliptical orbits,

$$TOF = t_B - t_A = \frac{T}{2\pi} (M_B - M_A) \quad (4)$$

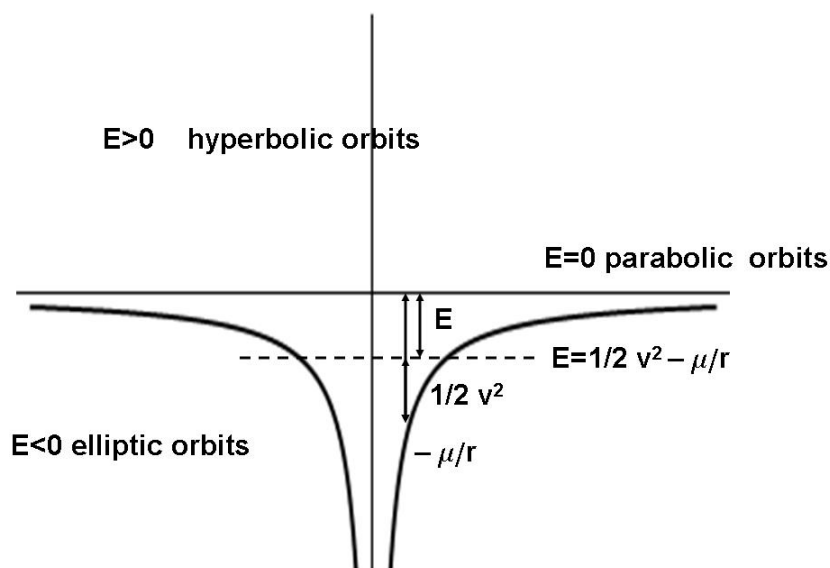
$$M = u - e \sin u \quad (5)$$

$$\cos u = \frac{e + \cos \theta}{1 + e \cos \theta}. \quad (6)$$

In this lecture, we will first derive an additional useful relationship expressing conservation of energy, and then examine different types of trajectories.

### Energy Integral

Since there are no dissipative mechanisms and the only force acting on  $m$  can be derived from a gravitational potential, the total energy for the orbit will be conserved. Recall that the gravitational potential per unit mass is given by  $-\mu/r$ . That is,  $\mathbf{F}/m = -\nabla(-\mu/r) = -(\mu/r^2)\mathbf{u}_r$ . Note that the origin (zero potential) for the gravitational potential is taken to be at infinity. Therefore, for finite values of  $r$ , the potential is negative. The kinetic energy per unit mass is  $v^2/2$ . Therefore, the total energy is the sum of the potential, which is a negative quantity, and the kinetic energy.



$$\frac{1}{2}v^2 - \frac{\mu}{r} = E \equiv \text{constant}.$$

The total specific energy,  $E$ , can be related to the parameters defining the trajectory by evaluating the total energy at the orbit's periapsis  $r_\pi, (\theta = 0)$ . From lecture 15, we have ,

$$r_\pi = (h^2/\mu)/(1 + e) \quad (7)$$

$$v_\pi^2 = h^2/r_\pi^2 = \mu(1 + e)/r_\pi, \quad (8)$$

since  $r_\pi$  and  $v_\pi$  are orthogonal at the periapsis. Thus,

$$E = \frac{1}{2}v_\pi^2 - \frac{\mu}{r_\pi} = \frac{\mu^2}{2h^2}(e^2 - 1). \quad (9)$$

We see that the value of the eccentricity determines the sign of  $E$ . In particular, for

$$\begin{aligned} e < 1 & \quad \text{the trajectory is closed (ellipse), and} & E < 0, \\ e = 1 & \quad \text{the trajectory is open (parabola), and} & E = 0, \\ e > 1 & \quad \text{the trajectory is open (hyperbola), and} & E > 0. \end{aligned}$$

Equations 1, 2, and 3, together with the energy integral 9, provide most of relationships necessary to solve basic engineering problems in orbital mechanics.

## Types of Orbits

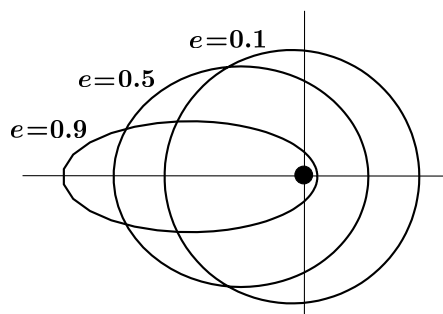
### Elliptic Orbits ( $e < 1$ )

When the trajectory is elliptical,  $h^2 = a\mu(1 - e^2)$  (see Lecture 15). Then, the total specific energy simplifies to  $E = -\mu/(2a)$ , and the conservation of energy can be expressed as

$$\frac{1}{2}v^2 - \frac{\mu}{r} = -\frac{\mu}{2a}. \quad (10)$$

This expression shows that the energy of an elliptical orbit depends only on the major semi-axis. We also see that for a fixed  $a$ , the value of  $h$  determines the eccentricity. (Recall that for fixed  $\mu$ , the period of the elliptical orbit also depended only upon the semimajor axis  $a$ .) There are two limiting cases:  $e \rightarrow 1$ , which gives  $h \rightarrow 0$ , which in turn implies that the minor semi-axis of the ellipse  $b \rightarrow 0$ ; and  $e = 0$  which corresponds to a circular orbit with  $h = \sqrt{a\mu}$ . In the first case, the maximum value of the eccentricity is limited by the size of the planet, since, for sufficiently large values of  $e$ , the trajectory will collapse onto the planet's surface.

Below we show three elliptical trajectories that have the same energy (same value of  $a$ ), but different eccentricities.



### Circular Orbits ( $e = 0$ )

This is a particular case of an elliptic orbit. The energy equation is given by equation 10. The radius is constant

$$r = \frac{h^2}{\mu} = \frac{v_c^2 r^2}{\mu}.$$

For orbits around the earth,  $\mu = gR^2$ , where  $g$  is the acceleration of gravity at the earth's surface, and  $R$  is the radius of the earth. Then,

$$v_c^2 = \frac{\mu}{r} = \frac{gR^2}{r}, \quad (11)$$

which shows that the square of the velocity of a circular orbit is inversely proportional to the radius. We now consider two particular orbits of interest:

- 1)  $r = R$

This corresponds to a hypothetical satellite orbiting the earth at a zero altitude above the earth's

surface. The orbit's velocity is

$$v_c = \sqrt{gR} = 7910 \text{ m/s},$$

and the period, from equation 3, is

$$\tau = \frac{2\pi}{\sqrt{gR^2}} R^{3/2} = 2\pi \sqrt{\frac{R}{g}} = 84.4 \text{ min}.$$

This period is called the “Schuler” period, and it is the minimum period that any free flight object can have in orbit around the earth.

## 2) Synchronous Orbits

These are orbits whose period is the same as the earth's sidereal rotational period (23.9 h). In addition, if the orbit is in the equatorial plane, the orbit is said to be *geostationary* because the satellite will stay fixed relative to an observer on the earth. Using equation 3,

$$a = \left( \frac{\tau^2 g R^2}{4\pi^2} \right)^{1/3} = 42042 \text{ km} \approx 6.6R,$$

which corresponds to an altitude above the surface of  $5.6R$ .

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### Example

### Elliptical Orbits

Consider a satellite launched from an altitude  $d$  above the earth's surface, with velocity  $v_c = \sqrt{\mu/(R+d)}$ , which includes any contribution from the rotation rate of the Earth. If the direction of the velocity is orthogonal to the position vector, the trajectory will clearly be a circular orbit of radius  $R+d$ . However, if the velocity is in any other direction, the trajectory will be an ellipse. The characteristics of the ellipse can easily be determined as follows: knowing  $\mathbf{r}$  and  $\mathbf{v}$ , we can determine  $h$ ; using equation (20), L15, we can determine  $e$ ; and from the trajectory equation, equation (10) L15, we can determine  $\theta$ , and hence the orientation of the ellipse.

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## Parabolic Orbits ( $e = 1$ )

From equation 1, we see that  $r \rightarrow \infty$  for  $\theta \rightarrow \pi$ . From the energy integral, with  $E = 0$ , we have that,

$$\frac{1}{2}v_e^2 - \frac{\mu}{r} = 0, \quad v_e^2 = \frac{2\mu}{r}. \quad (12)$$

Here,  $v_e$  is the escape velocity — the smallest velocity needed to escape the field of gravitational attraction. Comparing equations 11 and 12, we see that, for a given  $r$ , the escape velocity is a factor of  $\sqrt{2}$  larger than the velocity necessary to maintain a circular orbit. Thus, if a satellite is on a circular orbit with velocity  $v_c$ , the necessary  $\Delta v$  to escape is  $(\sqrt{2} - 1)v_c$ .

It should be noted that a satellite in a parabolic trajectory has a total specific energy,  $E$ , equal to zero. This means that when  $r$  increases, the kinetic energy is transformed to potential energy such that, at infinity, the residual velocity is equal to zero.

### Hyperbolic Trajectory ( $e > 1$ )

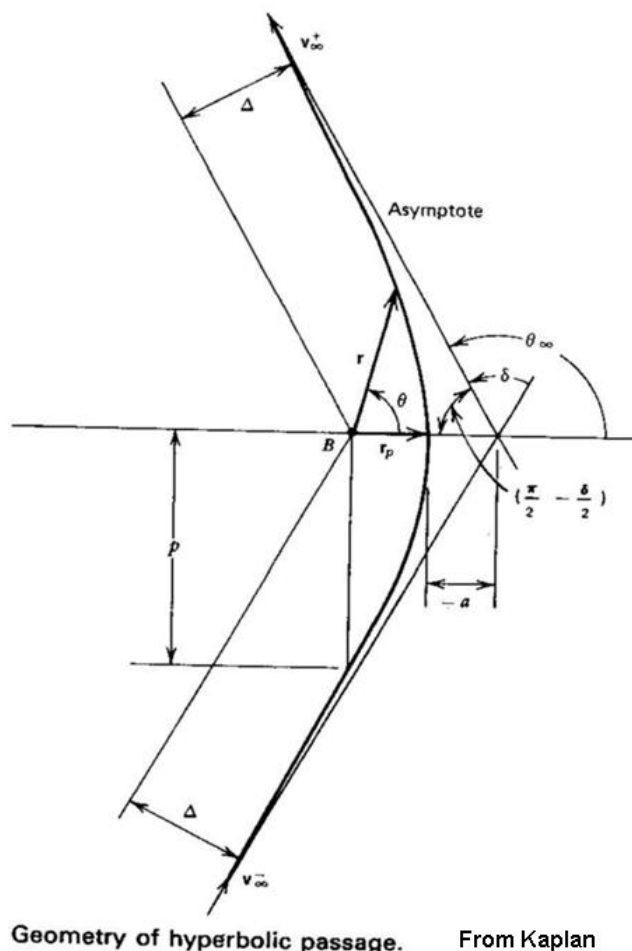
For a hyperbolic orbit,  $e > 1$ . The angular momentum is still given by equation (20) from lecture 15.

$$h^2 = a\mu(1 - e^2) \quad (13)$$

and the energy is constant, and given by

$$E = -\frac{\mu}{2a} = \frac{v^2}{2} - \frac{\mu}{r} = \frac{v_\infty^2}{2} \quad (14)$$

Therefore, for a hyperbolic orbit since both angular momentum and energy are positive, the semi-major axis  $a$  is negative. Since the potential goes to zero as  $r \rightarrow \infty$ , the magnitude of the velocity inbound is the same as the magnitude of the velocity outbound. The geometry of a given hyperbolic orbit is shown below (Figure taken from Kaplan). This basic orbit is used to describe a planetary flyby and/or a hyperbolic escape from a planetary orbit. For a given case, the periapsis  $r_\pi$ , equivalent to the point  $r_p$  in the figure, is taken as a point on a planetary orbit, and the velocity  $v_\pi$  is taken as the velocity of a satellite required to escape on a given hyperbolic trajectory from a circular orbit passing through the point  $r_\pi$ . From this, the trajectory is determined from  $v_\infty$  and  $\theta_\infty$ .



Geometry of hyperbolic passage. From Kaplan

This solution can be used to represent a flyby of a spacecraft past a planet. The direction of the orbit changes due to the flyby but the velocity (and energy) does not. However, this solution to the 2-body orbit problem is with respect to coordinates fixed at the **center of mass**; for a satellite-planet orbit, this is a coordinate system moving with the planet. Although the inbound and outbound velocities are the same **with respect to the moving planet**, when this is viewed in an inertial reference frame, considerable change in the satellite velocity can occur as a result of a flyby. This effect has been widely used in the design of planetary missions. We will discuss this in Lecture 17.

From the general orbit equation, valid for all values of  $e$ ,

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta} \quad (15)$$

when  $\theta \rightarrow \pm \cos^{-1}(1/e) = \theta_{\infty}$ , we have  $r \rightarrow \infty$ . Hence, the trajectories are open. Moreover, if the velocity  $v$  at a given  $r$  is known (such as near the planet), from conservation of energy, the velocity as  $r \rightarrow \infty$  is simply  $v_{\infty} = \sqrt{2E} = \sqrt{v^2 - 2\mu/r}$ . From conservation of energy,  $v_{\infty}$  is the same on both the inbound and outbound parts of the trajectory. From the conservation of angular momentum and the result  $v_{\infty} = \text{Constant}$ , the

final track of the orbit is a straight line offset by a constant distance  $\Delta$  from the line to the focus.

For a given angular momentum and energy level, the eccentricity of the orbit is determined by applying conservation of angular momentum

$$h = \sqrt{a\mu(1 - e^2)} = v_\infty \Delta = \sqrt{\frac{\mu^2(e^2 - 1)}{v_\infty^2}}, \quad (16)$$

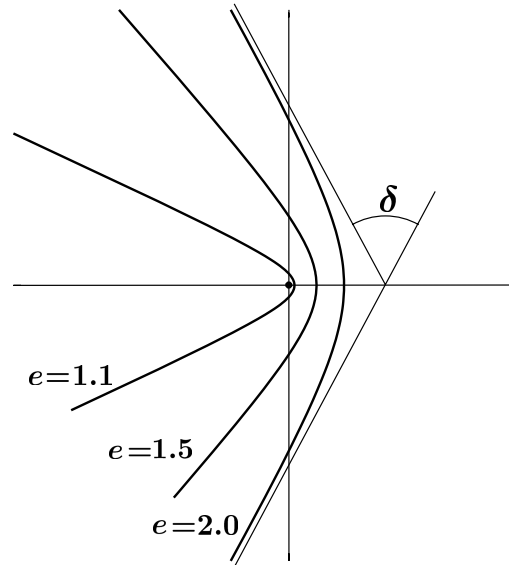
since  $a = \frac{-\mu}{v_\infty^2}$ , where  $h$  is a constant of the motion. This yields

$$e = \sqrt{1 + \frac{v_\infty^4 \Delta^2}{\mu^2}} \quad (17)$$

For application to orbit transfers, it is useful to express orbital parameters in term of the radius of the point of closest approach, the periapsis  $r_\pi$ , and the velocity at infinity  $v_\infty$ , resulting in

$$e = 1 + \frac{r_\pi v_\infty^2}{\mu} \quad (18)$$

An important parameter for hyperbolic orbits is the turning angle,  $\delta$ , which is the angle through which the velocity changes along the trajectory as the body travels from  $-\infty$  to  $\infty$ . The turning angle is given by  $\delta = 2\theta_\infty - \pi$ , or  $\delta = 2\sin^{-1}(-1/e)$ . Below we show several hyperbolic trajectories which have identical terminal velocities for different values of the eccentricity (and turning angles  $\delta$ ).



Hyperbolic orbits are the linkages between orbits about a given planet and interplanetary travel.

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### Example

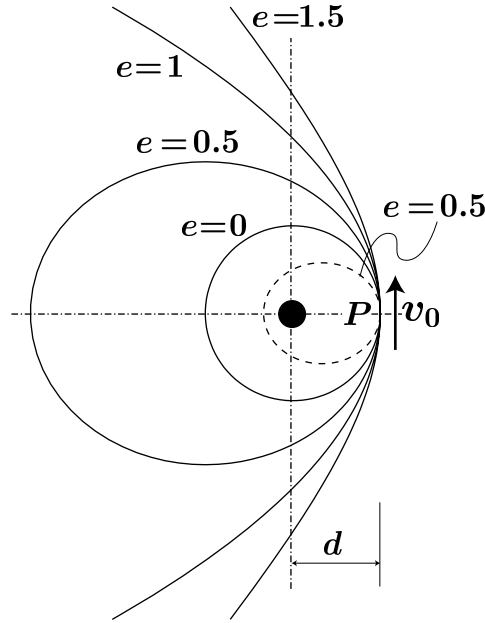
### Different orbits as a function of $v_0$

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We consider the problem of launching a satellite at an altitude  $d$  with an initial velocity  $v_0$ , along the

direction tangent to the earth's surface. We consider the different trajectories that are obtained as we vary the magnitude of  $v_0$ .

For  $v_0 = v_{0c} \equiv \sqrt{\mu/(R+d)}$ , the trajectory will be a circle ( $e = 0$ ). For  $v_0 = v_{0e} \equiv \sqrt{2\mu/(R+d)}$ , the trajectory will be parabola ( $e = 1$ ). For  $v_0 > v_{0e}$ , the trajectory will be a hyperbola, whereas for  $v_{0c} < v_0 < v_{0e}$  the trajectory will be elliptical. We note that, for all of these orbits, the launch point,  $P$ , is the orbit's perigee, or the closest point in the trajectory to the earth's center.



On the other hand, when the velocity  $v_0 < v_{0c}$ , the straightforward use of equations (1) and (2) gives a negative eccentricity! The eccentricity is negative because equation 1 assumes that the origin of  $\theta$  is taken to be at the orbit's perigee. It turns out that in this case, the orbit has a lower energy than the circular orbit, and, hence, the launch point is now the orbit's apogee. The proper use of equation 1 requires that  $\theta = \pi$ . In this case, we have  $d + R = (v_0^2(d + R)^2/\mu)/(1 + e \cos \pi)$ , which for  $v_0 < v_{0c}$  gives a positive eccentricity. In the picture above, we see one such trajectory depicted with a dotted line.



## Lecture 18 — Classical Orbital Elements

### Introduction

This lecture describes the Classical Orbital Elements that are commonly used to describe the orbital trajectory and location of a satellite. Before we do so, it is useful to describe the frame of reference that will be used and how it compares to other frames.

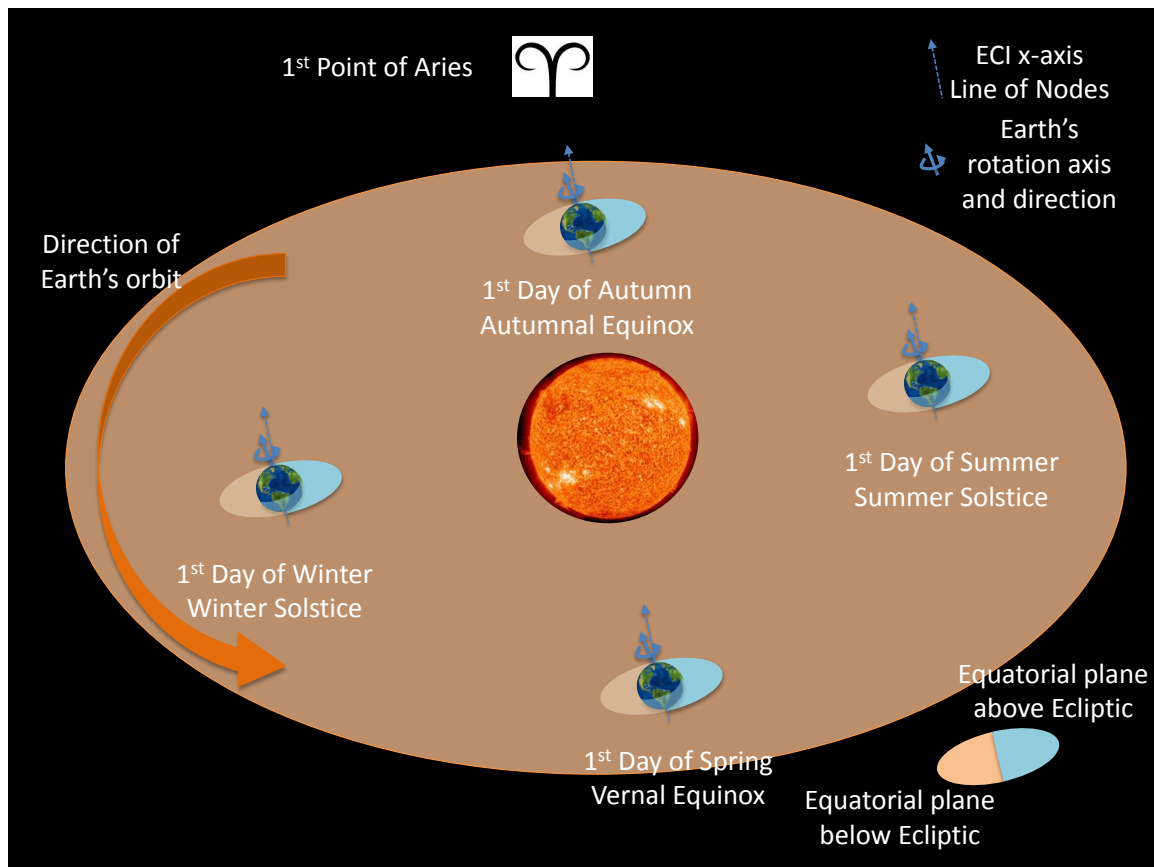
### Frames of Reference

All orbital frames share some common attributes. One axis aligns with the angular momentum of the dominant gravitational system and the other two axes define the plane within which the trajectories of the masses with significant gravitational parameters ( $\mu$ ) lie, forming a right hand, orthogonal frame. On a grand scale, the Milky Way galaxy, within which the Sun resides, has an angular momentum vector pointing in the direction of the galactic north pole. The stars tend to orbit the galactic center, around that pole in the right-hand direction, within a few degrees of the galactic plane. Zooming in on our Solar System, it has angular momentum that points in the direction of the ecliptic north pole and the planets orbit the Sun within a few degrees of the ecliptic plane. Zooming in further, the Earth's rotation has angular momentum which points in the direction of the North Pole, which is tilted about 23 degrees relative to the ecliptic north pole. This angle is the inclination of the Earth's polar axis to the angular momentum vector of its orbital plane, called the "obliquity of the ecliptic." The Moon's orbit around the Earth, on the other hand, has an angular momentum vector that is aligned within about one degree of the ecliptic north pole. For the rest of this Lecture, one axis of the Earth-centered reference frames will be aligned with the Earth's angular momentum vector associated with its rotation, not its orbit about the Sun.

The figure below shows that as the Earth orbits the Sun, in the counterclockwise direction when viewed from a point north of the ecliptic plane, the angular momentum vector associated with the Earth's sidereal rotation always points in the same inertial direction. This causes the equatorial plane to be inclined 23 degrees from the ecliptic plane, shown as the blue portion in the figure when it is above the ecliptic to the right of Earth and as tan when it is below the ecliptic to the left of Earth. Notice that this tilt creates a line of intersection between the two planes that always points in the same direction in inertial space as the Earth orbits the Sun. This is called the Line of Nodes. On the scale of our solar system, the First Point of Aries is very far away. Thus, the x-axis does not rotate as the Earth orbits the Sun.

## REFERENCES

The Earth Centered Inertial (ECI) reference frame has its origin at the center of mass of the Earth. As its name suggests, this frame does not rotate with the Earth. Instead, its orientation is fixed with respect to inertial space. However, its origin does travel with the Earth's center of mass while the Earth orbits the Sun. The positive x-axis of the ECI frame points along this Line of Nodes in the direction of the First Point of Aries. The ECI x-axis is easier to visualize on the Vernal Equinox when, on the first day of Spring (shown at the bottom of the figure), it points from the center of the Earth along the Line of Nodes, through the center of mass of the Sun, and towards the First Point of Aries shown by the  $\Upsilon$ ). Actually, it currently points to Pisces due to the precession of the Earth's rotation axis with a period of 26,000 years. Aries is the constellation in the zodiac that is behind the Sun at that time of year. It is one of the two points in inertial space (i.e., on the celestial sphere) at which the equator crosses the ecliptic plane, corresponding to the two directions of the Line of Nodes. The other is the First Point of Libra, which is in the opposite direction and is behind the Sun during the Autumnal Equinox. Note that the x-axis only points at the Sun on the first day of Spring because its direction is fixed in inertial space while the Earth orbits the Sun. The positive z-axis of the ECI frame points from the center of mass of the Earth towards the North Pole and is co-linear with the component of angular momentum associated with the Earth's rotation. The positive y-axis points from the Earth's center of mass towards the equator such that  $\vec{x} \times \vec{y} = \vec{z}$ . Neither the y-axis nor the z-axis lie in the ecliptic plane.



Another useful frame is the Earth Centered Fixed reference frame (ECF). Like the ECI frame, the z-axis

points towards the North Pole. Unlike the ECI frame, the ECF frame rotates with the Earth and therefore is a body-fixed frame. Its x-axis points to the equator at zero latitude and longitude off the west coast of Africa. In spherical coordinates, it is used to describe altitude in the  $\hat{e}_r$  direction, longitude in the  $\hat{e}_\theta$  direction, and latitude in the  $\hat{e}_\phi$  direction. The ECF frame will be useful when describing the satellite's location with respect to specific locations on the Earth's surface.

## Classical Orbital Elements (COEs)

Six Classical Orbital Elements are needed to uniquely describe a satellite's orbit and its location in that orbit. This corresponds to the six scalar parameters associated with initial position,  $\vec{r}$ , and velocity,  $\vec{v}$ , conditions. However, these six scalar parameters are not unique. In a previous lecture, we described a "closed" elliptical orbit using

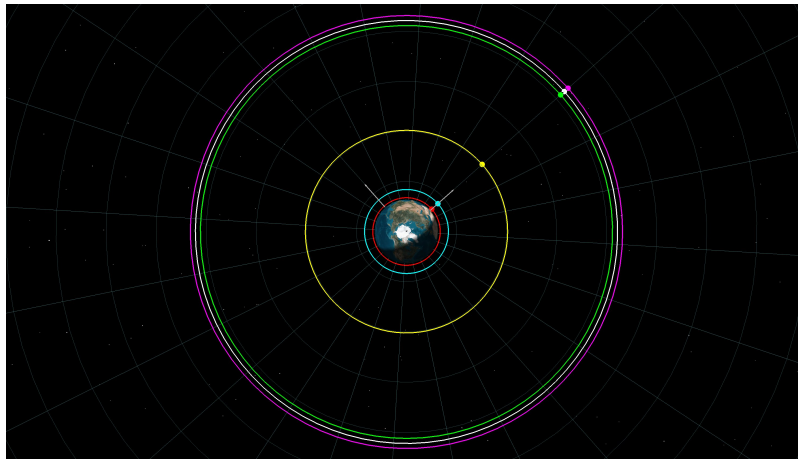
$$r = \frac{h^2/\mu}{1 + e \cos \theta} = \frac{a(1 - e^2)}{1 + e \cos \theta} \quad (1)$$

This equation has three parameters:  $a$  is the semi-major axis,  $e$  is the eccentricity of the ellipse, and  $\theta$  is the angular location of the satellite on that ellipse. However, this equation is from the perspective of the plane of the orbit. We need three more parameters to rotate the frame of the orbit into the ECI frame. The result is six parameters. When trying to remember the six Classical Orbital Elements, it is useful to think of the vowels (a, e, i, o, u, and sometimes y). As we will see in a moment, they are very similar to the actual variables used for the COEs, the first five of which completely describe the orbital trajectory while the sixth, like "sometimes y," is different from the others in that it describes the location of the satellite on that trajectory. The Classical Orbital Elements are:

| COE      | Units | Vowel | Name                              |
|----------|-------|-------|-----------------------------------|
| a        | m     | a     | Semi-major axis                   |
| e        | -     | e     | Eccentricity                      |
| i        | rad   | i     | Inclination                       |
| $\Omega$ | rad   | o     | Right ascension of ascending node |
| $\omega$ | rad   | u     | Argument of perigee               |
| v        | rad   | y     | True anomaly                      |

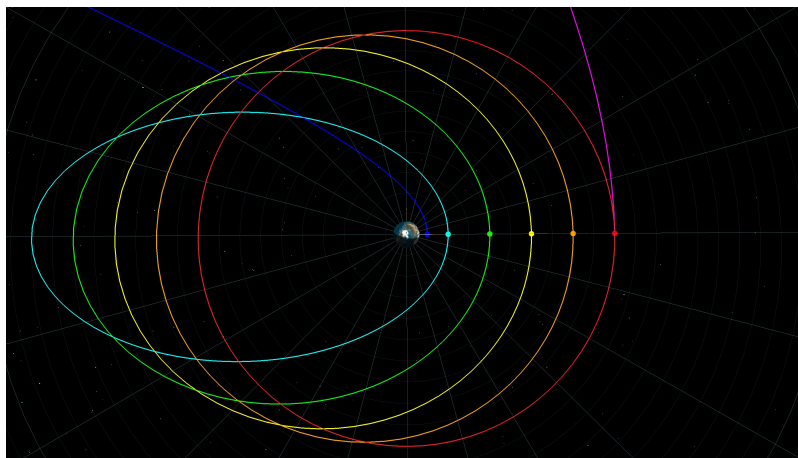
### Semi-Major Axis (a)

The semi-major axis ( $a$ ) is the distance from the center of a circular or elliptical orbit to the trajectory in the direction of maximum length. For a circular orbit, the direction of maximum length is undefined. Thus,  $a$  is the radius of a circular orbit. Altitude ( $h$ ) is the height above the surface so the orbital radius (6837 km) of Low Earth Orbit (LEO) equals the radius of Earth (6371 km) plus the altitude of LEO (466 km). The International Space Station (ISS) is in LEO. Due to its ballistic coefficient and the air drag in LEO, orbital energy is reduced, requiring frequent re-boosts of its orbital radius through in-track propulsive burns. The Global Positioning System (GPS) is at a semi-major axis of 20,200 km. Geosynchronous orbit is at 42,164 km. The figure below shows circular equatorial orbits with different semi-major axes.



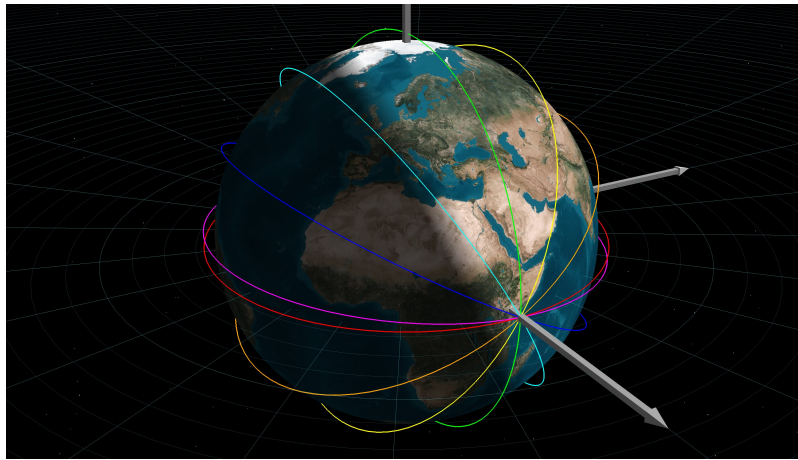
### Eccentricity ( $e$ )

The eccentricity describes how much a conic section deviates from a circle. For a "closed" orbit whose trajectory repeats over a finite amount of time,  $0 < e < 1$ . It defines the distance along the semi-major axis between the focus and the center of the ellipse as a fraction of the semi-major axis. The figure below shows equatorial orbits with different eccentricities.



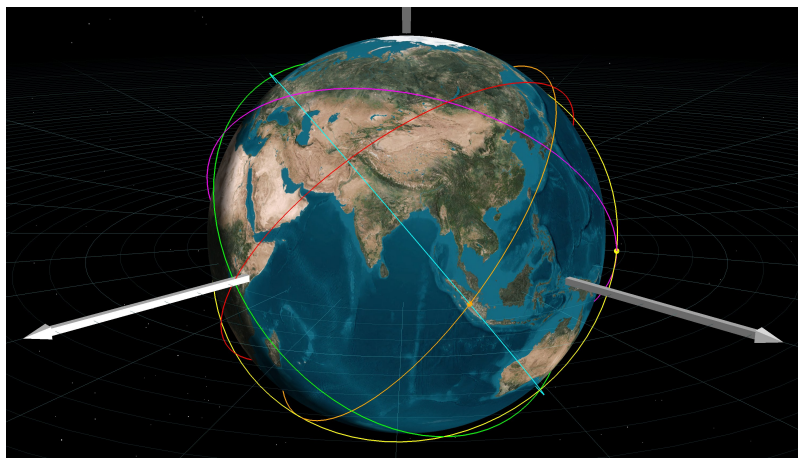
### Inclination ( $i$ )

The inclination is the angle between the equatorial plane and the plane of the orbit about the line of nodes vector. The line of nodes vector defines the line of intersection between the orbital plane and the equatorial plane. This is equivalent to the angle from the North Pole (ECI z-axis) to the orbit's angular momentum vector. The figure below shows a range of inclinations. Typically,  $0 < i < 180$  degrees. For  $i < 90$  degrees, the orbit is called prograde orbit. For  $i > 90$  degrees, it is called a retrograde orbit.



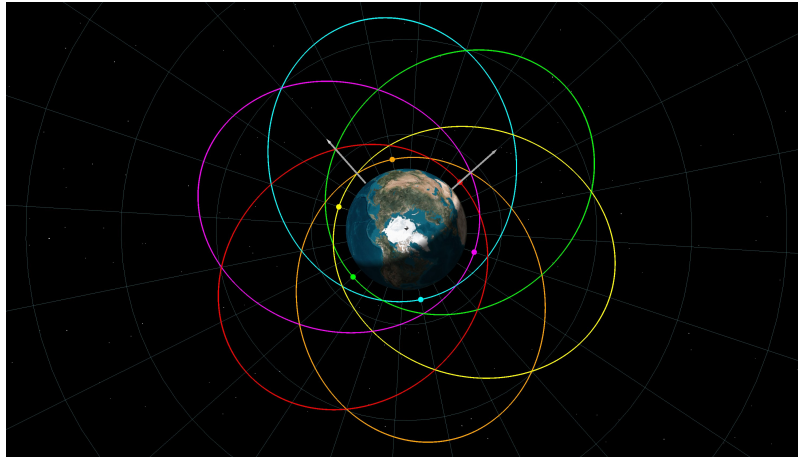
### Right Ascension of the Ascending Node ( $\Omega$ or RAAN)

The RAAN is the angle about the North Pole (ECI z-axis) in the equatorial plane starting from the ECI x-axis (Vernal Equinox) eastward to the point on the equator where the satellite's direction in the orbit crosses from the southern to the northern hemisphere. This intersection between the orbital plane and the equatorial plane is called the Line of Nodes.



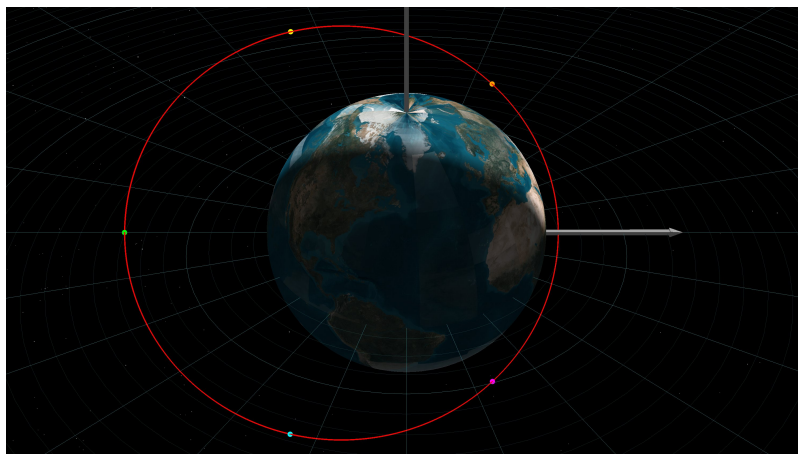
### Argument of Perigee ( $\omega$ )

The argument of perigee is the angle measured in the orbital plane, in the direction of the satellite's motion, from the ascending node to the satellite's closest approach to Earth (perigee). Apogee is the point in the orbit where the satellite is the furthest distance from Earth. In general, the terms are periapsis and apoapsis. For the Sun, they are perihelion and aphelion. For stars, they are periastron and apastron. For the Moon, they are pericynthion and apocynthion. If the RAAN is undefined, as for an equatorial orbit,  $\omega$  is measured eastward from the ECI x-axis (Vernal Equinox) to perigee and is called the true longitude of periapsis.



## True Anomaly ( $\nu$ )

The true anomaly is the angle measured in the orbital plane in the direction of satellite motion from the periapsis to the satellite's current position. For a circular and inclined orbit, use the "argument of latitude" instead, which is the angle measured from the ascending node. For circular and equatorial orbits, use the "true longitude" instead, which is the angle measured eastward from the ECI x-axis.



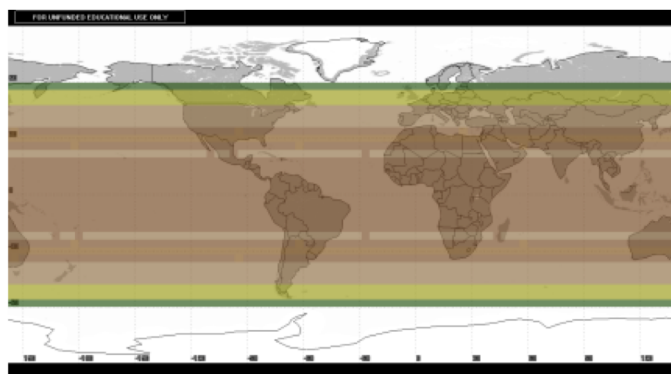
## Special Orbits

### Repeating Ground Track Orbits

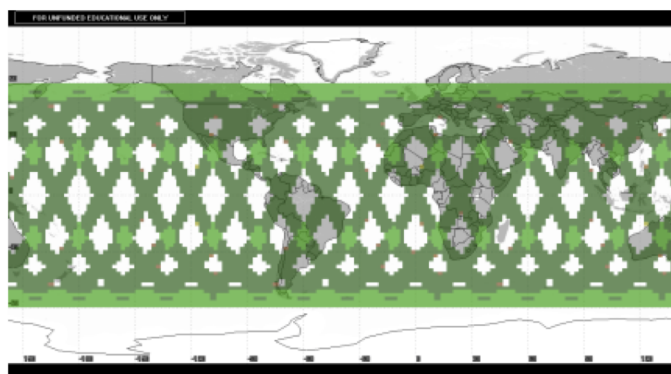
Without any external forces or torques to change the angular momentum of an orbit, the plane of that orbit, and its angular momentum vector orthogonal to that plane, will remain fixed in inertial space while the Earth rotates once every 23.9 hours (sidereal or inertial day). This will almost always cause the ground track of the satellite to shift across the Earth's surface during successive orbits, as shown in the figure below. The ground track is the point on the Earth's surface directly nadir to (below) the satellite. This is equivalent to the point on the inertial position vector from the Earth's center of mass to the satellite where it penetrates

## REFERENCES

the Earth's surface. Over the period of the orbit, one cycle of a distorted cosine function will lie over this latitude-longitude map of the Earth. Depending on the semi-major axis, successive orbits will show this cycle shifting westward or eastward on successive orbits. The revisit time is the amount of time it takes for a satellite, or any satellite in a constellation of satellites, to return to the same position over the ground. In the figure, green shows where the revisit time is short and brown shows where it is long. These are called Non-Repeating Ground Track orbits (NRGT).



When the period of the orbit is synchronized with the period of the Earth's sidereal day, the ground track will repeat. This is called a Repeating Ground Track orbit (RGT). An example ground track is shown in the figure below. A sidereal day is the time for the Earth to rotate, relative to an inertial frame, by  $2\pi$  radians. This is 23 hours, 56 minutes, and 4 seconds. Note that this is shorter than 24 hours because each year the Earth travels 360 degrees around the Sun over the course of 365 days, or about one degree per day. Therefore, the Sun travels eastward in the sky each sidereal day causing the Earth to experience one more sidereal day than solar day per year. Orbital period depends on semi-major axis, so there are discrete semi-major axes at which RGTs occur. Each RGT is defined by  $N_o/N_d$  where the ground track repeats every  $N_d$  days during which  $N_o$  orbits occur. Both  $N_o$  and  $N_d$  are integers.



RGTs are good for regional coverage and NRGTs are good for global coverage.

## Sun-Synchronous Orbits

The Earth is not a perfect sphere, it is oblate. The oblateness constant  $q$  is given by

$$q = \frac{r_e^3 \dot{\Phi}^2}{GM} = \frac{1}{289} \quad (2)$$

where  $r_e$  is the radius of the Earth at the equator,  $\dot{\Phi}$  is the angular rate of the Earth's rotation,  $G$  is the gravitational constant, and  $M$  is the Earth's mass.

Due to its spin, its radius in the equatorial plane (6384 km) is longer than through the poles (6353 km) by 31 km. This makes the tallest (Mauna Kea in Hawaii), highest (Everest in Nepal), and furthest (Chimborazo in Ecuador) mountains on Earth all different. Chimborazo is located one degree south of the equator taking advantage of this spin-induced bulge. Although the bulge dominates, there are other fluctuations in the Earth's distribution of mass (e.g., Himalayan Plateau) that can also great local variations in the strength and direction of Earth's gravitational force vector on a satellite that are not central forces. These variations are expanded in a series of spherical basis functions whose sum is the gravitational geopotential of the Earth. The second term ( $J_2$ ) is the Earth's oblateness. The non-central component can be considered as an external force which can change the angular momentum of the orbit. This is called precession. One type of orbit that takes advantage of this precession is called a Sun-Synchronous orbit. These are orbits of high inclination and retrograde ( $i = 98$  degrees) with altitudes in the range of 600-800 km. The altitude and inclination of the orbit can be chosen such that the angular momentum of the orbit precesses about one degree per day making it always point at the same angle relative to the Sun. If that angle is zero, the satellite will never be eclipsed, making its power system less dependent on batteries and therefore complex. That satellite will also pass over different parts of the Earth at the same local time (i.e., dusk-dawn or am-pm orbits). If the angle is 90 degrees, the local coverage is noon-midnight. Many Earth weather monitoring satellites take advantage of these types of orbits. The Afternoon-Train (or A-Train) is a constellation of weather monitoring satellites in the same orbit in a leader-following configuration that cross from south to north at 1 pm local time and take multiple measurements at the same time under the same solar illumination conditions.

## Molniya Orbits

A Molniya orbit is a type of highly elliptical orbit with an inclination of 63.4 degrees, an argument of perigee of -90 degrees and an orbital period of one half of a sidereal day. This orbit is named after the Soviet/Russian Molniya communications satellites. Due to its high eccentricity, the satellite spends most of its time near apogee over the Northern Hemisphere with its nadir point being at 63.4 degrees north latitude and altitude at 40,000 km. This inclination also places it in a frozen orbit where the Earth's oblateness ( $J_2$  term) will not perturb the orbit's angular momentum vector.

## Polar Constellations

To get global coverage for weather monitoring, communications, etc., without the need for a large antenna, optic or time delay associated with a geosynchronous satellite, a constellation of satellites in Low Earth Orbit (LEO) is used. Multiple orbital planes, inclined at  $i = 90$  degrees and evenly distributed in RAAN ( $\Omega$ ), with multiple satellites in each plane, is typically called a polar constellation because the orbital trajectories all cross at the north and south poles. The inclination of 90 degrees ensures that all latitudes can be observed (sometimes called coverage). The fact that the Earth rotates eastwards under these orbital trajectories



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ensures that all longitudes can be observed.

**Walker Constellations**

Polar Constellations have the highest coverage at the poles due to the convergence of the orbital trajectories at the poles. Also, launching into a polar orbit gets no  $\Delta v$  benefit from the Earth's rotation. Excess coverage in sparsely populated areas, along with high launch cost, have motivated the design of other constellations such as Walker Constellations. A Walker Constellation (such as that used for GPS), is characterized by its inclination (typically a little above 60 degrees), with multiple orbital planes with circular orbits evenly distributed in RAAN ( $\Omega$ ), or constant  $\Delta\Omega$ , with the same number of satellites per plane whose true anomalies have a fixed phase shift between adjacent planes.

## Lecture 19 — Orbit Transfers

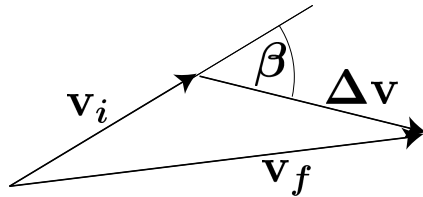
In this lecture, we will consider how to transfer from one orbit, or trajectory, to another. One of the assumptions that we shall make is that the velocity changes of the spacecraft, due to the propulsive effects, occur instantaneously. Although it obviously takes some time for the spacecraft to accelerate to the velocity of the new orbit, this assumption is reasonable when the burn time of the rocket is much smaller than the period of the orbit. In such cases, the  $\Delta v$  required to do the maneuver is simply the difference between the velocity of the final orbit minus the velocity of the initial orbit.

When the initial and final orbits intersect, the transfer can be accomplished with a single impulse. For more general cases, multiple impulses and intermediate transfer orbits may be required.

Given initial and final orbits, the objective is generally to perform the transfer with a minimum  $\Delta v$ . In some situations, however, the time needed to complete the transfer may also be an important consideration.

Most orbit transfers will require a change in the orbit's total specific energy,  $E$ . Let us consider the change in total energy obtained by an instantaneous impulse  $\Delta v$ . If  $\mathbf{v}_i$  is the initial velocity, the final velocity,  $\mathbf{v}_f$ , will simply be,

$$\mathbf{v}_f = \mathbf{v}_i + \Delta \mathbf{v} .$$



If we now look at the magnitude of these vectors, we have,

$$v_f^2 = v_i^2 + \Delta v^2 + 2v_i\Delta v \cos \beta ,$$

where  $\beta$  is the angle between  $\mathbf{v}_i$  and  $\Delta \mathbf{v}$ . The energy change will be

$$\Delta E = \frac{1}{2}\Delta v^2 + v_i\Delta v \cos \beta .$$

From this expression, we conclude that, for a given  $\Delta v$ , the change in energy will be largest when:

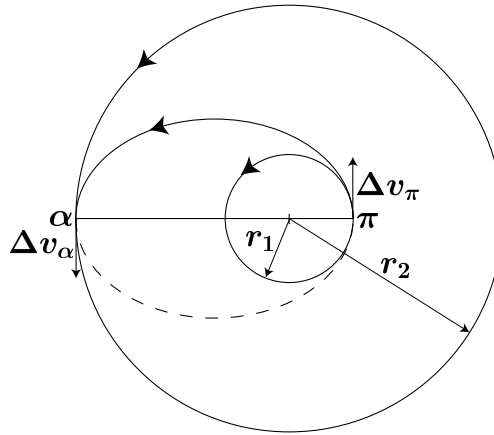
- $\mathbf{v}_i$  and  $\Delta \mathbf{v}$  are co-linear ( $\beta = 0$ ), and,

- $v_i$  is maximum.

For example, to transfer a satellite on an elliptical orbit to an escape trajectory, the most energy efficient impulse would be co-linear with the velocity and applied at the instant when the satellite is at the elliptical orbit's perigee, since at that point, the velocity is maximum. Of course, for many required maneuvers, the applied impulses are such that they cannot satisfy one or both of the above conditions. For instance, firing at the perigee in the previous example may cause the satellite to escape in a particular direction which may not be the required one.

## Hohmann Transfer

A Hohmann Transfer is a two-impulse elliptical transfer between two co-planar circular orbits. The transfer itself consists of an elliptical orbit with a perigee at the inner orbit and an apogee at the outer orbit.



It turns out that this transfer is usually optimal, as it requires the minimum  $\Delta v_T = |\Delta v_\pi| + |\Delta v_\alpha|$  to perform a transfer between two circular orbits. The exception for which Hohmann transfers are not optimal is for very large ratios of  $r_2/r_1$ , as discussed below.

The transfer orbit has a semi-major axis,  $a$ , which is

$$a = \frac{r_1 + r_2}{2} .$$

Hence, the energy of the transfer orbit is greater than the energy of the inner orbit ( $a = r_1$ ), and smaller than the energy of the outer orbit ( $a = r_2$ ). The velocities of the transfer orbit at perigee and apogee are given, from the conservation of energy equation, as

$$v_\pi^2 = \mu \left( \frac{2}{r_1} - \frac{2}{r_1 + r_2} \right) \quad (1)$$

$$v_\alpha^2 = \mu \left( \frac{2}{r_2} - \frac{2}{r_1 + r_2} \right) . \quad (2)$$

The velocities of the circular orbits are  $v_{c1} = \sqrt{\mu/r_1}$  and  $v_{c2} = \sqrt{\mu/r_2}$ . Hence, the required impulses at perigee and apogee are,

$$\begin{aligned}\Delta v_\pi &= v_\pi - v_{c1} = \sqrt{\frac{\mu}{r_1}} \left( \sqrt{\frac{2r_2}{r_1 + r_2}} - 1 \right) \\ \Delta v_\alpha &= v_{c2} - v_\alpha = \sqrt{\frac{\mu}{r_2}} \left( 1 - \sqrt{\frac{2r_1}{r_1 + r_2}} \right) .\end{aligned}$$

When the initial orbit has a radius larger than the final orbit, the same strategy can be followed but in this case, negative impulses will be required, first at apogee and then at perigee, to decelerate the satellite.

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**Example**
**Hohmann transfer [1]**


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A communication satellite was carried by the Space Shuttle into low earth orbit (LEO) at an altitude of 322 km and is to be transferred to a geostationary orbit (GEO) at 35,860 km using a Hohmann transfer. The characteristics of the transfer ellipse and the total  $\Delta v$  required,  $\Delta v_T$ , can be determined as follows:

For the inner orbit, we have,

$$\begin{aligned}r_1 &= R + d_1 = 6.378 \times 10^6 + 322 \times 10^3 = 6.70 \times 10^6 \text{ m} \\ v_{c1} &= \sqrt{\frac{\mu}{r_1}} = \sqrt{\frac{gR^2}{r_1}} = 7713 \text{ m/s} \\ E_1 &= -\frac{\mu}{2r_1} = -\frac{gR^2}{2r_1} = -2.975 \times 10^7 \text{ m}^2/\text{s}^2 \text{ (J/kg)}.\end{aligned}$$

Similarly, for the outer circular orbit,

$$\begin{aligned}r_2 &= R + d_2 = 42.24 \times 10^6 \text{ m} \\ v_{c2} &= 3072 \text{ m/s} \\ E_2 &= -4.718 \times 10^6 \text{ m}^2/\text{s}^2 \text{ (J/kg)}.\end{aligned}$$

For the transfer trajectory,

$$\begin{aligned}2a &= r_1 + r_2 = 48.94 \times 10^6 \text{ m} \\ E &= -\frac{\mu}{2a} = -8.144 \times 10^6 \text{ m}^2/\text{s}^2 \text{ (J/kg)},\end{aligned}$$

which shows that  $E_1 < E < E_2$ . The velocity of the elliptical transfer orbit at the perigee and apogee can be determined from equations 10 and 11, as,

$$\begin{aligned}v_\pi &= 10,130 \text{ m/s} , \\ v_\alpha &= 1,067 \text{ m/s} .\end{aligned}$$

Since the velocity at the perigee is orthogonal to the position vector, the specific angular momentum of the transfer orbit is,

$$h = r_1 v_\pi = 6.787 \times 10^{10} \text{ m}^2/\text{s} ,$$

and the eccentricity can be determined as,

$$e = \sqrt{1 + \frac{2Eh^2}{\mu^2}} = 0.7265 .$$

Finally, the impulses required are,

$$\begin{aligned} \Delta v_\pi &= v_\pi - v_{c1} = 10,130 - 7713 = 2414 \text{ m/s} \\ \Delta v_\alpha &= v_{c2} - v_\alpha = 3072 - 1607 = 1465 \text{ m/s}, \end{aligned}$$

The sign of the  $\Delta v$ 's indicates the direction of thrusting (whether the energy is to be increased or decreased) and the total  $\Delta v$  is the sum of the magnitudes. Thus,

$$\Delta v_T = |\Delta v_\pi| + |\Delta v_\alpha| = 2417 + 1465 = 3882 \text{ m/s}.$$

Since the transfer trajectory is one half of an ellipse, the time of flight (TOF) is simply half of the period,

$$TOF = \pi \sqrt{\frac{(24.47 \times 10^6)^3}{3.986 \times 10^{14}}} = 19,050 \text{ s} = 5.29 \text{ h} .$$

In order to illustrate the optimal nature of the Hohmann transfer, we consider now an alternative transfer in which we arbitrarily double the value of the semi-major axis of the Hohmann transfer ellipse, and find the characteristics and  $\Delta v_T$  of the resulting fast transfer. The semi-major axis of the transfer ellipse will be  $2a = 98 \times 10^6 \text{ m}$ , and  $E = -\mu/(2a) = -4.067 \times 10^6 \text{ m}^2/\text{s}^2 \text{ (J/kg)}$ . The velocity of the transfer orbit at departure will be

$$v_\pi = \sqrt{2 \left( -4.067 \times 10^6 + \frac{3.986 \times 10^{14}}{6.70 \times 10^6} \right)} = 10.530 \text{ m/s},$$

and,

$$\Delta v_\pi = 10,530 - 7713 = 2817 \text{ m/s} .$$

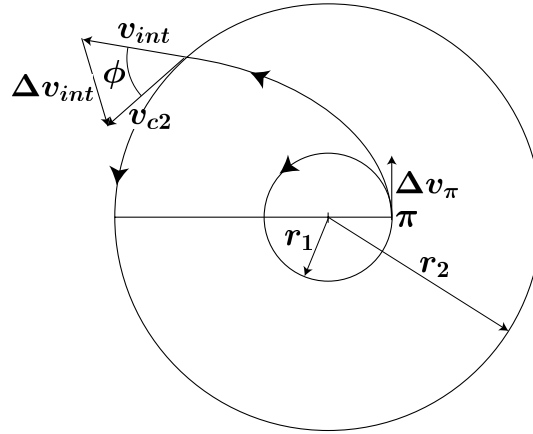
The specific angular momentum is,

$$h = 10,530 \times (6.70 \times 10^6) = 7.055 \times 10^{10} \text{ m}^2/\text{s},$$

and the orbit's eccentricity,  $e$ , is 0.863.

We can now calculate, from the energy conservation equation, the velocity of the transfer orbit at the point of interception with the outer orbit,  $v_{int}$ ,

$$v_{int} = \sqrt{2 \left( -4.067 \times 10^6 + \frac{3.986 \times 10^{14}}{42.24 \times 10^6} \right)} = 3277 \text{ m/s} .$$



Since the angular momentum,  $h$ , is conserved, we can determine the component of  $\mathbf{v}_{int}$  in the circumferential direction

$$(\mathbf{v}_{int})_{\theta} = \frac{h}{r_2} = 1670 \text{ m/s}$$

and the elevation angle,  $\phi$ , is thus,

$$\phi = \cos^{-1} \frac{(\mathbf{v}_{int})_{\theta}}{v_{int}} = 59.36^\circ$$

Finally, from geometrical considerations,

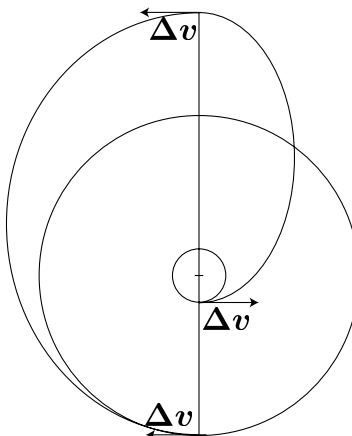
$$\Delta v_{int}^2 = v_{c2}^2 + v_{int}^2 - 2v_{c2}v_{int} \cos \phi,$$

which yields  $\Delta v_{int} = 3142 \text{ m/s}$ , so that

$$\Delta v_T = 2817 + 3142 = 5959 \text{ m/s}.$$

Comparing to the value of the Hohmann transfer  $\Delta v_T$  of 3875 m/s, we see that the fast transfer requires a  $\Delta v_T$  which is 54% higher. The analysis of elliptical trajectories which intersect the circular orbits at an angle is referred to as Lambert's problem. These trajectories can have faster transit times but at a greater cost in energy.

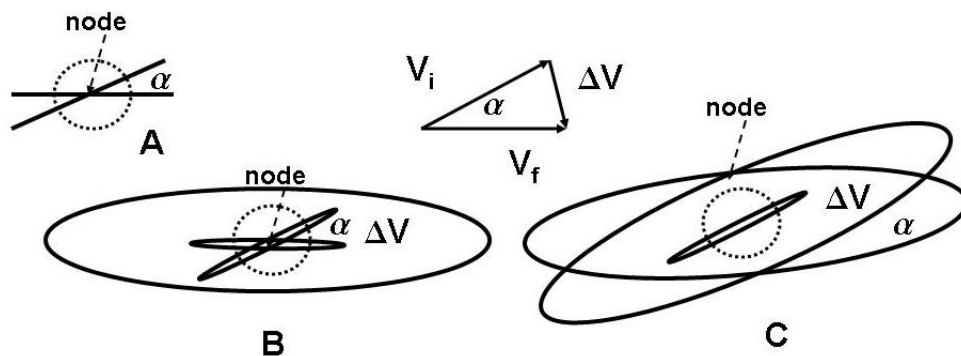
It can be shown that when the separation between the inner and outer orbits is very large ( $r_2 > 11.9r_1$ ) (a situation which rarely occurs), a three impulse transfer comprising of two ellipses can be more energy efficient than a two-impulse Hohmann transfer. This transfer is illustrated in the picture below. Notice that the distance from the origin at which the two transfer ellipses intersect is a free parameter, which can be determined to minimize the total  $\Delta v$ . Notice also that the final impulse is a  $\Delta v$  which opposes direction of motion, in order to decelerate from the large energy ellipse to the final circular orbit. Although this transfer may be more energy efficient relative to the two-impulse Hohmann transfer, it often involves much larger travel times. Try it!



## Orbital Inclination Change

A spacecraft launched from the Kennedy launch site located at  $28.58^\circ$  latitude, will arrive in a low-radius parking orbit with an orbital inclination, relative to the Earth's equatorial plane, of  $28.58^\circ$ , **A**. To change the orbit plane from the initial inclined orbit to that of the equatorial plane, without changing the orbital velocity, requires a  $\Delta V = V_c 2\sin(\alpha/2)$ , where  $\alpha$  is the orbital inclination and  $V_c$  is the orbital velocity, applied at the node, the point at which the two orbits intersect. If our goal is to place a satellite in a geosynchronous orbit in the Earth's equatorial plane, we have several options to complete this mission. We will examine two of these: options **B** and **C**.

Consider option **B**. In this option, we perform a plane change of the initial parking orbit to convert it into a parking orbit of the same radius in the Earth's equatorial plane. This inclination change is performed at the "node". This requires a  $\Delta V = V_{c_p} 2\sin(\alpha/2)$ , where  $V_{c_p}$  is the orbital velocity of the parking orbit. Having reached a orbit in the equatorial plane, we now perform a Hohmann transfer to the geosynchronous orbit.



In option **C**, we reverse the order of the transfers. We begin in the parking orbit at an inclination  $\alpha$ . We then perform a Hohmann transfer in this orbital plane to a geosynchronous orbit, which remains inclined at the angle  $\alpha$ . Since planetary orbits are planar, the  $\Delta V$ 's required to attain this orbit are identical to those required for the equatorial orbit. Having created a planar orbit with a geosynchronous radius, we now

## REFERENCES

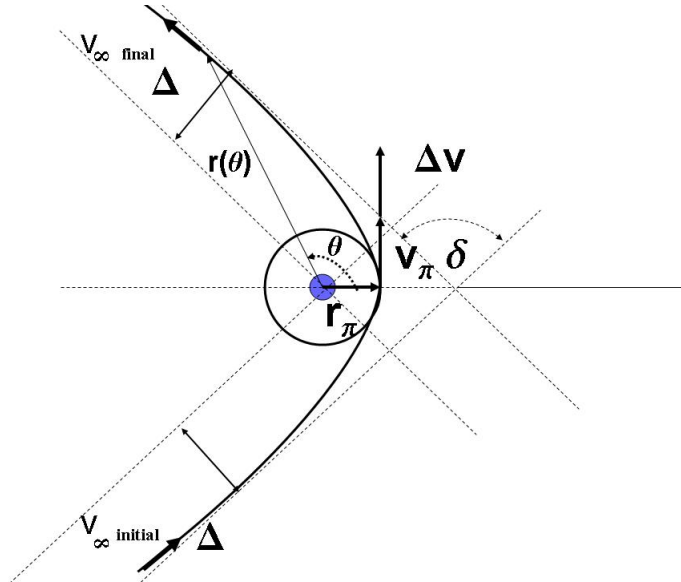
perform an inclination change at the node, which required a  $\Delta V = 2V_{c_g} \sin(\alpha/2)$  where  $V_{c_g}$  is the orbital velocity of the geosynchronous orbit.

We now examine which approach results in lower overall  $\Delta V$ . Since the  $\Delta V$ 's associated with the Hohmann transfer are the same for both options, we focus on the  $\Delta V$ 's associated with the plane changes. Since the velocity of the geosynchronous orbit  $V_{c_g} < V_{c_p}$ , the  $\Delta V$  required for the inclination change of the geosynchronous orbit is less than that from the parking orbit.

Thus in space missions that require an orbital plane change, it is more efficient to perform this maneuver at a low velocity orbit.

## Hyperbolic Orbit Transfer

In preparation for an examination of interplanetary transfer, we examine the issue of hyperbolic escape from a circular orbit where a  $\Delta V$  is applied to raise the velocity of a spacecraft from its circular orbit velocity  $v_c$  to a larger velocity  $v_{\pi_H}$  at the periapsis  $r_\pi$  of a desired hyperbolic orbit having a specified value of  $v_\infty$ . Given the resulting  $v_{\pi_H}$  and the radius of the periapsis  $r_\pi$ , we wish to determine the geometry of the hyperbola, the eccentricity, the angle between the orbit and the axis of symmetry. We can also run the problem backwards and ask what  $\Delta V$  is required at  $r_\pi$  to transfer a spacecraft on an inbound hyperbolic orbit, onto a circular orbit at radius  $r_\pi$ . These processes will be referred to as "hyperbolic escape" and "hyperbolic capture".



Consider a spacecraft in a circular orbit of radius  $r_\pi$ , often called a parking orbit. The properties of the hyperbolic orbit are completely determined by  $r_\pi$  and  $v_\infty$ , and of course  $\mu$ . It is desired to "escape" the influence of the planet and travel on a hyperbolic trajectory with velocity  $v_\infty$ . From the energy equation we determine the velocity  $v_{\pi_H}$  required at  $r_\pi$  to be on the hyperbolic orbit.

$$\frac{v_\infty^2}{2} = \frac{v_{\pi_H}^2}{2} - \frac{\mu}{r_\pi} \quad (3)$$



or

$$\frac{v_{\pi_H}^2}{2} = \frac{v_{\infty}^2}{2} + \frac{\mu}{r_{\pi}} \quad (4)$$

where  $\mu = \mu_{planet}$ . However if we are in a circular orbit at radius  $r_{\pi}$ , our velocity is that of a circular orbit with a radius  $r_{\pi}$ ,  $v = v_c = \sqrt{\mu/r_{\pi}}$ . This immediately determines the  $\Delta V$  required to "escape" from the parking orbit on to the hyperbolic orbit.

$$\Delta V = v_{\pi_H} - v_c, \quad (5)$$

From Lecture 16, we obtain the eccentricity of the resulting orbit as

$$\epsilon = 1 + \frac{r_{\pi} v_{\infty}^2}{\mu} \quad (6)$$

the turning angle  $\delta$  as

$$\delta = 2 \sin^{-1}(1/\epsilon) \quad (7)$$

and the offset  $\Delta$  as

$$\Delta = \frac{\sqrt{\mu^2(\epsilon^2 - 1)}}{v_{\infty}^2} \quad (8)$$

We can also run the problem backwards and ask what  $\Delta V$  is required at  $r_{\pi}$  to transfer a spacecraft on an inbound hyperbolic orbit, onto a circular orbit at radius  $r_{\pi}$ . Since from the energy equation  $v_{\infty}$  is the same for the inbound and outbound trajectories, and  $v_c$ , the velocity in the circular orbit at  $r_{\pi}$  is the same, the magnitude of  $\Delta V$  required is exactly the same but in the opposite direction, acting to slow down the spacecraft. Of course it's a bit more of concern thrusting oneself toward a moving planet, hoping to be captured and not impacted. Clearly the geometry of the offset distance  $\Delta$  plays a more important role in capture than in escape. We will discuss this shortly.

## Interplanetary Transfers: the Method of Patched Conics

The ideas of Hohmann transfer can be applied to interplanetary transfers with some modification. The Hohmann transfer for satellite orbits assumes the satellite is in orbit about a central body and that no other gravitational influence is nearby. When more than one planet is involved, the problem is no longer a two-body problem.

Nevertheless, it is common (at least to get a good approximation) to decompose the problem into a series of two body problems. Consider, for example, an interplanetary transfer in the solar system. For each planet we define the sphere of influence (SOI). Essentially, this is the region where the gravitational attraction due to the planet is larger than that of the sun.

The mission is broken into phases that are connected by patches where each patch is the solution of a two body problem. This is called the ***patched conic approach***. Consider, for instance, a mission to Mars. The first phase will consist of a geocentric hyperbola as the spacecraft escapes from earth SOI. The second phase would start at the edge of the earth's SOI, and would be an elliptical trajectory around the sun while the spacecraft travels to Mars. This orbit could be part of a Hohmann transfer sequence. The third phase would start at the edge of Mars' SOI, and would be a hyperbolic approach capture trajectory with the gravitational

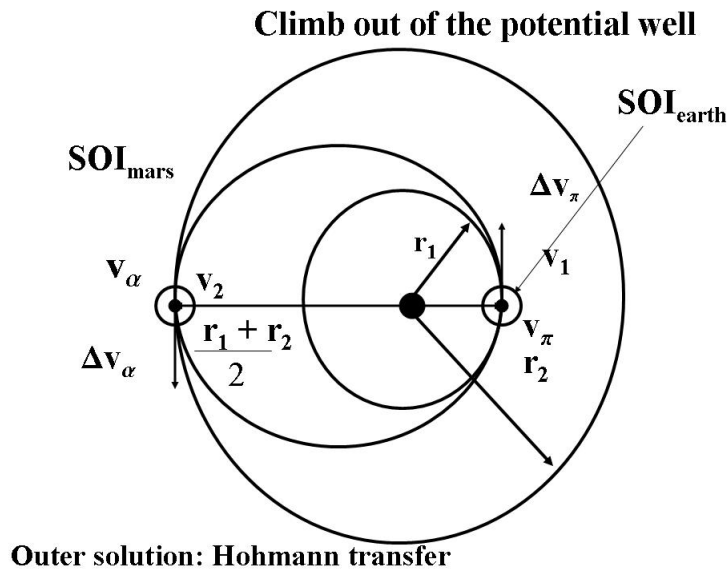
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field of Mars as the attracting force. This third phase can be thought of as a combination of a Hohmann transfer and a hyperbolic capture by the planet.

The time scales and length scales for the various phases of the mission are quite different. The time for a transfer to another planet is measured in months or years; the time scale for escape or capture from a planet is measured in days or hours. The length scale for planetary trajectories is measured in AU units where AU is the distance from the earth to the sun; the length scale for hyperbolic escape from a planet is measured in distances typical of planetary radii and orbits. This can be a challenge for an orbit calculation program since the step size must change dramatically near a planet. We will examine this problem analytically using the method of patched conics to get an approximate result. This approach does not work well for trajectories from earth to moon since the moon is in the SOI of the earth.

Thus the final position of the spacecraft on the escape trajectory from the planet, which occurs at  $r = \infty$  relative to the local (inner) planet coordinate system, occurs at the center of the planet itself in its orbit due to the gravitational field of the Sun.

It turns out that a Hohmann transfer orbit with hyperbolic escape and capture trajectories can be shown to be the minimum energy trajectory for a planetary mission just as the Hohmann transfer itself is an optimum solution for a specified satellite change of orbit. Of course a successful mission requires that the time of launch be selected so that the desired destination planet is at the destination when the spacecraft arrives. We assume that the proper launch time has been selected.



## Parabolic Escape

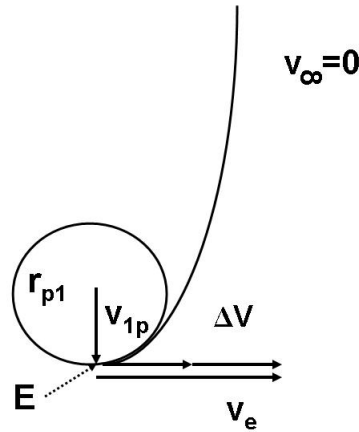
We first consider a spacecraft in a parking orbit about the Earth. One possible strategy for interplanetary transfer using the Hohmann transfer to create an elliptic transfer orbit about the Sun from Earth to another planet, would be to establish our spacecraft in a *co-orbit* with Earth about the Sun rather than an orbit about the Earth itself. That is, the spacecraft is moving in the same circular orbit with the same orbital velocity as the Earth but outside the Earth's gravitational field. In order to accomplish this, we will initiate

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a parabolic transfer orbit.

Orbits are defined in reference to their *moving gravitational center, in this case the moving planet*.

In a reference frame moving with the planet, the properties of the parabolic orbit are shown below.



A space craft is in a circular parking orbit of radius  $r_{1p}$  with circular velocity  $v_{1p} = \sqrt{\mu/r}$ . From the energy equation of a parabolic orbit, we obtain the relation between velocity and position at any point in the orbit.

$$\frac{v_e^2}{2} - \frac{\mu}{r} = 0 \quad (9)$$

$v_e$  is the velocity in the parabolic orbit, often referred to as "escape" velocity. If a  $\Delta V$  is applied, sufficient to cause the velocity at point  $E$  to equal  $v_e = \sqrt{\frac{2\mu}{r}}$ , the spacecraft will depart on a parabolic orbit in the direction shown. Its velocity at  $r = \infty$  **relative to the planet will be zero**; thus it will be in a **co-orbit** with the planet in its circular orbit about the sun. At this point, the Hohmann transfer orbit can be initiated. The  $\Delta V$  required is the difference between escape velocity and orbital circular velocity  $\Delta V = \sqrt{\frac{2\mu}{r}} - \sqrt{\frac{\mu}{r}}$ . In the method of patched conics, we ignore the actual separation between the planet center and the final position of the spacecraft based upon our earlier discussion of time scales and length scales: we locate our "escaped" spacecraft at the center of the Earth.

## Hyperbolic Escape

We now consider another strategy for interplanetary transfer, a hyperbolic escape from a parking orbit about a planet in orbit about the sun into an elliptical orbit about the sun, determined by a Hohmann transfer to the desired planet, combined with a hyperbolic capture into a parking orbit about the destination planet in orbit about the sun, for a complete interplanetary transfer. The details of the specific sequence we will discuss are appropriate for travel to an outer planet such as Mars.

We first consider the Hohmann transfer from a circular orbit about the sun coincident with the original planetary orbit to a circular orbit coincident with the destination planet's orbit about the sun. We neglect

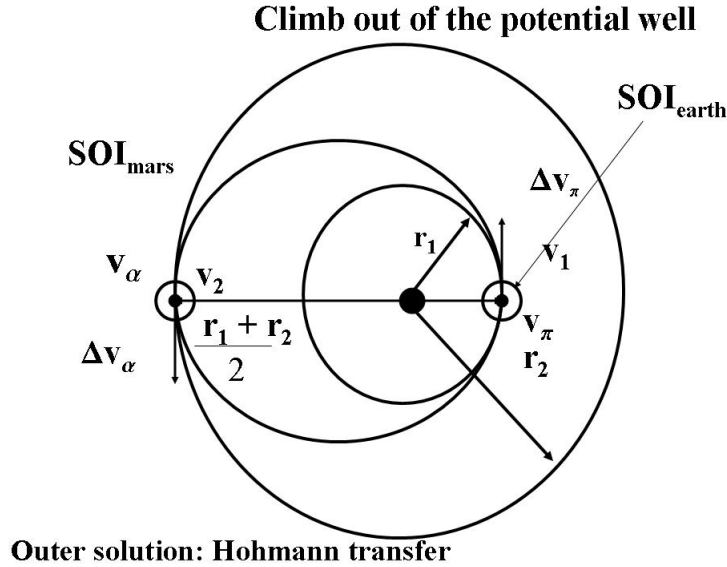
## REFERENCES

the gravitational fields of the two planets in this calculations. From Equations (1-2) we have  $v_\pi$  and  $v_\alpha$  for the elliptical connecting orbit as

$$v_\pi^2 = \mu \left( \frac{2}{r_1} - \frac{2}{r_1 + r_2} \right) \quad (10)$$

$$v_\alpha^2 = \mu \left( \frac{2}{r_2} - \frac{2}{r_1 + r_2} \right) \quad (11)$$

and orbital velocities  $v_1$  and  $v_2$  for the circular planetary orbits at  $r_1$  and  $r_2$  (Note that we have used the symbol  $\mu$  without designation. Obviously, the choice of  $\mu$  in a particular calculation depends upon the celestial body supporting the orbit.)  $v_1 = \sqrt{\mu/r_1}$  and  $v_2 = \sqrt{\mu/r_2}$  for the initial and final circular orbits about the Sun where  $\mu = \mu_{Sun}$ .



This determines the  $\Delta v_\pi$  and  $\Delta v_\alpha$  for the Hohmann interplanetary transfer. It also determines the initial and final conditions for a hyperbolic escape trajectory.

Consider conditions at the departure planet, denoted by subscript 1. The planet moves with the circular orbit velocity  $v_1$ . The hyperbolic trajectory is defined in a coordinate system **relative to the moving planet**, and the escape trajectory must end with the velocity appropriate for the Hohmann transfer ellipse in inertial space,  $v_\pi$  or  $v_\pi - v_1$  relative to the planet. The hyperbolic escape velocity  $v_\infty$  then equals exactly the  $\Delta v_\pi$  for a traditional Hohmann transfer. The only remaining unknown is the position of the spacecraft when it decides to escape.

The geometry of a hyperbolic escape comes directly from the general geometry of a hyperbolic orbit. In this case we desired to depart on a vertical trajectory ending with a velocity  $v_\infty = v_\pi - v_1$ . If the spacecraft is in a circular orbit around the earth at an orbital velocity  $v_{1C}$  and a radius  $r_1$ , this requires

$$\Delta v_{escape} = v_{1p} - v_{1C} \quad (12)$$

where  $v_{1p}$  is the velocity of the hyperbolic escape orbit at the periapsis  $r_p = r_1$ .

Since energy is constant for the hyperbolic orbit, the velocity in the circular orbit at  $r_p$ ,  $v_{1p}$ , required to reach  $v_\infty$  is given by

$$v_{1p}^2/2 = \mu/r_{1p} + v_{1\infty}^2/2 \quad (13)$$

where  $\mu = \mu_{Earth}$ . The parameters of the hyperbolic escape orbit follow from these conditions.

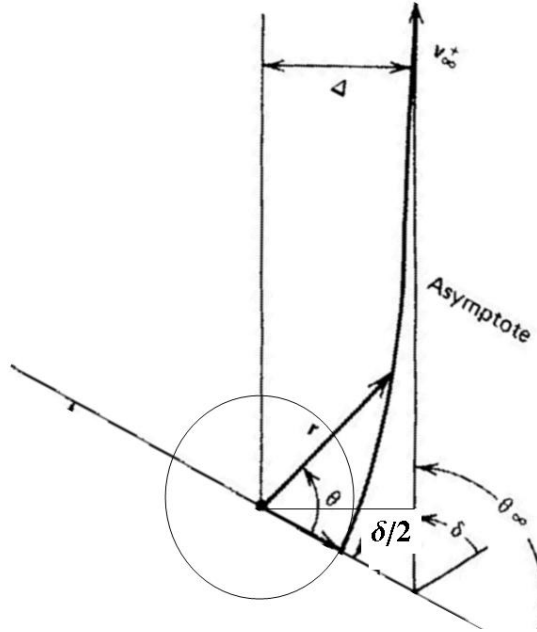
The eccentricity is given by

$$\epsilon_1 = 1 + \frac{r_1 v_\infty^2}{\mu} \quad (14)$$

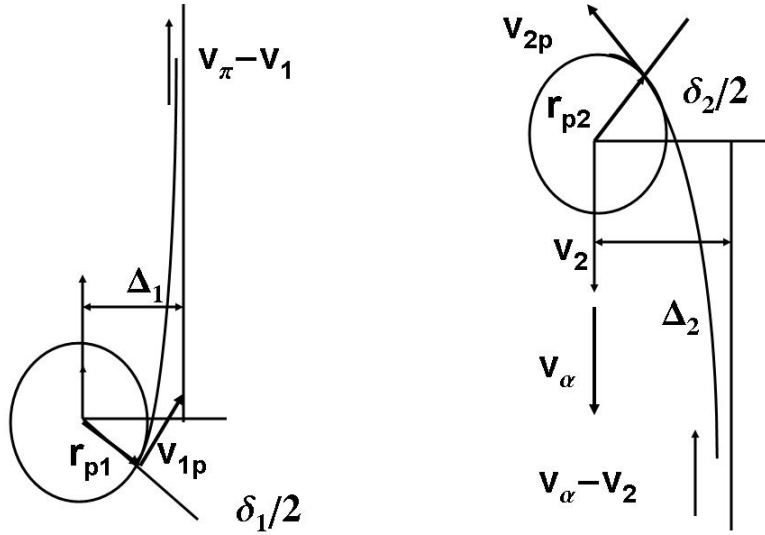
and the angular change from periapsis to infinity,  $\delta_1/2$ , is given by

$$\sin(\delta_1/2) = 1/\epsilon_1. \quad (15)$$

For the orbit geometry sketched below, this means that the launch—the point in the orbits where  $\Delta V$  is applied— from a circular orbit into a hyperbolic orbit with a final vertical direction as sketched occurs at an angle  $-\delta_1/2$  as shown in the figure. The  $\Delta V$  required is then  $v_{1p} - v_{1C}$ .



The trajectory shown is a counterclockwise escape trajectory. A clockwise escape trajectory is also possible but, because of the counterclockwise rotation of the earth, counterclockwise is preferred. One might worry about the sidewise displacement  $\Delta_1$  of the trajectory from the centerline of the planet orbit. However, this distance is very small compared to the interplanetary distances defined by the distance to the sun and is typical of the small errors inherent in the method of patched conics. Therefore, the hyperbolic escape trajectory has been defined as shown below. This enables the determination of the  $\Delta v_1$  for this portion of the mission. The hyperbolic escape trajectory is shown below.



Also shown is the hyperbolic capture trajectory at the final planetary destination. Both these trajectories are shown for the case of travel to an outer planet such as Mars.

We now consider the hyperbolic capture trajectory. From the Hohmann transfer calculation, we obtain the result that an additional  $\Delta v$  in the flight direction is required to circularize the orbit of the space craft into the destination planetary orbit, subscript 2. This means that the velocity  $v_\alpha$  of the elliptical Hohmann transfer orbit is less than the velocity of the planet in its circular orbit,  $v_2$ . Thus the planet will **overtake** the spacecraft resulting in a hyperbolic capture orbit. The velocity  $v_{2\infty}$  of the hyperbolic capture orbit as seen by the planet is  $v_\alpha - v_2$  and is directed towards the planet. We desire to capture the spacecraft into an orbit of radius  $r_{2p}$  about the planet and inquire what conditions are required to achieve this. From conservation of energy, we have  $v_{2p}^2/2 = \mu/r_{2p} + v_\infty^2/2$  where  $v_\infty = v_\alpha - v_2$ . The parameters  $r_{2p}$  and  $v_\infty$  will determine  $\Delta_2$ ,  $\epsilon_2 = 1 + r_{2p}v_{2\infty}^2/\mu$ ; and  $\sin(\delta_2/2) = 1/\epsilon_2$ , where, in this case,  $\mu = \mu_{planet}$ . Of course some mid-course correction may be required to set  $\Delta_2$  to achieve this capture. This will be ignored.

The final step in the interplanetary mission would be to insert the spacecraft into a circular orbit of radius  $r_{2p}$  about the planet. The spacecraft arrives tangent to the planetary circular orbit with a velocity  $v_{2p}$ . The required velocity in the circular planetary orbit is  $v_{2C} = \sqrt{\mu_p/r_{2p}}$ . Therefore the  $\Delta v$  required for this step is  $\Delta V = v_{2p} - \sqrt{\mu/r_{2p}}$  where  $\mu = \mu_{planet}$ .

This mission analysis is a bit arbitrary; the spacecraft could also descend to the planet surface or flyby to another destination. But if we desire to calculate the energy requirements for an earth to Mars mission, this is a reasonable formulation and allows a comparison between several options including the more complex trajectories using solutions to Lambert's problem. We can also complete a round-trip mission by remaining in orbit for a specified time and then retracing the process to return to earth. The  $\Delta v$ 's required to descend to and depart from the planet surface could also be included. These combined Hohmann, escape and capture trajectories are minimum energy trajectories for a given mission and serve as useful benchmarks.

For missions designed to visit several planets, the situation can become very complex as one often tries to

*REFERENCES*

take advantage of the gravitational fields of the planets encountered on the way, by entering into their SOI's with the objective of either changing direction or gaining additional impulse. This technique is often referred to as *gravity assist* or flyby.

**References**

- [1] F.J. Hale, *Introduction to Space Flight*, Prentice-Hall, 1994.
- [2] M.H. Kaplan, *Modern Spacecraft Dynamics and Control*, John Wiley and Sons, 1976.

## Lecture 20 — Exploring the Neighborhood: the Restricted Three-Body Problem

### The Three-Body Problem

In Lecture 15-17, we presented the solution to the two-body problem for mutual gravitational attraction. We were able to obtain this solution in closed form. For bodies of comparable mass, the solution showed that the bodies orbited about their center of mass, with orbital periods that could be obtained analytically by transforming the two-body problem into a "Kepler" problem with modified gravitational forces. For a body of negligible mass, such as a satellite orbiting the earth, we were able to place the earth at rest at the center of our coordinates, and obtain a simple expression for the orbit of the satellite.

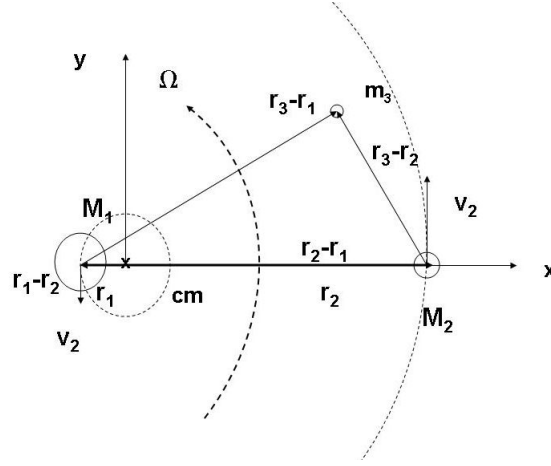
No such solution is available for the three-body problem. The general solution to the three-body problem must be obtained numerically, and even with these tools, the solution is *chaotic*: that is small changes in initial conditions lead to widely diverging behavior.

For analysis of space missions, such as a spacecraft moving between planets, or from earth to the moon, we may apply the model known as the restricted three-body problem, in which the mass of a small spacecraft is affected by the gravitational forces from two bodies, but the large bodies do not feel the influence of the spacecraft. We further restrict the motion of the two bodies to move in circular orbits, and we restrict the motion of the spacecraft to remain in the plane of the two bodies. This problem is referred to as the *circular planar restricted three-body problem*.

The circular planar restricted three-body problem is most easily analyzed/understood in a coordinate system rotating with the two primary bodies. Since the solution of the two-body problem of two bodies moving in circular orbits is that of the two primary masses rotating at constant angular velocity  $\Omega$  about their center of mass, in a coordinate system rotating with angular velocity  $\Omega$ , these two bodies are stationary. The third body moves in their gravitational fields and experiences the additional acceleration terms associated with the rotation of the coordinate system.

Consider a two-body system of mass  $M_1$  and  $M_2$  in an inertial reference frame. We add a small mass  $m_3$  which does not affect the large masses.





We consider planar x,y motion only. The equations of motion are

$$\ddot{x}_1 = -\frac{GM_2(x_1 - x_2)}{((x_1 - x_2)^2 + (y_1 - y_2)^2)^{3/2}} \quad (1)$$

$$\ddot{x}_2 = -\frac{GM_1(x_2 - x_1)}{((x_2 - x_1)^2 + (y_2 - y_1)^2)^{3/2}} \quad (2)$$

$$\ddot{x}_3 = -\frac{GM_1(x_3 - x_1)}{((x_3 - x_1)^2 + (y_3 - y_1)^2)^{3/2}} - \frac{GM_2(x_3 - x_2)}{((x_3 - x_2)^2 + (y_3 - y_2)^2)^{3/2}} \quad (3)$$

$$\ddot{y}_1 = -\frac{GM_2(y_1 - y_2)}{((x_1 - x_2)^2 + (y_1 - y_2)^2)^{3/2}} \quad (4)$$

$$\ddot{y}_2 = -\frac{GM_1(y_2 - y_1)}{((x_2 - x_1)^2 + (y_2 - y_1)^2)^{3/2}} \quad (5)$$

$$\ddot{y}_3 = -\frac{GM_1(y_3 - y_1)}{((x_3 - x_1)^2 + (y_3 - y_1)^2)^{3/2}} - \frac{GM_2(y_3 - y_2)}{((x_3 - x_2)^2 + (y_3 - y_2)^2)^{3/2}} \quad (6)$$

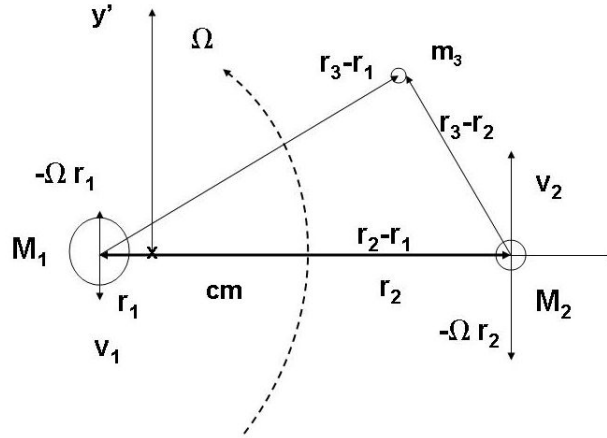
The solution to the motion of  $M_1$  and  $M_2$  is a Kepler problem, where the two bodies move in circular orbits about their center of mass with a frequency

$$\Omega = \sqrt{\frac{G(M_1 + M_2)}{R^3}}, \quad (7)$$

where  $R$  is the distance between them.  $R = -r_1 + r_2$  ( $r_1$  is negative). We can use these equations to solve numerically for the motion of the small mass  $m_3$  in the presence of the gravitational field of  $M_1$  and  $M_2$ ; in general no analytic solution is possible.

However, considerable insight can be gained about the behavior of the small mass  $m_3$  by moving into a coordinate system in which the two large masses  $M_1$  and  $M_2$  are stationary. Since the solution to the Kepler problem for these two masses is a rotation of angular velocity  $\Omega$  about the center of mass, if we place the origin of our coordinate system at the center of mass and rotate it with angular velocity  $\Omega$ , we will see that  $M_1$  and  $M_2$  are at rest; their velocities  $v_1$  and  $v_2$  which are obtained from the solution to the Kepler problem

are exactly canceled by the effect of rotation, in which the velocity at  $r_1$  and  $r_2$  are canceled since  $\Omega r_2$  and  $\Omega r_1$  are subtracted from  $v_2$  and  $v_1$  as the effect of the rotating coordinate system.



However, the acceleration in inertial space  $\ddot{x}$  and  $\ddot{y}$  must be modified to account for the rotating coordinate system. From Lecture 8, equations(11-13), the acceleration for planar motion  $x'(t)$  and  $y'(t)$  in a rotating coordinate system is given by

$$\ddot{x}(t) = \frac{d^2 x'}{dt^2} - x' \Omega^2 - 2\Omega \frac{dy'}{dt} \quad (8)$$

$$\ddot{y}(t) = \frac{d^2 y'}{dt^2} - y' \Omega^2 + 2\Omega \frac{dx'}{dt} \quad (9)$$

$$\ddot{z}(t) = \frac{d^2 z'}{dt^2} \quad (10)$$

Here we have included the acceleration in the  $z$  (out of plane) direction since it is unmodified by the rotation of the coordinate system. This will allow us to study out of plane motions if we wish.

We now modify the equations for the motion of the three masses in their mutual gravitational field by modifying the acceleration according to the formulae given in equation (8), (9), and (10). The gravitational terms are unmodified since they are unaffected by the rotation of the coordinate system. (The gravitational force have no time behavior and are unaffected by relative motion.) However, only the motion of the small mass  $m_3$  is of interest, since by the construction of the problem,  $M_1$  and  $M_2$  are stationary in our coordinate system. We locate them on the  $x$  axis at the points  $x'_1 = -r_1$ ,  $x'_2 = r_2$ ,  $y'_1 = 0$  and  $y'_2 = 0$ . We include the  $z(t)$  equation which is very straightforward;  $\ddot{z}(t)$  is unaffected by the rotation of the coordinate system; by our problem statement, the motion of the small mass does not affect the motion of the two large masses;

and the motion of the two large masses is planar, so that  $z_1$  and  $z_2$  are always zero.

$$\begin{aligned}\frac{d^2 x'_3}{dt^2} - x'_3 \Omega^2 - 2\Omega \frac{dy'_3}{dt} &= -\frac{GM_1(x'_3 - x'_1)}{((x'_3 - x'_1)^2 + (y'_3 - y'_1)^2 + z_3^2)^{3/2}} - \frac{GM_2(x'_3 - x'_2)}{((x'_3 - x'_2)^2 + (y'_3 - y'_2)^2 + z_3^2)^{3/2}} \\ \frac{d^2 y'_3}{dt^2} - y'_3 \Omega^2 + 2\Omega \frac{dx'_3}{dt} &= -\frac{GM_1(y'_3 - y'_1)}{((x'_3 - x'_1)^2 + (y'_3 - y'_1)^2 + z_3^2)^{3/2}} - \frac{GM_2(y'_3 - y'_2)}{((x'_3 - x'_2)^2 + (y'_3 - y'_2)^2 + z_3^2)^{3/2}} \\ \frac{d^2 z'_3}{dt^2} &= -\frac{GM_1(z'_3)}{((x'_3 - x'_1)^2 + (y'_3 - y'_1)^2 + (z'_3)^2)^{3/2}} - \frac{GM_2(z'_3)}{((x'_3 - x'_2)^2 + (y'_3 - y'_2)^2 + (z'_3)^2)^{3/2}}\end{aligned}\quad (11)$$

We now drop the  $'$ s and the subscript 3 for convenience and concentrate on the motion of  $m_3$  in the  $x, y$  plane. We and rewrite the equations **moving the centripetal acceleration to the right side**. The centripetal acceleration  $-x\Omega^2\vec{i} - y\Omega^2\vec{j}$  can be written as the gradient of a scalar:  $-x\Omega^2\vec{i} - y\Omega^2\vec{j} = -\nabla(\frac{1}{2}\Omega^2(x^2 + y^2)) = -\nabla(\frac{1}{2}\Omega^2 r^2)$ . If we do this, we can see that the entire right side of the equation can be written as the gradient of a scalar function of  $r$ , and thus could be thought of as a potential, related to a conservative force field as  $F_c = -\nabla V(r)$ . It contains both the gravitational potential and the potential formed from the centrifugal force. Because we have left the Coriolis acceleration on the left side of the equation, we do not have a simple equation for the behavior of a particle in a conservative force field in inertial space. However, if we are interested in the spatial behavior of the force field seen by a particle **at rest**, this formulation gives great insights. We write the equation for the motion of  $m_3$  in the  $x, y$  plane as

$$\ddot{\vec{r}} + 2(\vec{\Omega} \times \frac{d\vec{r}}{dt}) = -\nabla(V_g(r) - \frac{1}{2}\Omega^2 r^2) \quad (12)$$

where  $V_g(r)$  is the gravitational potential for the two bodies:  $M_1$  and  $M_2$  located at  $x = -r_1, y = 0$  and  $x = r_2, y = 0$ , as shown in the figures.

From Lecture 13, we have that the gravitational potential/per unit mass of  $m_3$  as

$$V_g(x, y) = -\frac{GM_1}{\sqrt{(x + r_1)^2 + y^2}} - \frac{GM_2}{\sqrt{(x - r_2)^2 + y^2}} \quad (13)$$

so that the total "potential" is

$$V(x, y) = -\frac{GM_1}{\sqrt{(x + r_1)^2 + y^2}} - \frac{GM_2}{\sqrt{(x - r_2)^2 + y^2}} - \frac{1}{2}\Omega^2(x^2 + y^2) \quad (14)$$

where the effect of centrifugal force has been written as a potential,  $V_\Omega = -\frac{1}{2}\Omega^2 r^2 = -\frac{1}{2}\Omega^2(x^2 + y^2)$ . We shall refer to the combination of gravitational and centrifugal potential as a pseudo-potential, since when the velocity is not zero, the additional Coriolis term must be included to determine the dynamics of the particle; in this case we do not have the simple picture of a particle exchanging energy between potential and kinetic. We will use the potential in a rotating coordinate system to investigate the neighborhood, and specifically to ask whether there are any points where a particle might be placed at rest and experience no force.

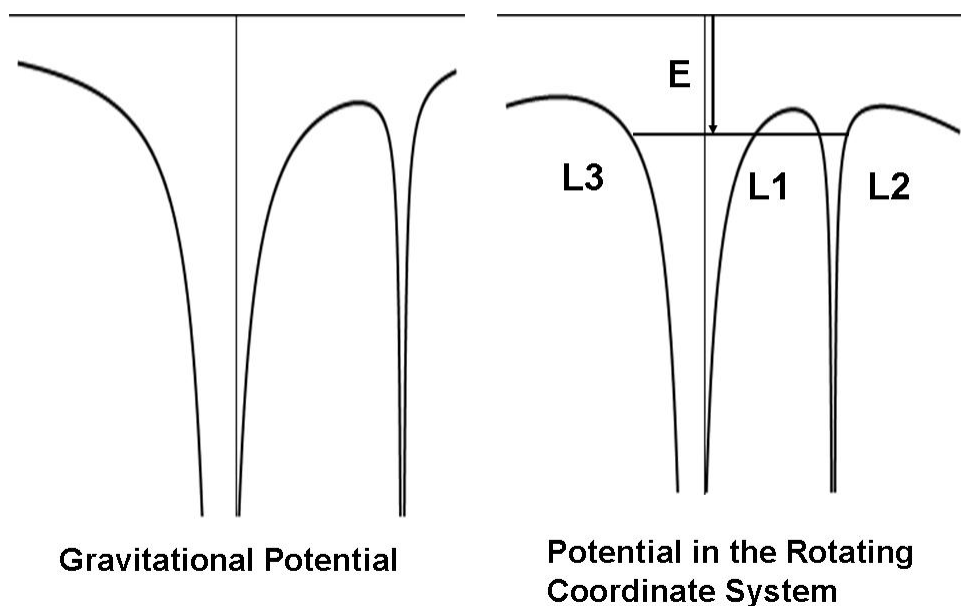
The governing equations can now be written in terms of the gradient of the total "potential"  $V(x, y, z)$  as

$$\frac{d^2x_3}{dt^2} - x_3\Omega^2 - 2\Omega\frac{dy_3}{dt} = -\frac{\partial V(x,y,z)}{\partial x} \quad (15)$$

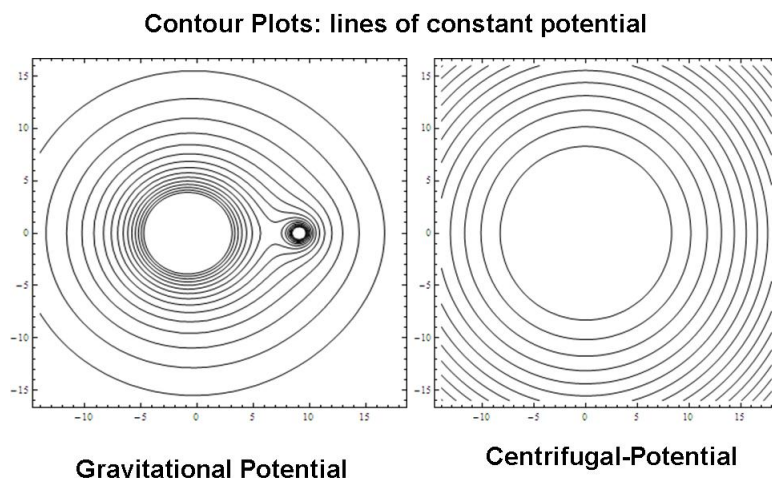
$$\frac{d^2y_3}{dt^2} - y_3\Omega^2 + 2\Omega\frac{dx_3}{dt} = -\frac{\partial V(x,y,z)}{\partial y} \quad (16)$$

$$\frac{d^2z_3}{dt^2} = -\frac{\partial V(x,y,z)}{\partial z} \quad (17)$$

We now examine the potentials numerically. For convenience, we set  $G = 1$ ,  $M_1 = 10$ ,  $M_2 = 1$ , and  $R = 10$ . This gives  $r_1 = 10/11$  and  $r_2 = 100/11$  and  $\Omega = \sqrt{11/1000}$ . We first examine the gravitational potential given by equation (13) along the  $x$  axis in comparison with the pseudo-potential. The results are shown in the figure. We recall, that at any point,  $\vec{F} = -\nabla V$ . Although we are only looking at the behavior along  $y = 0$ , we see an interesting difference between the behavior of the gravitational potential and the pseudo-potential.



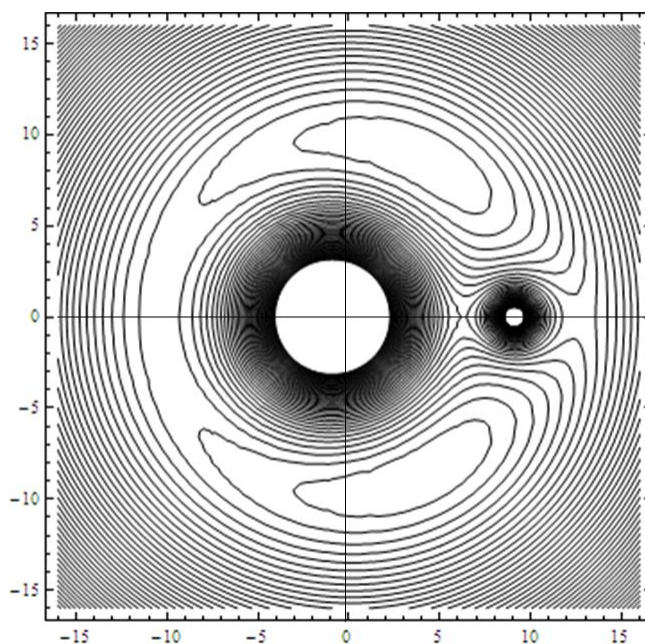
For the gravity potential, we do see a point where the gravitational pull of the two bodies is equal and opposite (Think of the earth-moon system.). However, for the pseudo-potential, we see **three points** where the gravitational plus centrifugal forces balance. To explore this further, we need to move into two dimensions in the plane. The result for the gravitational potential alone is not meaningful, since we are in a rotating coordinate system. The centrifugal force is a "real" force and must be included in any discussion of force; however the gravitational potential does give some physical insight into what's going on. We now move into two dimensions and examine the contour plot of the gravitational and the centrifugal potential; lines of constant potential in the  $x, y$  plane.



The figure shows lines of constant potential for the gravitational potential of the two bodies,  $M_1$  and  $M_2$ . We also show the potential for the centrifugal term which is a simple  $-\frac{1}{2}\Omega^2 r^2$  term; lines of constant potential are circles.

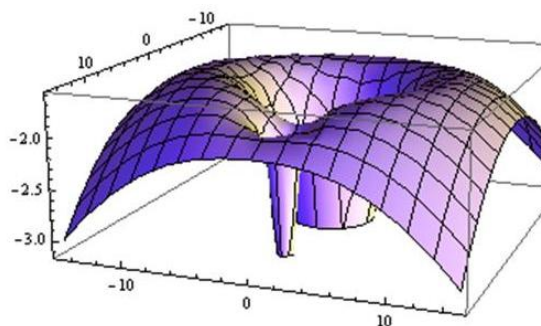
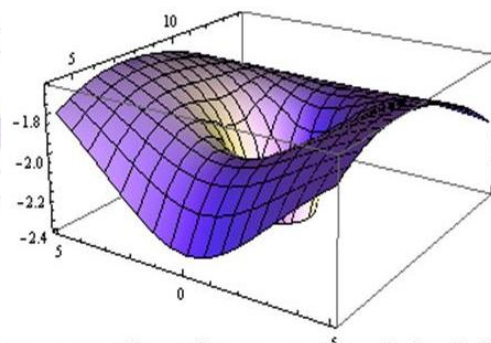
Now we combine these expressions to form the total potential for the two-body problem *in a rotating coordinate system* in which the two bodies are at rest. This will allow us to answer the question: for a small body  $m_3$  wandering about this neighborhood, are there any points where no force will be exerted on  $m_3$  and therefore points at which  $m_3$  might be able to remain at rest under the action of a balance of forces. There are five such points called the Lagrange points. Three of the Lagrange points are on the x axis at the points where the slope of the potential is zero; these are referred to (not always consistently) as  $L_1$ ,  $L_2$ , and  $L_3$ . The other two,  $L_4$  and  $L_5$ , are located off the x axis. As we shall see, the Lagrange points play an important role in the theory of planetary motion, and the design of space missions.

**Contour Plot  
of the Potential  
for the Two-Body  
problem in a  
Rotating  
Coordinate  
System**

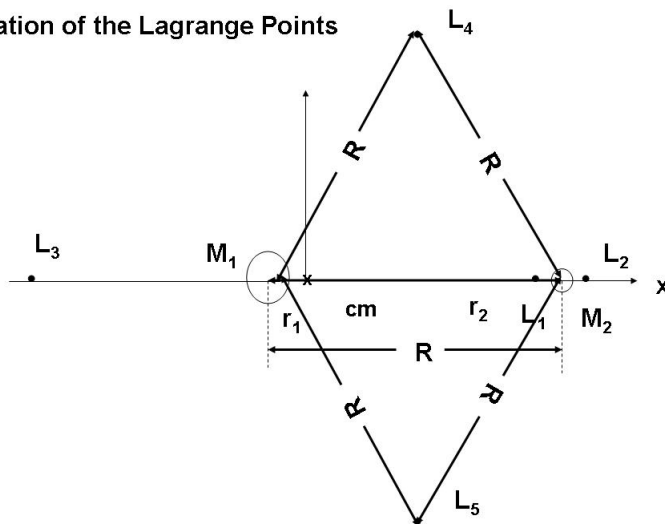


We see several regions of this figure which bear further examination. Recall from our examination of the behavior of the potential in the rotating coordinate system along the  $x$  axis, we identify three possible points where no force in the  $x$  direction would be exerted on a particle. In fact, due to the symmetry of the potential in the  $y$  direction, no  $y$  force will be felt on a particle located on the  $x$  axis. Therefore these three points are equilibrium points, in the sense that no force acts upon a particle at these points. Recall that following any discussion of equilibrium, is a discussion of stability: a ball balanced upon an overturned bowl is in equilibrium but not stable; a ball at the bottom and inside a bowl is at equilibrium and stable. These three points are visible in the contour plot, although the  $L_3$  point outboard of  $M_1$  is indistinct; it is there.

Also shown on the graph of the potential along the  $x$  axis in the rotating coordinate system is an indication of the energy of a particle. The value of  $E$  is negative and since the kinetic energy can never become negative, the value of the potential acts as a barrier to the motion of the particle. For a given energy, the contour shown in the two-dimensional contour plot acts as a barrier to the motion of the particle. From an examination of both the contour plot and the surface plot of the potential, we see the possibility of a "corridor of escape" near the lagrange points  $L_1$  and  $L_2$ . That is, a particle having a critical value of energy can travel through this corridor rather than going over the "hill" surrounding the central body. The surface plot presented shows the entire region as well as detail near the region of interest near the Lagrange point  $L_1$  and  $L_2$ . This "corridor of escape" makes the region about the Lagrange points of great interest for space missions.

**Energy Surface****Energy Surface near L1, L2**

Before completing our discussion, we turn our attention to a two-dimensional view of the closed contours off of the  $x$  axis. If this contour contains a "bowl" or a potential well, we have a possible point of stable equilibrium. For a closed contour of a continuous function, we are guaranteed a point of zero slope somewhere within the contour, a possible point of equilibrium; however, our dynamics must include the Coriolis forces which could affect the stability. We now analyze the location of these points in more detail and greater generality.

**Location of the Lagrange Points**

We now return to the two-body problem and do not choose particular numerical values for  $M_1$ ,  $M_2$  and  $R$ . We do take  $G = 1$  as before (you can think of this a particular non-dimensionalization of the equations; in any case from the form of  $V_g$  and the form of  $\Omega$ ,  $G$  drops out. We introduce

$$\vec{r} = x(t)\vec{i} + y(t)\vec{j} \quad (18)$$

$$r_1 = -\alpha R \quad (19)$$

$$r_2 = \beta R \quad (20)$$

where  $\alpha = \frac{M_2}{M_1+M_2}$ ;  $\beta = \frac{M_1}{M_1+M_2}$  and  $\Omega = \sqrt{\frac{G(M_1+M_2)}{R^3}}$  as required by the solution of the two-body problem.

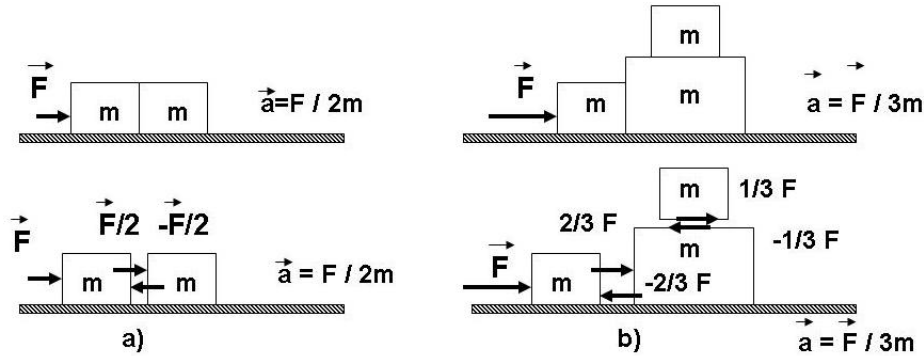
With this formulation, we first seek solutions for the points  $x$  along the  $y$  axis where the force on a particle would be zero. This is a complex problem, involving finding the roots of three fifth-order polynomials. We could do such a solution numerically, but for greater insight we seek an analytic solution. Such a solution can easily be found for small  $\alpha$ , where  $\alpha$  is the mass ratio. In fact the existence of such points requires that  $\alpha$  be small, so this is not a significant restriction on the utility of the result. Expanding the total  $x$  and  $y$  forces on the particle for small  $\alpha$  and setting the result equal to zero gives the following approximations for the location of these three equilibrium points along the  $x$  axis. They are called Lagrange points  $L_1$ ,  $L_2$  and  $L_3$ .

$$L_1 = R \left( 1 - \left( \frac{\alpha}{3} \right)^{1/3} \right) \quad (21)$$

$$L_2 = R \left( 1 + \left( \frac{\alpha}{3} \right)^{1/3} \right) \quad (22)$$

$$L_3 = -R \left( 1 + \left( \frac{5\alpha}{12} \right) \right) \quad (23)$$

To identify the remaining Lagrange point requires that we find points in  $x, y$  space for  $y \neq 0$  where both components of the force vector are zero. We choose to examine the components of force along a radial line from the origin.



### Internal Force Balance Constraints

With this choice, referring to the figure, we note that since the centrifugal force is aligned with vector  $\vec{r}$ , a force balance—if indeed there is one—would involve only gravitational terms. We express the force vector in components along a vectors parallel and perpendicular to  $\vec{r}$ :  $\mathbf{e}_{par} = x\mathbf{i} + y\mathbf{j}$  along the  $\mathbf{r}$  direction and  $\mathbf{e}_{per} = y\mathbf{i} - x\mathbf{j}$  perpendicular to this vector, to obtain  $F_{per}$  and  $F_{par}$ . Since  $\mathbf{e}_{par}$  and  $\mathbf{e}_{per}$  are not unit vectors, the resulting force will be proportional to the actual forces in these directions, but since we want to identify conditions where they are zero, this does not matter.

The components of total force in the  $\mathbf{i}$  and  $\mathbf{j}$  directions are

$$F_x = \Omega^2 \left( x - \frac{\beta(x + \alpha R)R^3}{((x + \alpha R)^2 + y^2)^{3/2}} - \frac{\alpha(x - \beta R)R^3}{((x - \beta R)^2 + y^2)^{3/2}} \right) \quad (24)$$

$$F_y = \Omega^2 \left( y - \frac{\beta y R^3}{((x + \alpha R)^2 + y^2)^{3/2}} - \frac{\alpha y R^3}{((x - \beta R)^2 + y^2)^{3/2}} \right) \quad (25)$$



We first examine the perpendicular component of force and solve for the point along  $r$  where it equals zero.

$$F_{per} \propto (F_x \mathbf{i} + F_y \mathbf{j}) \cdot (y \mathbf{i} - x \mathbf{j}) \quad (26)$$

resulting in

$$F_{per} \propto \alpha \beta \Omega^2 R^3 \left( \frac{1}{((x - R\beta)^2 + y^2)^{3/2}} - \frac{1}{((x + R\alpha)^2 + y^2)^{3/2}} \right) = 0 \quad (27)$$

For this force to equal zero, the two terms must be equal and opposite. Therefore, the magnitude of the two terms containing  $x$  must be equal;  $|x - \beta R| = |x + \alpha R|$ . This results in a solution  $x = \frac{1}{2}(\beta R - \alpha R)$ : that is the  $x$  location of the Lagrange point—if there is one—, lies exactly half way between the two masses. With this result, the equation for the forces parallel to  $\vec{r}$  reduce to

$$F_{par} \propto (F_x \mathbf{i} + F_y \mathbf{j}) \cdot (x \mathbf{i} + y \mathbf{j}) \quad (28)$$

$$F_{par} \propto \Omega^2 \frac{(x^2 + y^2)}{R} \left( \frac{1}{R^3} - \frac{1}{((x - R\beta)^2 + y^2)^{3/2}} \right) \quad (29)$$

Again, for the force to be zero, the two terms must be equal and opposite. Therefore, this result equates the distance between the point in question and  $M_2$  to the magnitude of  $R$ ; we conclude that the Lagrange point is at a distance  $R$  from  $M_2$  and by symmetry from  $M_1$ :  $L_4$  is at positive  $y$ ;  $L_5$  its mirror image. Since the masses are  $R$  apart on the  $x$  axis, the location of  $L_4$  is at the peak of an equilateral triangle:  $R$  on all three sides. We add these points to the figure showing location of the Lagrange points. This is a remarkable result. The location of the  $L_4$  and  $L_5$ , which have both  $x$  and  $y$  positions are

$$L_4 = \left( \frac{R(M_1 - M_2)}{2(M_1 + M_2)}, \frac{\sqrt{3}}{2}R \right) \quad (30)$$

$$L_5 = \left( \frac{R(M_1 - M_2)}{2(M_1 + M_2)}, -\frac{\sqrt{3}}{2}R \right) \quad (31)$$

## Stability and Application of the Lagrange Points

So much can be said about these results. In examining the stability of the mass placed at these equilibrium points, done as a note later in the section, we find that  $L_4$  and  $L_5$  are stable—due in part to the Coriolis force which acts on a particle in motion; and  $L_1$ ,  $L_2$ , and  $L_3$  are unstable. However, even though small perturbations to a mass located at these Lagrange points would produce an unstable motion, the amplification rate is small and it is far easier to locate a spacecraft near these points and apply an automatic control system for "station keeping" than to keep a spacecraft at rest or in an orbit near an arbitrary point in space. So, even though these points are points of unstable equilibrium, it requires only a small expenditure of fuel to keep spacecraft at or near these points. Current space missions place objects in orbits about these Lagrange points for various purposes, achieving very practical results.

Many space missions have used these points. For example the current plan is to locate the James Webb optical telescope in an orbit about the  $L_{earth-sun_2}$  point where it will be shielded from direct sunlight by the earth, and can be kept on station in a efficient manner.

In the natural world, groups of asteroids –called the Trojan asteroids–have been observed to cluster at the Sun-Jupiter  $L_4$  and  $L_5$  Lagrange points. Trojan asteroids have also been found at the  $L_4$  and  $L_5$  Lagrange point of Mars and Neptune.

**Note****Stability Analysis of the Lagrange Points**

We study the stability of the various Lagrange points by expanding the equations of motion about each equilibrium position,  $L_1$  to  $L_5$ . Setting

$$x = X_{L_i} + x_p \quad (32)$$

$$y = Y_{L_i} + y_p \quad (33)$$

$$z = z_p \quad (34)$$

where  $X_{L_i}$  and  $Y_{L_i}$  are the  $x$  and  $y$  locations of the  $i$ 'th Lagrange (equilibrium) points and  $x_p$ ,  $y_p$ , and  $z_p$  are small perturbations about these locations. We can, for a very small price, include the  $z$  direction, since the Lagrange points are located in the  $x, y$  plane, for which  $Z_{L_i} = 0$ . In studying the stability, we are essentially deriving the equations for small "oscillations" about the equilibrium point  $X_{L_i}, Y_{L_i}$ ; we intend to linearize these equations. Oscillatory or decaying solutions imply stability; exponentially diverging solutions imply instability. To obtain the equations governing small motions from equilibrium, we expand equations the set of equations (15-17) for small displacements  $x_p, y_p, z_p$  about the various Lagrange points. Since the gradient of the potential is zero at these equilibrium points, the resulting equations contain only second derivatives of the potential; in component form they become

$$\frac{d^2 x_p}{dt^2} - 2\Omega \frac{dy_p}{dt} = -V_{xx}x_p - V_{xy}y_p - V_{xz}z_p \quad (35)$$

$$\frac{d^2 y_p}{dt^2} + 2\Omega \frac{dx_p}{dt} = -V_{yx}x_p - V_{yy}y_p - V_{yz}z_p \quad (36)$$

$$\frac{d^2 z_p}{dt^2} = -V_{xz}x_p - V_{zy}y_p - V_{zz}z_p \quad (37)$$

The various second derivatives of the potential  $-V(x, y, z)$  are given in the table for the various Lagrange points. Many of them are zero. The second derivatives are symmetric such that  $V_{xy} = V_{yx}$ . This derivation allows the construction of the differential equations governing small displacement motion about the Lagrange points. From this we find that  $L_1, L_2$  and  $L_3$  are unstable, and  $L_4$  and  $L_5$  are stable.

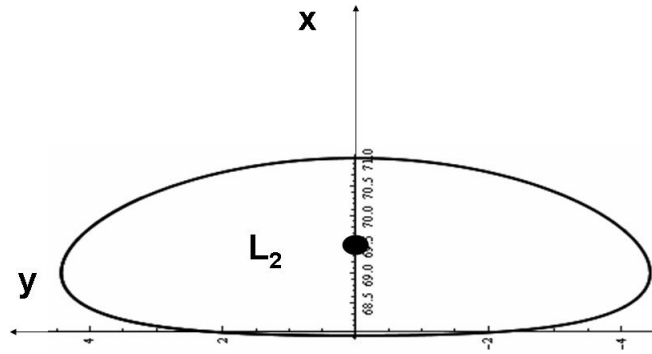
Table 1: Stability Coefficients for the Lagrange Points Expanded for  $\alpha \ll 1$

| point | $-V_{xx}$              | $-V_{yy}$                    | $-V_{zz}$    | $-V_{xy}$                            | $-V_{xz}$ | $-V_{yz}$ |
|-------|------------------------|------------------------------|--------------|--------------------------------------|-----------|-----------|
| L1    | $9\Omega^2$            | $-3\Omega^2$                 | $-4\Omega^2$ | 0                                    | 0         |           |
| L2    | $9\Omega^2$            | $-3\Omega^2$                 | $-4\Omega^2$ | 0                                    | 0         |           |
| L3    | $3\Omega^2$            | $-\frac{7}{8}\gamma\Omega^2$ | $-\Omega^2$  | 0                                    | 0         |           |
| L4    | $-\frac{3}{4}\Omega^2$ | $-\frac{9}{4}\Omega^2$       | $-\Omega^2$  | $-\frac{3\sqrt{3}}{4}\kappa\Omega^2$ | 0         |           |
| L5    | $-\frac{3}{4}\Omega^2$ | $-\frac{9}{4}\Omega^2$       | $-\Omega^2$  | $-\frac{3\sqrt{3}}{4}\kappa\Omega^2$ | 0         |           |

where  $\kappa = (M_1 - M_2)/(M_1 + M_2)$  and  $\gamma = M_2/M_1$ . These equations provide the foundation for the analysis

of station keeping of a spacecraft at a Lagrange point. Thus, as previously mentioned, even though the motions of a spacecraft located at one of the various co-linear Lagrange point would be unstable, the rate of amplification is small and the Lagrange points are very attractive locations for various space missions.

One of the solutions to the governing equations for both the linear and non-linear regime, if initial conditions are carefully set, are the so-called Lissajous orbits in the  $x, y$  plane and their three-dimensional extension into  $x, y, z$  called halo orbits. These orbits are natural solutions to the three-body problem. These are of practical importance for missions which place a spacecraft in such an orbit.



**Lissajous orbit about L2**

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**Note**      ***Solutions to the linearized disturbance equations at the various Lagrange points***

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In the linear equations for particle motion about the Lagrange point, equation (32-34), the  $x$  and  $y$  motions are coupled; the  $z$  motion is uncoupled. Therefore, we will consider only planar  $x, y$  motion; the motion  $z(t)$  can be included later. These equations have solutions of the form

$$x[t] = \sum_{i=1}^4 A_i \exp^{\lambda_i t} \quad (38)$$

$$y[t] = \sum_{i=1}^4 B_i \exp^{\lambda_i t} \quad (39)$$

where the  $\lambda_i$  are eigenvalues obtained from the solution to the linear equations. If the eigenvalues are real, the motion is unstable; if the eigenvalues are imaginary, constant amplitude oscillation occurs: the motion is stable. The  $x, y$  coupling produces four eigensolutions. The amplitude of each eigensolution is determined by initial conditions on velocity and position. However, if one of the solutions is unstable, a small error in the initial conditions will lead to a growing departure from the orbit.

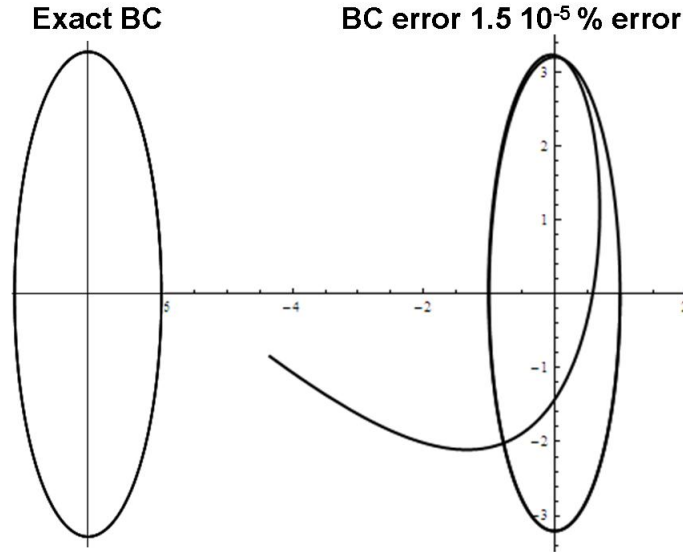
### **Stability of L1 and L2**

Using the expressions in Table 1 for the various second derivatives of  $V(x, y)$  at L1 or L2, we obtain the equations for  $x, y, z$  at L1 or L2. (The linear equations are the same at L1 and L2.)

$$\frac{d^2x}{dt^2} - 2\Omega \frac{dy}{dt} - 9\Omega^2 x = 0 \quad (40)$$

$$\frac{d^2y}{dt^2} + 2\Omega \frac{dx}{dt} + 3\Omega^2 y = 0 \quad (41)$$

Assuming a solution of the form  $x = A \exp^\lambda t$ ,  $y = B \exp^\lambda t$ , we obtain the four eigenvalues as  $\lambda_1 = \pm\Omega\sqrt{1+2\sqrt{7}}$  and  $\lambda_2 = \pm\Omega i\sqrt{2\sqrt{7}-1}$ . (We will study more about identifying eigenvalue solutions to systems of linear differential equations in Lecture 19 when we study multi-degree of freedom vibration.) This results in both an oscillatory solution and an exponential solution, one unstable and one decaying. The amplitudes of these various solutions within the total solution are determined by the initial conditions  $x(0), \dot{x}(0), y(0), \dot{y}(0)$ . If the initial conditions are carefully chosen to **precisely** eliminate the unstable solution, an orbit of constant amplitude results. However, since one of the solutions is unstable, a small error in the boundary conditions will lead to a growing departure from the orbit. This is shown below for a velocity error of  $1.5 \times 10^{-5}\%$ .

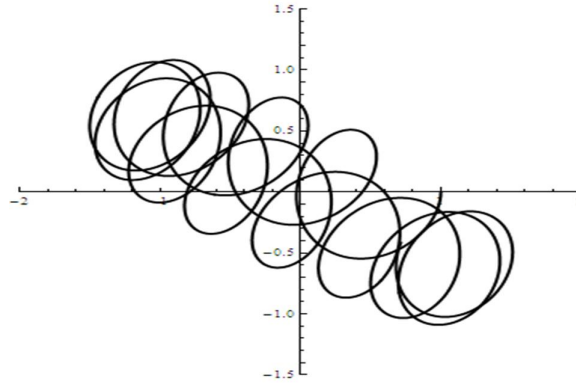


**Stability of L4 and L5** Careful analysis of the contours of potential about L4 and L5 reveals a hill not a bowl. Thus we might expect that orbits about L4 and L5 would be unstable, like a particle rolling off of an overturned bowl. However, the equations of motion contain an extra term beyond relating  $\ddot{x}$  to  $-\nabla V$ ; that term is the coriolis acceleration. Applying the relations for the various  $V'_{s_{xx}}$  from Table 1 to the general governing equations (32-34) results in the specific set of linear differential equations to determine the eigenvalues and modes about L4 and L5; since the equations are the same, we will examine only L4.

$$\frac{d^2x}{dt^2} - 2\Omega \frac{dy}{dt} + 3/4\Omega^2x + 3/4\sqrt{3}\kappa\Omega^2y = 0 \quad (42)$$

$$\frac{d^2y}{dt^2} + 2\Omega \frac{dx}{dt} + 3/4\sqrt{3}\kappa\Omega^2x + 9/4\Omega^2y = 0 \quad (43)$$

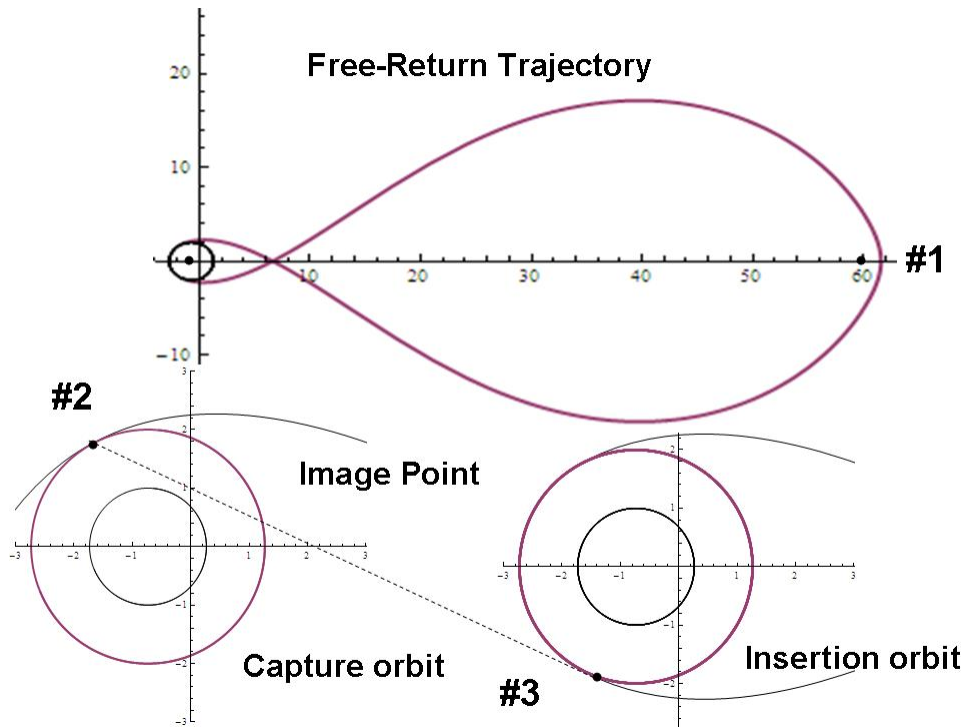
We again assume a solution of the form  $x = A \exp^\lambda t$ ,  $y = B \exp^\lambda t$ , and we obtain the four eigenvalues as  $\lambda_1 = \pm i \Omega/2\sqrt{2 - \sqrt{27\kappa^2 - 23}}$  and  $\lambda_2 = \pm i \Omega/2\sqrt{2 + \sqrt{27\kappa^2 - 23}}$ , where  $\kappa = (M_1 - M_2)/(M_1 + M_2)$ . The L4 point will be stable if the eigenvalues  $\lambda$  are purely imaginary. This will be true if  $\kappa^2 > 23/27$  which requires that  $M_1 > 25M_2 \frac{1+\sqrt{1-4/625}}{2}$ . The orbital motion about L4 is shown below; it contains two frequencies,  $\lambda_1$  and  $\lambda_2$  as expected.




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**Note**
**Free-Return Trajectories**

The early missions to the moon relied on a so-called free-return trajectory. This trajectory was designed such that an automatic return to earth was guaranteed in the event of a system failure. Such a trajectory is shown below. It fits nicely into our discussion of the restricted circular planar three-body problem formulated in a rotating coordinate system. The free return trajectory can be constructed by choosing the proper boundary conditions, initial position and initial velocity originating from an earth-centered parking orbit at point #3. However, the choice of these boundary conditions is perhaps not so obvious and an arbitrary choice of launch conditions will likely lead to a chaotic trajectory.

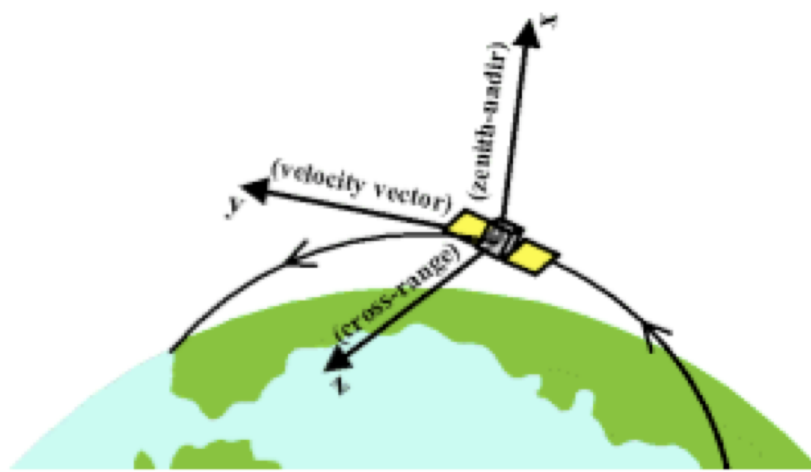


An easier way to determine these launch parameters is to go to the moon and figure out what orbit would return us to the neighborhood of earth, at which point a  $\Delta V$  could be used to place us in a parking orbit. We begin by choosing point #1 a distance from the moon, and an initial velocity  $V$  normal to the line between earth and moon. We then move along this orbit without further expenditure of fuel until we reach point #2. The characteristics of point #2 –found by trial and error–are that it is close enough to the earth to be captured in a parking orbit, as shown. We then apply the image theorem, flipping to point #3 at the same value of  $x$  but an equal and opposite value of  $y$ , at which point we apply a  $\Delta V$  of equal and opposite  $\Delta V_X$  and equal  $\Delta V_Y$ , as shown. With these initial conditions applied to launch us from the parking orbit, we will trace the free-return trajectory as shown.

## Lecture 21 — Clohessy-Wiltshire Dynamics

### Introduction

Clohessy-Wiltshire dynamics are a set of equations (often called Hill's Equations) describing the orbital motion of a body in close proximity to the origin of the local-vertical, local horizontal (LVLH) frame. To describe the LVLH frame, consider a satellite in a circular orbit around the Earth (see Figure). It has a radial direction from the center of the Earth that points in the zenith direction: upwards rather than downwards in the nadir direction. It also has a velocity direction, orthogonal to the zenith direction, associated with the orbit. Crossing the zenith direction with the orbital direction defines the third, orthogonal axis in this right handed frame which points in the direction of the orbit's angular momentum or cross-track direction. Notice that the orbital and cross-track directions define a plane which, from the perspective of the satellite, is the horizontal plane in the locality of the satellite: local horizontal (LH). The zenith direction defines the vertical in the locality of the satellite: local vertical (LV). Now, replace the satellite with the origin of the LVLH frame and define the x-axis as the zenith direction, the y-axis as the orbital direction, and the z-axis as the cross-track direction. The LVLH frame rotates  $2\pi$  radians about the positive z-axis once per orbit at an angular rate of  $(\vec{\Omega}_{h/i})_H = n\hat{k} = \sqrt{\mu/R_E^3}$ . Describing a satellite's motion relative to this LVLH frame is useful for analyzing the relative motion of two or more satellites in close proximity for the purpose of rendezvous and docking as well as inspection and sparse aperture synthesis using two or more formation flying satellites. Often, it is useful but not necessary to place the LVLH origin at the center of one of the satellites as long as it is understood that the origin of the LVLH frame follows a circular orbital trajectory.



## Equations of Motion Relative to an Orbiting Frame

We start by defining the inertial and relative vectors and substituting them into Kepler's Law

$$\ddot{\vec{r}}(t) = -\frac{\mu}{r^3(t)} \vec{r}(t) + \vec{f}(t)$$

where the force  $\vec{f}(t)$  is due to non-spherical gravity perturbations, thrusters, etc. In the inertial frame

$$(\vec{r}_s)_I = (\vec{r}_h)_I + (\vec{r}_{s/h})_I$$

where  $(\vec{r}_s)_I$  is the vector from the origin (center of the Earth) of the Earth-Centered Inertial (ECI) frame to the satellite,  $(\vec{r}_h)_I$  is the vector from the origin of the ECI frame to the origin of the LVLH (Hill's) frame, and  $(\vec{r}_{s/h})_I$  is the vector from the origin of the LVLH frame to the satellite. "Close proximity" is important because linearizing Kepler's Law is an approximation whose accuracy improves as  $(\vec{r}_{s/h})_I$  becomes shorter. Substituting into Kepler's Law, and dropping explicit time dependencies for simplicity, gives

$$(\ddot{\vec{r}}_s)_I = (\ddot{\vec{r}}_h)_I + (\ddot{\vec{r}}_{s/h})_I = -\frac{\mu}{\|(\vec{r}_h)_I + (\vec{r}_{s/h})_I\|^3} [(\vec{r}_h)_I + (\vec{r}_{s/h})_I] + \vec{f} \quad (1)$$

To linearize this equation, we need to perform a power expansion of

$$\begin{aligned} \|(\vec{r}_h)_I + (\vec{r}_{s/h})_I\|^{-3} &= [((\vec{r}_h)_I + (\vec{r}_{s/h})_I) \cdot ((\vec{r}_h)_I + (\vec{r}_{s/h})_I)]^{-3/2} \\ \|(\vec{r}_h)_I + (\vec{r}_{s/h})_I\|^{-3} &= \frac{1}{(r_h^3)_I} \left[ 1 + 2 \frac{(\vec{r}_h)_I \cdot (\vec{r}_{s/h})_I}{(r_h^2)_I} + \frac{(\vec{r}_{s/h})_I \cdot (\vec{r}_{s/h})_I}{(r_h^2)_I} \right]^{-3/2} \end{aligned}$$

Newton's generalized binomial theorem states that



$$(1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k \text{ for } (-1 < x < 1) \text{ where } \binom{\alpha}{k} = \frac{\alpha(\alpha-1)\dots(\alpha-k+1)}{k!}$$

Since the right two terms in the brackets are between  $\pm 1$ , because the first is of order  $O(1/r_h)$  and the second is of order  $O(1/r_h^2)$ , this theorem applies. Therefore, keeping first order terms gives

$$(r_s^{-3})_I = \|(\vec{r}_h)_I + (\vec{r}_{s/h})_I\|^{-3} = \frac{1}{(r_h^3)_I} \left[ 1 - \frac{3}{2} \left( 2 \frac{(\vec{r}_h)_I \cdot (\vec{r}_{s/h})_I}{(r_h^2)_I} + \frac{(\vec{r}_{s/h})_I \cdot (\vec{r}_{s/h})_I}{(r_h^2)_I} \right) + \dots \right]$$

$$(r_s^{-3})_I \approx (r_h^{-3})_I \left[ 1 - 3 \frac{(\vec{r}_h)_I \cdot (\vec{r}_{s/h})_I}{(r_h^2)_I} \right] \quad (2)$$

Now express the ECI accelerations in the LVLH (Hill's) frame using Coriolis' Theorem.

$$(\ddot{\vec{r}}_{s/h})_I = (\ddot{\vec{r}}_{s/h})_H + (\ddot{\vec{r}}_h)_I + 2 \left[ (\vec{\Omega}_{h/i})_H \times (\dot{\vec{r}}_{s/h})_H \right] + (\vec{\Omega}_{h/i})_H \times \left[ (\vec{\Omega}_{h/i})_H \times (\vec{r}_{s/h})_H \right] + \left[ (\dot{\vec{\Omega}}_{h/i})_H \times (\vec{r}_{s/h})_H \right]$$

In a circular, Keplerian orbit,  $(\ddot{\vec{r}}_{h/i})_I = 0$ ,  $(\dot{\vec{\Omega}}_{h/i})_H = 0$  and

$$(\ddot{\vec{r}}_{s/h})_I = (\ddot{\vec{r}}_{s/h})_H + 2 \left[ (\vec{\Omega}_{h/i})_H \times (\dot{\vec{r}}_{s/h})_H \right] + (\vec{\Omega}_{h/i})_H \times \left[ (\vec{\Omega}_{h/i})_H \times (\vec{r}_{s/h})_H \right] \quad (3)$$

Substituting Equations (2) and (3) into (1) gives

$$\begin{aligned} & (\ddot{\vec{r}}_h)_I + (\ddot{\vec{r}}_{s/h})_H + 2 \left[ (\vec{\Omega}_{h/i})_H \times (\dot{\vec{r}}_{s/h})_H \right] + (\vec{\Omega}_{h/i})_H \times \left[ (\vec{\Omega}_{h/i})_H \times (\vec{r}_{s/h})_H \right] \\ &= -\frac{\mu}{(r_h^3)_I} \left[ 1 - 3 \frac{(\vec{r}_h)_I \cdot (\vec{r}_{s/h})_I}{(r_h^2)_I} \right] [(\vec{r}_h)_I \cdot (\vec{r}_{s/h})_I] + (\vec{f})_I \end{aligned} \quad (4)$$

Substitute  $n = \sqrt{\mu/r_h^3}$ ,  $(\vec{r}_h)_H = (r_h \hat{i})_H$ , and  $(\ddot{\vec{r}}_h)_I = -(\mu/r_h^3)(\vec{r}_h)_I$  to get

$$(\ddot{\vec{r}}_{s/h})_H + 2 \left[ (\vec{\Omega}_{h/i})_H \times (\dot{\vec{r}}_{s/h})_H \right] + (\vec{\Omega}_{h/i})_H \times \left[ (\vec{\Omega}_{h/i})_H \times (\vec{r}_{s/h})_H \right] = -n^2 \left[ (\vec{r}_{s/h})_H - 3x_{s/h}(\hat{i})_H \right] + (\vec{f})_H \quad (5)$$

In the Hill's frame,  $(\vec{\Omega}_{h/i}) = n(\hat{k})_H$ . Using cartesian coordinates in the Hill's frame

$$\begin{Bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{Bmatrix} + \begin{Bmatrix} -2n\dot{y} \\ 2n\dot{x} \\ 0 \end{Bmatrix} + \begin{Bmatrix} -n^2x \\ -n^2y \\ 0 \end{Bmatrix} = \begin{Bmatrix} -n^2x + 3n^2x \\ -n^2y \\ -n^2z \end{Bmatrix} + \begin{Bmatrix} f_x \\ f_y \\ f_z \end{Bmatrix}$$

or

$$\ddot{x} - 2n\dot{y} - 3n^2x = f_x \quad (6)$$

$$\ddot{y} + 2n\dot{x} = f_y \quad (7)$$

$$\ddot{z} + n^2z = f_z \quad (8)$$

These equations were first derived by Hill in the 19th century in a study of lunar motions. They are also known as the Clohessy-Wiltshire equations in the astrodynamics literature.

## Relative Motion with no Thrust

The presence of terms in  $\dot{x}$  and  $\dot{y}$  in the Hill's equations give the impression that damping may occur, contrary to our understanding that we are dealing with a non-dissipative system. But notice that these velocity-dependent "forces" are perpendicular to the velocity, so that they do no work and dissipate no energy. However, they do turn the trajectory. Notice first that the out-of-plane motion, Equation (8), is decoupled from the in-plane motion, Equations (6) and (7). The "plane" to which we are referring is the orbital ( $x - y$ ) plane of the frame. In the absence of thrust, Equation (8) is just a harmonic oscillator equation, giving out-of-plane deviations

$$z = z_0 \cos(nt + \beta) \quad (9)$$

with sinusoidal variation at the orbital frequency,  $n$ . The meaning of this is clear from the figure. If body starts on the y-axis (in the orbital plane) of the orbiting LVLH frame, but with some cross-track velocity, it will be in a neighboring circular orbit slightly inclined with respect to that of the frame. Since both orbits have the same period, the relative motion is purely a cross-track oscillation with a period equal to that of the orbit. Let us now examine the in-plane motion. With  $f_y = 0$ , the in-track motion (Equation (7)) integrates directly

$$\dot{y} + 2nx = C \quad (10)$$

where  $C$  is a constant of integration. Substituting into Equation (6), describing zenith-direction motion, we obtain

$$\ddot{x} - 2n(C - 2nx) - 3n^2x = 0 \quad (11)$$

or

$$\ddot{x} + n^2x = 2nC \quad (12)$$

which is a forced harmonic oscillator, with general solution

$$x = x_0 \cos(nt + \alpha) + \frac{2C}{n} = x_0 \cos(nt + \alpha) + x_{off} \quad (13)$$

where  $x_{off} = 2C/n$  is a fixed position deviation in the zenith direction (altitude). Finally,  $x$  from Equation (13) can be substituted back into Equation (10), and we can integrate once more for  $y$

$$\dot{y} = C - 2nx_0 \sin(nt + \alpha) - 4C \quad (14)$$

$$y = -2x_0 \sin(nt + \alpha) - 3Ct + B = -2x_0 \sin(nt + \alpha) - \frac{3}{2}nx_{off}t + y_{off} \quad (15)$$

where  $y_{off}$  is a fixed deviation in the in-track direction. Combining these three equations gives the homogeneous, un-forced solution as

$$\begin{aligned} x &= x_0 \cos(nt + \alpha) + x_{off} \\ y &= -2x_0 \sin(nt + \alpha) - \frac{3}{2}nx_{off}t + y_{off} \\ z &= z_0 \cos(nt + \beta) \end{aligned} \tag{16}$$

The constants  $x_{off}$ ,  $y_{off}$ ,  $x_0$ ,  $z_0$ ,  $\alpha$ , and  $\beta$  can be evaluated in terms of the initial values of the deviations  $x(0)$ ,  $y(0)$ ,  $z(0)$  and their rates  $\dot{x}(0)$ ,  $\dot{y}(0)$ ,  $\dot{z}(0)$ . Alternately, we may want to rendezvous with the reference frame at some future time  $t_R$ , so that we impose  $x = y = \dot{x} = \dot{y} = 0$  at  $t_R$ , thereby obtaining the required position and velocity at the present time. Other combinations are possible. From Equation (15) we can see that unless  $x_{off} = 0$ , the offset in the zenith direction, there will be a secular (linearly increasing with time) deviation or drift in the in-track, orbital direction and, after a while, the body may be very far from the reference frame making the linearization, based on short distances, invalid. In fact, if we select  $x_0 = y_{off} = 0$ , we have the solution from Equations (13) and (15)

$$x = x_{off}; y = -\frac{3}{2}nx_{off}t \tag{17}$$

with a constant radial position deviation and a constant in-track velocity deviation (drift). This solution corresponds to two concentric, co-planar circular orbits of slightly different radii, hence slightly different orbital periods. Notice that if  $x_{off} > 0$ , which must represent a slightly elliptic orbit of the same period as the reference orbit (LVLH frame's orbit), i.e., with a semi-major axis equal to  $a$  (the semi-major axis of the LVLH frame's orbit), and perhaps with a slight inclination difference (if  $z \neq 0$ ). We then have for the in-plane motion

$$y = y_{off} - 2x_0 \sin(nt + \alpha) \tag{18}$$

$$x = x_0 \cos(nt + \alpha) \tag{19}$$

showing that the relative trajectories are ellipses elongated in the in-track direction by a 2:1 ratio, and centered at an arbitrary in-track position  $y_{off}$  (and at zero relative altitude in the zenith direction). The period of the elliptical relative motion is the same as that of the reference LVLH frame's orbit,  $T = 2\pi/n$ . Consider what happens if an astronaut standing on the tail of the Shuttle, which is orbiting the Earth tail first, pushes herself vertically upwards: as she drifts higher, her absolute angular velocity decreases, and the Shuttle overtakes her. Eventually, she reaches the apogee of her slightly elliptical orbit and accelerates back down. After half an orbit (about 45 minutes later), she crosses the original altitude (perhaps passing the nose of the Shuttle) and starts overtaking the Shuttle while underneath. Another half orbit later, she is back on the tail of the Shuttle (but on its underside). Also notice that when the astronaut is at perigee her velocity is higher and at apogee is lower than the velocity of the LVLH frame's orbit thereby causing the clockwise motion when viewed from a point on the positive cross-track axis.

## Eigenvalues of Hill's Equations

We can solve for the eigenvalues of the coupled set of Hill's equations, Equations (6)-(8), using Laplace Transforms. Transforming the homogeneous form of these equations gives

$$\begin{aligned} s^2 \bar{x} - 2ns\bar{y} - 3n^2 \bar{x} &= 0 \\ s^2 \bar{y} + 2ns\bar{x} &= 0 \\ s^2 \bar{z} + n^2 \bar{z} &= 0 \end{aligned} \tag{20}$$

giving the characteristic equation

$$s^2(s^2 + n^2)(s^2 + n^2) = 0$$

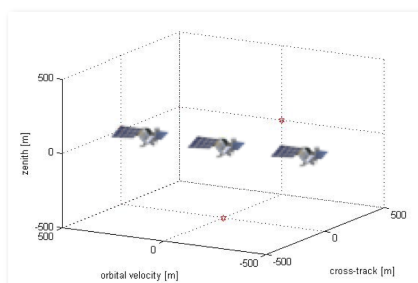
The body has six (6) poles since it has three (3) degrees of freedom (DOFs) each with two energy storage mechanisms (kinetic and potential). The roots (poles) of the characteristic equation are

$$a = \pm 0, \pm in, \pm in$$

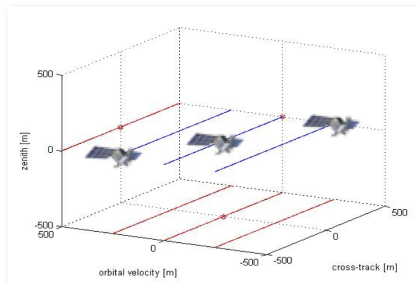
with two neutrally stable poles of zero frequency and four neutrally stable poles at the frequency of the LVLH frame's orbit.

## Dynamic Satellite Formations

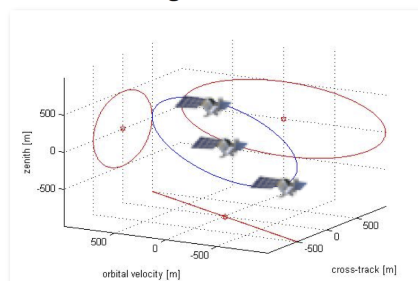
Hill's Equations can be used to design formations for sparse aperture synthesis: the combining of measurements from multiple satellites in close proximity, to synthesize an aperture (i.e., antenna) that has the angular resolution comparable to that of an antenna the size of the formation. The figure below shows some example formations.



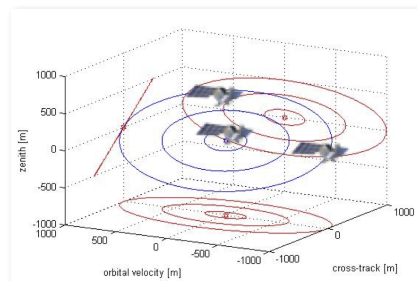
String of Pearls



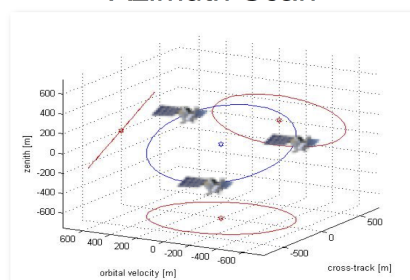
Cross-track Scan



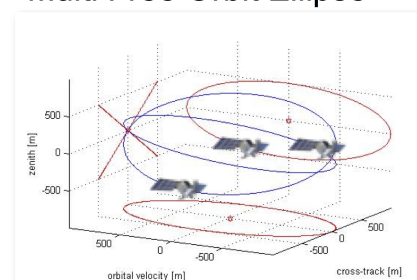
Azimuth Scan



Multi Free Orbit Ellipse



Nadir Circle



Dual Side Scan

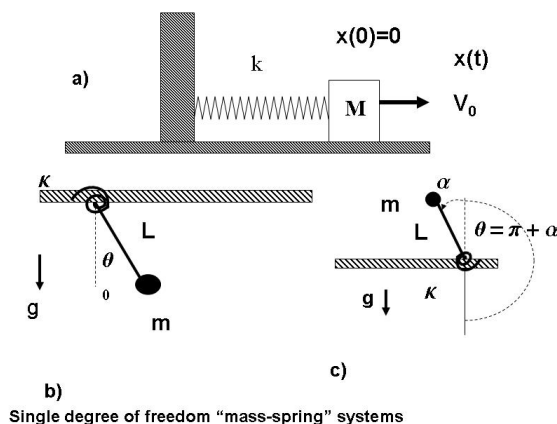
The blue lines are the satellite trajectories in the LVLH frame, where the zenith, x-direction is vertical, the in-track (orbital velocity) direction is to the left, and the cross-track direction is into the page. The red lines show the projection of the blue trajectory into the various planes. The orbital plane is in the back. Note that all of these trajectories project a 2x1 ellipse in the orbital plane. For some of these trajectories, this projection is a dot, which is a degenerate 2x1 ellipse. The in-track projection is on the left panel and the nadir projection is on the bottom panel. Notice that the "Nadir Circle" projects a circle in the direction of Earth. Several interesting, closed trajectory "clusters" of satellites can be made. Note that almost all of these formations are dynamic in nature since the satellites will move relative to each other. The three-dimensional nature of some of these formations is caused by proper phasing of the cross-track motion relative to the in-track motion. Remember that this solution is only valid for short distances and for a spherical Earth. The non-uniform, geo-potential terms, such as  $J_2$ , as well as slight offsets in altitude, will cause these formations to drift apart over time thereby requiring  $\Delta v$  to maintain their formations. However, this required  $\Delta v$  is substantially less than that required to maintain a static formation, with the exception of the "String of Pearls." Also remember that while the satellites will appear to orbit around each other, they are, in reality, orbiting the Earth with slightly different eccentricities and inclinations.

## Lecture 22 — Vibration, Normal Modes, Natural Frequencies, Instability

### Vibration, Instability

An important class of problems in dynamics concerns the free vibrations of systems. (The concept of free vibrations is important; this means that although an outside agent may have participated in causing an initial displacement or velocity—or both—of the system, the outside agent plays no further role, and the subsequent motion depends only upon the inherent properties of the system. This is in contrast to "forced" motion in which the system is continually driven by an external force.) We shall consider only undamped systems for which the total energy is conserved and for which the frequencies of oscillation are real. This forms the basis of the approach to more complex studies for forced motion of damped systems. We saw in Lecture 13, that the free vibration of a mass-spring system could be described as an oscillatory interchange between the kinetic and potential energy, and that we could determine the natural frequency of oscillation by equating the maximum value of these two quantities. (The natural frequency is the frequency at which the system will oscillate unaffected by outside forces. When we consider the oscillation of a pendulum, the gravitational force is considered to be an inherent part of the system.) The general behavior of a mass-spring system can be extended to elastic structures and systems experiencing gravitational forces, such as a pendulum. These systems can be combined to produce complex results, even for one-degree of freedom systems.

We begin our discussion with the solution of a simple mass-spring system, recognizing that this is a model for more complex systems as well.



In the figure, a) depicts the simple mass spring system: a mass  $M$ , sliding on a frictionless plane, restrained by a spring of spring constant  $k$  such that a force  $F(x) = -kx$  opposes the displacement  $x$ . (In a particular problem, the linear dependence of the force on  $x$  may be an approximation for small  $x$ .) In order to get a solution, the initial displacement and initial velocity must be specified. Common formulations are:  $x(0) = 0$ , and  $\frac{dx}{dt}(0) = V_0$  (The mass responds to an initial impulse.); or  $x(0) = X_0$  and  $\frac{dx}{dt}(0) = 0$  (The mass is given an initial displacement.). The general formulation is some combination of these initial conditions. From Newton's law, we obtain the governing differential equation

$$m \frac{d^2 x}{dt^2} = -kx \quad (1)$$

with  $x(0) = X_0$ , and  $\frac{dx}{dt}(0) = V_0$ .

The solution is of the general form,  $x(t) = \text{Re}(Ae^{i\omega t}) = \text{Re}(A_R + iA_i)(\cos(\omega t) + i\sin(\omega t))$ , where, at this point in the analysis, **both  $A$  and  $\omega$  are unknown**. That is, we assume a solution in which both  $A$  and  $\omega$  are unknown, and later when the solution is found and boundary conditions are considered, we will end up taking the real part of the expression. Depending upon whether  $A$  is purely real, purely imaginary, or some combination, we will in general get oscillatory behavior involving  $\sin$ 's and  $\cos$ 's since  $e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$ . For a system without damping,  $\omega^2 = \text{Real}(K)$  ( $K$  being some combination of system parameters.), so that  $\omega = \pm\sqrt{K}$ . For undamped systems, these two  $\pm$  values of  $\omega$ , are redundant; only one need be taken.

To solve the differential equation, equation(1) is rewritten

$$m \frac{d^2 x}{dt^2} + kx = 0. \quad (2)$$

With the assumed form of solution, this becomes

$$-\omega^2 m x(t) + kx(t) = 0 = x(t) * (-\omega^2 m + k) \quad (3)$$

Since the equation is linear, the amplitude  $A$  cannot be determined at this point; it will be determined by

satisfying initial conditions. We can however determine the frequency  $\omega$ . Since  $x = x(t)$ , for a valid solution, we require  $\omega = \sqrt{\frac{k}{m}}$ .

This approach works as well for systems b) and c). These two systems are pendulums restrained by torsion springs, which for small angles ( $\theta$  or  $\alpha$ ) produce a restoring torque proportional to the angular departure from equilibrium. Consider system b). Its equilibrium position is  $\theta = 0$ . The restoring torque from the spring is  $T_s = -\kappa\theta$ . The restoring torque from gravity is  $T_g = -mgL\sin\theta$  which for small angles becomes  $T_g = -mgL\theta$ . Writing Newton's law in a form appropriate for pendular motion, we obtain

$$mL \frac{d^2\theta}{dt^2} = \frac{1}{L}(T_g + T_s) \quad (4)$$

We assume a form of solution,  $\theta(t) = Ae^{i\omega t}$ , and rewrite the equation as before, moving all terms to the left-hand side.

$$\left(-mL\omega^2 + \frac{1}{L}(mgL + \kappa)\right)\theta(t) = 0. \quad (5)$$

Therefore for a solution we require

$$\omega = \sqrt{\frac{mgL + \kappa}{mL^2}} \quad (6)$$

Examining this result, we see that the combination of the spring and gravity acts to increase the natural frequency of the oscillation. Also if there is no spring,  $\kappa = 0$ , and the result becomes just the frequency of a pendulum  $\omega = \sqrt{\frac{g}{L}}$ .

System c) is perhaps a bit more interesting. In this case, we use the small angle  $\alpha$ . We take the equilibrium position of the spring to be  $\alpha = 0$  so that the restoring torque due to the spring is again  $T_s = -\kappa\alpha$ . But now in this case, we are expanding the gravitational potential about the point  $\alpha = 0$ . Since this is an unstable equilibrium point, this gives the restoring (–it doesn't restore, it keeps going!–) torque due to gravity as  $T_g = mgL\sin\alpha$  or for small  $\alpha$ ,  $T_g = mgL\alpha$ .

Writing the governing equation for this case, we obtain

$$\left(-mL\omega^2 + \frac{1}{L}(-mgL + \kappa)\right)\alpha(t) = 0. \quad (7)$$

Therefore for a solution we require

$$\omega = \sqrt{\frac{-mgL + \kappa}{mL^2}} \quad (8)$$

In this case, there is a critical value of  $\kappa$  for which  $\omega = 0$ . On either side of this point, we have  $\omega = \sqrt{\frac{\kappa - mgL}{mL^2}}$ , which gives  $\omega \rightarrow \text{real}$  for  $\kappa > mgL$ , and  $\omega \rightarrow \text{imag}$  for  $\kappa < mgL$ . Since we have assumed  $\alpha(t) = e^{i\omega t}$ , a real  $\omega$  will produce oscillatory motion; an imaginary  $\omega$  will produce exponentially diverging, or unstable, motions. We say that the pendulum for  $\kappa$  less than the critical value,  $\kappa = mgL$ , is unstable.

## General Solution for Damped 1-DOF System

In the last section, we assumed a solution of the form



$$x(t) = \text{Re}(Ae^{i\omega t}) = \text{Re}[(A_R + iA_I)(\cos \omega t + i \sin \omega t)] \quad (9)$$

In the more general case, when damping does exist, the solution is

$$x(t) = \text{Re}[(A_R + iA_I)e^{st}] = \text{Re}[(A_R + iA_I)e^{\sigma + i\omega t}] = \text{Re}[(A_R + iA_I)e^{\sigma t}(\cos \omega t + i \sin \omega t)] \quad (10)$$

This can be simplified to

$$x(t) = A_R e^{\sigma t} \cos(\omega t + \phi) \quad (11)$$

where

$$\phi = \tan^{-1} \frac{A_I}{A_R} \quad (12)$$

and

$$\dot{x}(t) = A_R e^{\sigma t} (\sigma \cos(\omega t + \phi) - \omega \sin(\omega t + \phi)) \quad (13)$$

where  $A_R$  and  $\phi$  depend on the initial conditions. Let's assume that  $x(0) = 0$  and  $\dot{x}(0) = v_o$ . Equation 11 gives  $\phi = \pi/2$  and Equation 13 gives  $A_R = -v_o/\omega$  giving

$$x(t) = \frac{v_o}{\omega} e^{\sigma t} \sin \omega t, \quad \dot{x}(t) = v_o e^{\sigma t} (\cos \omega t + \frac{\sigma}{\omega} \sin \omega t) \quad (14)$$

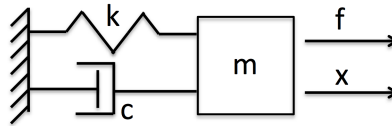
Now let  $x(0) = x_o$  and  $\dot{x}(0) = 0$ . We now get  $\phi = \tan^{-1}(\sigma/\omega)$  and  $A_R = x_o \sqrt{1 + (\sigma/\omega)^2}$  giving

$$x(t) = x_o \sqrt{1 + \left(\frac{\sigma}{\omega}\right)^2} e^{\sigma t} \cos\left(\omega t + \tan^{-1}\left(\frac{\sigma}{\omega}\right)\right) \quad (15)$$

and

$$\dot{x}(t) = x_o \sqrt{1 + \left(\frac{\sigma}{\omega}\right)^2} e^{\sigma t} \left[ \sigma \cos\left(\omega t + \tan^{-1}\left(\frac{\sigma}{\omega}\right)\right) - \omega \sin\left(\omega t + \tan^{-1}\left(\frac{\sigma}{\omega}\right)\right) \right] \quad (16)$$

### Example: Spring-Mass-Damper System



Referring to the figure, the dashpot ( $c$ ) creates a force opposing velocity ( $F = -c\dot{x} = -c\dot{x}$ ). It is a non-energy-conserving force because it dissipates energy and is not the gradient of a potential function. Newton's 2<sup>nd</sup> Law gives

$$m\ddot{x} = -c\dot{x} - kx + f \quad (17)$$

For a homogeneous solution ( $f = 0$ )

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (18)$$

Taking the Laplace transform gives

$$(ms^2 + cs + k)\bar{x} = 0 \quad (19)$$

with roots

$$s = -\frac{c}{2m} \left( 1 \mp \sqrt{1 - \frac{4mk}{c^2}} \right) \quad (20)$$

If

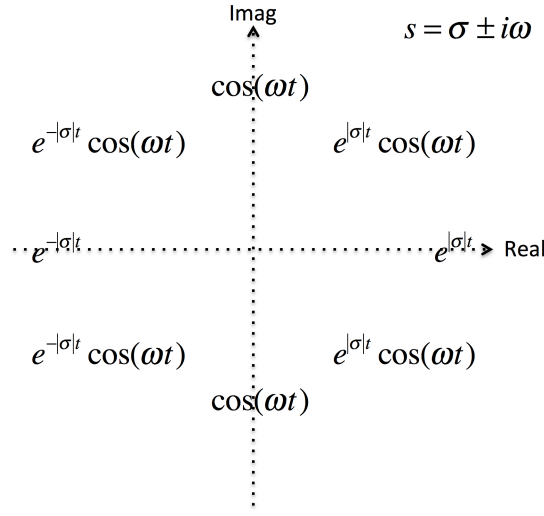
$$1 - \frac{4mk}{c^2} > 0, \quad c > 2\sqrt{mk}, \quad \text{and} \quad s = -\frac{c}{2m} \left( 1 \mp \sqrt{1 - \frac{4mk}{c^2}} \right) \quad (21)$$

and both poles are negative and real. If

$$1 - \frac{4mk}{c^2} < 0, \quad c < 2\sqrt{mk}, \quad \text{and} \quad s = -\frac{c}{2m} \left( 1 \mp i\sqrt{\frac{4mk}{c^2} - 1} \right) \quad (22)$$

and both poles are complex and conjugates of each other.

The figure below shows the behavior in the "s-plane," or Laplace Plane. The complex value for the pole  $s$  can be plotted in a two-dimensional plane where the horizontal axis is the real axis, corresponding to the real part ( $Real(s) = \sigma$ ), and the vertical axis is the imaginary axis, corresponding to the imaginary part ( $Imag(s) = \omega$ ). Poles always occur in complex conjugate pairs. Imaginary poles lie on the imaginary axis and result in oscillatory behavior whose amplitude does not change with time. Real poles lie on the real axis and exhibit exponentially growing or decaying behavior when the pole is in the right or left half plane, respectively. Complex poles in the left half plane exhibit oscillatory behavior whose amplitude decays exponentially with time (stable) while those in the right half plane grow exponentially with time (unstable). For the example above, the poles are on the imaginary axis when  $c = \zeta = 0$  and then migrate into the left half plane as  $c, \zeta > 0$  corresponding to the stabilizing effect of the damper.



Notice that

$$\sigma = -\frac{c}{2m} \quad \text{and} \quad \omega = \sqrt{\frac{k}{m} - \frac{c^2}{4m^2}} \quad (23)$$

These can be rewritten as

$$\sigma = -\frac{c}{2m} = \sqrt{\frac{k}{m}} \frac{c}{2\sqrt{mk}} = \zeta \omega_n \quad \text{and} \quad \omega = \sqrt{\frac{k}{m} - \frac{c^2}{4m^2}} = \sqrt{\frac{k}{m}} \sqrt{1 - \left(\frac{c}{2\sqrt{mk}}\right)^2} = \omega_n \sqrt{1 - \zeta^2} = \omega_d \quad (24)$$

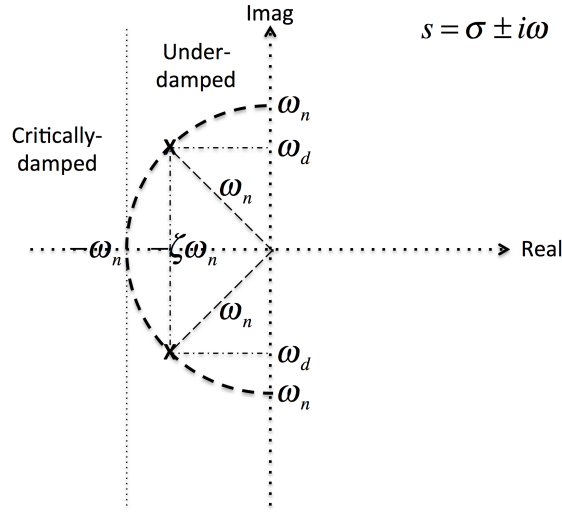
where  $\omega_n = \sqrt{k/m}$  is the natural (undamped) frequency,  $\omega_d$  is the damped frequency, and  $\zeta$  is the damping ratio. When  $c = 2\sqrt{mk}$ ,  $\zeta = 1$  and the system is critically damped. This is the value at which the complex conjugate poles transition to negative real poles. The general solution is

$$x(t) = Ae^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1 - \zeta^2} t + \phi) \quad (25)$$

and the equation of motion can be written as

$$\ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = \frac{1}{m} f \quad (26)$$

The figure below shows graphically the definitions of  $\omega_n, \omega_d, \zeta$  relative to the location of the poles in the s-plane.



As  $c, \zeta \Rightarrow \infty$ ,  $s \Rightarrow \infty, 0$ .

## Non-Homogeneous Forced Response

Let the force ( $f$ ) be

$$f(t) = kF(t) = kA \cos \omega t \quad (27)$$

where  $A$  is the amplitude and  $\omega$  is the driving frequency. Substituting

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \frac{k}{m}F(t) = \omega_n^2A \cos \omega t \quad (28)$$

Assume a steady-state solution (i.e., all homogeneous transients have died out) of the form

$$x(t) = \bar{X} \cos(\omega t + \phi) \quad (29)$$

Substituting

$$\bar{X}[(\omega_n^2 - \omega^2) \cos(\omega t + \phi) - 2\zeta\omega_n\omega \sin(\omega t + \phi)] = \omega_n^2A \cos \omega t \quad (30)$$

Remembering that

$$\cos(\omega t + \phi) = \cos \omega t \cos \phi - \sin \omega t \sin \phi, \quad \sin(\omega t + \phi) = \sin \omega t \cos \phi + \cos \omega t \sin \phi \quad (31)$$

Equating the  $\cos \omega t$  terms

$$\bar{X}[(\omega_n^2 - \omega^2) \cos \phi - 2\zeta\omega_n\omega \sin \phi] = \omega_n^2A \quad (32)$$

Equating the  $\sin \omega t$  terms

$$\bar{X}[-(\omega_n^2 - \omega^2) \sin \phi - 2\zeta\omega_n\omega \cos \phi] = 0 \quad (33)$$

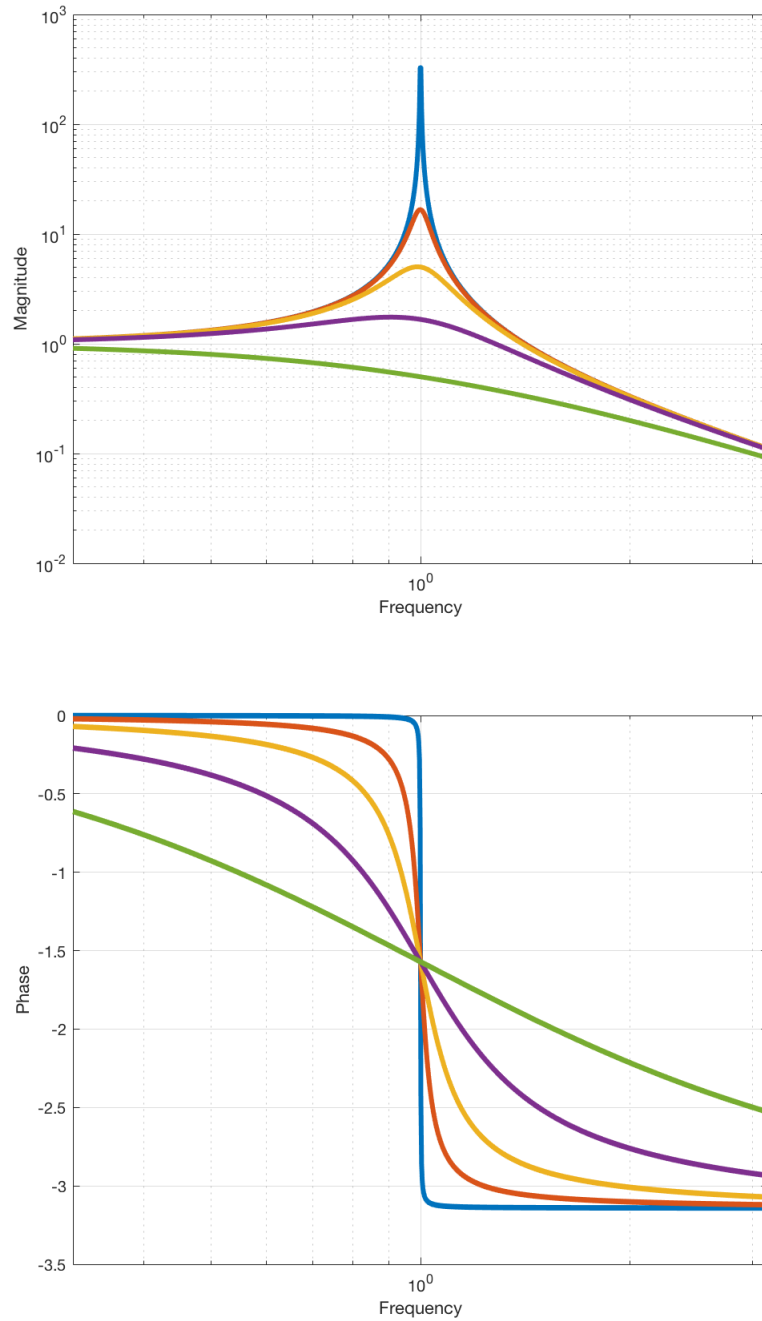
Solving gives the magnitude of the steady-state transfer function from the force to the displacement as well as the phase between the force and the resulting displacement

$$\left| \frac{\bar{X}}{A} \right| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left[\frac{2\zeta\omega}{\omega_n}\right]^2}} \quad \text{where} \quad \phi = \tan^{-1} \left[ -\frac{2\zeta\frac{\omega}{\omega_n}}{1 - \left(\frac{\omega}{\omega_n}\right)^2} \right] \quad (34)$$

The system has two components to its response: the transient homogeneous response and the steady-state non-homogeneous response. In this analysis, we derived the steady-state non-homogeneous response where we assumed that all transients have died out. Using the result above, we can introduce the concept of a steady-state transfer function as shown in the figure below. The upper figure plots the magnitude from Equation 34 and the lower plots the phase from Equation 34. At low frequency, the magnitude approaches one since the effect of the mass and damper are small and the result is the static deflection. As the frequency approaches resonance, for small  $\zeta$ , the denominator of the transfer function's magnitude becomes small resulting in a large peak whose ratio to the static deflection is approximated by  $1/(2\zeta)$ , often called the dynamic amplification. At high frequency, well above resonance, the effect of the mass dominates since the transfer function magnitude can be approximated as  $(\omega_n^2/\omega^2)$ , which decreases with a slope of -2 on a logarithmic plot ( $\log_{10}(\omega_n^2\omega^{-2}) = -2\omega_n^2 \log_{10} \omega$ ). Now look at the effect of  $\zeta$ . In both plots, five different values for  $\zeta$  are shown. As  $\zeta$  increases, the resonant peak decreases in magnitude and shifts lower in frequency, with the peak occurring at

$$\omega_{peak} = \omega_n \sqrt{1 - 2\zeta^2} \quad (35)$$

For the phase, at low frequency the displacement ( $x(t)$ ) is in phase ( $\phi = 0$ ) with the force. At the natural frequency ( $\omega_n$ ), the displacement lags the force by  $\pi/2$  ( $\phi = -\pi/2$ ). For light damping, this puts the force in phase with velocity allowing it to put energy into the system very efficiently, thereby causing large amplification of the motion relative to the static response. At high frequency, the phase drops further towards  $\phi = -\pi$ , causing the motion of the mass to be out of phase with the applied force. Care must be taken when evaluating the phase using Equation 34 since the four quadrant inverse tangent must be used to generate the plot as shown.



In the discussion above, we went directly to the magnitude of the transfer function by choosing a form of the solution that expressed the imaginary part as a phase angle. To find the actual complex transfer function, assume

$$f(t) = kAe^{i\omega t} \quad \text{and} \quad x(t) = \bar{X}e^{i\omega t} \quad (36)$$

Substituting into Equation 17 gives the following

$$\frac{\bar{X}}{A} = \frac{1}{\left(1 - \frac{\omega^2}{\omega_n^2}\right) + i \left(2\zeta \frac{\omega}{\omega_n}\right)} \quad (37)$$

which is a complex quantity. This has the form

$$\frac{\bar{X}}{A} = \frac{a + ib}{c + id} \quad (38)$$

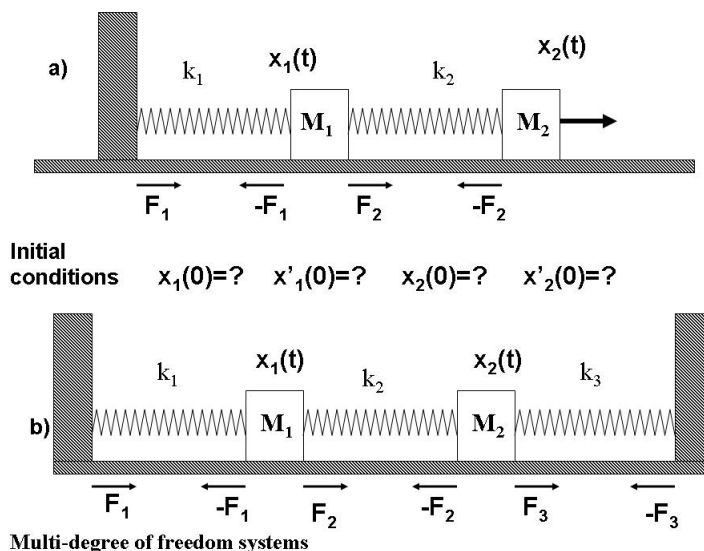
where the magnitude is

$$\left| \frac{\bar{X}}{A} \right| = \left| \frac{a + ib}{c + id} \right| = \sqrt{\frac{a^2 + b^2}{c^2 + d^2}} \quad (39)$$

which, for this problem, gives the magnitude in Equation 34.

## Vibration of Multi-Degree of Freedom Systems

We begin our treatment of systems with multiple degrees of freedom, by considering a two degree of freedom system. This system contains the essential features of multi-degree of freedom systems. Consider the *two* two-mass, two-spring systems shown in the figure.



In this case, there are two independent variables,  $x_1(t)$  and  $x_2(t)$ ; their motion is not independent, but is coupled by their attachments to the springs  $k_1$ ,  $k_2$  and for system b),  $k_3$ . The sketch shows the forces  $F_i$  acting on the masses as a result of the extension of the spring; these are equal and opposite at the ends of the springs. We consider both system a) and b). System b) is actually the simpler of the two systems because of its inherent symmetry.

The governing equations can be written as

for system a)

$$m_1 \frac{d^2 x_1}{dt^2} = -k_1 x_1 + k_2 (x_2 - x_1) \quad (40)$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k_2 (x_2 - x_1) \quad (41)$$

for system b)

$$m_1 \frac{d^2 x_1}{dt^2} = -k_1 x_1 + k_2 (x_2 - x_1) \quad (42)$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k_2 (x_2 - x_1) - k_3 x_2 \quad (43)$$

In both cases, as before, we assume a solution of the form  $x_1(t) = X_1 e^{i\omega t}$  and  $x_2(t) = X_2 e^{i\omega t}$ . However, as we will see, in this case, we will obtain two possible values for  $\omega^2$ ; both will be real; we will take only the positive value of  $\omega$  itself. These will be the two vibration modes of this two degree of freedom system. These results extend to  $N$   $\omega^2$ 's for an  $N$  degree of freedom system. Again, we will take only the positive value of  $\omega$ .

Consider first system b). With the assumed form of solution, and rewriting all terms on the left-hand side, we obtain

$$-\omega^2 m_1 X_1 + k_1 X_1 - k_2 (X_2 - X_1) = 0 \quad (44)$$

$$-\omega^2 m_2 X_2 + k_2 (X_2 - X_1) + k_3 X_2 = 0 \quad (45)$$

This equation can be written in matrix form as

$$\left( \begin{pmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{pmatrix} - \begin{pmatrix} m_1 \omega^2 & 0 \\ 0 & m_2 \omega^2 \end{pmatrix} \right) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (46)$$

or

$$\begin{pmatrix} k_1 + k_2 - m_1 \omega^2 & -k_2 \\ -k_2 & k_2 + k_3 - m_2 \omega^2 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (47)$$

This equation makes a very powerful statement. Since the right-hand side of both equations is zero, a condition for a solution is that the determinant of the matrix equals zero. This will give an algebraic equation with two solutions for  $\omega$ :  $\omega_1$  and  $\omega_2$ . These are the **"natural" frequencies** of the two degree of freedom system. In the general case, they are not equal; and both  $x_1$  and  $x_2$  participate in the oscillation at each frequency  $\omega_i$ . Also, as in the single degree of freedom system, the actual values of  $x_1(t)$  and  $x_2(t)$  are determined by initial conditions; in this case 4 initial conditions are required:  $x_1(0)$ ,  $x_2(0)$ ,  $\frac{dx_1}{dt}(0)$ , and  $\frac{dx_2}{dt}(0)$ . The actual values of  $X_1$  and  $X_2$  are of less interest than the relationships between them and the structure of the problem.

If the two masses are equal, a particularly simple form of a more general result follows from equation(16). We consider this as an introduction to the more general case. The more general case will be considered



shortly.

$$\begin{pmatrix} k_1/m + k_2/m - \omega^2 & -k_2/m \\ -k_2/m & k_2/m + k_3/m - \omega^2 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (48)$$

We determine the two values of  $\omega_i$  ( $\omega_1$  and  $\omega_2$ ) by setting the determinant equal to zero. We then substitute each value of  $\omega_i$  in turn into the matrix equation and determine for each  $\omega_i$  the coefficients  $X_{1i}$  and  $X_{2i}$ ; only their ratio can be determined. We write the coefficients  $X_{1i}$  and  $X_{2i}$  as vectors,  $\vec{X}_1 = (X_{11}, X_{21})$  and  $\vec{X}_2 = (X_{12}, X_{22})$ , where the subscript 1 refers to the mode associated with  $\omega_1$ , and 2 refers to the mode associated with  $\omega_2$ .

It is a remarkable property of the solution to the governing equations that these vectors are orthogonal: the dot product of  $\vec{X}_1 \cdot \vec{X}_2 = 0$  (We will follow with an example to amplify and clarify this.)

Consider the simplest case of system b) with both masses equal to  $m$  and all springs of stiffness  $k$ . In this case we have

$$\begin{pmatrix} 2k/m - \omega^2 & -k/m \\ -k/m & 2k/m - \omega^2 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (49)$$

Setting the determinant equal to zero gives two solutions for  $\omega$ :  $\omega_1 = \sqrt{k/m}$  and  $\omega_2 = \sqrt{3k/m}$ . The components of the  $x_1$  and  $x_2$  motion are:  $\vec{X}_1 = (1, 1)$  and  $\vec{X}_2 = (-1, 1)$ . The contribution of each normal mode to the final solution is determined by the initial conditions. This simple example gives great physical insight to the more general problem. The natural frequency is  $\omega_i$ ; the components  $\vec{X}_i = (X_{i1}, X_{i2})$  are called **"normal modes"**. In a given case, the contribution of each normal mode to the total solution is determined by the initial condition.

## Normal Modes of Multi-Degree of Freedom Systems

Examining the first "normal mode" of this system, we see an oscillation in which  $\vec{X}_1 = (1, 1)$  occurs at an oscillation frequency  $\omega_1 = \sqrt{k/m}$ . Since  $\vec{X}_1 = (1, 1)$ , the central spring does not deform, and the two masses oscillate, each on a single spring, thus giving a frequency  $\omega_1 = \sqrt{k/m}$ .

The second "normal mode" has a frequency  $\omega_2 = \sqrt{3k/m}$ , with  $\vec{X}_2 = (-1, 1)$ ; thus the masses move in opposite directions, and the frequency of oscillation is increased. It can be seen by inspection that the vector  $\vec{X}_1$  and  $\vec{X}_2$  are orthogonal (their dot product is zero.)

If such a system was at rest, and an initial impulse was given to **one** of the masses, **both** modes would be excited and a free oscillation would occur with each "mode" oscillating at "its" natural frequency.

The equation for general values of  $k_1$ ,  $k_2$  and  $k_3$  can be written

$$\left( \begin{pmatrix} k_1/m + k_2/m & -k_2/m \\ -k_2/m & k_2/m + k_3/m \end{pmatrix} - \omega^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (50)$$

## Characteristic Value Problem

This problem is called a characteristic-value or eigenvalue problem. Formulated in matrix notation it can be written

$$\left( \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (51)$$

The requirements on the simplest form of the characteristic value problem are that the matrix  $[A]$  is symmetric (This will always be true for combinations of masses and springs.), and that the characteristic value  $\lambda$  multiplies an identity matrix. (An identity matrix has all 1's on the diagonal and 0's off the diagonal; this will be true if all the masses are equal; if not, a more general form must be used yielding analogous results. This more general form will be considered shortly.)

For the form of the governing equation shown in equation(51), the characteristic value  $\lambda = \omega^2$ . The general solution to this problem will yield a set of solutions for  $\lambda$  equal to the size of the matrix : (i.e a  $4 \times 4$  matrix will result in 4  $\lambda$ 's). For each  $\lambda_i$ , a vector  $\vec{X}_i$  will be obtained. For this form of the characteristic value problem, the dot product between any two of these vectors is zero.

$$(X_{i,1}, X_{i,2}) \cdot (X_{j,1}, X_{j,2}) = 0 \quad (52)$$

for  $i \neq j$ . These are the **normal modes** of the system, and the  $\omega$ 's are the **natural frequencies**. Any numerical matrix method—such as MATLAB— will yield both the  $\lambda_i$ 's (called the eigenvalues) and the  $X_i$ 's, called the eigenvectors for a particular matrix  $[A]$ . A similar result is obtained for the modes of vibration of a continuous system such as a beam. The displacement of the various mode of vibration of a uniform beam are orthogonal.

The general solution for the motion of the masses is then given by an expansion in the **normal modes**,  $\vec{X}_i$ ,

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = A_1 e^{i\omega_1 t} \begin{pmatrix} X_{11} \\ X_{12} \end{pmatrix} + A_2 e^{i\omega_2 t} \begin{pmatrix} X_{21} \\ X_{22} \end{pmatrix}, \quad (53)$$

where  $\vec{X}_1 = (X_{11}, X_{21})$  and  $\vec{X}_2 = (X_{12}, X_{22})$  are the eigenvectors or normal modes from the solution of the characteristic-value problem, obtained by hand or numerically, and  $\omega_i$  is the natural frequency of that mode. Again, it is a remarkable and extremely useful property that the dot product of  $\vec{X}_i$  and  $\vec{X}_j$  is zero unless  $i = j$ . Good form would suggest that we normalize each  $X_i$  so that the magnitude of  $X_i \cdot X_i$  equals 1, but we usually don't and therefore need to define  $C_i = \vec{X}_i \cdot \vec{X}_i$ , the vector-magnitude-squared, for later use.

## Expansion in Normal Modes; Satisfaction of Initial Conditions

The general form of solution is given by equation (53). All that remains is to determine the coefficients  $A_1$  and  $A_2$ . This is done by satisfying the initial conditions on displacement and velocity. In the general case, since  $e^{i\omega t} = 1$  at  $t = 0$ , the initial displacement can be written

$$\begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = A_1 \begin{pmatrix} X_{11} \\ X_{12} \end{pmatrix} + A_2 \begin{pmatrix} X_{21} \\ X_{22} \end{pmatrix}. \quad (54)$$

and for an initial velocity, since  $i\omega e^{i\omega t} = i\omega$  at  $t = 0$ , the initial velocity can be written

$$\begin{pmatrix} v_1(0) \\ v_2(0) \end{pmatrix} = i\omega_1 A_1 \begin{pmatrix} X_{11} \\ X_{12} \end{pmatrix} + i\omega_2 A_2 \begin{pmatrix} X_{21} \\ X_{22} \end{pmatrix}. \quad (55)$$

It should be noted that in general  $A$  is complex; the real part relates to the initial displacement; the imaginary part to the initial velocity. If we consider  $A_i$  to be real, we are automatically assuming no initial velocity.

### Case 1: initial displacement non-zero; initial velocity zero

We first consider the case of an initial condition on the displacement, specifically  $x_1(0) = x_{10}$  and  $x_2(0) = x_{20}$ , with  $v_1(0) = 0$  and  $v_2(0) = 0$ . We define the initial-condition vector as  $\vec{X}_0 = (x_1(0), x_2(0))$ . To complete the solution, we need to obtain the values of  $A_1$  and  $A_2$  from equation(54). This is done by taking the dot product of both sides of equation(54) with the first mode  $\vec{X}_1$ . Finally we get our payoff for all our hard work. Since the dot product of  $\vec{X}_1$  with  $\vec{X}_2$  is zero; the dot product of  $\vec{X}_1$  with  $\vec{X}_1$  is  $C_1$ ; and the dot product of  $\vec{X}_0$  with  $\vec{X}_1$  is some  $G_1$ , we obtain

$$A_1 = \vec{X}_0 \cdot \vec{X}_1 / C_1 = \frac{G_1}{C_1} \quad (56)$$

and taking the dot product of  $\vec{X}_0$  with  $\vec{X}_2$  (which equals some  $G_2$ ), we obtain

$$A_2 = \vec{X}_0 \cdot \vec{X}_2 / C_2 = \frac{G_2}{C_2} \quad (57)$$

With  $A_1$  and  $A_2$  determined, we have a complete solution to the problem.

### Case 2: initial displacement zero; initial velocity non-zero

We now consider the case of an initial condition on the velocity, specifically  $\dot{x}_1(0) = v_{10}$  and  $\dot{x}_2(0) = v_{20}$ , with  $x_1(0) = 0$  and  $x_2(0) = 0$ . We define the initial-condition vector as  $\vec{V}_0 = (v_{10}, v_{20})$ . We use the coefficient  $B_i$  to define the solution for this case. The solution is again written as an expansion in normal modes oscillating at their natural frequency  $\omega_i$  of amplitude  $B_i$ , which is unknown at this point.

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = B_1 e^{i\omega_1 t} \begin{pmatrix} X_{11} \\ X_{12} \end{pmatrix} + B_2 e^{i\omega_2 t} \begin{pmatrix} X_{21} \\ X_{22} \end{pmatrix}. \quad (58)$$

As previously noted,  $B$  is complex; the real part relates to the initial displacement; the imaginary part to the initial velocity. If  $B_i$  is imaginary, there is no initial displacement. The velocity at  $t = 0$  is given by

$$\begin{pmatrix} v_1(0) \\ v_2(0) \end{pmatrix} = i\omega_1 B_1 \begin{pmatrix} X_{11} \\ X_{12} \end{pmatrix} + i\omega_2 B_2 \begin{pmatrix} X_{21} \\ X_{22} \end{pmatrix}. \quad (59)$$

To complete the solution, we need to obtain the values of  $B_1$  and  $B_2$  from equation(59). This is again done by taking the dot product of equation(59) with the first mode  $\vec{X}_1$ . Since the dot product of  $\vec{X}_1$  with  $\vec{X}_2$  is

zero, and the dot product of  $\vec{X}_1$  with  $\vec{X}_1$  is  $C_1$ , we obtain

$$B_1 = -\frac{Q_1 i}{C_1 \omega_1} \quad (60)$$

where  $Q_1 = \vec{V}_0 \cdot \vec{X}_1$  and taking the dot product of  $\vec{X}_0$  with  $\vec{X}_2$  we obtain

$$B_2 = -\frac{Q_2 i}{C_2 \omega_2} \quad (61)$$

where  $Q_2 = \vec{V}_0 \cdot \vec{X}_2$  and  $\vec{X}_2 \cdot \vec{X}_2 C_2 =$ . The fact that  $B_i$  is purely imaginary confirms our earlier observation that real coefficients imply a non-zero initial displacement while purely imaginary coefficients imply a non-zero initial velocity. A purely imaginary  $B_i$  simply implies that the displacement has a  $\sin(\omega t)$  behavior in contrast to a  $\cos(\omega t)$  behavior, since the real part of  $e^{i\omega t} = \cos(\omega t)$ , while the real part of  $ie^{i\omega t} = -\sin(\omega t)$ . A solution for general initial conditions on  $x_1(0)$ ,  $x_2(0)$ ,  $\dot{x}_1(0)$  and  $\dot{x}_2(0)$  would be a linear combination of these solutions.

## Solution for Unequal Masses

If the masses,  $m_i$ , are not equal we must use a more general form of the eigenvalue problem. Returning to Equation (19) for equal masses.

For the case of equal masses, from Equation (19), this can be written

$$\begin{pmatrix} k_1/m + k_2/m & -k_2/m \\ -k_2/m & k_2/m + k_3/m \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \omega^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (62)$$

For unequal masses, we rewrite the equation as

$$\begin{pmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \omega^2 \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \quad (63)$$

This equation is in the form of the generalized characteristic or eigenvalue problem. (See Hildebrand; Methods of Applied Mathematics.)

$$\left( \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} - \lambda \begin{pmatrix} b_1 & 0 \\ 0 & b_2 \end{pmatrix} \right) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (64)$$

where equation (20) has been rewritten as (33) in the extended formulation  $([A] - \lambda[B])(X) = 0$ . Following Hildebrand we note that both  $[A]$  and  $[B]$  are symmetric matrices. Moreover,  $[B]$  is a diagonal matrix. Following the solution of the generalized characteristic or eigenvalue problem, by numerical or other technique, the resulting orthogonality condition between the "normal" modes is modified as

$$(X_{i,1}, X_{i,2}) \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} \begin{pmatrix} X_{j,1} \\ X_{j,2} \end{pmatrix} = 0 \quad (65)$$

for  $i \neq j$ . Thus the vectors of displacement for the normal modes of vibration must be multiplied by the mass distribution to result in orthogonality. A similar result is obtained for the vibration of a continuous system, such as a beam with non-uniform mass distribution. The process of expanding the solution in term of normal modes goes through as before with the modification of the normality condition.

In more general configurations,  $([A] - \lambda[B])(\mathbf{X}) = 0$ , the matrix  $[B]$  may not be diagonal, that is

$$[B] = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \quad (66)$$

In this case, it is straightforward to find the inverse of the matrix  $[B]$ ,  $[B]^{-1}$  and convert the governing equation to standard form by multiplying both terms by  $[B]^{-1}$ .

$$([B]^{-1}[A] - \lambda[B]^{-1}[B])(X) = ([B]^{-1}[A] - \lambda[I])(X) = 0 \quad (67)$$

where  $[I]$  is the identity matrix. With this manipulation, we obtain the standard form of the characteristic value problem. Solution to this problem will yield the eigenvalues  $\lambda_i$  and the eigenvectors,  $X_i$ . For the generalized eigenvalue problem, the eigenvectors are not orthogonal, thus  $X_i \cdot X_j \neq 0$ . This might seem somewhat surprising since the form of the governing equation is similar to the classic eigenvalue problem; however, the matrix  $[B]^{-1}[A]$  is typically not symmetric. Normal eigenvectors are an outcome of a symmetric matrix. Thus in general  $X_i \cdot X_j \neq 0$ . However, the eigenvectors are orthogonal when multiplied through the  $[B]$  matrix:  $X_i \cdot [B] \cdot X_j = 0$  for  $i \neq j$ . Thus the more general problem can be formulated using these "extended" eigenvectors.

## Observations

The process we have outlined for finding the solution to the initial value problem to a multi-degree of freedom system, outlined from equation(20) on, works for system with degrees of freedom from 2 to 20,000 and beyond. This approach is of fundamental importance in analyzing vibrations in a wide variety of systems. The expansion in normal modes and its extension is also useful in more complex problems such as forced motions at frequencies other than  $\omega_i$ .

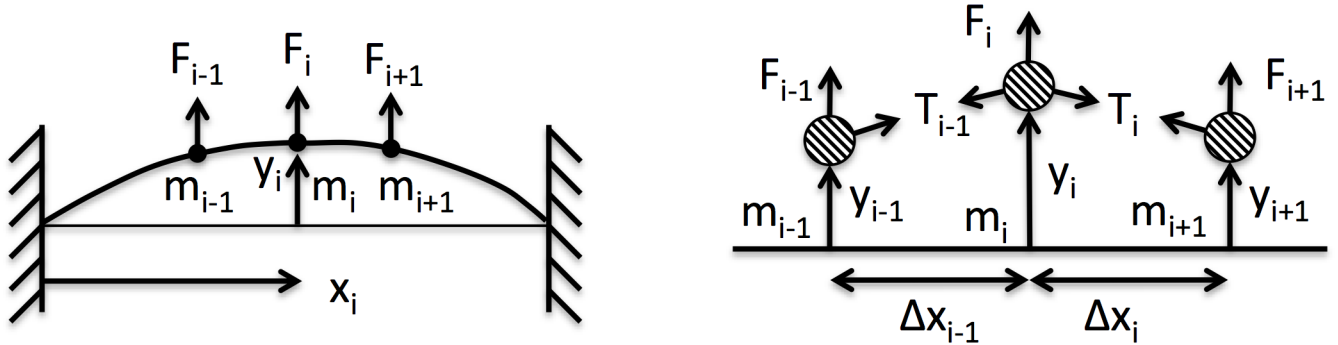
## References

- [1] Hildebrand: Methods of Applied Mathematics; for a discussion of the characteristic value problem, matrices and vectors.

## Lecture 23 — Continuous Dynamics

### Introduction

This lecture provides an introduction to the vibration dynamics of continuous systems such as tensioned strings, compression rods and bending beams. The approaches presented here can be extended to more complex structures such as solids, plates and shells. For complex geometries, the Finite Element Method (FEM) can be used, which is based upon the concepts presented herein. Think of continuous systems as similar to the discrete systems, described previously, with the exception that continuous systems have an infinite number of degrees of freedom (DOFs). To illustrate, consider a string under tension. We define a string as having distributed mass along its length and being able to support tension but not support an elastic bending motion. As shown in the figure below, discretize this string into a set of interconnected masses ( $m_i$ ) on which transverse forces ( $F_i$ ) can be applied and which interact with their neighboring masses through tension ( $T_i$ ) tangent to the string.



We can apply Newton's 2<sup>nd</sup> Law:

$$m_i \ddot{y}_i = T_i \frac{y_{i+1} - y_i}{\Delta x_i} - T_{i-1} \frac{y_i - y_{i-1}}{\Delta x_{i-1}} + F_i \quad (1)$$

for  $i = 2, 3, \dots, n-1$  assuming that  $\Delta x_{i-1}$  and  $\Delta x_i$  remain unchanged because  $y_i$  is small. In other words, we're invoking the small angle approximation. At the left and right ends of the string, we must enforce the boundary conditions ( $y_o = y_{n+1} = 0$  for a pinned condition). A pinned condition allows rotation but

no translation. Other boundary conditions could include springs, sliders, etc. as long as they act in the  $y$ -direction. We can rearrange this equation by defining  $y_{i+1} - y_i = \Delta y_i$  and  $y_i - y_{i-1} = \Delta y_{i-1}$ .

$$T_i \frac{\Delta y_i}{\Delta x_i} - T_{i-1} \frac{\Delta y_{i-1}}{\Delta x_{i-1}} + F_i = m_i \ddot{y}_i \quad (2)$$

for  $i = 1, 2, \dots, n$ . The first two terms give the incremental change in vertical tensile force between the left and right sides of  $m_i$ . Let's represent this incremental change with another  $\Delta$  and divide through by  $\Delta x_i$

$$\frac{1}{\Delta x_i} \left\{ \Delta \left( T_i \frac{\Delta y_i}{\Delta x_i} \right) + F_i = m_i \ddot{y}_i \right\} \quad (3)$$

for  $i = 1, 2, \dots, n$ . Rearranging

$$\frac{\Delta}{\Delta x_i} \left( T_i \frac{\Delta y_i}{\Delta x_i} \right) + \frac{F_i}{\Delta x_i} = \frac{m_i}{\Delta x_i} \ddot{y}_i \quad (4)$$

for  $i = 1, 2, \dots, n$ . Now let  $n \rightarrow \infty$ ,  $\Delta x_i \rightarrow 0$ , and  $m_i \rightarrow 0$ . In the limit,

$$\frac{\partial}{\partial x} \left[ T(x) \frac{\partial y(x, t)}{\partial x} \right] + f(x, t) = \rho(x) \frac{\partial^2 y(x, t)}{\partial t^2} \quad (5)$$

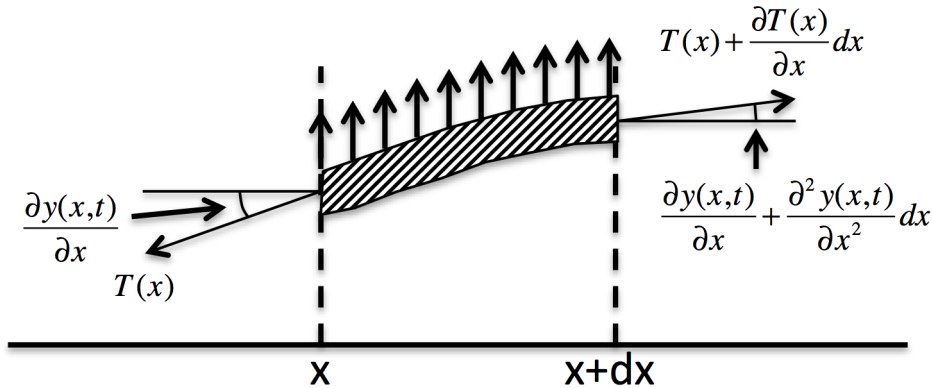
where

$$0 < x < L, f(x, t) = \lim_{\Delta x_i \rightarrow 0} \frac{F_i(t)}{\Delta x_i}, \rho(x) = \lim_{\Delta x_i \rightarrow 0} \frac{m_i}{\Delta x_i} \quad (6)$$

For this example, the boundary conditions are  $y(0, t) = y(L, T) = 0$ . This is known as the Boundary Value Problem where we have a partial differential equation and  $f(x, t)$  and  $\rho(x)$  are per unit length.

## Direct Derivation

To get a more direct derivation, we can use the figure below to recognize that the tension in the string and its angle relative to the  $x$ -axis changes across a differential element.



Summing the forces and applying Newton's 2<sup>nd</sup> Law,

$$\left[ T(x) + \frac{\partial T(x)}{\partial x} dx \right] \left[ \frac{\partial y(x, t)}{\partial x} + \frac{\partial^2 y(x, t)}{\partial x^2} dx \right] - T(x) \frac{\partial y(x, t)}{\partial x} + f(x, t) = \rho(x) dx \frac{\partial^2 y(x, t)}{\partial t^2} \quad (7)$$

Ignoring the second order terms (i.e., set  $dx^2 = 0$ ), we get Equation 5.

### Free Vibration: The Eigenvalue Problem

The free vibration problem assumes that the external force is zero (i.e.,  $f(x, t) = 0$ ). As we saw in the discrete free vibration problem, the vibration of a linear system can be separated into the vibration of distinct and orthogonal mode shapes which are then summed using the principle of superposition. Therefore, we seek a solution that separates the spatial shape  $Y(x)$  from the temporal motion  $Z(t)$ , called the Separation of Variables approach.

$$y(x, t) = Y(x)Z(t) \quad (8)$$

Substituting this solution into Equation 5 gives

$$\frac{1}{\rho(x)Y(x)} \frac{d}{dx} \left[ T(x) \frac{dY(x)}{dx} \right] = \frac{1}{Z(t)} \frac{d^2 Z(t)}{dt^2} \quad (9)$$

where the left side captures the spatial behavior while the right side captures the temporal behavior. Because each side captures a different behavior without coupling functions, they must equal the same constant. Since both sides have the units of frequency squared, let's use frequency ( $\omega$ ) as the constant.

$$\frac{d^2 Z(t)}{dt^2} + \omega^2 Z(t) = 0 \quad (10)$$

Is an harmonic oscillator and

$$-\frac{d}{dx} \left[ T(x) \frac{dY(x)}{dx} \right] = \omega^2 \rho(x) Y(x) \quad (11)$$

is valid over the range  $0 < x < L$ . The temporal solution is

$$Z(t) = C \cos(\omega t - \phi) \quad (12)$$

where  $C$  and  $\phi$  are determined by initial conditions. Assuming that  $\rho(x) = \rho$  is a constant and that  $T(x) = T$  is a constant, the spatial ODE is

$$\frac{d^2 Y(x)}{dx^2} + \beta^2 Y(x) = 0 \quad (13)$$

which is valid over the range  $0 < x < L$  with the boundary conditions  $Y(0) = Y(L) = 0$ . The Dispersion Relation

$$\beta^2 = \omega^2 \frac{\rho}{T} \quad (14)$$

relates the distinct spatial frequency (i.e., wavenumber, which is the inverse of wavelength) that can exist at a distinct temporal frequency. The solution is a spatial harmonic.

$$Y(x) = A \sin \beta x + B \cos \beta x \quad (15)$$



Satisfying the boundary conditions gives the infinite set of distinct eigenvalues as

$$\beta_r L = r\pi \text{ for } r = 1, 2, \dots \quad (16)$$

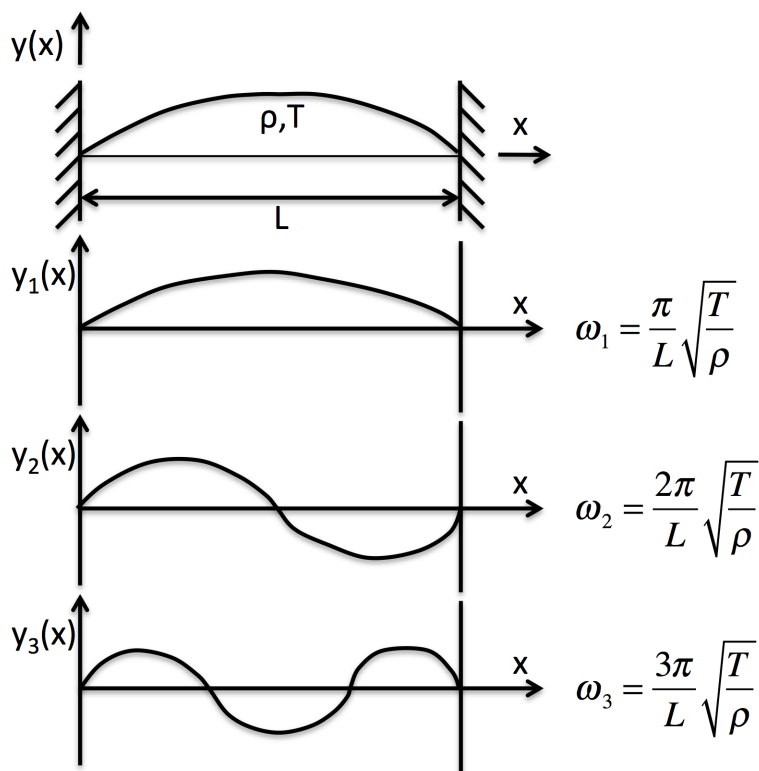
and eigenfunctions as

$$Y_r(x) = A_r \sin\left(\frac{r\pi x}{L}\right) \quad (17)$$

From Equation 14, the frequencies are

$$\omega_r = \beta_r \sqrt{\frac{T}{\rho}} = \frac{r\pi}{L} \sqrt{\frac{T}{\rho}} \quad (18)$$

where the term  $\sqrt{T/\rho}$  has units of velocity and is the transverse wave speed in the string. Notice that the wave speed is independent of frequency, making the string under tension a non-dispersive medium: complex shapes, made up by the superposition shapes with different spatial wavenumbers, preserve their shape as they propagate along the string. The figure below shows how the transverse displacement of the string can be decomposed into the sum of spatial eigenfunctions of wavelength  $\beta_r$  times temporal functions of frequency  $\omega_r$ . An equally valid way to view the eigenfunctions are as spatial wavelengths that, as they propagate along the string and reflect at the boundary conditions, constructively interfere.



An important attribute of an eigenfunction is that it is orthogonal to all other eigenfunctions

$$\int_0^L \rho(x) Y_r(x) Y_s(x) dx = 0 \quad (19)$$

for  $r \neq s$ , and

$$\int_0^L \rho(x) Y_r(x) Y_r(x) dx = \delta_r \quad (20)$$

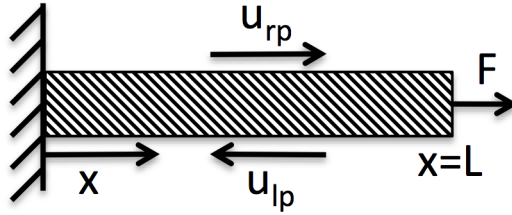
for  $r = 1, 2, \dots$  where  $\delta = 1$  is the Kronecker delta.

### Compression Rod: Alternate Solution

Consider the compression rod shown in the figure below. The motion we are going to consider involves compression waves that are in the  $x$ -direction that exhibit a deflection ( $u(x)$ ) and strain ( $du(x)/dx$ ). The governing partial differential equation of motion is

$$\frac{\partial}{\partial x} \left[ EA(x) \frac{\partial u}{\partial x} \right] = \rho A(x) \frac{\partial^2 u}{\partial t^2} \quad (21)$$

where  $E$  is the modulus of elasticity,  $A$  is the cross-sectional area, and  $\rho$  is the mass per unit volume. In this alternate solution, we are going to select a wave solution and assume that  $EA$  and  $\rho A$  are constants.



Consider the wave solution given by

$$u = U e^{ikx + i\omega t} \quad (22)$$

Substituting in to the governing EOM, we get the dispersion relation

$$-k^2 EA + \omega^2 \rho A = 0 \quad (23)$$

which gives a wave speed of

$$\frac{\omega}{k} = \sqrt{\frac{E}{\rho}} \quad (24)$$

As shown in the figure above, let's decompose the motion into two components: a leftward propagating wave ( $u_{lp}$ ) and a rightward propagating wave ( $u_{rp}$ ).

$$u = U_{rp} e^{-ikx + i\omega t} + U_{lp} e^{ikx + i\omega t} \quad (25)$$

To understand the direction of propagation, look at the exponent. For  $u_{lp}$ , to stay at a fixed point (or phase) on the wave requires that the exponent not change as time progresses forward. This requires that the location ( $x$ ) change in the negative, or leftward direction. For  $u_{rp}$ ,  $x$  must increase to the right to stay at the same point on the wave. Let's find a transformation from the wave amplitudes to the two energy storage states of position and strain in the governing differential equation

$$\begin{Bmatrix} u \\ \frac{\partial u}{\partial x} \end{Bmatrix} = \begin{bmatrix} 1 & 1 \\ ik & -ik \end{bmatrix} \begin{Bmatrix} u_{lp} \\ u_{rp} \end{Bmatrix} \quad (26)$$

We next need to define the boundary condition at  $x = L$  where we are going to attach a damper

$$F = -c\dot{u} = -i\omega cu \quad (27)$$

Inserting this boundary condition into the previous equation

$$\begin{bmatrix} 0 & EA \end{bmatrix} \begin{Bmatrix} u \\ \frac{\partial u}{\partial x} \end{Bmatrix} = F = \begin{bmatrix} -i\omega c & 0 \end{bmatrix} \begin{Bmatrix} u \\ \frac{\partial u}{\partial x} \end{Bmatrix} \quad (28)$$

Transforming into the wave domain

$$\begin{bmatrix} i\omega c & EA \end{bmatrix} \begin{bmatrix} 1 & 1 \\ ik & -ik \end{bmatrix} \begin{Bmatrix} u_{lp} \\ u_{rp} \end{Bmatrix} = 0 \quad (29)$$

At  $x = L$ , we can think of the rightward propagating wave as incoming to the boundary ( $u_{in} = u_{rp}$ ) and the leftward propagating wave as outgoing from the boundary ( $u_{out} = u_{lp}$ )

$$\begin{bmatrix} i\omega c + ikEA & i\omega c - ikEA \end{bmatrix} \begin{Bmatrix} u_{out} \\ u_{in} \end{Bmatrix} = 0 \quad (30)$$

Rearranging, we get the reflection coefficient ( $R$ ) of the boundary at  $x = L$ ,

$$u_{out} = \frac{kEA - \omega c}{kEA + \omega c} u_{in} = Ru_{in} \quad (31)$$

If the damper ( $c$ ) is zero,  $u_{out} = u_{in}$ . However, We can find a damper value that zeroes the reflection coefficient ( $R = 0$ ). The value is

$$c = A\sqrt{\rho E} \quad (32)$$

With  $R = 0$ , no waves reflect off of the boundary condition at  $x = L$ . This is called an Impedance Match because the damper of this specific value matches the impedance of the compression rod causing all of the energy to flow from the rod into the dashpot. This is also an important concept in the field of electrical power transmission where we do not want electrical power to reflect at power sub-stations.

**Thin Beam in Bending: Bernoulli-Euler Beams**

This problem entails the transverse displacement of a thin beam through bending. Again, we are going to use the separation of variables approach (i.e.,  $y(x, t) = Y(x)Z(t)$ ). The governing PDE is given by

$$\frac{\partial^2}{\partial x^2} \left[ EI(x) \frac{\partial^2 y(x, t)}{\partial x^2} \right] + \rho A(x) \frac{\partial^2 y(x, t)}{\partial t^2} = f(x, t) \quad (33)$$

where  $E$  is the modulus of elasticity,  $I$  is the area moment of inertia,  $\rho$  is the mass per unit volume, and  $A$  is the cross-sectional area of the beam. The spatial part of the problem becomes

$$\frac{d^2}{dx^2} \left[ EI(x) \frac{d^2 Y(x)}{dx^2} \right] = \omega^2 \rho A Y(x) \quad (34)$$

where  $0 < x < L$ ,  $EI$  and  $\rho A$  are assumed constant, and

$$\beta^4 = \omega^2 \sqrt{\frac{\rho A}{EI}} \quad (35)$$

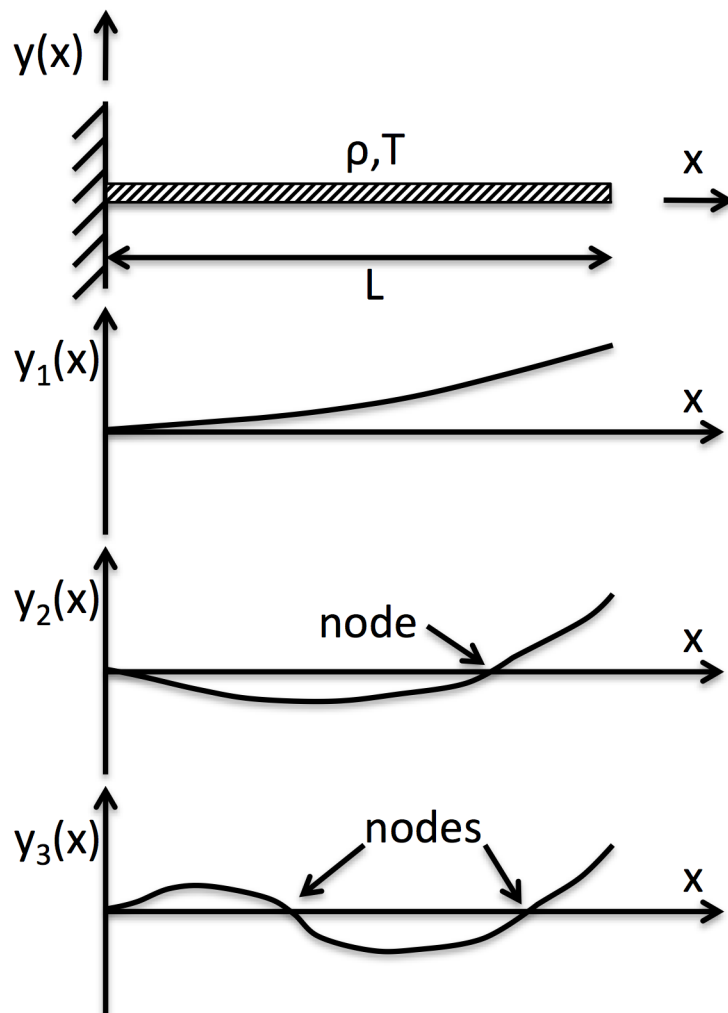
is the dispersion relation. The wave speed

$$\frac{\omega}{\beta} = \sqrt{\omega^4 \frac{EI}{\rho A}} \quad (36)$$

is dispersive: different wavelengths travel at different speeds. Notice that higher frequency, shorter wavelengths, have higher speeds. The eigenfunction solution now has the form

$$Y(x) = C_1 \sin \beta x + C_2 \cos \beta x + C_3 \sinh \beta x + C_4 \cosh \beta x \quad (37)$$

The figure below shows how the bending motion of a beam can be decomposed into the sum of specific shapes that vibrate at different frequencies. Each shape is made up of sinusoids and exponentials.



## Lecture 24 — Aeroelasticity: Classical Wing Flutter

### Introduction

The term “flutter” applies to a variety of self-excited vibrations in which aerodynamic forces act to energize structural modes of vibration. The structure can be a wing, a single panel of an aircraft fuselage, a turbine blade, a helicopter blade, or even a road sign in a gale force wind. Since flight vehicles are necessarily light, there is always the risk of encountering flutter phenomena in a new design, even if the usual static strength criteria are satisfied. When flutter occurs, its effects can be catastrophic. The science of Aeroelasticity is largely devoted to the prevention of flutter and related effects.

Here we treat a simplified form of torsion-bending wing flutter, also called “classical flutter.”

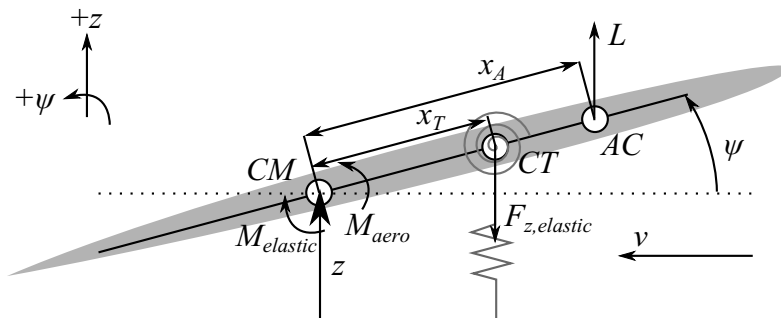
### Forces and Moments on a Deflected Wing

#### Elastic Forces

When an airplane wing is forced by aerodynamic loads, the result is a combination of bending and torsional deflections. In the theory of torsion-bending of beams of arbitrary cross-section it is shown that a point exists in the cross-section, called the Center of Torsion (CT), with the following two interrelated properties:

a) If the resultant of the applied loads goes through the CT, the beam deflects in pure bending, with no rotation.

b) If the applied load reduces to a pure moment, the resulting rotation of the cross-section is about the CT. Any set of transverse loads can be reduced to a force through the CT, plus a pure moment, and the net deformation can then be found from (a) and (b) above. Conversely, knowing the bending deflection of the CT, plus the section rotation, we can reconstruct the elastic net reaction force and torque of the beam.



Consider a wing section as shown in Figure 1, which is deflected from its equilibrium position (drawn here as being horizontal, although it may itself be rotated). The center of mass of the section (CM) is deflected by  $z$ , and the section is rotated through  $\psi$ , so that, for  $\psi \ll 1$ , the CT is deflected by  $z + x_T\psi$ , where  $x_T$  is the distance from the CT to the CM. Consequently, there will be a net downward elastic reaction force proportional to  $z + x_T\psi$ , and a net pitch-down elastic restoring torque about the CT proportional to  $\psi$ . Introducing the stiffness  $k_z$  and  $k_\psi$ , we write these reactions (force, plus torque about the CM) as

$$F_{z,elastic} = -k_z(z + x_T\psi) \quad (1)$$

$$M_{elastic} = -k_\psi\psi + F_{z,elastic}x_T = -k_\psi\psi - k_z(z + x_T\psi)x_T \quad (2)$$

Note that these are forces, moments and stiffnesses per unit span.

## Aerodynamic Forces

For purposes of deflection from an equilibrium setting, the change in lift of a wing section behaves as that of a flat plate rotated through the same angle. This lift is proportional to the relative wind's “dynamic head”  $q = (1/2)\rho V^2$ , to the planform area, and to the angle through which it is rotated. The planform area per unit span is the wing cord  $c$ , and we write for the lift per unit span

$$L = qca\psi \quad (3)$$

where  $a \approx 2\pi/\text{radian}$  is the lift-curve slope.

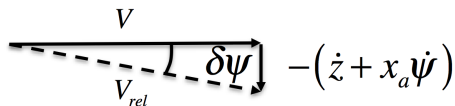


Figure 2: Rotation of relative wind.

This incremental lift acts at the profile's Aerodynamic Center (AC). In addition, the profile's curvature, or “camber” may produce a fixed lift component acting through some other point, but this should not concern us here (a more full discussion will follow in Lecture 34). The AC is near the point 25% of  $c$  from the leading

edge. In Equation 3, we wrote the angle of attack change due to the wing's rotation as  $\psi$ . But if the wing is flopping (i.e., if  $z$  and  $\psi$  vary in time), the relative wind has an additional rotation of its own. The AC moves vertically at a speed of  $\dot{z} + x_a \dot{\psi}$ , where  $x_a$  is the distance from the AC to the CM (Figure 1). Because of this, the incoming relative wind rotates (Figure 2) by

$$\delta\psi \approx -\frac{\dot{z} + x_a \dot{\psi}}{V} \quad (4)$$

and we should replace Equation 3 by

$$L = qca\left(\psi - \frac{\dot{z} + x_a \dot{\psi}}{V}\right) \quad (5)$$

In addition, this lift produces a pitch-up torque about the section CM of

$$M_{aero} = Lx_a \quad (6)$$

## Equations of Motion

Let  $m$  be the mass of the section per unit length, and  $I$  its moment of inertia, again per unit length. Assuming sections to act independent of each other (which is not really true), we can then collect the elastic and aerodynamic forces and moments together and write

$$m\ddot{z} = -k_z(z + x_T\psi) + qca\left(\psi - \frac{\dot{z} + x_a \dot{\psi}}{V}\right) \quad (7)$$

$$I\ddot{\psi} = -k_\psi\psi - k_z(z + x_T\psi)x_T + qca\left(\psi - \frac{\dot{z} + x_a \dot{\psi}}{V}\right)x_a \quad (8)$$

These are two coupled, second order, linear differential equations with constant coefficients. Let us seek solutions of the form

$$z = \bar{z}e^{st}; \psi = \bar{\psi}e^{st} \quad (9)$$

Substituting in Equations 7 and 8 and cancelling out  $e^{st}$ , we write the equations in the matrix form as

$$\begin{bmatrix} s^2 + \frac{qca}{mV}s + \frac{k_z}{m} & \frac{qca}{mV}x_a s + \frac{k_z}{m}x_T - \frac{qca}{m} \\ \frac{qca}{IV}x_a s + \frac{k_z}{I}x_T & s^2 + \frac{qca}{IV}x_a^2 s + \frac{k_\psi + k_z x_T^2}{I} - \frac{qca}{I}x_a \end{bmatrix} \begin{Bmatrix} \bar{z} \\ \bar{\psi} \end{Bmatrix} = 0 \quad (10)$$

For solutions other than  $\hat{z} = \hat{\psi} = 0$  to exist, the determinant of the matrix in Equation 10 must be zero. Expanding and collecting powers of  $s$  we find

$$s^4 + \frac{qca}{mV} \left(1 + \frac{mx_a^2}{I}\right) s^3 + \frac{1}{I} \left[k_\psi + k_z \left(x_T^2 + \frac{I}{m}\right) - qcax_a\right] s^2 + \frac{qca}{ImV} \left[k_\psi + k_z (x_a - x_T)^2\right] s + \frac{k_z}{mI} [k_\psi - qca(x_a - x_T)] = 0 \quad (11)$$

We are particularly interested in discovering whether the wing flapping/twisting motion will be stable or



not. The criteria developed in Lecture 28 are handy for this purpose. Recall that when the polynomial equation is  $s^4 + A_3s^3 + A_2s^2 + A_1s + A_0 = 0$ , the stability conditions are (a) all  $A_i$ 's positive and (b)  $A_1A_2A_3 > A_0A_3^2 + A_1^2$ . Here  $A_3$  and  $A_1$  are certainly positive, but at a high enough “dynamic head”, both  $A_2$  and  $A_0$  could become negative. Therefore, there are at least two critical speeds not to be exceeded (of which, of course, only the lower is relevant). Start with  $A_0$ ; we need

$$q < \frac{k_\psi}{ca(x_a - x_T)} \quad (12)$$

$A_0$  will always be positive if  $x_a < x_T$ , i.e., if the AC is placed behind the CT. Otherwise, there is stability below the speed given by Equation 12 and torsional divergence above it. The effect is easily understood; for instance if  $x_a < x_T$ , a pure positive torsion  $\psi$  introduces lift acting behind the center of torsion, which gives a restoring torque. If  $x_a > x_T$ , the lift acts ahead of the CT, and if the speed is great enough, this destabilizing torque cannot be countered by the wing's elastic reaction and the angle diverges (with unpleasant consequences). Turning now to  $A_2$ , the condition for stability demands

$$q < \frac{k_\psi + \left(\frac{I}{m} + x_T^2\right)k_z}{cax_a} \quad (13)$$

and if this dynamic head limitation is violated, the consequence is again an exponential divergence in both torsion and bending. The analysis of the constraint  $A_1A_2A_3 > A_0A_3^2 + A_1^2$  is more complicated, but it simplifies to a third limitation of the dynamic head, of the form

$$q < \frac{|k_\psi \frac{mx_a}{I} - k_z(x_a - x_T) \left(1 + \frac{m}{I}x_ax_T\right)|}{ca \left(1 + \frac{mx_a^2}{I}\right)} \quad (14)$$

which can be shown to be more restrictive than either of the other two (notice the smaller coefficient on  $k_\psi$  and the negative gain on  $k_z$ ). On the other hand, this restriction arises from the usually small  $A_1$  and  $A_3$  terms, and, as discussed in the previous lecture, might represent either a mild or strong oscillatory instability. To assess the degree of instability when Equation 14 is violated, it is useful to entirely neglect the aerodynamic damping effect represented by the additional relative wind angle of Equation 4. Tracing back the analysis, it is easy to see that this has the effect of eliminating the  $s^3$  and  $s$  terms in Equation 11, without affecting the others. The equation for  $s$  is now of the form

$$s^4 + A_2s^2 + A_0 = 0 \quad (15)$$

and, as shown in Lecture 28, a strong instability appears when the  $s^2$  roots become complex, i.e., when

$$\left(\frac{A_2}{2}\right)^2 < A_0 \quad (16)$$

This is a quadratic inequality in the dynamic head  $q$ . Rather than attempt a complete symbolic analysis, we will illustrate the behavior with a numerical example. But first, to facilitate calculations, we introduce non-dimensional variables as follows:

$$\begin{aligned}
\sigma &= \frac{s}{\sqrt{\frac{k_\psi}{I}}} \text{ (non-dimensional frequency)} \\
\beta &= \frac{\frac{k_z}{m}}{\frac{k_\psi}{I}} \text{ (ratio of bending to torsional frequency squared)} \\
\xi_a &= \frac{x_a}{\sqrt{\frac{I}{m}}}, \xi_T = \frac{x_T}{\sqrt{\frac{I}{m}}} \text{ (distances normalized by radius of gyration)} \\
Q &= \frac{qca\sqrt{\frac{I}{m}}}{k_\psi} \text{ (non-dimensional dynamic head)}
\end{aligned} \tag{17}$$

which reduces Equation 15 to

$$\sigma^4 + [1 + \beta(1 + \xi_T^2) - Q\xi_a]\sigma^2 + \beta[1 - Q(\xi_a - \xi_T)] = 0 \tag{18}$$

## Numerical Example

The usual high aspect-ratio wing has a rather low flapping frequency compared to its torsional frequency. We assume here that  $\beta = 0.1$ . The radius of gyration  $\sqrt{I/m}$  is perhaps about half of the cord, and  $x_a, x_T$  are fractions of that. Suppose  $\xi_a = 0.4$ ,  $\xi_T = 0.2$ . We then obtain

$$\sigma^4 + (1.104 - 0.4Q)\sigma^2 + 0.1(1 - 0.2Q) = 0 \tag{19}$$

The divergence limits are then

$$Q < 5 \text{ (for } A_0 > 0) \tag{20}$$

$$Q < 2.76 \text{ (for } A_2 > 0) \tag{21}$$

The condition  $(A_2/2)^2 > A_0$  gives, after collecting terms,

$$Q^2 - 5.02Q + 5.1176 > 0 \tag{22}$$

which requires that

$$Q < 1.423 \text{ or } Q > 3.758 \tag{23}$$

But, of course,  $Q > 3.397$  is ruled out by Equation 21. To assess the consequences of violating  $Q < 1.423$ , let us assume that  $Q = 2$  (below divergence limits). The equation is then  $\sigma^4 + 0.304 \sigma^2 + 0.06 = 0$ . The  $s^2$  roots are

$$s^2 = -0.152 \pm 0.192i$$

## REFERENCES

which have the square roots

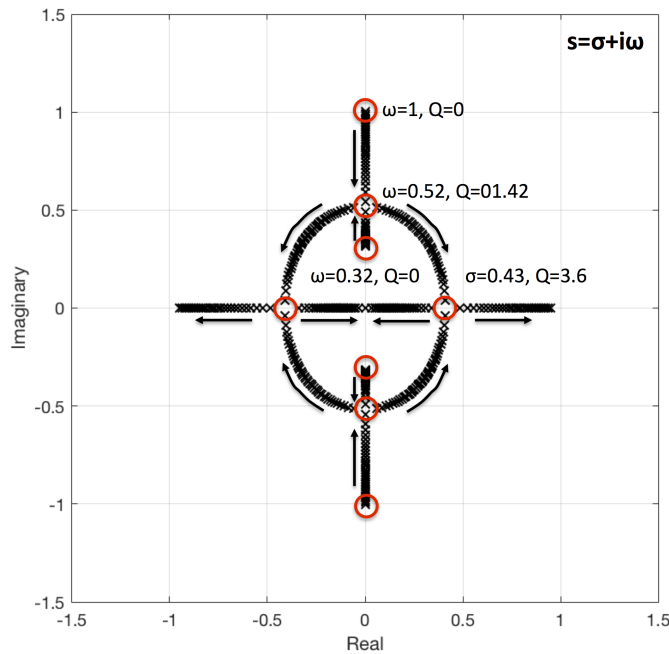
$$s_{1,2} = 0.216 \pm 0.446i$$

$$s_{2,3} = -0.216 \pm 0.446i \quad (24)$$

of which the first pair is a strongly unstable oscillation where the growth per cycle is

$$\exp\left(0.216 \times \frac{2\pi}{0.446}\right) = 20.91 \quad (25)$$

The roots of this quadratic equation are plotted as a function of  $Q$  in the figure below. Notice that they start at  $Q = 0$  on the imaginary axis, as four oscillatory poles, and coalesce at which point exponentially growing oscillatory flutter occurs. As  $Q$  gets higher, they again coalesce leading to exponential divergent flutter. An interesting feature of this “classical flutter” is revealed when  $\beta$  is allowed to vary over a range and the critical dynamic head  $Q_{cr}$  is calculated as above for each  $\beta$ . When the flapping and torsional frequencies are close to each other, the critical speed (or dynamic head) is very small, namely, the wing is the most susceptible to flutter. This might happen for all-control empennages or other short “wings,” and is due to the strong bending-torsion coupling occurring then.



## A Disclaimer

The theory presented here has many crudities in it, but perhaps the most important is the neglect of so-called “unsteady aerodynamics” effects. Since it takes a time of the order of  $V/c$  for the air to go over the wing

*REFERENCES*

profile, any oscillation happening faster than this might be influenced by this finite passage time, and the static lift and moment assumptions need to be revised. This is quite involved, however. The present theory is valid for slow oscillations only.

## Lecture 25 — Energy Methods: Lagrange's Equations

The motion of particles and rigid bodies is governed by Newton's law. In this section, we will derive an alternate approach, placing Newton's law into a form particularly convenient for multiple degree of freedom systems or systems in complex coordinate systems. This approach results in a set of equations called Lagrange's equations. They are the beginning of a complex, more mathematical approach to mechanics called analytical dynamics. In this course we will only deal with this method at an elementary level. Even at this simplified level, it is clear that considerable simplification occurs in deriving the equations of motion for complex systems. These two approaches—Newton's Law and Lagrange's Equations—are totally compatible. No new physical laws result for one approach vs. the other. Many have argued that Lagrange's Equations, based upon conservation of energy, are a more fundamental statement of the laws governing the motion of particles and rigid bodies. We shall not enter into this debate.

### Derivation of Lagrange's Equations in Cartesian Coordinates

We begin by considering the conservation equations for a large number ( $N$ ) of particles in a conservative force field using cartesian coordinates of position  $x_i$ . For this system, we write the total kinetic energy as

$$T = \sum_{i=1}^M \frac{1}{2} m_i \dot{x}_i^2. \quad (1)$$

where  $M$  is the number of degrees of freedom of the system.

For particles traveling only in one direction, only one  $x_i$  is required to define the position of each particle, so that the number of degrees of freedom  $M = N$ . For particles traveling in three dimensions, each particle requires 3  $x_i$  coordinates, so that  $M = 3 * N$ .

The momentum of a given particle in a given direction can be obtained by differentiating this expression with respect to the appropriate  $x_i$  coordinate. This gives the momentum  $p_i$  for this particular particle in this coordinate direction.

$$\frac{\partial T}{\partial \dot{x}_i} = p_i \quad (2)$$

The time derivative of the momentum is

$$\frac{d}{dt} \left( \frac{\partial T}{\partial \dot{x}_i} \right) = m_i \ddot{x}_i \quad (3)$$

For a conservative force field, the force on a particle is given by the derivative of the potential at the particle position in the desired direction.

$$F_i = -\frac{\partial V}{\partial x_i} \quad (4)$$

From Newton's law we have

$$F_i = \frac{dp_i}{dt} \quad (5)$$

Equating like terms from our manipulations on kinetic energy and the potential of a conservative force field, we write

$$\frac{d}{dt} \left( \frac{\partial T}{\partial \dot{x}_i} \right) = -\frac{\partial V}{\partial x_i} \quad (6)$$

Now we make use of the fact that

$$\frac{\partial T}{\partial x_i} = 0 \quad (7)$$

and

$$\frac{\partial V}{\partial \dot{x}_i} = 0 \quad (8)$$

Using these results, we can rewrite Equation (6) as

$$\frac{d}{dt} \left( \frac{\partial(T - V)}{\partial \dot{x}_i} \right) - \frac{\partial(T - V)}{\partial x_i} = 0 \quad (9)$$

We now define  $L = T - V$ :  $L$  is called the Lagrangian. Equation (9) takes the final form: Lagrange's equations in cartesian coordinates.

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{x}_i} \right) - \frac{\partial L}{\partial x_i} = 0 \quad (10)$$

where  $i$  is taken over all of the degrees of freedom of the system. Before moving on to more general coordinate systems, we will look at the application of Equation(10) to some simple systems.

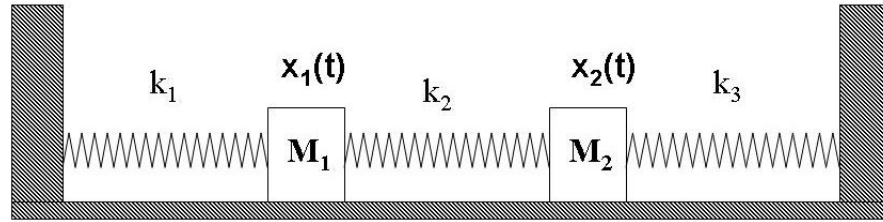
## Mass-spring System



We first consider a simple mass spring system. This is a one degree of freedom system, with one  $x_i$ . Its kinetic energy is  $T = (1/2)m\dot{x}^2$ ; its potential is  $V = (1/2)kx^2$ ; its Lagrangian is  $L = (1/2)m\dot{x}^2 - (1/2)kx^2$ . Applying Equation (10) to the Lagrangian of this simple system, we obtain the familiar differential equation for the mass-spring oscillator.

$$m \frac{d^2 x}{dt^2} + kx = 0 \quad (11)$$

Clearly, we would not go through a process of such complexity to derive this simple equation. However, let's consider a more complex system, governed by the same laws.



### Multi-degree of freedom systems

This is a two degree of freedom system, governed by two differential equations. There are three springs for this configuration. These governing equations could be obtained by applying Newton's Law to the force balance that exists at each mass due to the deflection of the springs as was done in Lecture 19. The deflection of springs 1 and 3 are influenced by the boundary condition at either end of the slot; in this case the deflection is zero.

The governing equations can also be obtained by direct application of Lagrange's Equation. This approach is quite straightforward. The expression for kinetic energy is

$$T = \sum_{n=1}^2 \frac{1}{2} m_i \dot{x}_i^2; \quad (12)$$

the expression for the potential is

$$V = \frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 (x_2 - x_1)^2 + \frac{1}{2} k_3 x_2^2 \quad (13)$$

Applying Lagrange's equation to  $T - V = L$

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{x}_1} \right) - \frac{\partial L}{\partial x_1} = 0 \quad (14)$$

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{x}_2} \right) - \frac{\partial L}{\partial x_2} = 0 \quad (15)$$

we obtain the governing equations as

$$m_1 \frac{d^2 x_1}{dt^2} = -k_1 x_1 + k_2 (x_2 - x_1) \quad (16)$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k_2 (x_2 - x_1) - k_3 x_2 \quad (17)$$

Clearly, for multi degree of freedom systems, this approach has advantages over the force balancing approach using Newton's law.

## Extension to General Coordinate Systems

A significant advantage of the Lagrangian approach to developing equations of motion for complex systems comes as we leave the cartesian  $x_i$  coordinate system and move into a general coordinate system. An example would be polar coordinates where for a two-dimensional position of a mass particle,  $x_1$  and  $x_2$  could be given by  $r$  and  $\theta$ . A two-degree of freedom system remains two-degree so that the number of coordinate variables required remains two.  $r$  and  $\theta$  and their counterparts in other coordinate systems will be referred to as **generalized coordinates**. We introduce quite general notation for the relationship between the  $n$  cartesian variables of position  $x_i$  and their description in **generalized coordinates**. (For some systems, the number of generalized coordinates is larger than the number of degrees of freedom and this is accounted for by introducing **constraints** on the system. This is an important part of the discussion of the Lagrange formulation. We shall not however develop these relations but will work directly with the number of variables equal to the number of degrees of freedom of the system.) We express the cartesian variable  $x_i$  using **generalized coordinates**  $q_j$ . (Polar coordinates  $r, \theta$  would be an example.)

$$x_i = x_i(q_1, \dots, q_j, \dots, q_M). \quad (18)$$

In the general case, each  $x_i$  could be dependent upon every  $q_j$ .

What is remarkable about the Lagrange formulation, is that (10) holds in a general coordinate system with  $x_i$  replaced by  $q_i$ .

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0 \quad (19)$$

Before showing how this result can be derived from Newton's Law, we show two applications in polar coordinates to demonstrate the power of the approach.

### Simple Pendulum by Lagrange's Equations

We first apply Lagrange's equation to derive the equations of motion of a simple pendulum in polar coordinates. This is a one degree of freedom system. However, it is convenient for later analysis of the double pendulum, to begin by describing the position of the mass point  $m_1$  with cartesian coordinates  $x_1$  and  $y_1$  and then express the Lagrangian in the polar angle  $\theta_1$ . Referring to a) in the figure below we have

$$x_1 = h_1 \sin \theta_1 \quad (20)$$

$$y_1 = -h_1 \cos \theta_1 \quad (21)$$

so that the kinetic energy is

$$T = \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2) = \frac{1}{2} m_1 h_1^2 \dot{\theta}_1^2 \quad (22)$$

The potential energy is

$$V = m_1 g y_1 = -m_1 g h_1 \cos \theta_1 \quad (23)$$



## REFERENCES

The Lagrangian is

$$L = T - V = \frac{1}{2}m_1h_1^2\dot{\theta}_1^2 + m_1gh_1 \cos \theta_1 \quad (24)$$

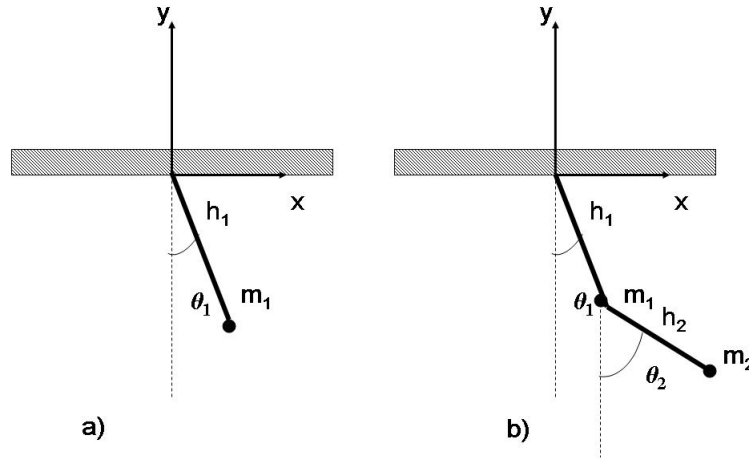
Applying (10) with  $q_1 = \theta_1$ , we obtain the differential equation governing the motion.

$$m_1h_1^2\ddot{\theta}_1 + m_1gh_1 \sin \theta_1 = 0 \quad (25)$$

Again, for such a simple system, we would typically not go through this formalism to obtain this result. However, this framework will enable us to derive the equations of motion for the more complex systems such as the double pendulum shown in b).

### Double Pendulum by Lagrange's Equations

Consider the double pendulum shown in b) consisting of two rods of length  $h_1$  and  $h_2$  with mass points  $m_1$  and  $m_2$  hung from a pivot. This system has two degrees of freedom:  $\theta_1$  and  $\theta_2$ .



To apply Lagrange's equations, we determine expressions for the kinetic energy and the potential as the system moves in angular displacement through the independent angles  $\theta_1$  and  $\theta_2$ . From the geometry we have

$$x_1 = h_1 \sin \theta_1 \quad (26)$$

$$y_1 = -h_1 \cos \theta_1 \quad (27)$$

$$x_2 = h_1 \sin \theta_1 + h_2 \sin \theta_2 \quad (28)$$

$$y_2 = -h_1 \cos \theta_1 - h_2 \cos \theta_2 \quad (29)$$

The kinetic energy is  $T = \frac{1}{2}m_1 (\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2}m_2 (\dot{x}_2^2 + \dot{y}_2^2)$ . Expressed in variables  $\theta_1$  and  $\theta_2$ , the kinetic energy of the system is

$$T = \frac{1}{2}m_1 h_1^2 \dot{\theta}_1^2 + \frac{1}{2}m_2 \left( h_1^2 \dot{\theta}_1^2 + h_2^2 \dot{\theta}_2^2 + 2h_1 h_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right) \quad (30)$$

The potential energy of the system is

$$V = m_1 g y_1 + m_2 g y_2 = -(m_1 + m_2) g h_1 \cos \theta_1 - m_2 g h_2 \cos \theta_2. \quad (31)$$

The Lagrangian is then

$$L = T - V = \frac{1}{2}(m_1 + m_2)h_1^2 \dot{\theta}_1^2 + \frac{1}{2}m_2 h_2^2 \dot{\theta}_2^2 + m_2 h_1 h_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) + (m_1 + m_2) g h_1 \cos \theta_1 + m_2 g h_2 \cos \theta_2 \quad (32)$$

Since the generalized coordinates are now  $\theta_1$  and  $\theta_2$ , Lagrange's equation becomes

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}_i} \right) - \frac{\partial L}{\partial \theta_i} = 0 \quad (33)$$

for both  $i = 1$  and  $i = 2$ .

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1} = 0 \quad (34)$$

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}_2} \right) - \frac{\partial L}{\partial \theta_2} = 0 \quad (35)$$

Working out the details, we have

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}_1} \right) = (m_1 + m_2) h_1^2 \ddot{\theta}_1 + m_2 h_1 h_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) - m_2 h_1 h_2 \dot{\theta}_2 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2) \quad (36)$$

$$\frac{\partial L}{\partial \theta_1} = -h_1 g (m_1 + m_2) \sin(\theta_1) - m_2 h_1 h_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \quad (37)$$

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}_2} \right) = m_2 h_2^2 \ddot{\theta}_2 + m_2 h_1 h_2 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) - m_2 h_1 h_2 \dot{\theta}_1 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2) \quad (38)$$

$$\frac{\partial L}{\partial \theta_2} = -h_2 g m_2 \sin(\theta_2) + m_2 h_1 h_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) \quad (39)$$

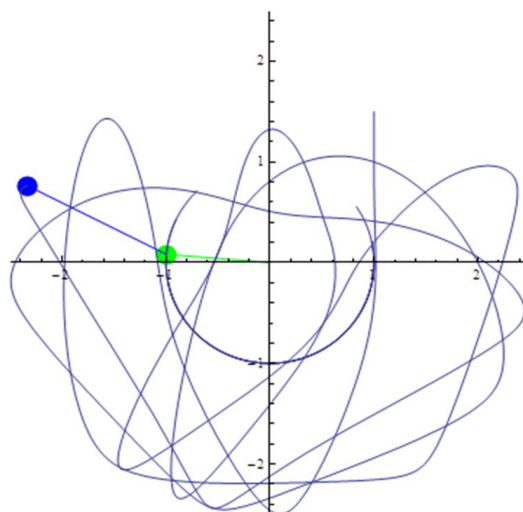
From Equations (35-39), we obtain the final form of the governing equations for the double pendulum

$$(m_1 + m_2) h_1 \ddot{\theta}_1 + m_2 h_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) + m_2 h_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + g(m_1 + m_2) \sin \theta_1 = 0 \quad (40)$$

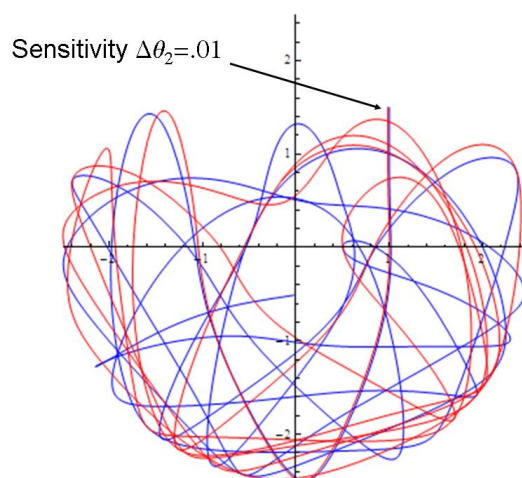
$$m_2 h_2 \ddot{\theta}_2 + m_2 h_1 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) - m_2 h_1 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + m_2 g \sin \theta_2 = 0 \quad (41)$$

The double pendulum is a system of great interest, displaying conventional linear multi degree of freedom system behavior for small  $\theta_1$  and  $\theta_2$ , but displaying chaotic behavior for large  $\theta$ . A chaotic system is a deterministic system that exhibits great sensitivity to the initial conditions: the "butterfly" effect. A simulation of the motion of a double pendulum is available on <http://scienceworld.wolfram.com/physics/DoublePendulum.html>. For a particular choice of initial conditions, the position of  $m_2$  with time is shown in the figure.

## REFERENCES



The sensitivity of the solution to initial conditions is shown below. In this case we compare two solutions that differ only by a  $\Delta\theta_2 = .01$ . One can see that initially the solutions are close; then gradually the solutions depart until they are far from coincident.



## Derivation of Lagrange's Equation for General Coordinate Systems

We now follow the earlier procedure we used to derive Lagrange's equation from Newton's law but using generalized coordinates instead of cartesian coordinates. (See additional reading: Slater and Frank and/or Marion and Thorton.)

The derivation of Lagrange's equation in cartesian coordinates was quite straightforward, since the acceleration of the  $x_i$  degree of freedom could be written  $a_i = d/dt(\partial T/\partial \dot{x}_i)$ . The force on the mass point due to the displacement  $x_i$  was directly  $F_i = -\partial V/\partial x_i$ , and the derivative of kinetic energy with respect to the displacement  $x_i$  was zero:  $\partial T/\partial x_i = 0$ . These simplifications are no longer true in generalized coordinates. The cartesian variables are expressed in terms of the generalized coordinates as

$$x_i = x_i(q_1, q_2, \dots, q_M). \quad (42)$$

To obtain the velocity  $\dot{x}_i$ , we take the time derivative of (42), applying the chain rule.

$$\dot{x}_i = \frac{dx_i}{dt} = \sum_{j=1}^M \frac{\partial x_i}{\partial q_j} \frac{dq_j}{dt} = \sum_{j=1}^M \frac{\partial x_i}{\partial q_j} \dot{q}_j \quad (43)$$

Taking the partial derivative of (43) with respect to  $\dot{q}_j$  we obtain a relation between these two derivatives,

$$\frac{\partial \dot{x}_i}{\partial \dot{q}_j} = \frac{\partial x_i}{\partial q_j}, \quad (44)$$

a result which we shall need shortly. In deriving equation (44), we take advantage of the fact that since the generalized coordinates  $q_i$  are independent,  $\frac{\partial q_i}{\partial q_j} = 0$  for  $i \neq j$ .

Following the approach leading to Equations (2-10), we define a **generalized momentum** as

$$p_i = \frac{\partial T}{\partial \dot{q}_i} \quad (45)$$

with  $T = \sum_{j=1}^M \frac{1}{2} m_j \dot{x}_j^2$  as given by Equation (1).

We are now ready to express Newton's law in the **generalized coordinates**  $q_i$  that we have introduced. For the **generalized momentum** we have

$$p_i = \frac{\partial T}{\partial \dot{q}_i} = \sum_{j=1}^M m_j \dot{x}_j \frac{\partial \dot{x}_j}{\partial \dot{q}_i} = \sum_{j=1}^M m_j \dot{x}_j \frac{\partial x_j}{\partial q_i}, \quad (46)$$

where we have made use of (44).

For a conservative force field, the work done during the displacements of all the  $dx_i$ 's is given by

$$dW = \sum_{i=1}^M F_i dx_i \quad (47)$$

or expressed in the generalized coordinates  $q_j$

$$dW = \sum_{i=1}^M F_i dx_i = \sum_{j=1}^M \sum_{i=1}^M F_i \frac{\partial x_i}{\partial q_j} dq_j \quad (48)$$

where we have used the identity

$$dx_i = \sum_{j=1}^M \frac{\partial x_i}{\partial q_j} dq_j. \quad (49)$$

We identify

$$\sum_{i=1}^M F_i \frac{\partial x_i}{\partial q_j} dq_j \quad (50)$$

as the work done by a "displacement" through  $dq_j$  and define a generalized force  $Q_j$  as

$$Q_j = \sum_{i=1}^M F_i \frac{\partial x_i}{\partial q_j} \quad (51)$$

so that the work is expressed as.

$$dW = \sum_{j=1}^M Q_j dq_j \quad (52)$$

For a conservative system, the work done by all the small displacements ( $dq_j$ 's) is

$$dW = -dV = - \sum_{j=1}^M \frac{\partial V}{\partial q_j} dq_j, \quad (53)$$

where  $V$  is the potential function expressed in the coordinate system of the generalized coordinates and  $\frac{\partial V}{\partial q_j}$  is the change in potential due to a change in the generalized coordinate  $q_j$ . So that the **generalized force** is

$$Q_j = - \frac{\partial V}{\partial q_j} \quad (54)$$

in analogy with (4). We now examine the time derivative of the **generalized momentum**,  $p_i$  (Equation(46)).

$$\frac{dp_i}{dt} = \frac{d}{dt} \left( \frac{\partial T}{\partial \dot{q}_i} \right) = \frac{d}{dt} \sum_{j=1}^M m_j \dot{x}_j \frac{\partial \dot{x}_j}{\partial \dot{q}_i} = \frac{d}{dt} \sum_{j=1}^M m_j \dot{x}_j \frac{\partial x_j}{\partial q_i} = \sum_{j=1}^M \left( m_j \ddot{x}_j \frac{\partial x_j}{\partial q_i} + m_j \dot{x}_j \frac{d}{dt} \left( \frac{\partial x_j}{\partial q_i} \right) \right). \quad (55)$$

Since  $\partial x_j / \partial q_i$  is a function of the  $q_k$ 's which are functions of time, we have by the chain rule

$$\frac{d}{dt} \left( \frac{\partial x_j}{\partial q_i} \right) = \sum_{k=1}^M \frac{\partial^2 x_j}{\partial q_i \partial q_k} \dot{q}_k \quad (56)$$

Therefore equation (57) becomes

$$\frac{dp_i}{dt} = \frac{d}{dt} \left( \frac{\partial T}{\partial \dot{q}_i} \right) = \sum_{j=1}^M \left( m_j \ddot{x}_j \frac{\partial x_j}{\partial q_i} + m_j \dot{x}_j \sum_{k=1}^M \frac{\partial^2 x_j}{\partial q_i \partial q_k} \dot{q}_k \right). \quad (57)$$

From Newton's law,  $m_j \ddot{x}_j = F_j$ , so that the first term in (55) is given by

$$\sum_{j=1}^M m_j \ddot{x}_j \frac{\partial x_j}{\partial q_i} = \sum_{j=1}^M F_j \frac{\partial x_j}{\partial q_i} = Q_i = - \frac{\partial V}{\partial q_i} \quad (58)$$

We write

$$\dot{x}_j = \frac{dx_j}{dt} = \sum_{k=1}^M \frac{\partial x_j}{\partial q_k} \frac{dq_k}{dt} = \sum_{k=1}^M \frac{\partial x_j}{\partial q_k} \dot{q}_k, \quad (59)$$

using  $k$  as the summation variable. We expand the quantity  $\frac{\partial T}{\partial q_i}$  as

$$\frac{\partial T}{\partial q_i} = \sum_{j=1}^N m_j \dot{x}_j \frac{\partial \dot{x}_j}{\partial q_i} = \sum_{j=1}^N m_j \dot{x}_j \frac{\partial}{\partial q_i} \sum_{k=1}^N \frac{\partial x_j}{\partial q_k} \dot{q}_k \quad (60)$$

This is precisely the second term in equation (57). We have now identified simpler forms for the two expressions in the equation for the time derivative of the **generalized momentum**. We may now write

$$\frac{dp_i}{dt} = \frac{d}{dt} \left( \frac{\partial T}{\partial \dot{q}_i} \right) = Q_i + \frac{\partial T}{\partial q_i} = -\frac{\partial V}{\partial q_i} + \frac{\partial T}{\partial q_i} \quad (61)$$

Since for a conservative system, the potential is independent of the velocities,  $\partial V / \partial \dot{q}_i = 0$ , Equation (61) states that

$$\frac{d}{dt} \left( \frac{\partial T}{\partial \dot{q}_i} \right) = -\frac{\partial V}{\partial q_i} + \frac{\partial T}{\partial q_i} \quad (62)$$

We can place this equation into final form by using the definition of the Lagrangian,  $L = T - V$ , to obtain the final form of Lagrange's equation as

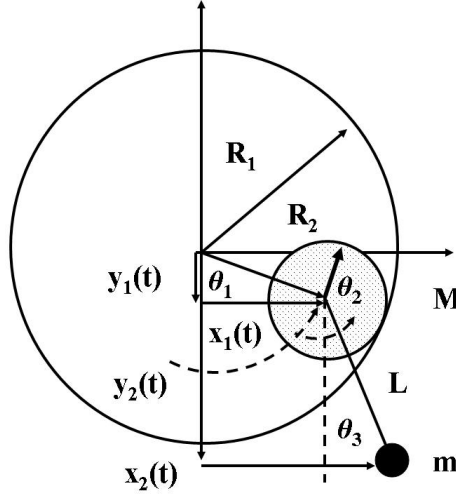
$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0 \quad (63)$$

in agreement with (19). Equation (63) is Lagrange's Equations in **generalized coordinates**. In our previous example, we applied this equation to simple and double pendulums in polar coordinates using  $q_1 = \theta_1$  and  $q_2 = \theta_2$ . What is significant about this equation, adding to its power, is that each  $i$  equation contains only derivatives with respect to that  $q_i$  and  $\dot{q}_i$ .

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### Example Problems

We consider three problems which provide examples of the power of Lagrange's equation to provide the governing equations for complex systems. In these cases, although we could obtain the full non-linear equations, except for the final example, we are interested only in the small amplitude motions for which the equations reduce to their linear form. These systems are ideal to demonstrate the power of the Lagrange approach having multiple degrees of freedom and complex relationships between the geometries and the kinetic and potential energies. We first consider a small solid cylinder of radius  $R_2$  and mass  $M$  rolling without slip in a larger cylinder of radius  $R_1$ . The larger cylinder is rolling on a flat, horizontal surface without slipping and gravity is down in the figure. This kinematic constraint provides a relationship between  $\theta_1$  and  $\theta_2$  such that  $\theta_2 = -(R_1/R_2)\theta_1$ ; in other words, the small cylinder rotates faster as it rolls around the larger cylinder. Because of this constraint, the small cylinder is not an additional degree of freedom but is coupled to its position within the larger cylinder. A pendulum of length  $L$  is pinned, yet free to rotate, at the CM of the smaller cylinder. The generalized coordinates characterizing the degrees of freedom are  $q_1 = \theta_1$  and  $q_2 = \theta_3$ .



The position of the center of mass of the small cylinder is given by

$$x_1(t) = (R_1 - R_2) \sin \theta_1 \quad (64)$$

$$y_1(t) = -(R_1 - R_2) \cos \theta_1 \quad (65)$$

The position of the pendulum mass is given by

$$x_2(t) = x_1(t) + L \sin \theta_3 = (R_1 - R_2) \sin \theta_1 + L \sin \theta_3 \quad (66)$$

$$y_2(t) = y_1(t) - L \cos \theta_3 = -(R_1 - R_2) \cos \theta_1 - L \cos \theta_3 \quad (67)$$

The Lagrangian is formed by taking the difference of the kinetic energy  $T$  and the potential energy  $V$ . We now form these terms. For kinetic energy we have dependence upon the velocity of the centers of mass of the two objects and the kinetic energy of rotation of the small cylinder about its center of mass. The moment of inertia of the small cylinder about its center of mass is  $I_G = (1/2)MR_2^2$ . Since its angular velocity  $\dot{\theta}_2$  is given by  $\dot{\theta}_2 = -\dot{\theta}_1(R_1/R_2)$ , its kinetic energy in rotation is given by  $T_R = (1/2)I_G\dot{\theta}_1^2(R_1/R_2)^2$ . Therefore, the total kinetic energy of the system is

$$T = \frac{1}{2}(I_G(R_1/R_2)^2)\dot{\theta}_1^2 + \frac{1}{2}M(\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2}m(\dot{x}_2^2 + \dot{y}_2^2) \quad (68)$$

with  $x_i(t)$  and  $y_i(t)$  expressed in terms of the variable  $\theta_i$  etc. We have

$$T = \frac{1}{2}((I_G(R_1/R_2)^2 + (M + m)(R_1 - R_2)^2)\dot{\theta}_1^2 + 2Lm(R_1 - R_2) \cos(\theta_1 - \theta_3)\dot{\theta}_1\dot{\theta}_3 + L^2m\dot{\theta}_3^2) \quad (69)$$

Although complex, one has a high degree of confidence that the right expression has been obtained. The expression for the gravitational potential is easily obtained as

$$V = Mgy_1(t) + mgy_2(t) = -(M + m)g(R_1 - R_2) \cos \theta_1 - mgL \cos \theta_3 \quad (70)$$

This system is very similar to the double pendulum; the new feature is the small rolling cylinder which adds to the kinetic energy but does not contribute an additional degree of freedom. Since we are only interested in the dynamics of the system for small angles  $\theta$ , we expand the kinetic and potential energies for small  $\theta$  before applying Lagrange's equation. Expanding  $\cos \theta \rightarrow 1 - \theta^2/2$  and  $\sin \theta \rightarrow \theta$  we obtain the expression for kinetic energy to  $O(\theta^2)$  as

$$T = \frac{1}{2}(I_G(R_1/R_2)^2 + (m + M)(R_1 - R_2)^2)\dot{\theta}_1^2 + 2Lm(R_1 - R_2)\dot{\theta}_1\dot{\theta}_3 + mL^2\dot{\theta}_3^2 \quad (71)$$

and the expression for  $V(\theta)$  as

$$V = -gLm - g(m + M)(R_1 - R_2) + \frac{1}{2}g(m + M)(R_1 - R_2)\theta_1^2 + \frac{1}{2}gLm\theta_3^2 \quad (72)$$

The Lagrangian formed from the kinetic and potential energies becomes

$$L = gLm + gm(R_1 - R_2) + gM(R_1 - R_2) - \frac{1}{2}g(M + m)(R_1 - R_2)\theta_1^2 - \frac{1}{2}gLm\theta_3^2 + \frac{1}{2}((m + M)(R_1 - R_2)^2 + I_G(R_1/R_2)^2)\dot{\theta}_1^2 + Lm(R_1 - R_2)\dot{\theta}_1\dot{\theta}_3 + \frac{1}{2}L^2m\dot{\theta}_3^2$$

Applying Lagrange's equation  $\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1} = 0$  and  $\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}_3} \right) - \frac{\partial L}{\partial \theta_3} = 0$  we obtain the governing differential equations as

$$(I_G(R_1/R_2)^2 + (M + m)(R_1 - R_2)^2)\ddot{\theta}_1 + Lm(R_1 - R_2)\ddot{\theta}_3 + (M + m)g(R_1 - R_2)\theta_1 = 0 \quad (73)$$

$$mL(R_1 - R_2)\ddot{\theta}_1 + L^2m\ddot{\theta}_3 + mgL\theta_3 = 0 \quad (74)$$

to obtain the governing linear equations for a two degree of freedom system. The solution for natural frequencies and vibration modes is found by applying the techniques introduced in Lecture 19. These equations can be put into the form for a generalized eigenvalue problem

$$[A] - \lambda[B] = 0 \quad (75)$$

where

$$A = \begin{bmatrix} g(M + m)(R_1 - R_2) & 0 \\ 0 & mgL \end{bmatrix} \quad (76)$$

$$B = \begin{bmatrix} I_G \left( \frac{R_1}{R_2} \right)^2 + (M + m)(R_1 - R_2)^2 & mL(R_1 - R_2) \\ mL(R_1 - R_2) & mL^2 \end{bmatrix} \quad (77)$$

To form the characteristic eigenvalue problem we multiply the equation by the inverse of the  $B$  matrix

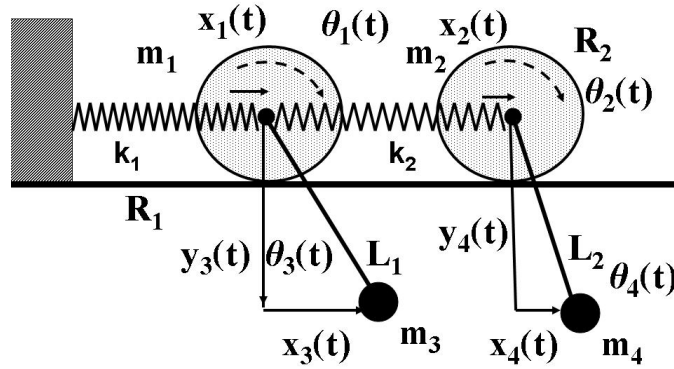
$$[B]^{-1}[A] - \lambda[I] = 0 \quad (78)$$



## REFERENCES

where  $[I]$  is the identity matrix. This will result in eigenvalues  $\lambda_1, \lambda_2$ , and 2 eigenvectors  $X_1, X_2$ . For the generalized eigenvalue problem, the eigenvectors are not orthogonal, thus  $X_i \cdot X_j \neq 0$ . This might seem somewhat surprising since the form of the governing equation is similar to the classic eigenvalue problem; however, the matrix  $[B]^{-1}[A]$  is typically not symmetric. Normal modes are an outcome of a symmetric matrix. Thus, in general,  $X_i \cdot X_j \neq 0$ . However, they are orthogonal when multiplied through by the  $[B]$  matrix:  $X_i \cdot [B] \cdot X_j = 0$  for  $i \neq j$ .

A second example is shown below. This is a four degree of freedom system. Two cylinders, restrained by springs, with freely-rotating pendula attached to their centers. The four degrees of freedom are the positions of the cylinders,  $x_1(t)$  and  $x_2(t)$ , and the angle of the two pendula  $\theta_1(t)$  and  $\theta_2(t)$ . Knowing these variables as functions of time gives us the position and velocity of every element of the system. The rolling without slip provides a relationship between  $\theta$  and  $x$  as  $\theta_1 = -x_1/R_1$  and  $\theta_2 = -x_2/R_2$ .



As before, the relation between the  $x$  and  $y$  positions of the pendula are given by

$$x_3(t) = x_1 + L_1 \sin \theta_3 \quad (79)$$

$$y_3(t) = -L_1 \cos \theta_3 \quad (80)$$

$$x_4(t) = x_2 + L_2 \sin \theta_4 \quad (81)$$

$$y_4(t) = -L_2 \cos \theta_4 \quad (82)$$

The kinetic energy is

$$T = \frac{1}{2} (I_{1G}(-\dot{x}_1/R_1)^2 + M_1 \dot{x}_1^2 + I_{2G}(-\dot{x}_2/R_2)^2 + M_2 \dot{x}_2^2 \quad (83)$$

$$+ m_3(\dot{x}_3^2 + \dot{y}_3^2) + m_4(\dot{x}_4^2 + \dot{y}_4^2)) \quad (84)$$

whereas the potential energy is

$$V = m_3 g y_3 + m_4 g y_4 + \frac{1}{2} k_1 x_1^2 + \frac{1}{2} k_2 (x_2 - x_1)^2 \quad (85)$$

Again, to obtain the small disturbance equations, we expand these expressions for small  $x$  and  $\theta$  to  $O(\theta^2)$ , for the Lagrangian, and apply Lagrange's equation. For small  $\theta_i$ , using  $\sin \theta \rightarrow \theta$  and  $\cos \theta \rightarrow 1 - \theta^2/2$ , the kinetic and potential energies expand to

$$T = \frac{1}{2}L_1^2 m_3 \dot{\theta}_3^2 + \frac{1}{2}L_2^2 m_4 \dot{\theta}_4^2 + L_1 m_3 \dot{\theta}_3 \dot{x}_1 + \frac{1}{2}M_1 \dot{x}_1^2 + \frac{1}{2}m_3 \dot{x}_1^2 + (I_{1G} \dot{x}_1^2)/(2R_1^2) \quad (86)$$

$$+ L_2 m_4 \dot{\theta}_4 \dot{x}_2 + \frac{1}{2}M_2 \dot{x}_2^2 + \frac{1}{2}m_4 \dot{x}_2^2 + (I_{2G} \dot{x}_2^2)/(2R_2^2) \quad (87)$$

$$V = -gL_1 m_3 - gL_2 m_4 + \frac{1}{2}gL_1 m_3 \theta_3^2 + \frac{1}{2}gL_2 m_4 \theta_4^2 \quad (88)$$

$$+ \frac{1}{2}k_1 x_1^2 + \frac{1}{2}k_2 x_1^2 - k_2 x_1 x_2 + \frac{1}{2}k_2 x_2^2 \quad (89)$$

The governing equations that result for this four degree of freedom system are

$$k_1 x_1 + k_2 x_1 - k_2 x_2 + L_1 m_3 \ddot{\theta}_3 + (M_1 + m_3) \ddot{x}_1 + I_{1G} \ddot{x}_1/R_1^2 = 0 \quad (90)$$

$$-k_2 x_1 + k_2 x_2 + L_2 m_4 \ddot{\theta}_4 + (M_2 + m_4) \ddot{x}_2 + I_{2G} \ddot{x}_2/R_2^2 = 0 \quad (91)$$

$$gL_1 m_3 \theta_3 + L_1^2 m_3 \ddot{\theta}_3 + L_1 m_3 \ddot{x}_1 = 0 \quad (92)$$

$$gL_2 m_4 \theta_4 + L_2^2 m_4 \ddot{\theta}_4 + L_2 m_4 \ddot{x}_2 = 0 \quad (93)$$

These equations can be put into the form for a generalized eigenvalue problem

$$[A] - \lambda[B] = 0 \quad (94)$$

where

$$A = \begin{bmatrix} k_1 + k_2 & -k_2 & 0 & 0 \\ -k_2 & k_2 & 0 & 0 \\ 0 & 0 & m_3 g L_1 & 0 \\ 0 & 0 & 0 & m_4 g L_2 \end{bmatrix} \quad (95)$$

$$B = \begin{bmatrix} M_1 + m_3 + I_{1G}/R_1^2 & 0 & m_3 L_1 & 0 \\ 0 & M_2 + m_4 + I_{2G}/R_2^2 & 0 & m_4 L_2 \\ m_3 L_1 & 0 & m_3 L_1^2 & 0 \\ 0 & m_4 L_2 & 0 & m_4 L_2^2 \end{bmatrix} \quad (96)$$

The solution proceeds by taking the inverse of the matrix  $[B]$  and forming the characteristic eigenvalue problem

$$[B]^{-1}[A] - \lambda[I] = 0 \quad (97)$$

where  $[I]$  is the identity matrix. This will result in 4 eigenvalues  $\lambda_1, \lambda_2, \lambda_3, \lambda_4$  and 4 eigenvectors  $X_1, X_2, X_3, X_4$ . For the generalized eigenvalue problem, the eigenvectors are not orthogonal, thus  $X_i \cdot X_j \neq 0$ . However, as previously discussed, they are orthogonal when multiplied through the  $[B]$  matrix:  $X_i \cdot [B] \cdot X_j = 0$  for  $i \neq j$ .

An additional example is a system of two pendula of equal length  $L$  connected by a linear spring with spring

constant  $k$ , referred to as a compound-spring pendulum; a two degree of freedom system. We first derive the equation governing the full non-linear equations of motion; we then develop the approximation for small amplitude oscillation. This is a two degree of freedom system with generalized coordinates  $\theta_1$  and  $\theta_2$ . The relation between the  $x$  and  $y$  displacements of the pendula are

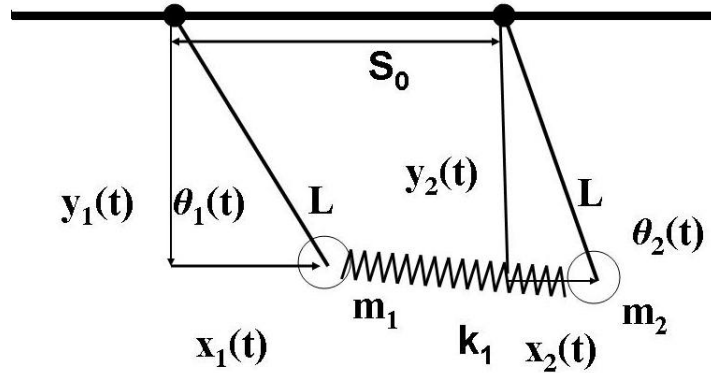
$$x_1(t) = L \sin \theta_1 \quad (98)$$

$$y_1(t) = -L \cos \theta_1 \quad (99)$$

$$x_2(t) = L \sin \theta_2 \quad (100)$$

$$y_2(t) = -L \cos \theta_2 \quad (101)$$

The pendula are located a distance  $S_0$  apart; we take the equilibrium position of the spring as  $S_0$ .



The potential is the sum of the gravitational potential due to the displacements of the pendula plus the potential energy stored in the spring. For the spring ( $V_{spring}$ ); and gravity ( $V_g$ ); with the total  $V_{total} = V_{spring} + V_g$

$$V_{spring} = \frac{1}{2}k(s - s_o)^2 = \frac{1}{2}k(\sqrt{(S_0 + x_2(t) - x_1(t))^2 + (y_2(t) - y_1(t))^2} - S_0)^2 \quad (102)$$

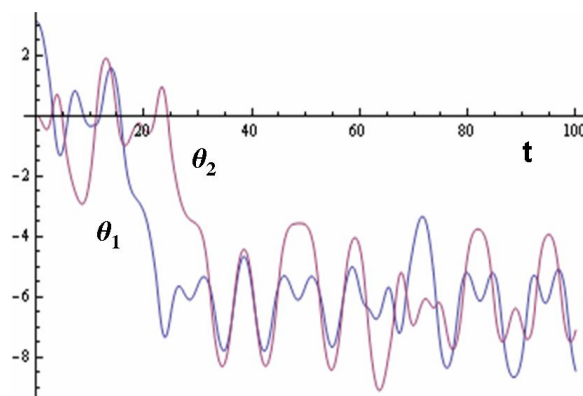
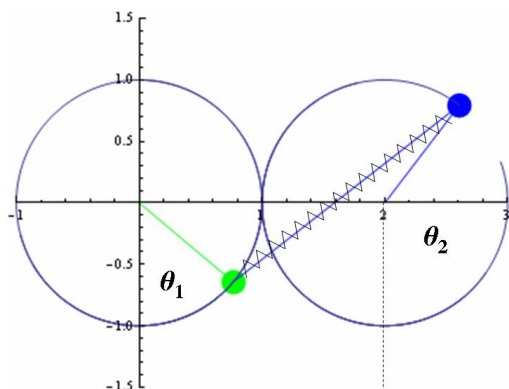
$$V_g = -gm_1L \cos \theta_1(t) - gm_2L \cos \theta_2(t) \quad (103)$$

The kinetic energy is

$$T = \frac{1}{2}m_1L^2\dot{\theta}_1(t)^2 + \frac{1}{2}m_2L^2\dot{\theta}_2(t)^2 \quad (104)$$

The two non-linear equations are obtained directly from Lagrange's equation. They are very complex and little insight would be gained by writing them down. They can however be solve numerically. Such a solution is shown below for initial values  $\theta_1 = \pi$ ,  $\dot{\theta}_1 = 0$ ,  $\theta_2 = 0$  and  $\dot{\theta}_2 = .1$ . Like the double pendulum problem, the compound pendulum solution is chaotic.

A typical instantaneous solution is shown along with the time behavior of  $\theta_1$  and  $\theta_2$ .



To obtain the governing equations for small amplitude motion, we expand the potential and kinetic energy for small  $\theta$  to  $O(\theta^2)$  using  $\sin \theta \rightarrow \theta$  and  $\cos \theta \rightarrow 1 - \theta^2/2$ .

$$V_{spring} = \frac{1}{2}kL^2(\theta_1^2 - 2\theta_1\theta_2 + \theta_2^2) \quad (105)$$

$$V_{gravity} = -(m_1 + m_2)gL + \frac{1}{2}m_1gL\theta_1^2 + \frac{1}{2}m_2gL\theta_2^2 \quad (106)$$

$$T = \frac{1}{2}(m_1L^2\dot{\theta}_1^2 + m_2L^2\dot{\theta}_2^2) \quad (107)$$

Applying Lagrange's equation we obtain the equations for small amplitude motion as

$$L^2m_1\ddot{\theta}_1 + kL^2\theta_1 - kL^2\theta_2 + gLm_1\theta_1 = 0 \quad (108)$$

$$L^2m_2\ddot{\theta}_2 + kL^2\theta_2 + gLm_2\theta_2 - kL^2\theta_1 = 0 \quad (109)$$

The eigenvalues and eigenvectors that form the solution can be found by applying the techniques introduced

## REFERENCES

in Lecture 19.

$$[B]^{-1}[A] - \lambda[I] = 0 \quad (110)$$

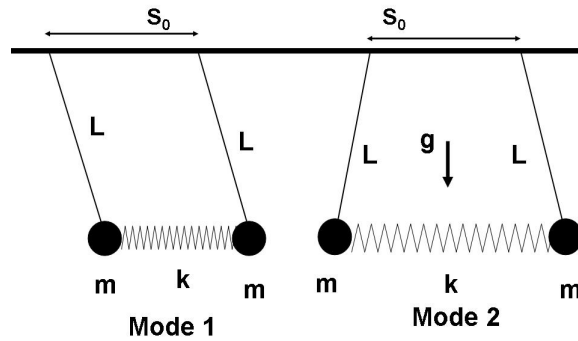
where

$$A = \begin{bmatrix} kL^2 + m_1gL & -kL^2 \\ -kL^2 & kL^2 + m_2gL \end{bmatrix} \quad (111)$$

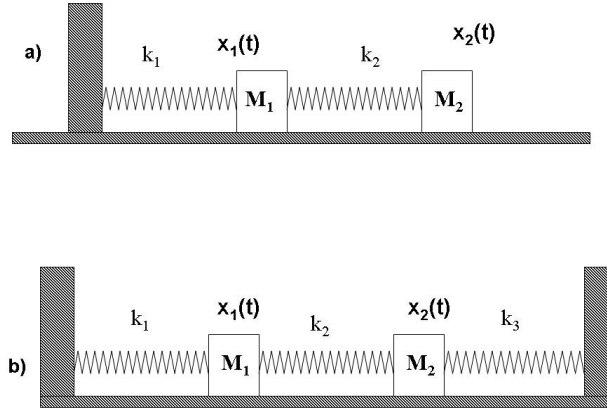
$$B = \begin{bmatrix} m_1L^2 & 0 \\ 0 & m_2L^2 \end{bmatrix} \quad (112)$$

The eigenvalues are  $g/L$  and  $(kL(m_1 + m_2) + gm_1m_2)/Lm_1m_2$ ; the vibration modes are  $\Theta_1 = \{1, 1\}$  and  $\Theta_2 = \{-m_2/m_1, 1\}$ . The modes are not orthogonal: that is  $\Theta_1 \cdot \Theta_2 \neq 0$ . As discussed in Lecture 19; the modes are normalized when multiplied through the matrix  $B = \{\{m_1, 0\}, \{0, m_2\}\}$  so that  $\Theta_1 \cdot [B] \cdot \Theta_2 = 0$ .

For equal masses  $m_1$  and  $m_2$ , the two modes of oscillation are shown. In the first mode, the pendula oscillate together and the spring is undeformed. The frequency is that of an ordinary single pendulum. In the second mode, the pendula oscillate in opposite directions; the spring is stretched; the frequency of the second mode is higher.

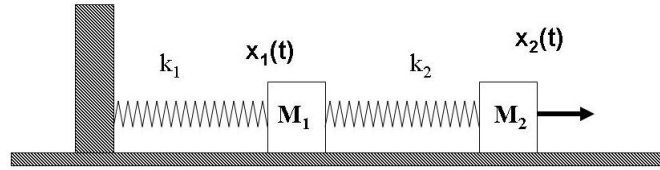


A final example of vibration is the mass spring system introduced in Lecture 19.



**Multi-degree of freedom systems**

In Lecture 19, we took advantage of the symmetry of system, considered equal masses  $m_1 = m_2 = m$ , and equal spring constants  $k_1 = k_2 = k$ , b) to obtain simple normal modes  $\{X_1\} = \{1, 1\}$ ,  $\{X_2\} = \{1, -1\}$ . We now consider a more general case a) where these simplifications are absent.



From Lecture 19, we have the equations of motion

$$m_1 \frac{d^2 x_1}{dt^2} = -k_1 x_1 + k_2 (x_2 - x_1) \quad (113)$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k_2 (x_2 - x_1) \quad (114)$$

This leads to an  $[A]$  matrix  $\{\{k_1 + k_2, -k_2\}, \{-k_2, k_2\}\}$  and a  $[B]$  matrix  $\{\{m_1, 0\}, \{0, m_2\}\}$ . This cannot be transformed into  $[A] - \lambda[I]$  and therefore we must use the generalized eigenvalue form  $[B]^{-1}[A] - \lambda[I] = [GA] - \lambda[I] = 0$  where  $[GA]$  is the generalized solution matrix,  $[GA] = \{\{(k_1 + k_2)/m_1, -k_2/m_1\}, \{-k_2/m_2, k_2/m_2\}\}$ . This matrix generated two eigenvalues and two eigenvectors; the eigenvectors are not orthogonal. However when multiplied through  $[B]$ , they are orthogonal. (The eigenvalues  $\lambda$  may be found by setting the determinant of the matrix  $[[GA] - \lambda[I]]$  equal to zero. For a two degree of freedom system, two values of  $\lambda$  are obtained. Each value of  $\lambda$  is then sequentially substituted into one of the governing equations, and a relationship between  $x_1$  and  $x_2$  is obtained. Normalizing the magnitude of this relation results in the mode shape for that frequency.)

Various levels of investigation can be carried out. We will assume  $k_1 = k_2$ , to obtain some simplification.

For this case the  $[A]$  matrix becomes  $[A] = \{\{2k, -k\}, \{-k, k\}\}$ ; the  $[B]$  matrix becomes  $[B] = \{\{m_1, 0\}, \{0, m_2\}\}$ ; and the  $[GA]$  matrix becomes  $[GA] = \{\{2k/m_1, -k/m_1\}, \{-k/m_2, k/m_2\}\}$ . For this case, the eigenvectors

are

$$X_1 = \{-(-km_1 + 2km_2 - \sqrt{-4k^2m_1m_2 + (-km_1 - 2km_2)^2})/(2km_1), 1\} \quad (115)$$

$$X_2 = \{-(-km_1 + 2km_2 + \sqrt{-4k^2m_1m_2 + (-km_1 - 2km_2)^2})/(2km_1), 1\} \quad (116)$$

with eigenvalues

$$\lambda_1 = \omega_1^2 = (km_1 + 2km_2 - \sqrt{-4k^2m_1m_2 + (-km_1 - 2km_2)^2})/(2m_1m_2) \quad (117)$$

$$\lambda_2 = \omega_2^2 = (km_1 + 2km_2 + \sqrt{-4k^2m_1m_2 + (-km_1 - 2km_2)^2})/(2m_1m_2) \quad (118)$$

These eigenvectors are not orthogonal, but become orthogonal when multiplied through  $[B]$ .  $\Theta_1.[B].\Theta_2 = 0$ .

The final simplification occurs when  $m_1 = m_2$ . In the case  $[A] = \{\{2k, -k\}, \{-k, k\}\}$  and  $[B] = \{\{m, 0\}, \{0, m\}\}$  leading to  $[GA] = \{\{2k/m, -k/m\}, \{-k/m, k/m\}\}$ . In this case, the normal modes are

$$X_1 = \{1/2(-1 + \sqrt{5}), 1\} \quad (119)$$

$$X_2 = \{-1/2(1 + \sqrt{5}), 1\} \quad (120)$$

with natural frequencies

$$\lambda_1 = \omega_1^2 = 3k/m - \sqrt{5}k/(2m) \quad (121)$$

$$\lambda_2 = \omega_2^2 = 3k/m + \sqrt{5}k/(2m) \quad (122)$$

In this case, the eigenvectors are orthogonal,  $X_1.X_2 = 0$  but because of the lack of symmetry in the problem, they are not the simple  $\{\{1, 1\}, \{1, -1\}\}$  of configuration a).

#### ADDITIONAL READING

Slater and Frank, *Mechanics*, Chapter IV

Marion and Thorton, Chapter 7

## Lecture 26 — 2D Rigid Body Dynamics: Impulse and Momentum; Pendulums

Last lecture, we saw the principle of impulse and momentum applied to particle motion. This principle was of particular importance when the applied forces were functions of time and when interactions between particles occurred over very short times, such as with impact forces. In this lecture, we extend these principles to two dimensional rigid body dynamics.

### Impulse and Momentum Equations

#### Linear Momentum

In lecture 11, we introduced the equations of motion for a two dimensional rigid body. The linear momentum for a system of particles is defined as

$$\mathbf{L} = m\mathbf{v}_G ,$$

where  $m$  is the total mass of the system, and  $\mathbf{v}_G$  is the velocity of the center of mass measured with respect to an inertial reference frame. Assuming the mass of the system to be constant, we have that the sum of the external applied forces to the system,  $\mathbf{F}$ , must equal the change in linear momentum,  $\dot{\mathbf{L}} = \mathbf{F}$ , or, integrating between times  $t_1$  and  $t_2$ ,

$$\mathbf{L}_2 - \mathbf{L}_1 = \int_{t_1}^{t_2} \mathbf{F} dt \quad (1)$$

Note that this is a vector equation and therefore must be satisfied for each component separately. Expression 1 is particularly useful when the precise time variation of the applied forces is unknown, but their total impulse can be calculated. Of course, when the impulse of the applied forces is zero, the momentum is conserved and we have  $(\mathbf{v}_G)_2 = (\mathbf{v}_G)_1$ .

#### Angular Momentum

A similar expression to 1 can be derived for the angular momentum if we start from the principle of conservation of angular momentum,  $\dot{\mathbf{H}}_G = \mathbf{M}_G$ . Here,  $\mathbf{H}_G = I_G\omega$  is the angular momentum about the center of



mass, and  $\mathbf{M}_G$  is the moment of all externally applied forces about the center of mass. Integrating between times  $t_1$  and  $t_2$ , we have,

$$(\mathbf{H}_G)_2 - (\mathbf{H}_G)_1 = \int_{t_1}^{t_2} \mathbf{M}_G dt . \quad (2)$$

In a similar manner, for rotation about a fixed point  $O$ , we can write,

$$(\mathbf{H}_O)_2 - (\mathbf{H}_O)_1 = \int_{t_1}^{t_2} \mathbf{M}_O dt , \quad (3)$$

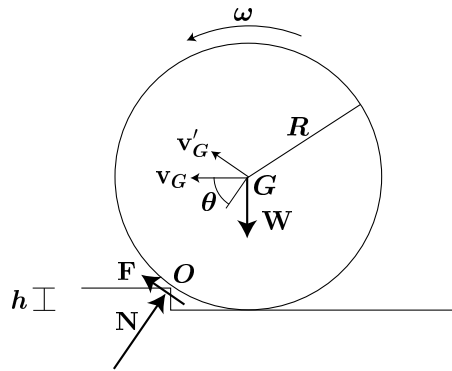
where  $\mathbf{H}_O = I_O \boldsymbol{\omega}$ , the moment of inertia,  $I_O$ , refers to the fixed point  $O$ , and the external moments are with respect to point  $O$ .

Finally, if the external applied moment is zero, then we have conservation of angular momentum, which implies  $\omega_2 = \omega_1$ .

### Example

### Cylinder rolling over a step

We consider a disk of radius  $R$  rolling over a flat surface, and striking a rough step of height  $h$ .



#### Initial Impact

First, we consider the situation an instant before and an instant after the impact with the step. Before the impact, the velocity of the center of mass,  $\mathbf{v}_G$ , is in the horizontal direction and has a magnitude of  $v_G = \omega R$ . After the cylinder hits the step, it starts turning around point  $O$ , (the notation  $O$  will be used throughout this discussion for the instantaneous point of contact) and therefore the velocity of the center of mass,  $\mathbf{v}_G$ , must be perpendicular to the segment  $OG$ . Clearly, this change in velocity direction happens over a very short time interval and is caused by the impulsive forces generated during the impact. Since we are assuming a rough step, the point on the cylinder which is in contact with the step at  $O$  has zero velocity (no slipping) during this short interval. Therefore, we can use equation 3 about point  $O$  to characterize the change in  $\mathbf{v}_G$ . The angular momentum of the cylinder about  $O$  before impact is

$$H_O = I_G \omega + m R \sin \theta v_G = \left[ \frac{I_G}{R} + m(R - h) \right] v_G .$$

The angular momentum just after impact is,

$$H'_O = I_O \omega' = I_O \frac{v'_G}{R} = \left[ \frac{I_G}{R} + mR \right] v'_G .$$

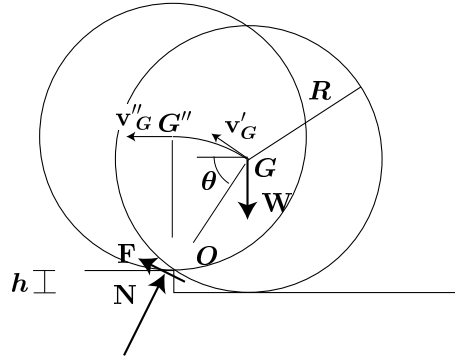
In these equations,  $\omega$  and  $\omega'$  are the angular velocities before and after the impact, respectively, and  $m$  is the mass of the cylinder. The change in angular momentum, according to equation 3, is equal to the angular impulse about  $O$ . Clearly, the normal and tangential forces,  $N$  and  $F$ , generate no moment about  $O$ . The only other force is the weight,  $\mathbf{W}$ , which is not an impulsive force. Hence, the impulse generated over a very short time interval,  $t_2 \rightarrow t_1$ , will be negligible. Thus,  $H_O = H'_O$ , which implies that,

$$v'_G = \frac{I_G + mR(R-h)}{I_G + mR^2} v_G = \frac{I_G + mR(R-h)}{I_O} v_G = \left(1 - \frac{Rh}{k_O^2}\right) v_G , \quad (4)$$

where  $k_O = \sqrt{I_O/m}$  is the radius of gyration of the cylinder about  $O$ .

#### Rolling about $O$

After the initial impact, the cylinder rolls about point  $O$ . During this process, since there are no dissipation mechanisms, energy will be conserved. Therefore, we can use the conservation of energy principle and require that the sum of the potential and kinetic energies remains constant.



The change in potential energy,  $mgh$ , is obtained by a decrease in the kinetic energy, thus

$$\frac{1}{2} I_O \omega'^2 = mgh + \frac{1}{2} I_O \omega''^2 ,$$

or,

$$v''_G^2 = v'_G^2 - 2gh \frac{R^2}{k_O^2} .$$

The residual velocity after the cylinder has climbed the step,  $v''_G$ , can be expressed in terms of the initial velocity,  $v_G$ , using 4, as,

$$v''_G^2 = \left(1 - \frac{Rh}{k_O^2}\right)^2 v_G^2 - 2gh \frac{R^2}{k_O^2} .$$

If we want to determine the minimum initial velocity,  $(v_G)_{min}$ , that would allow the cylinder to climb the

step, we set the residual velocity,  $v_G''$ , to zero and obtain,

$$(v_G)_{min}^2 = \frac{2gh \frac{R^2}{k_O^2}}{(1 - \frac{Rh}{k_O^2})^2} .$$

### Rebounding

So far, we have assumed that the cylinder pivots around point  $O$  without losing contact with the step. It is clear that if the velocity  $v_G'$  is very large, then, as soon as the turning starts, a force will be required to produce the centripetal acceleration. For small velocities, the weight should be sufficient, but, for larger velocities, it is possible that the normal reaction force,  $N$ , will become zero, and the cylinder will lose contact with the step. The normal force,  $N$ , can be calculated from the normal momentum equation. That is, the sum of the forces in the direction of  $OG$  should equal the centripetal acceleration,

$$W \sin \theta - N = m \frac{v_G'^2}{R} .$$

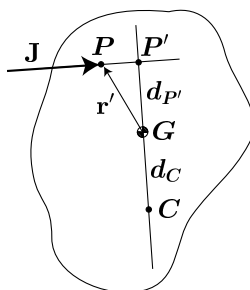
Setting  $W = mg$ ,  $\sin \theta = (R - h)/R$ , and  $N = 0$ , we can determine the minimum velocity,  $(v_G')_{rebound}$ , that will produce separation. The resulting expression can be combined with equation 4 to obtain an expression for the minimum initial velocity required for separation,

$$(v_G)_{rebound}^2 = \frac{g(R - h)}{(1 - \frac{Rh}{k_O^2})^2} .$$

We note that for  $h/(R - h) < k_O^2/(2R^2)$ ,  $(v_G)_{min} < (v_G)_{rebound}$ , and, in this case, there is a range of velocities for which it is possible for the cylinder to climb the step without rebounding.

## Center of Percussion Relative to an Instantaneous Center of Motion

In some situations, it is of interest to determine how a body should be impulsively set in motion such that a certain prescribed point—the point  $C$ —will be (at least momentarily) the instantaneous center of motion.



Consider a rigid body of mass  $m$  which is initially at rest. An impulse  $\mathbf{J}$  is applied at  $t = 0$  at point  $P$  in the body. The application of  $\mathbf{J}$  to the body initiates both translational and rotational motion. Thus,

$$\begin{aligned} m\mathbf{v}_G &= \mathbf{J} , \\ I_G\boldsymbol{\omega} &= \mathbf{r}' \times \mathbf{J} . \end{aligned}$$

The modulus of the initial angular velocity is  $\omega = d_{P'}J/I_G$ , and the modulus of the velocity of the center of mass is  $v_G = J/m$ . We now wish to find out the point  $P$  such that a prescribed point  $C$  in the body is the instantaneous center of motion. If  $C$  is the center of motion, then  $\mathbf{v}_G = \boldsymbol{\omega} \times \mathbf{r}_{CG}$ , or, in magnitude  $v_G = \omega d_C$ . Therefore,

$$d_C = \frac{v_G}{\omega} = \frac{J/m}{d_{P'}J/I_G} = \frac{I_G}{md_{P'}} ,$$

or,

$$d_P' = \frac{I_G}{md_C} = \frac{k_G^2}{d_C} .$$

or

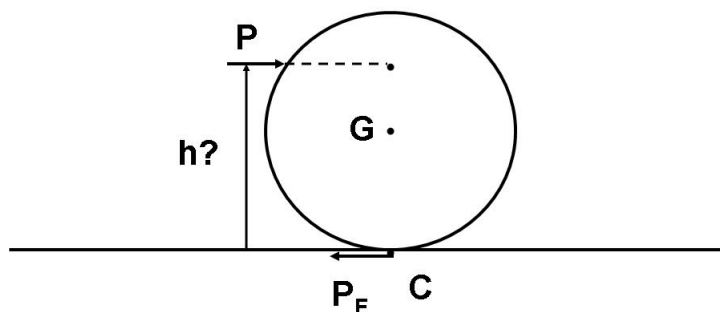
$$d_C d_P' = k_G^2$$

in other words, for any body, the product of  $d_P'$  the distance between the point of application of an impulse and  $d_C$ , the center of motion is a property of the body, the radius of gyration squared. Note that, since the impulsive moment is a vector, the same result would be obtained if the impulse  $\mathbf{J}$  were applied anywhere along the line through  $P'$  in the direction of  $\mathbf{J}$ . The point  $P'$  is called the *center of percussion* associated with the center of motion  $C$ .

### Example

### Striking a billiard ball

Consider a billiard ball resting on a table. In the general case, if we apply an impulse at a point  $h$  above the table, there will be a reaction friction impulse with the table and the ball will roll. However, this process is unreliable, in that the coefficient of friction might not be large enough to produce a smooth motion. For improved accuracy we want to hit the ball at a height  $h$  such that **no fiction force** is required. Therefore we want to know at what height above the table we have to hit a billiard ball so that no friction impulse occurs and the ball rolls on the table without slipping.



The moment of inertia of a homogeneous sphere about its center of mass is  $I_G = (2/5)mR^2$ . We take moments about the instantaneous center of rotation, the contact point  $C$  between the ball and the table. By the parallel axis theorem, the moment of inertia about  $C$  is  $I_C = 7/5MR_0^2$ . The governing equations are: 1) relate linear impulse to the change in linear momentum for the center of mass,  $MV_G = P - P_F$ ; 2) relate the moment of the impulse to the change in angular momentum taken about the point  $O$ ,  $Ph = I_O\omega$ ; 3) use the geometric relationship,  $v_G = -R_0\omega$ . We obtain the following result for the friction force required if the ball is to roll smoothly on the table.

$$P_F = P\left(1 - h\frac{5}{7R_0}\right) \quad (5)$$

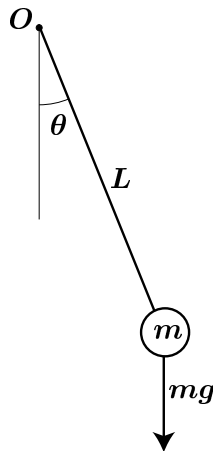
therefore, if  $h = \frac{7}{5}R_0$ , no friction force will occur and the ball will more reliably roll smoothly. For our earlier discussion of center of percussion, we see that the point  $h$  is the center of percussion relative to the contact (instantaneous center of rotation) point  $O$ .

## Pendulum

A pendulum is a rigid body suspended from a fixed point (hinge) which is offset with respect to the body's center of mass. If all the mass is assumed to be concentrated at a point, we obtain the idealized *simple pendulum*. Pendulums have played an important role in the history of dynamics. Galileo identified the pendulum as the first example of synchronous motion, which led to the first successful clock developed by Huygens. This clock incorporated a feedback mechanism that injected energy into the oscillations (the escapement, a mechanism used in timepieces to control movement and to provide periodic energy impulses to a pendulum or balance) to compensate for friction losses. In addition to horology (the science of measuring time), pendulums have important applications in gravimetry (the measurement of the specific gravity) and inertial navigation.

## Simple Pendulum

Consider a simple pendulum of mass  $m$  and length  $L$ .



The equation of motion can be derived from the conservation of angular momentum about the hinge point,  $O$ ,

$$I_O \ddot{\theta} = -mgL \sin \theta .$$

Since the moment of inertia is simply  $I_O = mL^2$ , we obtain the following non-linear equation of motion,

$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0 . \quad (6)$$

Multiplying this equation by  $\dot{\theta}$ , we can write,

$$\frac{d}{dt} \left( \frac{1}{2} \dot{\theta}^2 - \frac{g}{L} \cos \theta \right) = 0 ,$$

which implies that  $\dot{\theta}^2 - (2g/L) \cos \theta = \text{constant}$ . Setting  $\theta = \theta_{max}$ , when  $\dot{\theta} = 0$  we have,

$$\dot{\theta} = \pm \sqrt{\frac{2g}{L}} \sqrt{(\cos \theta - \cos \theta_{max})} .$$

This equation cannot be integrated further in an explicit manner. Its solution must be expressed in terms of, so called, elliptic functions. The period of the oscillation,  $T$ , is obtained by multiplying by four the time it takes for the pendulum to go from  $\theta = 0$  to  $\theta = \theta_{max}$ . Thus,

$$T = \frac{4}{\sqrt{(2g/L)}} \int_0^{\theta_{max}} \frac{d\theta}{\sqrt{\cos \theta - \cos \theta_{max}}} .$$

Again, this is an integral which cannot be evaluated explicitly, but can be approximated, assuming that  $\theta_{max}$  is not very large, as (the algebra is omitted here),

$$T \approx 2\pi \sqrt{\frac{L}{g}} \left( 1 + \frac{\theta_{max}^2}{16} \right) . \quad (7)$$

## Small amplitude approximation

If we assume that the amplitude of pendulum's oscillation is small, then  $\sin \theta \approx \theta$ , and the equation of motion, given by 6, becomes linear,

$$\ddot{\theta} + \frac{g}{L}\theta = 0 \quad . \quad (8)$$

This expression is much simpler than equation 6, and has solutions of the form,

$$\theta = A \sin \omega_n t + B \cos \omega_n t \quad ,$$

where  $\omega_n = \sqrt{g/L}$  is the natural frequency of oscillation. It is clear that these solutions are periodic, and the period is given by

$$T = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{L}{g}} \quad . \quad (9)$$

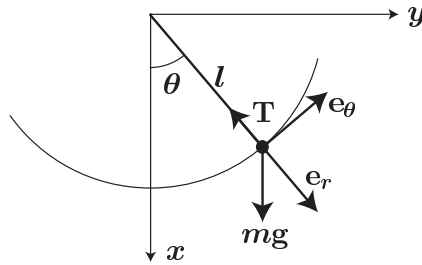
Setting  $\theta = 0$  for  $t = 0$ , and  $\theta = \theta_{max}$  for  $t = T/4$ , we obtain,

$$\theta = \theta_{max} \sin \omega_n t \quad .$$

We observe that, in the small amplitude case, the period is independent of  $\theta_{max}$ . This is called *synchronism* and is central to time-keeping functions in clocks. This means that, provided the amplitude is small, small changes in amplitude due to friction or other disturbances have little effect on the period. Comparing expression 9 with the approximate solution of the non-linear problem given by equation 7, we see that for amplitudes of, say,  $\theta_{max} = 30^\circ$ , the difference between the two expressions is only about 1.7%.

## Compound Pendulum

In the case of a compound pendulum, we can proceed in a similar manner.



Conservation of angular momentum about  $O$  gives,

$$I_O \ddot{\theta} = -mgr_G \sin \theta \quad .$$

Expressing  $I_O$  in terms of the radius of gyration,  $k_O$ ,

$$I_O = mk_O^2$$

we have,

$$\ddot{\theta} + \frac{gr_G}{k_O^2} \sin \theta = 0 . \quad (10)$$

We note that this equation is the same as equation 6 for the simple pendulum, if we identify the term  $g/L$  in equation 6 with the term  $gr_G/k_O^2$  in equation 10. This leads to the definition of an equivalent length,  $L_{equiv}$ , as,

$$L_{equiv} = \frac{k_O^2}{r_G} = \frac{k_G^2 + r_G^2}{r_G} . \quad (11)$$

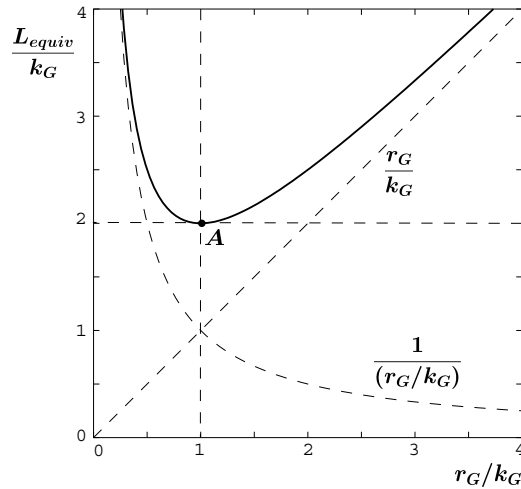
Here we have used the fact that  $I_O = I_G + mr_G^2$ , and therefore  $k_O^2 = k_G^2 + r_G^2$ . Thus, we have that the motion of a compound pendulum is identical to that of a simple pendulum of equivalent length  $L_{equiv}$ , given by equation 11. Using the small amplitude approximation, the period of the compound pendulum will be

$$T = 2\pi \sqrt{\frac{L_{equiv}}{g}} .$$

One question we may want to ask is whether, for a given body ( $k_G$  fixed), we can make the period (or  $L_{equiv}$ ) arbitrarily small by choosing  $r_G$  (or the hinge point  $O$ ) appropriately. From equation 11, we can write

$$\frac{L_{equiv}}{k_G} = \frac{r_G}{k_G} + \frac{1}{(r_G/k_G)} ,$$

which shows that, when we try to reduce  $L_{equiv}/k_G$  by reducing  $r_G$ , the term  $r_G/k_G$  is reduced, but the term  $1/(r_G/k_G)$  increases. This situation is shown in the graph, which also shows that the minimum value for  $L_{equiv}/k_G$  is 2, a value which is attained for  $r_G = k_G$ .



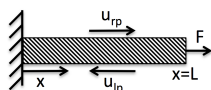
In conclusion, we have that for a given solid, the shortest equivalent length, and hence the fastest pendulum, occurs when it is suspended from a point which is at a distance  $r_G = k_G$  from the center of mass. In this case, the minimum equivalent length is

$$(L_{equiv})_{min} = 2k_G .$$

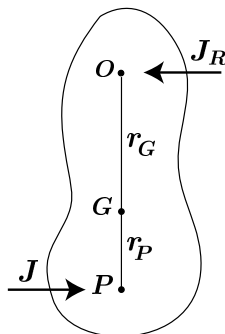


**Example****Minimum period pendulum**

Consider a uniform rod of length  $l$ . In this case,  $I_G = ml^2/12$ , and  $k_G = l/\sqrt{12} \approx 0.29l$ . Therefore, the fastest pendulum is obtained when the bar is suspended from a distance  $0.29l$  away from the center of mass. We note that if the suspension point is moved slightly, the period of the pendulum will increase. However, moving the suspension point will make practically no difference in the frequency, because the tangent at point  $A$  (see graph) is horizontal. This fact has been used in the construction of extremely accurate pendulums for clocks.

**Center of Percussion**

The center of percussion associated with a given center of motion was previously introduced. In the case of a pendulum, the desired center of motion is the hinge,  $O$ . The center of percussion is on the extension of the line that connects  $O$  with the center of mass.



From our earlier discussion, we know that the distance,  $r_P$ , between the center of percussion,  $P$ , and the center of mass,  $G$ , is given by,

$$r_P = \frac{k_G^2}{r_G} . \quad (12)$$

Recall that the center of percussion is the point at which we should strike the pendulum with a horizontal impulse so that  $O$  becomes the instantaneous center of motion. It is clear that, if  $O$  is the instantaneous center of motion, the horizontal reaction needed to keep the pendulum in place will be zero and, as a consequence, the impulse will not be felt at  $O$ . In fact, this can be verified directly in a straightforward manner. Let us assume that the pendulum is given an impulse  $J$  at a distance  $r_P$  below the center of mass,  $G$ . Conservation of momentum in the horizontal direction and conservation of angular momentum around

$G$  imply that,

$$\begin{aligned} mv_G &= J - J_R \\ I_G \omega &= Jr_P + J_R r_G . \end{aligned}$$

Solving for  $J_R$ , we have that,

$$J_R = J \frac{I_G - mr_G r_P}{I_G + mr_G^2} = J \frac{k_G^2 - r_G r_P}{I_O} ,$$

which clearly shows that the impulse reaction,  $J_R$ , is zero when  $r_P$  satisfies equation 12.

We note that, for a given body,  $r_P$  will be large when  $r_G$  is very small, and, as a consequence, the center of percussion will not be a material point in the body.

## Reversibility

It turns out that if the pendulum is suspended from point  $P$ , the roles of  $r_G$  and  $r_P$  are reversed. In particular, point  $O$  is the center of percussion relative to the instantaneous center of motion,  $P$ . Using equations 11 and 12, the equivalent length of a compound pendulum can be written as,

$$L_{equiv} = \frac{k_G^2 + r_G^2}{r_G} = \frac{r_P r_G + r_G^2}{r_G} = r_G + r_P ,$$

and the corresponding period is,

$$T = 2\pi \sqrt{\frac{L_{equiv}}{g}} = 2\pi \sqrt{\frac{r_G + r_P}{g}} .$$

It should be clear that when the pendulum is suspended from point  $P$ , the equivalent length does not change, since,

$$\frac{k_G^2 + r_P^2}{r_P} = \frac{r_P r_G + r_P^2}{r_P} = r_G + r_P \equiv L_{equiv} .$$

## The Kater Pendulum

The reversibility of point  $O$  and  $P$  described above is the principle of the reversible pendulum, invented by Kater to measure gravity with high accuracy. Kater's pendulum consists of a long bar, equipped with two fixed knife edges at an accurately known distance  $L$ , and with some moveable masses positioned along the bar. The positions of these masses on the bar are adjusted until the periods associated with suspension from either knife edge are precisely equal to, say,  $T$ . This guarantees that each point is the center of percussion relative to the other, and, thus,  $L = L_{equiv}$ . The local gravity is then given by,

$$g = L_{equiv} \left( \frac{2\pi}{T} \right)^2 .$$

### The Schuler Pendulum

Consider a pendulum suspended in vertical position from point  $O$ . If the support is suddenly accelerated in the horizontal direction, the pendulum will rotate due to the inertial forces acting on its center of mass. It is not difficult to show that the instantaneous center of motion in this case will be precisely the center of percussion associated with  $O$ .

Now, imagine a pendulum on the earth's surface whose center of percussion is at the earth's center. When the pendulum's support is accelerated, the pendulum rotates in a way such that it always points towards the center of the earth. Such a pendulum always keeps itself vertical regardless of the acceleration, and would be of obvious interest for applications such as inertial navigation.

It is interesting to see that, although conceptually correct, the construction of such a pendulum would pose serious technological challenges. If we assume an earth radius of  $R = 6370$  km, then  $r_P \approx R$ . Considering a pendulum with, say,  $k_G = 0.3$  m, the distance between the support and the center of mass would have to be,

$$r_G = \frac{0.3^2}{6.37 \times 10^6} = 1.5 \times 10^{-8} \text{ m} = 0.015 \mu\text{m}.$$

This distance approaches atomic dimensions and, in fact, such a pendulum has never been constructed using purely mechanical means. It is, however, routinely realized by replacing the physical pendulum with a combination of gyroscope and accelerometers having the same dynamics.

It is interesting to note that the period of the Schuler pendulum is given by

$$T_{Schuler} = 2\pi \sqrt{\frac{L_{equiv}}{g}} \approx 2\pi \sqrt{\frac{R}{g}} \approx 84.4 \text{ min.} ,$$

which is exactly the same period as that of a circular orbit around the earth at zero altitude.

### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

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## References

- [1] M. Martinez-Sanchez, *Unified Engineering Notes*, Course 95-96.

## Lecture 27 — 3D Rigid Body Kinematics

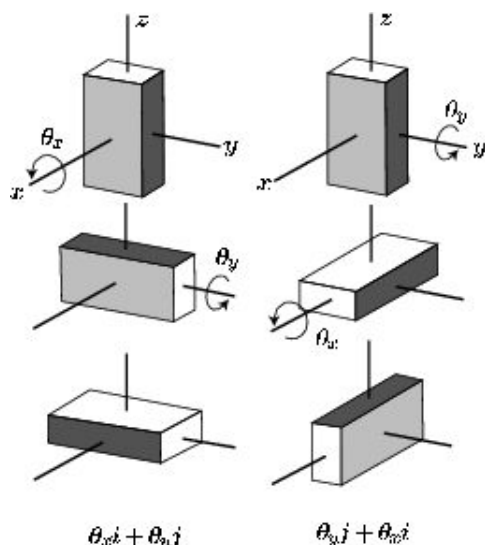
In this lecture, we consider the motion of a 3D rigid body. We shall see that in the general three-dimensional case, the angular velocity of the body can change in magnitude as well as in direction, and, as a consequence, the motion is considerably more complicated than that in two dimensions.

### Rotation About a Fixed Point

We consider first the simplified situation in which the 3D body moves in such a way that there is always a point,  $O$ , which is fixed. It is clear that, in this case, the path of any point in the rigid body which is at a distance  $r$  from  $O$  will be on a sphere of radius  $r$  that is centered at  $O$ . We point out that the fixed point  $O$  is not necessarily a point in the rigid body (the second example in these notes illustrates this point).

Euler's theorem states that the general displacement of a rigid body, with one fixed point, is a rotation about some axis. This means that any two rotations of arbitrary magnitude about different axes can always be combined into a single rotation about some axis.

At first sight, it seems that we should be able to express a rotation as a vector which has a direction along the axis of rotation and a magnitude that is equal to the angle of rotation. Unfortunately, if we consider two such rotation vectors,  $\theta_1$  and  $\theta_2$ , not only would the combined rotation  $\theta$  be different from  $\theta_1 + \theta_2$ , but in general  $\theta_1 + \theta_2 \neq \theta_2 + \theta_1$ . This situation is illustrated in the figure below, in which we consider a 3D rigid body undergoing two  $90^\circ$  rotations about the  $x$  and  $y$  axis.

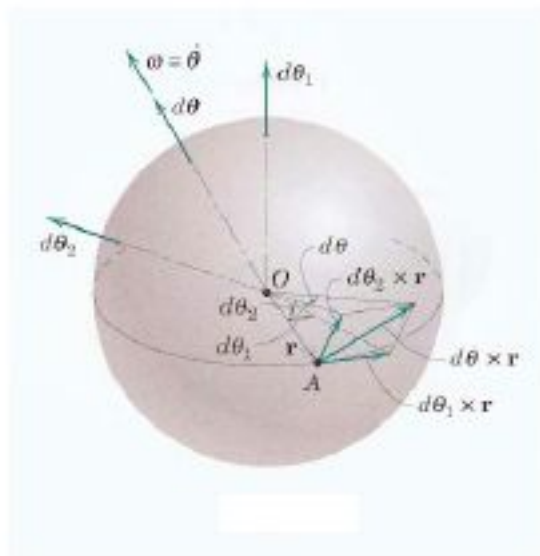


It is clear that the result of applying the rotation in  $x$  first and then in  $y$  is different from the result obtained by rotating first in  $y$  and then in  $x$ . Therefore, it is clear that *finite rotations* cannot be treated as vectors, since they do not satisfy simple vector operations such as the parallelogram vector addition law.

This result can also be understood by considering the rotation of axes by a coordinate transformation. Consider a transformation  $(x') = [T_1](x)$ , and a subsequent coordinate transformation  $(x'') = [T_2](x')$ . The  $(x'')$  coordinate obtained by the transformation  $(x'') = [T_1][T_2](x)$  will not be the same as the coordinates obtained by the transformation  $(x'') = [T_2][T_1](x)$ , in other words, order matters.

## Angular Velocity About a Fixed Point

On the other hand, if we consider *infinitesimal rotations* only, it is not difficult to verify that they do indeed behave as vectors. This is illustrated in the figure below, which considers the effect of two combined infinitesimal rotations,  $d\theta_1$  and  $d\theta_2$ , on point  $A$ .



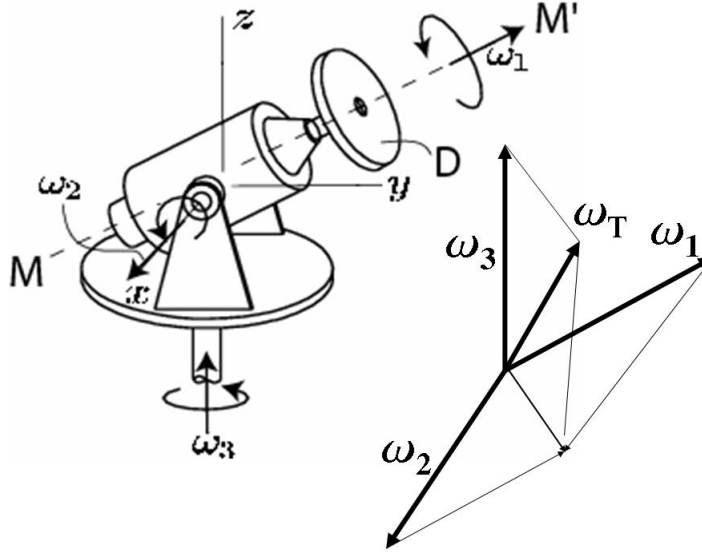
(figure reproduced from J.L. Meriam and K.L. Kraige, *Dynamics*, 5th edition, Wiley)

(figure reproduced from J.L. Meriam and K.L. Kraige, *Dynamics*, 5th edition, Wiley)

As a result of  $d\theta_1$ , point  $A$  has a displacement  $d\theta_1 \times r$ , and, as a result of  $d\theta_2$ , point  $A$  has a displacement  $d\theta_2 \times r$ . The total displacement of point  $A$  can then be obtained as  $d\theta \times r$ , where  $d\theta = d\theta_1 + d\theta_2$ . Therefore, it follows that angular velocities  $\omega_1 = \dot{\theta}_1$  and  $\omega_2 = \dot{\theta}_2$  can be added vectorially to give  $\omega = \omega_1 + \omega_2$ . This means that if at any instant the body is rotating about a given axis with angular velocity  $\omega_1$  and at the same time this axis is rotating about another axis with angular velocity  $\omega_2$ , the total angular velocity of the body will be simply  $\omega = \omega_1 + \omega_2$ . Therefore, although the finite rotations of a body about an axis are not vectors, the infinitesimal rotations are vectors. The angular velocity is thus a vector and for a complex configuration, the various components can be vectorially added to obtain the total angular velocity.

## REFERENCES

Consider the complex rotating configuration shown below. We want to determine the angular velocity of the disc D.



First, we note that the disc is rotating with angular velocity  $\omega_1$  about the axis  $MM'$ . In turn, this axis is rotating with angular velocity  $\omega_2$  about the horizontal axis, which is at this instant aligned with the  $x$  axis. At the same time, the whole assembly is rotating about the  $z$  axis with angular velocity  $\omega_3$ . Therefore, the total angular velocity of the disc  $\omega_{1_{total}}$  is vector sum of the individual angular velocity vectors. The resultant vector is shown in the figure:

$$\omega_{1_{total}} = \omega_1 + \omega_2 + \omega_3. \quad (1)$$

Expressed in the fixed  $x, y, z$  system for which the configuration is *instantaneously aligned* as shown, we have

$$\omega_{1_{total}} = \omega_2 \mathbf{i} + \omega_1 \cos \phi \mathbf{j} + (\omega_1 \sin \phi + \omega_3) \mathbf{k}. \quad (2)$$

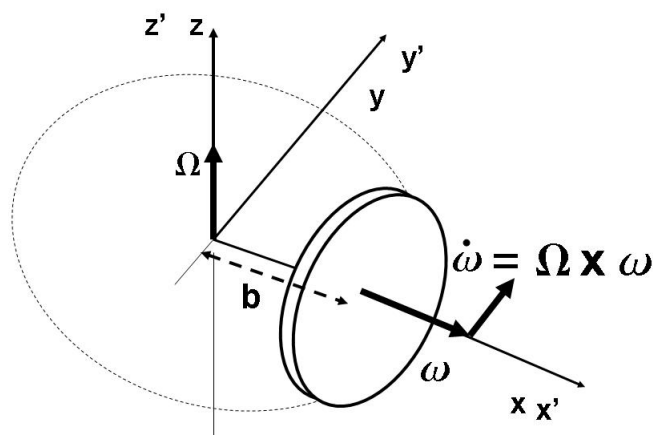
Here,  $\phi$  is the angle between  $MM'$  and the  $y$  axis at this instant. Also, if we wish to determine the total angular velocity at the axle  $\omega_2$ , we must account for the fact that the axle driving  $\omega_2$  is being driven by  $\omega_3$ . Thus the total angular velocity of the axle is  $\omega_{2_{total}} = \omega_3 + \omega_2$ .

## Angular Acceleration

In order to apply the principle of conservation of angular momentum, we need a general expression for the angular acceleration of the various components of a complex rotating configuration. We examine this problem with respect to inertial coordinate system,  $x, y, z$ .

We first consider a common situation in the study of rotating 3D objects, the rotating wheel with angular velocity  $\omega$  attached to a central hub which rotates with angular velocity  $\Omega$ , shown in the figure. The total angular velocity is the vector sum of  $\Omega$  and  $\omega$ .

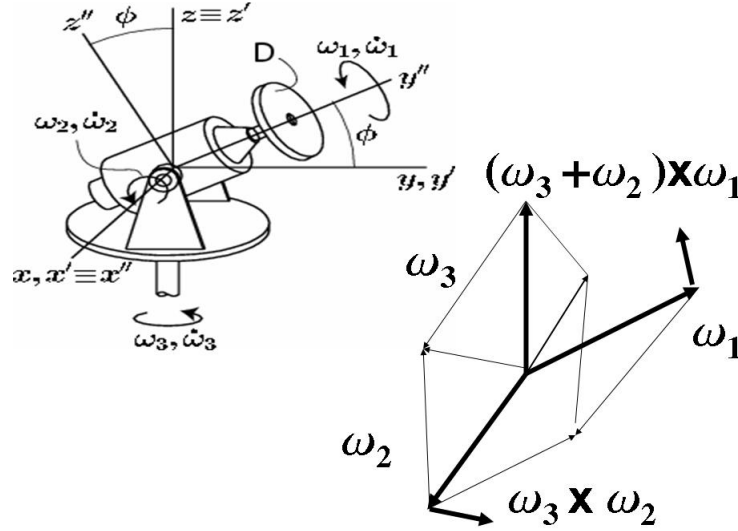
## REFERENCES



In this example, we take  $\omega$  constant in magnitude (but not direction) and  $\Omega$  as constant. It is clear that the rotation  $\Omega$  will rotate the vector  $\omega$ , changing its direction. The magnitude of  $\dot{\omega}$  is  $\Omega\omega$ , the direction is normal to  $\omega$ ; by Coriolis theorem, the result is  $\dot{\omega} = \Omega \times \omega$ . It is interesting to note that this result is independent of the distance  $b$  between the wheel and the axis of rotation for  $\Omega$ . This is a consequence of our earlier observation that in a rigid body rotating with angular velocity  $\omega$ , every point rotates with angular velocity  $\omega$ .

**Example****Multiple Observers**

In this example we illustrate a more systematic procedure for calculating the angular velocities and accelerations when several reference frames are involved. We want to determine the total angular acceleration of the disc D as a function of the various angular velocities and accelerations given in the diagram, at this instant of time. The disk is rotated by a drive at an angular velocity  $\omega_1$ . This device is rotating at an angular velocity  $\omega_2$  about an axle; and the entire system is mounted on a turntable which rotates with angular velocity  $\omega_3$ ; all of these angular velocities could be functions of time as well. The instantaneous angle of  $\omega_1$  with the horizontal is  $\phi$ .



The angular velocity vector for each component is the vector sum of the individual angular velocities of the components:

$$\omega_{3_{total}} = \omega_3 \quad (3)$$

$$\omega_{2_{total}} = \omega_3 + \omega_2 \quad (4)$$

$$\omega_{1_{total}} = \omega_1 + \omega_2 + \omega_3. \quad (5)$$

$\omega_3$  is unaffected by  $\omega_2$  or  $\omega_1$ ;  $\omega_{2_{total}}$  is "driven" by  $\omega_3$  but unaffected by  $\omega_1$ ; and  $\omega_{1_{total}}$  is "driven" by both  $\omega_1$  and  $\omega_2$ . The angular accelerations of the various components will now be worked through individually. Consider  $\omega_3$ . Since it is the "primary" rotor, the rotation  $\omega_2$  and  $\omega_1$  do not affect its motion. Therefore the angular acceleration is due solely to  $\dot{\omega}_3$ .

$$\frac{D}{Dt} \omega_{3T} = \alpha_{3T} = \dot{\omega}_3 \quad (6)$$

Consider  $\omega_2$ . This angular velocity vector will change with time, both due to  $\dot{\omega}_2$  as well as  $\dot{\omega}_3$  and the Coriolis term  $\omega_3 \times \omega_2$ , which changes the direction of  $\omega_2$  but not its magnitude. Therefore the total angular acceleration  $\alpha_{2T}$  equals

$$\frac{D}{Dt} \omega_{2T} = \alpha_{2T} = \dot{\omega}_2 + \dot{\omega}_3 + \omega_3 \times \omega_2 \quad (7)$$

And finally, to determine the **total** angular acceleration seen by the disk D,  $\alpha_{1T}$ , we combine the effects of the various rotations and their rates of change. Since the vector  $\omega_1$  is "driven" by  $\omega_2 + \omega_3$ , the total angular velocity  $\alpha_{1T}$  will be the sum of the time rates of change of the individual components and the Coriolis term  $(\omega_2 + \omega_3) \times \omega_1$ . Therefore

$$\alpha_{1T} = \frac{D}{Dt} \omega_{1T} = \dot{\omega}_1 + \dot{\omega}_2 + \dot{\omega}_3 + (\omega_2 + \omega_3) \times \omega_1 \quad (8)$$



We now resolve these vectors into the appropriate coordinate system, the fixed  $x, y, z$  system. The coordinate system  $x', y', z'$  is *instantaneously aligned* with the fixed coordinate system  $x, y, z$  so that  $\mathbf{i}' = \mathbf{i}$ ,  $\mathbf{j}' = \mathbf{j}$  and  $\mathbf{k}' = \mathbf{k}$ . Therefore we may write

$$\boldsymbol{\omega}_3 = \omega_3 \mathbf{k} \quad (9)$$

$$\boldsymbol{\omega}_2 = \omega_2 \mathbf{i} \quad (10)$$

$$\boldsymbol{\alpha}_3 = \dot{\omega}_3 \mathbf{k} \quad (11)$$

$$\boldsymbol{\alpha}_2 = \dot{\omega}_3 \mathbf{k} + \dot{\omega}_2 \mathbf{i} + \omega_3 \omega_2 \mathbf{j} \quad (12)$$

The  $x'', y'', z''$  coordinate system is *instantaneous aligned* with the  $x$  axis so that  $x'' = x$ ; however, it is rotated by an angle  $\phi$  from the  $y, z$  axis, such that expressed in the fixed  $x, y, z$  coordinate system, we have

$$\boldsymbol{\omega}_1 = \omega_1 \cos \phi \mathbf{j} + \omega_1 \sin \phi \mathbf{k} \quad (13)$$

Now that all of the angular velocities are expressed in the fixed coordinate system  $x, y, z$  we may collect the various terms to express the angular acceleration of the disk D,  $\boldsymbol{\alpha}_{1_{total}}$ , in the fixed  $x, y, z$  coordinate system *at this instant of time*.

$$\boldsymbol{\alpha}_{1_T} = \dot{\omega}_1 + \dot{\omega}_2 + \dot{\omega}_3 + (\boldsymbol{\omega}_2 + \boldsymbol{\omega}_3) \times \boldsymbol{\omega}_1 \quad (14)$$

$$\boldsymbol{\alpha}_{1_T} = \dot{\omega}_1 \cos \phi \mathbf{j} + \dot{\omega}_1 \sin \phi \mathbf{k} + \dot{\omega}_3 \mathbf{k} + \dot{\omega}_2 \mathbf{i} + (\omega_2 \mathbf{i} + \omega_3 \mathbf{k}) \times (\omega_1 \cos \phi \mathbf{j} + \omega_1 \sin \phi \mathbf{k}) \quad (15)$$

Putting it all together, we have in the  $x, y, z$  coordinate system *at this instant of time*

$$\boldsymbol{\alpha}_{1_T} = (\dot{\omega}_2 - \omega_1 \omega_3 \cos \phi) \mathbf{i} + (\dot{\omega}_1 \cos \phi + \omega_2 \omega_3 - \omega_1 \omega_2 \sin \phi) \mathbf{j} + (\dot{\omega}_3 + \dot{\omega}_1 \sin \phi + \omega_1 \omega_2 \cos \phi) \mathbf{k} \quad (16)$$

## Instantaneous Axis of Rotation

In two dimensions, we introduced the concept of instantaneous center of rotation. For a rotating body with one fixed point, we can extend this concept to an *instantaneous axis of rotation*.

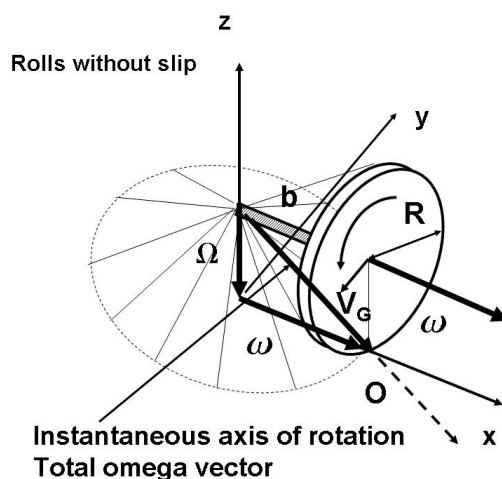
Consider a vertical disc rotating with a constant angular velocity  $\boldsymbol{\omega}$ , rolling without slip with an angular velocity about the vertical axis with angular velocity  $\boldsymbol{\Omega}$ . The no-slip condition requires that the velocity of the center of mass  $V_G = -\omega R$ . The disk rolling around a circle of radius  $b$  will have an angular velocity  $\Omega = \omega R/b$ . We note that just as in two-dimensions, point  $O$  is a fixed point, an instantaneous center of rotation. That is, at that instant, the motion of the disc is such that the distance from any point in the disc to the point  $O$  remains constant. The total instantaneous angular velocity is the vector sum of  $\boldsymbol{\Omega} + \boldsymbol{\omega}$ . The resultant vector is sketched and as would be expected, it is parallel to a line from the center of the hub to the point  $C$ .

Once the instantaneous angular velocity,  $\boldsymbol{\omega}_T = \boldsymbol{\Omega} + \boldsymbol{\omega}$ , has been determined, the velocity of any point in the rigid body is simply

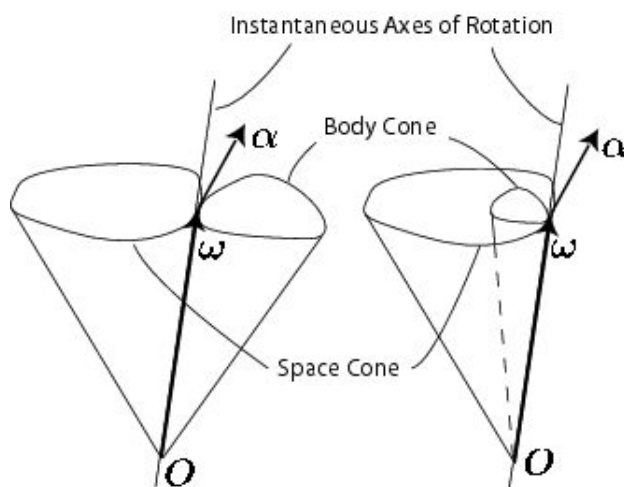
$$\mathbf{v} = \boldsymbol{\omega}_T \times \mathbf{r} , \quad (17)$$

## REFERENCES

where  $\mathbf{r}$  is the position vector of the point considered with respect to the fixed point  $O$ . It follows that for any point which is on the line passing through  $O$  and parallel to  $\boldsymbol{\omega}_T$ , the velocity will be zero. This line is therefore the *Instantaneous Axis of Rotation*.



We can now define two space curves, which are instantaneously tangent to the instantaneous axis of rotation. These curves are useful for more complex problems in the general rotation of a body about a fixed point. This simple example gives us a useful visualization of these space curves. The first is called the **body cone**: it is the *locus of points made by the instantaneous axis of rotation as the body traces its motion*. As can be seen, for this case, it is a cone of radius equal to the radius of the disc, extending to the hub at the  $z$  axis. (The term **body cone** is a bit of a misnomer; the body doesn't have to fit entirely inside the body cone.) The second space curve is called the **space cone**. It is defined as the *locus of points traced in space by the instantaneous axis of rotation*. At the instantaneous position of the body, these two cones are both tangent to the instantaneous axis of rotation. In this case, the space cone is the cone bounded by the curve traced by the instantaneous contact point and the fixed point, the center of the hub. The body cone and the space cone for more general three-dimensional motions are shown in the figure.



For a general motion, as the direction of the instantaneous axis of rotation (or the line passing through  $O$  parallel to  $\boldsymbol{\omega}$ ) changes in space, the locus of points defined by the axis generates a fixed *Space Cone*. If the change in this axis is viewed with respect to the rotating body, the locus of the axis generates a *Body Cone*. At any given instant, these two cones are tangent along the instantaneous axis of rotation. When the body is in motion, the body cone appears to roll either on the inside or the outside of the fixed space cone. This situation is illustrated in the figure below.

The acceleration of any point in the rigid body is obtained by taking the derivative of expression 17. Thus,

$$\mathbf{a} = \dot{\boldsymbol{\omega}}_T \times \mathbf{r} + \boldsymbol{\omega}_T \times \dot{\mathbf{r}} = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega}_T \times (\boldsymbol{\omega}_T \times \mathbf{r}) . \quad (18)$$

Here,  $\boldsymbol{\alpha}$  is the angular acceleration vector and is locally tangent to both the Space and the Body Cones.

## General Motion

In the general case, the displacement of a rigid body is determined by a translation plus a rotation about some axis. This result is a generalization of Euler's theorem, which is sometimes known as Chasles' theorem. In practice, this means that six parameters are needed to define the position of a 3D rigid body. For instance, we could choose three coordinates to specify the position of the center of mass, two angles to define the axis of rotation and an additional angle to determine the magnitude of the rotation.

Unlike the motion about a fixed point, it is not always possible to define an instantaneous axis of rotation. Consider, for instance, a body which is rotating with angular velocity  $\boldsymbol{\omega}$  and, at the same time, has a translational velocity parallel to  $\boldsymbol{\omega}$ . It is clear that, in this case, all the points in the body have a non-zero velocity, and therefore an instantaneous center of rotation cannot be defined.

It turns out that, in some situations, the motion of the center of mass of a 3D rigid body can be determined independent of the orientation. Consider, for instance, the motion of an orbiting satellite in free flight. In this situation, the sum of all external forces on the satellite does not depend on the satellite's attitude, and, therefore, it is possible to determine the position without knowing the attitude. In more complex situations, however, it may be necessary to solve simultaneously for both the position of the center of mass and the attitude.

The velocity,  $\mathbf{v}_P$ , and acceleration,  $\mathbf{a}_P$ , of a point,  $P$ , in the rigid body can be determined if we know the velocity,  $\mathbf{v}_{O'}$ , and acceleration,  $\mathbf{a}_{O'}$ , of a point in the rigid body,  $O'$ , as well as the body's angular velocity,  $\boldsymbol{\omega}$ , and acceleration,  $\boldsymbol{\alpha}$ .

$$\mathbf{v}_P = \mathbf{v}_{O'} + \boldsymbol{\omega} \times \mathbf{r}'_P \quad (19)$$

$$\mathbf{a}_P = \mathbf{a}_{O'} + \dot{\boldsymbol{\omega}} \times \mathbf{r}'_P + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}'_P) . \quad (20)$$

Here,  $\mathbf{r}'_P$  is the position vector of the point,  $P$ , relative to  $O'$ . We point out that the angular velocity and angular acceleration are the same for all the points in the rigid body.

## ADDITIONAL READING

*REFERENCES*

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition  
7/1, 7/2, 7/3, 7/4, 7/5, 7/6 (review)

## Lecture 28 — 3D Rigid Body Dynamics: The Inertia Tensor

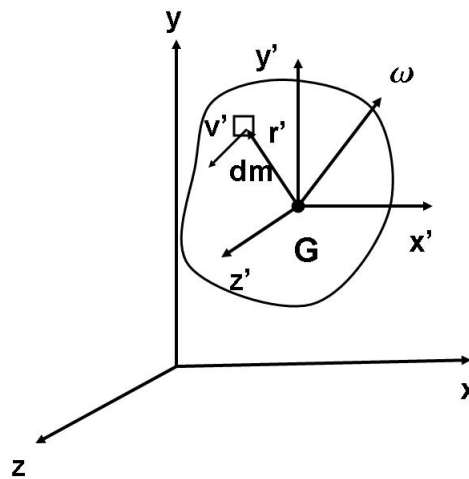
In this lecture, we will derive an expression for the angular momentum of a 3D rigid body. We shall see that this introduces the concept of the Inertia Tensor.

### Angular Momentum

We start with the angular momentum of a system of particles about its center of mass,

$$\mathbf{H}_G = \int_m \mathbf{r}' \times \mathbf{v}' dm . \quad (1)$$

Here,  $\mathbf{r}'$  is the position vector relative to the center of mass,  $\mathbf{v}'$  is the velocity relative to the center of mass. We note that, in the above expression, an integral is used instead of a summation, since we are now dealing with a continuum distribution of mass.



For a 3D rigid body, the distance between any particle and the center of mass will remain constant, and the particle velocity, relative to the center of mass, will be given by

$$\mathbf{v}' = \boldsymbol{\omega} \times \mathbf{r}' .$$

Thus, we have,

$$\mathbf{H}_G = \int_m \mathbf{r}' \times (\boldsymbol{\omega} \times \mathbf{r}') \, dm = \int_m [(\mathbf{r}' \cdot \mathbf{r}')\boldsymbol{\omega} - (\mathbf{r}' \cdot \boldsymbol{\omega})\mathbf{r}'] \, dm .$$

Here, we have used the vector identity  $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$ . We note that, for planar bodies undergoing a 2D motion in its own plane,  $\mathbf{r}'$  is perpendicular to  $\boldsymbol{\omega}$ , and the term  $(\mathbf{r}' \cdot \boldsymbol{\omega})$  is zero. In this case, the vectors  $\boldsymbol{\omega}$  and  $\mathbf{H}_G$  are always parallel. In the three-dimensional case however, this simplification does not occur, and as a consequence, the angular velocity vector,  $\boldsymbol{\omega}$ , and the angular momentum vector,  $\mathbf{H}_G$ , are in general, *not parallel*.

In cartesian coordinates, we have,  $\mathbf{r}' = x'\mathbf{i} + y'\mathbf{j} + z'\mathbf{k}$  and  $\boldsymbol{\omega} = \omega_x\mathbf{i} + \omega_y\mathbf{j} + \omega_z\mathbf{k}$ , and the above expression can be expanded to yield,

$$\begin{aligned} \mathbf{H}_G &= \left( \omega_x \int_m (x'^2 + y'^2 + z'^2) \, dm - \int_m (\omega_x x' + \omega_y y' + \omega_z z') x' \, dm \right) \mathbf{i} \\ &+ \left( \omega_y \int_m (x'^2 + y'^2 + z'^2) \, dm - \int_m (\omega_x x' + \omega_y y' + \omega_z z') y' \, dm \right) \mathbf{j} \\ &+ \left( \omega_z \int_m (x'^2 + y'^2 + z'^2) \, dm - \int_m (\omega_x x' + \omega_y y' + \omega_z z') z' \, dm \right) \mathbf{k} \\ &= (I_{xx}\omega_x - I_{xy}\omega_y - I_{xz}\omega_z) \mathbf{i} \\ &+ (-I_{yx}\omega_x + I_{yy}\omega_y - I_{yz}\omega_z) \mathbf{j} \\ &+ (-I_{zx}\omega_x - I_{zy}\omega_y + I_{zz}\omega_z) \mathbf{k} . \end{aligned} \quad (2)$$

The quantities  $I_{xx}$ ,  $I_{yy}$ , and  $I_{zz}$  are called *moments of inertia* with respect to the  $x$ ,  $y$  and  $z$  axis, respectively, and are given by

$$I_{xx} = \int_m (y'^2 + z'^2) \, dm , \quad I_{yy} = \int_m (x'^2 + z'^2) \, dm , \quad I_{zz} = \int_m (x'^2 + y'^2) \, dm .$$

We observe that the quantity in the integrand is precisely the square of the distance to the  $x$ ,  $y$  and  $z$  axis, respectively. They are analogous to the moment of inertia used in the two dimensional case. It is also clear, from their expressions, that the moments of inertia are always positive. The quantities  $I_{xy}$ ,  $I_{xz}$ ,  $I_{yx}$ ,  $I_{yz}$ ,  $I_{zx}$  and  $I_{zy}$  are called *products of inertia*. They can be positive, negative, or zero, and are given by,

$$I_{xy} = I_{yx} = \int_m x'y' \, dm , \quad I_{xz} = I_{zx} = \int_m x'z' \, dm , \quad I_{yz} = I_{zy} = \int_m y'z' \, dm .$$

They are a measure of the imbalance in the mass distribution. If we are interested in calculating the angular momentum with respect to a fixed point  $O$  then, the resulting expression would be,

$$\begin{aligned} \mathbf{H}_O &= (I_{xx})_O \omega_x - (I_{xy})_O \omega_y - (I_{xz})_O \omega_z \mathbf{i} \\ &+ (-I_{yx})_O \omega_x + (I_{yy})_O \omega_y - (I_{yz})_O \omega_z \mathbf{j} \\ &+ (-I_{zx})_O \omega_x - (I_{zy})_O \omega_y + (I_{zz})_O \omega_z \mathbf{k} . \end{aligned} \quad (3)$$

Here, the moments of products of inertia have expressions which are analogous to those given above but with  $x'$ ,  $y'$  and  $z'$  replaced by  $x$ ,  $y$  and  $z$ . Thus, we have that

$$(I_{xx})_O = \int_m (y^2 + z^2) dm, \quad (I_{yy})_O = \int_m (x^2 + z^2) dm, \quad (I_{zz})_O = \int_m (x^2 + y^2) dm,$$

and,

$$(I_{xy})_O = (I_{yx})_O = \int_m xy dm, \quad (I_{xz})_O = (I_{zx})_O = \int_m xz dm, \quad (I_{yz})_O = (I_{zy})_O = \int_m yz dm.$$

## The Tensor of Inertia

The expression for angular momentum given by equation (2), can be written in matrix form as,

$$\begin{pmatrix} H_{Gx} \\ H_{Gy} \\ H_{Gz} \end{pmatrix} = \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}. \quad (4)$$

or,

$$\mathbf{H}_G = [I_G] \boldsymbol{\omega}, \quad (5)$$

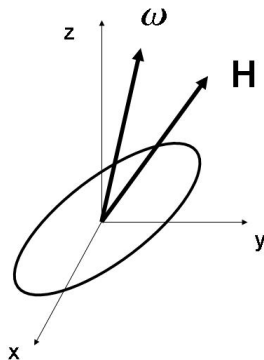
where  $[I_G]$  is the tensor of inertia (written in matrix form) about the center of mass  $G$  and with respect to the  $xyz$  axes. The tensor of inertia gives us an idea about how the mass is distributed in a rigid body.

Analogously, we can define the tensor of inertia about point  $O$ , by writing equation(3) in matrix form. Thus, we have

$$\mathbf{H}_O = [I_O] \boldsymbol{\omega},$$

where the components of  $[I_O]$  are the moments and products of inertia about point  $O$  given above.

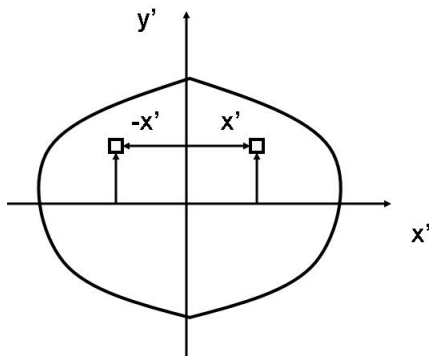
It follows from the definition of the products of inertia, that the tensors of inertia are always **symmetric**. The implications of equation (4) are that in many situations of importance, even for bodies of some symmetry, the angular momentum vector  $\vec{H}$  and the angular velocity vector  $\vec{\omega}$  are not parallel. This introduces considerable complexity into the analysis of the dynamics of rotating bodies in three dimensions.



## Principal Axes of Inertia

For a general three-dimensional body, it is *always* possible to find 3 mutually orthogonal axis (an  $x, y, z$  coordinate system) for which the products of inertia are zero, and the inertia matrix takes a diagonal form. In most problems, this would be the preferred system in which to formulate a problem. For a rotation about only one of these axis, the angular momentum vector is parallel to the angular velocity vector. For symmetric bodies, it may be obvious which axis are principle axis. However, for an irregular-shaped body this coordinate system may be difficult to determine by inspection; we will present a general method to determine these axes in the next section.

But, if the body has symmetries with respect to some of the axis, then some of the products of inertia become zero and we can identify the principal axes. For instance, if the body is symmetric with respect to the plane  $x' = 0$  then, we will have  $I_{x'y'} = I_{y'x'} = I_{x'z'} = I_{z'x'} = 0$  and  $x'$  will be a principal axis. This can be shown by looking at the definition of the products of inertia.

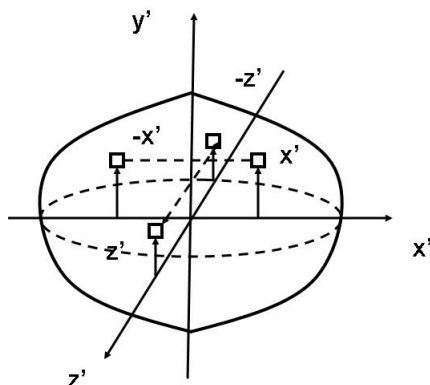


The integral for, say,  $I_{x'y'}$ , can be decomposed into two integrals for the two halves of the body at either side of the plane  $x' = 0$ . The integrand on one half,  $x'y'$ , will be equal in magnitude and opposite in sign to the integrand on the other half (because  $x'$  will change sign). Therefore, the integrals over the two halves will cancel each other and the product of inertia  $I_{x'y'}$  will be zero. (As will the product of inertia  $I_{x'z'}$ ).

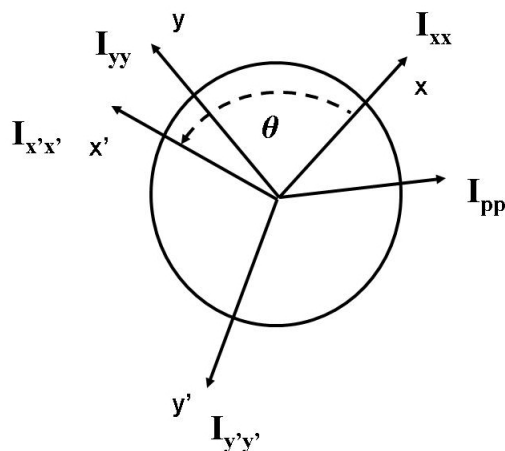


## REFERENCES

Also, if the body is symmetric with respect to two planes passing through the center of mass which are orthogonal to the coordinate axis, then the tensor of inertia is diagonal, with  $I_{x'y'} = I_{x'z'} = I_{y'z'} = 0$ .



Another case of practical importance is when we consider axisymmetric bodies of revolution. In this case, if one of the axis coincides with the axis of symmetry, the tensor of inertia has a simple diagonal form. For an axisymmetric body, the moments of inertia about the two axis in the plane will be equal. Therefore, the moment about any axis in this plane is equal to one of these. And therefore, any axis in the plane is a principal axis. One can extend this to show that if the moment of inertia is equal about two axis in the plane ( $I_{PP} = I_{xx}$ ), whether or not they are orthogonal, then all axes in the plane are principal axes and the moment of inertia is the same about all of them. In its inertial properties, the body behaves like a circular cylinder.

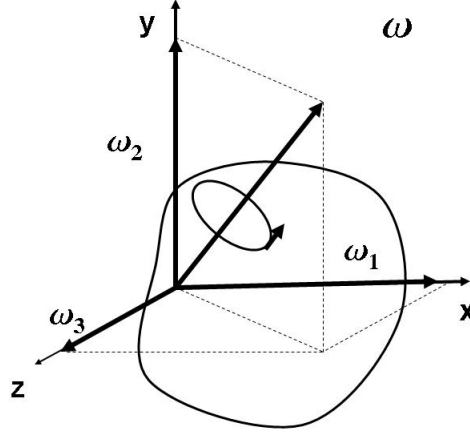


The tensor of inertia will take different forms when expressed in different axes. When the axes are such that the tensor of inertia is diagonal, then these axes are called the *principal axes of inertia*.

### The Search for Principal Axes and Moments of Inertia as an Eigenvalue Problem

Three orthogonal principal axes of inertia always exist even though in bodies without symmetries their directions may not be obvious. To find the principle axis of a general body consider the body shown in

the figure that rotates about an unknown principal axis. The total angular momentum vector is  $I\vec{\omega}$  in the direction of the principle axis. For rotation about the principal axis, the angular momentum and the angular velocity are in the same direction.



We seek a coordinate axes  $x$ ,  $y$  and  $z$ , about which a rotation  $\omega_x$ ,  $\omega_y$  and  $\omega_z$ , which is aligned with this coordinate direction, will be parallel to the angular momentum vector and related by the equation

$$\begin{pmatrix} H_{Gx} \\ H_{Gy} \\ H_{Gz} \end{pmatrix} = \begin{pmatrix} I\omega_x \\ I\omega_y \\ I\omega_z \end{pmatrix} = \begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}. \quad (6)$$

We then express the **general** form for angular momentum vector in components along the  $x$ ,  $y$  and  $z$  axis in terms of the components of  $\vec{\omega}$  along these axes using the **general** form of the inertia tensor in the  $x, y, z$  system, we have

$$\begin{pmatrix} H_{Gx} \\ H_{Gy} \\ H_{Gz} \end{pmatrix} = \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}. \quad (7)$$

To obtain the special directions of  $\omega$  that is aligned with a principal axis, we equate these two expressions.

$$\begin{pmatrix} H_{Gx} \\ H_{Gy} \\ H_{Gz} \end{pmatrix} = \begin{pmatrix} I\omega_x \\ I\omega_y \\ I\omega_z \end{pmatrix} = \begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} = \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}. \quad (8)$$

At this point in the process we know the inertia tensor in an arbitrary  $x$ ,  $y$ , and  $z$  system and are seeking the special orientation of  $\omega$  which will align the angular momentum  $H_G$  with the angular velocity  $\omega$ . Collecting terms from equation(10) on the left-hand side, we obtain

$$\begin{pmatrix} (I_{xx} - I) & -I_{xy} & -I_{xz} \\ -I_{yx} & (I_{yy} - I) & -I_{yz} \\ -I_{zx} & -I_{zy} & (I_{zz} - I) \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (9)$$

resulting in the requirement that

$$\left( \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{pmatrix} - I \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right) \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (10)$$

The structure of the solution for finding the principal axes of inertia and their magnitudes is a ***characteristic-value problem***. The three eigenvectors give the directions of the three principal axes, and the three eigenvalues give the moments of inertia with respect to each of these axis. Since the results of the eigenvalue problem produce three "vectors"  $\{\omega_x, \omega_y, \omega_z\}_{i=1,2,3}$  having the property  $\{\omega_x, \omega_y, \omega_z\}_i \cdot \{\omega_x, \omega_y, \omega_z\}_j = 0$  for  $i \neq j$ , these vectors give the coordinate directions of the three orthogonal principal axes.

In principal directions, the inertia tensor has the form

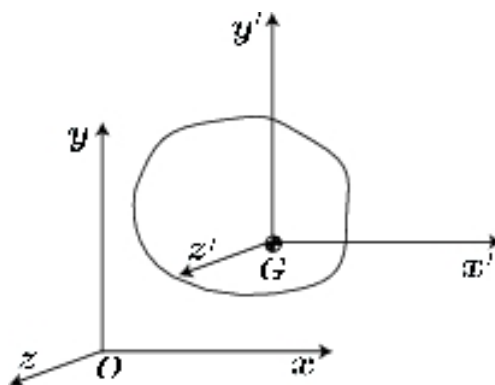
$$[I_G] = \begin{pmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{pmatrix}$$

where we will write  $I_x = I_{xx}$ ,  $I_y = I_{yy}$  and  $I_z = I_{zz}$ . Also, in principal axes we will then have

$$\mathbf{H}_G = I_x \omega_x \mathbf{i} + I_y \omega_y \mathbf{j} + I_z \omega_z \mathbf{k} .$$

## Parallel Axis Theorem

It will often be easier to obtain the tensor of inertia with respect to axis passing through the center of mass. In some problems however, we will need to calculate the tensor of inertia about different axes. The parallel axis theorem introduced in lecture 22 for the two dimensional moments of inertia can be extended and applied to each of the components of the tensor of inertia.



In particular we can write,

$$\begin{aligned}
 (I_{xx})_O &= \int_m (y^2 + z^2) dm = \int_m ((y_G + y')^2 + (z_G + z')^2) dm \\
 &= \int_m (y'^2 + z'^2) + 2y_G \int_m y' dm + 2z_G \int_m z' dm + (y_G^2 + z_G^2) \int_m dm \\
 &= I_{xx} + m(y_G^2 + z_G^2) .
 \end{aligned}$$

Here, we have use the fact that  $y'$  and  $z'$  are the coordinates relative to the center of mass and therefore their integrals over the body are equal to zero. Similarly, we can write,

$$(I_{yy})_O = I_{yy} + m(x_G^2 + z_G^2), \quad (I_{zz})_O = I_{zz} + m(x_G^2 + y_G^2),$$

and,

$$(I_{xy})_O = (I_{yx})_O = I_{xy} + mx_G y_G, \quad (I_{xz})_O = (I_{zx})_O = I_{xz} + mx_G z_G, \quad (I_{yz})_O = (I_{zy})_O = I_{yz} + my_G z_G .$$

## Rotation of Axes

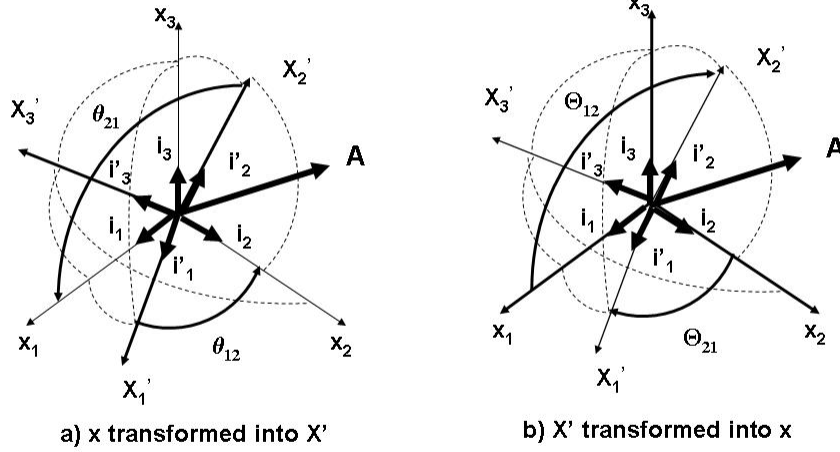
In some situations, we will know the tensor of inertia with respect to some axes  $xyz$  and, we will be interested in calculating the tensor of inertia with respect to a rotated set of axis  $x'y'z'$ . We denote by  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$  the unit vectors along the direction of  $xyz$  axes, and by  $\mathbf{i}'$ ,  $\mathbf{j}'$  and  $\mathbf{k}'$  the unit vectors along the direction of  $x'y'z'$  axes. The transformation of the inertia tensor can be accomplished by considering the transformation of the angular momentum vector  $\vec{H}$  and the angular velocity vector  $\vec{\omega}$ . We begin with the expression of  $\vec{H}$  and  $\vec{\omega}$  in the  $x_1, x_2, x_3$  system.

$$\vec{H} = [I]\vec{\omega} \tag{11}$$

The transformation of a vector from a coordinate system  $x_1, x_2, x_3$  into a coordinate system  $x'_1, x'_2, x'_3$  is given by

$$\begin{pmatrix} H'_1 \\ H'_2 \\ H'_3 \end{pmatrix} = \begin{pmatrix} \mathbf{i}'_1 \cdot \mathbf{i}_1 & \mathbf{i}'_1 \cdot \mathbf{i}_2 & \mathbf{i}'_1 \cdot \mathbf{i}_3 \\ \mathbf{i}'_2 \cdot \mathbf{i}_1 & \mathbf{i}'_2 \cdot \mathbf{i}_2 & \mathbf{i}'_2 \cdot \mathbf{i}_3 \\ \mathbf{i}'_3 \cdot \mathbf{i}_1 & \mathbf{i}'_3 \cdot \mathbf{i}_2 & \mathbf{i}'_3 \cdot \mathbf{i}_3 \end{pmatrix} \begin{pmatrix} H_1 \\ H_2 \\ H_3 \end{pmatrix} = [T] \begin{pmatrix} H_1 \\ H_2 \\ H_3 \end{pmatrix} ,$$

where  $[T]$  represents the transformation matrix.



Likewise, the transformation of  $\vec{\omega}$  into the  $x'_1, x'_2, x'_3$  system is given by

$$\begin{pmatrix} \omega'_1 \\ \omega'_2 \\ \omega'_3 \end{pmatrix} = \begin{pmatrix} i'_1 \cdot i_1 & i'_1 \cdot i_2 & i'_1 \cdot i_3 \\ i'_2 \cdot i_1 & i'_2 \cdot i_2 & i'_2 \cdot i_3 \\ i'_3 \cdot i_1 & i'_3 \cdot i_2 & i'_3 \cdot i_3 \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = [T] \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}.$$

To determine  $\vec{H}'$  (in the transformed system), we multiply both sides of equation (10) by  $[T]$ , and obtain

$$\vec{H}' = [T]\vec{H} = [T][I]\vec{\omega} \quad (12)$$

where  $[I]$  is the inertia matrix in the original coordinate system. But this equation is not quite in a useful form: we need to have the form  $\vec{H}' = [I']\vec{\omega}'$  to identify  $[I']$ , the inertia matrix in the new coordinate system. Taking advantage of the fact that  $[T]^T = [T]^{-1}$ , we may write

$$(\vec{H}') = [T]\vec{H} = [T][I][T]^T[T]\vec{\omega} = ([T][I][T]^T)([T]\vec{\omega}) = [I']\omega', \quad (13)$$

The terms are grouped so that it is obvious that the inertia tensor in the transformed coordinate system is

$$[I'] = [T][I][T]^T \quad (14)$$

Therefore, if we want to calculate the tensor of inertia with respect to axis  $x'y'z'$ , we can write in matrix

form

$$\begin{pmatrix} I_{x'x'} & -I_{x'y'} & -I_{x'z'} \\ -I_{y'x'} & I_{y'y'} & -I_{y'z'} \\ -I_{z'x'} & -I_{z'y'} & I_{z'z'} \end{pmatrix} = \begin{pmatrix} \mathbf{i}' \cdot \mathbf{i} & \mathbf{i}' \cdot \mathbf{j} & \mathbf{i}' \cdot \mathbf{k} \\ \mathbf{j}' \cdot \mathbf{i} & \mathbf{j}' \cdot \mathbf{j} & \mathbf{j}' \cdot \mathbf{k} \\ \mathbf{k}' \cdot \mathbf{i} & \mathbf{k}' \cdot \mathbf{j} & \mathbf{k}' \cdot \mathbf{k} \end{pmatrix} \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \mathbf{i} \cdot \mathbf{i}' & \mathbf{j} \cdot \mathbf{i}' & \mathbf{k} \cdot \mathbf{i}' \\ \mathbf{i} \cdot \mathbf{j}' & \mathbf{j} \cdot \mathbf{j}' & \mathbf{k} \cdot \mathbf{j}' \\ \mathbf{i} \cdot \mathbf{k}' & \mathbf{j} \cdot \mathbf{k}' & \mathbf{k} \cdot \mathbf{k}' \end{pmatrix},$$

where we have returned to the  $x, y, z$  notation.

Likewise, expressed in the direction cosines of the figure, the transformation is

$$[I'] = \begin{pmatrix} \cos(\theta_{11}) & \cos(\theta_{12}) & \cos(\theta_{13}) \\ \cos(\theta_{21}) & \cos(\theta_{22}) & \cos(\theta_{23}) \\ \cos(\theta_{31}) & \cos(\theta_{32}) & \cos(\theta_{33}) \end{pmatrix} \begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \cos(\Theta_{11}) & \cos(\Theta_{12}) & \cos(\Theta_{13}) \\ \cos(\Theta_{21}) & \cos(\Theta_{22}) & \cos(\Theta_{23}) \\ \cos(\Theta_{31}) & \cos(\Theta_{32}) & \cos(\Theta_{33}) \end{pmatrix} \dots \quad (15)$$

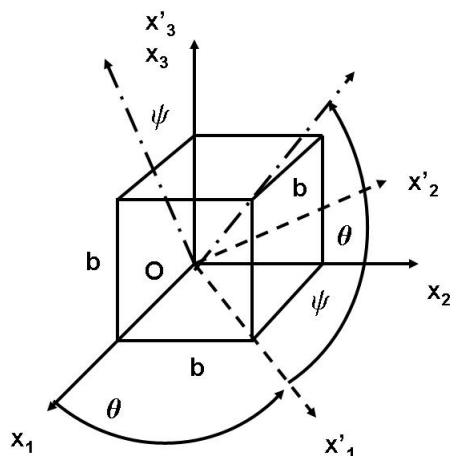
The special ***“principal axis”*** transformation matrix, which transforms a general inertia tensor into a diagonal inertia tensor, can be obtained directly from the solution to the eigenvalue problem since the eigenvectors, when normalized to unit length, give the three orthogonal directions, and thus the direction cosines, of the principal axes. The transformation matrix can be formed directly as

$$[T] = \begin{pmatrix} \omega_{11} & \omega_{12} & \omega_{13} \\ \omega_{21} & \omega_{22} & \omega_{23} \\ \omega_{31} & \omega_{32} & \omega_{33} \end{pmatrix} \quad (16)$$

where  $\omega_{ij}$  are the components of the normalized (unit length) eigenvector  $\boldsymbol{\omega}_i$ .

## Inertia Tensor of a Cube

We are going to consider and calculate the inertia tensor of a cube of equal length sides  $b$ . (This example is taken from Marion and Thornton.) For this choice of coordinates, it is obvious that the center of mass does not lie at the origin. It is also obvious that there is considerable symmetry in the geometry.



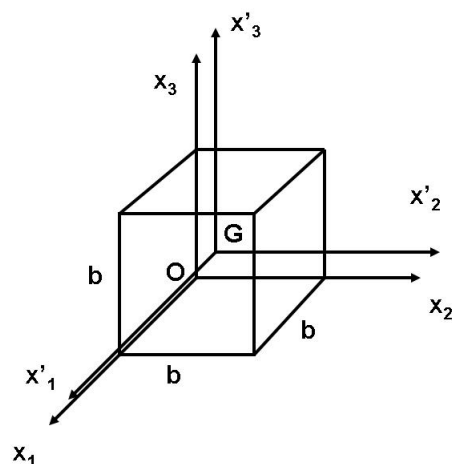
Evaluating the various integrals for the components of the inertia tensor, we have

$$[I] = \begin{pmatrix} \frac{2}{3}Mb^2 & -\frac{1}{4}Mb^2 & -\frac{1}{4}Mb^2 \\ -\frac{1}{4}Mb^2 & \frac{2}{3}Mb^2 & -\frac{1}{4}Mb^2 \\ -\frac{1}{4}Mb^2 & -\frac{1}{4}Mb^2 & \frac{2}{3}Mb^2 \end{pmatrix} \quad (17)$$

This result is perhaps a bit surprising; because of the strong symmetry of the cube we might have expected these coordinates to be principal axes. However, our origin of coordinates is not at the center of mass, and by the parallel axis theorem, we incur products of inertia the equal to total mass times the various distances of center of mass from the origin of coordinates.

There are several options to find a set of principal axes for this problem. One is to use the symmetry to identify a coordinate system in which the products of inertia vanish. The transformation sketched in the figure is a good candidate to obtain the principal axes; that is a rotation of  $\theta = \pi/4$  about the  $x_3$  axis, followed by a rotation of  $\psi = \sin^{-1} \frac{1}{\sqrt{3}}$  about the  $x'_1$  axis. This would result in the  $x'_1$  axis going through the upper point of the cube and through the center of mass as well. If we were to do this, we would find that indeed this transformation results in coordinate directions which are principal axis directions, and moreover that the inertias are equal about any axis in the plane perpendicular to the diagonal of the cube, so that all these axes are principal with equal moments of inertia. We now apply the parallel axis theorem to obtain the inertia tensor about the center of mass  $G$  located a distance  $\{b/2, b/2, b/2\}$  from the origin  $\{0, 0, 0\}$ .

## REFERENCES



We obtain

$$[I] = \begin{pmatrix} \frac{1}{6}Mb^2 & 0 & 0 \\ 0 & \frac{1}{6}Mb^2 & 0 \\ 0 & 0 & \frac{1}{6}Mb^2 \end{pmatrix} \quad (18)$$

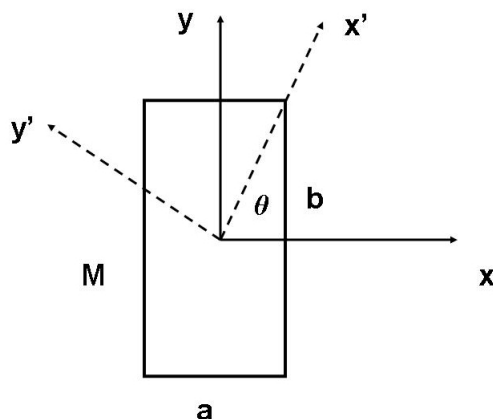
As expected, we obtain a diagonal inertia tensor about the center of mass. Of more significance, the inertias about all three principal axes are equal. Thus any axis that passes through the center of mass of a cube is a principal axis and every moment of inertia about this axis is equal to any other moment of inertia about any axis passing through the center of mass. Therefore, for moments taken about the center of mass, the cube is dynamically equivalent to a sphere.

Another more general method is to express the search for principal axes as an eigenvalue problem, as previously outlined. In a problem without strong and obvious symmetry, identifying an appropriate transformation is often a matter of guess work; therefore the eigenvalue formulation is more useful.

## Products of Inertia in a Plane

In many situations, we will know one principal axis from symmetry, and will be working with moments of inertia and inertia products in a plane. An example would be a flat plate of mass  $m$ , thickness  $t$ , width  $a$  and length  $b$ . It is clear from the sketch that the principal axes are the  $x, y$  coordinates. We may have reasons for wanting to express the moments of inertia and the products of inertia in the  $x', y'$  system (rotated from the  $x, y$  system by an angle  $\theta$ ) or to move back and forth between the  $x, y$  system, and the  $x', y'$  system.





The coordinate transformation that takes a vector in the  $x, y, z$  system into the  $x', y', z'$  system is

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (19)$$

or  $(\vec{x}') = [T](\vec{x})$ ; while the coordinate transformation that takes a vector in the  $x', y', z'$  system into the  $x, y, z$  system is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}. \quad (20)$$

or  $(\vec{x}) = [T]^T(\vec{x}')$  where we have taken advantage of the fact that the matrix for the reverse transformation ( $\vec{x}'$  to  $\vec{x}$ ) is the transpose of the matrix for the forward transformation ( $\vec{x}$  to  $\vec{x}'$ ). The inertia matrix for a rectangular plate with its center of mass at the origin in the  $x, y, z$  system, for which the principal axes line up with the coordinate directions is

$$I_{x,y,z} = M \begin{pmatrix} \frac{b^2}{12} & 0 & 0 \\ 0 & \frac{a^2}{12} & 0 \\ 0 & 0 & \frac{a^2+b^2}{12} \end{pmatrix}. \quad (21)$$

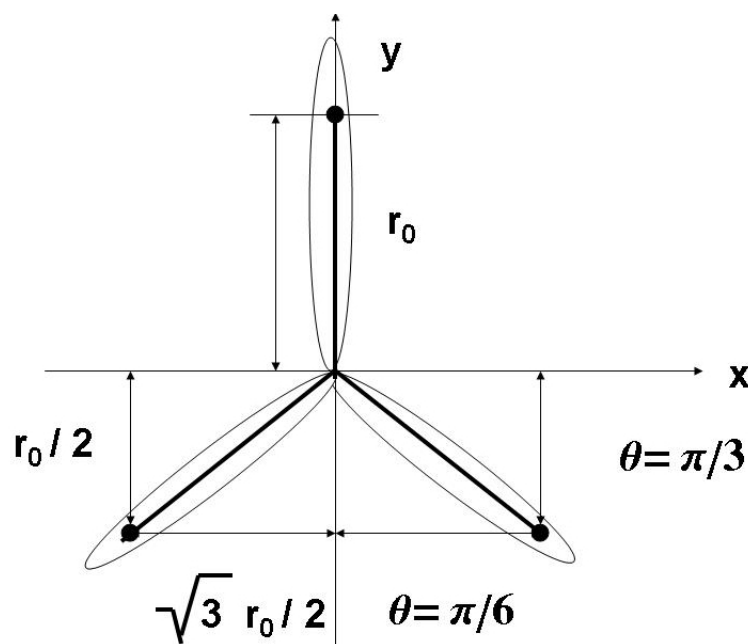
If we wish to obtain the inertia tensor in the  $x', y', z'$  system, we apply the transformation

$$I_{x',y',z'} = [T][I_{x,y,z}][T]^T \quad (22)$$

## The Symmetry of Three

An interesting and practical result for inertias in a plane comes from considering the moment of inertia of a body with tri-symmetry, such as a three-bladed propeller or a wind turbine. It is clear that the axis normal to the figure is a principal axis, with masses symmetrically balanced about the origin. It is also clear that both the  $x$  and  $y$  axis are principal axes, with inertia products going to zero by symmetry. What is perhaps

surprising, is that  $I_{xx}$  the moment of inertia about the x axis is equal to  $I_{yy}$  the moment of inertia about the y axis.



Consider the  $x$  axis; for a point  $r_0$  the contribution to the moment of inertia from all 3 mass points is  $I_{xx} = m(r_0^2 + 2(r_0/2)^2) = m\frac{3r_0^2}{2}$ . Consider the  $y$  axis; for a point  $r_0$  the contribution to the moment of inertia from all 3 mass points is  $I_{yy} = m(2(\sqrt{3}r_0/2)^2) = m\frac{3r_0^2}{2}$ . Therefore, the total moment of inertia about both the  $x$  and  $y$  axis will be an integral of the mass distribution in  $dm$  times  $\frac{3r_0^2}{2}$ . Since the moments of inertia are equal about the 2 principal axes  $x$  and  $y$ , the moment of inertia about **any axis** in the  $x,y$  plane is also equal to  $I_{xx}$ . The three-bladed propeller behaves as a uniform disk. This has important implications for the vibrations of propellers of historically-interesting high performance propeller fighter aircraft and wind turbines in yawing motion.

#### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition

7/7, Appendix B

S. T. Thorton and J. B. Marion, *Classical Dynamics of Particles and Systems*, 4th Edition, Chapter 11

## Lecture 29 — 3D Rigid Body Dynamics: Kinetic Energy; Example Equations of Motion

### 3D Rigid Body Dynamics

So far, we have 1) developed the framework for the description of changes in angular velocity due to a general motion of a three-dimensional rotating body, and 2) developed the framework for the effects of the distribution of mass of a three-dimensional rotating body on its motion, defining the principal axes of a body, the inertia tensor, and how to change from one reference coordinate system to another.

We now undertake the description of angular momentum, moments and motion of a general three-dimensional rotating body. We approach this very difficult general problem from **two points of view**.

The **first is to prescribe the motion** in term of given rotations about fixed axes and solve for the force system required to sustain this motion.

The **second is to study the "free" motions** of a body in a simple force field such as gravitational force acting through the center of mass or "free" motion such as occurs in a "zero-g" environment. The typical problems in this second category involve gyroscopes and spinning tops. The second set of problems is by far the more difficult.

Before we begin this general approach, we examine a case where kinetic energy can give us considerable insight into the behavior of a rotating body. This example has considerable practical importance and its neglect has been the cause of several system failures.

### Kinetic Energy for Systems of Particles

In Lecture 11, we derived the expression for the kinetic energy of a system of particles. Here, we derive the expression for the kinetic energy of a system of particles that will be used in the following lectures. A typical particle,  $i$ , will have a mass  $m_i$ , an absolute velocity  $\mathbf{v}_i$ , and a kinetic energy  $T_i = (1/2)m_i\mathbf{v}_i \cdot \mathbf{v}_i = (1/2)m_i v_i^2$ . The total kinetic energy of the system,  $T$ , is simply the sum of the kinetic energies for each particle,

$$T = \sum_{i=1}^n T_i = \sum_{i=1}^n \frac{1}{2} m_i v_i^2 = \frac{1}{2} m v_G^2 + \sum_{i=1}^n \frac{1}{2} m_i v_i'^2, \quad .$$

where  $v'$  is the velocity relative to the center of mass. Thus, we see that the kinetic energy of a system of particles equals the kinetic energy of a particle of mass  $m$  moving with the velocity of the center of mass, plus the kinetic energy due to the motion of the particles relative to the center of mass,  $G$ .

We have said nothing about the conservation of energy for a system of particles. As we shall see, that depends upon the details of internal interactions and the work done by the external forces. The same is true for a rigid body. Work done by internal stresses, or energy lost due to the complexities of a system described as a rigid body, but which in reality may have internal modes which drain energy, will act to decrease the kinetic energy. Thus, although we can confidently relate angular momentum to external forces, we have no such confidence in dealing with conservation of energy. As we shall see, this has important technological implications, especially for the stability of spacecraft. In this section we consider the particles to be components of a rigid body.

## Kinetic Energy for a 3D Rigid Body

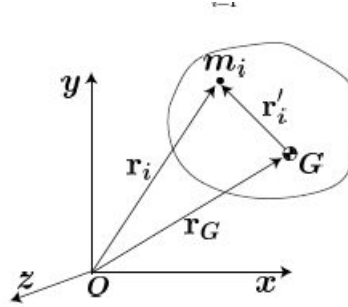
For a rigid body, the summation  $i = 1, n$  becomes an integral over the total mass  $M$ .

$$T = \int_m \frac{1}{2} v^2 dm = \frac{1}{2} M v_G^2 + \int_m \frac{1}{2} v'^2 dm .$$

For a rigid body, the velocity relative to the center of mass is written

$$\vec{v}' = \vec{\omega} \times \vec{r}' \quad (1)$$

where  $\vec{r}'$  is the vector to the mass  $dm$  for the center of mass  $G$ .



Using the vector identity

$$(\vec{A} \times \vec{B}) \cdot \vec{C} = \vec{A} \cdot (\vec{B} \times \vec{C}), \quad (2)$$

we have

$$v'^2 = \vec{v}' \cdot \vec{v}' = (\vec{\omega} \times \vec{r}') \cdot (\vec{\omega} \times \vec{r}') = (\vec{\omega} \cdot (\vec{r}' \times (\vec{\omega} \times \vec{r}')) = \vec{\omega} \cdot (\vec{r}' \times \vec{v}')$$

.

Therefore,

$$\int_m v'^2 dm = \vec{\omega} \cdot \int_m \vec{r}' \times \vec{v}' dm = \vec{\omega} \cdot \vec{H}_G, \quad (3)$$

and the total expression for kinetic energy of a rigid body can be written

$$T = \frac{1}{2}Mv_G^2 + \frac{1}{2}\vec{\omega} \cdot \vec{H}_G = \frac{1}{2}\vec{v}_G \cdot \vec{L}_G + \frac{1}{2}\vec{\omega} \cdot \vec{H}_G \quad (4)$$

where  $M\vec{L}_G$  is the linear momentum.

If there is a fixed point  $O$  about which the body rotates, then

$$T = \frac{1}{2}\vec{\omega} \cdot \vec{H}_O$$

where  $\vec{H}_O$  is the angular momentum about point  $O$ . Even if there is no fixed point about which the body rotates, there is an instantaneous center of rotation  $C$ . In some situations, it may be useful to write

$$T_C = \frac{1}{2}\vec{\omega} \cdot \vec{H}_C \quad (5)$$

If the angular velocity is expressed in principal axes, then the angular momentum about the center of mass can be written  $\vec{H}_G = I_{xx}\omega_x\vec{i} + I_{yy}\omega_y\vec{j} + I_{zz}\omega_z\vec{k}$  so that the kinetic energy can be written

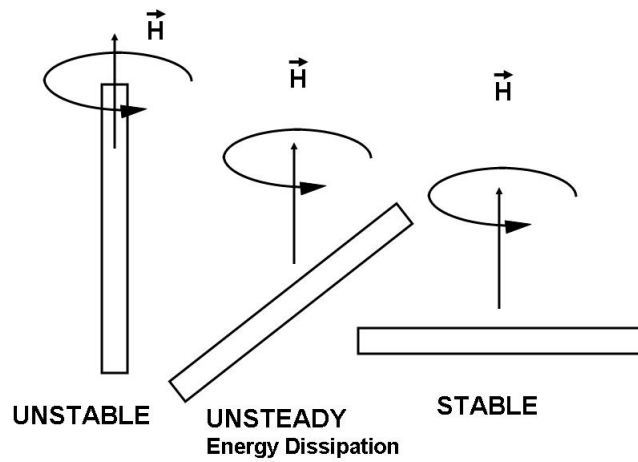
$$T = \frac{1}{2}Mv_G^2 + \frac{1}{2}(I_{xx}\omega_x^2 + I_{yy}\omega_y^2 + I_{zz}\omega_z^2) \quad (6)$$

The above expressions for kinetic energy are useful to apply the principle of work and energy. We see that the kinetic energy has two components: one is due to the translation,  $\frac{1}{2}Mv_G^2$ . The other is due to rotation. The rotational component can be written as

$$T_R = \frac{1}{2}\vec{\omega} \cdot \vec{H}_G = \frac{1}{2}(I_{xx}\omega_x^2 + I_{yy}\omega_y^2 + I_{zz}\omega_z^2) \quad (7)$$

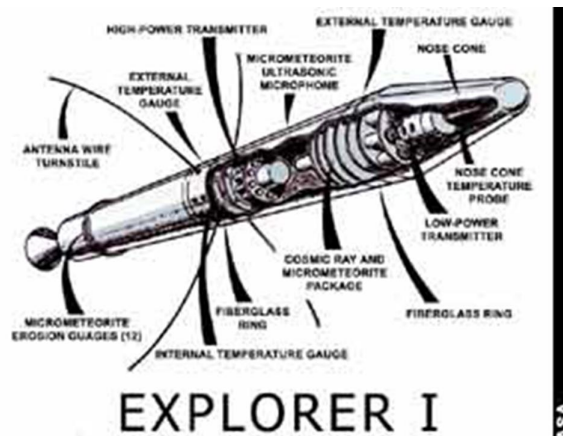
We have no assurance that energy will be conserved. Internal motions can dissipate energy. This is not true for angular momentum: internal forces and motions will not change angular momentum. The implications of this can be profound.

Consider a body spinning about the  $z$  axis of inertia with no external moments. Then the angular momentum will be constant and since it is a vector, it will always point in the  $z$  direction so that  $\vec{H}_G = H_z\vec{k}$  will be constant. Now let us assume that  $I_{zz} < I_{xx}$  and  $I_{zz} < I_{yy}$ . In this case, for a given angular momentum, **the kinetic energy is a maximum**, since  $T = \frac{1}{2}H_z\omega_z = \frac{1}{2}H_z^2/I_{zz}$ , and  $I_{zz}$  is the smallest moment of inertia. If there is any dissipation mechanism—structural damping, fuel sloshing, friction—then the angular momentum will stay constant in magnitude and direction while the energy decreases. The only way, that the energy can decrease while keeping the angular momentum constant is to change the axis of spin to one with a larger moment of inertia while keeping the vector direction and magnitude of the angular momentum vector constant. Therefore, if the body tumbles and begins to spin about the  $x$  or  $y$  body-fixed axis, the kinetic energy becomes  $T = \frac{1}{2}H_z^2/(I_{xx} \text{ or } I_{yy})$ . The kinetic energy will be reduced since both  $I_{xx}$  and  $I_{yy}$  are greater than  $I_{zz}$ .



The sketch shows the process by which this would occur. An initial spin about the body's  $z$  axis will transition over time to a spin about the body's  $x$  or  $y$  axis, while keeping the  $\vec{H}_G$  constant in magnitude and direction. The initial state is simple; the final state is simple; the intermediate states are complex in that the body axis is not aligned with the rotation vector, much like a spinning top performing a complex motion. Do we have to worry about this?

The first American satellite, Explorer 1, was built by the Jet Propulsion Laboratory in Pasadena, and launched from Canaveral on 31 January 1958 by a modified Jupiter-C missile into an orbit ranging between 354 kilometers and 2,515 kilometers at an angle of 33 degrees to the equator. The satellite was long and slender. It was spin-stabilized about its long axis—the one with the smallest moment of inertia. Spin stabilization has advantages in that the angular momentum remains fixed in direction in space under passive control, if it's stable! The satellite had flexible antenna which vibrated and dissipated energy. During the mission, it unexpectedly suffered an attitude failure in that it tipped over and began to spin about the axis with the maximum moment of inertia. Other space systems as well as re-entry vehicles are prone to this phenomena.



## Governing Equations for the Prescribed Rotational Motion of a Three-Dimensional Body

The governing equations are those of conservation of linear momentum  $\mathbf{L} = M\mathbf{v}_G$  and angular momentum,  $\mathbf{H} = [I]\boldsymbol{\omega}$ , where we have written the moment of inertia in matrix form to remind us that in general the direction of the angular momentum is not in the direction of the rotation vector  $\boldsymbol{\omega}$ . Conservation of linear momentum requires

$$\dot{\mathbf{L}} = \mathbf{F} \quad (8)$$

Conservation of angular momentum, about a fixed point  $O$ , requires

$$\dot{\mathbf{H}}_O = \mathbf{M} \quad (9)$$

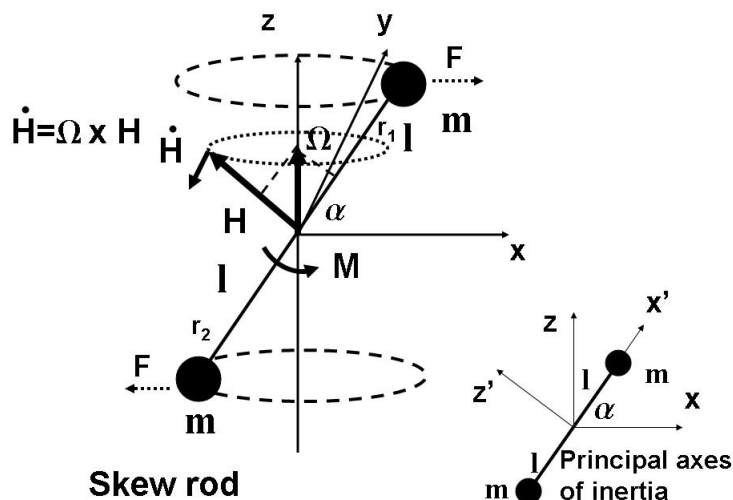
or about the center of mass  $G$

$$\dot{\mathbf{H}}_G = \mathbf{M}_G \quad (10)$$

We first examine cases in which the **motion is specified** and our task is to **determine the forces and moments required**.

### Example: Rotating Skew Rod

We consider a simple example which illustrate the effect of rotation about a non-principal axis. In this case, we consider two masses equal  $m_1$  and  $m_2$  attached to a massless rigid rod of length  $2l$  aligned from the horizontal by an angle  $\alpha$  and rotating about the  $z$  axis with *constant* angular velocity  $\Omega\mathbf{k}$ . We examine this problem at the **instant** when the rod is aligned with the  $x$  and  $z$  axis. Because of symmetry, the center of mass is at the origin. Therefore, we consider only the change in angular momentum. Because the magnitude of the angular velocity  $\Omega$  is constant, we are concerned only with its change in direction.



The angular momentum for this system is  $\mathbf{H} = \sum \mathbf{r} \times m_i \mathbf{v}_i$ . For this case, the velocity for mass  $m_1$  is  $\mathbf{v}_1 = \Omega l \cos \alpha \mathbf{j}$ , while for mass  $m_2$  the velocity is  $\mathbf{v}_2 = -\Omega l \cos \alpha \mathbf{j}$  while  $\mathbf{r}_1 = l \cos \alpha \mathbf{i} + l \sin \alpha \mathbf{k}$  and  $\mathbf{r}_2 = -l \cos \alpha \mathbf{i} - l \sin \alpha \mathbf{k}$ . At this instant of time, the angular momentum is in the  $x, z$  plane and of magnitude

$$\mathbf{H} = -2ml^2\Omega \cos \alpha \sin \alpha \mathbf{i} + 2ml^2\Omega \cos^2 \alpha \mathbf{k} \quad (11)$$

where  $\mathbf{i}$ ,  $\mathbf{j}$ , and  $\mathbf{k}$  are the unit vectors in the  $x, y, z$  directions. Since the axis about which the rotation takes place is not a principal axis, it is no surprise that the angular momentum vector is not aligned with the angular velocity vector. In this simple case, since we are dealing with two mass points, the rod has zero moment of inertia about this axis. Therefore the angular momentum vector is perpendicular to the rod. As the skew rod rotates about the  $z$  axis, the angular momentum changes as

$$\dot{\mathbf{H}} = \boldsymbol{\Omega} \times \mathbf{H} = -2m\Omega^2 l^2 \sin \alpha \cos \alpha \mathbf{j} \quad (12)$$

resulting in a "required" moment about the  $y$  axis to sustain the motion of

$$\mathbf{M} = -2m\Omega^2 l^2 \sin \alpha \cos \alpha \mathbf{j} \quad (13)$$

This moment has a physical interpretation in term of centripetal acceleration. As the rod rotates with angular velocity  $\boldsymbol{\Omega}$ , the individual mass points experience a centripetal acceleration directed towards the  $z$  axis of  $\mathbf{a} = -\Omega^2 l \cos \alpha \mathbf{i}$ . This requires an external force directed along the  $\mathbf{i}$  direction: on  $m_1$ , a force of  $\mathbf{F}_1 = -\Omega^2 ml \cos \alpha \mathbf{i}$ ; on  $m_2$ , a force of  $\mathbf{F}_2 = \Omega^2 ml \cos \alpha \mathbf{i}$ ; no net force is required because the body is supported at its center of mass. These forces can only be supplied/resisted by an external moment applied at the origin, where the rod is supported, a moment about the origin of  $\mathbf{M} = \mathbf{r}_1 \times \mathbf{F}_1 + \mathbf{r}_2 \times \mathbf{F}_2 = -2m\Omega^2 l^2 \sin \alpha \cos \alpha \mathbf{j}$ , which is the moment required to sustain the motion.

As the rod rotates about the  $z$  axis, the moment or torque required to sustain the motion also rotates. Thus any fixture that is used to attach the rotating rod to the axle must sustain this torque.

Observe that this force goes to zero for  $\alpha = 0$  and  $\alpha = \pi/2$ . For these  $\alpha$ 's, the rod would be rotating about a principal axis.

This result can also be obtained by resolving the angular velocity  $\boldsymbol{\omega}$  into its component along the principal axis  $x', z'$ . The moment of inertia about the  $x'$  axis is zero since a point mass has no moment of inertia. The moment of inertia about the  $z'$  axis is  $I_{z'z'} = 2l^2 m$ . The component of  $\Omega$  in the  $z'$  direction is  $\Omega'_z = \Omega \cos \alpha$  and the angular momentum in the  $z'$  direction is  $H'_z = 2ml^2 \Omega \cos \alpha$ . Resolved back into components along the  $x, z$  axis we obtain

$$\mathbf{H} = -2ml^2\Omega \cos \alpha \sin \alpha \mathbf{i} + 2ml^2\Omega \cos^2 \alpha \mathbf{k} \quad (14)$$

in agreement with (11).

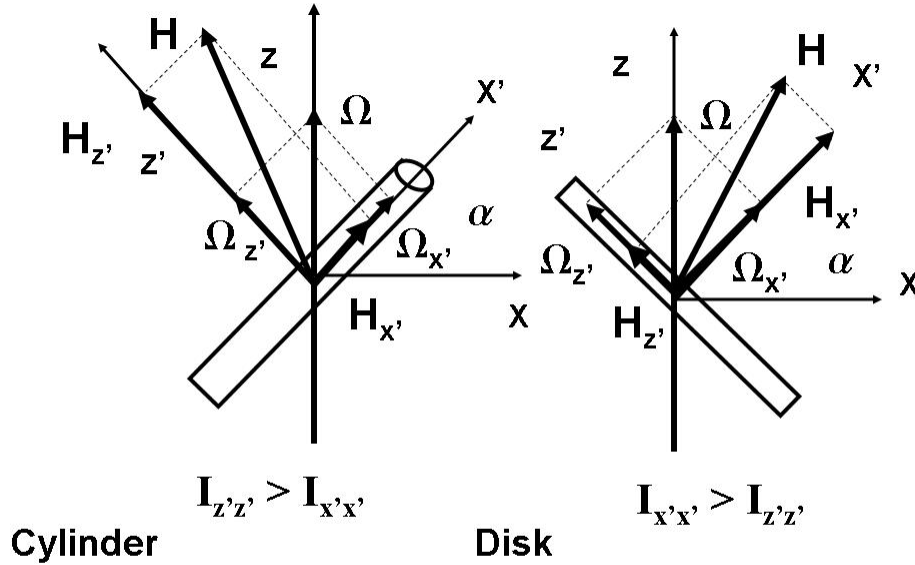
### Example: Rotating Cylinder

The previous discussion can easily be extended, replacing the rod and its mass points by a cylinder which rotates about the  $z$  axis with angular velocity  $\boldsymbol{\Omega}$ ; the  $z$  axis is not a principal axis. In this case, the moment of inertia  $I_{x'x'}$  is not equal to 0. Therefore, the angular momentum vector is not perpendicular to the cylinder



## REFERENCES

but is at an angle determined by the relative values of  $I_{z'z'}$  and  $I_{x'x'}$ . However for a "cylinder" as shown,  $I_{z'z'} = I_{y'y'}$ . As we shall show, the angular momentum is in the  $x, z$  plane, the plane formed by the cylinder axis and the  $z$  axis. As shown, we consider two "cylinders", one for which  $I_{z'z'} > I_{x'x'}$  and another, which is more like a disk, for which  $I_{z'z'} < I_{x'x'}$  where  $x', y', z'$  are principal axis as sketched. The axis of the cylinder makes an angle  $\alpha$  with the cylinder  $x'$  axis. We consider the motion at the **instant** at which the body lies in the  $x, z$  plane.



To determine the direction and magnitude of the angular momentum vector, we express the vector  $\mathbf{\Omega}$  in its components along principal axes  $\Omega_{x'}$  and  $\Omega_{z'}$ . The principal axis coordinate system  $x', z'$  is obtained from the  $x, z$  coordinate system by a rotation through the angle  $\alpha$ . The components of the angular momentum vector in the principal axis system can be immediately obtained as  $H_{x'} = I_{x'x'}\Omega_{x'}$  and  $H_{z'} = I_{z'z'}\Omega_{z'}$ . The total angular momentum vector formed from these components is sketched on the figure. For  $I_{z'z'} > I_{x'x'}$ , the angular momentum vector is tipped to the left of  $\mathbf{\Omega}$ ; for  $I_{z'z'} < I_{x'x'}$ , the angular momentum vector is tipped to the right of  $\mathbf{\Omega}$ . Since  $\dot{\mathbf{H}} = \mathbf{\Omega} \times \mathbf{H} = \mathbf{M}$ , the moment about the support point required to support this motion differ in sign in the two cases: out of or into the  $x, z$  plane.

Using the formalism of the inertia tensor in the following section, we shall show that the angular momentum expressed in the  $x, y, z$  coordinate system is

$$\begin{pmatrix} H_{Gx} \\ H_{Gy} \\ H_{Gz} \end{pmatrix} = \Omega \begin{pmatrix} \cos\alpha \sin\alpha I_{x'x'} - \cos\alpha \sin\alpha I_{z'z'} \\ 0 \\ \cos^2\alpha I_{z'z'} + \sin^2\alpha I_{x'x'} \end{pmatrix} \quad (15)$$

The angular momentum has only  $x$  and  $z$  components, and is thus in the  $x, z$  plane. If the angle  $\alpha$  is either 0 or  $\pi/2$  the angular momentum is aligned with the  $\mathbf{\Omega}$  vector and  $\dot{\mathbf{H}} = 0$ , since both cases, the rotation

occurs about a principal axis. The moment required to sustain this motion is normal to the  $x, z$  plane, in the  $y$  direction. It is given by

$$\mathbf{M} = \dot{\mathbf{H}} = \boldsymbol{\Omega} \times \mathbf{H} = -\Omega^2 (\cos \alpha \sin \alpha (I_{x'x'} - I_{z'z'})) \quad (16)$$

If  $I_{x'x'} > I_{z'z'}$ , the moment is in the negative  $y$  direction, out of the plane. If  $I_{x'x'} < I_{z'z'}$ , the moment is in the positive  $y$  direction, into the plane. If  $I_{x'x'} = I_{z'z'}$ , the moment is zero since the body is dynamically equivalent to a sphere. Also, the moment required disappears in  $\alpha = 0$  or  $\alpha = \pi/2$  since the rotation axis,  $x$  or  $z$ , becomes a principal axis.

If the body is not a "cylinder", i.e.  $I_{x'x'} \neq I_{y'y'} \neq I_{z'z'}$ , then the result is qualitatively similar but the angular momentum vector  $\mathbf{H}$  is not in the plane formed by the cylinder axis and  $\boldsymbol{\Omega}$ .

### Example: Use of the inertia tensor

The example of the skew rod may be treated using the more formal description in terms of the inertia tensor. The principal axis of this system is clearly along the rod joining the two masses, as shown in the figure. If we treat the masses as mass points, there is no moment of inertia about the  $x'$  axis, while the moments of inertia about the  $y'$  and  $z'$  axis are equal and equal to  $I_{z'z'} = I_{y'y'} = 2ml^2$ . The coordinate transformation from the principal axes  $x', y'$  to the  $x, y, z$  axis is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}. \quad (17)$$

The inertia tensor in the  $x, y, z$  system is obtained by the transformation introduced in Lecture 26:  $[I] = [T][I'][T]^T$ .

$$\begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{pmatrix} = \begin{pmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2ml^2 & 0 \\ 0 & 0 & 2ml^2 \end{pmatrix} \begin{pmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{pmatrix}. \quad (18)$$

resulting in  $[I]$  for the inertia matrix in the  $x, y, z$  system as

$$[I] = 2ml^2 \begin{pmatrix} \sin^2 \alpha & 0 & -\cos \alpha \sin \alpha \\ 0 & 1 & 0 \\ -\cos \alpha \sin \alpha & 0 & \cos^2 \alpha \end{pmatrix}. \quad (19)$$

This confirms that the  $x$  and  $z$  axis are not principal axes. The angular momentum vector is then

$$\begin{pmatrix} H_{Gx} \\ H_{Gy} \\ H_{Gz} \end{pmatrix} = 2ml^2 \begin{pmatrix} \sin^2 \alpha & 0 & -\cos \alpha \sin \alpha \\ 0 & 1 & 0 \\ -\cos \alpha \sin \alpha & 0 & \cos^2 \alpha \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \Omega \end{pmatrix} = 2ml^2 \Omega \begin{pmatrix} -\cos \alpha \sin \alpha \\ 0 \\ \cos^2 \alpha \end{pmatrix} \quad (20)$$

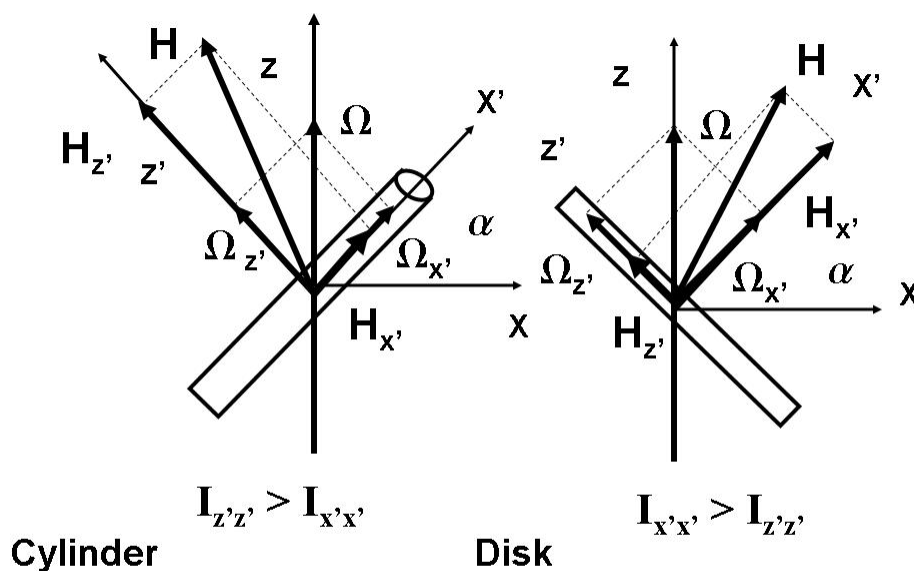
for the angular momentum. Leading to the required moment to sustain the motion as

$$\mathbf{M} = \dot{\mathbf{H}} = \boldsymbol{\Omega} \times \mathbf{H} = -2l^2 m \Omega^2 \cos \alpha \sin \alpha \mathbf{j} \quad (21)$$

in agreement with equation (13).

### Example: Rotating Cylinder using Inertia Tensor

These results can be extended to describe the angular momentum vector for a general body. We consider a body for which the moments of inertia about the principal axes,  $x', y', z'$  are  $I_{z'z'} = I_{y'y'}$  ie. a cylinder ( $I_{x'x'} < I_{z'z'}$ ), or a disk ( $I_{x'x'} > I_{z'z'}$ ). This body is fixed to an support and is spinning about the  $z$  axis with angular velocity  $\Omega$ . We desire to calculate the moment required at the attachment point to produce this motion.



Applying the previous transformation of the diagonal inertia tensor to the  $x, y, z$  coordinate system, we calculate the angular momentum in the  $x, y, z$  coordinate system using the same transformation matrix that we used for the rotating rod.

$$\begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{pmatrix} = \begin{pmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{pmatrix} \begin{pmatrix} I_{x'x'} & 0 & 0 \\ 0 & I_{z'z'} & 0 \\ 0 & 0 & I_{z'z'} \end{pmatrix} \begin{pmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{pmatrix}. \quad (22)$$

where the inertia tensor of the rotating rod has been replaced with the inertia tensor of the cylinder/disk with  $I_{y'y'} = I_{z'z'}$ . The angular momentum for a rotation only about the  $z$  axis is obtain as

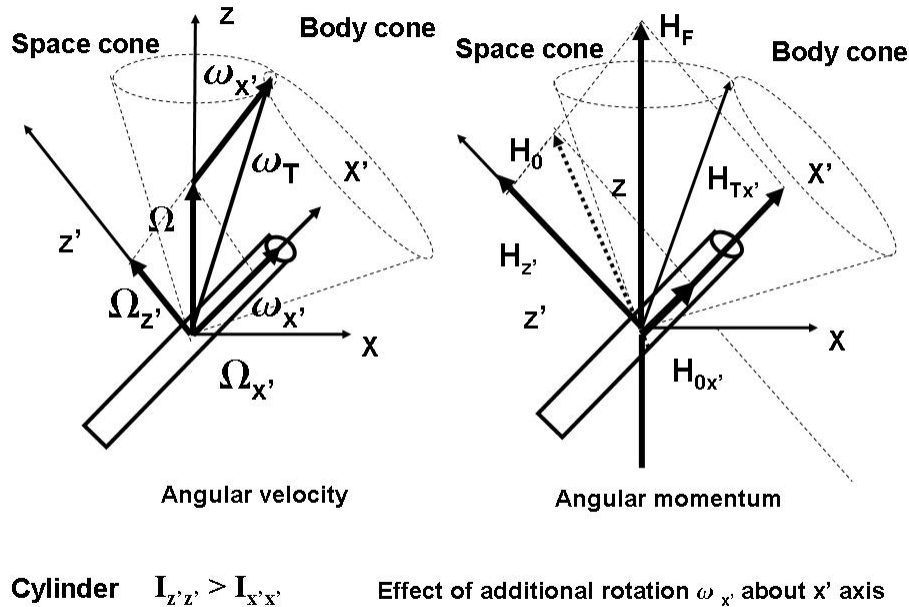
$$\begin{pmatrix} H_{Gx} \\ H_{Gy} \\ H_{Gz} \end{pmatrix} = \begin{pmatrix} \cos^2 \alpha I_{x'x'} + \sin^2 \alpha I_{z'z'} & 0 & \cos \alpha \sin \alpha I_{x'x'} - \cos \alpha \sin \alpha I_{z'z'} \\ 0 & I_{x'x'} & 0 \\ \cos \alpha \sin \alpha I_{x'x'} - \cos \alpha \sin \alpha I_{z'z'} & 0 & \cos^2 \alpha I_{z'z'} + \sin^2 \alpha I_{x'x'} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \Omega \end{pmatrix} \quad (23)$$

$$\begin{pmatrix} H_{Gx} \\ H_{Gy} \\ H_{Gz} \end{pmatrix} = \Omega \begin{pmatrix} \cos \alpha \sin \alpha I_{x'x'} - \cos \alpha \sin \alpha I_{z'z'} \\ 0 \\ \cos^2 \alpha I_{z'z'} + \sin^2 \alpha I_{x'x'} \end{pmatrix} \quad (24)$$

From this expression we can see that while the angular momentum vector always points in the positive  $z$  direction, the  $x$  component changes sign depending on the relative magnitude of  $I_{x'x'}$  and  $I_{z'z'}$ . Note also that if  $I_{x'x'} = I_{z'z'}$ , the  $x$  component of angular momentum goes to zero and the body is essentially a sphere.

### Example: Rotating Cylinder Including Additional Axial Rotation

We now examine the rotating "tipped" cylinder from our previous example, but we modify it by including an additional angular velocity  $\omega_{x'}$  about the  $x'$  axis; think of it as a rotating cylinder inside a sleeve attached to a fixed support. The geometry is the same as the previous example with the addition of an angular velocity  $\omega_{x'}$  about the  $x'$  axis. We show this figure for the special case we are aiming at but discuss it in general for arbitrary values of  $\omega_{x'}$ .



## REFERENCES

The total angular velocity vector is the vector sum of  $\boldsymbol{\Omega}$  and  $\boldsymbol{\omega}_{x'}$ . The vector  $\boldsymbol{\Omega}$  is in the  $z$  direction; the vector  $\boldsymbol{\omega}_{x'}$  is in the  $x'$  direction. The vector sum  $\boldsymbol{\omega}_T$  is in the direction shown. The vector  $\boldsymbol{\omega}_T$  is the instantaneous axis of rotation. As the cylinder rotates about the  $z$  axis, the "body cone" and "space cone" are sketched out as shown.

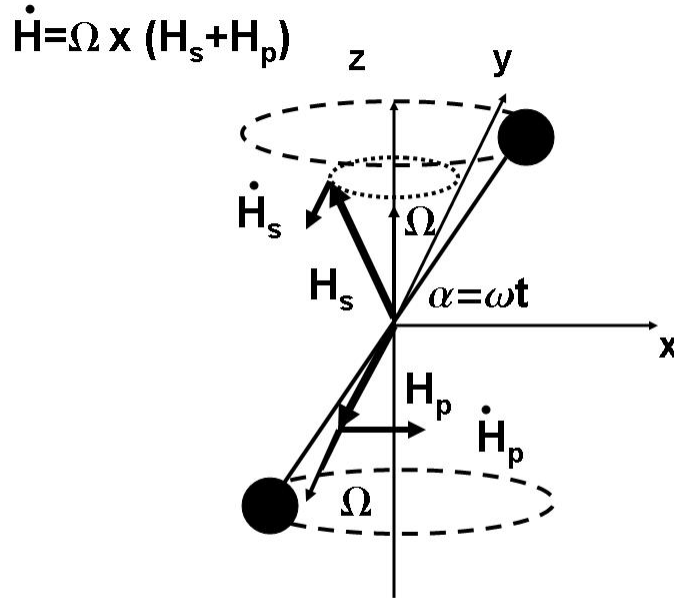
The total angular momentum is the vector sum of the angular momentum of the cylinder rotating with angular velocity  $\boldsymbol{\Omega}$ ,  $\mathbf{H}_0$ , plus the angular momentum  $\mathbf{H}_{x'}$  due to the rotation  $\boldsymbol{\omega}_{x'}$  about the  $x'$  axis,  $I_{x'x'}\boldsymbol{\omega}_{x'}$ .  $\mathbf{H}_F = \mathbf{H}_0 + \mathbf{H}_{x'}$ . In general, a positive rotation about the  $x'$  axis tips the resultant angular momentum vector  $\mathbf{H}_F$  towards the vertical  $z$  axis. We focus on the particular case where the  $\boldsymbol{\omega}_{x'}$  is chosen such that the angular momentum has been rotated to coincide with the  $z$  axis.

The implications of this are profound. Since the moment required to produce this motion is  $\mathbf{M} = \boldsymbol{\Omega} \times \mathbf{H}_F$ , **the moment required for this particular choice of  $\boldsymbol{\omega}_{x'}$  is zero, since  $\boldsymbol{\Omega}$  is aligned with  $\mathbf{H}_F$ .**

This particular solution is an example of "free" motion, that is the body would rotate with this particular combination of  $\boldsymbol{\Omega}$  and  $\boldsymbol{\omega}$ , as determined by the relative magnitudes of the inertia about the principal axis  $x'$  and  $z'$ ,  $I_{x'x'}$  and  $I_{z'z'}$ , and no moment would be required about the support to sustain this motion. We will return to this example when we consider "free" motions of rotating bodies.

### Example: Two-Bladed Wind Turbines

An exciting application of the analysis of the rotating skew rod is to the operation of a two-bladed wind turbine. Recall that the analysis predicted that a constant moment about a horizontal axis is required from the support structure if the skew rod is to rotate about the vertical axis with angular velocity  $\Omega$  aligned at an angle  $\alpha$ . **This analysis is also valid if  $\alpha$  is a function of time** if we include the angular momentum about the  $y$  axis as a result of the "propeller" rotating about its own axis. Let  $\alpha = \omega t$ , that is the rod rotates about the horizontal  $y$  axis through its center of mass and perpendicular to the plane containing the rod and masses, like a propeller. In addition, it rotates about the  $z$  axis following the previous analysis of the rotating skew rod. The total angular momentum is then the result from the previous analysis with  $\alpha = \omega t$ ; at an instant of time, the angular momentum of the skew rod becomes  $\mathbf{H}_s = -2ml^2\Omega \cos(\omega t)\sin(\omega t)\mathbf{i} + 2ml^2\Omega \cos^2(\omega t)\mathbf{k}$  plus angular momentum due to the rotation of the propeller of magnitude  $\mathbf{H}_p = -2\omega ml^2\mathbf{j}$ . The additional change in angular momentum due to the "propeller" rotation is simply  $\dot{\mathbf{H}}_p = \boldsymbol{\Omega} \times \mathbf{H}_p$ . This is constant in time in a coordinate system rotating with the plane of the propeller. This motion requires an applied steady moment about the  $z$  axis.



The total angular momentum of this configuration is then the sum of that for the skew rod plus that for the rotation of the propeller about the y axis. However, the additional change of angular momentum occurs only through the action of  $\Omega \mathbf{k}$  in rotating the angular momentum of the propeller. The angular velocity of the propeller,  $\omega$  does not rotate the angular momentum vector of the skew rod. Thus the time rate of change of angular momentum is due only to  $\Omega$ , the rotation about the vertical axis and not the rotation  $\omega$  of the propeller itself. From Equation (14), with  $\alpha$  replaced by  $\omega t$ , and the addition of the angular momentum of the propeller, the total angular momentum is

$$\mathbf{H} = \mathbf{H}_s + \mathbf{H}_p = -2l^2 m \Omega \cos(\omega t) \sin(\omega t) \mathbf{i} - 2ml^2 \omega \mathbf{j} + 2ml^2 \Omega \cos^2(\omega t) \mathbf{k} \quad (25)$$

Since only  $\Omega \mathbf{k}$  rotates the angular momentum vector, we have from Coriolis theorem that the rate of change of angular momentum is

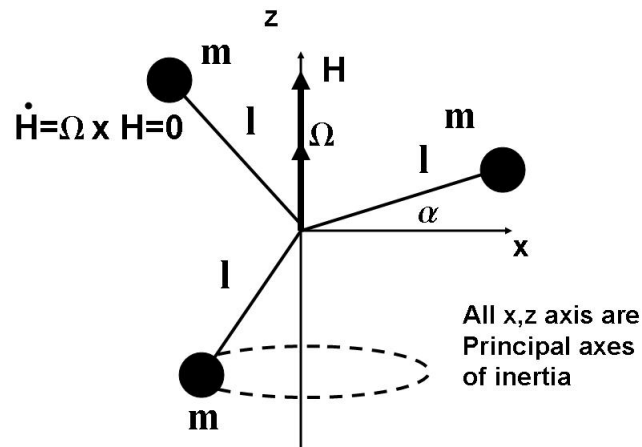
$$\dot{\mathbf{H}} = (\Omega \mathbf{k}) \times (\mathbf{H}_s + \mathbf{H}_p) \quad (26)$$

Then the moment required is

$$\mathbf{M} = 2l^2 m (\omega \Omega \mathbf{i} - \Omega^2 \cos(\omega t) \sin(\omega t) \mathbf{j}) + 0 \mathbf{k}. \quad (27)$$

Note the unsteady behavior of the required moment with time. The moment required to produce the propeller rotation about  $\mathbf{j}$ , and to move the direction of this rotation about the z axis with angular velocity  $\Omega$  is steady; the moment required to produce the rotation of the skew rod about  $\mathbf{k}$  is unsteady. This is due to the imbalance in inertia. At various times in its motion, the device is rotating about the z axis ( $\mathbf{k}$ ) which is not a principal axis for the configuration.

Consider this problem as a model for a two-bladed wind turbine. The wind turbine will happily rotate about its  $y$  axis with angular velocity  $\omega$ . But wind turbines must yaw to face into a new wind direction. This requires for a short time an angular velocity  $\Omega$  about the  $z$  axis. From our analysis, as the wind turbine rotates in yaw to face the new wind direction, an oscillatory moment of  $\mathbf{M} = -2l^2m\Omega^2 \cos(\omega t) \sin(\omega t)\mathbf{j}$  will be exerted on the support structure. This undesirable oscillatory load will be eliminated if the wind turbine has three blades so that all directions in the  $x, z$  plane are principal axis.



**3 Bladed Skew rod**

In this case, all axis in the  $z, x$  plane are principal axis with equal moments of inertia. For this case, the angular momentum vector  $\mathbf{H}$  due to the rotation  $\Omega$  is parallel to the angular velocity vector  $\Omega$ , so that  $\dot{\mathbf{H}} = 0$  and no unsteady moment will occur as the wind turbine rotates about the  $z$  axis to face the new on-coming wind direction. A steady moment in the  $\mathbf{j}$  direction is required to rotate the angular momentum vector due to the rotation of the propeller  $\omega$ .

The figure shows how this idea has caught on.



One can also see application of this principle to the design of early propeller driven fighter planes which must make rapid turns



#### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition  
5/1, 5/2, 5/3, 5/4 (review) , 5/5, 5/6 (review)



## Lecture 30— 3D Rigid Body Dynamics: Equations of Motion; Euler's Equations

### 3D Rigid Body Dynamics: Euler's Equations

We now turn to the task of deriving the general equations of motion for a three-dimensional rigid body. These equations are referred to as Euler's equations. The governing equations are those of conservation of linear momentum  $\mathbf{L} = M\mathbf{v}_G$  and angular momentum,  $\mathbf{H} = [I]\boldsymbol{\omega}$ , where we have written the moment of inertia in matrix form to remind us that in general the direction of the angular momentum is not in the direction of the rotation vector  $\boldsymbol{\omega}$ . Conservation of linear momentum requires

$$\dot{\mathbf{L}} = \mathbf{F} \quad (1)$$

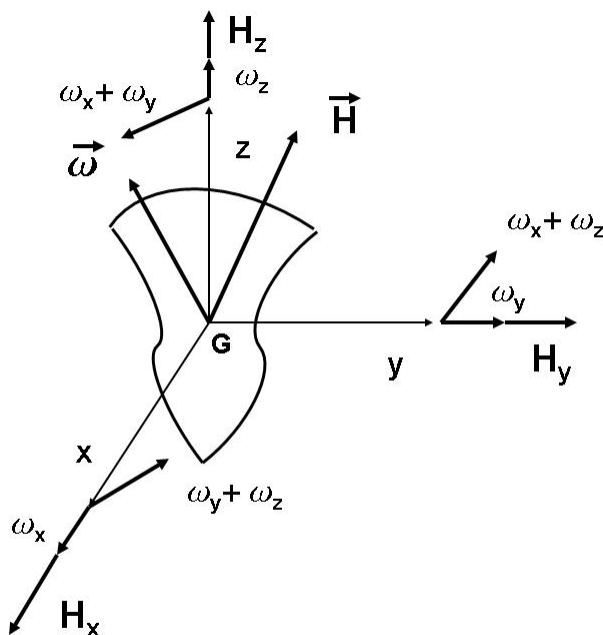
Conservation of angular momentum, about a fixed point  $O$ , requires

$$\dot{\mathbf{H}}_O = \mathbf{M} \quad (2)$$

or about the center of mass  $G$

$$\dot{\mathbf{H}}_G = \mathbf{M}_G \quad (3)$$

In our previous application of these equations, we specified the motion and used the equations to specify what moments would be required to produce the prescribed motion. In this more general formulation, we allow the body to execute free motions, possibly under the action of external moments. We consider the general motion of a body about its center of mass, first examining this in a inertial reference frame.



At an instant of time, we can calculate the angular momentum of the body as  $\mathbf{H} = [I]\boldsymbol{\omega}$ . One possible method to obtain the moments and the motion of the body is to perform our analysis in this inertial coordinate system. We would of course align our coordinate system initially with the principal axes of the body. We could then write

$$\mathbf{M}_G = \dot{\mathbf{H}}_G = d/dt([I]\boldsymbol{\omega}) = [\dot{I}]\boldsymbol{\omega} + [I]\dot{\boldsymbol{\omega}} \quad (4)$$

This would be a appropriate approach but the difficulty is keeping track of  $[\dot{I}]$  in the inertial coordinate system. The initial inertial axis, even if principal axis, will not remain principal axis, and the inertia "seen" in this coordinate system will vary with time. So unless we are considering the motion of a sphere, for which all axis are principal and the inertia tensor is constant about all axis, we cannot get very far with this approach.

## Body-Fixed Axis

We formulate the governing equations of motion in an axis system fixed to the body, paying the price for keeping track of the motion of the body in order to have the inertia tensor remain independent of time in our reference frame. Given our earlier discussion of terms added to the description of motion in a rotating and accelerating coordinate system, it may seem surprising that we can do this easily, but the statement of conservation of angular momentum about the center of mass, or about a fixed point of instantaneous rotation holds if we include the changes in angular momentum arising from Coriolis Theorem. The general body is shown in the figure. We fix the  $x, y, z$  axis to the body and instantaneously align them with  $x, y, z$ . Referring to the figure, we see the components of  $\boldsymbol{\omega}$ ,—  $\omega_1$ ,  $\omega_2$  and  $\omega_3$ — and the components of the angular momentum vector  $\mathbf{H}$ , which in general is not aligned with the angular velocity vector. We also see the vectors  $\omega_2\mathbf{j} + \omega_3\mathbf{k}$

applied to the  $x$  axis, with corresponding components of  $\boldsymbol{\omega}$  sketched at the  $y$  and  $z$  axes. At this instant of time, the change in  $\mathbf{H}$  will be due to actual time rate of change  $\dot{\mathbf{H}}$  plus the effect of the instantaneous rotation of the  $x$  axis due to  $\boldsymbol{\omega}$ : the change in  $\mathbf{H}$  due to the instantaneous rotation of the coordinate system is

$$\dot{\mathbf{H}} = [I]\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times \mathbf{H} \quad (5)$$

Equating the change in angular momentum to the external moments, we have the statement of conservation of angular momentum in body-fixed axes.

$$M_x = \dot{H}_x - H_y\omega_z + H_z\omega_y \quad (6)$$

$$M_y = \dot{H}_y - H_z\omega_x + H_x\omega_z \quad (7)$$

$$M_z = \dot{H}_z - H_x\omega_y + H_y\omega_x \quad (8)$$

This particular form of the equations of motion is valid for any set of body-fixed axes. If the axes chosen are principal axes, then we may express the conservation of angular momentum in terms of moments of inertia about the principle axes,

$$M_x = I_{xx}\dot{\omega}_x - (I_{yy} - I_{zz})\omega_y\omega_z \quad (9)$$

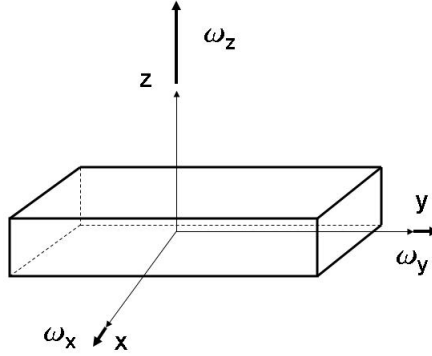
$$M_y = I_{yy}\dot{\omega}_y - (I_{zz} - I_{xx})\omega_z\omega_x \quad (10)$$

$$M_z = I_{zz}\dot{\omega}_z - (I_{xx} - I_{yy})\omega_x\omega_y \quad (11)$$

These equations are called Euler's equations. They provide several serious challenges to obtaining the general solution for the motion of a three-dimensional rigid body. First, they are non-linear (containing products of the unknown  $\omega$ 's). This means that elementary solutions cannot be combined to provide the solution for a more complex problem. But a more fundamental difficulty, is that ***we do not know the location of the axis system,  $x$ ,  $y$  and  $z$ .*** Recall that since the axes are fixed to the body, we are committed to follow the body as it rotates in order to use these equations to obtain a solution. Thus, we must develop a method to follow the changes in axis location as the body rotates. Before turning to this problem, we examine a situation where we know the location of the axis, at least approximately.

## Stability of Free Motion about a Principal Axis

Consider a body rotating about the  $z$  axis—a principal axis—with angular velocity  $\omega_z$ . Without loss of generality, we may consider this to be a rectangular block. We take advantage of the fact that we know to a good approximation the axis of rotation, at least initially. We examine the question of stability to small perturbations, rotations about the  $x$  and  $y$  axis of magnitude  $\omega_x$  and  $\omega_y$  where  $\omega_x \ll \omega_z$  and  $\omega_y \ll \omega_z$ . The moments of inertia about the  $x$ ,  $y$  and  $z$  axes are  $I_{xx}$ ,  $I_{yy}$  and  $I_{zz}$ ; we say nothing about the magnitudes of these inertias at this point.



Assume a small impulsive moment that initiates a small rotation about the  $x$  and  $y$  axes and thereafter the motion proceeds with no applied external moments. For this case, Euler's equations become

$$0 = I_{xx}\dot{\omega}_x - (I_{yy} - I_{zz})\omega_y\omega_z \quad (12)$$

$$0 = I_{yy}\dot{\omega}_y - (I_{zz} - I_{xx})\omega_z\omega_x \quad (13)$$

$$0 = I_{zz}\dot{\omega}_z - (I_{xx} - I_{yy})\omega_x\omega_y \quad (14)$$

The  $z$  equation contains the product of two small terms in contrast to the other two equations where the size of the terms is comparable—as far as we can tell. Therefore  $\dot{\omega}_z \ll \dot{\omega}_x$  and  $\dot{\omega}_z \ll \dot{\omega}_y$ . We therefore note that since  $I_{zz}\dot{\omega}_z = (I_{xx} - I_{yy})\omega_x\omega_y$  with both  $\omega_x$  and  $\omega_y$  being small quantities, we may take  $\omega_z$  as constant, and equal to some  $\omega$ , since  $\dot{\omega}_z \ll \dot{\omega}_x$  and  $\dot{\omega}_z \ll \dot{\omega}_y$ . We now differentiate equation (12) with time, and substitute  $\dot{\omega}_y$  from equation (13) to obtain an equation for  $\omega_x$ . (We could do this as well for  $\omega_y$  with the same result.)

$$I_{xx}\ddot{\omega}_x - \frac{(I_{yy} - I_{zz})(I_{zz} - I_{xx})}{I_{yy}}\omega^2\omega_x = 0 \quad (15)$$

or

$$\ddot{\omega}_x - A\omega_x = 0 \quad (16)$$

The solution to this differential equation for  $\omega_x(t)$  is

$$\omega_x(t) = Be^{\sqrt{A}t} + Ce^{-\sqrt{A}t} \quad (17)$$

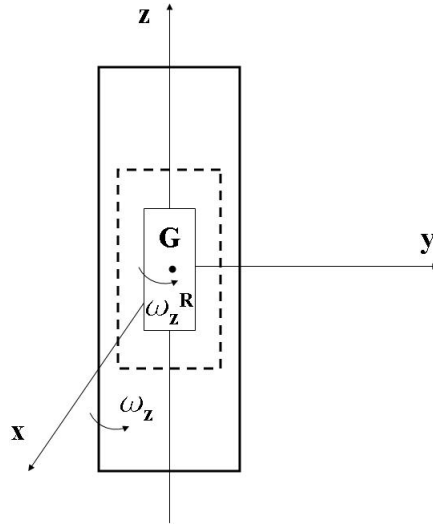
The stability of the motion is determined by the sign of  $A$ . If  $A$  is positive, an exponential divergence will result, and the initial small perturbation in  $\omega_x$  will grow without bound, as least as predicted by small perturbation analysis. If the sign of  $A$  is negative, oscillatory motion of  $\omega_x$  (and  $\omega_y$ ) will result, and the motion is stable.

## REFERENCES

Examining  $A = \frac{(I_{yy} - I_{zz})(I_{zz} - I_{xx})}{I_{yy}}$ , we see that the condition for  $A$  to be positive is that  $I_{zz}$  is intermediate between  $I_{xx}$  and  $I_{yy}$ . That is  $I_{xx} < I_{zz} < I_{yy}$  or  $I_{yy} < I_{zz} < I_{xx}$ . We conclude that a body rotating about an axis where the moment of inertia is intermediate between the other two inertias, is unstable. Also, that rotation about either the largest or smallest inertias is stable. This consideration relates to stability of a rotating body as predicted from Euler's equation; we have already examined the stability of rotation about the smallest inertia axis and concluded that if energy is dissipated, that motion is unstable.

## Stability of a Gyrostat

The analysis of the proceeding section can easily be extended to a more important, more useful and more complex configuration: the gyrostat satellite. A gyrostat consists of a spinning body which contains within itself another spinning body, referred to as the rotor. Gyrostat, or dual-spin, satellites are used when the external body of the satellite must spin slowly to accomplish its mission while it needs the stabilization provided by faster rotation. This is accomplished by placing a rotor inside the satellite. The angular momentum of the rotor is driven by a motor attached to the satellite; no net momentum increase occurs when an adjustment in rotor speed is made. The examination of this device is quite straightforward if the rotor principal axes are aligned with the satellite principal axes, the rotor is axisymmetric so that  $I_{xx}^R = I_{yy}^R$ , and the center of mass of the rotor is placed at the center of mass of the satellite. Thus the rotor is constrained to move with the rotational motion of the satellite and principal axes remain principal.



The moment of inertia of the rotor about its principal axes is

$$[I^R] = \begin{pmatrix} I_{xx}^R & 0 & 0 \\ 0 & I_{xx}^R & 0 \\ 0 & 0 & I_{zz}^R \end{pmatrix} \quad (18)$$

while the moment of inertia of the satellite platform is

$$[I^P] = \begin{pmatrix} I_{xx}^P & 0 & 0 \\ 0 & I_{yy}^P & 0 \\ 0 & 0 & I_{zz}^P \end{pmatrix} \quad (19)$$

Because of the geometric constraints on the system,  $\omega_x$  and  $\omega_y$  are equal for both platform and rotor while  $\omega_z$  differs for platform,  $\omega_z^P$  and rotor,  $\omega_z^R$ . Therefore the total angular momentum is

$$[H] = \begin{pmatrix} I_{xx}^P & 0 & 0 \\ 0 & I_{yy}^P & 0 \\ 0 & 0 & I_{zz}^P \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z^P \end{pmatrix} + \begin{pmatrix} I_{xx}^R & 0 & 0 \\ 0 & I_{xx}^R & 0 \\ 0 & 0 & I_{zz}^R \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z^R \end{pmatrix} \quad (20)$$

Because of the constraints on  $\omega$ , we can write the angular momentum as

$$[H] = \begin{pmatrix} I_{xx}^P + I_{xx}^R & 0 & 0 \\ 0 & I_{yy}^P + I_{xx}^R & 0 \\ 0 & 0 & I_{zz}^P + I_{zz}^R \omega_z^R / \omega_z^P \end{pmatrix} \begin{pmatrix} \omega_x^P \\ \omega_y^P \\ \omega_z^P \end{pmatrix} \quad (21)$$

where  $\omega$  has been identified as  $\omega^P$ .

This is a remarkable result. It indicates that the effect of the rotation of the rotor on the angular momentum of the gyrostator can be incorporated by a modification of the moment of inertia of the platform about the  $z$  axis. We will call this the "effective" moment of inertia,

$$I_{zz_{eff}}^P = I_{zz}^P + I_{zz}^R \omega_z^R / \omega_z^P \quad (22)$$

where,  $\omega_z^R$  is the rotational velocity of the rotor.

With this result, and the identification of the "effective" moment of inertia in  $z$ , the previous analysis of the stability of rotation of a body with unequal moments of inertia goes through with the replacement of  $I_{zz}$  by  $I_{zz_{eff}}$ . Thus a satellite with  $I_{zz}$  a minimum inertia, which would be unstable due to dissipation, can be stabilized by the addition of the angular momentum of a rotor oriented along the  $z$  axis.

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J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition 7.9

J. B. Marion, S. T. Thornton, *Classical Dynamics of Particles and Systems*, 11.8

W.T. Thompson, *Introduction to Space Dynamics*, Chapter 5

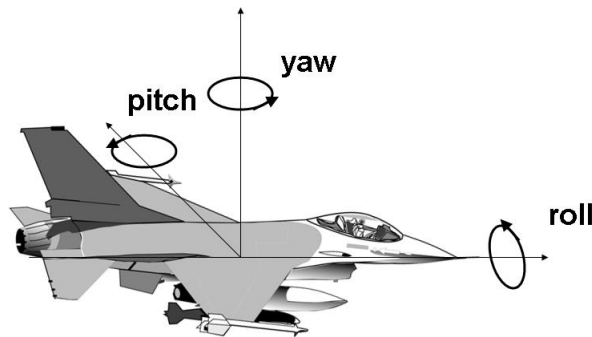
F. P. J. Rimrott, *Introductory Attitude Dynamics*, Chapter 11

## Lecture 31 — 3D Rigid Body Dynamics: Euler Angles

### Euler Angles

The difficulty of describing the positions of the body-fixed axis of a rotating body is approached through the use of Euler angles: spin  $\psi$ , nutation  $\theta$ , and precession  $\phi$ , shown below. In this case we surmount the difficulty of keeping track of the principal axes fixed to the body by making their orientation the **unknowns** in our equations of motion; then the angular velocities and angular accelerations which appear in Euler's equations are expressed in terms of these fundamental unknowns, the positions of the principal axes expressed as angular deviations from some initial positions.

Unfortunately, there is no standard formulation nor standard notation for Euler angles. We choose to follow one typically used in physics textbooks. However, for aircraft and spacecraft motion a slightly different one is used; the primary difference is in the definition of the “pitch” angle. For aircraft motion, we usually refer the motion to a horizontal rather than to a vertical axis. In a description of aircraft motion,  $\psi$  would be the roll angle;  $\phi$  the yaw angle; and  $\theta$  the pitch angle. The pitch angle  $\theta$  would be measured from the horizontal rather than from the vertical, as is customary and useful to describe a spinning top.



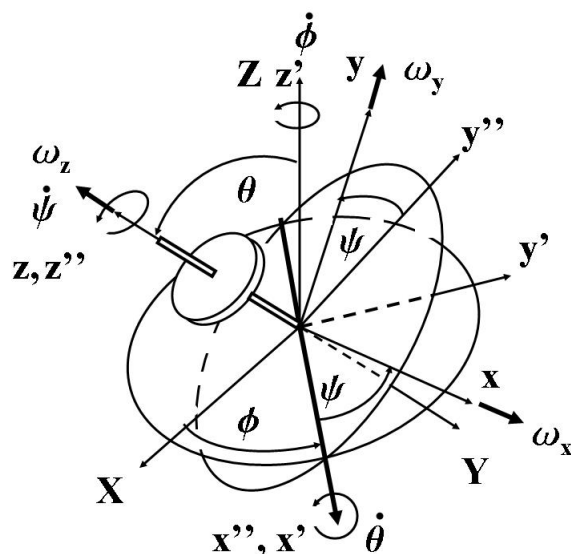
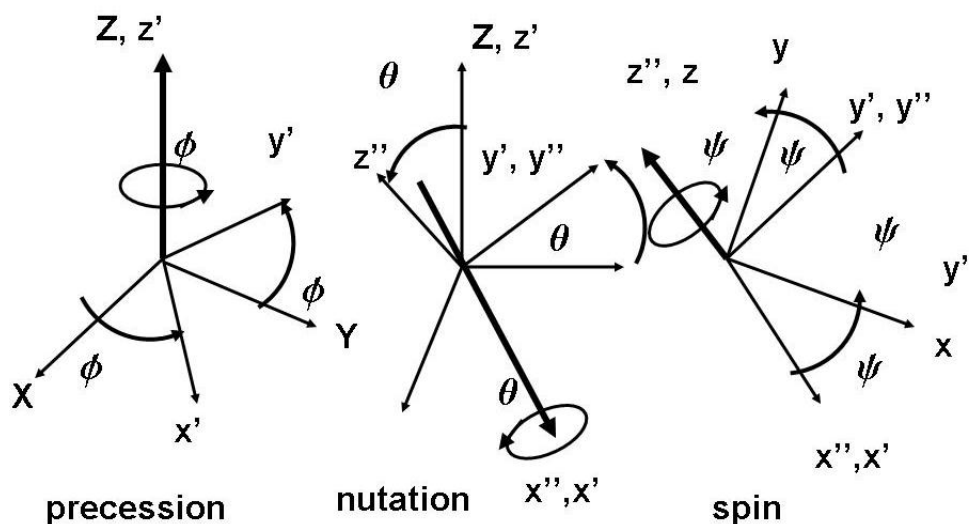


Figure 1: Euler Angles

In order to describe the angular orientation and angular velocity of a rotating body, we need three angles. As shown on the figure, we need to specify the rotation of the body about its “spin” or  $z$  body-fixed axis, the angle  $\psi$  as shown. This axis can also “nutate” through an angle  $\theta$ , and “precess” through an angle  $\phi$ . To develop the description of this motion, we use a series of transformations of coordinates, as we did in Lecture 3. The final result is shown below. This is the coordinate system used for the description of motion of a general three-dimensional rigid body described in body-fixed axis.

To identify the new positions of the principal axes as a result of angular displacement through the three Euler angles, we go through a series of coordinate rotations, as introduced in Lecture 3.



We first rotate from an initial  $X, Y, Z$  system into an  $x', y', z'$  system through a rotation  $\phi$  about the  $Z, z'$



axis. The angle  $\phi$  is called the angle of precession.

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = [T_1] \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}.$$

The resulting  $x', y'$  coordinates remain in the  $X, Y$  plane. Then, we rotate about the  $x'$  axis into the  $x'', y'', z''$  system through an angle  $\theta$ . The  $x''$  axis remains coincident with the  $x'$  axis. The axis of rotation for this transformation is called the "line of nodes". The plane containing the  $x'', y''$  coordinate is now tipped through an angle  $\theta$  relative to the original  $X, Y$  plane. The angle  $\theta$  is called the angle of nutation.

$$\begin{pmatrix} x'' \\ y'' \\ z'' \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = [T_2] \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}.$$

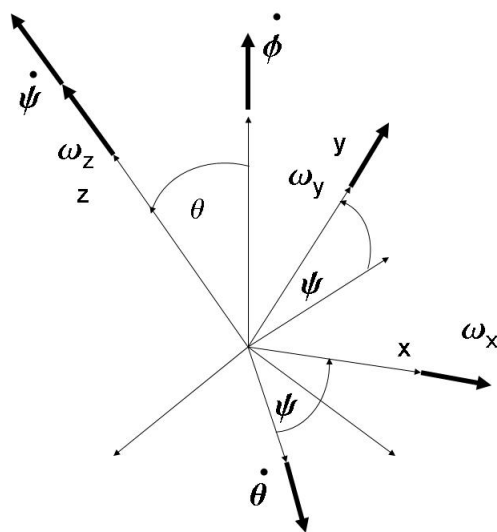
And finally, we rotate about the  $z'', z$  system through an angle  $\psi$  into the  $x, y, z$  system. The  $z''$  axis is called the spin axis. It is coincident with the  $z$  axis. The angle  $\psi$  is called the spin angle; the angular velocity  $\dot{\psi}$  the spin velocity.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x'' \\ y'' \\ z'' \end{pmatrix} = [T_3] \begin{pmatrix} x'' \\ y'' \\ z'' \end{pmatrix}.$$

The final "Euler" transformation is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = [T_3][T_2][T_1] \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} \cos \psi \cos \phi - \cos \theta \sin \phi \sin \psi & \cos \psi \sin \phi + \cos \theta \cos \phi \sin \psi & \sin \theta \sin \psi \\ -\sin \psi \cos \phi - \cos \theta \sin \phi \cos \psi & -\sin \psi \sin \phi + \cos \theta \cos \phi \cos \psi & \sin \theta \cos \psi \\ \sin \phi \sin \theta & -\cos \phi \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}.$$

This is the final  $x, y, z$  body-fixed coordinate system for the analysis, with angular velocities  $\omega_x, \omega_y, \omega_z$  as shown. The individual coordinate rotations  $\dot{\phi}$ ,  $\dot{\theta}$  and  $\dot{\psi}$  give us the angular velocities. However, these vectors do not form an orthogonal set:  $\dot{\phi}$  is along the original  $Z$  axis;  $\dot{\theta}$  is along the line of nodes or the  $x'$  axis; while  $\dot{\psi}$  is along the  $z$  or spin axis.



This is easily reorganized by taking the components of these angular velocities about the final  $x, y, z$  coordinate system using the Euler angles, giving

$$\omega_x = \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi \quad (1)$$

$$\omega_y = \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi \quad (2)$$

$$\omega_z = \dot{\phi} \cos \theta + \dot{\psi} \quad (3)$$

We could press on, developing formulae for angular momentum, and changes in angular momentum in this coordinate system, applying these expressions to Euler's equations and develop the complete set of governing differential equations. In general, these equations are very difficult to solve. We will gain more understanding by selecting a few simpler problems that are characteristic of the more general motions of rotating bodies.

### 3D Rigid Body Dynamics: Free Motions of a Rotating Body

We consider a rotating body in the absence of applied/external moments. There could be an overall gravitational force acting through the center of mass, but that will not affect our ability to study the rotational motion about the center of mass independent of such a force and the resulting acceleration of the center of mass. (Recall that we may equate moments to the rate of change of angular momentum about the center of mass even if the center of mass is accelerating.) Such a body could be a satellite in rotational motion in orbit. The rotational motion about its center of mass as described by the Euler equations will be independent of its orbital motion as defined by Kepler's laws. For this example, we consider that the body is symmetric such that the moments of inertia about two axis are equal,  $I_{xx} = I_{yy} = I_0$ , and the moment of inertia about  $z$  is  $I$ . The general form of Euler's equations for a free body (no applied moments) is

$$0 = I_{xx}\dot{\omega}_x - (I_{yy} - I_{zz})\omega_y\omega_z \quad (4)$$

$$0 = I_{yy}\dot{\omega}_y - (I_{zz} - I_{xx})\omega_z\omega_x \quad (5)$$

$$0 = I_{zz}\dot{\omega}_z - (I_{xx} - I_{yy})\omega_x\omega_y \quad (6)$$

For the special case of a symmetric body for which  $I_{xx} = I_{yy} = I_0$  and  $I_{zz} = I$  these equations become

$$0 = I_0\dot{\omega}_x - (I_0 - I)\omega_y\omega_z \quad (7)$$

$$0 = I_0\dot{\omega}_y - (I - I_0)\omega_z\omega_x \quad (8)$$

$$0 = I\dot{\omega}_z \quad (9)$$

We conclude that for a symmetric body,  $\omega_z$ , the angular velocity about the spin axis, is constant. Inserting this result into the two remaining equations gives

$$I_0\dot{\omega}_x = ((I_0 - I)\omega_z)\omega_y \quad (10)$$

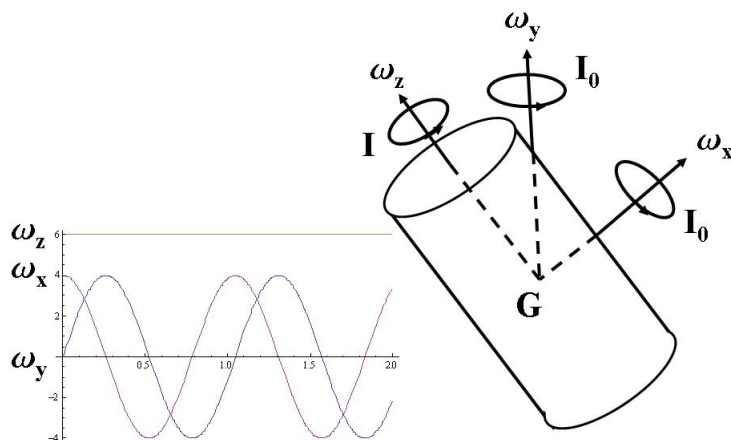
$$I_0\dot{\omega}_y = -((I_0 - I)\omega_z)\omega_x. \quad (11)$$

Since  $\omega_z$  is **constant**, this gives two linear equations for the unknown  $\omega_x$  and  $\omega_y$ . Assuming a solution of the form  $\omega_x = A_x e^{i\Omega t}$  and  $\omega_y = A_y e^{i\Omega t}$ , whereas before we intend to take the real part of the assumed solution, we obtain the following solution for  $\omega_x$  and  $\omega_y$

$$\omega_x = A \cos \Omega t \quad (12)$$

$$\omega_y = A \sin \Omega t \quad (13)$$

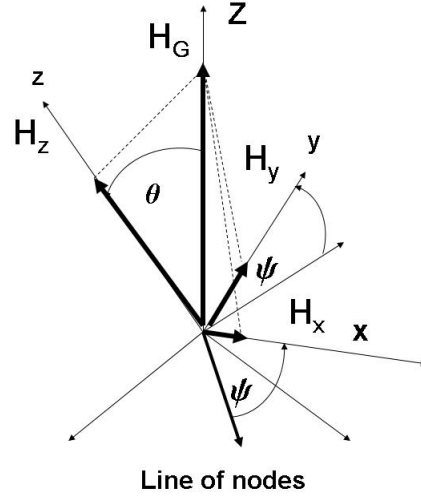
where  $\Omega = \omega_z(I - I_0)/I_0$  and  $A$  is determined by initial conditions. Since  $\omega_z$  is constant, the total angular velocity  $\omega = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2} = \sqrt{A^2 + \omega_z^2}$  is constant. The example demonstrates the direct use of the Euler equations. Although the components of the  $\boldsymbol{\omega}$  vector can be found from the solution of a linear equation, additional work must be done to find the actual position of the body. The body motion predicted by this solution is sketched below.



The  $x, y, z$  axis are body fixed axis, rotating with the body; the solutions for  $\omega_x(t), \omega_y(t)$  and  $\omega_z$  give the components of  $\boldsymbol{\omega}$  following these moving axis. If angular velocity transducers were mounted on the body to measure the components of  $\boldsymbol{\omega}$ ,  $\omega_x(t), \omega_y(t)$  and  $\omega_z$  from the solution to the Euler equations would be obtained, shown in the figure as functions of time. Clearly, as seen from a fixed observer this body undergoes a complex spinning and tumbling motion. We could work out the details of body motion as seen by a fixed observer. (See Marion and Thornton for details.) However, this is most easily accomplished by reformulating the problem expressing Euler's equation using Euler angles.

### Description of Free Motions of a Rotating Body Using Euler Angles

The motion of a free body, no matter how complex, proceeds with an angular momentum vector which is constant in direction and magnitude. For body-fixed principal axis, the angular momentum vector is given by  $\mathbf{H}_G = I_{xx}\omega_x + I_{yy}\omega_y + I_{zz}\omega_z$ . It is convenient to align the constant angular momentum vector with the  $Z$  axis of the Euler angle system introduced previously and express the angular momentum in the  $i, j, k$  system. The angular momentum in the  $x, y, z$  system,  $\mathbf{H}_G = \{H_x, H_y, H_z\}$  is obtained by applying the Euler transformation to the angular momentum vector expressed in the  $X, Y, Z$  system,  $\mathbf{H}_G = \{0, 0, H_G\}$ .



$$\mathbf{H}_G = H_G \sin \theta \sin \psi \mathbf{i} + H_G \sin \theta \cos \psi \mathbf{j} + H_G \cos \theta \mathbf{k} \quad (14)$$

Then the relationship between the angular velocity components and the Euler angles and their time derivative given in Eq.(1-3) is used to express the angular momentum vector in the Euler angle coordinate system.

$$H_G \sin \theta \sin \psi = I_{xx} \omega_x = I_{xx} (\dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi) \quad (15)$$

$$H_G \sin \theta \cos \psi = I_{yy} \omega_y = I_{yy} (\dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi) \quad (16)$$

$$H_G \cos \theta = I_{zz} \omega_z = I_{zz} (\dot{\phi} \cos \theta + \dot{\psi}) \quad (17)$$

where  $H_G$  is the magnitude of the  $\mathbf{H}_G$  vector.

The first two equations can be alternatively multiplied by  $\sin \psi$  or  $\cos \psi$ , then added or subtracted to give expressions for  $\dot{\phi}$  and  $\dot{\theta}$ . Then a final form for spin rate  $\dot{\psi}$  can be found, resulting in

$$\dot{\phi} = H_G \left( \frac{\cos^2 \psi}{I_{yy}} + \frac{\sin^2 \psi}{I_{xx}} \right) \quad (18)$$

$$\dot{\theta} = H_G \left( \frac{1}{I_{xx}} - \frac{1}{I_{yy}} \right) \sin \theta \sin \psi \cos \psi \quad (19)$$

$$\dot{\psi} = H_G \left( \frac{1}{I_{zz}} - \frac{\cos^2 \psi}{I_{yy}} - \frac{\sin^2 \psi}{I_{xx}} \right) \cos \theta \quad (20)$$

For constant  $H_G$ , these equations constitute a first order set of non-linear equations for the Euler angle  $\phi, \theta$  and  $\psi$  and their time derivatives  $\dot{\phi}, \dot{\theta}$  and  $\dot{\psi}$ . In the general case, these equations must be solved numerically. Considerable simplification and insight can be gained for axisymmetric bodies for which  $I_{xx} = I_{yy} = I_0$  and  $I_{zz} = I$ . In this case, we have

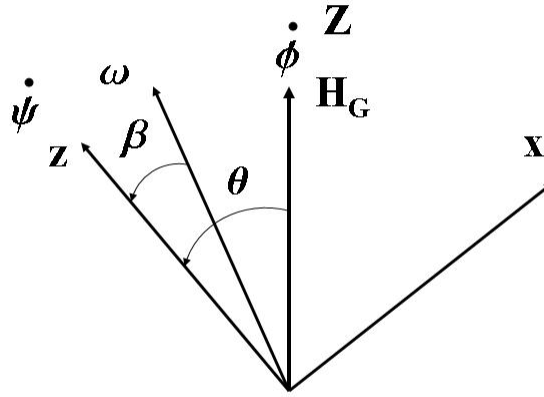
$$\dot{\phi} = H_G/I_0 \quad (21)$$

$$\dot{\theta} = 0 \quad (22)$$

$$\dot{\psi} = H_G\left(\frac{1}{I} - \frac{1}{I_0}\right)\cos\theta \quad (23)$$

Thus, in this case, the nutation angle  $\theta$  is constant; the spin velocity  $\dot{\psi}$  is constant, and the precession velocity  $\dot{\phi}$  is constant. Note that if  $I_0$  is greater than  $I$ ,  $\dot{\phi}$  and  $\dot{\psi}$  are of the same sign; and if  $I_0$  is less than  $I$ ,  $\dot{\phi}$  and  $\dot{\psi}$  are of opposite signs; for  $I_0 = I$ , the problem falls apart since we are now dealing with the inertial equivalent of a sphere which will not exhibit precession.

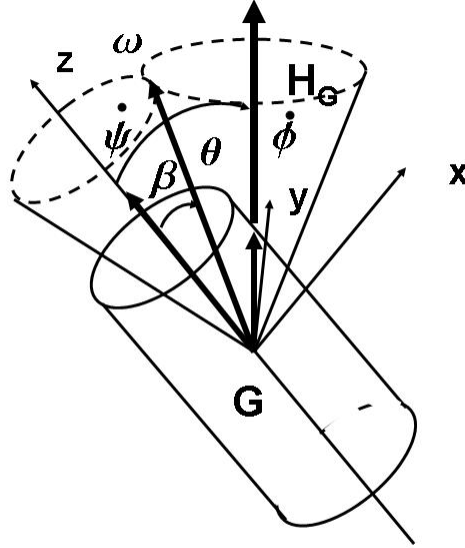
We now examine the geometry of the solution in detail. We assume that the body has some initial angular momentum that could have arisen from an earlier impulsive moment applied to the body or from a set of initial conditions set by an earlier motion. In this case, we would know both the magnitude and direction of the angular momentum  $H_G$  and its angle  $\theta$  from the spin axis. The geometry of the solution is shown below. We align the angular momentum vector with the  $Z$  axis.



For general motion of an axisymmetric body, the angular momentum  $\mathbf{H}_G$  and the angular velocity  $\boldsymbol{\omega}$  vectors are not parallel. Since the kinetic energy of the system for free motion is constant and equal to  $T = 1/2\boldsymbol{\omega} \cdot \mathbf{H}_G$ , the angle between  $\boldsymbol{\omega}$  and  $\mathbf{H}_G$  is constant. Using the spin axis  $z$  as a reference, the angular momentum  $\mathbf{H}_G$  makes an angle  $\theta$  with the spin axis; the angular velocity  $\boldsymbol{\omega}$  makes an angle  $\beta$  with the spin axis. For this body, although the angular momentum is not aligned with the angular velocity, they must be in the  $x, z$  plane; this is a consequence of symmetry: that  $I_{yy} = I_{xx} = I_0$ . Therefore both  $\omega_y$  and  $H_{Gy}$  must be zero. We examine the system in the plane formed by  $\mathbf{H}_G$  and  $\boldsymbol{\omega}$ , as shown in the figure. The relation between the

angular velocities  $\dot{\psi}$  and  $\dot{\phi}$  must be such that the vectors  $I_0\omega_y\mathbf{j} + I\omega_z\mathbf{k} = \mathbf{H}_G$ . Since  $\mathbf{H}_G$  makes an angle  $\theta$  with the  $z$  axis, and  $\boldsymbol{\omega}$  makes an angle  $\beta$  with the  $Z$  axis, we have

$$\tan \theta = H_{Gx}/H_{Gz} = (I_0\omega_x)/(I\omega_z) = (I_0/I) \tan \beta. \quad (24)$$



What we would see in this motion is a body spinning about the  $z$  axis with  $\dot{\psi}$  and “precessing” about the  $Z$  axis with  $\dot{\phi}$  – this is essentially a definition of precession. The  $\boldsymbol{\omega}$  vector, which is the instantaneous axis of rotation, is in the  $x, z$  plane.

We now focus on the  $Z$  axis, and investigate the motion that can exist for a freely rotating body with the given parameters when it has an angular momentum in the  $Z$  direction. We see that a body rotating about its own axis with angular velocity  $\dot{\psi}$  at an angle  $\theta$  from the  $Z$  axis, will also undergo a steady precession of  $\dot{\phi}$  about the  $Z$  axis, keeping the angular momentum  $\mathbf{H}_G$  directed along the  $Z$  axis. The instantaneous axis of rotation maintains a fixed angle  $\beta$  from the spin axis  $z$ . The  $\boldsymbol{\omega}$  vector maintains a fixed angle  $\theta - \beta$  from the vertical. The motion is that of the body cone rolling around the space cone.

A variety of useful and general relations can be written between the various components of angular velocity  $\boldsymbol{\omega}$  and angular momentum  $\mathbf{H}$ , the Euler angles  $\phi$  and  $\psi$  and the angles  $\theta$  and  $\beta$ . Using the notation  $H_G$  and

$\omega$  for the magnitude of the  $\mathbf{H}_G$  and  $\boldsymbol{\omega}$  vectors, we have

$$\omega_x = \omega \sin \beta \quad (25)$$

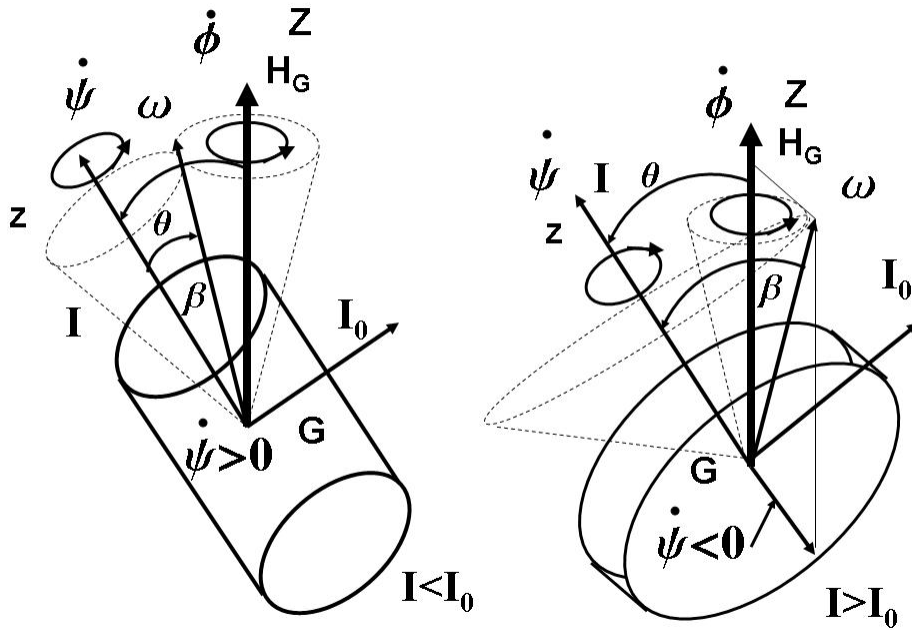
$$\omega_z = \omega \cos \beta \quad (26)$$

$$H_G = \sqrt{H_x^2 + H_z^2} = \sqrt{(\omega_x I_0)^2 + (\omega_z I)^2} = \omega \sqrt{I_0^2 \sin^2 \beta + I^2 \cos^2 \beta} \quad (27)$$

$$\dot{\phi} = \omega \sqrt{(\sin^2 \beta + (I/I_0)^2 \cos^2 \beta)} \quad (28)$$

$$\dot{\psi} = (1 - (I/I_0)) \omega \cos \beta \quad (29)$$

$$(30)$$



Two different solution geometries are shown: one for which  $I_0 > I$ ; one for which  $I_0 < I$ . Consider first the case where  $I < I_0$ . This would be true for a long thin body such as a spinning football, a reentering slender missile or an F-16 in roll.

In this case, the spin  $\dot{\psi}$  and the precession  $\dot{\phi}$  are of the same sign. The body precesses about the angular momentum vector while spinning. The vector  $\boldsymbol{\omega}$  is the instantaneous axis of rotation. The instantaneous axis of rotation is the instantaneous tangent between the body cone and the space cone. The **outside** of the body cone, shown in the figure as a cone of half-angle  $\beta$  aligned along the body axis, rotates about the **outside** of the space cone, shown in the figure as a cone of half-angle  $\theta - \beta$  with its axis aligned along the angular momentum vector. This is called direct precession. For direct precession,  $\beta < \theta$ .

Now consider the case where  $I > I_0$ ; this would be true for a frisbee, a flat spinning satellite, or a silver dollar tossed into the air.



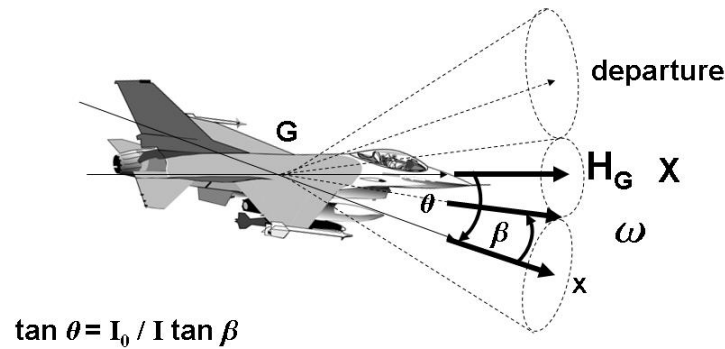
In this case, the spin  $\dot{\psi}$  and the precession  $\dot{\phi}$  are of opposite signs. This means that the  $\omega$  vector is on the other side of the angular momentum vector relative to the spin axis  $z$ . The body still precesses about the angular momentum vector while spinning. The vector  $\omega$  is still the instantaneous axis of rotation. But now, the **inside** of the body cone rotates about the **outside** of the space cone. The angle  $\beta$  is greater than the angle  $\theta$ . The body cone shown in the figure as a cone of half-angle  $\beta$  aligned along the body axis; the space cone has a half-angle  $\beta - \theta$  and is aligned along the angular momentum vector. This is called retrograde precession; the rotations are in opposite directions. Depending upon body geometry, one of these solutions would be obtained whenever the angular momentum vector is not directed along a single principal axis. This would occur if an impulsive moment is applied to a body along any axis which is not a principal axis. An example of this is discussed below.

## Extreme Aircraft Dynamics

The dynamics of aircraft have traditionally been dominated by aerodynamic forces. The location of the center of mass relative to the aerodynamic center was an important consideration as was the question of whether the vertical tail provided enough yaw moment to keep the vehicle flying straight. The details of the response of aerodynamic forces to small disturbances of the vehicle in pitch, roll and yaw, determined the stability of the aircraft and the frequency of the various longitudinal and lateral stability modes.

With the introduction of high-performance fighter aircraft, whose moments of inertia about all three axes were comparable, which had the ability to initiate rapid rolling, and whose roll axis was not a principal axis, we entered into a new flight regime where, at the limit, dynamics dominated aerodynamics.

Consider this limit, where dynamics dominates aerodynamics. We have a F-16 at high altitude where atmospheric density is small, and we have a test pilot giving a strong roll input about the aircraft roll axis which is not a principal axis. For simplicity, we take the  $y$  and  $z$  inertias to be equal and equal to  $I_0$ . Low aspect ratio aircraft have moments of inertia in roll that are less than that in pitch or yaw so that this example is given by the case  $I_0 > I$  and we may apply the free-body spinning solution just discussed. We consider that an impulsive moment/torque is applied about the roll axis and inquire what free-body motion this would set up. At this limit, we are neglecting aerodynamics forces. In agreement with our previous analysis, the moment about the  $x$  axis would produce an angular momentum about the  $x$  axis. But since the  $x$  axis is not a principal axis, we would initiate a coning motion as shown.



This could come as quite surprise to a pilot. The question of what happens next is dependent upon the details of the aerodynamic forces, but just to comment on the historical record: in the first days of testing high-performance fighter aircraft, several test pilots lost control of their aircraft, in some cases with fatal results. This phenomenon, called roll coupling, is now well understand and incorporated into the design and testing of new fighter aircraft.

#### ADDITIONAL READING

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition 7/9

W.T. Thompson, *Introduction to Space Dynamics*, Chapter 5

J. H. Ginsberg, *Advanced Engineering Dynamics*, Second Edition, Chapter 8

J.B Marion and S.T. Thornton, *Classical Dynamics*, Chapter 10

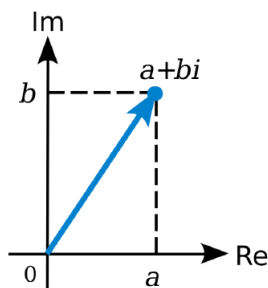
J.C. Slater and N.H. Frank, *Mechanics*, Chapter 6

## Lecture 32 — Quaternions

### 1 What are quaternions?

Quaternions are a type of complex number that has one real element and three imaginary elements. They are most often used as attitude parameters due to their desirable properties that make them quite effective and easy to work with, although they are not as intuitive as other attitude parameters such as Euler angles (roll/pitch/yaw rotation sequences). Nevertheless, this handout will attempt to give a solid conceptual basis for understanding quaternions as they relate to the description of attitude dynamics.

The inventor of the quaternion was Irish mathematician Sir William Rowan Hamilton. He is purported to have conceived of the quaternion while strolling along a bridge with his wife, and was so inspired that he carved the definition into the stone. A chiseled plaque on Broom Bridge commemorates the occasion. At the time, Hamilton was trying to think of a way to extend complex numbers to higher dimensional spaces. Recall that a complex number of the form  $a + bi$  can be expressed as a two-dimensional vector on the real-imaginary plane. If the vector has unit length, i.e.  $\sqrt{a^2 + b^2} = 1$ , then the point of the vector lies on a unit circle drawn in the real-imaginary plane. Now, imagine how this could be useful for describing the attitude of something like the tuning knob on a radio: a unit-length complex number of the form  $a + bi$  can exactly specify the angular position of the knob, because it gives us information about the angle through which it was turned with respect to some starting point.



A tuning knob on a radio has only 1 rotational degree of freedom, that is, it can only turn about one axis. Something like a satellite in orbit, however, has a full 3 rotational degrees of freedom – it is free to rotate about any axis! In much the same way as a complex number with one imaginary element can specify the angular orientation of an object with one rotational degree of freedom, a complex number with three

imaginary elements can specify the angular orientation of an object with three rotational degrees of freedom. The result is a complex number of the form  $a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ , where the  $\mathbf{i}$ ,  $\mathbf{j}$ , and  $\mathbf{k}$  imaginary axes are all orthogonal to each other. The governing formula, and what Hamilton carved into Broom Bridge, is as follows:

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1 \quad (1)$$

This special type of complex number is called a quaternion, because it has 4 elements: one real, and three imaginary. One of the difficulties with quaternions is that it is difficult to visualize them: Just as a unit complex number  $a + b\mathbf{i}$  specifies a point on the circumference of a unit circle, a unit quaternion specifies a point on the hypersurface of a 4-dimensional hypersphere!

Quaternions consist of one scalar part and one vector part, and are written as 4-vectors, sometimes with the scalar part first, and sometimes with the scalar part last. In fact, this is just one of many differences in the way that quaternions differ by various conventions. For example, some quaternion conventions use a left-handed coordinate system, and are thus very different than quaternions that use a typical right-handed coordinate system, which are more common. The dangerous part about this is that there is no way to tell just by looking at the quaternion itself what convention is being used – which is why the most important thing to learn from this lecture is:

**IF YOU EVER WORK WITH QUATERNIONS, BE ABSOLUTELY EXPLICIT ABOUT THE CONVENTION YOU ARE USING! IF YOU DON'T KNOW THE CONVENTION, DROP EVERYTHING AND LEARN IT! OTHERWISE, YOU WILL ONLY CONFUSE YOURSELF AND OTHERS!**

The quaternion convention used can be specified by answering 4 yes-or-no questions, meaning that there are  $2^4 = 16$  different ways to define a quaternion. The worst part is that many of the differences in conventions simply involve an inversion of the quaternion, meaning that if there are two differences, the quaternion is inverted twice, and appears to be numerically identical to the original. This is not an “elegant” property. The quaternions still mean different things. It is beyond the scope of this class to enumerate all of the different quaternion conventions – all you need to know is that there are many, and it is important to know which one you are using. From here on out, this class will be using the following convention:

- Scalar element is first in the matrix.
- Coordinate system is right-handed.
- Quaternion operation is passive (works like a DCM or rotation matrix).
- Directionality is global-to-local by default.

$$\mathbf{q} = \begin{bmatrix} q_0 \\ \vec{\mathbf{q}} \end{bmatrix} = \begin{bmatrix} q_0 \\ q_1 \\ q_3 \\ q_4 \end{bmatrix} = \begin{bmatrix} q_w \\ q_x \\ q_y \\ q_z \end{bmatrix} \quad (2)$$

This is the one of the most common quaternion conventions used, and is well-supported by scientific and engineering tools such as MATLAB. Don't worry too much about the last two bullets for now; they will be better explained below. The definition of the quaternion as a rotation operator in this convention is defined as follows: for some Euler axis (3-dimensional unit vector  $\vec{\mathbf{e}}$ ), and some displacement angle  $\theta$ :

$$\mathbf{q} = \begin{bmatrix} \cos \frac{\theta}{2} \\ \vec{\mathbf{e}} \sin \frac{\theta}{2} \end{bmatrix} \quad (3)$$

The quaternion itself being simply a rotation operator, it can be expressed as a direction cosine matrix (DCM, or rotation matrix) as follows:

$$\mathbf{R} = \begin{bmatrix} q_w^2 + q_x^2 - q_y^2 - q_z^2 & 2(q_x q_y - q_w q_z) & 2(q_x q_z + q_w q_y) \\ 2(q_x q_y + q_w q_z) & q_w^2 - q_x^2 + q_y^2 - q_z^2 & 2(q_y q_z - q_w q_x) \\ 2(q_x q_z - q_w q_y) & 2(q_y q_z + q_w q_x) & q_w^2 - q_x^2 - q_y^2 + q_z^2 \end{bmatrix} \quad (4)$$

As far as notation, a written quaternion should unambiguously indicate two pieces of information in order: coordinate frame from which it is transforming (A), and coordinate to which it is transforming (B):

$$\mathbf{q}_{BA} \equiv \mathbf{q}_A^B \equiv \mathbf{q}_{B \leftarrow A} \quad (5)$$

## 2 Why do we use quaternions?

Scientists and engineers are fond of using quaternions to describe attitude dynamics because they are straightforward to work with mathematically. They are continuous on the entire domain of possible attitudes that an object can have in three-dimensional space, and thus have no singularities, which cannot be said for sequences of Euler angle rotations. The term for this singularity in Euler angle sequences is called “gimbal lock,” which is a loss of a degree of freedom which two out of the three gimbal axes “line up,” that is, two of three concentric orthogonal rings can be made to be coplanar by specifying a certain rotation. The resultant configuration after this effective loss of basis can no longer span the space of possible 3D orientations. No such peculiarity is to be feared when using quaternions!

From an algorithmic perspective, quaternion math involves less computation in general than Euler angles, especially in the realm of state propagation – instead of integrating products of trigonometric functions, you need only carry out a sequence of multiplication operations (no need for lookup tables/Taylor series expansions or numerical integration). This is of great utility to optimize the efficiency of computer simulations and real-time control systems, which can even perform state feedback using the concept of the “error quaternion.”

## 2.1 Mathematical properties of the unit quaternion

- Unit quaternions by definition have unit length, thus  $|q| = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} = 1$ .
- Quaternion multiplication is denoted by the operator  $\otimes$ .
- Associative:  $q_1 \otimes (q_2 \otimes q_3) = (q_1 \otimes q_2) \otimes q_3$ .
- NOT commutative, however:  $q_1 \otimes q_2 \neq q_2 \otimes q_1$  (order matters!)
- To perform quaternion multiplication:

$$\mathbf{q} \otimes \mathbf{r} = \mathbf{n} = n_0 + n_1 \hat{\mathbf{i}} + n_2 \hat{\mathbf{j}} + n_3 \hat{\mathbf{k}} \quad (6)$$

$$n_0 = (r_0 q_0 - r_1 q_1 - r_2 q_2 - r_3 q_3) \quad (7)$$

$$n_1 = (r_0 q_1 + r_1 q_0 - r_2 q_3 + r_3 q_2) \quad (8)$$

$$n_2 = (r_0 q_2 + r_1 q_3 + r_2 q_0 - r_3 q_1) \quad (9)$$

$$n_3 = (r_0 q_3 - r_1 q_2 + r_2 q_1 + r_3 q_0) \quad (10)$$

(this is consistent with MATLAB's `quatmultiply()` function).

- To represent a series of successive rotations:  $\mathbf{q}_A^D = \mathbf{q}_A^B \otimes \mathbf{q}_B^C \otimes \mathbf{q}_C^D$  (Careful... convention matters!)
- Quaternions have conjugates. The conjugate of a quaternion effectively performs the inverse rotation operation. It is found by simply negating only the vector part of the quaternion.

$$\text{conj}(\mathbf{q}) = \mathbf{q}^* = \begin{bmatrix} q_0 \\ -\vec{\mathbf{q}} \end{bmatrix}$$

$$\text{conj}(\mathbf{q}_A^B) = \mathbf{q}_A^{B*} = \mathbf{q}_B^A$$

- Since a quaternion is a rotation operation, it can be used to resolve a vector from one frame into another. This is similar to using a DCM or rotation matrix on a vector.

Option 1: form the DCM  $\mathbf{R}$  from the quaternion as described above, and resolve  $\vec{\mathbf{v}}_A$  into the B frame as follows:  $\vec{\mathbf{v}}_B = \mathbf{R} \vec{\mathbf{v}}_A$

Option 2: perform quaternion multiplication to resolve  $\vec{\mathbf{v}}_A$  into the B frame as follows:  $\vec{\mathbf{v}}_B = \mathbf{q}_A^B \otimes \vec{\mathbf{v}}_A \otimes \mathbf{q}_A^{B*}$

Note that in order to use Option 2, you need to perform quaternion multiplication on a 3-dimensional vector. This is done by simply representing  $\vec{\mathbf{v}}_A$  in “quaternion form” as a 4-vector with a scalar component equal to zero (it also doesn't have to be normalized to unit length).

- The quaternion representation of attitude is a “double-cover” of rotation, meaning that for every possible orientation there are exactly two distinct unit quaternions that correspond to it. This is an artifact of its dimensionality and is not usually a problem in any sense, but can be a confusing factor for beginners not aware of this property. The “double-cover” counterpart of any quaternion is its negative

(which is NOT the same as its conjugate, i.e. all four elements are negated instead of just the vector part).

For example, the exact orientation specified by  $\mathbf{q}$  is also specified by  $-\mathbf{q}$ .

This is not true of  $\mathbf{q}$  and  $\mathbf{q}^*$ , however. Instead,  $\mathbf{q}^*$  specifies the resultant opposite orientation with respect to some initial orientation.

- The rate of change of the attitude unit quaternion can be expressed as follows:

$$\dot{\mathbf{q}} = -\frac{1}{2}\boldsymbol{\omega} \otimes \mathbf{q} = -\frac{1}{2}\boldsymbol{\Omega}\mathbf{q}$$

In this equation,  $\boldsymbol{\omega}$  is the body angular rate of the object for which  $\mathbf{q}$  describes the angular position of. Again, you must augment the 3-vector with a zero in the “scalar” place to perform quaternion multiplication, or just use matrix multiplication with  $\boldsymbol{\Omega}$  defined as follows:

$$\boldsymbol{\Omega} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 & \omega_1 \\ \omega_3 & 0 & -\omega_1 & \omega_2 \\ -\omega_2 & \omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix} \quad (11)$$

The above equation is actually a special case of a more general quaternion kinematic equation:

$$\dot{\mathbf{q}}_A^B = \frac{1}{2} (\mathbf{q}_A^B \otimes [\boldsymbol{\omega}_A]_A - [\boldsymbol{\omega}_B]_B \otimes \mathbf{q}_A^B) \quad (12)$$

In this equation, a passive quaternion (like the one described by this document) specifying the transformation from frame A to B is used.  $[\boldsymbol{\omega}_A]_A$  is the angular rate of the A frame expressed in its own coordinate system, and  $[\boldsymbol{\omega}_B]_B$  is the angular rate of the B frame expressed in its own coordinate system. The first equation above simply assumes that the A frame is inertial, so only the second term remains.

### 3 Sources

<http://www.iri.upc.edu/people/jsola/JoanSola/objectes/notes/kinematics.pdf>

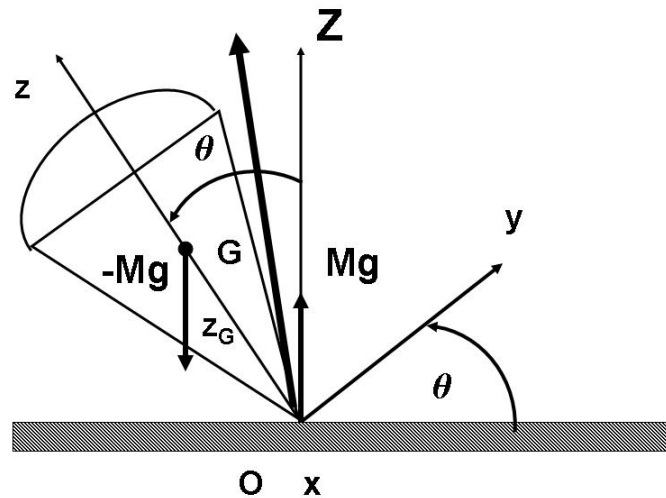
<http://www.tu-berlin.de/fileadmin/fg169/miscellaneous/Quaternions.pdf>

## Lecture 33 — 3D Rigid Body Dynamics: Tops and Gyroscopes

### 3D Rigid Body Dynamics: Euler Equations in Euler Angles

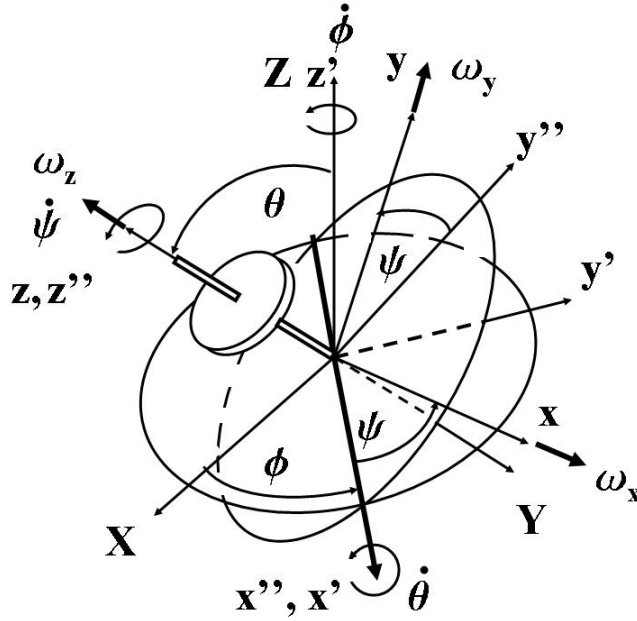
In lecture 26, we introduced the Euler angles as a framework for formulating and solving the equations for conservation of angular momentum. We applied this framework to the free-body motion of a symmetrical body whose angular momentum vector was not aligned with a principal axis. The angular momentum was however constant. We now apply Euler angles and Euler's equations to a slightly more general case, a top or gyroscope in the presence of gravity.

We consider a top rotating about a fixed point  $O$  on a flat plane in the presence of gravity. Unlike our previous example of free-body motion, the angular momentum vector is not aligned with the  $Z$  axis, but precesses about the  $Z$  axis due to the applied moment. Whether we take the origin at the center of mass  $G$  or the fixed point  $O$ , the applied moment about the  $x$  axis is  $M_x = Mg z_G \sin \theta$ , where  $z_G$  is the distance to the center of mass..



Initially, we shall not assume steady motion, but will develop Euler's equations in the Euler angle variables  $\psi$  (spin),  $\phi$  (precession) and  $\theta$  (nutation).





Referring to the figure showing the Euler angles, and referring to our study of free-body motion, we have the following relationships between the angular velocities along the  $x, y, z$  axes and the time rate of change of the Euler angles. The angular velocity vectors for  $\dot{\theta}$ ,  $\dot{\phi}$  and  $\dot{\psi}$  are shown in the figure. Note that these three angular velocity vectors are not orthogonal, giving rise to some cross products when the angular velocities  $\omega_i$  are calculated about the three principal axes.

$$\omega_x = \dot{\phi} \sin \theta \sin \phi + \dot{\theta} \cos \psi \quad (1)$$

$$\omega_y = \dot{\phi} \sin \theta \cos \phi - \dot{\theta} \sin \psi \quad (2)$$

$$\omega_z = \dot{\phi} \cos \theta + \dot{\psi} \quad (3)$$

### 3D Rigid Body Dynamics: Euler's Equations

We consider a symmetric body, appropriate for a top, for which the moments of inertia  $I_{xx} = I_{yy} = I_0$  and  $I_{zz} = I$ . The angular momentum is then

$$H_x = I_0 \omega_x \quad (4)$$

$$H_y = I_0 \omega_y \quad (5)$$

$$H_z = I \omega_z \quad (6)$$

For the general motion of a three-dimensional body, we have Euler's equations in **body-fixed axes** which rotate with the body so that the moment of inertia is constant in time. In this body-fixed coordinate system,

the conservation of angular momentum is

$$\dot{\mathbf{H}} = \frac{d}{dt} ([I]\{\omega\}) = \text{Applied Moments} \quad (7)$$

Since we have chosen to work in a rotating coordinate system so that  $\frac{d}{dt}I = 0$ , **we must pay the price**, applying Coriolis theorem to obtain the time derivative of the angular velocity vector in the rotating coordinate system

$$\dot{\mathbf{H}} = \frac{d}{dt}\mathbf{H} + \boldsymbol{\Omega} \times \mathbf{H}, \quad (8)$$

resulting in the "Euler" equations expressed in the  $x, y, z$  coordinate system moving with the body. In general, we must rotate with the total angular velocity  $\boldsymbol{\omega}$  of the body, so that the governing equation for the conservation of angular momentum become, with  $\boldsymbol{\Omega} = \boldsymbol{\omega}$ .

$$M_x = \dot{H}_x - H_y\Omega_z + H_z\Omega_y \quad (9)$$

$$M_y = \dot{H}_y - H_z\Omega_x + H_x\Omega_z \quad (10)$$

$$M_z = \dot{H}_z - H_x\Omega_y + H_y\Omega_x \quad (11)$$

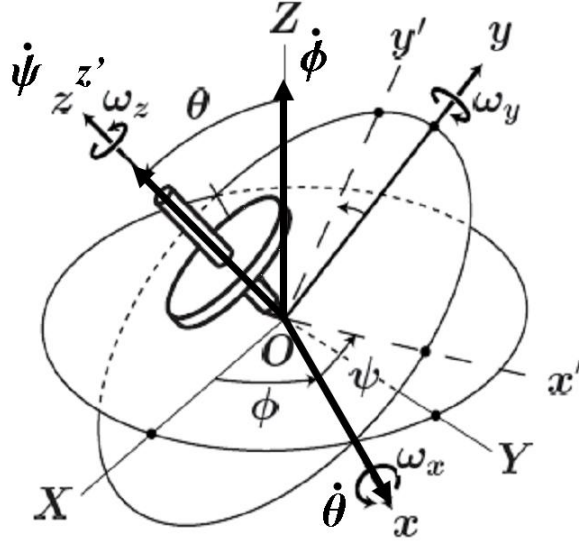
where  $\Omega$  is the rotational angular velocity of our axis system. In this case because of the symmetry of the body, we are free to **choose** to allow the spin velocity to rotate relative to our body-fixed axes system and fasten our axis system to  $\dot{\phi}$  and  $\dot{\theta}$ . That is, we follow the motion of the top in  $\theta$  and  $\phi$  but allow it to spin with angular velocity  $\dot{\psi}$  relative to our coordinate system rotating with angular velocity  $\boldsymbol{\Omega}$ . So that

$$\Omega_x = \dot{\theta} \quad (12)$$

$$\Omega_y = \dot{\phi} \sin\theta \quad (13)$$

$$\Omega_z = \dot{\phi} \cos\theta. \quad (14)$$

For this choice of coordinate system, we have essentially performed a rotation in  $\phi$  and  $\theta$  only, leading to the geometry shown.



In this coordinate system, since the  $\psi$  rotation did not occur, the angular velocity of the body is

$$\omega_x = \dot{\theta} \quad (15)$$

$$\omega_y = \dot{\phi} \sin \theta \quad (16)$$

$$\omega_z = \dot{\phi} \cos \theta + \dot{\psi}. \quad (17)$$

For this choice of coordinate system, we have

$$H_x = I_0 \dot{\theta} \quad (18)$$

$$H_y = I_0 \dot{\phi} \sin \theta \quad (19)$$

$$H_z = I(\dot{\phi} \cos \theta + \dot{\psi}) \quad (20)$$

resulting in

$$\dot{H}_x = I_0 \ddot{\theta} \quad (21)$$

$$\dot{H}_y = I_0(\ddot{\phi} \sin \theta + \dot{\phi} \dot{\theta} \cos \theta) \quad (22)$$

$$\dot{H}_z = I(\ddot{\phi} \cos \theta - \dot{\phi} \dot{\theta} \sin \theta + \ddot{\psi}) \quad (23)$$

and  $\mathbf{\Omega} \times \mathbf{H}$  in components is

$$(I - I_0) \cos \theta \sin \theta \dot{\phi}^2 + I \sin \theta \dot{\phi} \dot{\psi} \quad (24)$$

$$(I_0 - I) \cos \theta \dot{\phi} \dot{\theta} - I \dot{\psi} \dot{\theta} \quad (25)$$

$$0. \quad (26)$$

Note that the  $z$  component is zero. Since the axes chosen are principal axes, the final form of Euler's equations becomes,

$$M_x = I_0(\ddot{\theta} - \dot{\phi}^2 \sin \theta \cos \theta) + I \dot{\phi} \sin \theta (\dot{\phi} \cos \theta + \dot{\psi}) \quad (27)$$

$$M_y = I_0(\ddot{\phi} \sin \theta + 2\dot{\phi}\dot{\theta} \cos \theta) - I\dot{\theta}(\dot{\phi} \cos \theta + \dot{\psi}) \quad (28)$$

$$M_z = I(\ddot{\psi} + \ddot{\phi} \cos \theta - \dot{\phi}\dot{\theta} \sin \theta) \quad (29)$$

For a top,  $M_x = Mgz_G \sin \theta$ ,  $M_y = 0$  and  $M_z = 0$ . These equations are unsteady and non-linear. We can gain insight by examining the character of some special solutions and constants of the motion.

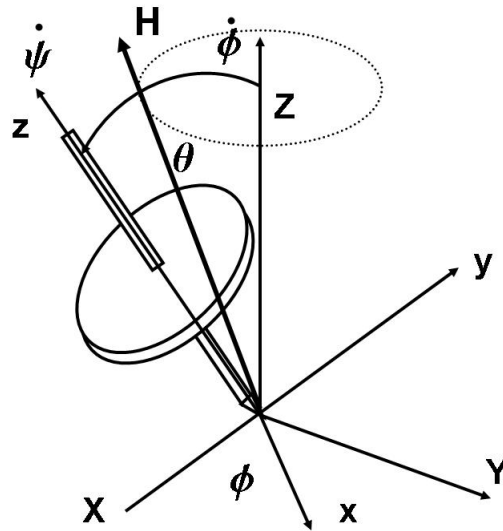
### Steady Precession: Gyroscopic Motion

We now consider the steady precession of a top about the  $Z$  axis. In terms of the variables we have defined, the top rotates with spin velocity  $\dot{\psi}$  about its principal axis, and precesses with angular velocity  $\dot{\phi}$  while maintaining a constant angle  $\theta$  with the vertical axis  $Z$ . We then have

$$\dot{\phi} = \text{constant} = \dot{\phi}_0 \quad \ddot{\phi} = 0 \quad (30)$$

$$\theta = \text{constant} = \theta_0 \quad \dot{\theta} = \ddot{\theta} = 0 \quad (31)$$

$$\dot{\psi} = \text{constant} \quad \ddot{\psi} = 0 \quad (32)$$



For a steady precession of a top, Euler's equations reduce to

$$\dot{\phi} \sin \theta (I(\dot{\phi} \cos \theta + \dot{\psi}) - I_0 \dot{\phi} \cos \theta) = M_x = Mgz_G \sin \theta \quad (33)$$

Note that  $\sin\theta$  cancels, resulting in

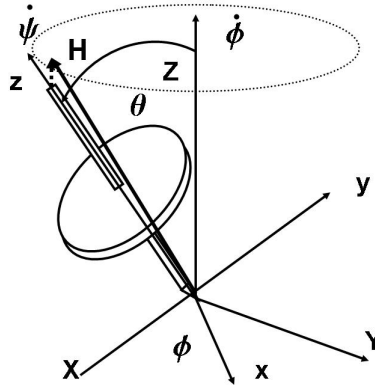
$$I\dot{\phi}\dot{\psi} - (I_0 - I)\dot{\phi}^2 \cos\theta = Mgz_G \quad (34)$$

In the usual case for tops or gyroscopes, we have  $\dot{\psi} \gg \dot{\phi}$  so that  $\dot{\phi}^2$  may be ignored. Therefore, for steady precession, the relationship between the precession angular velocity and the spin angular velocity is

$$\dot{\phi} = Mgz_G / (I\dot{\psi}) \quad (35)$$

where  $I = I_{zz}$ , the moment of inertia of the gyroscope about its spin axis. This result is also true if  $\theta = 0$ . For this motion, the angular momentum vector is not aligned with the  $Z$  axis as for free-body motion, but is in the plane of  $z, Z$ , and rotates around the  $Z$  axis according to the applied external moment which is constant and in the  $x$  direction. In the limit as  $\dot{\psi} \gg \dot{\phi}$  the angular momentum vector is **essentially** along the axis of rotation,  $z$ , with a **slight** component due to  $\dot{\phi}$  which can be calculated **after**  $\dot{\psi}$  has been determined. Some attempt to sketch this is shown in the figure, but even this small correction is an approximation for  $\dot{\psi} \gg \dot{\phi}$ . For  $\dot{\psi} \gg \dot{\phi}$ , this same result can be obtained by less complex approach, but it is important to realize that this is an approximation. For the spin velocity much greater than the precessional velocity, we may take the angular momentum vector as directed **along** the spin or  $z$  axis,  $\mathbf{H} = I\omega\mathbf{k}$ . Then the applied moment will precess the vector with rotation rate  $\boldsymbol{\Omega} \times \mathbf{H}$ . Since  $\boldsymbol{\Omega} = \Omega\mathbf{K} = \dot{\phi}\mathbf{K}$ , (where  $\mathbf{K}$  is a unit vector in the  $Z$  direction), this give a result for  $\Omega$  in agreement with the previous result

$$\Omega = Mgz_G / (I\omega) \quad (36)$$



Since  $\theta$  dropped out of the previous equation, for  $\theta = 0$ , we again have  $\Omega = Mgz_G / (I\omega)$ .

The limit  $\dot{\psi} \gg \dot{\phi}$  is the "gyroscopic" limit where the device behaves as a gyroscope rather than as the more general case of a top. The difference is that, for a gyroscope,  $\omega$  is larger than any other rotation rate in the system, such as the angular velocity of an aircraft or spacecraft. This makes the gyroscope a useful basis for many instruments. We shall return to this issue.

## Unsteady Precession of a Top: Integrals of Motion

For the general case of a top in a gravitational field, we have Euler's Equations. In the most general case, we will have spin  $\dot{\psi}$ , precession  $\dot{\phi}$  and **nutation**  $\dot{\theta}$ , all varying with time.

$$M_x = I_0(\ddot{\theta} - \dot{\phi}^2 \sin \theta \cos \theta) + I\dot{\phi} \sin \theta (\dot{\phi} \cos \theta + \dot{\psi}) = Mgz_G \sin \theta \quad (37)$$

$$M_y = I_0(\ddot{\phi} \sin \theta + 2\dot{\phi}\dot{\theta} \cos \theta) - I\dot{\theta}(\dot{\phi} \cos \theta + \dot{\psi}) = 0 \quad (38)$$

$$M_z = I(\ddot{\psi} + \ddot{\phi} \cos \theta - \dot{\phi}\dot{\theta} \sin \theta) = 0 \quad (39)$$

The consequences of  $M_y = 0$  and  $M_z = 0$  provide two quantities that must be constant/conserved during the motion. (We will later show that these results can also be obtained by applying Lagrange's equation to this system.)

$$d/dt(I_0 \sin^2 \theta \dot{\phi} + I \cos \theta (\dot{\psi} + \dot{\phi} \cos \theta)) = \frac{dp_\phi}{dt} = 0 \quad (40)$$

$$d/dt(I(\dot{\psi} + \dot{\phi} \cos \theta)) = \frac{dp_\psi}{dt} = 0 \quad (41)$$

It can easily be seen by inspection that differentiating  $p_\psi$  and setting it equal to 0 yields the equation  $M_z = 0$ . The equation from  $\frac{dp_\phi}{dt}$  is more complex; to obtain  $M_y$  requires combinations of  $\frac{dp_\phi}{dt}$  and  $\frac{dp_\psi}{dt}$ .

But the conclusion is that  $p_\phi$  and  $p_\psi$  are constants of the motion; we will show later that these can be considered generalized momenta, derived by application of Lagrange's equation.

Therefore, the spin angular velocity and the precession angular velocity are instantaneously related to the nutation angle  $\theta$  through

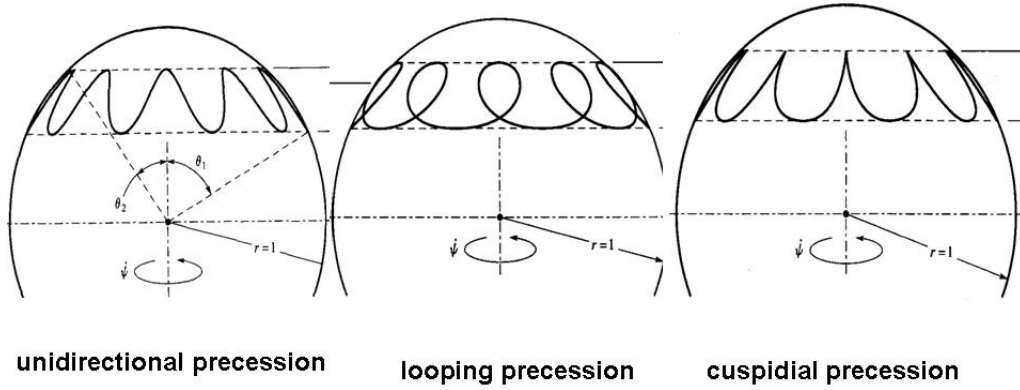
$$\dot{\phi} = \frac{p_\phi - p_\psi \cos \theta}{I_0 \sin^2 \theta} \quad (42)$$

$$\dot{\psi} = \frac{p_\psi}{I_0} - \frac{p_\phi - p_\psi \cos \theta}{I \sin^2 \theta} \quad (43)$$

We may therefore write the remaining governing equation as

$$\ddot{\theta} + \left( \frac{\frac{p_\phi}{I_0} - \frac{p_\psi}{I} \cos \theta}{\sin^3 \theta} \right) \left( \frac{p_\psi}{I} - \frac{p_\phi \cos \theta}{I_0} \right) - \frac{gMz_G}{I_0} = 0 \quad (44)$$

This reduces the problem to the motion of a single variable  $\theta$ ; once  $\theta(t)$  has been determined, the constancy of  $p_\phi/I_0$  and  $p_\psi/I$  during the motion, give the relations between the nutation angle  $\theta(t)$ , the spin angle  $\psi(t)$ , and precession angle  $\phi(t)$ . For this more complex motion, all three angles change with time, and the tip of the top traces out a motion, inscribed on the surface of a sphere for visualization, as shown in the figure. These various motions are referred to as unidirectional precession, looping precession and cuspidal motion.



For more discussion of these solutions see J.B Marion and S.T. Thornton, *Classical Dynamics*, Chapter 11.

### Spinning Top by Lagrange's Equation

The constancy of two momenta obtained by application of Euler's equation can be found perhaps more directly by application of Lagrange's equation. We write the kinetic energy of a spinning top as

$$T = 1/2I_0(\omega_x^2 + \omega_y^2) + 1/2I\omega_z^2 \quad (45)$$

We use the full form of the angular velocities in the Euler system moving with the spinning top and obtain

$$\omega_x^2 = (\dot{\phi} \sin \theta \sin \phi + \dot{\theta} \cos \psi)^2 \quad (46)$$

$$\omega_y^2 = (\dot{\phi} \sin \theta \cos \phi - \dot{\theta} \sin \psi)^2 \quad (47)$$

$$\omega_z^2 = (\dot{\phi} \cos \theta + \dot{\psi})^2 \quad (48)$$

resulting in

$$\omega_x^2 + \omega_y^2 = \dot{\phi}^2 \sin^2 \theta + \dot{\theta}^2 \quad (49)$$

$$\omega_z^2 = (\dot{\phi} \cos \theta + \dot{\psi})^2. \quad (50)$$

Therefore the kinetic energy is

$$T = 1/2I_0(\dot{\phi}^2 \sin^2 \theta + \dot{\theta}^2) + 1/2I(\dot{\phi} \cos \theta + \dot{\psi})^2 \quad (51)$$

For a top, the potential energy is

$$V = Mgz_G \cos \theta \quad (52)$$

so that the Lagrangian is

$$L = 1/2I_0(\dot{\phi}^2 \sin^2 \theta + \dot{\theta}^2) + 1/2I(\dot{\phi} \cos \theta + \dot{\psi})^2 - Mgz_G \cos \theta. \quad (53)$$

We consider  $\theta$ ,  $\phi$  and  $\psi$  to be our generalized coordinates,  $q_i$ . And we move forward applying Lagrange's equation,  $d/dt(\partial L/\partial \dot{q}_i) - \partial L/\partial q_i = 0$ . However, since  $L$  does not contain either  $\phi$  or  $\psi$  but only their

derivatives  $\dot{\phi}$  and  $\dot{\psi}$ , we obtain  $\partial L/\partial q_i = 0$  for both  $q_i = \phi$  and  $q_i = \psi$ . Therefore we conclude that  $d/dt(\partial L/\partial \dot{q}_i) = 0$  for both  $q_i = \phi$  and  $q_i = \psi$ . Thus we have identified a quantity  $\partial L/\partial \dot{q}_i$  that is independent of time. Consistent with our earlier formulation of Lagrange's equation we identify these quantities as "generalized" momenta  $p_\phi$  and  $p_\psi$ .

$$\frac{\partial L}{\partial \dot{\phi}} = (I_0 \sin^2 \theta + I \cos^2 \theta) \dot{\phi} + I \dot{\psi} \cos \theta = p_\phi \quad (54)$$

$$\frac{\partial L}{\partial \dot{\psi}} = I(\dot{\psi} + \dot{\phi} \cos \theta) = p_\psi \quad (55)$$

Since  $\frac{\partial L}{\partial \phi} = 0$  and  $\frac{\partial L}{\partial \psi} = 0$ , the generalized momenta  $p_\phi$  and  $p_\psi$  are constants of the motion.

$$\frac{dp_\phi}{dt} = 0 \quad (56)$$

$$\frac{dp_\psi}{dt} = 0 \quad (57)$$

We define these generalized momenta are of course the same constants identified using Euler's equations in equations (38-39). Considering the complexity of the two derivations, **Lagrange's equations win!** The final governing equation is obtained from

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0 \quad (58)$$

resulting in

$$\ddot{\theta} + \left( \frac{\frac{p_\phi}{I_0} - \frac{p_\psi}{I} \cos \theta}{\sin^3 \theta} \right) \left( \frac{p_\psi}{I} - \frac{p_\phi \cos \theta}{I_0} \right) - \frac{gMz_G}{I_0} = 0, \quad (59)$$

in agreement with equation (44).

We also observe that total energy is a constant of the motion.

$$E = 1/2 * I_0(\dot{\phi}^2 \sin^2 \theta + \dot{\theta}^2) + 1/2 I_0 \dot{\psi}^2 \cos^2 \theta + MH z_G \cos \theta \quad (60)$$

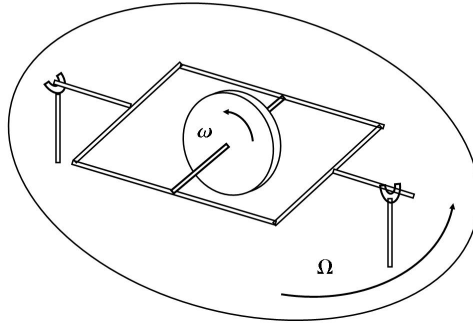
For more discussion of these solutions see J.B Marion and S.T. Thornton, *Classical Dynamics*, Chapter 11.

## Extreme Tops: Gyroscopes

Considerable simplification and practical application results if the angular velocity  $\omega$  is an order of magnitude larger than angular velocity associated with the motions of a system of interest, such as an aircraft or a spacecraft. In this case, the gyroscope can be used as an instrument to measure some quantity of interest in a vehicle.

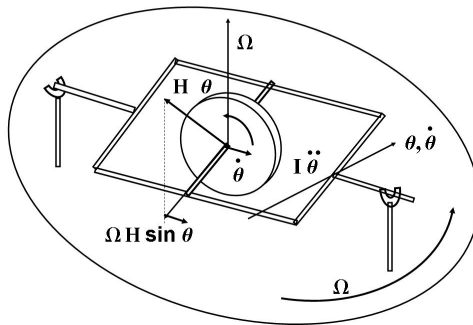
In order to use a gyroscope as an instrument, we first have to consider how it is mounted and affixed to the vehicle. The classic mount is called a gimbal. It permits free rotation of the gyroscope about its own axis, and depending on how complex the gimbal system is, free rotation about other axis as well.





Consider the gyroscope shown in the figure. It is free to spin about its axis with angular velocity  $\omega$ . It is attached to a gimbal which is free to rotate about the support as sketched. The entire apparatus is mounted on a turntable rotating with angular velocity  $\Omega$ , where  $\omega \gg \Omega$ . If we can measure the orientation of the axis of rotation of the disk relative to the turntable mount, we could possibly use this as an instrument. What does it measure?

Consider that the initial angular momentum vector of the disk is in the horizontal plane. The axis of the disk is free to rotate about the horizontal axis through an angle  $\theta$ . As the turntable rotates with angular velocity  $\Omega$ , the axis of the disk will rotate in  $\theta$ —its only free direction— to keep its total rate of change of angular momentum zero, since it can experience no moments/torques from its mounting in the gimbals—that is why we use gimbals. There are two source of changes in angular momentum, the unsteady component  $I_0\ddot{\theta}$  if  $\theta$  changes with time; and the component of the angular momentum of the disk in the horizontal plane that is precessed by the turntable rotation  $\Omega$ . Both of these term are vectors aligned with the gimbals; therefore their sum must be zero.



The unsteady term is  $I_0\ddot{\theta}$ , where  $I_0$  is the moment of inertia of the disk about the axis perpendicular to its spin axis; this vector is in the horizontal direction. The horizontal component of the angular momentum of the disk is  $I\omega\sin\theta$ , where  $I$  is the moment of inertia of the disk about its axis of rotation. The rate of change is also in the horizontal direction, perpendicular to  $\Omega$  and of magnitude  $I\omega\Omega\sin\theta$ . The sum of these two terms, which must be zero, is

$$I_0\ddot{\theta} + I\omega\Omega\sin\theta = 0 \quad (61)$$

This is the familiar equation for the oscillation of a pendulum. If there is a small amount of friction in the support, the system will settle at  $\theta = 0$  and will indicate the direction of rotation of the turntable axis.

This is perhaps not too interesting, since we already know the direction of rotation of the turntable. But if we have this device on a rotating body, such as the earth, it will again indicate the direction of the axis of rotation of the body, in this case the earth. We call this direction "North". Thus, this application of the gyroscope could be used as a compass, often called a gyrocompass.

**ADDITIONAL READING**

J.L. Meriam and L.G. Kraige, *Engineering Mechanics, DYNAMICS*, 5th Edition  
7/9

J.B Marion and S.T. Thornton, *Classical Dynamics*, Chapter 11

W.T. Thompson, *Introduction to Space Dynamics*, Chapter 5

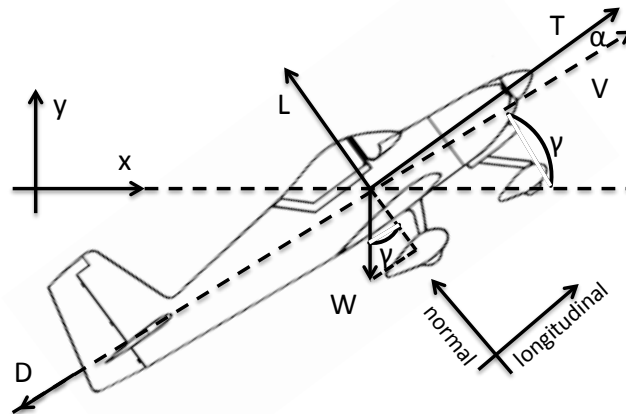
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## Lecture 34 – Longitudinal Aircraft Dynamics

### 4 Introduction

We now build upon the purely static concepts introduced earlier, and consider the response of the aircraft to disturbances and/or control inputs. Figure 1 shows the forces acting on the aircraft. In addition to the lift and drag, there is also engine thrust,  $T$ .



**Figure 1: Forces acting on an aircraft in a flight inclined at a flight path  $\gamma$  to the horizon**

In this analysis, we will idealize  $T$  as being insensitive to aircraft angle of attack and velocity variations, and also aligned with the aircraft axis. The reality is somewhat more complex, but this level of modeling should

suffice to bring out the major dynamical effects. The trajectory is along the velocity vector  $\vec{V}$ , making an angle  $\gamma$  to the horizontal. The zero-lift line of the aircraft is further rotated by the angle of attack  $\alpha$ , so that the absolute inclination of the aircraft is

$$\mu = \gamma + \alpha \quad (1)$$

The lift is drawn through the CM for simplicity, but, as we know, it acts through the instantaneous CP. Equivalently, we can place it at the aircraft AC, if we add the zero-lift moment  $M_{ac}$ . The forces and moments do not have to be in balance: the imbalances will give rise to linear and angular accelerations, according to the dynamical equations of motion.

#### 4.1 Center-of-Mass Equations of Motion

We use intrinsic coordinates (along the trajectory and perpendicular to it).

a) Equation along  $\vec{V}$ :

$$m \frac{dV}{dt} = T \cos \alpha - mg \sin \gamma - D \quad (2)$$

b) Equation perpendicular to  $\vec{V}$ : Let  $R$  be the trajectory radius of curvature, given by

$$\frac{1}{R} = \frac{d\gamma}{ds} = \frac{1}{V} \frac{d\gamma}{dt} \quad (3)$$

Then the centripetal acceleration  $V^2/R$  can be written as  $V^2/R = V(d\gamma/dt)$ , and the normal equation of motion is

$$mV \frac{d\gamma}{dt} = L - mg \cos \gamma + T \sin \alpha \quad (4)$$

Notice that we are assuming moderate speeds only. If we were dealing with a hypersonic vehicle, such as the re-entering Shuttle, we would need to work in polar coordinates, with the main result that we would add an upwards "inertial force" of  $mV^2/R_g$  due to the Earth's curvature (the trajectory "hugs" the Earth, on average).

## 4.2 Motion about the Center of Mass

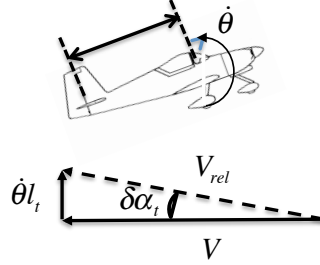


Figure 2: Lift induced on tail.

We introduced most of the torques acting previously. But, as we anticipated, there are damping torques when the vehicle is rotating. The most important of these arises from the extra lift induced on the tail due to a pitch-up motion  $\psi$ . The tail has then an extra downwards velocity  $\dot{\psi}l_t$  and "sees an upwards relative air velocity  $\dot{\psi}l_t$ ". Superimposing this on the wind speed  $\bar{V}$ , this amounts to an extra angle of attack

$$\delta\alpha_t = \tan^{-1} \left( \frac{\dot{\psi}l_t}{V} \right) \approx \frac{\dot{\psi}l_t}{V} \quad (5)$$

and hence a pitch-down torque

$$M_{\dot{\psi}} = -qA_t a_t \frac{\dot{\psi}l_t}{V} l_t \quad (6)$$

The remaining (static) torques can be represented as

$$M_{cm} = (qA_w c) C_{M_{cm}} = (qA_w c) \left( C_{M_0} + \frac{\partial C_{M_{cm}}}{\partial \alpha} \alpha \right) \quad (7)$$

where the zero-lift moment coefficient is

$$C_{M_0} = C_{M_{ac}} + \frac{A_t l_t}{A_w c} a_t (i_w - i_t) \quad (8)$$

and the sensitivity to angle of attack is (assuming  $l_t - x_a \approx l_t$ )

$$\frac{\partial C_{M_{cm}}}{\partial \alpha} = - \left[ \frac{A_t l_t}{A_w c} a_t (1 - \epsilon_\alpha) - \frac{x_a}{c} a_w \right] = - \left[ \left( \frac{x_a}{c} \right)_{max} - \left( \frac{x_a}{c} \right) \right] a_w \quad (9)$$

To write the angular equation of motion, about the aircraft's y-axis (the wing tip to wing tip axis, or span), let  $I_y$  be the corresponding moment of inertia and  $k_y$  be the radius of gyration, defined by

$$I_y = m k_y^2 \quad (10)$$

Then we obtain

$$mk_y^2 \frac{d^2\psi}{dt^2} = qA_w c \left( C_{M_0} + \frac{\partial C_{M_{cm}}}{\partial \alpha} \alpha \right) - qA_t a_t \frac{l_t^2}{V} \frac{d\psi}{dt} \quad (11)$$

## 5 Linearized Equations

Some of the above equations have already made use of the small-angle assumptions. We now formalize this approximation further by assuming small deviations from uniform, level flight. Specifically, we separate the variables  $(V, \alpha, \gamma, \psi)$  into their mean values  $(\bar{V}, \bar{\alpha}, \bar{\gamma} = 0, \bar{\psi} = \bar{\alpha})$  and their temporary deviations  $(V', \alpha', \gamma', \psi')$ . Note that  $\bar{\gamma} = 0$ , i.e., we perturb about level flight. We also set  $\cos \alpha \approx 1, \sin \alpha \approx \alpha$  and  $\cos \gamma \approx 1, \sin \gamma \approx \gamma$ . For the uniform (mean) conditions, Equations 25, 27, and 34 read

$$T - \bar{D} = 0 \quad (12)$$

$$\bar{L} - mg + T\bar{\alpha} = 0 \quad (13)$$

$$C_{M_0} + \frac{\partial C_{M_{cm}}}{\partial \alpha} \bar{\alpha} = 0 \quad (14)$$

We next substitute  $V = \bar{V} + V', \alpha = \bar{\alpha} + \alpha'$ , etc, and then subtract away the mean Equations 35-37. We obtain

$$m \frac{dV'}{dt} = -mg\gamma' - \text{neglect} \quad (15)$$

$$m\bar{V} \frac{d\gamma'}{dt} = L' + T\alpha' \quad (16)$$

$$mk_y^2 \frac{d^2\psi'}{dt^2} = \bar{q}A_w c \frac{\partial C_{M_{cm}}}{\partial \alpha} \alpha' + q'A_w c \left( C_{M_0} + \frac{\partial C_{M_{cm}}}{\partial \alpha} \bar{\alpha} \right) - \bar{q}A_t a_t \frac{l_t^2}{\bar{V}} \frac{d\psi'}{dt} \quad (17)$$

The middle term on the right hand side is zero leaving

$$mk_y^2 \frac{d^2\psi'}{dt^2} = \bar{q}A_w c \frac{\partial C_{M_{cm}}}{\partial \alpha} \alpha' - \bar{q}A_t a_t \frac{l_t^2}{\bar{V}} \frac{d\psi'}{dt} \quad (18)$$

Note the formal analogy between linear small perturbations and differentiation. For instance, since

$$L = \frac{1}{2} \rho V^2 A_w a \alpha \quad (19)$$

we can use logarithmic differentiation (take the log, then differentiate) to obtain

$$\frac{L'}{L} = 2 \frac{V'}{V} + \frac{\alpha'}{\alpha} \quad (20)$$

We now express  $\alpha'$  in terms of  $\gamma', \psi' (\alpha' = \psi' - \gamma')$ . The Equations 38-40 with 41 substituted, become a coupled set for  $V', \gamma', \psi'$ :

$$\frac{dV'}{dt} = -g\gamma' \quad (21)$$

$$m\bar{V}\frac{d\gamma'}{dt} = \bar{L}\left(2\frac{V'}{\bar{V}} + \frac{\alpha'}{\bar{\alpha}}\right) + T\alpha' = 2\frac{\bar{L}}{\bar{V}}V' + \left(\frac{\bar{L}}{\bar{\alpha}} + T\right)(\psi' - \gamma') \quad (22)$$

$$mk_y^2\frac{d^2\psi'}{dt^2} + \bar{q}A_t a_t \frac{l_t^2}{\bar{V}}\frac{d\psi'}{dt} - \bar{q}A_w c \frac{\partial C_{M_{cm}}}{\partial \alpha}(\psi' - \gamma') = 0 \quad (23)$$

Before we go on, notice that  $\bar{L}/\bar{V} = mg/\bar{\alpha}$ . Also notice that we have included no moments or forces from any control surface, such as the elevator or the flaps. This is a "stick fixed" situation, where all controls are assumed to be locked in place. In reality, of course, either the pilot responds to the disturbances by some deliberate control action, or, in the absence of such actions, the control surfaces themselves respond in some manner, depending on their own dynamics ("stick free" response). The "stick free" analysis is more complex, because the response of the controls needs to be carefully modeled, depending on their specific geometry and mass properties, as well as their aerodynamic effectiveness. The pilot response (controlled flight) can be modeled if the specific feedback laws are known, i.e., how much elevator-up does the pilot apply per perceived degree of pitch down, etc.

## 6 The Formal Treatment

We start by re-writing the three equations of "stick-fixed" longitudinal motion:

$$\frac{dV'}{dt} = -g\gamma' \quad (24)$$

$$m\bar{V}\frac{d\gamma'}{dt} = 2\frac{\bar{L}}{\bar{V}}V' + \left(\frac{mg}{\bar{\alpha}}\right)(\psi' - \gamma') \quad (25)$$

$$mk_y^2\frac{d^2\psi'}{dt^2} + \bar{q}A_t a_t \frac{l_t^2}{\bar{V}}\frac{d\psi'}{dt} - \bar{q}A_w c \frac{\partial C_{M_{cm}}}{\partial \alpha}(\psi' - \gamma') = 0 \quad (26)$$

Since these are linear, constant coefficient equations, they admit solutions of the type

$$V' = \text{Re}\left(\hat{V}e^{i\omega t}\right), \text{etc.} \quad (27)$$

where  $\text{Re}(\ )$  stands for the real part of a complex number, and  $\hat{V}$  is the complex amplitude of the velocity perturbation. To interpret this type of solution, suppose  $\hat{V}$  has a magnitude  $|\hat{V}|$  and a phase  $\phi_v$ :  $\hat{V} = |\hat{V}|e^{i\phi_v}$ . Then

$$V' = \text{Re}\left(\bar{V}e^{i(\omega t - \phi_v)}\right) \quad (28)$$

The quantity *omega* is as yet undetermined, and is in general a complex number

$$\omega = \omega_r + i\omega_i \quad (29)$$

So that

$$V' = \text{Re} \left( \hat{V} e^{-\omega_i t} e^{i(\omega_r t + \phi_v)} \right) = \bar{V} e^{-\omega_i t} \cos(\omega_r t + \phi_v) \quad (30)$$

We are therefore assuming solutions which are damped (or amplifying, if  $\omega_i < 0$ ) sinusoidal oscillations, each of the independent variables having in general a different amplitude ( $|\hat{V}|, |\hat{\gamma}|, |\hat{\psi}|$ ) and phase ( $\phi_v, \phi_\gamma, \phi_\psi$ ). The exponential form (Equation 27) is clearly a very compact and convenient notation for this relatively complicated type of function. We next substitute Equation 27 and the related expressions for  $\gamma'$  and  $\psi'$  into Equations 24-26. In so doing, note that

$$\frac{dV'}{dt} = \frac{d}{dt} \text{Re} \left( \hat{V} e^{i\omega t} \right) = \text{Re} \left( i\omega \hat{V} e^{i\omega t} \right), \text{etc.} \quad (31)$$

and so each equation will be of the form

$$\text{Re} \left( (\dots) e^{i\omega t} \right) = 0 \quad (32)$$

This will clearly be satisfied if we make the inner parenthesis be zero. The results are:

$$i\omega \bar{V} = -g\hat{\gamma} \quad (33)$$

$$m\bar{V}i\omega\hat{\gamma} = 2\frac{\bar{L}}{\bar{V}}\hat{V} + \frac{mg}{\alpha}(\hat{\psi} - \hat{\gamma}) \quad (34)$$

$$-mk_y^2\omega^2\hat{\psi} + i\omega\bar{q}A_t a_t \frac{l_t^2}{\bar{V}}\hat{\psi} - \bar{q}A_w c \frac{\partial C_{M_{cm}}}{\partial \alpha}(\hat{\psi} - \hat{\gamma}) = 0 \quad (35)$$

which is a set of linear, homogeneous equations for  $\hat{V}, \hat{\gamma}, \hat{\psi}$ . If these are to have a solution other than  $\hat{V} = 0, \hat{\gamma} = 0, \hat{\psi} = 0$ , the determinant of the coefficients must be made zero, and this is what provides an equation for  $\omega$ . Before proceeding, we rearrange Equations 31-33 in matrix form and simplify the notation by defining the quantities

$$\omega_f^2 = \frac{\bar{q}A_w c}{mk_y^2} \left( -\frac{\partial C_{M_{cm}}}{\partial \alpha} \right) \quad (36)$$

and

$$\zeta_f = \frac{1}{2\omega_f} \frac{\bar{q}A_t a_t l_t^2}{\bar{V}mk_y^2} \quad (37)$$

We obtain

$$\begin{bmatrix} i\frac{\omega\bar{V}}{g} & 1 & 0 \\ -\frac{2\bar{L}\bar{\alpha}}{mg} & 1 + \frac{i\omega\bar{V}\bar{\alpha}}{g} & -1 \\ 0 & 1 & \frac{\omega^2}{\omega_f^2} - 1 - 2i\zeta_f \frac{\omega}{\omega_f} \end{bmatrix} \begin{pmatrix} \frac{\hat{V}}{\bar{V}} \\ \hat{\gamma} \\ \hat{\psi} \end{pmatrix} = 0 \quad (38)$$

We should now set the determinant of the matrix in Equation 36 equal to zero and solve the resulting 4th order equations for  $\omega$ . We shall come back to this systematic method of solution in a particular numerical



example, but before we do that, it is useful to stop and see whether we can take a shortcut based upon physical reasoning that will yield useful solutions with less mathematical apparatus.

## 6.1 The High Frequency Mode

Airplanes are designed to me longitudinally stable, namely, that they develop a pitch-down moment in response to an increase of the angle of attack. The negative quantity  $\partial C_{M_{cm}}/\partial\alpha$  is a measure of this effect, which is a kind of weathervaning action, brought about mainly by the tail. Therefore, if the airplane rotates by  $\psi$ , we expect the restoring torque to set up an oscillation with a frequency which should be proportional to the square root of  $(-\partial C_{M_{cm}}/\partial\alpha)$ , and inversely proportional to  $I_y = mk_y^2$ . The complicating factor is that a pitch rotation  $\psi'$  must lead to a deviation  $\gamma'$  of the trajectory, but if the pitch oscillation is relatively fast, there will not be time enough for the trajectory to deviate significantly from the horizontal, and we may be able to set  $\gamma' = 0$  for analyzing pitch motion. The result is a second-order differential equation for  $\psi'$ , which is much easier to analyze than the coupled 4th order set we started from. In fact, using the definitions of  $\omega_f, \zeta_f$  in Equations 34 and 35, we can write Equation 33 in the form

$$\frac{d^2\psi'}{dt^2} + 2\zeta_f\omega_f\frac{d\psi'}{dt} + \omega_f^2\psi' = 0 \quad (39)$$

which is the standard form for a 2nd order oscillator with a natural frequency  $\omega_f$  and damping ratio  $\zeta_f$ . Looking at Equation 34, the frequency  $\omega_f$  is indeed of the form we expected. The damping ratio  $\zeta_f$  (Equation 35) is from the extra tail lift due to a pitch rate  $\dot{\psi}$ .

Numerical Example: Return to the generic light aircraft. We have

$$c_L \approx \frac{mg}{\bar{q}A_w} = 0.549$$

$$-\frac{\partial C_{M_{cm}}}{\partial\alpha} = a_w \left( \left( \frac{x_a}{c} \right)_{max} - \left( \frac{x_a}{c} \right) \right) = 5(0.264 - 0.132) = 0.66$$

$$k_y = \sqrt{\frac{1600}{1050}} = 1.234m, c = 1.6m$$

$$\bar{V} = 50 \frac{m}{sec}, \bar{q} = \frac{1}{2} \times 1 \times 50^2 = 1250 N/m^2, A_t = 2.3m^2, l_t = 4.0m, a_t = 4/rad$$

And so we calculate (Equation 34)

$$\omega_f = \sqrt{\frac{g}{C_L} \frac{c}{k_y^2} \left( -\frac{\partial C_{M_{cm}}}{\partial\alpha} \right)} = \sqrt{\frac{9.81}{0.549} \frac{1.6}{(1.234)^2} \times 0.66} = 3.52 rad/sec$$

which corresponds to an oscillation period of

$$T_f = \frac{2\pi}{\omega_f} = 1.79 sec$$

We also find from Equation 34,

$$\zeta_f = \frac{1}{2 \times 3.52} \frac{1250 \times 2.3 \times 4 \times 4^2}{50 \times 1600} = 0.327$$

The solution to Equation 37 with  $\psi'(0) = \psi'_0$  and  $(d\psi'/dt)_0$  is

$$\psi' = \psi'_0 e^{-1(\zeta_f \omega_f)t} \cosh \left( \sqrt{1 + |\zeta_f|^2} |\omega_f| t \right) \quad (40)$$

where  $\cosh x = (1/2)(e^x + e^{-x})$ . Clearly, the angle diverges exponentially like

$$\exp \left[ \left( \sqrt{1 + |\zeta_f|^2} - |\zeta_f| \right) |\omega_f| t \right]$$

For our example, suppose the CM were moved back to  $1.5(x_a)_{max}$ , which has the effect of reversing the sign of  $\partial C_{M_{cm}}/\partial \alpha$  without changing its magnitude. The positive exponential is then

$$\exp \left[ \left( \sqrt{1 + (0.327)^2} - 0.327 \right) 3.52t \right] = e^{2.55t}$$

The initial perturbation will grow by a factor of  $\exp(2.718)$  in approximately  $t = (1/2.55) = 0.39sec$ , and by a factor of 10 in  $t = (\ln(10))/2.55 = 0.9sec$ . This is a very rapid divergence speed indeed. Airplanes (especially very fragile fighters) are sometimes designed to be statically unstable, and this is compensated through active control (fly-by-wire). This example should serve to emphasize that the controller must then be very fast (and very reliable). Of course, the analysis has been based on linearization for small angles. In a real divergence situation, the wings will stall when the angle of attack exceeds perhaps 20-30 degrees, which (a) invalidates the solution, and (b) provides some limitation to the actual divergence. It would still be no fun to try and pilot such an aircraft.

## 7 The Slow Mode

In the previous discussion we identified a "fast mode" in which the aircraft, while flying in a straight line, executes damped (hopefully) oscillations in pitch. This is not, however, the only possible mode. Since the pitch (and angle of attack) oscillations are fairly fast, there may be an underlying, slower motion, in which the trajectory departs from level flight (i.e.,  $\gamma' \neq 0$ ), and on which the rapid  $\alpha'$  oscillations exert little influence, because they average out. We are thus led to investigate the effect of setting  $\alpha' = 0$  (no angle of attack variations) in the governing equations (Equations 24-26). Since  $\alpha' = \psi' - \gamma'$ , we obtain from the first pair of equations (governing the center of mass motion):

$$\frac{dV'}{dt} = -g\gamma' \quad (41)$$

$$m\bar{V} \frac{d\gamma'}{dt} \approx 2 \frac{\bar{L}}{\bar{V}} V' \quad (42)$$

To obtain a single, second order equation for  $\gamma'$ , differentiate Equation 42 and substitute from Equation 41:

$$m\bar{V} \frac{d^2\gamma'}{dt^2} + 2 \frac{\bar{L}}{\bar{V}} g\gamma' = 0 \quad (43)$$

or

$$\frac{d^2\gamma'}{dt^2} + 2\frac{\bar{L}g}{m\bar{V}^2}\gamma' = 0 \quad (44)$$

Ignoring some small contribution from engine thrust to the vertical balance, we have  $mg \approx \bar{L}$ , so that

$$\frac{d^2\gamma'}{dt^2} + 2\left(\frac{g}{\bar{V}}\right)^2\gamma' = 0 \quad (45)$$

This describes an undamped harmonic oscillation with a frequency

$$\omega_{PH} = \sqrt{2}\frac{g}{\bar{V}} \quad (46)$$

where

$$T_{PH} = \frac{2\pi}{\omega_{PH}} = \frac{2\pi}{\sqrt{2}}\frac{\bar{V}}{g} \quad (47)$$

is the time it takes for a falling object to reach a speed  $\bar{V}$ . The variable  $\omega_{PH}$  is called the "phugoid" frequency. For the example of a light aircraft ( $\bar{V} = 50$  m/sec),

$$\omega_{PH} = 0.277 \text{ rad/sec}$$

$$T_{PH} = \frac{2\pi}{\omega_{PH}} = 22.6 \text{ sec}$$

This is indeed much slower than the  $T_f = 1.79$  sec we found for the "fast mode." Notice that an identical equation can be obtained for  $V'$  (for example, differentiate Equation 41 and substitute for  $d\gamma'/dt$  from Equation 42). Thus, in this "phugoid mode," both the trajectory angle  $\gamma$  and the velocity  $V$  fluctuate (although not in phase with each other). Of course, if  $\gamma$  is varying, so must be the flight attitude, because

$$\tan \gamma \approx \gamma = \frac{1}{\bar{V}} \frac{dh}{dt} \quad (48)$$

To see more clearly what is happening, assume the aircraft, for some reason, develops an instantaneous small upwards flight path slope ( $\gamma'(t=0) = \gamma'_0$ ), while its velocity remains constant ( $V'_0 = 0$ ). The solution to Equation 45 satisfying  $\gamma'(0) = \gamma'_0$  and (because of  $V'(0) = 0$ ),  $(d\gamma'/dt)_0 = 0$ , is

$$\gamma' = \gamma'_0 \cos(\omega_{PH}t) \quad (49)$$

From Equation 42 we then have

$$V' = -\frac{m\bar{V}^2}{2\bar{L}} \frac{d\gamma'}{dt} = -\frac{\bar{V}^2}{2g} \omega_{PH} \gamma'_0 \sin(\omega_{PH}t) \quad (50)$$

or, using  $\omega_{PH} = \sqrt{2}g/\bar{V}$ ,

$$V' = -\frac{\bar{V}}{\sqrt{2}} \gamma'_0 \sin(\omega_{PH}t) \quad (51)$$

Also, from Equation 48, with  $h = \bar{h} + h'$

$$\frac{dh'}{dt} = \bar{V}\gamma'_0 \cos(\omega_{PH}t) \quad (52)$$

and integrating, with  $h'(0) = 0$ ,

$$h' = \frac{\bar{V}\gamma'_0}{\omega_{PH}} \sin(\omega_{PH}t) = \frac{1}{\sqrt{2}} \frac{\bar{V}^2}{g} \gamma'_0 \sin(\omega_{PH}t) \quad (53)$$

These results are sketched in Figure 1.

Because the aircraft starts with some "uphill" heading  $\gamma_0$ , we see a subsequent loss of speed (lagging 90 degrees). We also see the gain of altitude as the  $\gamma'$  accumulates (also lagging 90 degrees with respect to  $\gamma'$ ). So, the velocity and altitude vary in counterphase: when the aircraft is higher, it goes slower, and vice versa. In fact, this is a consequence of energy conservation, because the thrust and drag are assumed to continuously cancel each other, and hence do no work. The total energy per unit mass is than

$$\frac{V^2}{2} + gh = \text{Constant} \quad (54)$$

and its perturbation is

$$\bar{V}V' + gh' = 0 \quad (55)$$

which is indeed satisfied by Equations 51 and 53. It is of interest to find out what should be seen by the pilot of a second aircraft flying in formation with the first one, but suffering no disturbance. The altitude variation seen will be in  $h'$  (Equation 53). In addition, because of the velocity variations, the disturbed aircraft will lead or lag the chase plane by  $x'(dx'/dt = V')$ .

$$x' = -\frac{\bar{V}\gamma'_0}{\sqrt{2}\omega_{PH}}(1 - \cos(\omega_{PH}t)) \quad (56)$$

$$h' = -\frac{\bar{V}^2\gamma'_0}{2g}(1 - \cos(\omega_{PH}t)) \quad (57)$$

A cross-plot of  $h'$  versus  $x'$ , which is the trajectory of the oscillating aircraft as seen from the chase plane, is shown in Figure 2.

The perturbed aircraft tends to lag by the horizontal oscillation amplitude, which is  $1/\sqrt{2}$  of the vertical amplitude.

## 8 The Exact Frequencies

Having obtained useful results from the approximate decoupling of the equations of motion into a fast pitch mode and a slow velocity - altitude mode, we still need to verify their accuracy by returning to the fully coupled formulation embodied in the set of equations, which we repeat here for reference (using  $\bar{L} = mg$  to simplify):

$$\begin{bmatrix} i\frac{\omega\bar{V}}{g} & 1 & 0 \\ -2\bar{\alpha} & 1 + i\frac{\omega\bar{V}}{g}\bar{\alpha} & -1 \\ 0 & 1 & \left(\frac{\omega}{\omega_f}\right)^2 - 1 - 2i\zeta_f\frac{\omega}{\omega_f} \end{bmatrix} \begin{Bmatrix} \hat{\bar{V}} \\ \hat{\gamma} \\ \hat{\psi} \end{Bmatrix} = 0 \quad (58)$$

We will only solve these equations for the particular example we have been working with, for which,  $(\bar{V}/g = 5.1 \text{ sec}$ ,  $\bar{\alpha} = 0.11 \text{ rad}$ ,  $\omega_f = 3.52 \text{ rad/sec}$  and  $\zeta_f = 0.327$ . The determinant is then

$$\begin{vmatrix} 5.100i\omega & 1 & 0 \\ -0.220 & 1 + 0.561i\omega & -1 \\ 0 & 1 & 0.081\omega^2 - 1 - 0.186i\omega \end{vmatrix} \quad (59)$$

and we equate it to zero in order to calculate  $\omega$ . Let  $i\omega = x$  and expand

$$x^4 + 4.087x^3 + 16.57x^2 + 0.177x + 0.952 = 0 \quad (60)$$

Solving this quartic by hand is not easy but can be found using a computer, where  $i\omega = x$

$$i\omega_{PH} = -2.0453 \pm 3.5134i, 0.0018 \pm 0.2400i \quad (61)$$

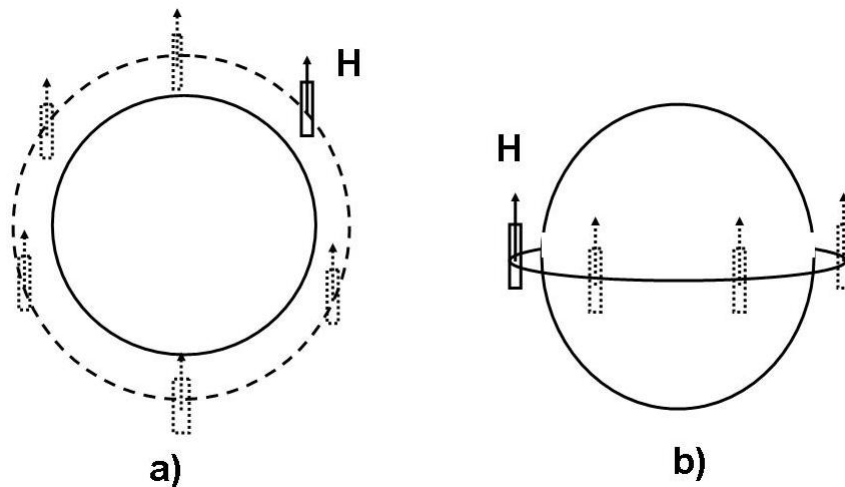
where the natural frequencies are  $\omega_n$ , the damped frequencies are  $\omega_d$ , and the damping ratios are  $\zeta =$  for the phugoid and fast mode, respectively.

## Lecture 35 — Satellite Attitude Dynamics

This analysis of satellite attitude dynamics for a satellite in orbit brings together orbital mechanics and rigid body dynamics. We first examine a highly simplified case. Later we will examine the effects of the non-uniform gravitational field, the effects of spin and effects of the relative moments of inertia about the principal axes.

### Satellite Attitude in the Absence of External Moments

We examine a highly simplified model of a satellite in orbit about the Earth. Assuming that the satellite has an initial angular momentum  $\mathbf{H}$  oriented with respect to its orbit about the Earth as shown in a), and that the attitude is stable in the "rigid-body" sense, its progress in orbit would be as sketched: **angular momentum is conserved**. Whether this is appropriate or not depends on the mission of the satellite and the accuracy of the assumption of no external moments.



For an astronomical-observation satellite, a constant orientation would be desirable; for a communications satellite or an Earth observation satellite, the desired orientation would maintain a fixed orientation with respect to the center of the Earth.

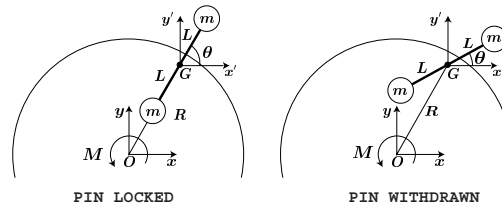
A second possible geometry is to orient the satellite such that the angular momentum vector is perpendicular to the orbital plane, as in b). If in addition, the angular velocity  $\boldsymbol{\omega}$  is chosen as the orbital rotation rate,

the satellite would be continually "looking" at the earth, as would be desired for a communications or other Earth-fixed satellite.

However, the assumption of no external moments is not adequate for a number of reasons. External moments exist: solar pressure, aerodynamics forces, forces and moments from on-board propulsive control systems. However, for satellites in Earth orbit, the most important external moment comes from the non-uniformity of the gravitational field.

## Gravity-Gradient Stabilization

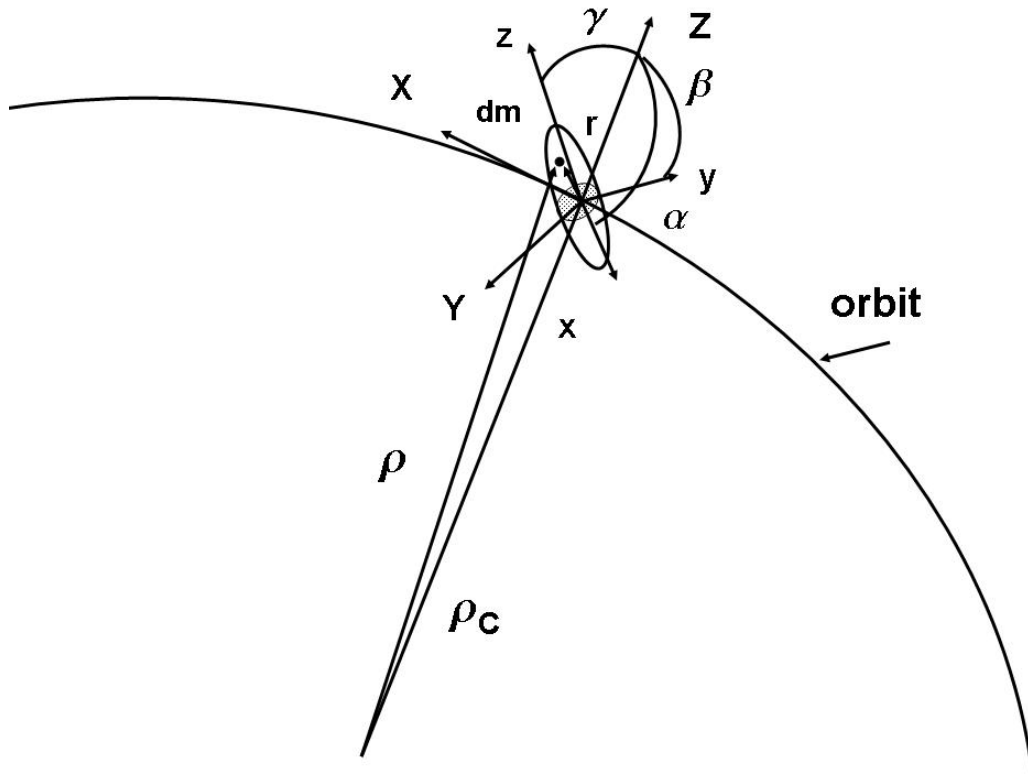
The simplest model for gravity gradient stabilization of a long skinny satellite in the Earth's gravitational field comes from considering simple dumbbell, two mass points separated by a distance  $R$  orbiting about the Earth with its center of mass at an orbital radius  $R_0$ .



Since the gravitation force on each mass is given by  $\mathbf{F} = -\frac{\mu m}{r^2} \mathbf{e}_i$ , where  $\mathbf{e}_i$  is the unit normal in the  $r$  direction at mass point  $i$ , the gravitational force on the lower mass is higher than the force on the farther mass. Moreover, the direction of the force is towards the center of the Earth. Thus when the dumbbell is tipped through an angle  $\theta$ , a restoring moment on the dumbbell occurs. For small angles  $\theta$ , a linear restoring force is created and the dumbbell acts like a pendulum in a uniform gravitation field. This additional force acts to stabilize the motion of the satellite and keep it "pointed" towards the center of the Earth. We now consider the moment provided by the nonuniform gravity field on a general body represented as an ellipsoid.

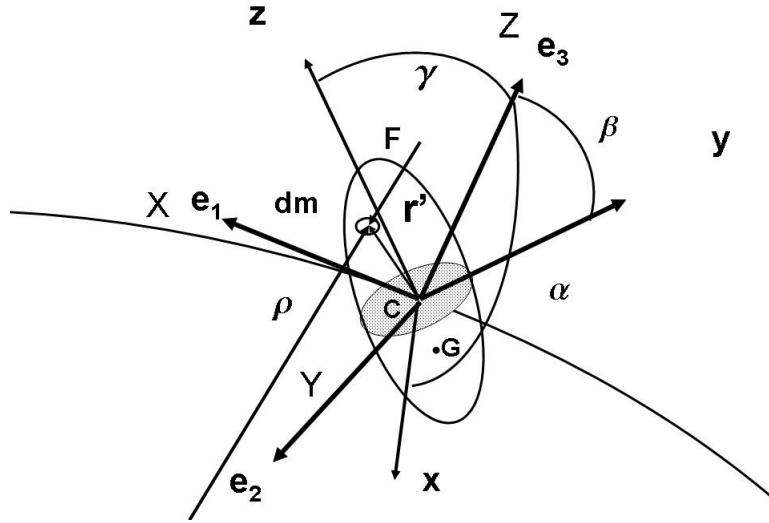
## Gravity-Gradient Moment

We now examine the moments exerted at the center of mass of a body of finite size in orbit about a fixed gravitational center. The orbital coordinate system is shown: the  $Z$  coordinate points outward from the gravitational center; the  $X$  coordinate point along the direction of motion in the orbit, and the  $Y$  coordinate is normal to the orbital plane. This coordinate system is located at the center of mass of the body. The body is described by a body-fixed coordinate system, aligned with the principal axes of the body,  $x, y, z$ . In the  $X, Y, Z$  coordinate system, the vector to the center of mass from the gravitational center is  $\mathbf{r}_{hoC}$ ; the vector to an arbitrary mass point  $dm$  is  $\mathbf{r}_{ho}$



The local coordinate system of the body in the center gravitational field is shown below: the  $X, Y, Z$  coordinate system with unit vectors  $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ ; the body principal axes  $x, y, z$ ; the center of mass  $C$ . The gravitational force vector to an arbitrary mass point  $dm$  is  $\rho$ ; the gravitational force  $\mathbf{F}$  on a mass point  $dm$  acts in this direction. Also shown is the gravitational center  $G$ : this is the point at which the total gravitational force on the body acts. The fact that the total gravitational force on the body acts at a point other than the center of mass produces an external "gravity-gradient" moment about the center of mass.





The coordinate transformation between  $X, Y, Z$  and  $x, y, z$  can be expressed through the direction cosines between the coordinate axes. It turns out that only the  $Z$  coordinate is important; we write  $x = Z \cos \alpha$ ,  $y = Z \cos \beta$  and  $z = Z \cos \gamma$  where  $\alpha, \beta$  and  $\gamma$  are shown.

To determine the moment about the center of mass due to the non-uniform gravitational field, we begin with the potential for the gravitational force acting at a point in the  $X, Y, Z$  coordinate system

$$V(X, Y, Z) = -\frac{\mu}{\sqrt{(\rho_C + Z)^2 + X^2 + Y^2}}, \quad (1)$$

where  $\rho_C$  is the distance between the center of mass of the satellite and the gravitational center. Since  $\rho_C \gg X, Y, \text{ or } Z$ , we expand this potential for  $\rho_C \gg X, Y, Z$  and obtain to second order in  $X, Y, Z$

$$V(X, Y, Z) = -\frac{\mu}{\rho_C} + \frac{\mu Z}{\rho_C^2} - \frac{\mu Z^2}{\rho_C^3} + \frac{\mu}{2\rho_C}(Y^2 + X^2) \quad (2)$$

We add and subtract  $\frac{\mu}{2\rho_C}(Z^2)$  to form a term containing  $X^2 + Y^2 + Z^2$  and obtain

$$V(X, Y, Z) = -\frac{\mu}{\rho_C} + \frac{\mu Z}{\rho_C^2} - \frac{3\mu Z^2}{2\rho_C^3} + \frac{\mu}{2\rho_C}(Y^2 + X^2 + Z^2) \quad (3)$$

At this point we observe that the term  $X^2 + Y^2 + Z^2$  forms a spherical surface about the center of mass; thus it is incapable of creating a force which produces a moment about the center of mass since the gradient points directly towards the center of mass; we will drop it.

We now express the potential in the principal axis system  $x, y, z$ , expressing  $Z$  in terms of the direction cosines to the  $x, y, z$  body-fixed axes.

$$V(x, y, z) = -\frac{\mu}{\rho_C} + \frac{\mu(x \cos \alpha + y \cos \beta + z \cos \gamma)}{\rho_C^2} - \frac{3\mu}{2\rho_C^3}(x \cos \alpha + y \cos \beta + z \cos \gamma)^2 \quad (4)$$

We then determine the forces acting on the mass point  $dm$  as

$$d\mathbf{F} = \mathbf{F}dm = -\nabla_{xyz}V(x, y, z)dm \quad (5)$$

To calculate the moment on the body we write

$$\mathbf{M} = \int \int \int (\mathbf{r}' \times \mathbf{F})dm \quad (6)$$

where  $\mathbf{r}' = \{x, y, z\}$ . The cross product term contains many products of  $x, y, z$ . However, since our coordinate system is along principal axes, all cross-product integrals are zero, being cross products of inertia, such as  $\int \int \int xydm = 0$ . Also, since our coordinate system is at the center of mass, integrals such as  $\int \int \int zdm = 0$ . We therefore drop all terms which integrate to zero. The result is

$$\mathbf{M} = -\frac{3\mu}{\rho_C^3} \int \int \int ((z^2 - y^2) \cos \beta \cos \gamma \mathbf{i} + (x^2 - z^2) \cos \alpha \cos \gamma \mathbf{j} + (y^2 - x^2) \cos \alpha \cos \beta \mathbf{k})dm. \quad (7)$$

The terms in  $\mathbf{M}$  are not quite moments of inertia about principal axes, but if we add and subtract various terms, to form moments of inertia, we obtain

$$\mathbf{M} = -\frac{3\mu}{\rho_C^3} \int \int \int ((x^2 + z^2 - y^2 - x^2) \cos \beta \cos \gamma \mathbf{i} + (x^2 + y^2 - z^2 - y^2) \cos \alpha \cos \gamma \mathbf{j} + (z^2 + y^2 - x^2 - z^2) \cos \alpha \cos \beta \mathbf{k})dm, \quad (8)$$

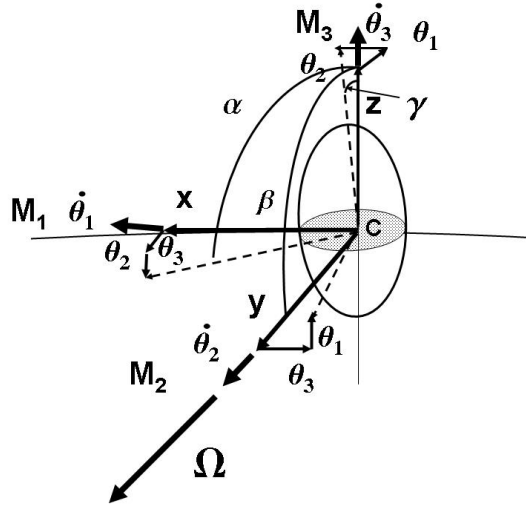
resulting in

$$\mathbf{M} = 3\Omega^2 \begin{pmatrix} (I_{zz} - I_{yy}) \cos \beta \cos \gamma \\ (I_{xx} - I_{zz}) \cos \alpha \cos \gamma \\ (I_{yy} - I_{xx}) \cos \alpha \cos \beta \end{pmatrix} \quad (9)$$

This is the full non-linear result for the moments about the center of mass due to the non-uniform gravitational field. The grouping  $\mu/\rho_C^3$  has been replaced by  $\Omega^2$ , where  $\Omega$  is the angular velocity of the spacecraft in a circular orbit of radius  $\rho_C$ .

## Linear Stability Analysis

We now develop the governing equations for small angles. As shown in the figure, we align the body-fixed principal axes coordinates  $x, y, z$  with the orbital coordinates  $X, Y, Z$ . We apply small rotations  $\theta_1, \theta_2, \theta_3$  about these axes. The angles  $\alpha, \beta$ , and  $\gamma$  which appear in the full non-linear result for the moment on the body in principal axes due to the gravity gradient are shown.



For small rotation angles, we have

$$\cos \alpha = \cos(\pi/2 + \theta_2) = -\theta_2 \quad (10)$$

$$\cos \beta = \cos(\pi/2 - \theta_1) = \theta_1 \quad (11)$$

$$\cos \gamma = 1 \quad (12)$$

resulting in an expression for the moment for small angle rotations

$$\mathbf{M} = 3\Omega^2 \begin{pmatrix} -(I_{yy} - I_{zz})\theta_1 \\ -(I_{xx} - I_{zz})\theta_2 \\ 0 \end{pmatrix} \quad (13)$$

To determine the stability of the satellite when subjected to gravity gradient moments, we rewrite Euler's equations with the gravity moments as the applied moments in the  $x, y, z$  directions.

$$M_x = I_{xx}\dot{\omega}_x - (I_{yy} - I_{zz})\omega_y\omega_z = -3\Omega^2(I_{yy} - I_{zz})\theta_1 \quad (14)$$

$$M_y = I_{yy}\dot{\omega}_y - (I_{zz} - I_{xx})\omega_z\omega_x = 3\Omega^2(I_{zz} - I_{xx})\theta_2 \quad (15)$$

$$M_z = I_{zz}\dot{\omega}_z - (I_{xx} - I_{yy})\omega_x\omega_y = 0 \quad (16)$$

The orbital angular velocity  $\mathbf{\Omega}$  in the original  $X, Y, Z$  system is shown in the figure; it is in the  $\mathbf{j}$  direction, normal to the orbital plane; this adds to  $\omega_y$ . In body-fixed coordinates, tipped by angles  $\theta_1, \theta_2, \theta_3$  there are small components of  $\Omega$  aligned and additive to  $\dot{\theta}_1$  and  $\dot{\theta}_2$ . Thus the angular velocity about the body-fixed  $x, y, z$  axes is  $\boldsymbol{\omega} = \{\omega_x, \omega_y, \omega_z\} = \{\dot{\theta}_1 + \theta_3\Omega, \dot{\theta}_2 + \Omega, \dot{\theta}_3 - \theta_1\Omega\}$ . Rewriting the equations, keeping only linear

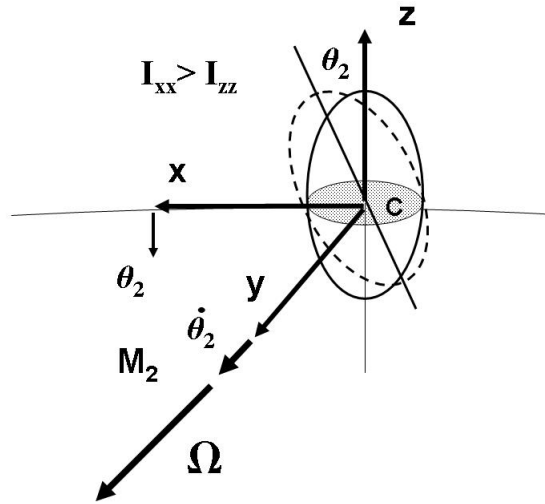
terms in  $\theta_1, \theta_2, \theta_3$ , we obtain the stability equations for an orbiting body in a gravitational field.

$$I_{xx}\ddot{\theta}_1 + \Omega(I_{xx} + I_{zz} - I_{yy})\dot{\theta}_3 + 4\Omega^2(I_{yy} - I_{zz})\theta_1 = 0 \quad (17)$$

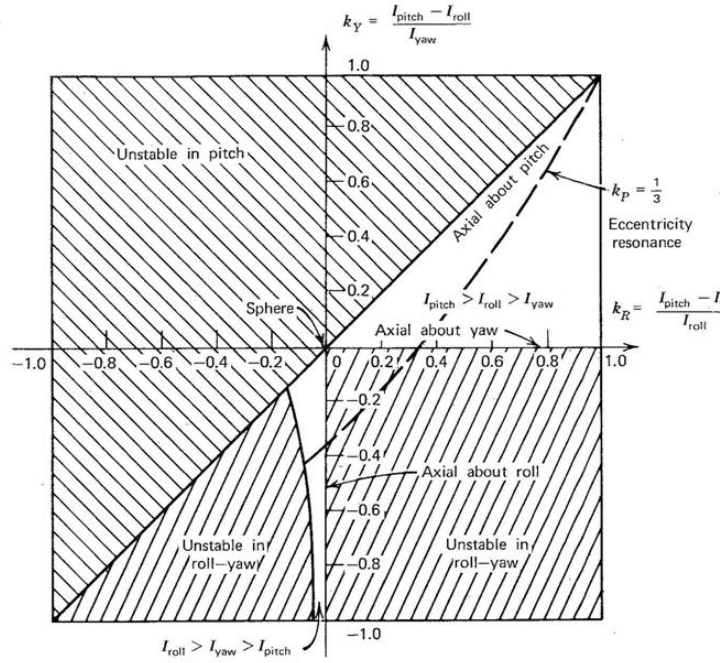
$$I_{yy}\ddot{\theta}_2 + 3\Omega^2(I_{xx} - I_{zz})\theta_2 = 0 \quad (18)$$

$$I_{zz}\ddot{\theta}_3 + \Omega(-I_{xx} + I_{yy} - I_{zz})\dot{\theta}_1 + \Omega^2(I_{yy} - I_{xx})\theta_3 = 0 \quad (19)$$

The pitch motion,  $\theta_2$  normal to the orbit plane, is uncoupled from the other motions; it is stable if  $I_{xx} > I_{zz}$ . The body oscillates with a frequency  $\Omega\sqrt{\frac{3(I_{xx}-I_{zz})}{I_{yy}}}$ , behaving as a pendulum in the non-uniform gravitational field.



The stability of the roll-yaw motion,  $\theta_1, \theta_3$  places rather severe and complex requirements on the ratios of  $(I_{yy} - I_{xx})/I_{zz}$  and  $(I_{yy} - I_{zz})/I_{yy}$ ; the results of this stability analysis are discussed in Kaplan, where intricacies of the stability figure shown below are discussed.



Gravity-gradient stability regions.

## Spin Stabilization of Attitude Against Gravity Gradient Torque

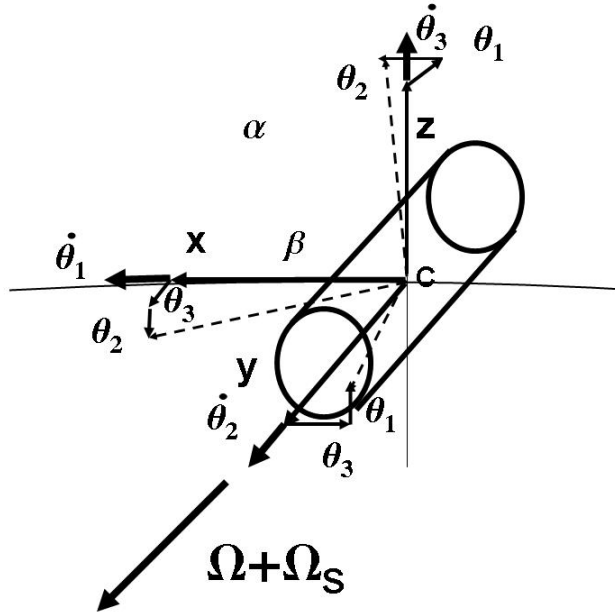
Another application of this approach is to investigate the spin-stabilization of the satellite spinning about the  $y$  axis in the non-uniform gravity-gradient field; the spin axis  $y$  is normal to the orbital plane; the spin angular velocity is  $\Omega_S$ . In this analysis, an asymmetric body was assumed, for which  $I_{xx} = I_{zz}$ . We use the body-fixed coordinate system previously introduced for which  $\boldsymbol{\omega} = \{\omega_x, \omega_y, \omega_z\} = \{\dot{\theta}_1 + \theta_3\Omega, \dot{\theta}_2 + \Omega, \dot{\theta}_3 - \theta_1\Omega\}$  where  $\Omega$  is the angular velocity due to orbital rotation about the  $j$  axis. The angular velocity due to the spin of the body adds to the angular momentum in the  $j$  direction. Since  $\Omega_S$  adds **only** to the angular velocity in the  $j$  direction, it is not necessary to rotate the body-fixed axis about  $j$  with this additional angular velocity; we do however include  $\Omega_S$  in the angular momentum. The angular momentum in this coordinate system is

$$\mathbf{H} = \{I_{xx}\omega_1, I_{yy}(\omega_2 + \Omega_S), I_{zz}\omega_3\}. \quad (20)$$

where  $\Omega_S$  is the angular spin velocity of the satellite in the  $j$  direction, normal to the orbital plane. The governing Euler equations become

$$\mathbf{M} = \dot{\mathbf{H}} + \boldsymbol{\omega} \times \mathbf{H} \quad (21)$$

where  $\mathbf{M} = -3\Omega^2(I_{yy} - I_{zz})\theta_1\mathbf{i}$ .



When the components of Euler's equations are expanded and only terms linear in  $\theta_1, \theta_2, \theta_3$  are kept, we obtain the stability equations.

(22)

$$I_{zz}\ddot{\theta}_1 - \Omega(I_{yy} - 2I_{zz})\dot{\theta}_3 - I_{yy}\Omega_S\dot{\theta}_3 + 4\Omega^2(I_{yy} - I_{zz})\theta_1 + I_{yy}\Omega\Omega_S\theta_1 = 0 \quad (23)$$

$$I_{yy}\ddot{\theta}_2 = 0 \quad (24)$$

$$I_{zz}\ddot{\theta}_3 + \Omega(I_{yy} - I_{zz})\dot{\theta}_1 + \Omega_S I_{yy}\dot{\theta}_1 + \Omega^2(I_{yy} - I_{zz})\theta_3 + \Omega\Omega_S I_{yy}\theta_3 = 0 \quad (25)$$

The angular motion normal to the orbit plane,  $\theta_2$  is uncoupled from the motions in  $\theta_1$  and  $\theta_3$ . With the assumption of  $I_{xx} = I_{zz}$  the gravitational gradient restoring moment about  $\mathbf{j}$  vanishes and the angular velocity remains constant. There are two independent parameters in these equations that determine the stability of the satellite spinning about the  $y$  axis in the gravity gradient field; these are the ratio of the satellite spin to the orbital angular velocity,  $\bar{\Omega} = \Omega_S/\Omega$ ; and the ratio of the moments of inertia  $R = I_{yy}/I_{zz}$ .

### Details of Stability Analysis

To investigate the linear stability analysis of this system, we assume small oscillations of the form  $\theta_i = \Theta_i \exp^{\alpha t}$ , for  $i = 1, 2, 3$ . Since the motion of  $\theta_2$  is uncoupled and stable, we will not further consider it. The remaining equations are coupled linear equations between  $\theta_1$  and  $\theta_3$ . With the assumed form of solution, we obtain

$$(I_{zz}\alpha^2 + 4(I_{yy} - I_{zz})\Omega^2 + I_{yy}\Omega_S\Omega)\Theta_1 - (-2I_{zz}\Omega + I_{yy}\Omega + I_{yy}\Omega_S)\alpha\Theta_3 = 0 \quad (26)$$

$$(I_{zz}\alpha^2 + (I_{yy} - I_{zz})\Omega^2 + I_{yy}(\Omega + \Omega_S)\Omega)\Theta_3 + (I_{yy}(\Omega + \Omega_S) - 2I_{zz}\Omega)\alpha\Theta_1 = 0 \quad (27)$$

This leads to a quartic equation for  $\alpha$ . If  $\alpha$  is real and positive, the system is unstable: if  $\alpha$  is imaginary, the system is stable. Following Thomson, the quartic equation can be written in terms of  $R$  and  $\bar{\Omega}$ .

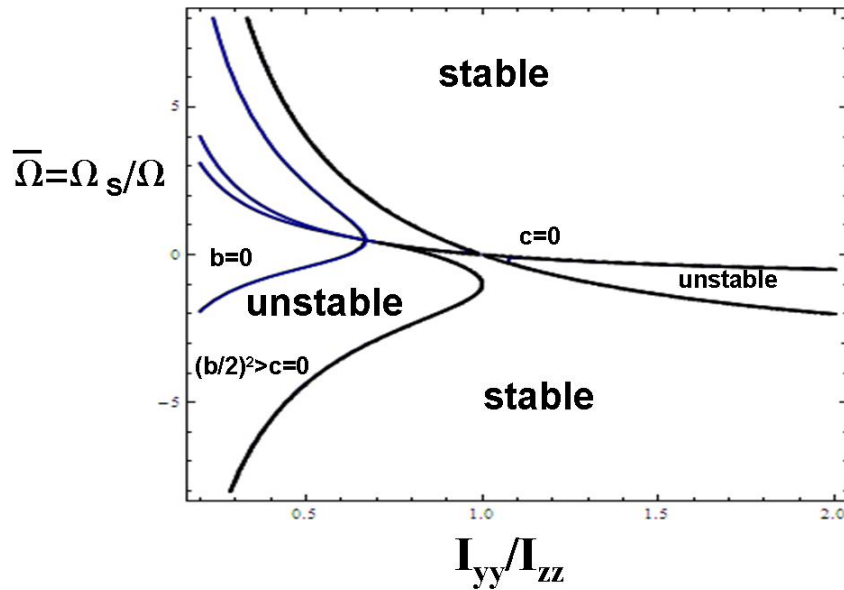
$$\bar{\alpha}^4 - b\bar{\alpha}^2 + c = 0 \quad (28)$$

where  $\bar{\alpha} = \alpha/\Omega$ ,  $R = I_{yy}/I_{zz}$  and  $\bar{\Omega} = \Omega_s/\Omega$ . The coefficients  $b$  and  $c$  are

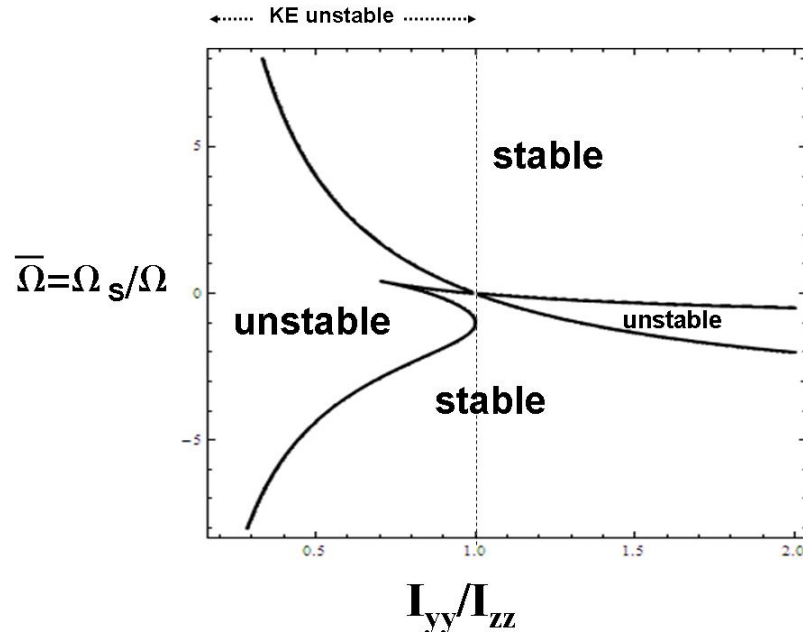
$$b = 1 - R + 2\bar{\Omega}R - R^2 - 2\bar{\Omega}R^2 - \bar{\Omega}^2R^2 \quad (29)$$

$$c = 4 - 8R - 5\bar{\Omega}R + 4R^2 + 5\bar{\Omega}R^2 + \bar{\Omega}^2R^2 \quad (30)$$

As Thomson (1962) points out, in order to have no positive real solutions for  $\alpha$ , several things must be true about  $b$  and  $c$ . In particular:  $b/2$  must be negative;  $c$  must be positive; and  $(b/2)^2 > c$ . Numerical satisfaction of these condition is shown in the figure; stability for the system requires that system parameters remain outside of all of these bounding curves, as shown.



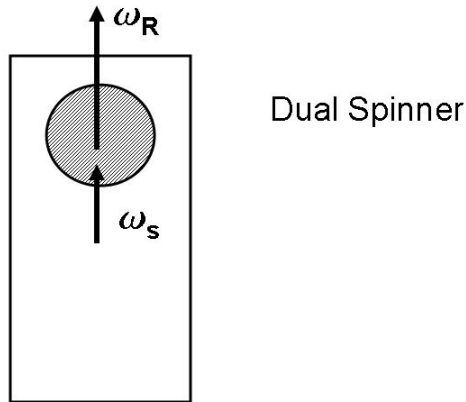
The final stability regions are shown below: the regions where all required conditions are met. The value  $\bar{\Omega} = -1$  corresponds to a satellite with no spin in an inertial frame. The results show that an inertial spin of magnitude  $|\bar{\Omega}| > 1$  will allow the stability of a satellite with  $I_{yy}/I_{zz} < 1$ . However, dissipation can lead to loss of kinetic energy, resulting in instability for  $I_{yy}/I_{zz} < 1$  over time.



## Dual Spinner

One of the important results for Lecture 25 was the concept of a gyrostat. In its simplest form, it is a body which contains within it another body which is spinning at a different angular velocity. In its simplest form, both bodies would be asymmetric (the moments of inertia about axes normal to the spin axis would be equal) and spinning about the same principal axis. With this simplification it was shown that the "effective" moment of inertia of the body about the axis of spin of the second body can be increased by the "spinning" of the second body. Thus in a situation where the desired spin axis of the first body might be at the minimum moment of inertia, and thus unstable due to energy dissipation, the presence of the second body, usually spinning at a much faster rate, can create a situation where the desired axis of spin becomes stable, becoming effectively the axis of maximum "effective" moment of inertia. Satellites which employ this principle are called dual spinners.

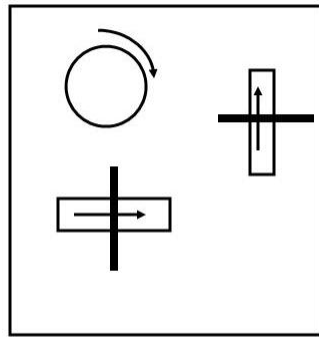




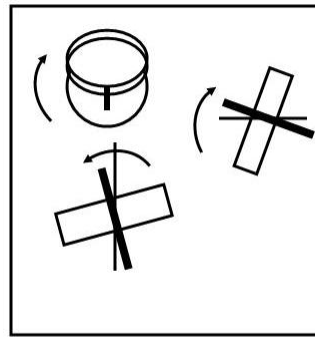
An example of the use of such a satellite consider a satellite whose axis is normal to the orbital plane. In order to keep the satellite pointed at the earth, a rotation rate of  $\Omega$ , equal to the rotation rate of the satellite in Earth orbit, is required. If the satellite rotates about its minimum moment of inertia, the satellite would be unstable. Also, a satellite in this orientation would be unstable to gravity gradient moments. However, if the satellite is a gyrostat, dual spinner, containing a rotating device which increases the "effective" moment of inertia about the axis normal to the orbital plane, the satellite will become stable. If this technique were accomplished with a single rotating body, the satellite would not be able to keep its orientation pointing at the Earth. But with a dual spinner, part of the satellite can remain pointing at the Earth, while the "spinner" provide stability.

## Satellite Attitude Control

We have concentrated thus far on the natural stability of the satellite. Given the scarcity of power and mass in a typical satellite application, the basic design should be stable to minimize resources necessary for control. However, the pointing accuracy required of a satellite cannot typically be managed by passive stability. Thus other mechanisms are required to fine-tune the attitude of the satellite. For the most part, this is accomplished by thrusting devices or momentum control devices, or a combination of these. We shall not review these device in detail other than to point out that momentum control devices fit within the framework of angular momentum of three-dimensional rigid body motion.



**Momentum Wheels**



**Control Moment Gyros**

The simplest of these is the momentum wheel. A momentum wheel is a rotating mass fixed to a shaft which is fixed to the satellite body. Imagine that three momentum wheels are located on the three principal axes of the vehicle; the wheels are controlled by torque motors attached to the frame of the vehicle. Since the system is isolated, no net moment is created when the wheel is turned by action of the motor. Thus, conserving angular momentum, a turning of the wheel turns the vehicle in the opposite direction by an amount proportional to the moments of inertias of the satellite and the wheel. A more complex device is the control moment gyro. In this configuration a wheel spins about an axis. When a change in angular momentum of the vehicle is required, rather than increasing the speed of the wheel with the axis fixed to the vehicle, the wheel spin is constant but the axis of the wheel is rotated, transferring angular momentum to the vehicle, resulting in a change in attitude. There is also a combined device where both the angular velocity and direction of rotation are changed. We shall not discuss the strengths and weakness of each approach; our purpose is to place these devices within the framework of angular momentum conservation for three-dimensional rigid bodies.

#### **ADDITIONAL READING**

W. T. Thompson, *Introduction to Space Dynamics*,

J. P. C. Hughes, *Spacecraft Attitude Dynamics*, Dover publications, 2004

F. P. J. Rimrott, *Introductory Attitude Dynamics*, Springer-Verlag, 1989

M. H. Kaplan, *Modern Spacecraft Dynamics and Control*, Wiley, 1976

W. T. Thomson, *Spin Stabilization of Attitude Against Gravity Torque*, *Jour. Astronautical Sciences*, 1964